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Van Allen et al.

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(54) **BIOMARKERS PREDICTIVE OF
ANTI-IMMUNE CHECKPOINT RESPONSE**

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2018, now Pat. No. 11,377,697.

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11, 2017.

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G01N 33/574 (2006.01)

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CPC **C12Q 1/6886** (2013.01); **G01N 33/5011**
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2600/106 (2013.01); **C12Q 2600/156**
(2013.01); **C12Q 2600/158** (2013.01)

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USPC 424/174.1
See application file for complete search history.

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(57) **ABSTRACT**

The present invention is based on the identification of novel
biomarkers predictive of responsiveness to anti-immune
checkpoint therapies.

18 Claims, 22 Drawing Sheets

Specification includes a Sequence Listing.

Figure 1

Included samples:
Passed whole exome sequencing quality control

Cancer Type	Number of Pre-Treatment Tumor Samples	Source
Anal	1	DFCI (1)
Bladder	13	DFCI (9) DFCI CCPM (4)
HNSCC	14	DFCI (14)
Lung	67	DFCI SU2C (31) CANSEQ (1) Rizvi (34 – MAF only) PD-L1 (1)
Melanoma	176	BroadNext10 (5) Schadendorf (97) Rizwan Haq (2) Snyder (34 – MAF only) Hugo (34 – MAF only) Zaretsky (4)
Sarcoma	1	DFCI (1)
Total	272	

Figure 2

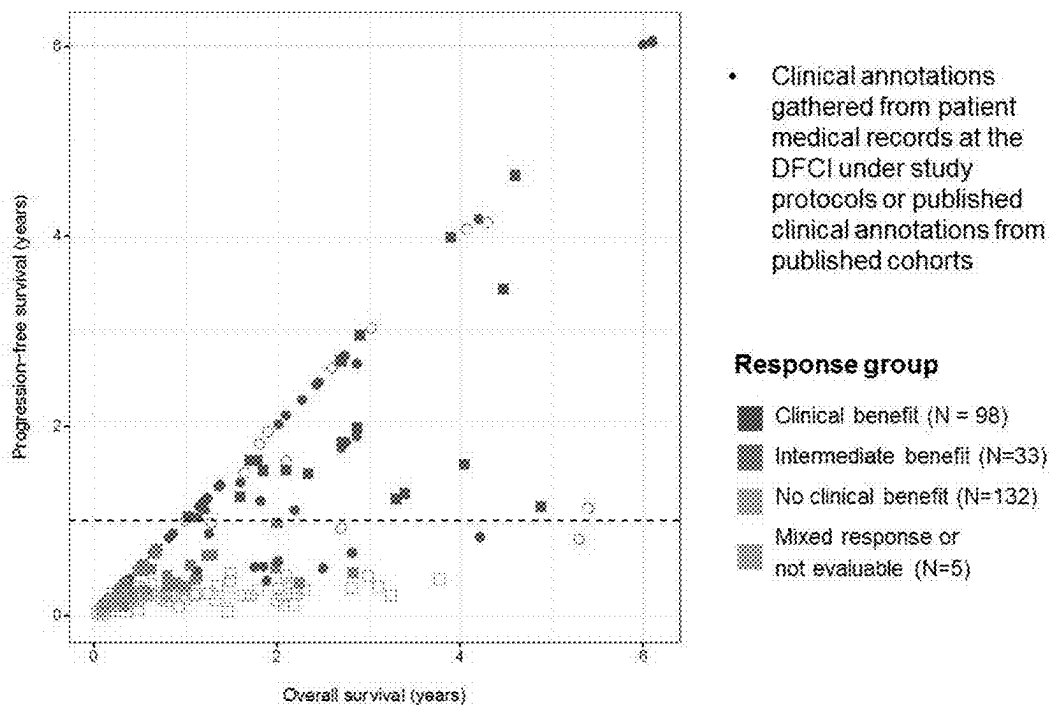
Quality Control

- **Exclusion Criteria:**
 - Limited contamination with non-patient DNA:
 - Contamination Estimation (ContEst) < 5%
 - Limited admixture between tumor and normal DNA:
 - CopyNumberQC < 5 mix-ups
 - Mean target coverage:
 - Tumor: <25x
 - Normal: <15x
 - Estimated purity (ABSOLUTE) <10%
- **Inclusion Criteria:**
 - At least 1 clonal driver mutation (Tamborero et al. 2013)
 - RCC-CA209009_5_73 included despite not having any clonal driver mutations
 - Sequencing from pre-treatment patient tumor: Excluded on-treatment and post-treatment tumors from Snyder et al. and Hugo et al.
 - Patient received immune checkpoint therapy without combination targeted therapy (e.g. vemurafenib, dacarbazine, carboplatin, etc.)
 - Patients receiving combination immune checkpoint therapy (e.g. anti-PD1 blockade with anti-CTLA4 blockade) were included

Figure 3

Clinical benefit from immune checkpoint therapy

N=268 patients *36 not shown due to incomplete survival data



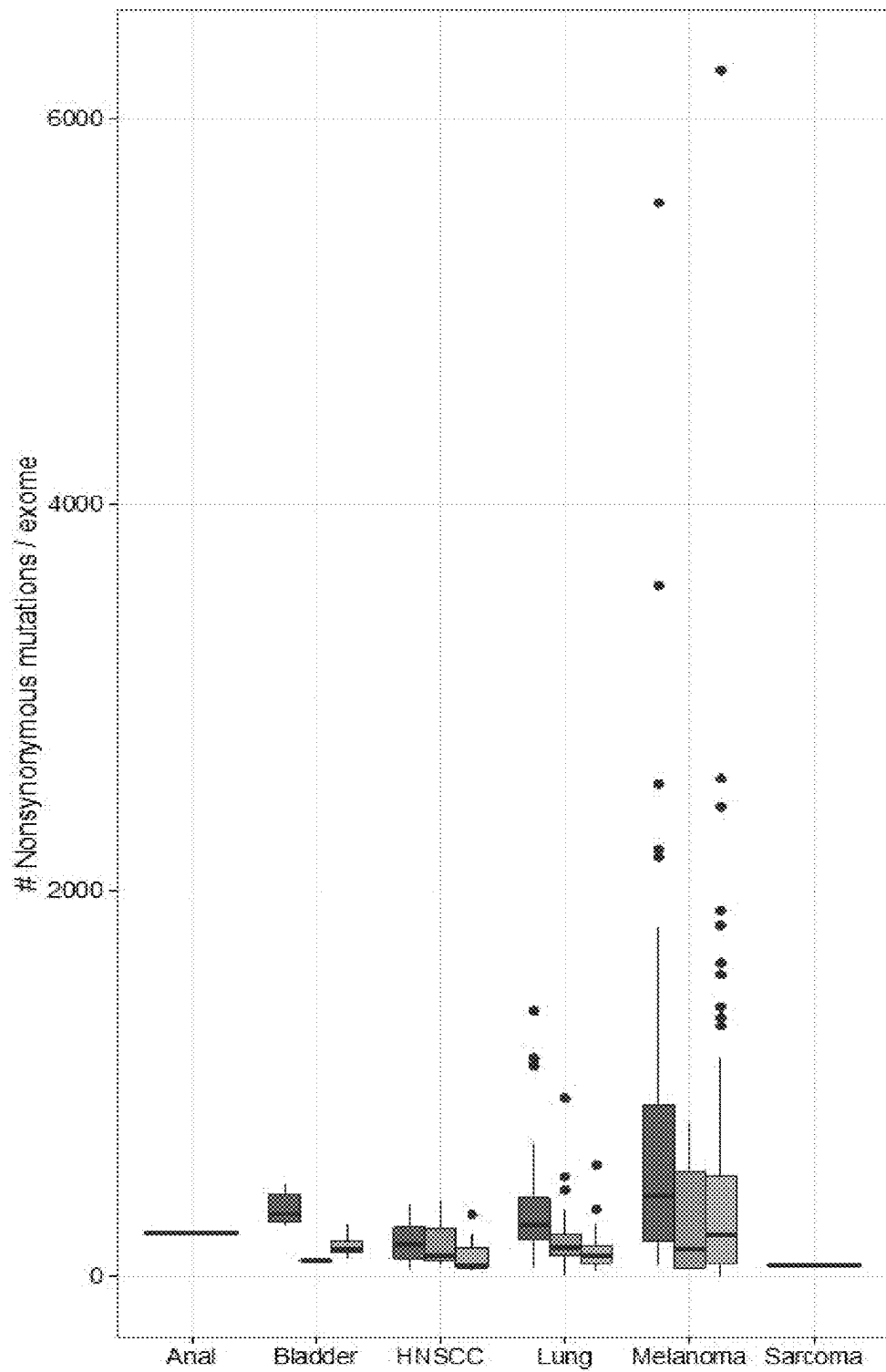


Figure 4

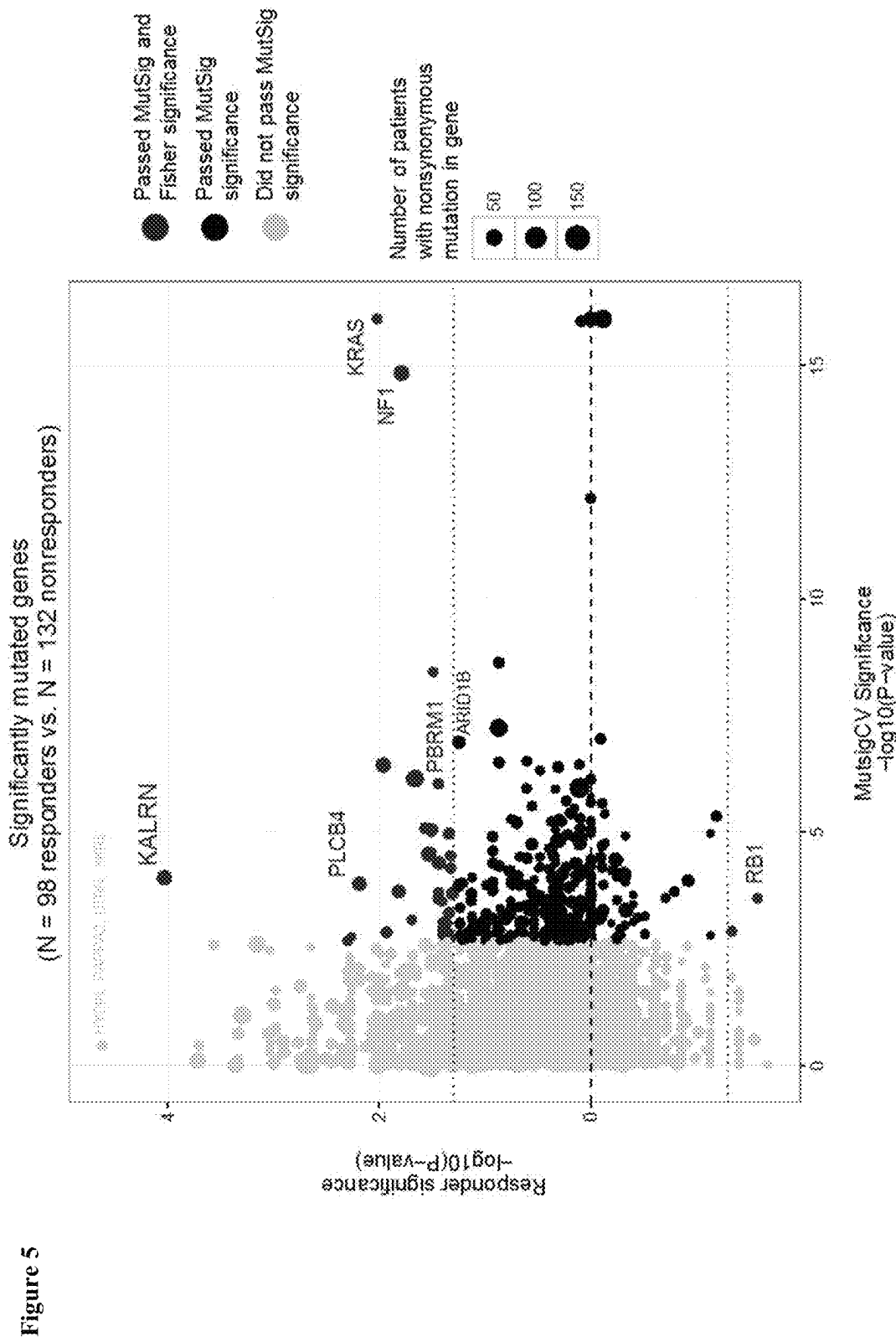
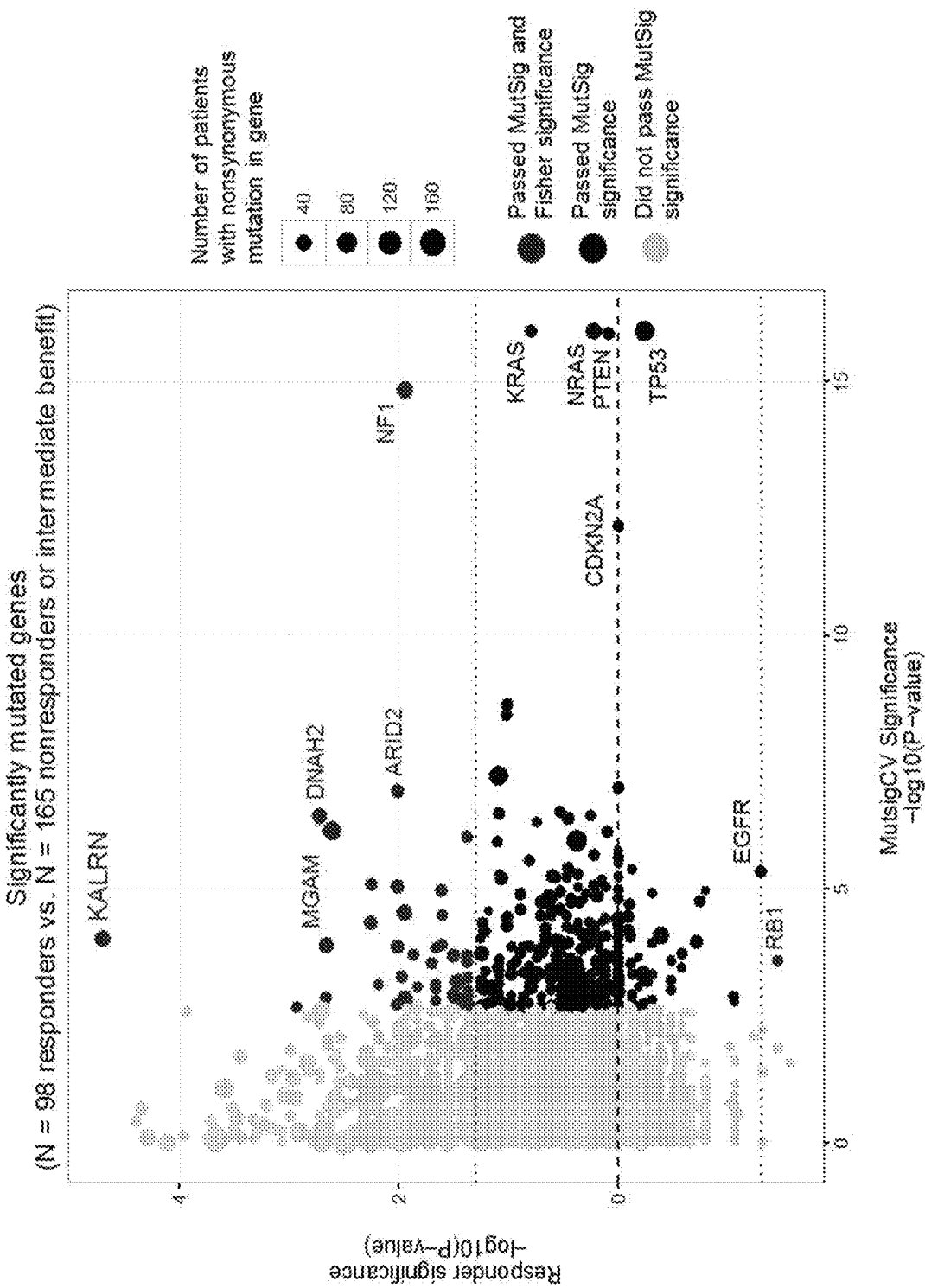


Figure 6



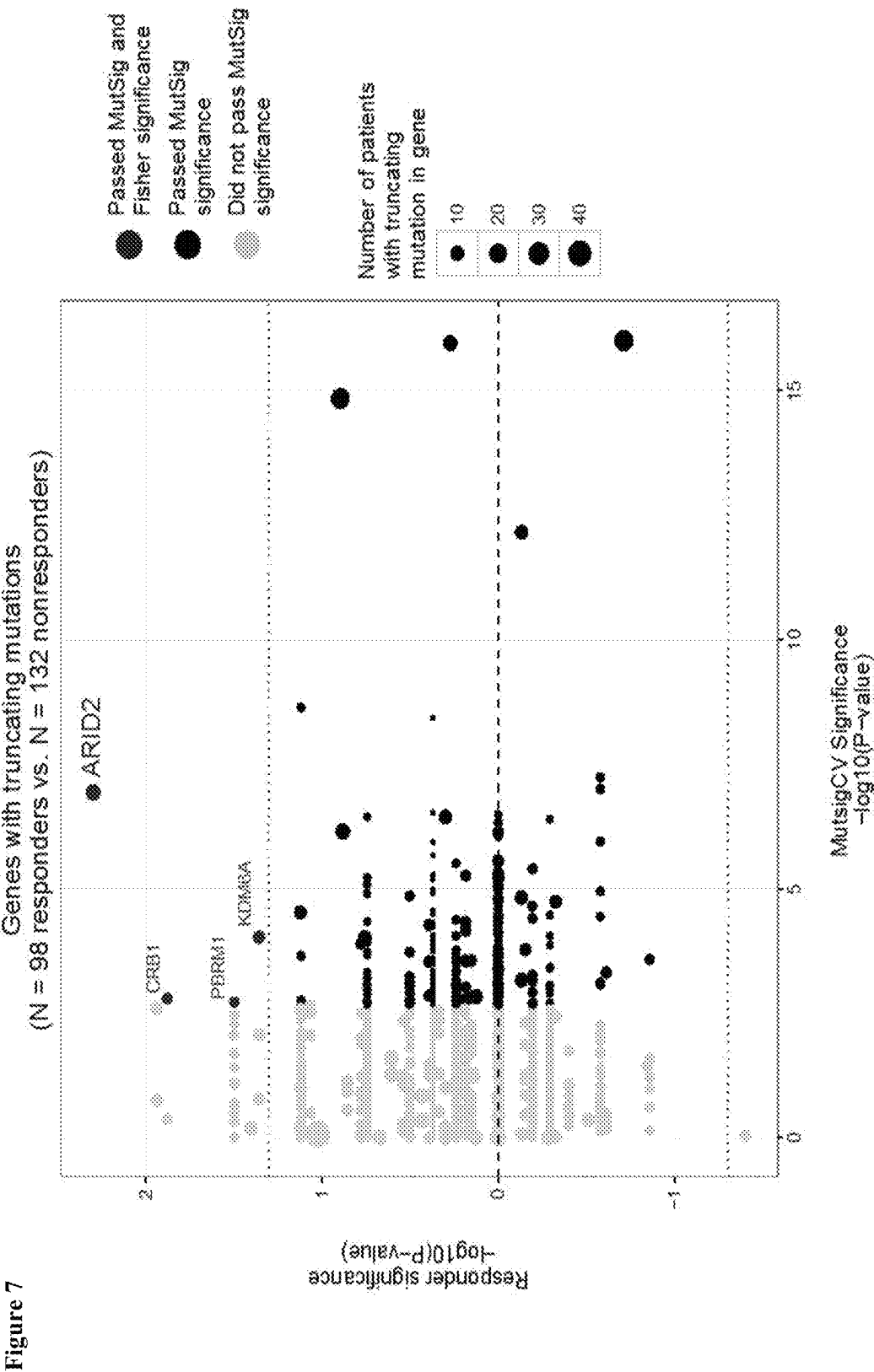


Figure 8

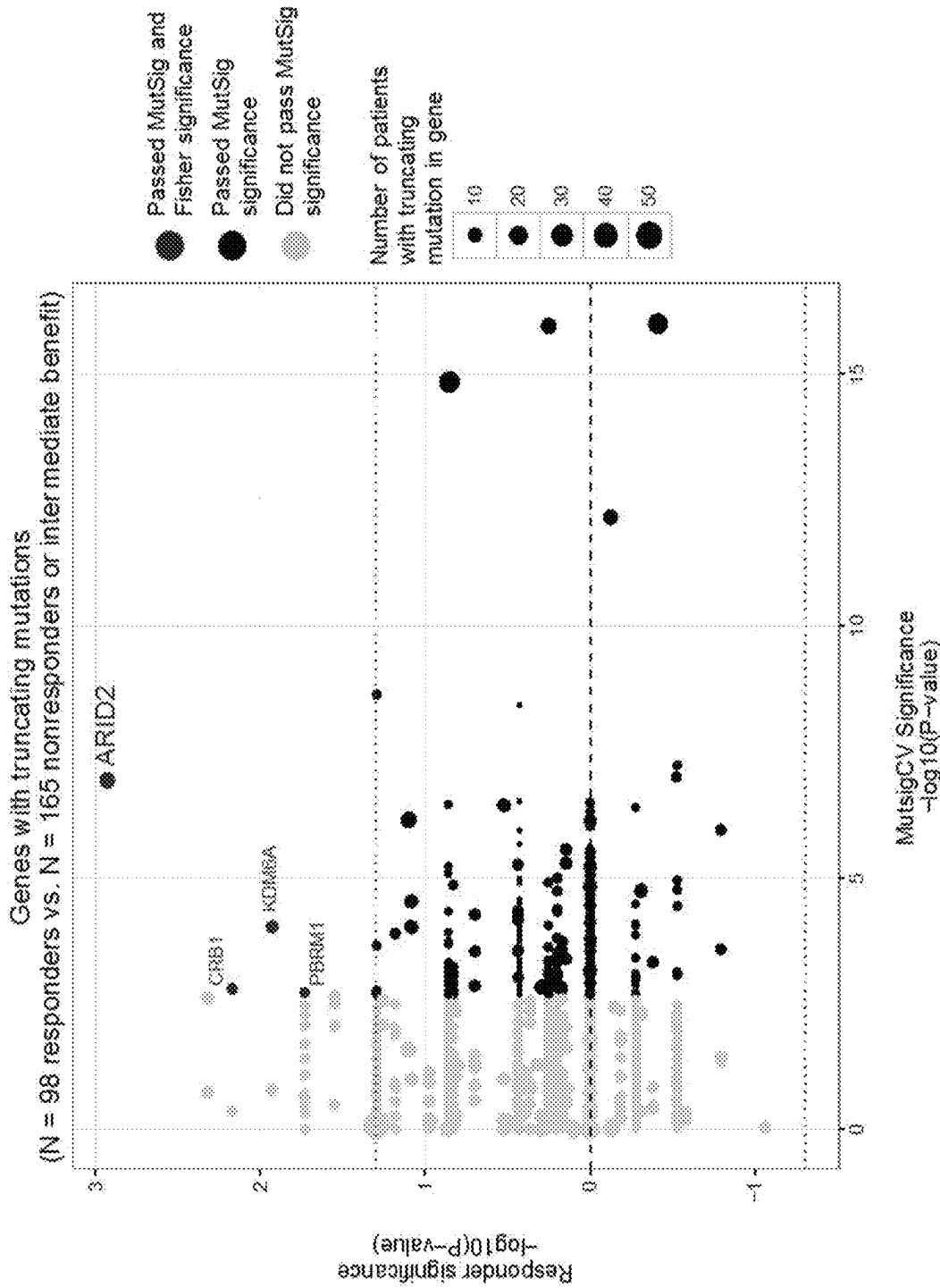


Figure 9

SWI/SNF Complex

- SWI/SNF complex: ATP-dependent chromatin remodeling complexes mutated in >20% of human cancers

- BAF:

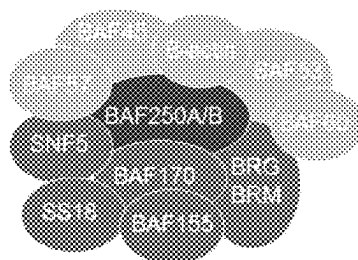
- BAF250A (ARID1A) • BAF190/BRM (SMARCA2)
- BAF250B (ARID1B) • BAF47 (SMARCB1)
- BAF57 (SMARCE1) • BAF53A (ACTL6A)

- PBAF:

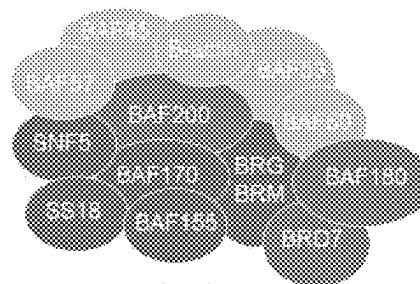
- **BAF200 (ARID2)**
- **BAF180 (PBRM1)**
- BRD7
- BAF45A (PHF10)

- Shared:

- BRG1/BAF190 (SMARCA4)
- BAF155 (SMARCC1)
- BAF170 (SMARCC2)



BAF



PBAF

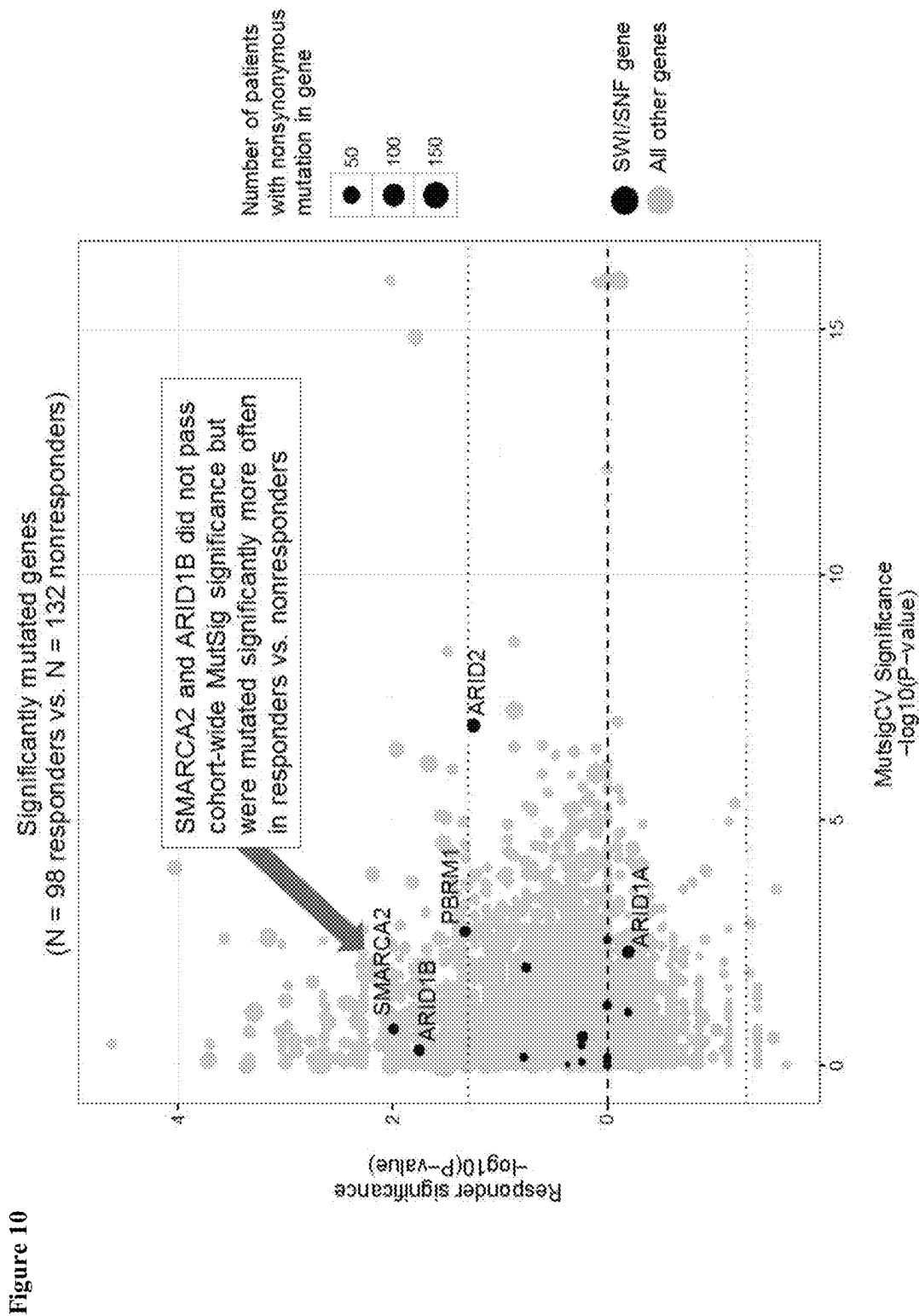


Figure 10

Figure 11

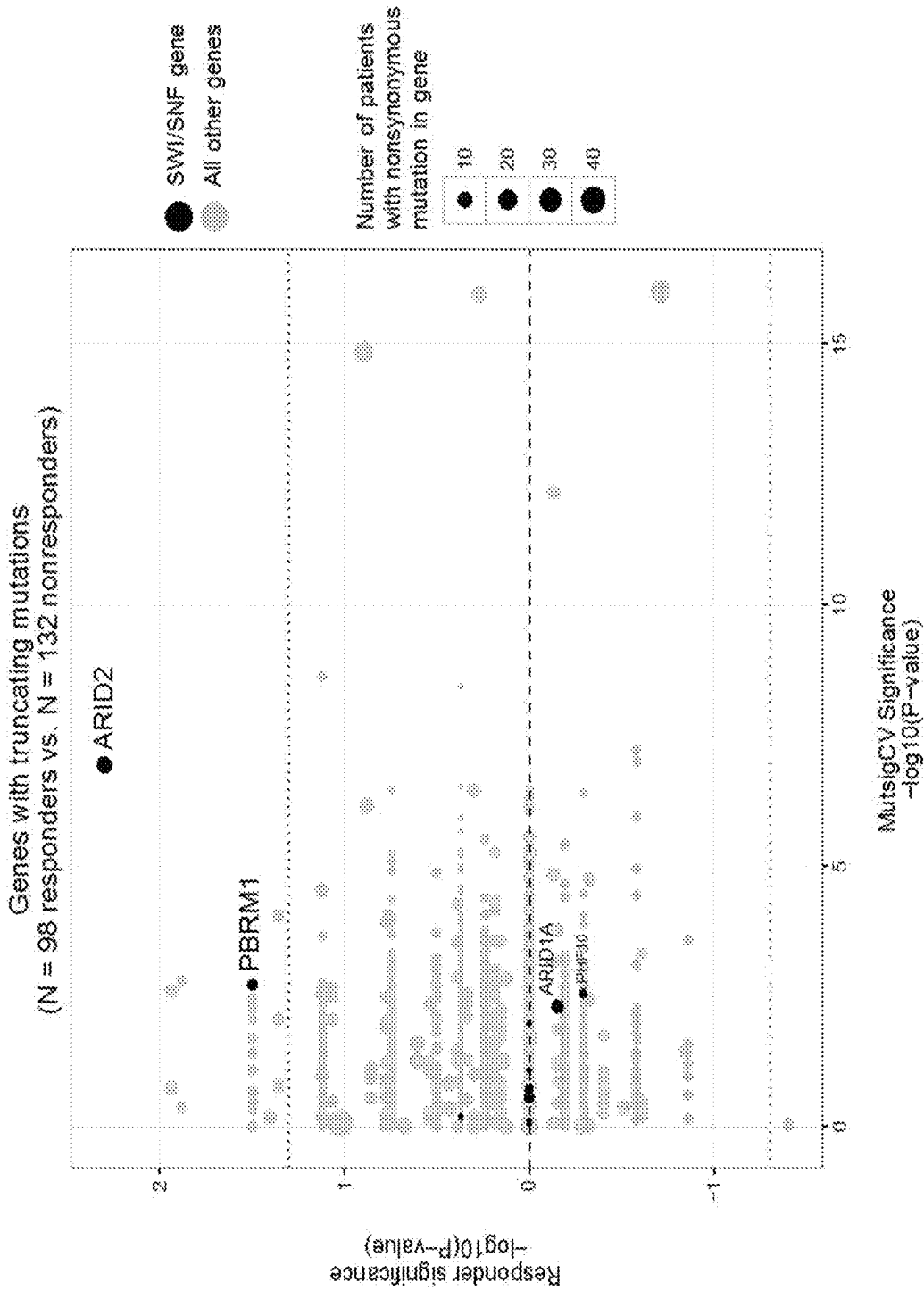


Figure 12

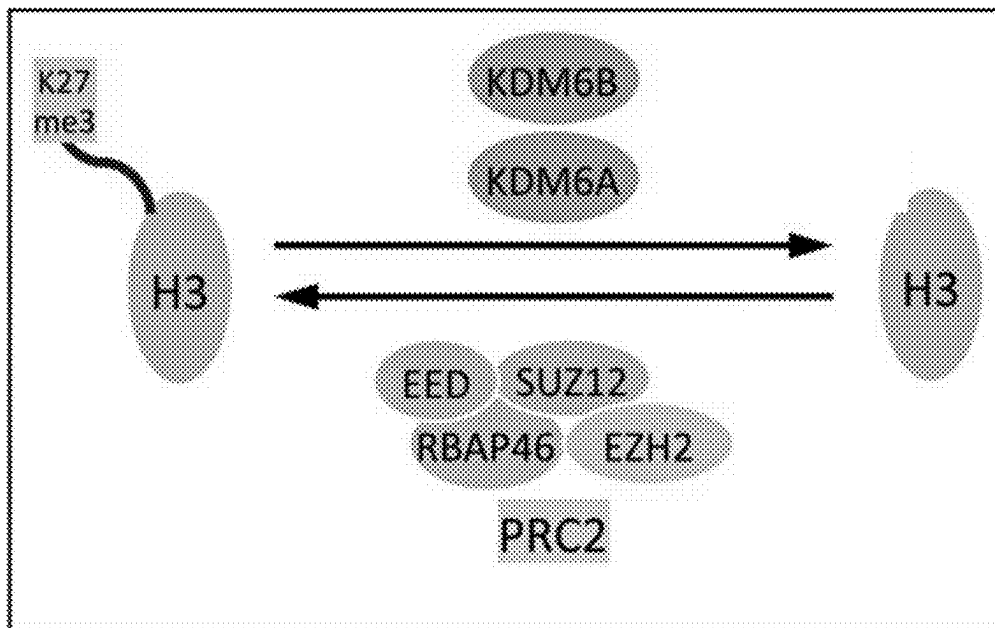
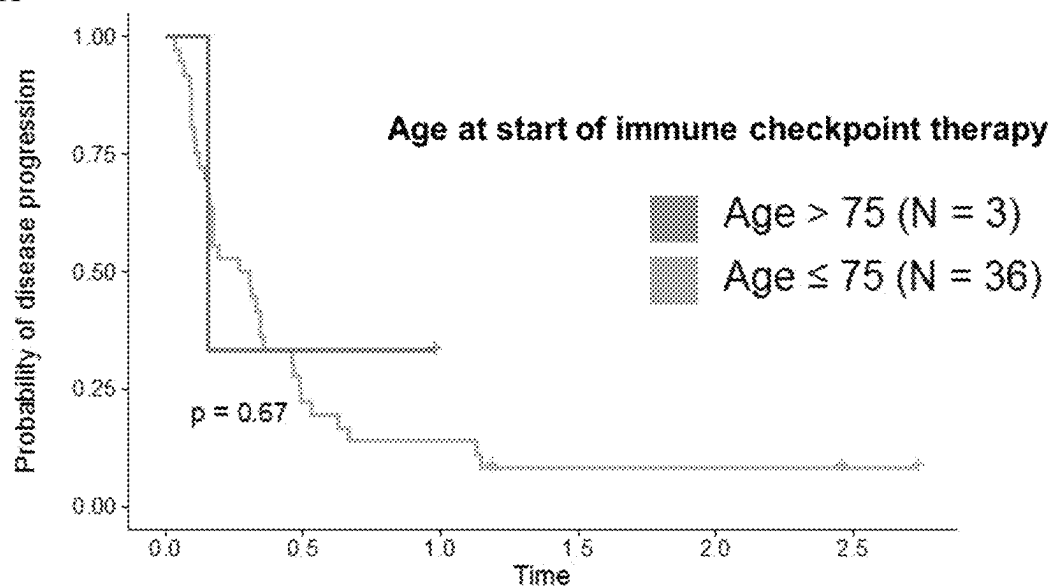


Figure 13

A



B

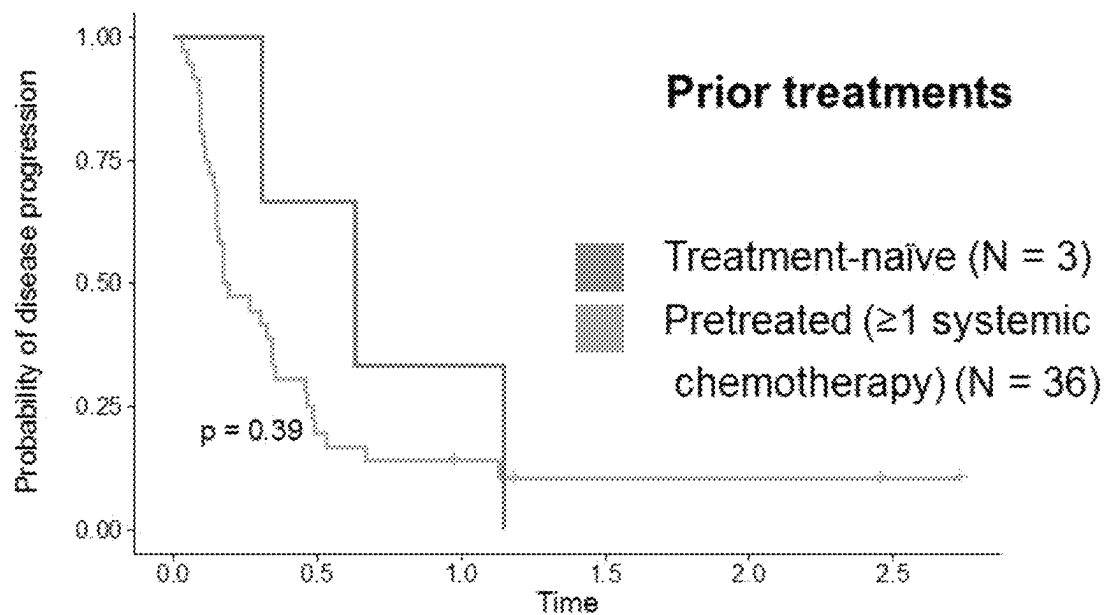
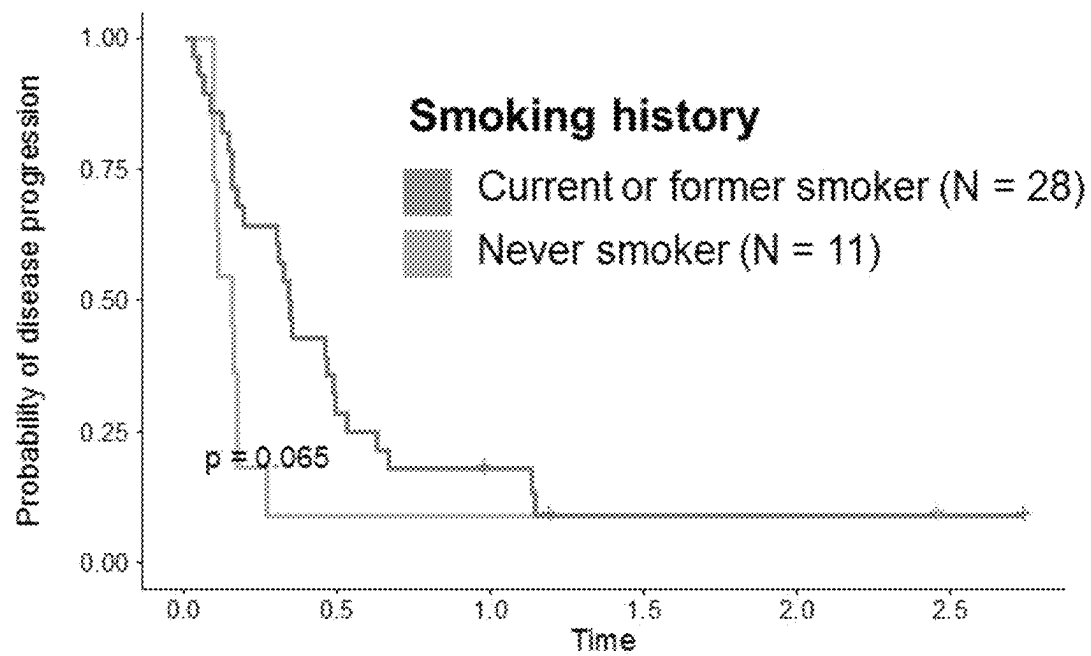


Figure 13 (cont.)

C



D

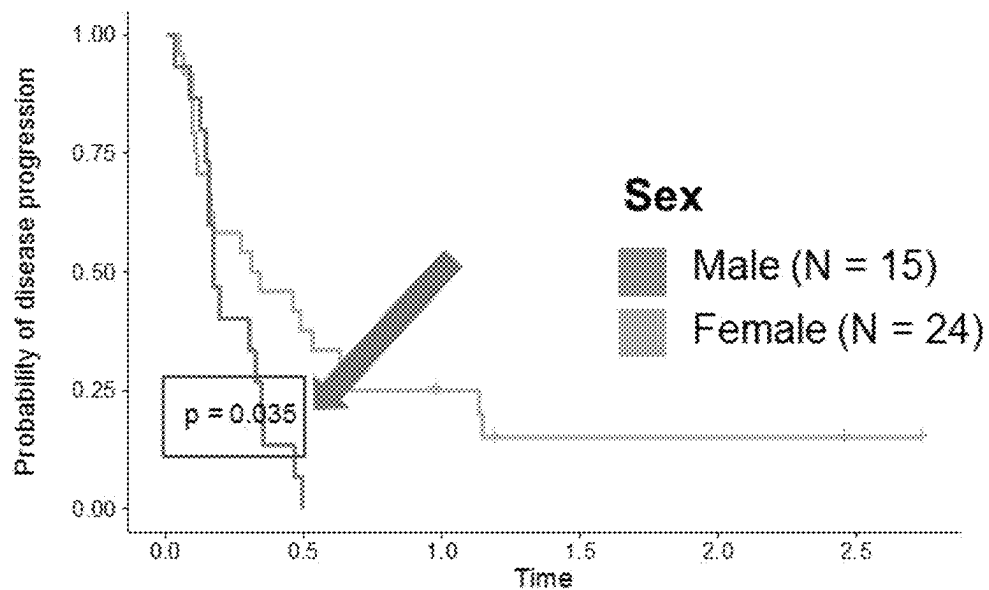


Figure 14

Quality Control

39 pre-treatment tumors underwent whole exome sequencing



ContEst: Evaluation for sample contamination with non-patient genomic material (<10%)

- 1 patient excluded: SU2C-1009 (SD)

CopyNumberQC: Evaluation for normal contamination in tumor sample (<5 mix-ups)

- 1 patient excluded: LUAD-BS-12-M17368 (PR)

Mean target coverage: Evaluation for adequate sequencing coverage to detect mutations (MTC > 30x for tumor and 15x for normal)

- 2 patients excluded: SU2C-1012 (PR), SU2C-1014 (PD)
 - 2nd pre-treatment tumor from LUAD-1011 also excluded

ABSOLUTE: Evaluation for adequate sample purity to detect somatic mutations (>10% tumor cells)

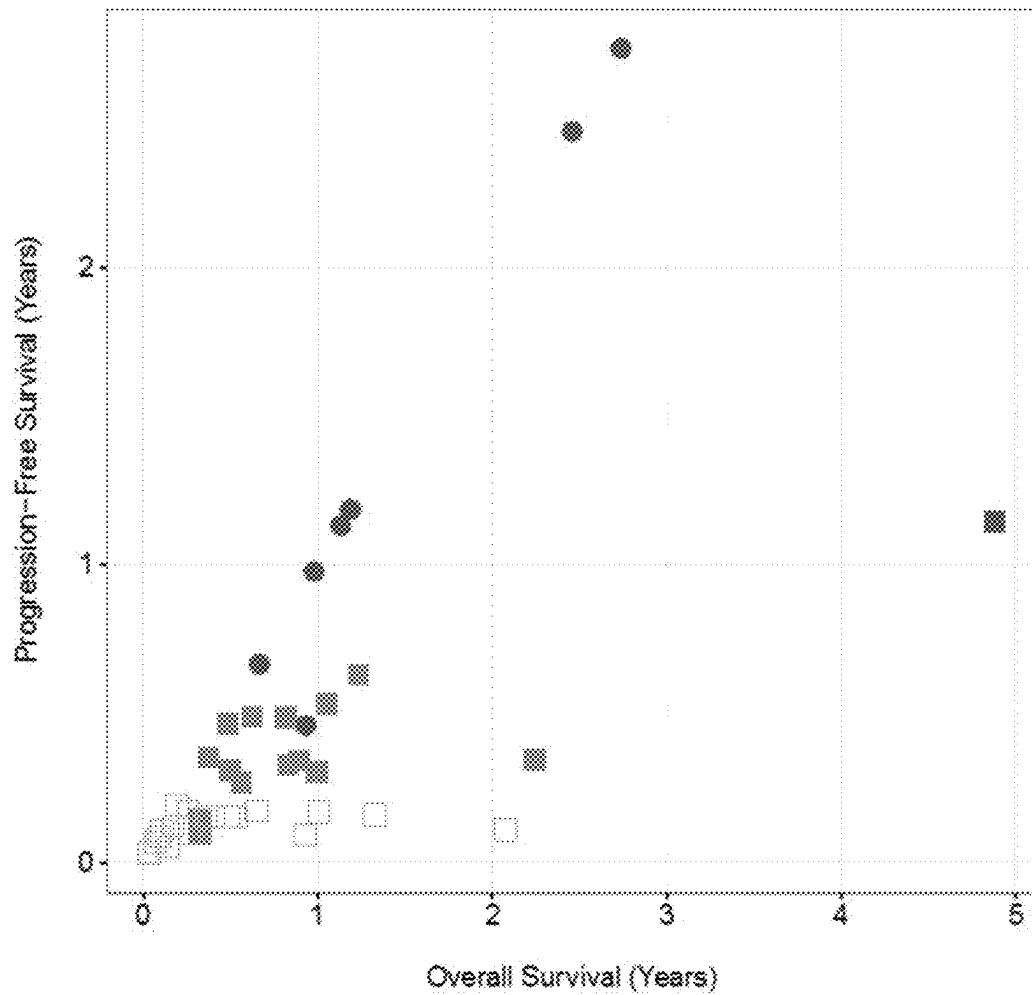
- 4 patients excluded: SU2C-1019 (PD), SU2C-1015 (PD), SU2C-1008 (PD), SU2C-1001 (SD)



31 pre-treatment tumors included in final analysis

- Plus 1 additional pre-treatment tumor from LUAD-1007 and 3 additional post-progression tumors from LUAD-1020

Figure 15

**Response group**

- Clinical benefit
- ▨ Stable disease
- ▩ No clinical benefit

Best RECIST

- PR
- SD
- PD

Figure 16

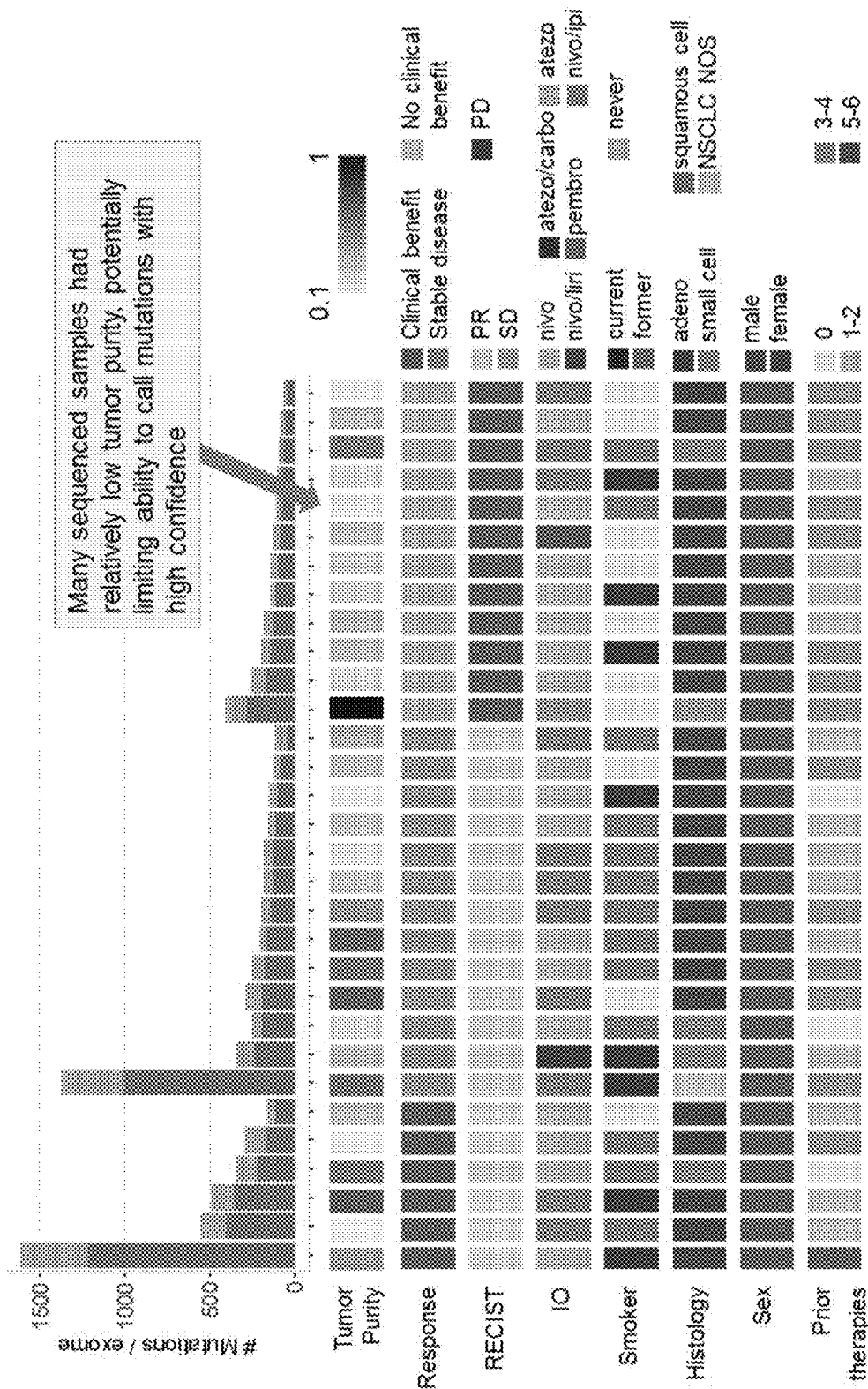
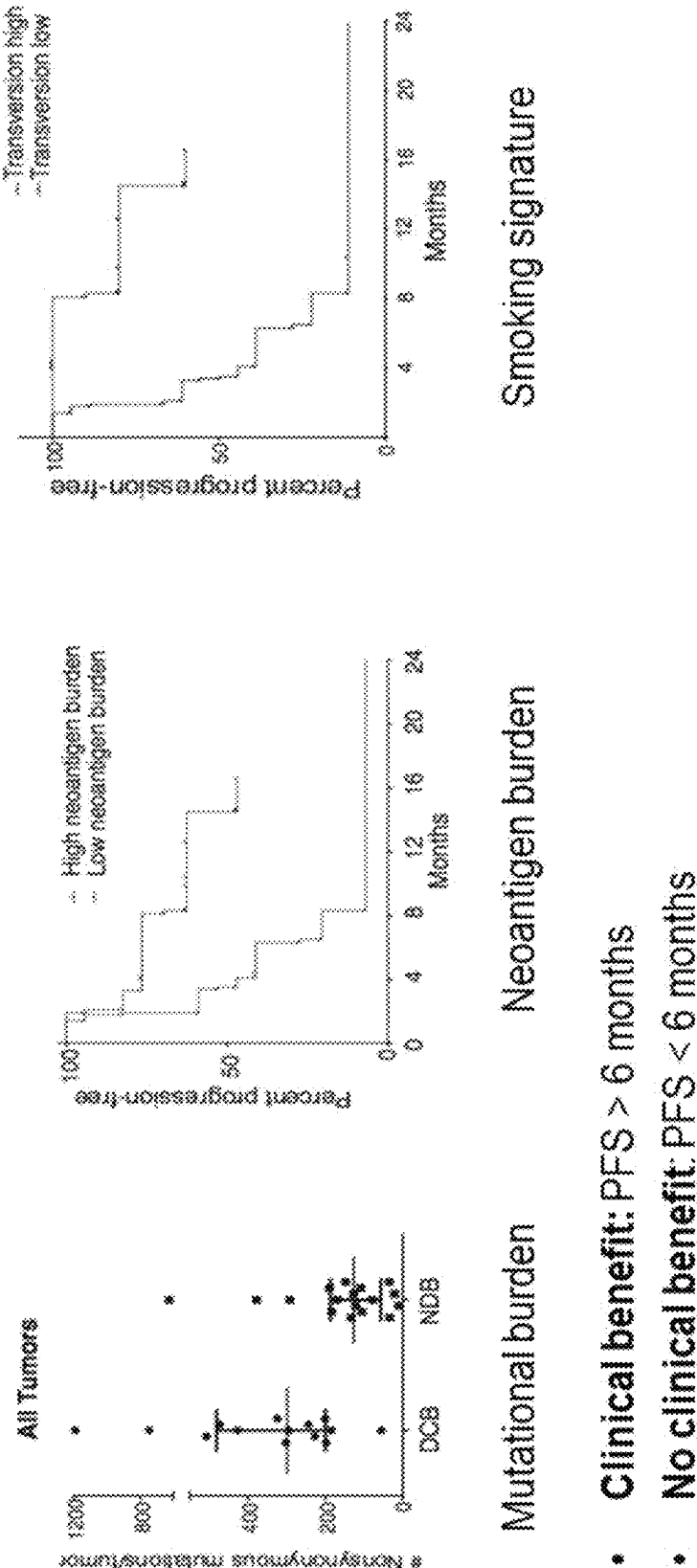
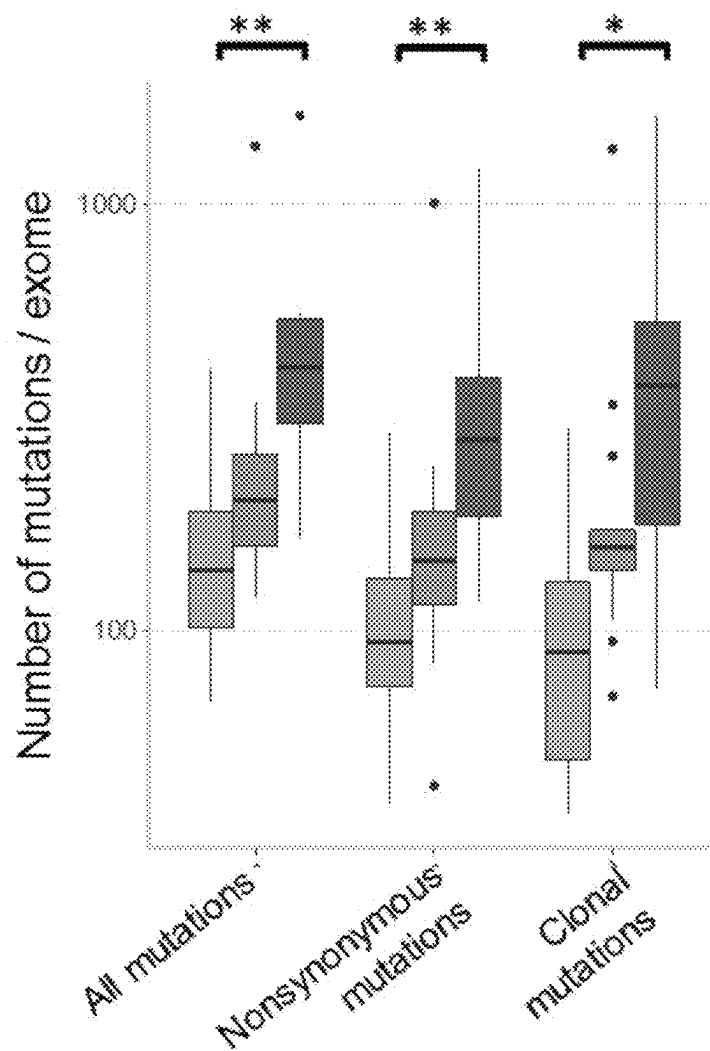


Figure 17



- **Clinical benefit:** PFS > 6 months
- **No clinical benefit:** PFS < 6 months

Figure 18

**Response to IO**

- Clinical benefit (N = 6)
- ▨ Stable disease (N = 13)
- ░ No clinical benefit (N = 12)

*p<0.05; **p <0.005
Wilcoxon rank-sum test

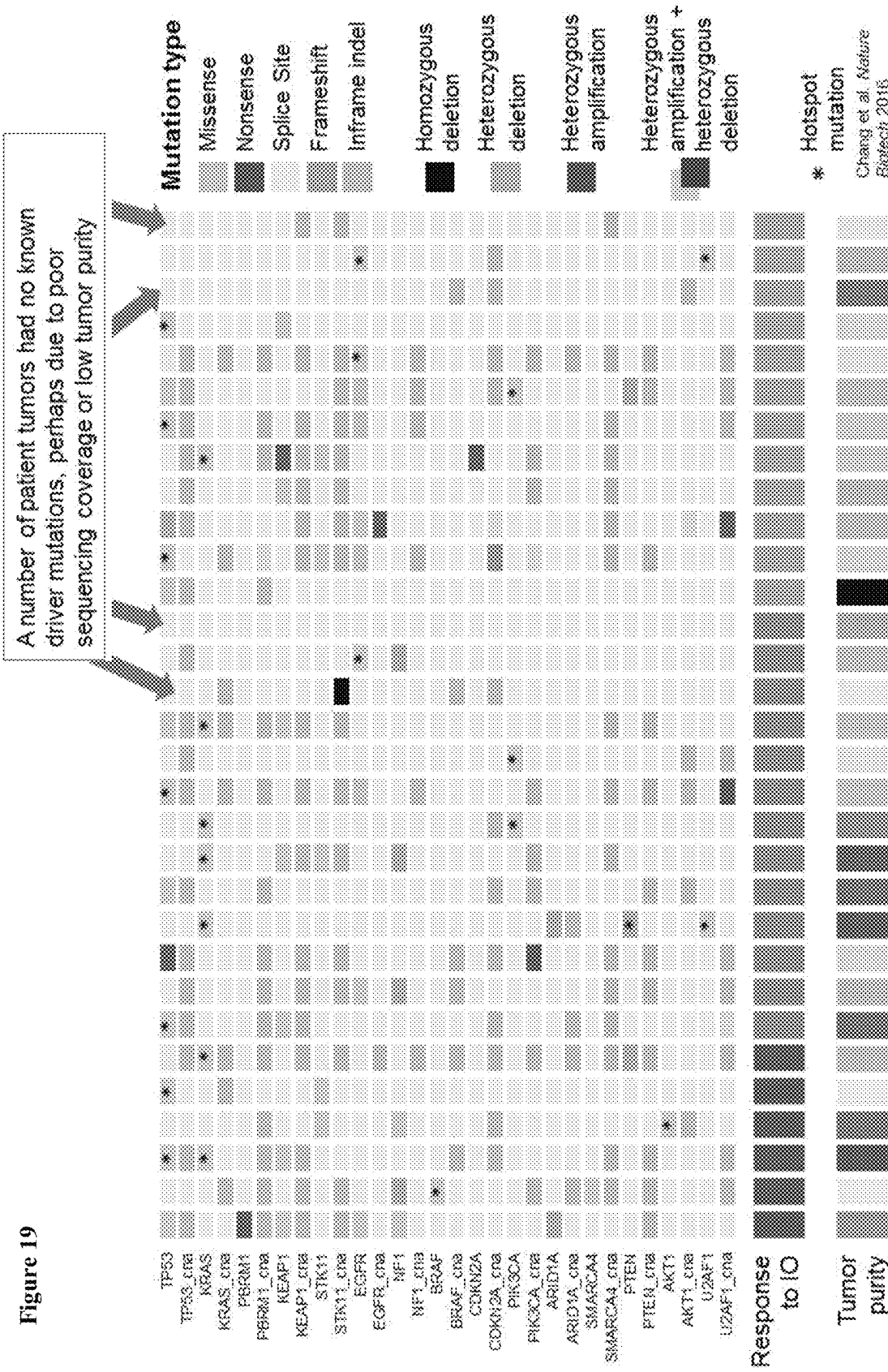


Figure 20

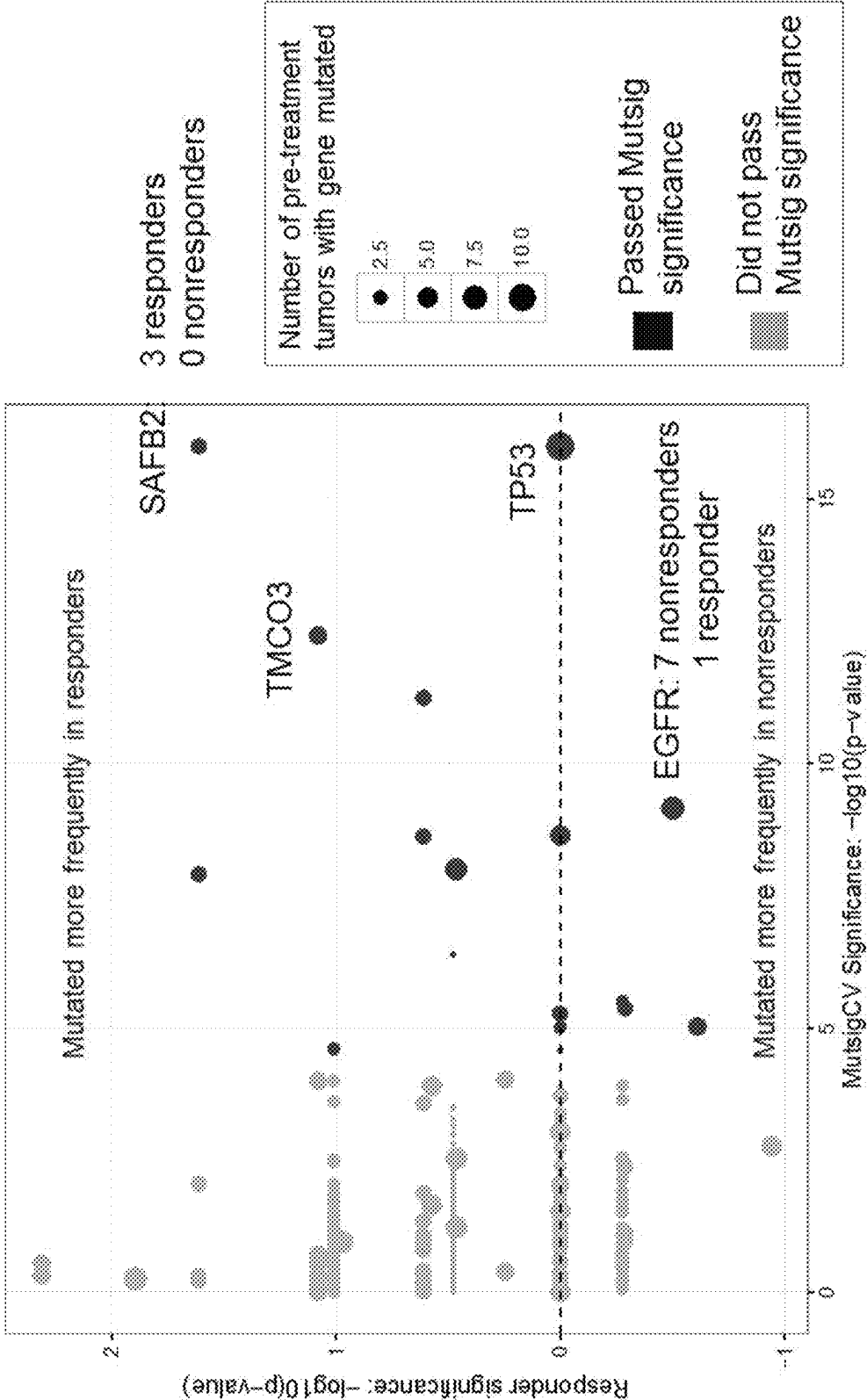
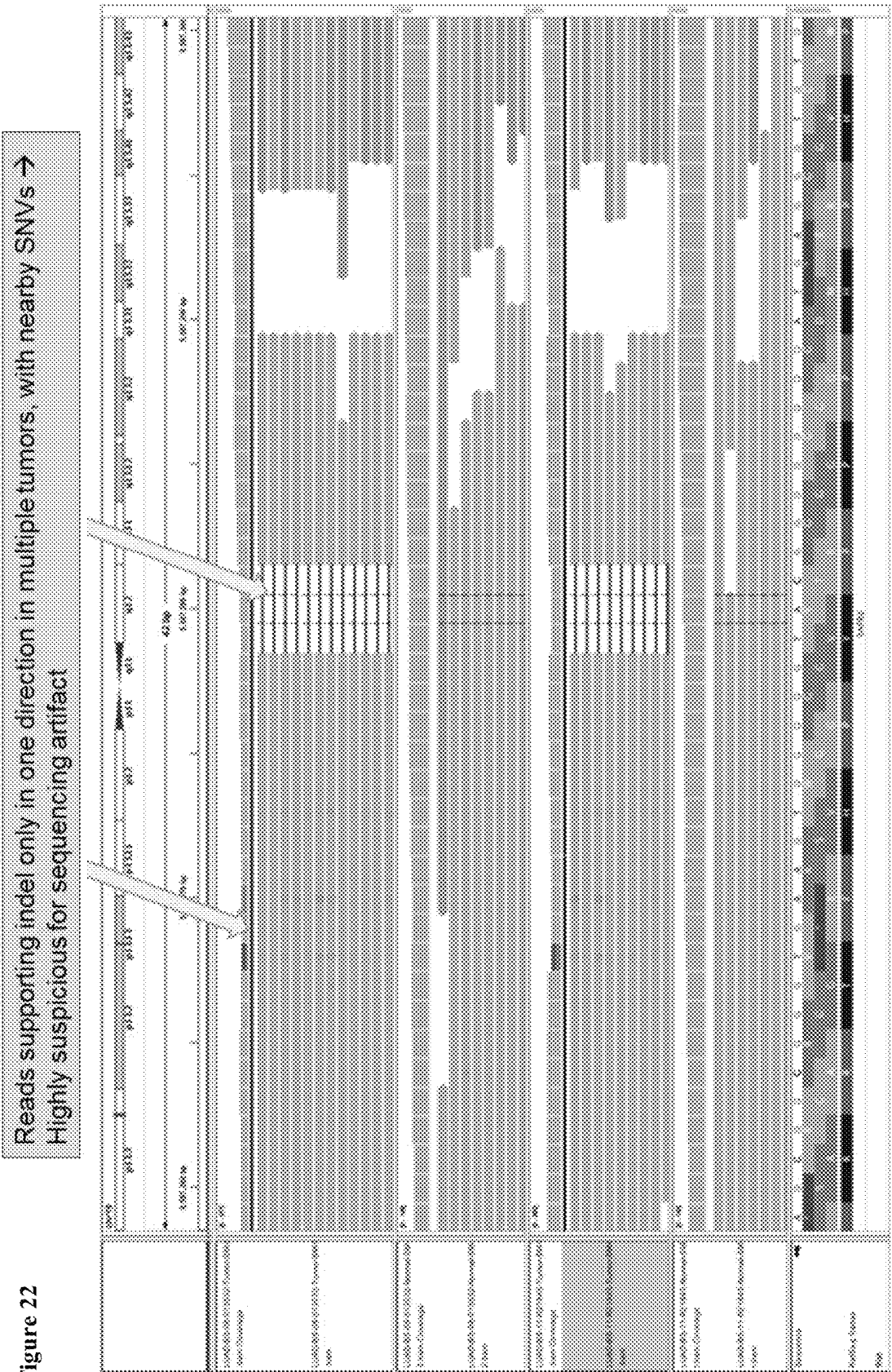


Figure 22



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BIOMARKERS PREDICTIVE OF ANTI-IMMUNE CHECKPOINT RESPONSE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 62/445,105, filed on 11 Jan. 2017; the entire contents of said application are incorporated herein in their entirety by this reference.

BACKGROUND OF THE INVENTION

Immune checkpoint therapies can yield durable responses and long-lasting survival benefit across some cancer types (Topalian et al. (2015) *Cancer Cell* 27:450-461). Indeed, checkpoint therapies have been approved for use in metastatic melanoma, non-small cell lung cancer, bladder cancer, and renal cell carcinoma, including as a first-line therapy for non-small cell lung cancer. However, many subjects among a population of subjects having the same cancer type do not exhibit a therapeutic benefit or relapse despite being treated with the same immune checkpoint therapy. It is presently unclear which factors associated with a cancer or type thereof, such as mutational load, neoantigen presentation, transcriptomic signatures, microbiome features, immune cell infiltration, or other indicators, are predictive of response to immune checkpoint therapies. Accordingly, there remains a great need in the art to identify biomarkers predictive of immune checkpoint therapy in order to better treat cancer of subjects in need thereof.

SUMMARY OF THE INVENTION

The present invention is based, at least in part, on the discovery that alterations in multiple oncogenic signaling pathways, including SWI/SNF pathway but also other chromatin modifiers, such as KDM6A, and EGFR signaling, predict response or resistance to immune checkpoint therapies, including (but not limited to) monoclonal antibodies targeting PD-1, PD-L1, and CTLA-4, across multiple cancer types. The SWI/SNF chromatin remodeling complex, which contains ARID1A, ARID1B, ARID2, SMARCA2, SMARCA4, SMARCB1, and PBRM1 subunits, among other subunits, plays a role in replication, transcription, DNA repair, and control of cell proliferation and differentiation. Although alterations in SWI/SNF subunits are known to play a role in the pathogenesis of ~20% of human cancers, including clear cell renal cell carcinoma, lung cancer, squamous cell carcinomas, hepatocellular carcinoma, small cell lung cancer, colorectal cancer, and pancreatic cancer (Kadoch and Crabtree (2015) *Sci. Adv.* 1:e150047), it was heretofore unknown that a mutation in one or more subunits of the SWI/SNF complex (e.g., mutations in one or more subunits of the PBAF complex, such as PBRM1 and ARID2), is predictive of response to immune checkpoint inhibitors. The same lack of predictive response applies to mutations in certain chromatin modifiers, such as KDM6A, and certain EGFR signaling components described herein. Since mutations in certain SWI/SNF complex subunits, chromatin modifiers, and/or EGFR signaling components described herein are found within a variety of cancers and types thereof, including bladder cancer, renal cell carcinoma, lung cancer, and head and neck squamous cell carcinoma, these biomarkers have wide-ranging impli-

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cations for patient stratification for immune checkpoint therapy across a wide variety of hyperproliferative disorders.

In one aspect, a method of identifying the likelihood of a cancer in a subject to be responsive to an immune checkpoint therapy, the method comprising a) obtaining or providing a subject sample from a patient having cancer; b) measuring the amount or activity of at least one biomarker listed in Table 1 in the subject sample; and c) comparing said amount or activity of the at least one biomarker listed in Table 1 in a control sample, wherein the absence of or a significantly decreased amount or activity of the at least one biomarker listed in Table 1 in the subject sample and/or the presence of or a significantly increased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being more likely to be responsive to the immune checkpoint therapy; and wherein the presence of or a significantly increased amount or activity of the at least one biomarker listed in Table 1 in the subject sample and/or the absence of or a decreased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being less likely to be responsive to the immune checkpoint therapy, is provided.

In another aspect, a method of identifying the likelihood of a cancer in a subject to be responsive to immune checkpoint therapy, the method comprising a) obtaining or providing a subject sample from a patient having cancer, wherein the sample comprises nucleic acid molecules from the subject; b) determining the copy number of at least one biomarker listed in Table 1 in the subject sample; and c) comparing said copy number to that of a control sample, wherein a decreased copy number of the at least one biomarker listed in Table 1 in the in the subject sample and/or an increased copy number of the at least one biomarker listed in Table 1 having a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being more likely to be responsive to the immune checkpoint therapy; and wherein a wild type or increased copy number of the biomarker in the subject sample and/or or a decreased copy number of the at least one biomarker listed in Table 1 having a loss of function mutation in the sample relative to the control sample identifies the cancer as being less likely to be responsive to the immune checkpoint therapy, is provided.

Numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the method provided herein further comprises recommending, prescribing, or administering the immune checkpoint therapy if the cancer is determined likely to be responsive to the immune checkpoint therapy or administering an anti-cancer therapy other than the immune checkpoint therapy if the cancer is determined be less likely to be responsive to the immune checkpoint therapy. The anti-cancer therapy may be, for example, selected from the group consisting of targeted therapy, chemotherapy, radiation therapy, and/or hormonal therapy. In another embodiment, the control sample described herein is determined from a cancerous or non-cancerous sample from either the patient or a member of the same species to which the patient belongs. In still another embodiment, the control sample is a cancerous or non-cancerous sample from the patient obtained from an earlier point in time than the patient sample. In yet another embodiment, the control

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sample is obtained before the patient has received immune checkpoint therapy and the patient sample is obtained after the patient has received immune checkpoint therapy. In another embodiment, the control sample described herein comprises cells or does not comprise cells. In still another embodiment, the control sample comprises cancer cells known to be responsive or non-responsive to the immune checkpoint therapy.

In another aspect, a method of assessing the efficacy of an agent for treating a cancer in a subject that is unlikely to be responsive to an immune checkpoint therapy, comprising a) detecting in a first subject sample and maintained in the presence of the agent the amount or activity of at least one biomarker listed in Table 1; b) detecting the amount or activity of the at least one biomarker listed in Table 1 in a second subject sample and maintained in the absence of the test compound; and c) comparing the amount or activity of the at least one biomarker listed in Table 1 from steps a) and b), wherein the presence of or a significantly increased amount or activity of the at least one biomarker listed in Table 1 in the first subject sample and/or the absence of or a decreased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the first subject sample, relative to at least one subsequent subject sample, indicates that the agent treats the cancer in the subject, is provided.

In another aspect, a method of assessing the efficacy of an agent for treating a cancer in a subject or prognosing progression of a cancer in a subject, comprising a) detecting in a subject sample at a first point in time the amount or activity of at least one biomarker listed in Table 1; b) repeating step a) during at least one subsequent point in time after administration of the agent; and c) comparing the expression and/or activity detected in steps a) and b), wherein the presence of or a significantly increased amount or activity of the at least one biomarker listed in Table 1 in the first subject sample and/or the absence of or a decreased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the first subject sample, relative to at least one subsequent subject sample, indicates that the cancer is unlikely to progress or that the agent treats the cancer in the subject, is provided. In one embodiment, between the first point in time and the subsequent point in time, the subject has undergone treatment, completed treatment, and/or is in remission for the cancer. In another embodiment, the first and/or at least one subsequent sample is selected from the group consisting of ex vivo and in vivo samples. In still another embodiment, the first and/or at least one subsequent sample is obtained from an animal model of the cancer. In yet another embodiment, the first and/or at least one subsequent sample is a portion of a single sample or pooled samples obtained from the subject.

In another aspect, a cell-based assay for screening for agents that have a cytotoxic or cytostatic effect on a cancer cell that is unresponsive to an immune checkpoint therapy comprising, contacting the cancer cell with a test agent, and determining the ability of the test agent to decrease the amount or activity of at least one biomarker listed in Table 1 in the subject sample and/or increase the amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation, is provided. In one embodiment, the step of contacting occurs in vivo, ex vivo, or in vitro. In another embodiment, the subject sample and/or the control sample has not been contacted with any anti-cancer treatment or inhibitor of an immune checkpoint. In still another embodiment, the subject has not been administered any anti-cancer treatment or inhibitor of an immune checkpoint.

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In yet another embodiment, the method or the cell-based assay provided herein further comprises recommending, prescribing, or administering at least one additional anti-cancer therapeutic agent. In another embodiment, the at least one additional anti-cancer therapeutic agent comprises an anti-PD-1 antibody and/or an anti-CTLA4 antibody.

As described above, numerous embodiments are contemplated for any aspect of the present invention described herein. For example, in one embodiment, the subject sample is selected from the group consisting of serum, whole blood, plasma, urine, cells, cell lines, and biopsies. In another embodiment, the amount of the at least one biomarker listed in Table 1 is detected using a reagent which specifically binds with the protein. For example, the reagent may be selected from the group consisting of an antibody, an antibody derivative, and an antibody fragment. In still another embodiment, the at least one biomarker listed in Table 1 is assessed by detecting the presence in the sample of a transcribed polynucleotide or portion thereof. For example, the transcribed polynucleotide may be an mRNA or a cDNA. The transcribed polynucleotide can be detected by identifying a nucleic acid that anneals with the biomarker nucleic acid, or a portion thereof, under stringent hybridization conditions. In yet another embodiment, the step of detecting further comprises amplifying the transcribed polynucleotide. In another embodiment, the at least one biomarker listed in Table 1 is human PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, or EGFR, or a fragment thereof. In still another embodiment, the immune checkpoint therapy described herein comprises at least one antibody selected from the group consisting of anti-PD-1 antibodies, anti-CTLA-4 antibodies, anti-PD-L1 antibodies, anti-PD-L2 antibodies, and combinations thereof. For example, the immune checkpoint therapy may comprise an anti-PD-1 antibody and/or an anti-CTLA4 antibody. In yet another embodiment, the likelihood of the cancer in the subject to be responsive to immune checkpoint therapy is the likelihood of at least one criteria selected from the group consisting of cellular proliferation, tumor burden, m-stage, metastasis, progressive disease, clinical benefit rate, survival until mortality, pathological complete response, semi-quantitative measures of pathologic response, clinical complete remission, clinical partial remission, clinical stable disease, recurrence-free survival, metastasis free survival, disease free survival, circulating tumor cell decrease, circulating marker response, and RECIST criteria. In another embodiment, the cancer is a solid tumor. In still another embodiment, the cancer is selected from the group consisting of melanoma, lung cancer, head and neck squamous cell carcinoma (HNSCC), sarcoma, bladder cancer, and renal cell cancer. In another embodiment, the cancer is melanoma. In still another embodiment, the cancer is metastatic. In still another embodiment, the subject described herein is a mammal. In yet another embodiment, the mammal is an animal model of cancer. In another embodiment, the mammal is a human.

BRIEF DESCRIPTION OF FIGURES

FIG. 1 summarizes the different types of cancer samples and their sources for analysis.

FIG. 2 depicts two criteria (exclusion and inclusion) for selecting quality controls for analysis.

FIG. 3 depicts that different patients had different degrees of clinical benefit from immune checkpoint therapy.

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FIG. 4 compares the amount of nonsynonymous mutations in patients having different degrees of clinical benefit from immune checkpoint therapy.

FIG. 5 shows genes significantly mutated in responders vs. non-responders.

FIG. 6 shows genes significantly mutated in responders vs. non-responders or intermediate responders (such as those having intermediate clinical benefit).

FIG. 7 shows genes significantly mutated (such as those having truncating mutations) in responders vs. non-responders.

FIG. 8 shows genes significantly mutated (such as those having truncating mutations) in responders vs. non-responders or intermediate responders (those having intermediate clinical benefit).

FIG. 9 depicts protein subunits of the SWI/SNF protein complex.

FIG. 10 shows SWI/SNF-relevant genes significantly mutated in responders vs. non-responders.

FIG. 11 shows SWI/SNF-relevant genes significantly mutated (such as those having truncating mutations) in responders vs. non-responders.

FIG. 12 depicts an enzymatic function scheme of KDM6A.

FIG. 13 includes 4 panels, identified as panels A, B, C, and D, which show the Kaplan-Meier analysis result for baseline clinical variables as predictors of PFS for SU2C cohort (N=39).

FIG. 14 shows the quality control processes for analyzing the SU2C cohort.

FIG. 15 depicts the different responses of 39 SU2C lung cancer patients to anti-PD-1/PD-L1 therapy.

FIG. 16 shows the mutational burden and response to immune checkpoint therapies of each patient (N=31).

FIG. 17 shows the relationship between clinical burden and clinical benefit in a cohort in Rizvi et al. (2015) Science 348:124-128. RECIST was not taken into account (such that 2 patients with PR and PFS of ~4 months were considered nonresponders).

FIG. 18 shows that pre-treatment tumor mutational load was a strong predictor of response to immune checkpoint therapy in anti-PD1/PD-L1-treated lung cancer. All mutations: CB vs. NCB; $p=0.003$. All mutations: CB or SD vs. NCB; $p=0.004$. Nonsyns: CB vs. NCB; $p=0.0047$. Nonsyns: CB or SD vs. NCB; $p=0.0064$. Clonal: CB vs. NCB; $p=0.024$. Clonal: CB or SD vs. NCB; $p=0.007$. If dropping two large outliers (highest mutational load CB and SD), p -values for all mutations go to 0.009 and 0.011.

FIG. 19 shows commonly mutated genes in lung cancer. NF1 alterations were more frequent in responders (3/6 clinical benefit, 3/13 stable disease, 0/12 NCB). EGFR hotspot alterations were seen more frequently in nonresponders. KRAS hotspot alterations seen more frequently in responders (1/6 clinical benefit, 4/13 SD, 1/12 NCB). SU2C-1006: splice site mutation in MET; missense mutation in LTBP1. SU2C-1066: 3 missense mutations in LEPR. SU2C-1068: 2 missense mutations in LEPR. SU2C-1067: Missense mutations in STAG2 and SRCAP. EGFR hotspot is L858. SU2C-1066 may be excluded, since its Purity=0.36.

FIG. 20 shows significantly mutated genes (N=6 clinical benefit vs. 12 no clinical benefit).

FIG. 21 shows that patients with hotspot mutations in EGFR uniformly did not respond to immune checkpoint therapy.

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FIG. 22 shows that SAFB2 indels were likely caused by sequencing artifact.

DETAILED DESCRIPTION OF THE INVENTION

It has been determined herein that certain SWI/SNF complex subunits (e.g., PBRM1, ARID2, and other SWI/SNF complex subunits described herein, such as in the Tables and Examples), additional chromatin modifiers (e.g., such as KDM6A), and EGFR signaling components are specific biomarkers for predicted clinical outcome in a wide variety of cancers afflicting patients who have received anti-immune checkpoint-based therapy (e.g., anti-PD1 and/or anti-CTLA4 agents). Accordingly, the present invention relates, in part, to methods for stratifying patients and predicting response of a cancer in a subject to immune checkpoint therapy based upon a determination and analysis of mutations, described herein, of biomarkers, compared to a control. In addition, such analyses can be used in order to provide useful anti-immune checkpoint treatment regimens (e.g., based on predictions of clinical response, subject survival or relapse, timing of adjuvant or neoadjuvant treatment, etc.).

I. Definitions

The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element.

The term “altered amount” or “altered level” refers to increased or decreased copy number (e.g., germline and/or somatic) of a biomarker nucleic acid, e.g., increased or decreased expression level in a cancer sample, as compared to the expression level or copy number of the biomarker nucleic acid in a control sample. The term “altered amount” of a biomarker also includes an increased or decreased protein level of a biomarker protein in a sample, e.g., a cancer sample, as compared to the corresponding protein level in a normal, control sample. Furthermore, an altered amount of a biomarker protein may be determined by detecting posttranslational modification such as methylation status of the marker, which may affect the expression or activity of the biomarker protein.

The amount of a biomarker in a subject is “significantly” higher or lower than the normal amount of the biomarker, if the amount of the biomarker is greater or less, respectively, than the normal level by an amount greater than the standard error of the assay employed to assess amount, and preferably at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 150%, 200%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or than that amount. Such “significance” can be assessed from any desired or known point of comparison, such as a particular post-treatment versus pre-treatment biomarker measurement ratio (e.g., 1-fold, 1.1-fold, 1.2-fold, 1.3-fold, 1.4-fold, 1.5-fold, and the like) or a particular pre-treatment serum biomarker protein measurement (e.g., 2,500 pg/ml, 2,750 pg/ml, 3,000 pg/ml, 3,175 pg/ml, 3,250 pg/ml, 3,500 pg/ml, and the like). Alternately, the amount of the biomarker in the subject can be considered “significantly” higher or lower than the normal amount if the amount is at least about two, and preferably at least about three, four, or five times, higher or lower, respectively, than the normal amount of the biomarker. Such “significance” can also be applied to any other measured parameter

described herein, such as for expression, inhibition, cytotoxicity, cell growth, and the like.

The term “altered level of expression” of a biomarker refers to an expression level or copy number of the biomarker in a test sample, e.g., a sample derived from a patient suffering from cancer, that is greater or less than the standard error of the assay employed to assess expression or copy number, and is preferably at least twice, and more preferably three, four, five or ten or more times the expression level or copy number of the biomarker in a control sample (e.g., sample from a healthy subjects not having the associated disease) and preferably, the average expression level or copy number of the biomarker in several control samples. The altered level of expression is greater or less than the standard error of the assay employed to assess expression or copy number, and is preferably at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 150%, 200%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more times the expression level or copy number of the biomarker in a control sample (e.g., sample from a healthy subjects not having the associated disease) and preferably, the average expression level or copy number of the biomarker in several control samples.

The term “altered activity” of a biomarker refers to an activity of the biomarker which is increased or decreased in a disease state, e.g., in a cancer sample, as compared to the activity of the biomarker in a normal, control sample. Altered activity of the biomarker may be the result of, for example, altered expression of the biomarker, altered protein level of the biomarker, altered structure of the biomarker, or, e.g., an altered interaction with other proteins involved in the same or different pathway as the biomarker or altered interaction with transcriptional activators or inhibitors.

The term “altered structure” of a biomarker refers to the presence of mutations or allelic variants within a biomarker nucleic acid or protein, e.g., mutations which affect expression or activity of the biomarker nucleic acid or protein, as compared to the normal or wild-type gene or protein. For example, mutations include, but are not limited to substitutions, deletions, or addition mutations. Mutations may be present in the coding or non-coding region of the biomarker nucleic acid.

The term “SWI/SNF complex” refers to SWItch/Sucrose Non-Fermentable, a nucleosome remodeling complex found in both eukaryotes and prokaryotes (Neigeborn Carlson (1984) *Genetics* 108:845-858; Stem et al. (1984) *J Mol. Biol.* 178:853-868). The SWI/SNF complex was first discovered in the yeast, *Saccharomyces cerevisiae*, named after yeast mating types switching (SWI) and sucrose nonfermenting (SNF) pathways (Workman and Kingston (1998) *Annu Rev Biochem.* 67:545-579; Sudarsanam and Winston (2000) *Trends Genet.* 16:345-351). It is a group of proteins comprising, at least, SWI1, SWI2/SNF2, SWI3, SWI5, and SWI6, as well as other polypeptides (Pazin and Kadonaga (1997) *Cell* 88:737-740). A genetic screening for suppressive mutations of the SWI/SNF phenotypes identified different histones and chromatin components, suggesting that these proteins were possibly involved in histone binding and chromatin organization (Winston and Carlson (1992) *Trends Genet.* 8:387-391). Biochemical purification of the SWI/SNF2p in *S. cerevisiae* demonstrated that this protein was part of a complex containing an additional 11 polypeptides, with a combined molecular weight over 1.5 MDa. The SWI/SNF complex contains the ATPase Swi2/Snf2p, two actin-related proteins (Arp7p and Arp9) and other subunits involved in DNA and protein-protein interactions. The purified SWI/SNF complex was able to alter the nucleosome

structure in an ATP-dependent manner (Workman and Kingston (1998), supra; Vignali et al. (2000) *Mol Cell Biol.* 20:1899-1910). The structures of the SWI/SNF and RSC complexes are highly conserved but not identical, reflecting an increasing complexity of chromatin (e.g., an increased genome size, the presence of DNA methylation, and more complex genetic organization) through evolution. For this reason, the SWI/SNF complex in higher eukaryotes maintains core components, but also substitute or add on other components with more specialized or tissue-specific domains. Yeast contains two distinct and similar remodeling complexes, SWI/SNF and RSC (Remodeling the Structure of Chromatin). In *Drosophila*, the two complexes are called BAP (Brahma Associated Protein) and PBAP (Polybromo-associated BAP) complexes. The human analogs are BAF (Brg1 Associated Factors, or SWI/SNF-A) and PBAF (Polybromo-associated BAF, or SWI/SNF-B). As shown in FIG. 9, the BAF complex comprises, at least, BAF250A (ARID1A), BAF250B (ARID1B), BAF57 (SMARCE1), BAF190/BRM (SMARCA2), BAF47 (SMARCB1), BAF53A (ACTL6A), BRG1/BAF190 (SMARCA4), BAF155 (SMARCC1), and BAF170 (SMARCC2). The PBAF complex comprises, at last, BAF200 (ARID2), BAF180 (PBRM1), BRD7, BAF45A (PHF10), BRG1/BAF190 (SMARCA4), BAF155 (SMARCC1), and BAF170 (SMARCC2). As in *Drosophila*, human BAF and PBAF share the different core components BAF47, BAF57, BAF60, BAF155, BAF170, BAF45 and the two actins b-Actin and BAF53 (Mohrmann and Verrijzer (2005) *Biochim Biophys Acta.* 1681:59-73). The central core of the BAF and PBAF is the ATPase catalytic subunit BRG1/hBRM, which contains multiple domains to bind to other protein subunits and acetylated histones. For a summary of different complex subunits and their domain structure, see Tang et al. (2010) *Prog Biophys Mol Biol.* 102:122-128 (e.g., FIG. 3), Hohmann and Vakoc (2014) *Trends Genet.* 30:356-363 (e.g., FIG. 1), and Kadoch and Crabtree (2015) *Sci. Adv.* 1:e1500447. For chromatin remodeling, the SWI/SNF complex use the energy of ATP hydrolysis to slide the DNA around the nucleosome. The first step consists in the binding between the remodeler and the nucleosome. This binding occurs with nanomolar affinity and reduces the digestion of nucleosomal DNA by nucleases. The 3-D structure of the yeast RSC complex was first solved and imaged using negative stain electron microscopy (Asturias et al. (2002) *Proc Natl Acad Sci USA* 99:13477-13480). The first Cryo-EM structure of the yeast SWI/SNF complex was published in 2008 (Dechassa et al. 2008). DNA footprinting data showed that the SWI/SNF complex makes close contacts with only one gyre of nucleosomal DNA. Protein crosslinking showed that the ATPase SWI2/SNF2p and Swi5p (the homologue of In1p in human), Snf6, Swi29, Snf11 and Sw82p (not conserved in human) make close contact with the histones. Several individual SWI/SNF subunits are encoded by gene families, whose protein products are mutually exclusive in the complex (Wu et al. (2009) *Cell* 136: 200-206). Thus, only one paralog is incorporated in a given SWI/SNF assembly. The only exceptions are BAF155 and BAF170, which are always present in the complex as homo- or hetero-dimers. Combinatorial association of SWI/SNF subunits could in principle give rise to hundreds of distinct complexes, although the exact number has yet to be determined (Wu et al. (2009), supra). Genetic evidence suggests that distinct subunit configurations of SWI/SNF are equipped to perform specialized functions. As an example, SWI/SNF contains one of two ATPase subunits, BRG1 or BRM/SMARCA2, which share 75% amino acid sequence

identity (Khavari et al. (1993) *Nature* 366:170-174). While in certain cell types BRG1 and BRM can compensate for loss of the other subunit, in other contexts these two ATPases perform divergent functions (Strobeck et al. (2002) *J Biol Chem.* 277:4782-4789; Hoffman et al. (2014) *Proc Natl Acad Sci USA.* 111:3128-3133). In some cell types, BRG1 and BRM can even functionally oppose one another to regulate differentiation (Flowers et al. (2009) *J Biol Chem.* 284:10067-10075). The functional specificity of BRG1 and BRM has been linked to sequence variations near their N-terminus, which have different interaction specificities for transcription factors (Kadam and Emerson (2003) *Mol Cell.* 11:377-389). Another example of paralogous subunits that form mutually exclusive SWI/SNF complexes are ARID1A/BAF250A, ARID1B/BAF250B, and ARID2/BAF200. ARID1A and ARID1B share 60% sequence identity, but yet can perform opposing functions in regulating the cell cycle, with MYC being an important downstream target of each paralog (Nagl et al. (2007) *EMBO J.* 26:752-763). ARID2 has diverged considerably from ARID1A/ARID1B and exists in a unique SWI/SNF assembly known as PBAF (or SWI/SNF-B), which contains several unique subunits not found in ARID1A/B-containing complexes. The composition of SWI/SNF can also be dynamically reconfigured during cell fate transitions through cell type-specific expression patterns of certain subunits. For example, BAF53A/ACTL6A is repressed and replaced by BAF53B/ACTL6B during neuronal differentiation, a switch that is essential for proper neuronal functions in vivo (Lessard et al. (2007) *Neuron* 55:201-215). These studies stress that SWI/SNF in fact represents a collection of multi-subunit complexes whose integrated functions control diverse cellular processes, which is also incorporated in the scope of definitions of the instant disclosure. Two recently published meta-analyses of cancer genome sequencing data estimate that nearly 20% of human cancers harbor mutations in one (or more) of the genes encoding SWI/SNF (Kadoch et al. (2013) *Nat Genet.* 45:592-601; Shain and Pollack (2013) *PLoS One.* 8:e55119). Such mutations are generally loss-of-function, implicating SWI/SNF as a major tumor suppressor in diverse cancers. Specific SWI/SNF gene mutations are generally linked to a specific subset of cancer lineages: SNF5 is mutated in malignant rhabdoid tumors (MRT), PBRM1/BAF180 is frequently inactivated in renal carcinoma, and BRG1 is mutated in non-small cell lung cancer (NSCLC) and several other cancers. In the instant disclosure, the scope of "SWI/SNF complex" may cover at least one fraction or the whole complex (e.g., some or all subunit proteins/other components), either in the human BAF/PBAF forms or their homologs/orthologs in other species (e.g., the yeast and *drosophila* forms described herein). Preferably, a "SWI/SNF complex" described herein contains at least part of the full complex bio-functionality, such as binding to other subunits/components, binding to DNA/histone, catalyzing ATP, promoting chromatin remodeling, etc.

The term "BAF complex" refers to at least one type of mammalian SWI/SNF complexes. Its nucleosome remodeling activity can be reconstituted with a set of four core subunits (BRG1/SMARCA4, SNF5/SMARCB1, BAF155/SMARCC1, and BAF170/SMARCC2), which have orthologs in the yeast complex (Phelan et al. (1999) *Mol Cell.* 3:247-253). However, mammalian SWI/SNF contains several subunits not found in the yeast counterpart, which can provide interaction surfaces for chromatin (e.g. acetylation recognition by bromodomains) or transcription factors and thus contribute to the genomic targeting of the complex (Wang et al. (1996) *EMBO J* 15:5370-5382; Wang

et al. (1996) *Genes Dev.* 10:2117-2130; Nie et al. (2000)). A key attribute of mammalian SWI/SNF is the heterogeneity of subunit configurations that can exist in different tissues and even in a single cell type (e.g., as BAF, PBAF, neural progenitor BAF (npBAF), neuron BAF (nBAF), embryonic stem cell BAF (esBAF), etc.). In some embodiments, the BAF complex described herein refers to one type of mammalian SWI/SNF complexes, which is different from PBAF complexes.

The term "PBAF complex" refers to one type of mammalian SWI/SNF complexes originally known as SWI/SNF-B. It is highly related to the BAF complex and can be separated with conventional chromatographic approaches. For example, human BAF and PBAF complexes share multiple identical subunits (such as BRG, BAF170, BAF155, BAF60, BAF57, BAF53, BAF45, actin, SS18, and hSNF5/INI1, as illustrated in FIG. 9). However, while BAF contains BAF250 subunit, PBAF contains BAF180 and BAF200, instead (Lemon et al. (2001) *Nature* 414:924-998; Yan et al. (2005) *Genes Dev.* 19:1662-1667). Moreover, they do have selectivity in regulating interferon-responsive genes (Yan et al. (2005), supra, showing that BAF200, but not BAF180, is required for PBAF to mediate expression of IFITM1 gene induced by IFN- α , while the IFITM3 gene expression is dependent on BAF but not PBAF). Due to these differences, PBAF, but not BAF, was able to activate vitamin D receptor-dependent transcription on a chromatinized template in vitro (Lemon et al. (2001), supra). The 3-D structure of human PBAF complex preserved in negative stain was found to be similar to yeast RSC but dramatically different from yeast SWI/SNF (Leschziner et al. (2005) *Structure* 13:267-275).

The term "BRG" or "BRG1/BAF190 (SMARCA4)" refers to a subunit of the SWI/SNF complex, which can be found in either BAF or PBAF complex. It is an ATP-dependent helicase and a transcription activator, encoded by the SMARCA4 gene. BRG1 can also bind BRCA1, as well as regulate the expression of the tumorigenic protein CD44. BRG1 is important for development past the pre-implantation stage. Without having a functional BRG1, exhibited with knockout research, the embryo will not hatch out of the zona pellucida, which will inhibit implantation from occurring on the endometrium (uterine wall). BRG1 is also crucial to the development of sperm. During the first stages of meiosis in spermatogenesis there are high levels of BRG1. When BRG1 is genetically damaged, meiosis is stopped in prophase 1, hindering the development of sperm and would result in infertility. More knockout research has concluded BRG1's aid in the development of smooth muscle. In a BRG1 knockout, smooth muscle in the gastrointestinal tract lacks contractility, and intestines are incomplete in some cases. Another defect occurring in knocking out BRG1 in smooth muscle development is heart complications such as an open ductus arteriosus after birth (Kim et al. (2012) *Development* 139:1133-1140; Zhang et al. (2011) *Mol. Cell. Biol.* 31:2618-2631). Mutations in SMARCA4 were first recognized in human lung cancer cell lines (Medina et al. (2008) *Hum. Mut.* 29:617-622). Later it was recognized that mutations exist in a significant frequency of medulloblastoma and pancreatic cancers among other tumor subtypes (Jones et al. (2012) *Nature* 488:100-105; Shain et al. (2012) *Proc Natl Acad Sci USA* 109:E252-E259; Shain and Pollack (2013), supra). Mutations in BRG1 (or SMARCA4) appear to be mutually exclusive with the presence of activation at any of the MYC-genes, which indicates that the BRG1 and MYC proteins are functionally related. Another recent study demonstrated a causal role of BRG1 in the control of retinoic

acid and glucocorticoid-induced cell differentiation in lung cancer and in other tumor types. This enables the cancer cell to sustain undifferentiated gene expression programs that affect the control of key cellular processes. Furthermore, it explains why lung cancer and other solid tumors are completely refractory to treatments based on these compounds that are effective therapies for some types of leukemia (Romero et al. (2012) *EMBO Mol. Med.* 4:603-616). The role of BRG1 in sensitivity or resistance to anti-cancer drugs had been recently highlighted by the elucidation of the mechanisms of action of darinaparsin, an arsenic-based anti-cancer drugs. Darinaparsin has been shown to induce phosphorylation of BRG1, which leads to its exclusion from the chromatin. When excluded from the chromatin, BRG1 can no longer act as a transcriptional co-regulator. This leads to the inability of cells to express HO-1, a cytoprotective enzyme. BRG1 has been shown to interact with proteins such as ACTL6A, ARID1A, ARID1B, BRCA1, CTNNB1, CBX5, CREBBP, CCNE1, ESR1, FANCA, HSP90B1, ING1, Myc, NR3C1, P53, POLR2A, PHB, SIN3A, SMARCB1, SMARCC1, SMARCC2, SMARCE1, STAT2, STK11, etc.

The term “BRG” or “BRG1/BAF190 (SMARCA4)” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BRG1 (SMARCA4) cDNA and human BRG1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, seven different human BRG1 isoforms are known. Human BRG1 isoform A (NP_001122321.1) is encodable by the transcript variant 1 (NM_001128849.1), which is the longest transcript. Human BRG1 isoform B (NP_001122316.1 or NP_003063.2) is encodable by the transcript variant 2 (NM_001128844.1), which differs in the 5' UTR and lacks an alternate exon in the 3' coding region, compared to the variant 1, and also by the transcript variant 3 (NM_003072.3), which lacks an alternate exon in the 3' coding region compared to variant 1. Human BRG1 isoform C (NP_001122317.1) is encodable by the transcript variant 4 (NM_001128845.1), which lacks two alternate in-frame exons and uses an alternate splice site in the 3' coding region, compared to variant 1. Human BRG1 isoform D (NP_001122318.1) is encodable by the transcript variant 5 (NM_001128846.1), which lacks two alternate in-frame exons and uses two alternate splice sites in the 3' coding region, compared to variant 1. Human BRG1 isoform E (NP_001122319.1) is encodable by the transcript variant 6 (NM_001128847.1), which lacks two alternate in-frame exons in the 3' coding region, compared to variant 1. Human BRG1 isoform F (NP_001122320.1) is encodable by the transcript variant 7 (NM_001128848.1), which lacks two alternate in-frame exons and uses an alternate splice site in the 3' coding region, compared to variant 1. Nucleic acid and polypeptide sequences of BRG1 orthologs in organisms other than humans are well known and include, for example, chimpanzee BRG1 (XM_016935029.1 and XP_016790518.1, XM_016935038.1 and XP_016790527.1, XM_016935039.1 and XP_016790528.1, XM_016935036.1 and XP_016790525.1, XM_016935037.1 and XP_016790526.1, XM_016935041.1 and XP_016790530.1, XM_016935040.1 and XP_016790529.1, XM_016935042.1 and XP_016790531.1, XM_016935043.1 and XP_016790532.1, XM_016935035.1 and XP_016790524.1, XM_016935032.1 and XP_016790521.1, XM_016935033.1 and XP_016790522.1, XM_016935030.1 and XP_016790519.1,

XM_016935031.1 and XP_016790520.1, and XM_016935034.1 and XP_016790523.1), Rhesus monkey BRG1 (XM_015122901.1 and XP_014978387.1, XM_015122902.1 and XP_014978388.1, XM_015122903.1 and XP_014978389.1, XM_015122906.1 and XP_014978392.1, XM_015122905.1 and XP_014978391.1, XM_015122904.1 and XP_014978390.1, XM_015122907.1 and XP_014978393.1, XM_015122909.1 and XP_014978395.1, and XM_015122910.1 and XP_014978396.1), dog BRG1 (XM_014122046.1 and XP_013977521.1, XM_014122043.1 and XP_013977518.1, XM_014122042.1 and XP_013977517.1, XM_014122041.1 and XP_013977516.1, XM_014122045.1 and XP_013977520.1, and XM_014122044.1 and XP_013977519.1), cattle BRG1 (NM_001105614.1 and NP_001099084.1), mouse BRG1 (NM_001174078.1 and NP_001167549.1, NM_001174079.1 and NP_001167550.1, and NM_011417.3 and NP_035547.2), rat BRG1 (NM_134368.1 and NP_599195.1), chicken BRG1 (NM_205059.1 and NP_990390.1), and zebrafish BRG1 (NM_181603.1 and NP_853634.1).

Anti-BRG1 antibodies suitable for detecting BRG1 protein are well-known in the art and include, for example, MABE1118, MABE121, MABE60, and 07-478 (poly- and mono-clonal antibodies from EMD Millipore, Billerica, MA), AM26021PU-N, AP23972PU-N, TA322909, TA322910, TA327280, TA347049, TA347050, TA347851, and TA349038 (antibodies from OnGene Technologies, Rockville, MD), NB100-2594, AF5738, NBP2-22234, NBP2-41270, NBP1-51230, and NBP1-40379 (antibodies from Novus Biologicals, Littleton, CO), ab110641, ab4081, ab215998, ab108318, ab70558, ab118558, ab133257, ab92496, ab196535, and ab196315 (antibodies from AbCam, Cambridge, MA), Cat #: 720129, 730011, 730051, MA1-10062, PA5-17003, and PA5-17008 (antibodies from ThermoFisher Scientific, Waltham, MA), GTX633391, GTX32478, GTX31917, GTX16472, and GTX50842 (antibodies from GeneTex, Irvine, CA), antibody 7749 (ProSci, Poway, CA), Brg-1 (N-15), Brg-1 (N-15) X, Brg-1 (H-88), Brg-1 (H-88) X, Brg-1 (P-18), Brg-1 (P-18) X, Brg-1 (G-7), Brg-1 (G-7) X, Brg-1 (H-10), and Brg-1 (H-10) X (antibodies from Santa Cruz Biotechnology, Dallas, TX), antibody of Cat. AF5738 (R&D Systmes, Minneapolis, MN), etc. In addition, reagents are well-known for detecting BRG1 expression. Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing BRG1 Expression can be found in the commercial product lists of the above-referenced companies. PFI 3 is a known small molecule inhibitor of polybromo 1 and BRG1 (e.g., Cat. B7744 from APExBIO, Houston, TX). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRG1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an BRG1 molecule of the present invention.

The term “BRM” or “BRM/BAF190 (SMARCA2)” refers to a subunit of the SWI/SNF complex, which can be found in either BAF or PBAF complexes. It is an ATP-dependent helicase and a transcription activator, encoded by the SMARCA2 gene. The catalytic core of the SWI/SNF complex can be either of two closely related ATPases, BRM or BRG1, with the potential that the choice of alternative subunits is a key determinant of specificity. Instead of impeding differentiation as was seen with BRG1 depletion, depletion of BRM caused accelerated progression to the differentiation phenotype. BRM was found to regulate genes

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different from those as BRG1 targets and be capable of overriding BRG1-dependent activation of the osteocalcin promoter, due to its interaction with different ARID family members (Flowers et al. (2009), supra). The known binding partners for BRM include, for example, ACTL6A, ARID1B, CEBPB, POLR2A, Prohibitin, SIN3A, SMARCB1, and SMARCC1.

The term “BRM” or “BRM/BAF190 (SMARCA2)” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BRM (SMARCA2) cDNA and human BRM protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, seven different human BRM isoforms are known. Human BRM isoform A (NP_003061.3 or NP_001276325.1) is encodable by the transcript variant 1 (NM_003070.4), which is the longest transcript, or the transcript variant 3 (NM_001289396.1), which differs in the 5' UTR, compared to variant 1. Human BRM isoform B (NP_620614.2) is encodable by the transcript variant 2 (NM_139045.3), which lacks an alternate in-frame exon in the coding region, compared to variant 1. Human BRM isoform C (NP_001276326.1) is encodable by the transcript variant 4 (NM_001289397.1), which uses an alternate in-frame splice site and lacks an alternate in-frame exon in the 3' coding region, compared to variant 1. Human BRM isoform D (NP_001276327.1) is encodable by the transcript variant 5 (NM_001289398.1), which differs in the 5' UTR, lacks a portion of the 5' coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Human BRM isoform E (NP_001276328.1) is encodable by the transcript variant 6 (NM_001289399.1), which differs in the 5' UTR, lacks a portion of the 5' coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Human BRM isoform F (NP_001276329.1) is encodable by the transcript variant 7 (NM_001289400.1), which differs in the 5' UTR, lacks a portion of the 5' coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Nucleic acid and polypeptide sequences of BRM orthologs in organisms other than humans are well known and include, for example, chimpanzee BRM (XM_016960529.1 and XP_016816018.1), dog BRG1 (XM_005615906.2 and XP_005615963.1, XM_845066.4 and XP_850159.1, XM_005615905.2 and XP_005615962.1, XM_005615904.2 and XP_005615961.1, XM_005615903.2 and XP_005615960.1, and XM_005615902.2 and XP_005615959.1), cattle BRM (NM_001099115.2 and NP_001092585.1), mouse BRM (NM_001347439.1 and NP_001334368.1, NM_011416.2 and NP_035546.2, and NM_026003.2 and NP_080279.1), rat BRM (NM_001004446.1 and NP_001004446.1), chicken BRM (NM_205139.1 and NP_990470.1), tropical clawed frog BRM (XM_012952601.1 and XP_012808055.1, XM_012952608.2 and XP_012808062.1, XM_012952597.2 and XP_012808051.1, XM_012952613.2 and XP_012808067.1, and XM_002941009.4 and XP_002941055.2), and zebrafish BRM (NM_001044775.2 and NP_001038240.1).

Anti-BRM antibodies suitable for detecting BRM protein are well-known in the art and include, for example, antibody MABE89 (EMD Millipore, Billerica, MA), antibody TA351725 (OnGene Technologies, Rockville, MD), NBPI-90015, NBP1-80042, NB100-55308, NB100-55309, NB100-55307, and H00006595-M06 (antibodies from Novus Biologicals, Littleton, CO), ab15597, ab12165, ab58188, and ab200480 (antibodies from AbCam, Cam-

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bridge, MA), Cat #: 11966 and 6889 (antibodies from Cell Signaling, Danvers, MA), etc. In addition, reagents are well-known for detecting BRM expression. Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing BRM Expression can be found in the commercial product lists of the above-referenced companies. For example, BRM RNAi product H00006595-R02 (Novus Biologicals), CRISPER gRNA products from GenScript, Piscataway, NJ, and other inhibitory RNA products from Origene, ViGene Biosciences (Rockville, MD), and Santa Cruz. It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRM molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an BRM molecule of the present invention.

The term “BAF200” or “ARID2” refers to AT-rich interactive domain-containing protein 2, a subunit of the SWI/SNF complex, which can be found in PBAF but not BAF complexes. It facilitates ligand-dependent transcriptional activation by nuclear receptors. The ARID2 gene, located on chromosome 12q in humans, consists of 21 exons; orthologs are known from mouse, rat, cattle, chicken, and mosquito (Zhao et al. (2011) *Oncotarget* 2:886-891). A conditional knockout mouse line, called *Arid2^{tm1a(EUCOMM)Wtsi}* was generated as part of the International Knockout Mouse Consortium program, a high-throughput mutagenesis project to generate and distribute animal models of disease (Skames et al. (2011) *Nature* 474:337-342). Human ARID2 protein has 1835 amino acids and a molecular mass of 197391 Da. The ARID2 protein contains two conserved C-terminal C2H2 zinc fingers motifs, a region rich in the amino acid residues proline and glutamine, a RFX (regulatory factor X)-type winged-helix DNA-binding domain (e.g., amino acids 521-601 of SEQ ID NO:8), and a conserved N-terminal AT-rich DNA interaction domain (e.g., amino acids 19-101 of SEQ ID NO:8; Zhao et al. (2011), supra). Mutation studies have revealed ARID2 to be a significant tumor suppressor in many cancer subtypes. ARID2 mutations are prevalent in hepatocellular carcinoma (Li et al. (2011) *Nature Genetics*. 43:828-829) and melanoma (Hodis et al. (2012) *Cell* 150:251-263; Krauthammer et al. (2012) *Nature Genetics*. 44:1006-1014). Mutations are present in a smaller but significant fraction in a wide range of other tumors (Shain and Pollack (2013), supra). ARID2 mutations are enriched in hepatitis C virus-associated hepatocellular carcinoma in the U.S. and European patient populations compared with the overall mutation frequency (Zhao et al. (2011), supra). The known binding partners for ARID2 include, e.g., Serum Response Factor (SRF) and SRF cofactors MYOCD, NKX2-5 and SRFBP1.

The term “BAF200” or “ARID2” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human ARID2 cDNA and human ARID2 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ARID2 isoforms are known. Human ARID2 isoform A (NP_689854.2) is encodable by the transcript variant 1 (NM_152641.3), which is the longer transcript. Human ARID2 isoform B (NP_001334768.1) is encodable by the transcript variant 2 (NM_001347839.1), which differs in the 3' UTR and 3' coding region compared to isoform A. The encoded isoform B has a shorter C-terminus compared to isoform A. Nucleic acid and polypeptide sequences of ARID2 orthologs in organisms other than humans are well known and include, for example, chimpanzee ARID2

(XM_016923581.1 and XP_016779070.1, and XM_016923580.1 and XP_016779069.1), Rhesus monkey ARID2 (XM_015151522.1 and XP_015007008.1), dog ARID2 (XM_003433553.2 and XP_003433601.2; and XM_014108583.1 and XP_013964058.1), cattle ARID2 (XM_002687323.5 and XP_002687369.1; and XM_015463314.1 and XP_015318800.1), mouse ARID2 (NM_175251.4 and NP_780460.3), rat ARID2 (XM_345867.8 and XP_345868.4; and XM_008776620.1 and XP_008774842.1), chicken ARID2 (XM_004937552.2 and XP_004937609.1, XM_004937551.2 and XP_004937608.1, XM_004937554.2 and XP_004937611.1, and XM_416046.5 and XP_416046.2), tropical clawed frog ARID2 (XM_002932805.4 and XP_002932851.1, XM_018092278.1 and XP_017947767.1, and XM_018092279.1 and XP_017947768.1), and zebrafish ARID2 (NM_001077763.1 and NP_001071231.1, and XM_005164457.3 and XP_005164514.1). ReRepresentative sequences of ARID2 orthologs are presented below in Table 1.

Anti-ARID2 antibodies suitable for detecting ARID2 protein are well-known in the art and include, for example, antibodies ABE316 and 04-080 (EMD Millipore, Billerica, MA), antibodies NBP1-26615, NBP2-43567, and NBP1-26614 (Novus Biologicals, Littleton, CO), antibodies ab51019, ab166850, ab113283, and ab56082 (AbCam, Cambridge, MA), antibodies Cat #: PA5-35857 and PA5-51258 (ThermoFisher Scientific, Waltham, MA), antibodies GTX129444, GTX129443, and GTX632011 (GeneTex, Irvine, CA), ARID2 (H-182) Antibody, ARID2 (H-182) X Antibody, ARID2 (5-13) Antibody, ARID2 (5-13) X Antibody, ARID2 (E-3) Antibody, and ARID2 (E-3) X Antibody (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting ARID2 expression. Multiple clinical tests of PBRM1 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000541481.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #SR316272, shRNA products #TR306601, TR505226, TG306601, SR420583, and CRISPER products #KN212320 and KN30154 from Origene Technologies (Rockville, MD), RNAi product H00196528-R01 (Novus Biologicals), CRISPER gRNA products from GenScript (Cat. #KN301549 and KN212320, Piscataway, NJ) and from Santa Cruz (sc-401863), and RNAi products from Santa Cruz (Cat #sc-96225 and sc-77400). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID2 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an ARID2 molecule of the present invention.

The term “loss-of-function mutation” for BAF200/ARID2 refers to any mutation in a ARID2-related nucleic acid or protein that results in reduced or eliminated ARID2 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frame-shift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of ARID2. Such

mutations reduce or eliminate ARID2 protein amounts and/or function by eliminating proper coding sequences required for proper ARID2 protein translation and/or coding for ARID2 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a reRepresentative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID2 protein amounts and/or function is described in the Tables and the Examples.

The term “BRD7” refers to Bromodomain-containing protein 7, a subunit of the SWI/SNF complex, which can be found in PBAF but not BAF complexes. BRD7 is a transcriptional corepressor that binds to target promoters (e.g., the ESR1 promoter) and down-regulates the expression of target genes, leading to increased histone H3 acetylation at Lys-9 (H3K9ac). BRD7 can recruit other proteins such as BRCA1 and POU2F1 to, e.g., the ESR1 promoter for its function. BRD7 activates the Wnt signaling pathway in a DVL1-dependent manner by negatively regulating the GSK3B phosphotransferase activity, while BRD7 induces dephosphorylation of GSK3B at Tyr-216. BRD7 is also a coactivator for TP53-mediated activation of gene transcription and is required for TP53-mediated cell-cycle arrest in response to oncogene activation. BRD7 promotes acetylation of TP53 at Lys-382, and thereby promotes efficient recruitment of TP53 to target promoters. BRD7 also inhibits cell cycle progression from G1 to S phase. For studies on BRD7 functions, see Zhou et al. (2006) *J. Cell. Biochem.* 98:920-930; Harte et al. (2010) *Cancer Res.* 70:2538-2547; Drost et al. (2010) *Nat. Cell Biol.* 12:380-389. The known binding partners for BRD7 also include, e.g., Tripartite Motif Containing 24 (TRIM24), Protein Tyrosine Phosphatase, Non-Receptor Type 13 (PTPN13), Disheveled Segment Polarity Protein 1 (DVL1), interferon regulatory factor 2 (IRF2) (Staal et al. (2000) *J. Cell. Physiol.* 185:269-279) and heterogeneous nuclear ribonucleoprotein U-like protein 1 (HNRPUL1) (Kzhyshkowska et al. (2003) *Biochem. J. England.* 371:385-393). Human BRD7 protein has 651 amino acids and a molecular mass of 74139 Da, with a N-terminal nuclear localization signal (e.g., amino acids 65-96 of SEQ ID NO:14), a Bromo-BRD7-like domain (e.g., amino acids 135-232 of SEQ ID NO:14), and a DUF3512 domain (e.g., amino acids 287-533 of SEQ ID NO:14).

The term “BRD7” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human BRD7 cDNA and human BRD7 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human BRD7 isoforms are known. Human BRD7 isoform A (NP_001167455.1) is encodable by the transcript variant 1 (NM_001173984.2), which is the longer transcript. Human BRD7 isoform B (NP_037395.2) is encodable by the transcript variant 2 (NM_013263.4), which uses an alternate in-frame splice site in the 3' coding region, compared to variant 1. The resulting isoform B lacks one internal residue, compared to isoform A. Nucleic acid and polypeptide sequences of BRD7 orthologs in organisms other than humans are well known and include, for example, chimpanzee BRD7 (XM_009430766.2 and XP_009429041.1, XM_016929816.1 and XP_016785305.1, XM_016929815.1 and XP_016785304.1, and XM_003315094.4 and XP_003315142.1), Rhesus monkey BRD7 (XM_015126104.1 and XP_014981590.1,

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XM_015126103.1 and XP_014981589.1, XM_001083389.3 and XP_001083389.2, and XM_015126105.1 and XP_014981591.1), dog BRD7 (XM_014106954.1 and XP_013962429.1), cattle BRD7 (NM_001103260.2 and NP_001096730.1), mouse BRD7 (NM_012047.2 and NP_036177.1), chicken BRD7 (NM_001005839.1 and NP_001005839.1), tropical clawed frog BRD7 (NM_001008007.1 and NP_001008008.1), and zebrafish BRD7 (NM_213366.2 and NP_998531.2). Representative sequences of BRD7 orthologs are presented below in Table 1.

Anti-BRD7 antibodies suitable for detecting BRD7 protein are well-known in the art and include, for example, antibody TA343710 (Origene), antibody NBP1-28727 (Novus Biologicals, Littleton, CO), antibodies ab56036, ab46553, ab202324, and ab114061 (AbCam, Cambridge, MA), antibodies Cat #: 15125 and 14910 (Cell Signaling), antibody GTX118755 (GeneTex, Irvine, CA), BRD7 (P-13) Antibody, BRD7 (T-12) Antibody, BRD7 (H-77) Antibody, BRD7 (H-2) Antibody, and BRD7 (B-8) Antibody (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting BRD7 expression. A clinical test of BRD7 is available in NIH Genetic Testing Registry (GTR®) with GTR Test ID: GTR000540400.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing BRD7 expression can be found in the commercial product lists of the above-referenced companies, such as shRNA product #TR100001 and CRISPER products #KN302255 and KN208734 from Origene Technologies (Rockville, MD), RNAi product H00029117-R01 (Novus Biologicals), and small molecule inhibitors BI 9564 and TP472 (Tocris Bioscience, UK). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRD7 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an BRD7 molecule of the present invention.

The term “loss-of-function mutation” for BRD7 refers to any mutation in a BRD7-related nucleic acid or protein that results in reduced or eliminated BRD7 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of BRD7. Such mutations reduce or eliminate BRD7 protein amounts and/or function by eliminating proper coding sequences required for proper BRD7 protein translation and/or coding for BRD7 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a reRepresentative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated BRD7 protein amounts and/or function is described in the Tables and the Examples.

The term “BAF45A” or “PHF10” refers to PHD finger protein 10, a subunit of the PBAF complex having two zinc finger domains at its C-terminus. PHF10 belongs to the neural progenitors-specific chromatin remodeling complex

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(npBAF complex) and is required for the proliferation of neural progenitors. During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. PHF10 gene encodes at least two types of evolutionarily conserved, ubiquitously expressed isoforms that are incorporated into the PBAF complex in a mutually exclusive manner. One isoform contains C-terminal tandem PHD fingers, which in the other isoform are replaced by the consensus sequence for phosphorylation-dependent SUMO 1 conjugation (PDSM) (Brechalov et al. (2014) *Cell Cycle* 13:1970-1979). PBAF complexes containing different PHF10 isoforms can bind to the promoters of the same genes but produce different effects on the recruitment of Pol II to the promoter and on the level of gene transcription. PHF10 is a transcriptional repressor of caspase 3 and impairs the programmed cell death pathway in human gastric cancer at the transcriptional level (Wei et al. (2010) *Mol Cancer Ther.* 9:1764-1774). Knockdown of PHF10 expression in gastric cancer cells led to significant induction of caspase-3 expression at both the RNA and protein levels and thus induced alteration of caspase-3 substrates in a time-dependent manner (Wei et al. (2010), supra). Results from luciferase assays by the same group indicated that PHF10 acted as a transcriptional repressor when the two PHD domains contained in PHF10 were intact. Human PHF10 protein has 498 amino acids and a molecular mass of 56051 Da, with two domains essential to induce neural progenitor proliferation (e.g., amino acids 89-185 and 292-334 of SEQ ID NO:20) and two PHD finger domains (e.g., amino acids 379-433 and 435-478 of SEQ ID NO:20). By similarity, PHF 10 binds to ACTL6A/BAF53A, SMARCA2/BRM/BAF190B, SMARCA4/BRG1/BAF190A and PBRM1/BAF180.

The term “BAF45A” or “PHF10” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human PHF10 cDNA and human PHF10 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human PHF10 isoforms are known. Human PHF10 isoform A (NP_060758.2) is encodable by the transcript variant 1 (NM_018288.3), which is the longer transcript. Human PHF10 isoform B (NP_579866.2) is encodable by the transcript variant 2 (NM_133325.2), which uses an alternate splice junction which results in six fewer nt when compared to variant 1. The isoform B lacks 2 internal amino acids compared to isoform A. Nucleic acid and polypeptide sequences of PHF10 orthologs in organisms other than humans are well known and include, for example, chimpanzee PHF10 (XM_016956680.1 and XP_016812169.1, XM_016956679.1 and XP_016812168.1, and XM_016956681.1 and XP_016812170.1), Rhesus monkey PHF10 (XM_015137735.1 and XP_014993221.1, and XM_015137734.1 and XP_014993220.1), dog PHF10

(XM_005627727.2 and XP_005627784.1, XM_005627726.2 and XP_005627783.1, XM_532272.5 and XP_532272.4, XM_014118230.1 and XP_013973705.1, and XM_014118231.1 and XP_013973706.1), cattle PHF10 (NM_001038052.1 and NP_001033141.1), mouse PHF10 (NM_024250.4 and NP_077212.3), rat PHF10 (NM_001024747.2 and NP_001019918.2), chicken PHF10 (XM_015284374.1 and XP_015139860.1), tropical clawed frog PHF10 (NM_001030472.1 and NP_001025643.1), zebrafish PHF10 (NM_200655.3 and NP_956949.3), and *C. elegans* PHF10 (NM_001047648.2 and NP_001041113.1, NM_001047647.2 and NP_001041112.1, and NM_001313168.1 and NP_001300097.1). Representative sequences of PHF10 orthologs are presented below in Table 1.

Anti-PHF10 antibodies suitable for detecting PHF10 protein are well-known in the art and include, for example, antibody TA346797 (Origene), antibodies NBP1-52879, NBP2-19795, NBP2-33759, and H00055274-B01P (Novus Biologicals, Littleton, CO), antibodies ab154637, ab80939, and ab68114 (AbCam, Cambridge, MA), antibody Cat #PA5-30678 (ThermoFisher Scientific), antibody Cat #26-352 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting PHF10 expression. A clinical test of PHF10 for hereditary disease is available with the test ID no. GTR000536577 in NIH Genetic Testing Registry (GTR*), offered by Fulgent Clinical Diagnostics Lab (Temple City, CA). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing PHF10 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #sc-95343 and sc-152206 and CRISPER products #sc-410593 from Santa Cruz Biotechnology, RNAi products H00055274-R01 and H00055274-R02 (Novus Biologicals), and multiple CRISPER products from GenScript (Piscataway, NJ). Human PHF10 knockout cell (from HAP1 cell line) is also available from Horizon Discovery (Cat #HZGH002778c011, UK). It is to be noted that the term can further be used to refer to any combination of features described herein regarding PHF10 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an PHF10 molecule of the present invention.

The term “loss-of-function mutation” for BAF45A/PHF10 refers to any mutation in a PHF10-related nucleic acid or protein that results in reduced or eliminated PHF10 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frame-shift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of PHF10. Such mutations reduce or eliminate PHF10 protein amounts and/or function by eliminating proper coding sequences required for proper PHF10 protein translation and/or coding for PHF10 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a reRepresentative list describing a wide variety of structural mutations correlated

with the functional result of reduced or eliminated PHF10 protein amounts and/or function is described in the Tables and the Examples.

The term “PBRM1” or “BAF180” refers to protein Polybromo-1, which is a subunit of ATP-dependent chromatin-remodeling complexes. PBRM1 functions in the regulation of gene expression as a constituent of the evolutionary-conserved SWI/SNF chromatin remodelling complexes (Euskirchen et al. (2012) *J Biol. Chem.* 287:30897-30905). Beside BRD7 and BAF200, PBRM1 is one of the unique components of the SWI/SNF-B complex, also known as polybromo/BRG1-associated factors (or PBAF), absent in the SWI/SNF-A (BAF) complex (Xue et al. (2000) *Proc Natl Acad Sci USA.* 97:13015-13020; Brownlee et al. (2012) *Biochem Soc Trans.* 40:364-369). On that account, and because it contains bromodomains known to mediate binding to acetylated histones, PBRM1 has been postulated to target PBAF complex to specific chromatin sites, therefore providing the functional selectivity for the complex (Xue et al. (2000), supra; Lemon et al. (2001) *Nature* 414:924-928; Brownlee et al. (2012), supra). Although direct evidence for PBRM1 involvement is lacking, SWI/SNF complexes have also been shown to play a role in DNA damage response (Park et al. (2006) *EMBO J.* 25:3986-3997). In vivo studies have shown that PBRM1 deletion leads to embryonic lethality in mice, where PBRM1 is required for mammalian cardiac chamber maturation and coronary vessel formation (Wang et al. (2004) *Genes Dev.* 18:3106-3116; Huang et al. (2008) *Dev Biol.* 319:258-266). PBRM1 mutations are most predominant in renal cell carcinomas (RCCs) and have been detected in over 40% of cases, placing PBRM1 second (after VHL) on the list of most frequently mutated genes in this cancer (Varela et al. (2011) *Nature* 469:539-542; Hakimi et al. (2013) *Eur Urol.* 63:848-854; Pena-Llopis et al. (2012) *Nat Genet.* 44:751-759; Pawlowski et al. (2013) *Int J Cancer.* 132:E11-E17). PBRM1 mutations have also been found in a smaller group of breast and pancreatic cancers (Xia et al. (2008) *Cancer Res.* 68:1667-1674; Shain et al. (2012) *Proc Natl Acad Sci USA.* 109:E252-E259; Numata et al. (2013) *Int J Oncol.* 42:403-410). PBRM1 mutations are more common in patients with advance stages (Hakimi et al. (2013), supra) and loss of PBRM1 protein expression has been associated with advanced tumour stage, low differentiation grade and worse patient outcome (Pawlowski et al. (2013), supra). In another study, no correlation between PBRM1 status and tumour grade was found (Pena-Llopis et al. (2012), supra). Although PBRM1-mutant tumours are associated with better prognosis than BAP1-mutant tumours, tumours mutated for both PBRM1 and BAP1 exhibit the greatest aggressiveness (Kapur et al. (2013) *Lancet Oncol.* 14:159-167). PBRM1 is ubiquitously expressed during mouse embryonic development (Wang et al. (2004), supra) and has been detected in various human tissues including pancreas, kidney, skeletal muscle, liver, lung, placenta, brain, heart, intestine, ovaries, testis, prostate, thymus and spleen (Xue et al. (2000), supra; Horikawa and Barrett (2002) *DNA Seq.* 13:211-215).

PBRM1 protein localises to the nucleus of cells (Nicolas and Goodwin (1996) *Gene* 175:233-240). As a component of the PBAF chromatin-remodelling complex, it associates with chromatin (Thompson (2009) *Biochimie.* 91:309-319), and has been reported to confer the localisation of PBAF complex to the kinetochores of mitotic chromosomes (Xue et al. (2000), supra). Human PBRM1 gene encodes a 1582 amino acid protein, also referred to as BAF180. Six bromodomains (BD1-6), known to recognize acetylated lysine residues and frequently found in chromatin-associated pro-

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teins, constitute the N-terminal half of PBRM1 (e.g., six BD domains at amino acid residue no. 44-156, 182-284, 383-484, 519-622, 658-762, and 775-882 of SEQ ID NO:2). The C-terminal half of PBRM1 contains two bromo-adjacent homology (BAH) domains (BAH1 and BAH2, e.g., at amino acid residue no. 957-1049 and 1130-1248 of SE ID NO:2), present in some proteins involved in transcription regulation. High mobility group (HMG) domain is located close to the C-terminus of PBRM1 (e.g., amino acid residue no. 1328-1377 of SEQ ID NO:2). HMG domains are found in a number of factors regulating DNA-dependent processes where HMG domains often mediate interactions with DNA.

The term "PBRM1" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human PBRM1 cDNA and human PBRM1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human PBRM1 isoforms are known. Human PBRM1 transcript variant 2 (NM_181042.4) represents the longest transcript. Human PBRM1 transcript variant 1 (NM_018313.4, having a CDS from the 115-4863 nucleotide residue of SEQ ID NO:1) differs in the 5' UTR and uses an alternate exon and splice site in the 3' coding region, thus encoding a distinct protein sequence (NP_060783.3, as SEQ ID NO:2) of the same length as the isoform (NP_851385.1) encoded by variant 2. Nucleic acid and polypeptide sequences of PBRM1 orthologs in organisms other than humans are well known and include, for example, chimpanzee PBRM1 (XM_009445611.2 and XP_009443886.1, XM_009445608.2 and XP_009443883.1, XM_009445602.2 and XP_009443877.1, XM_016941258.1 and XP_016796747.1, XM_016941256.1 and XP_016796745.1, XM_016941249.1 and XP_016796738.1, XM_016941260.1 and XP_016796749.1, XM_016941253.1 and XP_016796742.1, XM_016941250.1 and XP_016796739.1, XM_016941261.1 and XP_016796750.1, XM_009445605.2 and XP_009443880.1, XM_016941252.1 and XP_016796741.1, XM_009445603.2 and XP_009443878.1, XM_016941263.1 and XP_016796752.1, XM_016941262.1 and XP_016796751.1, XM_009445604.2 and XP_009443879.1, XM_016941251.1 and XP_016796740.1, XM_016941257.1 and XP_016796746.1, XM_016941255.1 and XP_016796744.1, XM_016941254.1 and XP_016796743.1, XM_016941265.1 and XP_016796754.1, XM_016941264.1 and XP_016796753.1, XM_016941248.1 and XP_016796737.1, XM_009445617.2 and XP_009443892.1, XM_009445616.2 and XP_009443891.1, XM_009445619.2 and XP_009443894.1 XM_009445615.2 and XP_009443890.1, XM_009445618.2 and XP_009443893.1, and XM_016941266.1 and XP_016796755.1), rhesus monkey PBRM1 (XM_015130736.1 and XP_014986222.1, XM_015130739.1 and XP_014986225.1, XM_015130737.1 and XP_014986223.1, XM_015130740.1 and XP_014986226.1, XM_015130727.1 and XP_014986213.1, XM_015130726.1 and XP_014986212.1, XM_015130728.1 and XP_014986214.1, XM_015130743.1 and XP_014986229.1, XM_015130731.1 and XP_014986217.1, XM_015130745.1 and XP_014986231.1, XM_015130741.1 and XP_014986227.1, XM_015130734.1 and XP_014986220.1, XM_015130744.1 and XP_014986230.1, XM_015130748.1 and XP_014986234.1, XM_015130746.1 and XP_014986232.1,

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XM_015130742.1 and XP_014986228.1, XM_015130747.1 and XP_014986233.1, XM_015130730.1 and XP_014986216.1, XM_015130732.1 and XP_014986218.1, XM_015130733.1 and XP_014986219.1, XM_015130735.1 and XP_014986221.1, XM_015130738.1 and XP_014986224.1, and XM_015130725.1 and XP_014986211.1), dog PBRM1 (XM_005632441.2 and XP_005632498.1, XM_014121868.1 and XP_013977343.1, XM_005632451.2 and XP_005632508.1, XM_014121867.1 and XP_013977342.1, XM_005632440.2 and XP_005632497.1, XM_005632446.2 and XP_005632503.1, XM_533797.5 and XP_533797.4, XM_005632442.2 and XP_005632499.1, XM_005632439.2 and XP_005632496.1, XM_014121869.1 and XP_013977344.1, XM_005632448.1 and XP_005632505.1, XM_005632449.1 and XP_005632506.1, XM_005632452.1 and XP_005632509.1, XM_005632445.1 and XP_005632502.1, XM_005632450.1 and XP_005632507.1, XM_005632453.1 and XP_005632510.1, XM_014121870.1 and XP_013977345.1, XM_005632443.1 and XP_005632500.1, XM_005632444.1 and XP_005632501.1, and XM_005632447.2 and XP_005632504.1), cow PBRM1 (XM_005222983.3 and XP_005223040.1, XM_005222979.3 and XP_005223036.1, XM_015459550.1 and XP_015315036.1, XM_015459551.1 and XP_015315037.1, XM_015459548.1 and XP_015315034.1, XM_010817826.1 and XP_010816128.1, XM_010817829.1 and XP_010816131.1, XM_010817830.1 and XP_010816132.1, XM_010817823.1 and XP_010816125.1, XM_010817824.2 and XP_010816126.1, XM_010817819.2 and XP_010816121.1, XM_010817827.2 and XP_010816129.1, XM_010817828.2 and XP_010816130.1, XM_010817817.2 and XP_010816119.1, and XM_010817818.2 and XP_010816120.1), mouse PBRM1 (NM_001081251.1 and NP_001074720.1), chicken PBRM1 (NM_205165.1 and NP_990496.1), tropical clawed frog PBRM1 (XM_018090224.1 and XP_017945713.1), zebrafish PBRM1 (XM_009305786.2 and XP_009304061.1, XM_009305785.2 and XP_009304060.1, and XM_009305787.2 and XP_009304062.1), fruit fly PBRM1 (NM_143031.2 and NP_651288.1), and worm PBRM1 (NM_001025837.3 and NP_001021008.1 and .NM_001025838.2 and NP_001021009.1). ReRepresentative sequences of PBRM1 orthologs are presented below in Table 1.

Anti-PBRM1 antibodies suitable for detecting PBRM1 protein are well-known in the art and include, for example, ABE70 (rabbit polyclonal antibody, EMD Millipore, Billerica, MA), TA345237 and TA345238 (rabbit polyclonal antibodies, OriGene Technologies, Rockville, MD), NBP2-30673 (mouse monoclonal) and other polyclonal antibodies (Novus Biologicals, Littleton, CO), ab196022 (rabbit mAb, AbCam, Cambridge, MA), PAH437Hu01 and PAH437Hu02 (rabbit polyclonal antibodies, Cloud-Clone Corp., Houston, TX), GTX100781 (GeneTex, Irvine, CA), 25-498 (ProSci, Poway, CA), sc-367222 (Santa Cruz Biotechnology, Dallas, TX), etc. In addition, reagents are well-known for detecting PBRM1 expression (see, for example, PBRM1 Hu-Cy3 or Hu-Cy5 SmartFlare™ RNA Detection Probe (EMD Millipore). Multiple clinical tests of PBRM1 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000537378.2 which is offered by Fulgent Clinical

Diagnostics Lab (Temple City, CA)). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing PBRM1 expression can be found in the commercial product lists of the above-referenced companies. Ribavirin and PFI 3 are known PBRM1 inhibitors. It is to be noted that the term can further be used to refer to any combination of features described herein regarding PBRM1 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an PBRM1 molecule of the present invention.

The term “PBRM1 loss-of-function mutation” refers to any mutation in a PBRM1-related nucleic acid or protein that results in reduced or eliminated PBRM1 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of PBRM1. Such mutations reduce or eliminate PBRM1 protein amounts and/or function by eliminating proper coding sequences required for proper PBRM1 protein translation and/or coding for PBRM1 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated PBRM1 protein amounts and/or function is described in the Tables and the Examples.

The term “BAF250A” or “ARID1A” refers to AT-rich interactive domain-containing protein 1A, a subunit of the SWI/SNF complex, which can be found in BAF but not PBAF complex. In humans there are two BAF250 isoforms, BAF250A/ARID1A and BAF250B/ARID1B. They are thought to be E3 ubiquitin ligases that target histone H2B (Li et al. (2010) *Mol. Cell. Biol.* 30:1673-1688). ARID1A is highly expressed in the spleen, thymus, prostate, testes, ovaries, small intestine, colon and peripheral leukocytes. ARID1A is involved in transcriptional activation and repression of select genes by chromatin remodeling. It is also involved in vitamin D-coupled transcription regulation by associating with the WINAC complex, a chromatin-remodeling complex recruited by vitamin D receptor. ARID1A belongs to the neural progenitors-specific chromatin remodeling (npBAF) and the neuron-specific chromatin remodeling (nBAF) complexes, which are involved in switching developing neurons from stem/progenitors to post-mitotic chromatin remodeling as they exit the cell cycle and become committed to their adult state. ARID1A also plays key roles in maintaining embryonic stem cell pluripotency and in cardiac development and function (Lei et al. (2012) *J Biol. Chem.* 287:24255-24262; Gao et al. (2008) *Proc. Natl. Acad. Sci. U.S.A.* 105:6656-6661). Loss of BAF250a expression was seen in 42% of the ovarian clear cell carcinoma samples and 21% of the endometrioid carcinoma samples, compared with just 1% of the high-grade serous carcinoma samples. ARID1A deficiency also impairs the DNA damage checkpoint and sensitizes cells to PARP inhibitors (Shen et al. (2015) *Cancer Discov.* 5:752-767). Human ARID1A protein has 2285 amino acids and a molecular mass of 242045 Da, with at least a DNA-binding

domain that can specifically bind an AT-rich DNA sequence, recognized by a SWI/SNF complex at the beta-globin locus, and a C-terminus domain for glucocorticoid receptor-dependent transcriptional activation. ARID1A has been shown to interact with proteins such as SMARCB1/BAF47 (Kato et al. (2002) *J. Biol. Chem.* 277:5498-505; Wang et al. (1996) *EMBO J.* 15:5370-5382) and SMARCA4/BRG1 (Wang et al. (1996), supra; Zhao et al. (1998) *Cell* 95:625-636), etc.

The term “BAF250A” or “ARID1A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BAF250A (ARID1A) cDNA and human BAF250A (ARID1A) protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ARID1A isoforms are known. Human ARID1A isoform A (NP_006006.3) is encodable by the transcript variant 1 (NM_006015.4), which is the longer transcript. Human ARID1A isoform B (NP_624361.1) is encodable by the transcript variant 2 (NM_139135.2), which lacks a segment in the coding region compared to variant 1. Isoform B thus lacks an internal segment, compared to isoform A. Nucleic acid and polypeptide sequences of ARID1A orthologs in organisms other than humans are well known and include, for example, chimpanzee ARID1A (XM_016956953.1 and XP_016812442.1, XM_016956958.1 and XP_016812447.1, and XM_009451423.2 and XP_009449698.2), Rhesus monkey ARID1A (XM_015132119.1 and XP_014987605.1, and XM_015132127.1 and XP_014987613.1), dog ARID1A (XM_847453.5 and XP_852546.3, XM_005617743.2 and XP_005617800.1, XM_005617742.2 and XP_005617799.1, XM_005617744.2 and XP_005617801.1, XM_005617746.2 and XP_005617803.1, and XM_005617745.2 and XP_005617802.1), cattle ARID1A (NM_001205785.1 and NP_001192714.1), mouse ARID1A (NM_001080819.1 and NP_001074288.1), rat ARID1A (NM_001106635.1 and NP_001100105.1), chicken ARID1A (XM_015297557.1 and XP_015153043.1, XM_015297556.1 and XP_015153042.1, and XM_417693.5 and XP_417693.5), tropical clawed frog ARID1A (XM_002934639.4 and XP_002934685.2), and zebrafish ARID1A (XM_009294131.2 and XP_009292406.1, and XM_009294132.2 and XP_009292407.1).

Anti-ARID1A antibodies suitable for detecting ARID1A protein are well-known in the art and include, for example, antibody Cat #04-080 (EMD Millipore, Billerica, MA), antibodies TA349170, TA350870, and TA350871 (OriGene Technologies, Rockville, MD), antibodies NBP1-88932, NB100-55334, NBP2-43566, NB100-55333, and H00008289-Q01 (Novus Biologicals, Littleton, CO), antibodies ab182560, ab182561, ab176395, and ab97995 (AbCam, Cambridge, MA), antibodies Cat #: 12354 and 12854 (Cell Signaling Technology, Danvers, MA), antibodies GTX129433, GTX129432, GTX632013, GTX12388, and GTX31619 (GeneTex, Irvine, CA), etc. In addition, reagents are well-known for detecting ARID1A expression. For example, multiple clinical tests for ARID1A are available at NIH Genetic Testing Registry (GTR©) (e.g., GTR Test ID: GTR000520952.1 for mental retardation, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID1A expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00008289-R01, H00008289-R02, and H00008289-R03 (Novus Biologicals) and CRISPR products KN301547G1 and KN301547G2 (Origene). Other CRISPR products include sc-400469

(Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID1A molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an ARID1A molecule of the present invention.

The term “loss-of-function mutation” for BAF250A/ARID1A refers to any mutation in an ARID1A-related nucleic acid or protein that results in reduced or eliminated ARID1A protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of ARID1A. Such mutations reduce or eliminate ARID1A protein amounts and/or function by eliminating proper coding sequences required for proper ARID1A protein translation and/or coding for ARID1A proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID1A protein amounts and/or function is described in the Tables and the Examples.

The term “BAF250B” or “ARID1B” refers to AT-rich interactive domain-containing protein 1B, a subunit of the SWI/SNF complex, which can be found in BAF but not PBAF complex. ARID1B and ARID1A are alternative and mutually exclusive ARID-subunits of the SWI/SNF complex. Germline mutations in ARID1B are associated with Coffin-Siris syndrome (Tsurusaki et al. (2012) *Nat. Genet.* 44:376-378; Santen et al. (2012) *Nat. Genet.* 44:379-380). Somatic mutations in ARID1B are associated with several cancer subtypes, suggesting that it is a tumor suppressor gene (Shai and Pollack (2013) *PLoS ONE* 8:e55119; Sausen et al. (2013) *Nat. Genet.* 45:12-17; Shain et al. (2012) *Proc. Natl. Acad. Sci. U.S.A.* 109:E252-E259; Fujimoto et al. (2012) *Nat. Genet.* 44:760-764). Human ARID1A protein has 2236 amino acids and a molecular mass of 236123 Da, with at least a DNA-binding domain that can specifically bind an AT-rich DNA sequence, recognized by a SWI/SNF complex at the beta-globin locus, and a C-terminus domain for glucocorticoid receptor-dependent transcriptional activation. ARID1B has been shown to interact with SMARCA4/BRG1 (Hurlstone et al. (2002) *Biochem. J.* 364:255-264; Inoue et al. (2002) *J. Biol. Chem.* 277:41674-41685 and SMARCA2/BRM (Inoue et al. (2002), *supra*).

The term “BAF250B” or “ARID1B” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BAF250B (ARID1B) cDNA and human BAF250B (ARID1B) protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human ARID1B isoforms are known. Human ARID1B isoform A (NP_059989.2) is encodable by the transcript variant 1 (NM_017519.2). Human ARID1B isoform B (NP_065783.3) is encodable by the transcript variant 2 (NM_020732.3). Human ARID1B isoform C

(NP_001333742.1) is encodable by the transcript variant 3 (NM_001346813.1). Nucleic acid and polypeptide sequences of ARID1B orthologs in organisms other than humans are well known and include, for example, Rhesus monkey ARID1B (XM_015137088.1 and XP_014992574.1), dog ARID1B (XM_014112912.1 and XP_013968387.1), cattle ARID1B (XM_010808714.2 and XP_010807016.1, and XM_015464874.1 and XP_015320360.1), mouse ARID1B (NM_001085355.1 and NP_001078824.1), rat ARID1B (XM_017604567.1 and XP_017460056.1), chicken ARID1B (XM_015284235.1 and XP_015139721.1, XM_015284233.1 and XP_015139719.1, XM_015284238.1 and XP_015139724.1, XM_015284230.1 and XP_015139716.1, XM_015284234.1 and XP_015139720.1, XM_015284231.1 and XP_015139717.1, XM_015284232.1 and XP_015139718.1, XM_015284236.1 and XP_015139722.1, and XM_015284237.1 and XP_015139723.1), tropical clawed frog ARID1B (XM_004914629.3 and XP_004914686.1, XM_004914631.3 and XP_004914688.1, XM_004914630.3 and XP_004914687.1, XM_004914634.3 and XP_004914691.1, XM_002931507.4 and XP_002931553.2, XM_004914632.3 and XP_004914689.1, XM_004914635.3 and XP_004914692.1, XM_004914633.3 and XP_004914690.1, XM_004914636.3 and XP_004914693.1, and XM_004914637.3 and XP_004914694.1), and zebrafish ARID1B (XM_009294544.2 and XP_009292819.1, XM_009294545.2 and XP_009292820.1, XM_005160356.3 and XP_005160413.1, XM_005160355.3 and XP_005160412.1, XM_005160354.3 and XP_005160411.1, and XM_692987.8 and XP_698079.4).

Anti-ARID1B antibodies suitable for detecting ARID1B protein are well-known in the art and include, for example, antibody Cat #ABE316 (EMD Millipore, Billerica, MA), antibody TA315663 (OriGene Technologies, Rockville, MD), antibodies H00057492-M02, H00057492-M01, NB100-57485, NBP1-89358, and NB100-57484 (Novus Biologicals, Littleton, CO), antibodies ab57461, ab69571, ab84461, and ab163568 (AbCam, Cambridge, MA), antibodies Cat #: PA5-38739, PA5-49852, and PA5-50918 (ThermoFisher Scientific, Danvers, MA), antibodies GTX130708, GTX60275, and GTX56037 (GeneTex, Irvine, CA), ARID1B (KMN1) Antibody and other antibodies (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting ARID1B expression. For example, multiple clinical tests for ARID1B are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000520953.1 for mental retardation, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID1B expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00057492-R03, H00057492-R01, and H00057492-R02 (Novus Biologicals) and CRISPR products KN301548 and KN214830 (Origene). Other CRISPR products include sc-402365 (Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID1B molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an ARID1B molecule of the present invention.

The term “loss-of-function mutation” for BAF250B/ARID1B refers to any mutation in an ARID1B-related nucleic acid or protein that results in reduced or eliminated ARID1B protein amounts and/or function. For example,

nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of ARID1B. Such mutations reduce or eliminate ARID1B protein amounts and/or function by eliminating proper coding sequences required for proper ARID1B protein translation and/or coding for ARID1B proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID1B protein amounts and/or function is described in the Tables and the Examples.

The term “CRB1” refers to Crumbs homolog 1, a protein similar to the *Drosophila* crumbs protein and localizes to the inner segment of mammalian photoreceptors. In *Drosophila* crumbs localizes to the stalk of the fly photoreceptor and may be a component of the molecular scaffold that controls proper development of polarity in the eye. CRB1 gene is involved in the Hippo signaling pathway. Mutations in this gene are associated with a severe form of retinitis pigmentosa, RP12, and with Leber congenital amaurosis. One study suggests that mutations in this gene are associated with keratoconus in patients that already have Leber’s congenital amaurosis (McMahon et al. (2009) *Invest. Ophthalmol. Vis. Sci.* 50:3185-3187). CRB1 mutation is also related to lung squamous cell carcinoma (SQCC) (Li et al. (2015) *Sci. Rep.* 5:Article 14237) and retinal dystrophy (Li et al. (2014) *Int J Mol Med* 33:913-918). The human CRB1 protein has 1406 amino acids and a molecular mass of 154183 Da.

The term “CRB1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human CRB1 cDNA and human CRB1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, four different human CRB1 isoforms are known. Human CRB1 isoform A (NP_957705.1) is the longest isoform and is encodable by the transcript variant 1 (NM_201253.2). Human CRB1 isoform B (NP_001180569.1) is encodable by the transcript variant 2 (NM_001193640.1), which lacks two in-frame exons compared to variant 1. The resulting isoform B has the same N- and C-termini but is shorter compared to isoform A. Human CRB1 isoform C (NP_001244894.1) is encodable by the transcript variant 3 (NM_001257965.1), which contains three noncoding exons in place of the first exon and contains an alternate in-frame exon compared to variant 1. The resulting isoform C is shorter at the N-terminus and contains an alternate internal segment compared to isoform A. Human CRB1 isoform D (NP_001244895.1) is encodable by the transcript variant 4 (NM_001257966.1), which lacks an alternate in-frame segment of two coding exons and most of a third compared to variant 1. The resulting isoform D has the same N- and C-termini but lacks an alternate internal segment compared to isoform A. Nucleic acid and polypeptide sequences of CRB1 orthologs in organisms other than humans are well known and include, for example, chimpanzee CRB1 (XM_009440300.2 and XP_009438575.1, XM_009440289.2 and XP_009438564.1,

XM_009440291.2 and XP_009438566.1, XM_016934908.1 and XP_016790397.1, XM_016934919.1 and XP_016790408.1, XM_016934927.1 and XP_016790416.1, XM_525009.5 and XP_525009.2, and XM_016934898.1 and XP_016790387.1), Rhesus monkey CRB1 (XM_015120817.1 and XP_014976303.1, XM_001110878.3 and XP_001110878.2, XM_001110912.3 and XP_001110912.2, XM_015120808.1 and XP_014976294.1, and XM_015120812.1 and XP_014976298.1), dog CRB1 (XM_014115056.1 and XP_013970531.1, XM_014115058.1 and XP_013970533.1, XM_005622293.2 and XP_005622350.1, and XM_014115057.1 and XP_013970532.1), cattle CRB1 (XM_010813559.2 and XP_010811861.1), mouse CRB1 (NM_133239.2 and NP_573502.2), rat CRB1 (NM_001107182.1 and NP_001100652.1), chicken CRB1 (XM_015290380.1 and XP_015145866.1, and XM_003641670.3 and XP_003641718.2), tropical clawed frog ARID1B (XM_018093205.1 and XP_017948694.1), and zebrafish CRB1 (NM_001038408.1).

Anti-CRB1 antibodies suitable for detecting CRB1 protein are well-known in the art and include, for example, antibody Cat #MABN1572 and ABE553 (EMD Millipore, Billerica, MA), antibody TA319859 (OnGene Technologies, Rockville, MD), antibody NBP2-41201 (Novus Biologicals, Littleton, CO), antibody ab156282 (AbCam, Cambridge, MA), antibody GTX32103 (GeneTex, Irvine, CA), CRB1 (H-14) Antibody (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting CRB1 expression. For example, multiple clinical tests for CRB1 are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000515886.2 for retinitis pigmentosa type 12, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing CRB1 Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00023418-R01 and H00023418-R02 (Novus Biologicals) and CRISPR products KN303799 and KN212347 (Origene). Other CRISPR products include sc-418097 (Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding CRB1 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an CRB1 molecule of the present invention.

The term “loss-of-function mutation” for CRB1 refers to any mutation in a CRB1-related nucleic acid or protein that results in reduced or eliminated CRB1 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of CRB1. Such mutations reduce or eliminate CRB1 protein amounts and/or function by eliminating proper coding sequences required for proper CRB1 protein translation and/or coding for CRB1 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction

in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated CRB1 protein amounts and/or function is described in the Tables and the Examples.

The term "EGFR" refers to the epidermal growth factor receptor, a transmembrane glycoprotein that is a member of the ErbB family of receptors, a subfamily of four closely related receptor tyrosine kinases: EGFR (ErbB-1), HER2/c-neu (ErbB-2), Her 3 (ErbB-3) and Her 4 (ErbB-4). This protein is a receptor for members of the epidermal growth factor family. Binding of the protein to a ligand induces receptor homo- and/or heterodimerization and tyrosine autophosphorylation on key cytoplasmic residues. The activated EGFR then recruits adapter proteins like GRB2 which in turn activates complex downstream signaling cascades, leading to cell proliferation. Known ligands of EGFR include EGF, TGFA/TGF-alpha, amphiregulin, epigen/EPGN, BTC/betacellulin, epiregulin/EREG and HBEGF/heparin-binding EGF. While being activated, autophosphorylation of several tyrosine (Y) residues in the C-terminal domain of EGFR occurs. These include Y992, Y1045, Y1068, Y1148 and Y1173, as shown in the adjacent diagram (Downward et al. (1984) *Nature* 311:483-485). This autophosphorylation elicits downstream activation and signaling by several other proteins that associate with the phosphorylated tyrosines through their own phosphotyrosine-binding SH2 domains. These downstream signaling proteins initiate several signal transduction cascades, principally the MAPK, Akt and JNK pathways, leading to DNA synthesis and cell proliferation (Oda et al. (2005) *Mol. Sys. Biol.* 1:2005.0010). Such proteins modulate phenotypes such as cell migration, adhesion, and proliferation. Activation of the receptor is important for the innate immune response in human skin. The kinase domain of EGFR can also cross-phosphorylate tyrosine residues of other receptors it is aggregated with, and can itself be activated in that manner. EGFR activates at least 4 major downstream signaling cascades including the RAS-RAF-MEK-ERK, PI3 kinase-AKT, PLCgamma-PKC and STATs modules. EGFR may also activate the NF-kappa-B signaling cascade and other proteins like RGS16, by activating its GTPase activity, and probably coupling the EGF receptor signaling to the G protein-coupled receptor signaling. EGFR also phosphorylates MUC1 and increases its interaction with SRC and CTNNB1/beta-catenin. Mutations that lead to EGFR overexpression (i.e., upregulation) or overactivity have been associated with a number of cancers, including squamous-cell carcinoma of the lung (80% of cases), anal cancers (Walker et al. (2009) *Hum. Pathol.* 40:1517-1527), glioblastoma (50%) and epithelial tumors of the head and neck (80-100%) (Kumar et al. (2013) *Robbins basic pathology*. Philadelphia: Elsevier/Saunders. p. 179). These somatic mutations involving EGFR lead to its constant activation, which produces uncontrolled cell division. In glioblastoma a more or less specific mutation of EGFR, called EGFRvIII is often observed (Kuan et al. (2001) *Endocr. Relat. Cancer.* 8:83-96). Aberrant EGFR signaling has been implicated in psoriasis, eczema and atherosclerosis (Jost et al. (2000) *Eur. J Dermatol.* 10:505-510; Dreux et al. (2006) *Atherosclerosis* 186:38-53). However, its exact roles in these conditions are ill-defined. Human EGFR protein has 1210 amino acids and a molecular mass of 134277 Da, with at least a receptor L domain (amino acid no. 57-168 of SEQ ID NO:92), a Furin-like domain (amino acid no. 185-335 of SDEQ ID NO:92), another receptor L domain (amino acid no. 361-481

of SEQ ID NO:92), a growth factor receptor domain IV (amino acid no. 505-637 of SEQ ID NO: 92), a transmembrane region (amino acid no. 646-668 of SEQ ID NO:92), and a catalytic domain of the protein tyrosine kinase family (amino acid no. 704-1016 of SEQ ID NO:92). The structure and domains of human EGFR may be found at the World Wide Web address of www.uniprot.org/uniprot/P00533#structure and www.ebi.ac.uk/interpro/protein/P00533. EGFR has been shown to interact with proteins such as AR, ARF4, CAV1, CAV3, CBL, CBLB, CBLC, CD44, CDC25A, CRK, CTNNB1, DCN, EGF, GRB14, Grb2, JAK2, MUC1, NCK1, NCK2, PKC alpha, PLCG1, PLSR1, PTPN1, PTPN11, PTPN6, PTPRK, SH2D3A, SH3KBP1, SHC1, SOS1, Src, STAT1, STAT3, STAT5A, UBC, and WAS.

The term "EGFR" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human EGFR cDNA and human EGFR protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, nine different human EGFR isoforms are known. Human EGFR isoform A (NP_005219.2), the longest isoform, is encodable by the transcript variant 1 (NM_005228.4). Human EGFR isoform B (NP_958439.1) is encodable by the transcript variant 2 (NM_201282.1), which uses a different 3' terminal exon when compared to variant 1. The resulting isoform B has a shorter and distinct C-terminus. Human EGFR isoform C (also known as ErbB1-S, NP_958440.1) is encodable by the transcript variant 3 (NM_201283.1), which uses a different 3' terminal exon when compared to variant 1. The resulting isoform C has a shorter and distinct C-terminus. Only the extracellular domain is present in isoform C. Human EGFR isoform D (NP_958441.1) is encodable by the transcript variant 4 (NM_201284.1), which uses a different 3' terminal exon when compared to variant 1. The resulting isoform D has a shorter and distinct C-terminus. Only the extracellular domain is present in isoform D. Human EGFR isoform E (NP_001333826.1) is encodable by the transcript variant 5 (NM_001346897.1), which lacks an in-frame exon in the 5' coding region and its 3' terminal exon extends past a splice site that is used in variant 1. The encoded isoform E is shorter and has a distinct C-terminus compared to isoform A. Human EGFR isoform F (NP_001333827.1) is encodable by the transcript variant 6 (NM_001346898.1), which has a 3' terminal exon that extends past a splice site that is used in variant 1. The encoded isoform F has a shorter and distinct C-terminus compared to isoform A. Human EGFR isoform G (NP_001333828.1) is encodable by the transcript variant 7 (NM_001346899.1), which lacks an in-frame exon in the 5' coding region, compared to variant 1. Human EGFR isoform H (NP_001333829.1) is encodable by the transcript variant 8 (NM_001346900.1), which uses a novel 5' terminal exon compared to variant 1. The encoded isoform H has a shorter and distinct N-terminus compared to isoform A. Human EGFR isoform I (a.k.a. EGFRvIII, delta-EGFR, and de2-7EGFR; NP_001333870.1) is encodable by the transcript variant 9 (NM_001346941.1), which has an in-frame deletion of six exons in the 5' coding region, compared to variant 1. The encoded isoform I has a shorter extracellular domain compared to isoform A. This variant is considered to be tumorigenic and the encoded protein lacks normal ligand binding ability and is constitutively active. Nucleic acid and polypeptide sequences of EGFR orthologs in organisms other than humans are well known and include, for example, chimpanzee EGFR (XM_519102.6 and XP_519102.3, and XM_001156264.5 and XP_001156264.1), Rhesus monkey

EGFR (XM_015133436.1 and XP_014988922.1, and XM_015133437.1 and XP_014988923.1), dog EGFR (XM_014120756.1 and XP_013976231.1), cattle EGFR (XM_002696890.4 and XP_002696936.2, and XM_592211.8 and XP_592211.4), mouse EGFR (NM_007912.4 and NP_031938.1, and NM_207655.2 and NP_997538.1), rat EGFR (NM_031507.1 and NP_113695.1, XM_008770416.2 and XP_008768638.1, XM_008770418.2 and XP_008768640.1, and XM_017599073.1 and XP_017454562.1), chicken EGFR (NM_205497.2 and NP_990828.2), tropical clawed frog EGFR (XM_002939914.4 and XP_002939960.2), and zebrafish EGFR (NM_194424.1 and NP_919405.1).

Anti-EGFR antibodies suitable for detecting EGFR protein are well-known in the art and include, for example, antibody Cat #06-847 (EMD Millipore, Billerica, MA), antibodies AM00029BT-N, AM00029PU-N, and others (OnGene Technologies, Rockville, MD), antibodies Cat #MAB8967, AF231, AF1095, and others (R&D Systems, Minneapolis, MN), antibodies NB120-10414, NBP1-84814, and others (Novus Biologicals, Littleton, CO), antibodies ab52894, ab40815, and others (AbCam, Cambridge, MA), antibodies Cat #: 4267, 2244, 48685, and others (Cell Signaling Technology, Danvers, MA), antibodies GTX121919, GTX628887, and others (GeneTex, Irvine, CA), etc. In addition, reagents are well-known for detecting EGFR expression. For example, multiple clinical tests for EGFR are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000514557.2 for EGFR mutation by Sanger Sequencing, offered by Cancer Genetics, Inc. (Rutherford, NJ), GTR Test ID: GTR000510455.1 for lung cancer, offered by Centogene AG, Germany, and other tests). Commercial ELISA kits for detecting EGFR are available, at least, from R&D Systems (Cat #DYC1095B-2, DYC1854-2, DEGFRO, DYC3570-2, etc.). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing EGFR Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products SC-29301, SC-44340, and others and CRISPR products sc-400015 (Santa Cruz Biotechnology). Other similar products include TG320326, TR320326, TG509941, and others shRNA products, as well as KN214877, KN204201, and others CRISPR products (Origene). Small molecule compounds are known to regulate EGFR expression, such as Cat. #A8197 and other inhibitors (ApexBio, Houston, TX), CAS 879127-07-8 and other inhibitors or activators (EMD Millipore). Known EGFR inhibitory drugs include, at least, Iressa™ (gefitinib), Tarceva™ (erlotinib), Tykerb™ (lapatinib), Erbitux™ (cetuximab), Vectibix™ (panitumumab), Caprelsa™ (vandetanib), Tagrisso™ (osimertinib), Portrazza™ (necitumumab), etc. It is to be noted that the term can further be used to refer to any combination of features described herein regarding EGFR molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an EGFR molecule of the present invention.

The term “loss-of-function mutation” for EGFR refers to any mutation in an EGFR-related nucleic acid or protein that results in reduced or eliminated EGFR protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and com-

binations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of EGFR. Such mutations reduce or eliminate EGFR protein amounts and/or function by eliminating proper coding sequences required for proper EGFR protein translation and/or coding for EGFR proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated EGFR protein amounts and/or function is described in the Tables and the Examples. In some embodiments, the term “hotspot mutation” for EGFR refers to a mutation that is commonly known to be mutated in EGFR associated with cancer. In some instances, such “hotspot mutations” can be those known to cause resistance to anti-EGFR therapies such as those described in Example 4.

Unless otherwise specified here within, the terms “antibody” and “antibodies” broadly encompass naturally-occurring forms of antibodies (e.g. IgG, IgA, IgM, IgE) and recombinant antibodies such as single-chain antibodies, chimeric and humanized antibodies and multi-specific antibodies, as well as fragments and derivatives of all of the foregoing, which fragments and derivatives have at least an antigenic binding site. Antibody derivatives may comprise a protein or chemical moiety conjugated to an antibody.

The term “antibody” as used herein also includes an “antigen-binding portion” of an antibody (or simply “antibody portion”). The term “antigen-binding portion”, as used herein, refers to one or more fragments of an antibody that retain the ability to specifically bind to an antigen (e.g., a biomarker polypeptide or fragment thereof). It has been shown that the antigen-binding function of an antibody can be performed by fragments of a full-length antibody. Examples of binding fragments encompassed within the term “antigen-binding portion” of an antibody include (i) a Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CHI domains; (ii) a F(ab')₂ fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the VH and CHI domains; (iv) a Fv fragment consisting of the VL and VH domains of a single arm of an antibody, (v) a dAb fragment (Ward et al. (1989) *Nature* 341:544-546), which consists of a VH domain; and (vi) an isolated complementarity determining region (CDR). Furthermore, although the two domains of the Fv fragment, VL and VH, are coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the VL and VH regions pair to form monovalent polypeptides (known as single chain Fv (scFv); see e.g., Bird et al. (1988) *Science* 242:423-426; and Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; and Osbourn et al. 1998, *Nature Biotechnology* 16: 778). Such single chain antibodies are also intended to be encompassed within the term “antigen-binding portion” of an antibody. Any VH and VL sequences of specific scFv can be linked to human immunoglobulin constant region cDNA or genomic sequences, in order to generate expression vectors encoding complete IgG polypeptides or other isotypes. VH and VL can also be used in the generation of Fab, Fv or other fragments of immunoglobulins using either protein chemistry or recombinant DNA technology. Other forms of single chain antibodies, such as diabodies are also encompassed. Diabodies are

bivalent, bispecific antibodies in which VH and VL domains are expressed on a single polypeptide chain, but using a linker that is too short to allow for pairing between the two domains on the same chain, thereby forcing the domains to pair with complementary domains of another chain and creating two antigen binding sites (see e.g., Holliger, P., et al. (1993) *Proc. Natl. Acad. Sci. USA* 90:6444-6448; Poljak, R. J., et al. (1994) *Structure* 2:1121-1123).

Still further, an antibody or antigen-binding portion thereof may be part of larger immunoadhesion polypeptides, formed by covalent or noncovalent association of the antibody or antibody portion with one or more other proteins or peptides. Examples of such immunoadhesion polypeptides include use of the streptavidin core region to make a tetrameric scFv polypeptide (Kipriyanov, S. M., et al. (1995) *Human Antibodies and Hybridomas* 6:93-101) and use of a cysteine residue, biomarker peptide and a C-terminal polyhistidine tag to make bivalent and biotinylated scFv polypeptides (Kipriyanov, S. M., et al. (1994) *Mol. Immunol.* 31:1047-1058). Antibody portions, such as Fab and F(ab')₂ fragments, can be prepared from whole antibodies using conventional techniques, such as papain or pepsin digestion, respectively, of whole antibodies. Moreover, antibodies, antibody portions and immunoadhesion polypeptides can be obtained using standard recombinant DNA techniques, as described herein.

Antibodies may be polyclonal or monoclonal; xenogeneic, allogeneic, or syngeneic; or modified forms thereof (e.g. humanized, chimeric, etc.). Antibodies may also be fully human. Preferably, antibodies of the present invention bind specifically or substantially specifically to a biomarker polypeptide or fragment thereof. The terms "monoclonal antibodies" and "monoclonal antibody composition", as used herein, refer to a population of antibody polypeptides that contain only one species of an antigen binding site capable of immunoreacting with a particular epitope of an antigen, whereas the term "polyclonal antibodies" and "polyclonal antibody composition" refer to a population of antibody polypeptides that contain multiple species of antigen binding sites capable of interacting with a particular antigen. A monoclonal antibody composition typically displays a single binding affinity for a particular antigen with which it immunoreacts.

Antibodies may also be "humanized", which is intended to include antibodies made by a non-human cell having variable and constant regions which have been altered to more closely resemble antibodies that would be made by a human cell. For example, by altering the non-human antibody amino acid sequence to incorporate amino acids found in human germline immunoglobulin sequences. The humanized antibodies of the present invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs. The term "humanized antibody", as used herein, also includes antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

The term "assigned score" refers to the numerical value designated for each of the biomarkers after being measured in a patient sample. The assigned score correlates to the absence, presence or inferred amount of the biomarker in the sample. The assigned score can be generated manually (e.g., by visual inspection) or with the aid of instrumentation for image acquisition and analysis. In certain embodiments, the assigned score is determined by a qualitative assessment, for

example, detection of a fluorescent readout on a graded scale, or quantitative assessment. In one embodiment, an "aggregate score," which refers to the combination of assigned scores from a plurality of measured biomarkers, is determined. In one embodiment the aggregate score is a summation of assigned scores. In another embodiment, combination of assigned scores involves performing mathematical operations on the assigned scores before combining them into an aggregate score. In certain, embodiments, the aggregate score is also referred to herein as the "predictive score."

The term "biomarker" refers to a measurable entity of the present invention that has been determined to be predictive of immune checkpoint therapy effects on a cancer. Biomarkers can include, without limitation, nucleic acids and proteins, including those shown in Table 1, the Examples, and the Figures.

A "blocking" antibody or an antibody "antagonist" is one which inhibits or reduces at least one biological activity of the antigen(s) it binds. In certain embodiments, the blocking antibodies or antagonist antibodies or fragments thereof described herein substantially or completely inhibit a given biological activity of the antigen(s).

The term "body fluid" refers to fluids that are excreted or secreted from the body as well as fluids that are normally not (e.g. amniotic fluid, aqueous humor, bile, blood and blood plasma, cerebrospinal fluid, cerumen and earwax, cowper's fluid or pre-ejaculatory fluid, chyle, chyme, stool, female ejaculate, interstitial fluid, intracellular fluid, lymph, menses, breast milk, mucus, pleural fluid, pus, saliva, sebum, semen, serum, sweat, synovial fluid, tears, urine, vaginal lubrication, vitreous humor, vomit).

The terms "cancer" or "tumor" or "hyperproliferative" refer to the presence of cells possessing characteristics typical of cancer-causing cells, such as uncontrolled proliferation, immortality, metastatic potential, rapid growth and proliferation rate, and certain characteristic morphological features. In some embodiments, such cells exhibit such characteristics in part or in full due to the expression and activity of immune checkpoint proteins, such as PD-1, PD-L1, and/or CTLA-4. Cancer cells are often in the form of a tumor, but such cells may exist alone within an animal, or may be a non-tumorigenic cancer cell, such as a leukemia cell. As used herein, the term "cancer" includes premalignant as well as malignant cancers. Cancers include, but are not limited to, B cell cancer, e.g., multiple myeloma, Waldenstrom's macroglobulinemia, the heavy chain diseases, such as, for example, alpha chain disease, gamma chain disease, and mu chain disease, benign monoclonal gammopathy, and immunocytic amyloidosis, melanomas, breast cancer, lung cancer, bronchus cancer, colorectal cancer, prostate cancer, pancreatic cancer, stomach cancer, ovarian cancer, urinary bladder cancer, brain or central nervous system cancer, peripheral nervous system cancer, esophageal cancer, cervical cancer, uterine or endometrial cancer, cancer of the oral cavity or pharynx, liver cancer, kidney cancer, testicular cancer, biliary tract cancer, small bowel or appendix cancer, salivary gland cancer, thyroid gland cancer, adrenal gland cancer, osteosarcoma, chondrosarcoma, cancer of hematologic tissues, and the like. Other non-limiting examples of types of cancers applicable to the methods encompassed by the present invention include human sarcomas and carcinomas, e.g., fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteogenic sarcoma, chordoma, angiosarcoma, endotheliosarcoma, lymphangiosarcoma, lymphangioendotheliosarcoma, synovialoma, mesothelioma, Ewing's tumor, leiomyosarcoma,

rhabdomyosarcoma, colon carcinoma, colorectal cancer, pancreatic cancer, breast cancer, ovarian cancer, prostate cancer, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat gland carcinoma, sebaceous gland carcinoma, papillary carcinoma, papillary adenocarcinomas, cystadenocarcinoma, medullary carcinoma, bronchogenic carcinoma, renal cell carcinoma, hepatoma, bile duct carcinoma, liver cancer, choriocarcinoma, seminoma, embryonal carcinoma, Wilms' tumor, cervical cancer, bone cancer, brain tumor, testicular cancer, lung carcinoma, small cell lung carcinoma, bladder carcinoma, epithelial carcinoma, glioma, astrocytoma, medulloblastoma, craniopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, meningioma, melanoma, neuroblastoma, retinoblastoma; leukemias, e.g., acute lymphocytic leukemia and acute myelocytic leukemia (myeloblastic, promyelocytic, myelomonocytic, monocytic and erythroleukemia); chronic leukemia (chronic myelocytic (granulocytic) leukemia and chronic lymphocytic leukemia); and polycythemia vera, lymphoma (Hodgkin's disease and non-Hodgkin's disease), multiple myeloma, Waldenstrom's macroglobulinemia, and heavy chain disease. In some embodiments, cancers are epithelial in nature and include but are not limited to, bladder cancer, breast cancer, cervical cancer, colon cancer, gynecologic cancers, renal cancer, laryngeal cancer, lung cancer, oral cancer, head and neck cancer, ovarian cancer, pancreatic cancer, prostate cancer, or skin cancer. In other embodiments, the cancer is breast cancer, prostate cancer, lung cancer, or colon cancer. In still other embodiments, the epithelial cancer is non-small-cell lung cancer, nonpapillary renal cell carcinoma, cervical carcinoma, ovarian carcinoma (e.g., serous ovarian carcinoma), or breast carcinoma. The epithelial cancers may be characterized in various other ways including, but not limited to, serous, endometrioid, mucinous, clear cell, Brenner, or undifferentiated.

The term "coding region" refers to regions of a nucleotide sequence comprising codons which are translated into amino acid residues, whereas the term "noncoding region" refers to regions of a nucleotide sequence that are not translated into amino acids (e.g., 5' and 3' untranslated regions).

The term "complementary" refers to the broad concept of sequence complementarity between regions of two nucleic acid strands or between two regions of the same nucleic acid strand. It is known that an adenine residue of a first nucleic acid region is capable of forming specific hydrogen bonds ("base pairing") with a residue of a second nucleic acid region which is antiparallel to the first region if the residue is thymine or uracil. Similarly, it is known that a cytosine residue of a first nucleic acid strand is capable of base pairing with a residue of a second nucleic acid strand which is antiparallel to the first strand if the residue is guanine. A first region of a nucleic acid is complementary to a second region of the same or a different nucleic acid if, when the two regions are arranged in an antiparallel fashion, at least one nucleotide residue of the first region is capable of base pairing with a residue of the second region. Preferably, the first region comprises a first portion and the second region comprises a second portion, whereby, when the first and second portions are arranged in an antiparallel fashion, at least about 50%, and preferably at least about 75%, at least about 90%, or at least about 95% of the nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion. More preferably, all nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion.

The terms "conjoint therapy" and "combination therapy," as used herein, refer to the administration of two or more therapeutic substances, e.g., combinations of anti-immune checkpoint therapies, multiple inhibitors of an immune checkpoint of interest, combinations of immune checkpoint therapy with an inhibitor of PBRM1 (ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, EGFR, and the like), and combinations thereof. The different agents comprising the combination therapy may be administered concomitant with, prior to, or following the administration of one or more therapeutic agents.

The term "control" refers to any reference standard suitable to provide a comparison to the expression products in the test sample. In one embodiment, the control comprises obtaining a "control sample" from which expression product levels are detected and compared to the expression product levels from the test sample. Such a control sample may comprise any suitable sample, including but not limited to a sample from a control cancer patient (can be stored sample or previous sample measurement) with a known outcome; normal tissue or cells isolated from a subject, such as a normal patient or the cancer patient, cultured primary cells/tissues isolated from a subject such as a normal subject or the cancer patient, adjacent normal cells/tissues obtained from the same organ or body location of the cancer patient, a tissue or cell sample isolated from a normal subject, or a primary cells/tissues obtained from a depository. In another preferred embodiment, the control may comprise a reference standard expression product level from any suitable source, including but not limited to housekeeping genes, an expression product level range from normal tissue (or other previously analyzed control sample), a previously determined expression product level range within a test sample from a group of patients, or a set of patients with a certain outcome (for example, survival for one, two, three, four years, etc.) or receiving a certain treatment (for example, standard of care cancer therapy). It will be understood by those of skill in the art that such control samples and reference standard expression product levels can be used in combination as controls in the methods of the present invention. In one embodiment, the control may comprise normal or non-cancerous cell/tissue sample. In another preferred embodiment, the control may comprise an expression level for a set of patients, such as a set of cancer patients, or for a set of cancer patients receiving a certain treatment, or for a set of patients with one outcome versus another outcome. In the former case, the specific expression product level of each patient can be assigned to a percentile level of expression, or expressed as either higher or lower than the mean or average of the reference standard expression level. In another preferred embodiment, the control may comprise normal cells, cells from patients treated with combination chemotherapy, and cells from patients having benign cancer. In another embodiment, the control may also comprise a measured value for example, average level of expression of a particular gene in a population compared to the level of expression of a housekeeping gene in the same population. Such a population may comprise normal subjects, cancer patients who have not undergone any treatment (i.e., treatment naive), cancer patients undergoing standard of care therapy, or patients having benign cancer. In another preferred embodiment, the control comprises a ratio transformation of expression product levels, including but not limited to determining a ratio of expression product levels of two genes in the test sample and comparing it to any suitable ratio of the same two genes in a reference standard; determining expression product levels of the two or more genes in the test sample

and determining a difference in expression product levels in any suitable control; and determining expression product levels of the two or more genes in the test sample, normalizing their expression to expression of housekeeping genes in the test sample, and comparing to any suitable control. In particularly preferred embodiments, the control comprises a control sample which is of the same lineage and/or type as the test sample. In another embodiment, the control may comprise expression product levels grouped as percentiles within or based on a set of patient samples, such as all patients with cancer. In one embodiment a control expression product level is established wherein higher or lower levels of expression product relative to, for instance, a particular percentile, are used as the basis for predicting outcome. In another preferred embodiment, a control expression product level is established using expression product levels from cancer control patients with a known outcome, and the expression product levels from the test sample are compared to the control expression product level as the basis for predicting outcome. As demonstrated by the data below, the methods of the present invention are not limited to use of a specific cut-point in comparing the level of expression product in the test sample to the control.

The "copy number" of a biomarker nucleic acid refers to the number of DNA sequences in a cell (e.g., germline and/or somatic) encoding a particular gene product. Generally, for a given gene, a mammal has two copies of each gene. The copy number can be increased, however, by gene amplification or duplication, or reduced by deletion. For example, germline copy number changes include changes at one or more genomic loci, wherein said one or more genomic loci are not accounted for by the number of copies in the normal complement of germline copies in a control (e.g., the normal copy number in germline DNA for the same species as that from which the specific germline DNA and corresponding copy number were determined). Somatic copy number changes include changes at one or more genomic loci, wherein said one or more genomic loci are not accounted for by the number of copies in germline DNA of a control (e.g., copy number in germline DNA for the same subject as that from which the somatic DNA and corresponding copy number were determined).

The "normal" copy number (e.g., germline and/or somatic) of a biomarker nucleic acid or "normal" level of expression of a biomarker nucleic acid or protein is the activity/level of expression or copy number in a biological sample, e.g., a sample containing tissue, whole blood, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, and bone marrow, from a subject, e.g., a human, not afflicted with cancer, or from a corresponding non-cancerous tissue in the same subject who has cancer.

As used herein, the term "costimulate" with reference to activated immune cells includes the ability of a costimulatory molecule to provide a second, non-activating receptor mediated signal (a "costimulatory signal") that induces proliferation or effector function. For example, a costimulatory signal can result in cytokine secretion, e.g., in a T cell that has received a T cell-receptor mediated signal. Immune cells that have received a cell-receptor mediated signal, e.g., via an activating receptor are referred to herein as "activated immune cells."

The term "determining a suitable treatment regimen for the subject" is taken to mean the determination of a treatment regimen (i.e., a single therapy or a combination of different therapies that are used for the prevention and/or treatment of the cancer in the subject) for a subject that is started, modified and/or ended based or essentially based or

at least partially based on the results of the analysis according to the present invention. One example is determining whether to provide targeted therapy against a cancer to provide immunotherapy that generally increases immune responses against the cancer (e.g., immune checkpoint therapy). Another example is starting an adjuvant therapy after surgery whose purpose is to decrease the risk of recurrence, another would be to modify the dosage of a particular chemotherapy. The determination can, in addition to the results of the analysis according to the present invention, be based on personal characteristics of the subject to be treated. In most cases, the actual determination of the suitable treatment regimen for the subject will be performed by the attending physician or doctor.

The term "diagnosing cancer" includes the use of the methods, systems, and code of the present invention to determine the presence or absence of a cancer or subtype thereof in an individual. The term also includes methods, systems, and code for assessing the level of disease activity in an individual.

A molecule is "fixed" or "affixed" to a substrate if it is covalently or non-covalently associated with the substrate such that the substrate can be rinsed with a fluid (e.g. standard saline citrate, pH 7.4) without a substantial fraction of the molecule dissociating from the substrate.

The term "expression signature" or "signature" refers to a group of two or more coordinately expressed biomarkers. For example, the genes, proteins, metabolites, and the like making up this signature may be expressed in a specific cell lineage, stage of differentiation, or during a particular biological response. The biomarkers can reflect biological aspects of the tumors in which they are expressed, such as the cell of origin of the cancer, the nature of the non-malignant cells in the biopsy, and the oncogenic mechanisms responsible for the cancer. Expression data and gene expression levels can be stored on computer readable media, e.g., the computer readable medium used in conjunction with a microarray or chip reading device. Such expression data can be manipulated to generate expression signatures.

"Homologous" as used herein, refers to nucleotide sequence similarity between two regions of the same nucleic acid strand or between regions of two different nucleic acid strands. When a nucleotide residue position in both regions is occupied by the same nucleotide residue, then the regions are homologous at that position. A first region is homologous to a second region if at least one nucleotide residue position of each region is occupied by the same residue. Homology between two regions is expressed in terms of the proportion of nucleotide residue positions of the two regions that are occupied by the same nucleotide residue. By way of example, a region having the nucleotide sequence 5'-AT-TGCC-3' and a region having the nucleotide sequence 5'-TATGGC-3' share 50% homology. Preferably, the first region comprises a first portion and the second region comprises a second portion, whereby, at least about 50%, and preferably at least about 75%, at least about 90%, or at least about 95% of the nucleotide residue positions of each of the portions are occupied by the same nucleotide residue. More preferably, all nucleotide residue positions of each of the portions are occupied by the same nucleotide residue.

The term "immune cell" refers to cells that play a role in the immune response. Immune cells are of hematopoietic origin, and include lymphocytes, such as B cells and T cells; natural killer cells; myeloid cells, such as monocytes, macrophages, eosinophils, mast cells, basophils, and granulocytes.

The term “immune checkpoint” refers to a group of molecules on the cell surface of CD4+ and/or CD8+ T cells that fine-tune immune responses by down-modulating or inhibiting an anti-tumor immune response. Immune checkpoint proteins are well known in the art and include, without limitation, CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, 2B4, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, and A2aR (see, for example, WO 2012/177624). The term further encompasses biologically active protein fragment, as well as nucleic acids encoding full-length immune checkpoint proteins and biologically active protein fragments thereof. In some embodiment, the term further encompasses any fragment according to homology descriptions provided herein.

“Immune checkpoint therapy” refers to the use of agents that inhibit immune checkpoint nucleic acids and/or proteins. Inhibition of one or more immune checkpoints can block or otherwise neutralize inhibitory signaling to thereby upregulate an immune response in order to more efficaciously treat cancer. Exemplary agents useful for inhibiting immune checkpoints include antibodies, small molecules, peptides, peptidomimetics, natural ligands, and derivatives of natural ligands, that can either bind and/or inactivate or inhibit immune checkpoint proteins, or fragments thereof; as well as RNA interference, antisense, nucleic acid aptamers, etc. that can downregulate the expression and/or activity of immune checkpoint nucleic acids, or fragments thereof. Exemplary agents for upregulating an immune response include antibodies against one or more immune checkpoint proteins block the interaction between the proteins and its natural receptor(s); a non-activating form of one or more immune checkpoint proteins (e.g., a dominant negative polypeptide); small molecules or peptides that block the interaction between one or more immune checkpoint proteins and its natural receptor(s); fusion proteins (e.g. the extracellular portion of an immune checkpoint inhibition protein fused to the Fc portion of an antibody or immunoglobulin) that bind to its natural receptor(s); nucleic acid molecules that block immune checkpoint nucleic acid transcription or translation; and the like. Such agents can directly block the interaction between the one or more immune checkpoints and its natural receptor(s) (e.g., antibodies) to prevent inhibitory signaling and upregulate an immune response. Alternatively, agents can indirectly block the interaction between one or more immune checkpoint proteins and its natural receptor(s) to prevent inhibitory signaling and upregulate an immune response. For example, a soluble version of an immune checkpoint protein ligand such as a stabilized extracellular domain can binding to its receptor to indirectly reduce the effective concentration of the receptor to bind to an appropriate ligand. In one embodiment, anti-PD-1 antibodies, anti-PD-L1 antibodies, and anti-CTLA-4 antibodies, either alone or in combination, are used to inhibit immune checkpoints.

“Ipilimumab” is a representative example of an immune checkpoint therapy. Ipilimumab (previously MDX-010; Medarex Inc., marketed by Bristol-Myers Squibb as YERVOY™) is a fully human anti-human CTLA-4 monoclonal antibody that blocks the binding of CTLA-4 to CD80 and CD86 expressed on antigen presenting cells, thereby, blocking the negative down-regulation of the immune responses elicited by the interaction of these molecules (see, for example, WO 2013/169971, U.S. Pat. Publ. 2002/0086014, and U.S. Pat. Publ. 2003/0086930).

The term “immune response” includes T cell mediated and/or B cell mediated immune responses. Exemplary immune responses include T cell responses, e.g., cytokine production and cellular cytotoxicity. In addition, the term immune response includes immune responses that are indirectly effected by T cell activation, e.g., antibody production (humoral responses) and activation of cytokine responsive cells, e.g., macrophages.

The term “immunotherapeutic agent” can include any molecule, peptide, antibody or other agent which can stimulate a host immune system to generate an immune response to a tumor or cancer in the subject. Various immunotherapeutic agents are useful in the compositions and methods described herein.

The term “inhibit” includes the decrease, limitation, or blockage, of, for example a particular action, function, or interaction. In some embodiments, cancer is “inhibited” if at least one symptom of the cancer is alleviated, terminated, slowed, or prevented. As used herein, cancer is also “inhibited” if recurrence or metastasis of the cancer is reduced, slowed, delayed, or prevented.

The term “interaction”, when referring to an interaction between two molecules, refers to the physical contact (e.g., binding) of the molecules with one another. Generally, such an interaction results in an activity (which produces a biological effect) of one or both of said molecules.

An “isolated protein” refers to a protein that is substantially free of other proteins, cellular material, separation medium, and culture medium when isolated from cells or produced by recombinant DNA techniques, or chemical precursors or other chemicals when chemically synthesized. An “isolated” or “purified” protein or biologically active portion thereof is substantially free of cellular material or other contaminating proteins from the cell or tissue source from which the antibody, polypeptide, peptide or fusion protein is derived, or substantially free from chemical precursors or other chemicals when chemically synthesized. The language “substantially free of cellular material” includes preparations of a biomarker polypeptide or fragment thereof, in which the protein is separated from cellular components of the cells from which it is isolated or recombinantly produced. In one embodiment, the language “substantially free of cellular material” includes preparations of a biomarker protein or fragment thereof, having less than about 30% (by dry weight) of non-biomarker protein (also referred to herein as a “contaminating protein”), more preferably less than about 20% of non-biomarker protein, still more preferably less than about 10% of non-biomarker protein, and most preferably less than about 5% non-biomarker protein. When antibody, polypeptide, peptide or fusion protein or fragment thereof, e.g., a biologically active fragment thereof, is recombinantly produced, it is also preferably substantially free of culture medium, i.e., culture medium represents less than about 20%, more preferably less than about 10%, and most preferably less than about 5% of the volume of the protein preparation.

The term “KDM6A” refers to a particular lysine demethylase containing a JmjC-domain that catalyzes the demethylation of tri-/di-methylated histone H3. The term “KDM6A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human KDM6A cDNA and human KDM6A protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, the nucleic acid and amino acid sequences of a representative human KDM6A biomarker (also known as UTX or MGC141941 or bA386N14.2 or

DKFZp686A03225) is available to the public at the GenBank database under NM_021140.2 and NP_006696.2. Nucleic acid and polypeptide sequences of KDM6A orthologs in organisms other than humans are well known and include, for example, mouse KDM6A (NM_009483.1 and NP_033509.1), rat KDM6A (XM_002730185.2 and XP_002730231.1), chimpanzee KDM6A (XM_002806207.1 and XP_002806253.1), chicken KDM6A (XM_416762.3 and XP_416762.3), fruit fly KDM6A (NM_001201844.1 and NP_001188773.1), and worm KDM6A (NM_077049.3 and NP_509450.1). Representative sequences of KDM6A orthologs are presented below in Table 1.

Anti-KDM6A antibodies suitable for detecting KDM6A protein are well-known in the art and include, for example, antibody ab36938 (Abcam), 16F9.1 (EMD Millipore), PA5-31828 (ThermoFisher), NBP1-80628 and H00007403-M05 (Novus Biologicals), etc. Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing KDM6A expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #sc-76881 and sc-76882 and CRISPR products #sc-514859 from Santa Cruz Biotechnology, as well as multiple RNAi products and CRISPR products from Origene and GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding KDM6A molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an KDM6A molecule of the present invention.

The term "loss-of-function mutation" for KDM6A refers to any mutation in a KDM6A-related nucleic acid or protein that results in reduced or eliminated KDM6A protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a "nonsynonymous mutation," meaning that the mutation alters the amino acid sequence of KDM6A. Such mutations reduce or eliminate KDM6A protein amounts and/or function by eliminating proper coding sequences required for proper KDM6A protein translation and/or coding for KDM6A proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of subcellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated KDM6A protein amounts and/or function is described in the Tables and the Examples.

A "kit" is any manufacture (e.g. a package or container) comprising at least one reagent, e.g. a probe or small molecule, for specifically detecting and/or affecting the expression of a marker of the present invention. The kit may be promoted, distributed, or sold as a unit for performing the methods of the present invention. The kit may comprise one or more reagents necessary to express a composition useful in the methods of the present invention. In certain embodiments, the kit may further comprise a reference standard, e.g., a nucleic acid encoding a protein that does not affect or regulate signaling pathways controlling cell growth, divi-

sion, migration, survival or apoptosis. One skilled in the art can envision many such control proteins, including, but not limited to, common molecular tags (e.g., green fluorescent protein and beta-galactosidase), proteins not classified in any of pathway encompassing cell growth, division, migration, survival or apoptosis by GeneOntology reference, or ubiquitous housekeeping proteins. Reagents in the kit may be provided in individual containers or as mixtures of two or more reagents in a single container. In addition, instructional materials which describe the use of the compositions within the kit can be included.

The term "neoadjuvant therapy" refers to a treatment given before the primary treatment. Examples of neoadjuvant therapy can include chemotherapy, radiation therapy, and hormone therapy. For example, in treating breast cancer, neoadjuvant therapy can allow patients with large breast cancer to undergo breast-conserving surgery.

The "normal" level of expression of a biomarker is the level of expression of the biomarker in cells of a subject, e.g., a human patient, not afflicted with a cancer. An "over-expression" or "significantly higher level of expression" of a biomarker refers to an expression level in a test sample that is greater than the standard error of the assay employed to assess expression, and is preferably at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more higher than the expression activity or level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples. A "significantly lower level of expression" of a biomarker refers to an expression level in a test sample that is at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more lower than the expression level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples.

An "over-expression" or "significantly higher level of expression" of a biomarker refers to an expression level in a test sample that is greater than the standard error of the assay employed to assess expression, and is preferably at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more higher than the expression activity or level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples. A "significantly lower level of expression" of a biomarker refers to an expression level in a test sample that is at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more lower than the expression level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples.

The term "pre-determined" biomarker amount and/or activity measurement(s) may be a biomarker amount and/or activity measurement(s) used to, by way of example only,

evaluate a subject that may be selected for a particular treatment, evaluate a response to a treatment such as an anti-immune checkpoint inhibitor therapy, and/or evaluate the disease state. A pre-determined biomarker amount and/or activity measurement(s) may be determined in populations of patients with or without cancer. The pre-determined biomarker amount and/or activity measurement(s) can be a single number, equally applicable to every patient, or the pre-determined biomarker amount and/or activity measurement(s) can vary according to specific subpopulations of patients. Age, weight, height, and other factors of a subject may affect the pre-determined biomarker amount and/or activity measurement(s) of the individual. Furthermore, the pre-determined biomarker amount and/or activity can be determined for each subject individually. In one embodiment, the amounts determined and/or compared in a method described herein are based on absolute measurements. In another embodiment, the amounts determined and/or compared in a method described herein are based on relative measurements, such as ratios (e.g., serum biomarker normalized to the expression of a housekeeping or otherwise generally constant biomarker). The pre-determined biomarker amount and/or activity measurement(s) can be any suitable standard. For example, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from the same or a different human for whom a patient selection is being assessed. In one embodiment, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from a previous assessment of the same patient. In such a manner, the progress of the selection of the patient can be monitored over time. In addition, the control can be obtained from an assessment of another human or multiple humans, e.g., selected groups of humans, if the subject is a human. In such a manner, the extent of the selection of the human for whom selection is being assessed can be compared to suitable other humans, e.g., other humans who are in a similar situation to the human of interest, such as those suffering from similar or the same condition(s) and/or of the same ethnic group.

The term "predictive" includes the use of a biomarker nucleic acid and/or protein status, e.g., over- or under-activity, emergence, expression, growth, remission, recurrence or resistance of tumors before, during or after therapy, for determining the likelihood of response of a cancer to anti-immune checkpoint treatment (e.g., therapeutic antibodies against CTLA-4, PD-1, PD-L1, and the like). Such predictive use of the biomarker may be confirmed by, e.g., (1) increased or decreased copy number (e.g., by FISH, FISH plus SKY, single-molecule sequencing, e.g., as described in the art at least at *J. Biotechnol.*, 86:289-301, or qPCR), overexpression or underexpression of a biomarker nucleic acid (e.g., by ISH, Northern Blot, or qPCR), increased or decreased biomarker protein (e.g., by IHC), or increased or decreased activity, e.g., in more than about 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 100%, or more of assayed human cancers types or cancer samples; (2) its absolute or relatively modulated presence or absence in a biological sample, e.g., a sample containing tissue, whole blood, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, or bone marrow, from a subject, e.g. a human, afflicted with cancer; (3) its absolute or relatively modulated presence or absence in clinical subset of patients with cancer (e.g., those responding to a particular immune checkpoint therapy or those developing resistance thereto).

The term "pre-malignant lesions" as described herein refers to a lesion that, while not cancerous, has potential for

becoming cancerous. It also includes the term "pre-malignant disorders" or "potentially malignant disorders." In particular this refers to a benign, morphologically and/or histologically altered tissue that has a greater than normal risk of malignant transformation, and a disease or a patient's habit that does not necessarily alter the clinical appearance of local tissue but is associated with a greater than normal risk of precancerous lesion or cancer development in that tissue (leukoplakia, erythroplakia, erythroleukoplakia lichen planus (lichenoid reaction) and any lesion or an area which histological examination showed atypia of cells or dysplasia.

The terms "prevent," "preventing," "prevention," "prophylactic treatment," and the like refer to reducing the probability of developing a disease, disorder, or condition in a subject, who does not have, but is at risk of or susceptible to developing a disease, disorder, or condition.

The term "probe" refers to any molecule which is capable of selectively binding to a specifically intended target molecule, for example, a nucleotide transcript or protein encoded by or corresponding to a biomarker nucleic acid. Probes can be either synthesized by one skilled in the art, or derived from appropriate biological preparations. For purposes of detection of the target molecule, probes may be specifically designed to be labeled, as described herein. Examples of molecules that can be utilized as probes include, but are not limited to, RNA, DNA, proteins, antibodies, and organic molecules.

The term "prognosis" includes a prediction of the probable course and outcome of cancer or the likelihood of recovery from the disease. In some embodiments, the use of statistical algorithms provides a prognosis of cancer in an individual. For example, the prognosis can be surgery, development of a clinical subtype of cancer (e.g., solid tumors, such as lung cancer, melanoma, and renal cell carcinoma), development of one or more clinical factors, development of intestinal cancer, or recovery from the disease.

The term "response to immune checkpoint therapy" relates to any response of the hyperproliferative disorder (e.g., cancer) to an immune checkpoint therapy, such as immune checkpoint therapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant chemotherapy. Hyperproliferative disorder response may be assessed, for example for efficacy or in a neoadjuvant or adjuvant situation, where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or palpation. Responses may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or in a qualitative fashion like "pathological complete response" (pCR), "clinical complete remission" (cCR), "clinical partial remission" (cPR), "clinical stable disease" (cSD), "clinical progressive disease" (cPD) or other qualitative criteria. Assessment of hyperproliferative disorder response may be done early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed. This is typically three months after initiation of neoadjuvant therapy. In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of

patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6 months out from the end of therapy. The shorthand for this formula is $CBR=CR+PR+SD$ over 6 months. In some embodiments, the CBR for a particular cancer therapeutic regimen is at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more. Additional criteria for evaluating the response to cancer therapies are related to "survival," which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); "recurrence-free survival" (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence. For example, in order to determine appropriate threshold values, a particular cancer therapeutic regimen can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any cancer therapy. The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following cancer therapy for whom biomarker measurement values are known. In certain embodiments, the doses administered are standard doses known in the art for cancer therapeutic agents. The period of time for which subjects are monitored can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of a cancer therapy can be determined using well-known methods in the art, such as those described in the Examples section.

The term "resistance" refers to an acquired or natural resistance of a cancer sample or a mammal to a cancer therapy (i.e., being nonresponsive to or having reduced or limited response to the therapeutic treatment), such as having a reduced response to a therapeutic treatment by 25% or more, for example, 30%, 40%, 50%, 60%, 70%, 80%, or more, to 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more. The reduction in response can be measured by comparing with the same cancer sample or mammal before the resistance is acquired, or by comparing with a different cancer sample or a mammal who is known to have no resistance to the therapeutic treatment. A typical acquired resistance to chemotherapy is called "multidrug resistance." The multidrug resistance can be mediated by P-glycoprotein or can be mediated by other mechanisms, or it can occur when a mammal is infected with a multi-drug-resistant microorganism or a combination of microorganisms. The determination of resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician, for example, can be measured by cell proliferative assays and cell death assays as described herein as "sensitizing." In some embodiments, the term "reverses resistance" means that the use of a second agent in combination with a primary cancer therapy (e.g., chemotherapeutic or radiation therapy) is able to produce a significant decrease in

tumor volume at a level of statistical significance (e.g., $p<0.05$) when compared to tumor volume of untreated tumor in the circumstance where the primary cancer therapy (e.g., chemotherapeutic or radiation therapy) alone is unable to produce a statistically significant decrease in tumor volume compared to tumor volume of untreated tumor. This generally applies to tumor volume measurements made at a time when the untreated tumor is growing log rhythmically.

The terms "response" or "responsiveness" refers to an anti-cancer response, e.g. in the sense of reduction of tumor size or inhibiting tumor growth. The terms can also refer to an improved prognosis, for example, as reflected by an increased time to recurrence, which is the period to first recurrence censoring for second primary cancer as a first event or death without evidence of recurrence, or an increased overall survival, which is the period from treatment to death from any cause. To respond or to have a response means there is a beneficial endpoint attained when exposed to a stimulus. Alternatively, a negative or detrimental symptom is minimized, mitigated or attenuated on exposure to a stimulus. It will be appreciated that evaluating the likelihood that a tumor or subject will exhibit a favorable response is equivalent to evaluating the likelihood that the tumor or subject will not exhibit favorable response (i.e., will exhibit a lack of response or be non-responsive).

An "RNA interfering agent" as used herein, is defined as any agent which interferes with or inhibits expression of a target biomarker gene by RNA interference (RNAi). Such RNA interfering agents include, but are not limited to, nucleic acid molecules including RNA molecules which are homologous to the target biomarker gene of the present invention, or a fragment thereof, short interfering RNA (siRNA), and small molecules which interfere with or inhibit expression of a target biomarker nucleic acid by RNA interference (RNAi).

"RNA interference (RNAi)" is an evolutionally conserved process whereby the expression or introduction of RNA of a sequence that is identical or highly similar to a target biomarker nucleic acid results in the sequence specific degradation or specific post-transcriptional gene silencing (PTGS) of messenger RNA (mRNA) transcribed from that targeted gene (see Coburn, G. and Cullen, B. (2002) *J. of Virology* 76(18):9225), thereby inhibiting expression of the target biomarker nucleic acid. In one embodiment, the RNA is double stranded RNA (dsRNA). This process has been described in plants, invertebrates, and mammalian cells. In nature, RNAi is initiated by the dsRNA-specific endonuclease Dicer, which promotes processive cleavage of long dsRNA into double-stranded fragments termed siRNAs. siRNAs are incorporated into a protein complex that recognizes and cleaves target mRNAs. RNAi can also be initiated by introducing nucleic acid molecules, e.g., synthetic siRNAs or RNA interfering agents, to inhibit or silence the expression of target biomarker nucleic acids. As used herein, "inhibition of target biomarker nucleic acid expression" or "inhibition of marker gene expression" includes any decrease in expression or protein activity or level of the target biomarker nucleic acid or protein encoded by the target biomarker nucleic acid. The decrease may be of at least 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 99% or more as compared to the expression of a target biomarker nucleic acid or the activity or level of the protein encoded by a target biomarker nucleic acid which has not been targeted by an RNA interfering agent.

The term "sample" used for detecting or determining the presence or level of at least one biomarker is typically whole blood, plasma, serum, saliva, urine, stool (e.g., feces), tears,

and any other bodily fluid (e.g., as described above under the definition of “body fluids”), or a tissue sample (e.g., biopsy) such as a small intestine, colon sample, or surgical resection tissue. In certain instances, the method of the present invention further comprises obtaining the sample from the individual prior to detecting or determining the presence or level of at least one marker in the sample.

The term “sensitize” means to alter cancer cells or tumor cells in a way that allows for more effective treatment of the associated cancer with a cancer therapy (e.g., anti-immune checkpoint, chemotherapeutic, and/or radiation therapy). In some embodiments, normal cells are not affected to an extent that causes the normal cells to be unduly injured by the immune checkpoint therapy. An increased sensitivity or a reduced sensitivity to a therapeutic treatment is measured according to a known method in the art for the particular treatment and methods described herein below, including, but not limited to, cell proliferative assays (Tanigawa N, Kern D H, Kikasa Y, Morton D L, *Cancer Res* 1982; 42: 2159-2164), cell death assays (Weisenthal L M, Shoemaker R H, Marsden J A, Dill P L, Baker J A, Moran E M, *Cancer Res* 1984; 94: 161-173; Weisenthal L M, Lippman M E, *Cancer Treat Rep* 1985; 69: 615-632; Weisenthal L M, In: Kaspers G J L, Pieters R, Twentymann P R, Weisenthal L M, Veerman A J P, eds. *Drug Resistance in Leukemia and Lymphoma*. Langhorne, P A: Harwood Academic Publishers, 1993: 415-432; Weisenthal L M, *Contrib Gynecol Obstet* 1994; 19: 82-90). The sensitivity or resistance may also be measured in animal by measuring the tumor size reduction over a period of time, for example, 6 month for human and 4-6 weeks for mouse. A composition or a method sensitizes response to a therapeutic treatment if the increase in treatment sensitivity or the reduction in resistance is 25% or more, for example, 30%, 40%, 50%, 60%, 70%, 80%, or more, to 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more, compared to treatment sensitivity or resistance in the absence of such composition or method. The determination of sensitivity or resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician. It is to be understood that any method described herein for enhancing the efficacy of a cancer therapy can be equally applied to methods for sensitizing hyperproliferative or otherwise cancerous cells (e.g., resistant cells) to the cancer therapy.

The term “synergistic effect” refers to the combined effect of two or more anti-immune checkpoint agents can be greater than the sum of the separate effects of the anticancer agents alone.

“Short interfering RNA” (siRNA), also referred to herein as “small interfering RNA” is defined as an agent which functions to inhibit expression of a target biomarker nucleic acid, e.g., by RNAi. An siRNA may be chemically synthesized, may be produced by in vitro transcription, or may be produced within a host cell. In one embodiment, siRNA is a double stranded RNA (dsRNA) molecule of about 15 to about 40 nucleotides in length, preferably about 15 to about 28 nucleotides, more preferably about 19 to about 25 nucleotides in length, and more preferably about 19, 20, 21, or 22 nucleotides in length, and may contain a 3' and/or 5' overhang on each strand having a length of about 0, 1, 2, 3, 4, or 5 nucleotides. The length of the overhang is independent between the two strands, i.e., the length of the overhang on one strand is not dependent on the length of the overhang on the second strand. Preferably the siRNA is capable of promoting RNA interference through degradation or specific post-transcriptional gene silencing (PTGS) of the target messenger RNA (mRNA).

In another embodiment, an siRNA is a small hairpin (also called stem loop) RNA (shRNA). In one embodiment, these shRNAs are composed of a short (e.g., 19-25 nucleotide) antisense strand, followed by a 5-9 nucleotide loop, and the analogous sense strand. Alternatively, the sense strand may precede the nucleotide loop structure and the antisense strand may follow. These shRNAs may be contained in plasmids, retroviruses, and lentiviruses and expressed from, for example, the pol III U6 promoter, or another promoter (see, e.g., Stewart, et al. (2003) *RNA* 9:493-501 incorporated by reference herein).

RNA interfering agents, e.g., siRNA molecules, may be administered to a patient having or at risk for having cancer, to inhibit expression of a biomarker gene which is overexpressed in cancer and thereby treat, prevent, or inhibit cancer in the subject.

The term “subject” refers to any healthy animal, mammal or human, or any animal, mammal or human afflicted with a cancer, e.g., lung, ovarian, pancreatic, liver, breast, prostate, and colon carcinomas, as well as melanoma and multiple myeloma. The term “subject” is interchangeable with “patient.”

The term “survival” includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g. time of diagnosis or start of treatment) and end point (e.g. death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.

The term “therapeutic effect” refers to a local or systemic effect in animals, particularly mammals, and more particularly humans, caused by a pharmacologically active substance. The term thus means any substance intended for use in the diagnosis, cure, mitigation, treatment or prevention of disease or in the enhancement of desirable physical or mental development and conditions in an animal or human. The phrase “therapeutically-effective amount” means that amount of such a substance that produces some desired local or systemic effect at a reasonable benefit/risk ratio applicable to any treatment. In certain embodiments, a therapeutically effective amount of a compound will depend on its therapeutic index, solubility, and the like. For example, certain compounds discovered by the methods of the present invention may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

The terms “therapeutically-effective amount” and “effective amount” as used herein means that amount of a compound, material, or composition comprising a compound of the present invention which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment. Toxicity and therapeutic efficacy of subject compounds may be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ and the ED₅₀. Compositions that exhibit large therapeutic indices are preferred. In some embodiments, the LD₅₀ (lethal dosage) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%,

400%, 500%, 600%, 700%, 800%, 900%, 1000% or more reduced for the agent relative to no administration of the agent. Similarly, the ED₅₀ (i.e., the concentration which achieves a half-maximal inhibition of symptoms) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. Also, Similarly, the IC₅₀ (i.e., the concentration which achieves half-maximal cytotoxic or cytostatic effect on cancer cells) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. In some embodiments, cancer cell growth in an assay can be inhibited by at least about 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100%. In another embodiment, at least about a 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100% decrease in a solid malignancy can be achieved.

A “transcribed polynucleotide” or “nucleotide transcript” is a polynucleotide (e.g. an mRNA, hnRNA, a cDNA, or an analog of such RNA or cDNA) which is complementary to or homologous with all or a portion of a mature mRNA made by transcription of a biomarker nucleic acid and normal post-transcriptional processing (e.g. splicing), if any, of the RNA transcript, and reverse transcription of the RNA transcript.

As used herein, the term “unresponsiveness” includes refractivity of immune cells to stimulation, e.g., stimulation via an activating receptor or a cytokine. Unresponsiveness can occur, e.g., because of exposure to immunosuppressants or exposure to high doses of antigen. As used herein, the term “anergy” or “tolerance” includes refractivity to activating receptor-mediated stimulation. Such refractivity is generally antigen-specific and persists after exposure to the tolerizing antigen has ceased. For example, anergy in T cells (as opposed to unresponsiveness) is characterized by lack of cytokine production, e.g., IL-2. T cell anergy occurs when T cells are exposed to antigen and receive a first signal (a T cell receptor or CD-3 mediated signal) in the absence of a second signal (a costimulatory signal). Under these conditions, reexposure of the cells to the same antigen (even if reexposure occurs in the presence of a costimulatory polypeptide) results in failure to produce cytokines and, thus, failure to proliferate. Anergic T cells can, however, proliferate if cultured with cytokines (e.g., IL-2). For example, T cell anergy can also be observed by the lack of IL-2 production by T lymphocytes as measured by ELISA or by a proliferation assay using an indicator cell line. Alternatively, a reporter gene construct can be used. For example, anergic T cells fail to initiate IL-2 gene transcription induced by a heterologous promoter under the control of the 5' IL-2 gene enhancer or by a multimer of the AP1 sequence that can be found within the enhancer (Kang et al. (1992) *Science* 257:1134).

There is a known and definite correspondence between the amino acid sequence of a particular protein and the nucleotide sequences that can code for the protein, as defined by the genetic code (shown below). Likewise, there is a known and definite correspondence between the nucleotide sequence of a particular nucleic acid and the amino acid sequence encoded by that nucleic acid, as defined by the genetic code.

GENETIC CODE

Alanine (Ala, A)	GCA, GCC, GCG, GCT
Arginine (Arg, R)	AGA, ACG, CGA, CGC, CGG, CGT
Asparagine (Asn, N)	AAC, AAT
Aspartic acid (Asp, D)	GAC, GAT
Cysteine (Cys, C)	TGC, TGT
Glutamic acid (Glu, E)	GAA, GAG
Glutamine (Gln, Q)	CAA, CAG
Glycine (Gly, G)	GGA, GGC, GGG, GGT
Histidine (His, H)	CAC, CAT
Isoleucine (Ile, I)	ATA, ATC, ATT
Leucine (Leu, L)	CTA, CTC, CTG, CTT, TTA, TTG
Lysine (Lys, K)	AAA, AAG
Methionine (Met, M)	ATG
Phenylalanine (Phe, F)	TTC, TTT
Proline (Pro, P)	CCA, CCC, CCG, CCT
Serine (Ser, S)	AGC, AGT, TCA, TCC, TCG, TCT
Threonine (Thr, T)	ACA, ACC, ACG, ACT
Tryptophan (Trp, W)	TGG
Tyrosine (Tyr, Y)	TAC, TAT
Valine (Val, V)	GTA, GTC, GTG, GTT
Termination signal (end)	TAA, TAG, TGA

An important and well known feature of the genetic code is its redundancy, whereby, for most of the amino acids used to make proteins, more than one coding nucleotide triplet may be employed (illustrated above). Therefore, a number of different nucleotide sequences may code for a given amino acid sequence. Such nucleotide sequences are considered functionally equivalent since they result in the production of the same amino acid sequence in all organisms (although certain organisms may translate some sequences more efficiently than they do others). Moreover, occasionally, a methylated variant of a purine or pyrimidine may be found in a given nucleotide sequence. Such methylations do not affect the coding relationship between the trinucleotide codon and the corresponding amino acid.

In view of the foregoing, the nucleotide sequence of a DNA or RNA encoding a biomarker nucleic acid (or any portion thereof) can be used to derive the polypeptide amino acid sequence, using the genetic code to translate the DNA or RNA into an amino acid sequence. Likewise, for polypeptide amino acid sequence, corresponding nucleotide sequences that can encode the polypeptide can be deduced from the genetic code (which, because of its redundancy, will produce multiple nucleic acid sequences for any given amino acid sequence). Thus, description and/or disclosure herein of a nucleotide sequence which encodes a polypeptide should be considered to also include description and/or disclosure of the amino acid sequence encoded by the nucleotide sequence. Similarly, description and/or disclosure of a polypeptide amino acid sequence herein should be considered to also include description and/or disclosure of all possible nucleotide sequences that can encode the amino acid sequence.

Finally, nucleic acid and amino acid sequence information for the loci and biomarkers of the present invention (e.g., biomarkers listed in Table 1) are well known in the art and readily available on publicly available databases, such as the National Center for Biotechnology Information (NCBI). For example, exemplary nucleic acid and amino acid sequences derived from publicly available sequence databases are provided below.

SEQ ID NO: 1 Human PBRM1 Transcript Variant 1 cDNA Sequence
(NM_018313.4)

1	cgccgcgcgg	cggaggagc	aatagcagca	cgctgtgcgg	ccacggggcg	gggcgcgcgg
61	gtcgcgtgac	gcggcgcggg	ctcgacggcg	cggagcgctg	ggaagtgtga	ttccatgggt
121	tccaagagaa	gaagagctac	ctccccttc	agcagtgcta	gcggggactt	tgtgatgggt
181	caccattctg	tgtcaacacc	aggcccaagc	agggaaaagga	ggagactttc	caatcttcca
241	actgtagatc	ctatttccgt	gtgccattgaa	ctctataata	ccatccgaga	ctataaggat
301	gaacaggcca	gacttctctg	tgaagtcttc	ataatggcac	caaaagcgag	aaatacaacca
361	gactattatg	aagtgtgttc	tcagcccat	gacttgatga	aaatccaaca	gaaactaaaa
421	atggaagagt	atgatgatgt	ttaatttgetg	actgctgaat	tcagctttct	ttttaacaat
481	gcaaagtctc	atataaagcc	agatttctct	gaatataaag	ccgcttgcaa	actctgggat
541	ttgtaccttc	gaaccaagaaa	tgagtttgtt	cagaaggag	aagcagatga	cgaagatgat
601	gatgaagatg	ggcaagacaa	tcagggcaca	gtgactgaag	gatctttctc	agcttacttg
661	aaggagatcc	tggcagcagt	ttcttgaagc	atagtttgat	ctacaaactc	ctacaggcgt
721	ctcattagcg	aactttttca	gaaactgtct	tctaaagtgc	aatatccaga	ttattatgca
781	ataatttagg	agcctataga	ttccaagacc	atgtcccgca	ggatcacaga	tgggaagcta
841	aaagtgattc	atgcaatgtg	caaaagtata	gatctctctg	caaaaaatgc	caaaactcta
901	aatgagcctg	gctctcaagt	attcaaggat	gc aaattcaa	ttaaaaaaat	attttatatg
961	aaaaaggctg	aaattgaaca	tcatgaatg	gctaagtcaa	gtcttcgaat	gaggactcca
1021	tccaacttgg	ctgcagccag	actgcaggat	ccttcacaca	gtaaaggcag	ccttggtgaa
1081	gagagaaatc	ccactagcaa	gtattaccgt	aataaaagag	cagtacaagg	aggtcgttta
1141	tcagacaatta	caatggcact	tcaaatatgg	tcgaaaagtg	aagaagatgc	tgcctttagt
1201	gctgcacgtg	atgaaggagg	agagtgcaga	ccagaagaca	tcactctct	tatgatgttt
1261	tcaaatcctt	tttatcagct	ttatgacaca	gttaggagtt	gtcggaataa	ccaagggcag
1321	ctaatagctg	aaccttttta	ccatttgect	tcaaaagaaa	aatacctgca	ttattaccag
1381	caaatataaa	tgcctatctc	actacaaag	atccgaacaa	caactgaaga	tcaagaatat
1441	gaaactttag	atcatttgga	gtgtgatctg	aatttaagt	ttgaaaaatg	caaacgctat
1501	aatgtgccca	attcagccat	ctacaagcga	gttctaaaat	tgcagcaagt	tatgcaggca
1561	aagaagaaga	agcttgcag	gagagacgat	atcgagacg	gagacagct	gatctctcca
1621	gccacctctg	atactggtag	tgc caaaga	aaaagtaaaa	agaacataag	aaagcagcga
1681	atgaaattct	tattcaattg	tgtcttgtaa	cgctcagagc	caggttcagg	cagaagactt
1741	tgtcacctat	ttatggttaa	acctccaaa	aggactatc	ctgattatta	taaaatcatc
1801	ttggagccaa	tggactttgaa	aataattgag	cataacatcc	gcaatgacaa	atatgctggt
1861	gaagagggaa	tgtatagaaga	ctagaagtgt	atgttccgga	atgccaggca	ctataatgag
1921	gagggctccc	agbtttataa	tgatgcacat	atcctcgaga	aggtatactca	ggagaaaagg
1981	aaagagctgg	gcccaactgcc	tgatgatgat	gacatggctt	ctcccaact	caagctgagt
2041	aggaagagtg	gcattttctc	taaaaaacta	aaatacatga	ctccaatgca	gcgaaactca
2101	aatgaggtct	atgaagctgt	aaaaaactat	actgataaga	ggggctgcgc	ctcagtgccc
2161	atatttctga	ggcttccctc	tagatctgag	ttgcttgact	actatctgac	tattaaaaag
2221	cccatggaca	tggaaaaaat	tccaagtca	atgatggcca	acaagtacca	agatattgac
2281	tctatgggtg	aggactttgt	catgatgttt	aaataatgc	gtacatacca	tgagccggag
2341	tctttgatct	acaaagatgc	tctttgtcta	cacaaagtc	tgcttgaac	acgcagagac
2401	ctggagggag	atagggactc	tcatgtccca	aatgtgactt	tgctgatcca	agagcttacc
2461	cacaatcttt	ttgtgtcagt	catgagtcat	caggatgatg	agggaaagatg	ctacagcat
2521	tctttagcag	aaattcctgc	tgttgatccc	actatttcta	acaaaccacc	ccttcaactt
2581	gacataatta	ggaagaatgt	tgaaaatgt	cgctaccgtc	gggttgattt	atttcaagag
2641	catatgtttg	aaagtattgga	acagacaa	aggatgaatc	ggacagattt	agaaatatat
2701	gaagatgcag	tagaagactt	cgagttttt	attaaaatct	gtgatgaact	ctgcaaaaaa
2761	ggagagatgc	ttctttccac	ggcactcagc	tataccacaa	aacatttgca	taatgtatgt
2821	gagaaagaga	gaaggaaaaa	attgcca aaa	gaaatagaga	aagataaact	aaaacgagaa
2881	gaagaaaaaa	gagaagctga	aaagagtgaa	gattctctcg	gtgctgcagg	cctctcaggc
2941	ttacatcgca	cataacgcga	ggactgtagc	tttaaaacaa	gcatgtacca	tgttggagat
3001	tatgctctat	tgaaactctg	agaggccaa	ctacaaccac	atatcgctgt	tattgaaaga
3061	ctgtgggagg	attcagctga	aaaagaagtt	tttaagagtg	actattacaa	caaagttcca
3121	gttagataaa	ttctaggcaa	gtgtgtgtgt	atgtttgtca	aggaataact	taagttatgt
3181	gcagaaaaat	tcagagatga	ggatgtttt	gttctgtaat	cacggtattc	tgcgaaaaac
3241	aaatctttta	agaaaattaa	actgtggacc	atgcccatca	gctcagctag	gtttgtccct
3301	cgggatgtgc	ctctgctgtg	ggcttcgctg	gctctgtat	ttgcaaatgc	agataaaaggt
3361	gatgatgaga	agaaatcaga	ccactcagag	gcagctcgag	ctgaagacaa	ttttaaactt
3421	gaaaaggaaa	aagaagatgt	ccctgtgga	atgtccaatg	gtgaagcag	ttgcactac
3481	tttgagcaga	tccattacaa				

TABLE 1-continued

4681	acactggcag	ctcgaagacg	cgacgtccat	ttgtcgaaag	aacaggagag	cgcctaccc
4741	tctcactggc	tgaagacaa	aggggccac	accaccatgg	cagatgccct	ctggcgctt
4801	cgagatttga	tgctccggga	cacctcaac	attcgccaag	catacaacct	agaaaaatgtt
4861	taatcacatc	attacgatttc	ttttatatag	aagcataaag	agttgtggat	cagtagccat
4921	tttagttact	gggggtgggg	ggaaggaaca	aaggaggata	atttttattg	cattttactg
4981	tacatcacia	ggccattttt	atatacggac	acttttaata	agctatttca	atttgtttgt
5041	tatattaagt	tgactttatc	aaatacacia	agattttttt	gcataatggt	ccttcgttta
5101	aaaccagttt	cataattggt	tgtatatgta	gacttggagt	tttatctttt	tacttgttgc
5161	catggaactg	aaaccattag	agggttttgt	cttggcttgg	ggtttttgtt	ttcttggttt
5221	tgggtttttt	tatataata	tataaaagaa	caaaatgaaa	aaaaacacac	acacacaaga
5281	gtttacagat	tagtttaaat	tgataatgaa	atgtgaagtt	tgctcagtt	tacatcttag
5341	agaggggagt	atacttgtgt	ttgtttcatg	tgccatgaata	tcttaagcca	ctttctgcaa
5401	aagctgtttc	ttacagatga	agtgctttct	ttgaaaggtg	gttatttagg	ttttagatgt
5461	ttaatagaca	cagcacattt	gctctattaa	ctcagaggct	cactacagaa	atatgtaatc
5521	agtgtgtgtc	atctgtctgc	agctaagtga	cctcctggac	accaggaggg	gaaaaagcac
5581	tttttcaatt	tgctgtgagt	agacatctgt	gagttagact	atgggtctag	tgatttttgc
5641	agaacacgtg	cacaacctg	aggatgttt	aatctaggca	ggtacgttta	aggatatttt
5701	gatctattta	taatgaattc	acaatttatg	cctataaatt	tcagatgatt	taaaatttta
5761	aacctgtttc	tgcgaaaaac	attgaagttc	gtcttgaaga	aagcattaag	gtatgcattg
5821	aggtgattta	tttttaaa	taacacctaa	cctaaccatg	gtaagagagt	atggaactag
5881	atatgagctg	tataaagaag	ataattgtga	acaagtagat	tgattgcctt	catatacaag
5941	tatgtttttg	tattccttat	ttccttatta	tcagatgtat	tttttctttt	aagtttcaat
6001	gttgtttata	ttctcaacca	gaaatttaat	actttctaaa	atatttttta	aatttagctt
6061	gtgtttttga	attacaggag	aagggatca	taattttaata	aaacgcttac	tagaagacc
6121	attacagatc	ccaaacactt	gggtttgtgt	accctgtctt	tcttatatga	cctacaata
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6241	tttttgaaac	ctgctttata	gtaactgggt	gattgccatt	gtggtagagc	ttccactgct
6301	gtttataatc	tgagagagtt	aatctcagag	gatgcttttt	tccttttaat	ctgctagaa
6361	tcagtaccca	gatgtttaat	tactgtactt	attaatcat	gagggcaaaa	gagtgtagaa
6421	tggaaaaaag	tctctgtgat	ctagatactt	taaatatggg	aggcccttta	acttaattgc
6481	ctttagtcaa	ccactggatt	tgaatttgca	tcaagtattt	taaaataat	tgaatttaaa
6541	aaaatgtatt	gcagtagtgt	gtcagtacct	tattgttaaa	gtgagtcaga	taaatcttca
6601	attcctggct	atttgggcaa	ttgaatcatc	atggactgta	taatgcaatc	agattatttt
6661	gtttctagac	atccttgaat	tacaccaaa	aacatgaaat	ttagtgtggt	ttaaatattt
6721	tatttatttc	atgcattcat	tttatttccc	ttaaggtctg	gatgagactt	ctttggggag
6781	cctctaaaaa	aatttttcac	tgggggccac	gtgggtcatt	agaagccaga	gctctcctcc
6841	aggctccttc	ccagtgccca	gaggtgctat	aggaacata	gatccagcca	ggggcttccc
6901	taaagcagtg	cagcaccggc	ccagggcac	actagacagg	ccctaattaa	gtttttttta
6961	aaaagcctgt	gtatttattt	tagaatcatg	tttttctgta	tattaaactg	ggggatctgc
7021	ttaatattta	ttgataaaga	tttgaggcca	gccatcttca	aaaaagaaaa	aaaaattgac
7081	tcaagaaagt	acaagtaaac	tatacacctt	tttttcataa	gttttaggaa	ctgtagtaat
7141	gtggcttaga	aagtataatg	gcctaaatgt	tttcaaaatg	taagttcctg	tggagaagaa
7201	ttgtttatat	tgcataacgg	gggactgagg	ggaaacctga	ggtttaaaac	agtatgtttg
7261	tcagccaaat	gattttaaag	gcctttaact	gttttggttg	ttgttttttt	tttaagccac
7321	tctccccctc	ctatgaggaa	gaattgagag	gggcacctat	tctgttaaaa	tccccaaatt
7381	gggtgttgat	attttgagct	tgaatgtttt	catacctgat	taaaacttgg	tttattctaa
7441	ttttctgtatc	atatcatctg	aggtttacgt	ggttaactagt	cttataacat	gtatgtatct
7501	tttttttgtt	gttcctctaa	agctttttta	tccaaataaa	tacagagttt	gcaagtgat
7561	ttggattaac					

SEQ ID NO: 2 Human PBRM1 Variant 1 Amino Acid Sequence (NP_060783.3)

1	mgskrrrats	psssvsgdfd	dghhsvstpg	psrkrrrlsn	lptvdpiavc	helyntirdy
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121	nnaksyykpd	speykaackl	wdlylrtrne	fvqkgeadde	dddedggdnq	gtvtgsspa
181	ylkeileql	eaivvatnps	grliselfqk	lpskvqypdy	yaiikepidl	ktiaqriqng
241	syksihamak	di1llaknak	tynepgsqvf	kdansikkif	ymkkaeiehh	emaksslrnr
301	tpsnlaaarl	tgpsshskgs	geernptsky	yrnkraavgg	rlsaitmalq	ygseesedaa
361	laaaryeege	seasitsfsm	dvsnpfygly	dtvrscrnnq	ggliapfyh	ipskkkypdy
421	yqqikmpisl	qqirtklknq	eyetldhle	dlnlmfenak	rynvpnsaiy	krvlklqqvm
481	qakkkelarr	ddiedgdsml	ssatsdtgsa	krkskknirk	qrmkilfnvv	learepgsgr
541	rlcdlfmvpk	skkdydpdyk	iilepmdlki	iehnirndky	ageegmiedm	klmfrnarhy
601	neegsqvynd	ahileklle	krkelgplpd	dddmaspklk	lsrksgispk	kskymtpmqg
661	klnevyeavk	nytdkrgrrl	saiflrlpsr	selpdyylti	kkpmdmekir	shmmankyqd
721	idsmvedfvm	mfnnaactyne	pesliykda	vlhkvlletr	rdlegdedsh	vpnvntllige
781	lihnlfsvsm	shqdegrcy	sds1aeipav	dnpfnkpppl	tfdiirknve	nnryrrldlf
841	qehmfefler	arrrnrttse	iyedavelqq	ffikirdelc	kngeillspa	lsyttkhlhn
901	dvekerkeal	pkeieedklk	reeekreaek	sedssgaagl	sglhrtyeqd	csfknsmyhf
961	gdyvyvepae	anlqphivci	erlwedsaek	evfksdyynk	vpvskilgk	vvmfvkeyfk
1021	lcpenfdrdd	vfvcesrysa	ktksfkklkl	wtmpissvrf	vprdvplpvv	rvasvfanad
1081	kgddekntdn	sedsraednf	nlekekedvp	vemsngpgc	hyfeqlhynd	mwlvvgdcvf
1141	ikshglvrpr	vgriekvwvr	dgaayfygpi	fihpheete	ptkmfykkev	flsnleetc
1201	mtcilgkav	lsfkdf1scr	pteipendil	icesrynesd	qkmkfkglk	rfs1sakvvd
1261	deiyfyrkpi	vpqkpspl1	ekkiqlleak	faeleggddd	ieemgeedse	stpk1sakgsa
1321	kkegskrkln	msgy1lfss	mravikaqhp	dysfgelsrl	vgtewrnllet	akkaeyegmm
1381	ggyppglppl	qgpvdglvsm	gsmqplhpgg	ppphlpppgv	pgipgippgg	vmnqgvapmv
1441	gtpaggsy	gqgv1lgpp	gqgapppyg	phpagppviq	qpttpmfvap	pk1tqrllhs
1501	eaylkyiegl	saesnsiskw	dq1laarrdd	vh1skeqesr	lpshwlkskg	ahttmadalw
1561	rlrd1m1rdd	ln1rqaynle	nv			

TABLE 1-continued

SEQ ID NO: 3 Human PBRM1 Transcript Variant 2 cDNA Sequence
(NM_181042.4)

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121	agggggcagt	tgggcaggca	tgccctattc	ctggaattga	cagtcattcc	taataagttg
181	gattccatgg	gttccaagag	aagaagagct	acctcccctt	ccagcagtg	cagcggggac
241	tttgcagatg	ggcaccattc	tgtgtcaaca	ccaggcccaa	gcaggaaaag	gaggagactt
301	tccaatcttc	caactgtaga	tcctattgcc	gtgtgccatg	aactctataa	taccatccga
361	gactataaag	atgaacaggg	cagacttctc	tgtgagctct	tcattagggc	accaaagcga
421	agaaatcaac	cagactatta	tgaagtgggt	tctcagccca	ttgacttgat	gaaaatccaa
481	cagaaactaa	aaatggaaga	gtatgatgat	gttaatttgc	tgactgctga	cttcagctt
541	ctttttaaca	atgcaaaagt	ctattataag	ccagattctc	ctgaatataa	agccgcttgc
601	aaactctggg	atttgtacct	tcgaacaaga	aatgagtttg	ttcagaaaag	agaagcagat
661	gacgaagatg	atgatgaaga	tgggcaagac	aatcagggca	cagtgaactga	aggatcttct
721	ccagcttact	tgaaggagat	cctggagcag	cttcttgaag	ccatagttgt	agctacaaat
781	ccatcaggag	gtctccattg	cgaacttttt	cagaaactgc	cttctaaagt	gcaatatcca
841	gattattatg	caataattaa	ggagcctata	gatctcaaga	ccattgcccc	gaggatacag
901	aatgggaagt	acaaaagtat	tcatgcaatg	gccaaagata	tagatctcct	cgcaaaaaat
961	gccaaaactt	ataatgagcc	tggctctcaa	gtattcaagg	atgcaaattc	aattaaaaaa
1021	atattttata	tgaaaaaggc	tgaattgaa	catcatgaaa	tggctaagtc	aagtcttcga
1081	atgaggactc	catccaactt	ggctgcagcc	agactgcagc	gtccttcaca	cagtaaaaggc
1141	agccttggtg	aagagagaaa	tcccactagc	aagtattacc	gtataaaaag	agcagtacaa
1201	ggaggtcggt	tatcagcaat	tacaatggca	cttcaatgat	gctcagaaaag	tgaagaagat
1261	gctgctttag	ctgctgcacg	ctatgaagag	ggagagtcag	aagcagaaaag	catcacttcc
1321	tttatggatg	tttcaaatcc	tttttatcag	ctttatgaca	cagttaggag	ttgtcggaat
1381	aaccaagggc	agctaatagc	tgaacctttt	taccatttgc	cttcaaaaga	aaaataccct
1441	gattattacc	agcaaatata	aatgcccata	tcactacaac	agatccgaac	aaaactgaag
1501	aatcaagaat	atgaaccttt	agatcatttg	gagtggtatc	tgaatttaat	gtttgaaaat
1561	gccaaacgct	ataatgtgcc	caattcagcc	atctacaagc	gagttctaaa	attgcagcaa
1621	gttatgcagg	caaagaagaa	agagcttgcc	aggagagacg	atatcgagga	cggagacagc
1681	atgatctctt	cagccacctc	tgatactggt	agtgccaaaa	gaaaaagtaa	aaagaacata
1741	agaaagcagc	gaatgaaaat	cttattcaat	gttgttcttg	aagctcgaga	gccagggttca
1801	ggcagaagac	tttgtgacct	atttatggtt	aaaccatcca	aaaaggacta	tcttgattat
1861	tataaaatca	ttctggagcc	aatggacttg	aaaataattg	agcataacat	ccgcgaatgac
1921	aaatatgctg	gtgaagaggg	aatgatagaa	gacatgaagc	tgatgttccg	gaatgccagg
1981	cactataaat	aggaggggtc	ccaggtttat	aatgatgcac	atatcctgga	gaagttactc
2041	aaggagaaaa	ggaaagagct	gggcccactg	cctgatgatg	atgacatggc	ttctcccaaa
2101	ctcaagctga	gtagggaagag	tggcatttct	cctaaaaaat	caaaatacat	gactccaatg
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2221	cgctcagtg	ccatatttct	gaggcttccc	tctagatctg	agttgcttga	ctactctctg
2281	actattaaaa	agcccatgga	catggaaaaa	attcgaagtc	acatgatggc	caacaagtac
2341	caagatattg	actctatggt	tgaggacttt	gtcatgatgt	ttataaatgc	ctgtacatcc
2401	aatgagccgg	agtctttgat	ctacaaagat	gctcttggtc	tacacaaagt	cctgcttgaa
2461	acacgcagag	acctggaggg	agatgaggac	tctcatgtcc	caaatgtgac	tttgctgatt
2521	caagagctta	tccacaatct	ttttgtgtca	gtcatgagtc	atcaggatga	tgagggaaga
2581	tgctacagcg	atcttttagc	agaaaattcc	gctgtggatc	ccaaatttcc	taacaaacca
2641	ccccttacat	ttgacataat	taggaagaat	gttgaaaata	atcgctaccg	tcggcttgat
2701	ttattttcaag	agcatatggt	tgaagtattg	gaacgagcaa	gaaggatgaa	tcggacagat
2761	tcagaaaat	atgaagatgc	agtagaactt	cagcagtttt	ttattaaaa	tcgtgatgaa
2821	ctctgcaaaa	atggagagat	tcttctttca	ccggcactca	gctataccac	aaaacatttg
2881	cataatgatg	tggagaaaag	gagaaaaggaa	aaattgccaa	aagaaataga	ggaagataaaa
2941	ctaaaacgag	aagagaagct	gaaaagagtg	aagattcttc	tgggtgctgca	acatgtgtgc
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3061	catgttggag	attacgtcta	tgtggaacct	gcagaggcca	acctacaacc	acatatcgct
3121	tgtattgaaa	gactgtggga	ggattcagct	gggtgaaaaa	gggtgtatgg	ctgttggttt
3181	taccgaccaa	atgaaacatt	ccacctggct	acacgaaaat	ttctagaaaa	agaagttttt
3241	aagagtgtg	attacaacaa	agttccagtt	agtaaaattc	taggcaagtg	tgtggtcatg
3301	tttgtcaagg	aatactttaa	gttatgcccc	gaaaacttcc	gagatgagga	tgtttttgtc
3361	tgtgaatcac	ggtattctgc	caaaacccaa	tcttttaaga	aaattaaact	gtggaccatg
3421	cccatcagct	cagtcagggt	tgtccctcgg	gatgtgcttc	tgctgtggtg	tcgcgtggcc
3481	tctgtatttg	caaatgcaga	taaagggtgat	gatgagaaga	atcacagaca	ctcagaggac
3541	agtcgagctg	aagacaattt	taacttggaa	aaggaaaaag	aagatgtccc	tgtggaaatg
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3721	attgaaaaag	tatgggttcg	agatggagct	gcatattttt	atggccccat	cttcattcac
3781	ccagaagaaa	cagagcatga	gcccacaaaa	atgttctaca	aaaaagaagt	atttctgagt
3841	aatctggcag	aaaccttgcc	catgacatgt	attctcgga	agtgctgagt	gttgctatct
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3961	agccgctaca	atgagagcga	caagcagatg	aagaaattca	aaggattgaa	gaggttttca
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4321	cggaaaaatca	acatgagttg	ctacatctcg	ttcagcagtg	agatgagggc	tgtgatatag
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4561	cctccagggc	agcaggcacc	acctccatat	cccggccccc	atccagctgg	acccctgttc
4621	atacagcagc	caacaacacc	catgtttgta	gctccccccc	caaagacca	gcggcttctt

TABLE 1-continued

4681	cactcagagg	cctacctgaa	atacattgaa	ggactcagtg	cggagtccaa	cagcattagc
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4801	agccgctac	cctctcactg	gctgaaaagc	aaaggggccc	acaccacccat	ggcagatgcc
4861	ctctggcgcc	ttcgagatgtt	gatgctccgg	gacaccccca	acattcgcca	agcatataac
4921	ctagaaaatg	tttaatacaca	tcattacgtt	tcttttatat	agaagcataa	agagttgtgg
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5221	tttacttggt	gccatgggac	tgaaccattt	agagggtttt	gtcttggtt	gggggttttg
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5341	acacacacaa	gagtttacag	attagtttaa	attgataatg	aaatgtgaag	tttgctctag
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5641	gggaaaaagc	actttttcaa	ttgtgctgag	ttagacatct	gtgagttaga	ctatgggtgc
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5761	taaggatatt	ttgatctatt	tataatgaat	tcacaattta	tgctataaaa	tttcagatga
5821	tttaaaattt	taaacctggt	acattgaaaa	acattgaagt	tcgtcttgaa	gaaagcatta
5881	aggtatgcat	qqaqqtgatt	tatttttttaa	cataaacact	aaactaacat	gggttaagaga
5941	gtatggaaact	agatatgagc	tgtataagaa	gcataattgt	gaacaagtag	attgattgcc
6001	ttcatataca	agtatgtttt	agtattcctt	atttccttat	tatcagatgt	attttttctt
6061	ttaagtttca	atgttgttat	aattctcaac	cagaaattta	atactttcta	aaatattttt
6121	taaatattgc	ttgtgctttt	gaattacagg	agaagggaat	cataatttaa	taaacgcttt
6181	actagaagaa	ccattacaga	tcccaaacac	ttgggtttgg	tgaccctgtc	tttcttatat
6241	gacctacaa	taaacatttg	aaggcagcat	aggatggcag	acagtaggaa	cattgtttca
6301	cttggcgccg	tgtttttgaa	acctgcttta	tagtaactgg	gtgattgcca	ttgtggtaga
6361	gcttccactg	ctgtttataa	tctgagagag	ttaatctcag	aggatgcttt	tttcttttta
6421	atctgctatg	aatcagtagc	cagatgttta	attactgtac	ttatataatc	atgaggggcaa
6481	aagagtgtag	aatggaaaaa	agtctctgtg	atctagatac	tttaaatatg	ggaggccctt
6541	taacttaatt	gccttttagtc	aaccactgga	tttgaatttg	catcaagtat	tttaataaat
6601	attgaattta	aaaaaatgta	ttgcagtagt	gtgtcagtag	ccttattgtta	aagttagtca
6661	gataaatctt	caattctctg	ctatttgggc	aattgaatca	tcattggactg	tataatgcaa
6721	tcagattatt	ttgtttctag	acatccttga	attacaccaa	agaacatgaa	atttagttgt
6781	gggttaatta	tttatttatt	tcattgcattc	attttatttc	ccttaaggctc	tggtatgagac
6841	ttctttgggg	agcctctaaa	aaaatttttc	actgggggccc	acgtgggtca	ttagaagcca
6901	gagctctcct	ccaggctcct	tcccagtgcc	tagagggtct	ataggaaaca	tagatccagc
6961	caggggcttc	cctaaagcag	tgcagcaccc	gcccaggggc	tcactagaca	ggccctaatt
7021	aagttttttt	taaaaagcct	gtgtatttat	tttagaatca	gtttttctg	tatatattct
7081	tggtgggat	cgttaataat	taggatataa	gatttgaggt	cagccatctt	caaaaagaa
7141	aaaaaattg	actcaagaaa	gtacaagtaa	actatacacc	tttttttcat	aagtttttagg
7201	aactgtagta	atgtgtctta	gaaagtataa	tggcctaaat	gttttcaaaa	tgtaagttcc
7261	tgtggagaag	aattgtttat	attgcaaacg	gggggactga	ggggaaacctg	taggtttaaa
7321	acagtatggt	tgtcagccaa	ctgattttaa	aggcctttaa	ctgttttgg	tggtgttttt
7381	tttttaagcc	acctctccct	tcctatgagg	aagaattgag	aggggcacct	atttctgtaa
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7501	gggttattct	aatttctgta	tcataatcat	tgagggtttac	gtggtaacta	gtcttataac
7561	atgtatgtat	cttttttttg	ttgttcactc	aaagcttttt	aatccaaat	

SEQ ID NO: 4 Human PBRM1 Variant 2 Amino Acid Sequence (NP_851385.1)

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121	nnaksyykpd	speykaackl	wdlylrtrne	fvqkgeadde	dddedggdnq	gtvtegsspa
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241	syksihamak	di1llaknak	tynepgsqvf	kdansikkif	ymkkaeiehh	emaksslrnr
301	tpsnlaaarl	tgpsksgsl	geernptsky	yrnkravggg	rlsaitmalq	ygseesedaa
361	laaaryeege	seasitsfm	dvsnpfygly	dtvrscrnnq	gqliaepfyh	lpskkkypdy
421	yqqikmpisl	qqirtklknq	eyetldhle	dlnlmfenak	rynvpnsaiy	krvlklqvm
481	qakkkelarr	ddiedgdsml	ssatsdtgsa	krkskknirk	qrmkilfnvv	learepgsgr
541	rlcdlrmvvp	skkdypdyk	iilepmdlki	iehnirndky	ageegmiedm	klmfrnarhy
601	neegsqvynd	ahileklle	krkelgplpd	dddmaspkkl	lsrksgispk	kskymtpmqg
661	klnevyeavk	nytdkgrrrl	saifrlrpsr	selptyliti	kkpmdmekir	shmmankyqd
721	idsmvedfvm	mfnnaactyne	pesliykda	vlhkvllctr	rdlegdedsh	vnpnvtllige
781	lihnlfvsvm	shqdegrcy	sds1aeipav	dpnfpnkppl	tfdiirknve	nnryrrldlf
841	qehmfefvler	arrrmrttse	iyedavelqq	ffikirdelc	kngeillspa	lsyttkhlhn
901	dvekerkekl	pkeieedklk	reeekreaek	sedssgaagl	sglhrtyeqd	csfknsmyhv
961	gdyvyvpeae	anlqphivci	erlwedsage	kwlygcwfyf	pnetfhlatr	kflekevfks
1021	dyynkvpsvk	ilgkcvvmfv	keyfklcpen	frdedvfvce	srysaktksf	kkiklwtmpi
1081	ssvrfvprdv	plpvrvvasv	fanadkgdde	kntdnsedsr	aednfnleke	kedvpvemsn
1141	gepgchfyef	lhndmwlkv	gdcvfikshg	lvprprvgrie	kwvrvdgaay	fygpfihpe
1201	eteheptkmf	ykkevfslsl	eetcpmtcil	gkcavlsfkd	flscrpteip	endillcesr
1261	ynesdkgmkk	fkg1krfs1s	akvvddeiyy	frkpivpqke	pspllekkiq	lleakfaele
1321	ggdddieemg	eedseviepp	slpqlqtpla	seldimpytp	pqstpsakg	sakkegskrk
1381	inmsgyilfs	semravikaq	hpdysfgels	rlvgtewrnl	etakkaeyeg	vmnqgvapmv
1441	gtpapggspy	ggqgv1lgpp	ggqapppypg	phpagppviq	qpttpmfvap	ppktqrl1hs
1501	eaylkyiegl	saesnsiskw	dqtlalarrd	vhlskeqesr	lps1hw1kskg	ahttmadalw
1561	rlrslm1rdr	lnirqaynle	nv			

TABLE 1-continued

SEQ ID	NO: 5 Mouse PBRM1 cDNA Sequence (NM_001081251.1)
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61	gcccgggcca gaggaagcgt agcagccgtg gcggccacgg ggccggggctc ggccgtcggg
121	gaccgcagcc ggggctgcag gcggcggagc ggccggccttg ccaacacttg gtgtcacatg
181	tgagcctccc acatgtgtgc actctccatt ccagctctgt gattgaactc tgcctcttatt
241	gactaggggg cacttgggca ggcattgcttc attcctggag ttgacagtca tttcataaga
301	agttggattc catgggttcc aagagaagaa gagccacctc tcctccagc agtgtcagtg
361	gagactttga tgacgggac cattctgtgc ctacaccagg cccaagcagg aaaaggagaa
421	gactgtccaa tcttccaact gtatgacctt ttgctgtgtg ccatgaactc tataaaccca
481	tccgagacta taaggatgaa cagggcagac tcctctgtga gctgttcatt agggctccaa
541	agcggagaaa tcaaccagac tattatgaag tggtttctca gccattgac ttgatgaaaa
601	tccaacagaa acttaaaatg gaagagtatg atgatgttaa tctactgact gctgacttcc
661	agctgctttt taacaatgca aaggcctact ataagccaga ttccctgag tataaagctg
721	cttgtaactc ctgggatttg taccttcgaa caagaaatga gtttgttcag aaaggagaag
781	cagacgatga agatgatgac gaagatgggc aagacaatca aggcacactg gctgacggct
841	cttctccagg ttatctgaag gagatcctgg agcagcttct tgaagccata gttgtagcca
901	caaatccatc aggcaggctc atcagtgaac ttttctcaga actgccttcc aaagtgaat
961	atccagacta ttatgcaata attaaaggac ctatagatct caagaccatt gctcagagga
1021	tacagaatgg aagctacaaa agtatcacgc caatggccaa agatatagat cttctagcaa
1081	aaaatgccaa aacatacaat gagcctgggt ctcaagtatt caaggatgcc aattcgatta
1141	aaaaaatatt ttatatgaaa aaggcagaaa ttgaacatca tgaatgact aatatcaagt
1201	ttcgaataag gactgcatca aatttggctg cagccaggct gacaggtcct tcgcacaata
1261	aaagcagcct tggatgaagaa agaaacccca ctagcaagta ttaccgtaat aaaagagcag
1321	tccaaagggg tcgcttgcga gcaattacca tggcacttca gtatggatca gagagtgaag
1381	aggacgctgc tttagctgct gcacgctatg aagaagggga atctgaagca gagagcatca
1441	cttctctcat ggaactgttc aacctcttct atcagcttta cgacacagtt aggagctgta
1501	ggaatcacca agggcagctc atagctgaac ctttcttcca tttgccttca aagaaaaaat
1561	accagatgta ttatcagcaa attaaaatgc ccatatcact tcaacagatc agaaacaaagc
1621	taaagaacca agaatatgaa actttagatc atttggagtg tgatctgaat ttaatgtttg
1681	aaaatgccaa acgttataac gttcccaatt cagccatcta taagcgagtt ctaaaactgc
1741	agcaagtcat gacggcaaa gaaagaggag ttgcgaggag agatgacatt gaggcggag
1801	acagcatgat ctctcagcc acttctgaca ctggtagtgc caaaaggaaa aggaatactc
1861	atgacagtga gatgtgggt ctccaggaggc tatccagtaa aaagaacata agaaaaacgc
1921	gaatgaaaaa ttattccaat ttgttcttgg aagctcgaga gccaggttca ggcagaagac
1981	tttgcgatct atttatgggt aagccatcca agaaggacta tcctgattat tataaaatca
2041	tcttagagcc aatggacctg aaaataattg agcataacat ccgaaatgac aaatatgcag
2101	gtgaagaagg aatgatggaa gacatgaaac tcatgttccg caatgccagg cactacaatg
2161	aggaggggct ccaggtatatac aatgatgccc atatcctgga gaagttactc aaagataaaa
2221	ggaaagagct gggccctctg cctgatgatg atgacatggc ttctcccaaa cttaaattga
2281	gtaggaaagc ttgtgtttct cctaagaaat caaagtacat gactccaatg cagcagaaac
2341	tgaatgaagt gtatgaagct gtaaaagaact atactgataa gaggggtcgc cgccttagtg
2401	ctatatcttc aagactcccc tctagatcag agctgcctga ctactacctg accattaaaa
2461	agcccatgga cttatgaaaa attcgaaagtc acatgatggc aaacaagtac caagcatag
2521	attctatggt agaggacttt gtcatgatgt ttaataatgc ctgtacctac aatgaaccag
2581	agtctttgat ctacaagat gcccttgtac tgcataaagt cctccttgag actcggagag
2641	acctggaggg agatggagat tctcatgtcc ctaatgtgac gttgtgatt caagagctca
2701	tccataacct ttttgtgtca gtcatgagtc atcaggatga cgaaggaggg tgttacagcg
2761	actccttagc agaaattcct gctgtggatc ccaactctcc caataaacct ccccttacct
2821	ttgacattat caggaaaaat gttgaaagta atcggatcgc gcgacttgat ttatttcagg
2881	agcatatggt tgaagtattg gaacgggcaa gaaggatgaa ccggacagat tccgaaatat
2941	atgaggatgc tgtagaactt cagcagtttt ttattagaat tctgtgatga ctctgcaaaa
3001	atggagagat ccttctttct ccagcactca gctataccac aaaaactctg cataacgatg
3061	tggaaaaaga aaaaaggaaa aaattgccta aagaaataga ggaagataaa ctaaaacgcg
3121	aagaagaaaa aagagaagct gaaaaaagtg aagattcttc aggtaccaca ggcctctcag
3181	gcttacatct gacgttcagc caggactgca gctttaaaga cagcatgcat catgtcggag
3241	attatgtcta tgttgaacct gcggaggcca atctacaacc acatatagtg tgtattgaga
3301	gactgtggga ggattcagct ggtgaaaaat ggttgtacgg ctgttgggtt tatcggccaa
3361	atgaaacatt ccatttggct acacgaaaaa ttctagaaaa agaagttttt aagagtgaat
3421	actacaataa agtacctgtt agtaaaattc taggcaaatg tgtagtcatg tttgtcaagg
3481	aatactttaa attatgtcca gaaaacttct gcgatgagga tgtttttgtc tgtgaatcga
3541	ggtattctgc caaaacccaa tcttttaaga aaattaaact gtggaccatg cccatcagtt
3601	cagtttagatt tgcctctcgg gatgtgcctt tgctgtgggt ccgagtgagg tctgtgtttg
3661	caaatgcaga taaaggggat gatgagaaga atacagacaa ctacagatgac aatagagctg
3721	aagacaattt taacttggaa aaggaaaaag aagatgttcc tgtggagatg tccaatgggt
3781	agccagggtg ccactacttt gagcagcttc ggtacaatga catgtggctg aaggttgggt
3841	attgtgtctt catcaaaatc cacggcttgg tgcgccctcg tgtgggcaga attgagaaag
3901	tatgggtccg agatggagct gcatattttt atggccctat cttcattcat ccagaagaaa
3961	cagaacatga gccccaaaaa atgttctaca aaaaagaagt gtttctgagt aatctggaag
4021	agacctgccc tatgatttgt attctgggga aatgtgcagt gctgtcattc aaggacttcc
4081	tctctgcag gccaatctga ataccagaaa atgacattct gctttgtgag agccgtata
4141	atgagagtga caagcagatg aagaagttca agggtttgaa gaggttttca ctctctgcta
4201	aagttttaga tgatgaaatc tactacttca gaaaaccaat cattctcag aaggaaacct
4261	caccttgggt agaatctgaa atacaattgc tagaagctaa atttgcagag ttagaaggag
4321	gagatgatga tattgaggag atgggagaa aggatagtga agtcatgaa gctccatctc
4381	tacctcaact gcagacaccc ctggccaatg agttggacct catgcctat acacccccac
4441	agtctacccc aaagcagtg caaagaagga aagttctaaa cgaaaaatca cgaagagag
4501	acatgagtggt ctacattttt ttcagcagtg aaatgagagc tgtgattaaa gccacgacc
4561	cagactactc ttttggggag ctacgacagc tgggtggggc agaattggaga aaccttga
4621	cagcccaagaa agcagaatat gaagagcggg cagctaaagt tgctgagcag caggagagag
4681	agcgagcagc acagcaaacg cagccgagtg cttctccccg agcaggcacc cctgtggggg

TABLE 1-continued

4741	ctctcatggg	gggtgggtcca	ccaccaaacac	caatgggggat	gctcaatcag	cagttgacac
4801	ctgttgccagg	catgatgggt	ggctatccgc	caggccttcc	acctttgcag	ggcccagttg
4861	atggccttgt	tagcatgggc	agcatgcagc	cacttcaccc	tggggggcct	ccacctcacc
4921	atcttcgcc	aggtgtgcct	ggcctcccag	gcattcccacc	accgggtgtg	atgaatcaag
4981	gagtagcccc	catggtaggg	actccagcac	caggtggaag	tccgtatgga	caacaggtag
5041	gagttttggg	acctccaggg	cagcaggcac	cacctccata	tctgtgtcct	catccagctg
5101	gccccctgt	catcacgacg	ccaacaacgc	ccatggttgt	ggctcccca	ccaaagaccc
5161	aaaggcttct	ccactcagag	gcctacctga	aatacattga	aggactcagt	gctgaatcca
5221	acagcatttag	caagtgggac	caaaactttg	cagctcgaa	acgggatgtc	catttgtcca
5281	aagaacagga	gagccgcta	ccttctcact	ggctcaaaag	taaaggggca	cacaccacca
5341	tggcagatgc	cctctggcgc	ctacgggatt	taatgcttcg	agacactctc	aacatccgac
5401	aggcatacaa	cctagaaaat	gtttaatcac	atcactgttt	cttctgtgga	agcaaaaggt
5461	tgtggagcgg	tagccatttt	agttactggg	gtgggaggga	ggaacaaagg	atgataattt
5521	ttattgcatt	ttattgtaca	tcacacagcc	atttttatat	aaggacactt	ttataaagct
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5701	ctttacttgt	taaaggactt	aaatcatcaa	aggttttggc	ttggcttagg	gttttcgttt
5761	tcttttttat	aaatatatat	tatatatata	tacacatata	aaagaaaaaa	tgaaaaaaa
5821	gtttacaaat	ttaagttgac	aatgaaatgt	gaagttgggtc	ctagtttaca	tcttagagga
5881	atgtatatgt	atgttttaca	tgctaaata	tctgcagggtt	ttcttacagg	taaagcgaag
5941	tgctttgaaa	agtttagatt	atcacatgtg	gacagatgcg	gcatttttgc	tctattaaca
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6061	aagggaatg	taactttgtg	ctgagttaga	catctgtatt	gtcagtgatt	cttgtagaat
6121	atgtgctcag	atctgagtta	tatttagttt	tggaaagtaa	gttgaagagt	acttttgatc
6181	agttttatgt	tcagtttatg	attttagttt	ttgcttcat	gttatacat	tatgatttga
6241	aactgtacat	ctgttacctt	gaaaaacatt	gaagaaagta	ctgaagtgtg	catggagggtg
6301	gtttaagcat	aatacttaac	ccaagaaaag	gtgtaagtgg	acacaaagctg	tgctctgcaca
6361	tagctgtgca	gggtagactg	cctacataca	catggccggg	attctttatt	tcttggttat
6421	caatttatagt	gctttgtttg	tttcagggtt	ggaattctca	accagaaata	atactttcta
6481	aaatattttta	aaattcagct	tgtgctttgg	attatagaag	gaaattatac	tttaagaaaa
6541	tgttcacaaa	aaaaaaagac	tattacagat	ccaataact	ggattttggtg	
6601	accttgtctt	tctttctttt	cttgagacat	ggctcacta	ccaacctcgg	ctggactgga
6661	gctcagtgta	tagaccaggc	tagtctcaaa	ctctgcctct	tcttcccaag	tgctgggatt
6721	aaggggcagg	accatagtgc	tcagcaacca	caacctgtc	tttccaacac	ggccctagcg
6781	taagcactga	ggcagtgctg	agtgctcagg	cagcagcaaa	catttcccg	gggtgggttt
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6901	tgattgttaa	atgacactct	tcttttagaa	tataacggat	cagtaactcat	gtttaattgc
6961	catgcttaat	aaatcatgag	aacaaaagag	tatagaatgg	aaagcattcc	ctggtagcta
7021	ctttaaatac	aggagccctg	taacttaata	ccagtagtca	accactggat	ctcagttttc
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7201	tcatcatgga	ctatataatg	taataagatt	cttttctctt	aaggatcct	tgaattacac
7261	caaagaacca	gaaacttaac	tttggttaaa	ttattttatt	atttcatgca	ttatatttct
7321	ttttcttttt	aaaggttttag	atgaggctcc	ttagggagtc	tctaaaaccg	cttctactatc
7381	agcaaccagg	agtactagaa	gccagagcac	tcttctcctc	ggctcctccc	cagtgctcta
7441	gtgctgtagg	aaccaagagc	cagcccccag	ttccccagg	cagtaaaaaa	ccagcacagg
7501	gggctgtgtc	cctaaggcaa	gccttgattt	cctttaaaaa	aaacaaaaaa	aaacaaacaaa
7561	aaaaaaaaac	ctaattaact	aaagcattta	aggcactatt	tatttttagaa	tcatgctttt
7621	gaagagcatc	agtgattact	taggggttaa	tatgtaaaag	tcagacatct	ccaaaaacag
7681	aaaaagtaca	agtaaacac	acactttctc	atgactttta	agaactgtag	taatgtggct
7741	taggaaatat	aattggcctaa	ttgttttcaa	aatgtaagtt	cctgtgaaga	attttgttta
7801	tattgggttg	gggagcctata	ggtttaaaat	agaatgtcag	tcagctgact	taaaaaacat
7861	tggttttact	aagtctgcct	tccccttcta	aggaagaact	gagtgggtaa	gggacagggtg
7921	tgtaaaatct	ccaaattggat	gttacagctt	tcagcttgaa	cgtttgtttc	cagacctgat
7981	taaaatttgg	tttattctaa	tttctgtact	atatcatctg	aggttttaag	tggttaactgg
8041	ttctatacca	tgtatgtatc	atatgtttgt	tcatacaagc	tttttaatcc	aaataaaaaac
8101	aacagtttgc	aaagtga				

SEQ ID NO: 6 Mouse PBRM1 Amino Acid Sequence (NP_001074720.1)

1	mgskrrrats	psssvsgdfd	dghhsvptpg	pskrrrrlsn	lptvdpiavc	helyntirdy
61	kdeggrllce	lfirapkrnn	qpdyevvsq	pidlmkiqk	lkmeeyddvn	lltadfqllf
121	nnakayypkd	speykaackl	wdlylrtrne	fvqkgeadde	dddedggdnq	gtladgsspg
181	ylkeileqil	eaivvatnps	grliselfqk	lpskvqpydy	yaiikepidl	ktiaqriqng
241	syksihamak	diillaknak	tynepgsqvf	kdansikkif	ymkkaeiehh	emtksslrir
301	tasnlaaarl	tgpsnhksl	geernptsy	yrnkravggg	rlsaitmalq	ygseseedaa
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421	yqgikmpisl	qqirtklknq	eyetldhle	dlnlmfenak	rynvpnsaiy	krvlklqgvm
481	qakkkelarr	ddiedgdsmi	ssatsdtgss	krkrnthdse	mlglrrlsk	knirkqrmki
541	lfnvvleare	pgsgrrlcdl	fmvkspskdy	pdyykiilep	mdlkiielni	rndkyageeg
601	mmedmklmfr	naryhneegs	qvyndahile	kllkdkrkel	gplpddddma	spklklrks
661	gvspkkskym	tpmqgklnev	yeavknytdk	rgrrlsailf	rlpsrselpd	yyltikkpmd
721	mekirshmma	nkyqddismv	edfvmmfnda	ctynepesli	ykdaivlhkv	lletrrdleg
781	dedshvpnvt	lliqelihnl	fvsvmshqdd	egrcysdsila	eipavdpnsp	nkppltdfii
841	rknvesnrly	rlldlfqehmf	evlerarrmn	rtdeieyeda	velqqffiri	rdelckngei
901	llspalsytt	khlnhdveke	kkeklpkeie	edklkreec	rsaeksedss	gttglsglhr
961	tsyqdcsfkn	smhyvgdyvy	vepaeanalqp	hivcierlwe	dsagekwlyg	cwfyrypnetf
1021	hlatrkflek	evfksdyynk	vpvskilgkc	vvmfvkeyfk	lcpenfrded	vfvcesrysa
1081	ktksfkki	wtmipissvrf	vprdvplpvv	rvasvfanad	kgddekntdn	sddnraednf
1141	nlekedvyp	vemsngpgc	hyfeqlrynd	mwlkvgdvfv	ikshglvrpr	vgriekvwvr
1201	dgaayfygpi	fihpheethe	ptkmfykkev	flsnleetc	mscilgkcv	isfkdfllscr

TABLE 1-continued

1261	pteipendil	lcesrynesd	kqmkfkfklk	rfslsakvvd	deiiyyfrkpi	ipqkepspl1
1321	ekkiqlleak	faeleggddd	ieemgeedse	vieapslpql	qtplaneidl	mpytpqgstp
1381	ksakgsakke	sskrkinmsg	yilfssemra	vikahpdyd	fgelsrlvgt	ewrnlatakk
1441	aeeyeraakv	aeqgereraa	qqqqpsaspr	agtpvgalmg	vvpptpmmgm	lnqqitpvag
1501	mmggypplp	plggpvdglv	smgsmqplhp	ggppphhlpp	gvpglpigipp	pgvmnqgvap
1561	mvgtpppggs	pygqgvglg	ppgqgapppy	pgphpagppv	iqqpttpmfv	apppktqrll
1621	hseaylkyie	glsaesnsis	kwdqtlarr	rdvhliskee	srlpshwlks	kgahttmada
1681	iwrlrdlmlr	dtlnirqayn	lenv			

SEQ ID NO: 7 Human ARID2 cDNA Sequence Vairant 1 (NM_152641.3, CDS:
from 129 to 5636)

1	ggcccatgac	tgagccccgc	cgccgcccgc	cgaggaacgg	gtcccgggct	ctggtaggaa
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121	aaaaataaat	ggcaaatcgc	acggggaagg	cgctctccga	cgagcggaga	aagggaactcg
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421	aaaagtagca	gaaagttcat	cattttgggg	aggatgatga	tgaggtacca	ccaggcaatc
481	caaagccaca	gcttcctatt	ggtgcaattc	catcttccta	caattaccag	caacacagtg
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901	aaggtacatc	aggagaatgg	atttgggagt	ctttatttca	tccacctcga	aagctgggca
961	ttaacgatat	tgaaggacag	cgggtacttc	agattgcagt	gattttgaga	aatctttcct
1021	ttgaggaggg	caatgttaag	ctcttgccag	ctaactgtac	ctgtcttcgt	ttcctattac
1081	tttctgcaca	tagtcatctt	atttctttta	ggcaattagg	ccttgacaca	ttagaaaata
1141	ttgcagctga	gcttttactg	gacctgtgtg	atttcaaaac	tactcatctg	atgtttcata
1201	ctgttacaaa	atgttctaag	tcaagggata	gattttttaa	gatgagaggc	atggaaattt
1261	tgggaaatct	ttgcaaaagc	gaagataatg	gtgttttaat	ttgtgaatat	gtggatcagg
1321	attcctacag	agagatcatt	tgatcatctc	ctttacctga	tggtgtgctt	gtaatctcaa
1381	cactcgagggt	gctatacatg	ctcacggaaa	tgggagatgt	tgcttgacaca	aaaattgcaa
1441	aagtagaaaa	gagcatagac	atgttagtgt	gtctgggttc	tatggatatt	cagatgtttg
1501	gccctgatgc	actagctgcg	gtaaaactca	ttgaacaccc	aagttccagt	catcaaatgt
1561	tatctgaaat	taggcaccaa	gctatagagc	aagtcacaa	ccagactcat	gtagcatctg
1621	ccccagcttc	cagagcaggt	gtagcgcagc	atgttgcttc	acctccagga	atagtggaag
1681	tagatagtga	gaagtttgct	tgctcagtg	ttaaatgctc	ttttgaagta	aatccagatt
1741	gttctgtttc	tgcagagcaa	atgtattctg	aatacctctc	gacttgcaag	aaattagctc
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1921	gagtaaaacg	gagggctata	ccacttccca	ttcagatgta	ctatcagcag	caaccagttt
1981	ctactttctg	tggtctgtgt	gattctgttc	ctgatgtatc	tctgtctcct	tcacctgcag
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2101	aatcttcaaa	tctgactgca	acacaaatgt	cttttctctg	acaaggtggt	catactgtgg
2161	cacaaaatgt	ttcaagaatt	ccacaaaatc	cttcacctca	tacccaccag	caacaaaatg
2221	ctccagtgac	tgctattcaa	agtaaaagctc	caattccttg	tgaagtgtgt	aaggctacag
2281	ttatccagaa	ttccatcccc	cagacaggag	ttcctgttag	tattgtctgt	ggaggaggac
2341	ctccacagag	ttctgttgtt	cagaatcata	gtacagggcc	acaacctgtt	acagttgtga
2401	attctcagag	attgcttcac	catccatctg	taattccaca	gcagctctca	ttacacacag
2461	tggtaccagg	acagatccct	tcaggcactc	ctgttacagt	aattcaacaa	gctgtcccac
2521	agagtcatat	gtttggcaga	gtacagaaca	taccagcatg	tactttctca	gtttcacagg
2581	gtcaacagtt	aatcacccaca	tcaccccaac	ctgtgcaaac	ttactctcaa	cagacatcag
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2761	ttactattgc	tggtgtccca	agtcacaca	cctcaagggt	aggggttcag	aacattgcac
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2881	cacagcagcc	aacccaacaa	agcgtagtga	ttgtaagcca	gccagctcaa	caaggtcaaa
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3241	tgcaagtcca	acctcaacag	tcgaatgcag	gagttgggtca	gctgcctct	ggtgagtcga
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4021	agtacagtga	ctcaagctca	cctccttcaa	actcagggaa	aattcaaggt	gagactaatc
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TABLE 1-continued

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4261	atcataaaac	ttccaatcat	gtaggaaatg	gtgagatc	tccaatggaa	ccacaaggga
4321	cttttagatat	cactcagcaa	gatactgcca	aaggtgatca	actagaaga	atttctaattg
4381	gacctgtatt	aactttgggt	ggttcacatg	tgagcagtat	acaggaggct	tcaaatgagg
4441	caacacagca	atttagtggg	actgatttgc	ttaatggacc	tctagcttca	agtttgaatt
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4561	tacagggaca	tcaaatcata	gcagttcccg	actcaggatc	aaaagtatcc	catttctcctg
4621	ccctatcatc	tgacgttccg	tctacaaatg	gcacagcaga	atgcaaaact	gtaaagaggc
4681	cagcagagga	tactgatagg	gaacacagtc	caggaattcc	aaataaagta	ggagttagaa
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5161	agcgggtttc	ttttattacc	cacttgccag	ataagcactg	ttcaaaaggt	gccttacttg
5221	caggattaaa	acaagatgaa	ccaggacaa	caggaagtca	gaagtcttct	accaagcagc
5281	caactgtagg	gggcacaaagc	tcaactccta	gagcacaaaa	ggccattgtg	aatcatccca
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6241	tagatcttgt	aaatatttgt	gaatagaatg	aatacctttc	atgccactgc	agccactgga
6301	aatacattct	gcgggtcctc	agaagcatca	ttggtagggt	ctaaagtttt	ctagacttct
6361	ctgtcaattg	taagttaattg	tgatatattc	tatgcagtg	atgaatgttc	tttaaatattg
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6661	tgtacaagct	aatctctagg	acagaatttt	ttgtatttgt	aaattgggtt	aaatacatgg
6721	aattttatc	aggtttttct	ctgtgttata	tatgcattat	gtgcagggtat	gatattttct
6781	tcactacttt	ttctatctta	atatagtgtg	gaatttttat	gtattattct	tccattctta
6841	atactgtacc	acattctctg	tcagaaactg	ctcacttctc	taaattgtct	tttttcccc
6901	agcgtgaaat	gtatccattt	ataactgcct	attgcctgtt	ctattagcat	ccaaaaatgt
6961	ggaaggcctc	ccaaccacca	tttctgctgt	gtccttagga	tgtgcagtaa	aaaaataga
7021	cctaaccagt	tattgttatag	aatggcttta	tttactttgg	tgactgttta	tagtttttaa
7081	ataaaagact	gaacattttc	ttgagtcctt	catttctgag	tatgcttaag	acatcttaaa
7141	aataatagaga	gaattctaaa	ttcagctgaa	ggcaagggtat	aacggtcacc	tacctatttg
7201	attatatggt	gattgataac	atattaaata	gagaacaaat	aagagaggct	ctttacatga
7261	caaatttgca	tgaataaagc	agattaaacca	agtatttatt	tttcatcttg	ttataatgca
7321	gagcaaatgt	agagaacagc	aaatgattga	tgcagttaaa	gctcaaatatg	ccttttttta
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7441	tattcagaat	cacagtgtat	ccaaatcttc	agcttgaatt	tggaggcaga	ttcttagagt
7501	gaaaaagcct	cagtttccat	attaaaaatg	ttttaaatat	tttgattgaa	ttagtaccaa
7561	tgtaaaaact	agttttcttc	tgaaggaggga	tccctggcgc	tgtcctgcca	tgtctcaaa
7621	gaatgtttga	gaaacttcat	ctaatattag	ttataagggt	gtggaattta	tgtctggccc
7681	accttccaag	actggcactg	cccaacagac	accgctgaaa	tcatgtgggt	atccctagga
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7801	ctattccacc	tacttatcca	aaaatctctt	ttaaaaagta	aatttgtgca	acaacgttta
7861	tttgaaagat	aatgtcttct	caaaatcaga	aactgcagtg	gtaattaaat	taatagaaaa
7921	gagaacaaac	tgacagttta	gaaaaatggg	tttcatattc	accattcttc	cacctctatg
7981	aattgcattgc	tgtagttcta	gcttttctgc	tataatatgt	aaatatgact	gtagcctttt
8041	aagcttcagt	ctcagcagag	aatttcctaa	atgcgtttga	cctaataaaa	ctgatcatgg
8101	cttccacttt	tcttatagct	ttatagaact	atataataat	atgagcttgc	atgagcttgc
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8221	aaagggtgctc	cttctagac	agaaaactgc	tgtatttacag	tattgttatt	tttgtctaaa
8281	gttctgtaaa	tacatctctt	aatgttatct	ttgagaaatc	tatgtaataa	atatagtcta
8341	caacatagag	actgtataat	tctgtgttat	atatgtgcct	agtgtctgt	tggcactcaa
8401	taaaattttta	gtaacaaaat	tgataatcat	atagcgaagg	catatttttc	ttccaagctc
8461	aagtcaggat	tgtgactata	tattaatgag	actcagtaat	ccaaaccaca	cctgagaact
8521	cgtctcatta	ctttatagtc	atgtcatgta	tgttttttta	accatgaaat	gacaataaaa
8581	tgattttttta	aatgagaaaa	aaaaaaaaaa	aaaaaaaaaa		

SEQ ID NO: 8 Human ARID2 Amino Acid Sequence Isoform A (NP_6X9854.2)

1 manstgkapp derrickflaf delrqfhhsr gspfkkipav ggkeldlhgl ytrvttlggf
61 akvsekmgwg eiweenfnpr scsnaafalk qylyrlyky ekvhhfgedd devppgnpkp
121 qlpigaipss ynyqqhsvsd ylrqsyglsm dfnspndynk lvsllsglp nevdfainvc

TABLE 1-continued

181	tllesneskhv	mglekdpkii	tlllanagvf	ddtlgststv	fgeewkekrd	rdtvkfwkdi
241	vddnevrldi	sdrnkshegt	sgewiweslf	hprprklgind	iegqrvlqia	vilrnlsfee
301	gnvkllaanr	tcrlrlllisa	hshfislrlq	gltdlgniaa	ellldpvdfk	tthlmfhtvt
361	kclmsrdrlf	kmrgmeilgn	lckaedngvl	iceyvddqsy	reichltltp	dvlvlvistle
421	vlymlremgd	vactkiakve	ksidmlvclv	smdiqmtgpd	alaavklich	pssshqmlse
481	irpqaieqvq	tqthvasapa	sravvaqhva	pppgiveids	ekfacqwlva	hfevnpdcsv
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721	nsipqtgvpv	siavgggppp	ssvvqnhstg	ppqvtvvnsg	tlhhpsvip	qqsplhtvvp
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841	qsqdtviiap	pqvvttsasn	ivsatsvqnf	qvattgqmti	agvpspqasr	vgfqniapkp
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961	qlrgqpnipt	sspspvpvat	nnqvptamss	sstpqsggpp	ptvsqmlsvk	rqqqqghspa
1021	pppqvqvqv	qppqvvqmv	qpqqsnagvg	qpasgessli	kqlllpkrp	stpggklilp
1081	apqippppna	rapspqvvyq	vasnqaagfg	vqggtpaqql	lvqqqnvqlv	psampsggv
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1381	tsnhvngnei	spmegqgtld	itqddtakgd	qlerisngpv	ltlgsssvss	iqeasnaatq
1441	qfsgtdllng	plassinndv	pqqrpsvsvs	phsttsviqg	hqiaivpdsq	skvshspals
1501	sdvrstngta	ecttkvkrpae	dtretvagi	pnkvgrivrt	isdppnagcs	atmvavpaga
1561	dpstvkvvai	esavqkqkqh	pptyvqnvvp	qntpmpspsa	vqvqggpnss	qpspfsqssq
1621	pgdpmrkpgq	gkfqtspqvf	yhaatehgsk	dvypggclwe	gcepfrqrf	kaivnhpsaa
1681	sfithlqdkh	cskdallagl	kqdepqgags	qksstkqptv	ggtsstpraq	kaivnhpsaa
1741	lmalrrgsrn	lvfrdftdek	egpitkhir	taaliknig	kysecgrlll	krhennlsvl
1801	aisnmeasst	lakcllyelnf	tvqskeqekd	semql		

SEQ ID NO: 9 Human ARID2 cDNA Sequence Vairant 2 (NM_001347839.1. CDS:
from 129 to 5495)

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121	aaaaaataat	ggcaaaactcg	acgggggaag	cgccctccgga	cgagcggaga	aagggaactcg
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481	caagccacac	gcttctctatt	ggtgcaattc	catcttctca	caattaccag	caacacagtg
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1861	atacagtgaa	gagagtggtg	gattccagta	gcaatgggca	ggcacatatt	catgtggtg
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2521	agagtcatat	gtttggcaga	gtacagaaca	taccagcatg	tacttctaca	gtttcacagg
2581	gtcaacagct	aatcaccaca	tcaccccaac	ctgtgcaaac	ttcatctcaa	cagacatcag
2641	ctggtagcca	gtcacaagat	actgttatca	tagcaccccc	acagtatgta	acaacttctg
2701	catccaatat	tgtctcagca	acttcagtac	agaattttca	ggtagctaca	ggacaaatgg
2761	ttactattgc	tgtgttccca	agtcacaca	cctcaagggt	agggtttcag	aacattgcac
2821	caaaacctct	cccttctcag	caagtttcat	ctacagtggt	acagcagcct	attcaacaa
2881	cacagcagcc	aacccaacaa	agcgtagtga	ttgtaagcca	gccagctcaa	caaggtcaaa

TABLE 1-continued

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3061	cagctactaa	taaccaagtc	cctactgcca	tgctgctgct	ctctacccct	caatcacagg
3121	gaccacctcc	tactgtcagt	caaatgttat	ctgtgaaaag	gcagcaacag	cagcaacatt
3181	caccagcacc	cccaccacag	caggtacaag	tacaagtcca	gcagcccca	caagtacaga
3241	tgcaagtcca	acctcaacag	tcgaatgcag	gagttggtca	gcctgcctct	ggtgagtcga
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3361	ttctccagc	ttccacagatt	cctcccccta	ataatgcaag	agctcctagc	cctcaggtgg
3421	tctatcaggt	ggccagtaac	caagccgcag	gttttgaggt	gcaggggcaa	actccagctc
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3541	ggggagtaca	aactgtgtccc	atttcgaact	tacaaatatt	gccaggtcca	ctgatctcaa
3601	atagcccagc	aaccatttttc	caagggaact	ctggcaacca	ggtaaccata	acagttgtgc
3661	caaatacgag	ttttgacact	gcaactgtga	gtcaggga	tgcaactcag	ctcattgctc
3721	cagcaggat	taccatgagc	ggaacgcaga	caggagttgg	acttcagta	caaacgcttc
3781	cagccactca	agcatctcct	gctggacaat	catcatgtac	tactgtact	ccccattca
3841	aaggtgataa	aataattttgc	caaaaggagg	aggaagcaaa	ggaagcaaca	ggtttacatg
3901	ttcatgaacg	taaaattgaa	gtcatggaga	acccgtcctg	ccgacgagga	gccacaaaca
3961	ccagcaatgg	ggatacaaa	gaaaaatgaa	tgcatgtggg	aagtctttta	aatgggagaa
4021	agtacagtga	ctcaagctta	cctccttcaa	actcaggga	aattcaaatg	gagactaatc
4081	agtgctcact	aatcagtaat	gggcatcat	tgggaattagg	tgagaatgga	gcactctggga
4141	aacagaactc	agaacaaata	gacatgcaag	atatcaaaa	tgatttgaga	aaaccgctag
4201	ttaatggagt	tttgtatttc	gataaaggag	acggtttctca	tttaagcaaa	aacattccaa
4261	atcataaaac	ttccaatcat	gtaggaaatg	gcgagatatt	tccaatggaa	ccacaaggga
4321	ccttagatat	cactcagcaa	gatactgcca	aaggtgatca	actagaaa	atttctaagt
4381	gacctgtatt	aactttgggt	gggtccatctg	tgagcagtat	acaggaggct	tcaaatgcgg
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4621	ccctatcatc	tgacgttcgg	tctacaaatg	gcacagcaga	atgcaaaact	gtaaagaggc
4681	cagcagagga	tactgatagg	gaaacagtcg	caggaaatcc	aaataaagta	ggagttagaa
4741	ttgttacaat	cagtgacccc	aacaatgctg	gctgcagcgc	aacaatgggt	cgtgtgccag
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4921	caccagctgc	caagctgcag	ggccagccta	acagttctca	gcctctctca	ttcagcggat
4981	ccagtcagcc	tgagatcca	atgagaaaac	ctggacagaa	cttcattgtgt	ctgtggcagt
5041	cttgtaaaaa	gtgggttcag	acaccctcac	aggttttcta	ccatgcagca	actgaacatg
5101	gaggaaaaa	tgatatcca	gggcagcgtc	tttgggaagg	ttgtgagcct	tttcagcgag
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5221	caggattaaa	acaagatgaa	ccaggacaag	caggaaagtc	gaagtcttct	accaagcagc
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5401	atgaaaaaga	gggaccaata	actaaacaca	tccgactaac	agctgcctta	atattaaaaa
5461	atattggtta	atattcagaa	tgtggtcgca	ggcgagtaat	atgttttctg	tagccaaagt
5521	gaatttagtt	tatttttatt	ttacatataa	gttaataaaa	ttagataaact	gtattttctt
5581	cattgttttt	ctcaccaatt	ttgcaaatat	atccaaaagt	ttatgcctag	gtcaggccat
5641	gatgagctct	taaaagtcaa	aaataaatag	aagttaaaac	aacaaaaaaa	aaaaaaaaaa
5701	aaa					

SEQ ID NO: 10 Human ARID2 Amino Acid Sequence Isoform B (NP_001334768.1)

1	manstgkapp	derrkglaf1	delrqfhhsr	gspfkkipav	gkeldlhl	ytrvtltggf
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121	qlpigaipss	ynyqghsvsd	ylrsgyglsm	dfnspndynk	lvslslsglp	nevdfainvc
181	tllesneskhv	mqlekdpkii	tlflanagvf	ddtlgsfstv	fgeewektd	rdfvkfwkdi
241	vddnevrdli	sdnrkshegt	sgewiweisf	hprprklgind	iegqrvlqia	vilrnlsfee
301	gnvkllaanr	tcrlflllisa	hshfislrql	glrtlgniaa	ellldpvdfk	tthlmfhtvt
361	kclmsrdrfl	krmrgmeilgn	lckaedngvl	iceyvvdqdsy	reilchltlp	dvlllvistle
421	vlymltemgd	vactkiakve	ksidmlvclv	smdiqmfgpd	alaaavklieh	pssshqmlse
481	irpqaieqvq	tqthvasapa	sravvaqhva	pppgiveids	ekfacqwlna	hfevnpdcsv
541	sraemyseyl	stcsklargg	iltstgfykc	lrvfpnhtv	krvedssng	qahihvvgvk
601	rraiplpiqm	yyqqqpvtst	vrvdsvpdv	spapspagip	hgsqtignhf	qtrtpvanqss
661	nltatqmsfp	vqgvhtvaqt	vsripqnpvp	hthqqgnapv	tvigskapip	cevvkatviq
721	nsipqtgvvp	siavgggppq	ssvvqnhstg	pqpvtvnsq	tlhhpsvip	qqsplhtvvp
781	gqipsqtpvt	viqqavpqs	mfgrvqnipa	ctstvsqggq	littspqpvt	tssqqtssags
841	qsqdtvliap	pqvvttsasn	ivsatsvqnf	qvattqmvti	agvpspqasr	vgfqniapkp
901	lpsqgvsvst	vqqpiggqqq	ptqqsvvivs	qpaggqgtya	paihqvlan	paalpaggtv
961	qltgqpnitp	sspspvpap	nnqvptamss	sstpqsggpp	ptvsmqlsvk	rqqqqqghspa
1021	pppqgvqvq	qppqgvqmqv	qpqgsnagvg	qpasgessli	kqlllpkrp	stpggklilp
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1201	itmsgtgtgv	glpvqtlpat	gaspagqssc	ttatppfkqd	kiicqkeee	keatglhvhe
1261	rkievmenps	crrgatntsn	gdtkenemhv	gsllngrkys	dsslpnsng	kisqsetnqcs
1321	lisngpslel	gensagsgkn	seqidmqdik	sdlrkplvng	icdfdkgdgs	hlsknipnhk
1381	tsnhvngie	spmegqgtld	itqgdakgd	qlerisngpv	ltlpgssvss	iqeasnaatq
1441	qfsgtdllng	plasslndsv	pqqrpssvvs	phsttsviqg	hqiaavpdsg	skvshspals
1501	sdvrstngta	ecktvkrpae	dodretvagi	pnkvgvrvit	isdppnnagcs	atmvavpaga
1561	dpstvkvai	esavvgkqgh	pptyvqnvvp	qntpmppspa	vqvggpnss	qpspfsqssq
1621	pgdpmrkpgq	nmclwqsc	kwfqtspqv	yhaatehggk	dvypgqclwe	gcpefqrgrf
1681	sfithlqdkh	cskdallagl	kqdepqgags	qkssttkqptv	ggtssstpraq	kaivnhpsaa
1741	lmalrrgrsn	lvfrdftdek	egpitkhir1	taalilkniq	kysecgrr	

SEQ ID NO 11 Mouse ARID2 cDNA Sequence (NM_175251.4. CDS: from 129 to 5495)

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121	ccgctgttga	accgcgactc	gggttttttc	gggagacctc	ctctggcaaa	aatatggcaa
181	actcgcacgg	gaaggcgccct	cgcgcagcgc	ggaggaaggg	actggcttcc	ctggacgagc
241	tgcggcagtt	ccaccacagg	agagggtcgc	cggttaagaa	gatccctcgc	gtgggtggga
301	aggagtctga	tcttcacggt	ctctacacca	gactcactac	tttaggcggg	ttcgcgaagg
361	tttctgagaa	gaatcagttg	ggagaaattg	ttgaagagt	caactttccc	agaaattggt
421	ccaacgcgtc	ctttgcttta	aaacagttat	acttcgctta	tctagaaagg	tacgagaaag
481	ttcatcattt	tggggaagat	gatgatgagg	taccaccagg	caatccaaag	ccacagcttc
541	ctattggtgc	aatcccatct	tctacaatt	accagcaaca	cagcgtgtca	gattatctac
601	gccaaagtta	tgggttatct	atggatttta	atctgcgcta	tgtataaat	aaactggtgc
661	ttcactgttt	atctggaact	ccaagtgaag	tggacttcgc	tattaatagt	tgcactctcc
721	tatcaaatga	aagcaagcac	gtcatgcagc	ttgagaagga	tcccaaaatc	atcactttta
781	tgctgcctaa	tgcggggggt	ctctagtgaca	ctttaggact	atctctctct	gtctttggag
841	aagatggcgc	agagaagact	gatagacctt	ttgttaagtt	ttggaaagac	atttgttgat
901	acaatgaagt	gggagatctc	atttctgaca	gaaacaaggc	tcatgaagat	acaccaggag
961	aatggatttg	ggaattcttta	ctctatccac	ctcgaaagct	gggcattaat	gacatccga
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TABLE 1-continued

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6721	cgcattcgct	acgtgcaagc	tcagcttggt	cagggttttt	tgattttgta	aattggttta
6781	aatacatgga	attttataca	ggttttctcc	tggtttatat	atgcattatg	tgacggtagt
6841	atattttctt	cactactttt	tctatcttaa	tatagtgtgg	aattttattg	tattatctt
6901	ccatttctaa	tactgtacca	cattcctgct	cagaaactgc	tcacttctct	aaattgtctt
6961	ttcccccag	cgtgaaatgt	atccacttat	aactgcccct	tgctgttctt	attagcatcc
7021	aaaaatgttg	agggcctccc	aaccaccatt	tctgctgtgt	ccttaggatg	tgacgtaaaa
7081	aaatatagac	ctgacagttt	atggtataga	atggccttat	ttacttttgt	gactgtttat
7141	agtttttaaa	taaaagactg	aacattttct	tgagtctctt	atctctgagt	atgcttaaga
7201	catctaaaa	ttgaaatgct	agctgaaggc	aaggtcaaac	ggtaacctac	ttactttata
7261	ctttgtgatt	gtagagaaca	gaaagggtga	tcattgtgata	ggacacatg	gtcacggtag
7321	gaaggagacc	aggagaccaa	atgttttgtt	tacagtagta	tgagttagtag	ccccagagag
7381	cgagagacag	ttagggtctg	gttgcttac	tggtgtgccc	gcactatct	gactgagagc
7441	tttgtttacc	attcgactct	aggtttcagt	ttaactaatt	caggggcagc	ttcttggtcaa
7501	tgagcttcag	tctggacagt	tcaaacatct	tgattaaatt	agtacccaaa	agtaattctt
7561	cccaggggt	cctgtgctc	tcagctctaa	ctgtaaagaa	tggtggcgca	caccagaaac
7621	ttggtattct	cagggttggt	gcgtttgact	tcttcgctct	agcctggggc	tgcccagcag
7681	acaccctgag	tccaggtacc	ttactgtatc	cctcaaatat	cgccagacta	aaggtttcta
7741	agggcagata	gtgtgagaaa	tttatattca	ctgtgtttat	ctaaaaaat	tgaggttttt
7801	gaaataattt	ttgtaacatc	actgtttgct	tgccctcaag	gtaccttttt	ccttccaaaag
7861	caggaaatta	ccatggtggt	tagcctttag	tagcagaaac	gacaggctta	agaaagtggc
7921	ttccatagtc	accatctgtg	cacctcactg	aattgcaccc	tgtagatgta	gatttttgtg
7981	ttaaaatgta	taaatgtgtc	tttagtgctt	ttaagcaatg	gtctcagcag	aattttctaa
8041	atgtatctga	cctgacgaaa	ccaatttcta	gcccccttta	ggcttccctt	ccggcagctt
8101	tactgtgctt	atggataaaga	cttggtgggt	aacgcggttg	aagtgtctct	gcagtcacag
8161	gcctgcagaa	ccctgcagct	cacgaaaagg	tgctccttgc	tagacagaaa	cttgctgact
8221	tccagtattg	ttatttttgt	ctaaagtctt	gtaaatataa	gctttaatgt	tatcttttag
8281	agatctatgt	aaataatagt	caagaacata	gagactgtac	aattctgtgt	tatatatgtg
8341	cctagtgcct	tggtggcaat	taataaattt	taagtaacaa	aactgatgat	catatagtga
8401	aggcatattt	ttcttccgac	ttgagacagg	atatgactat	atattaatga	gactcaataa
8461	accaagccac	acatgaaaac	ttgtctcatt	actttatagc	catgccaatg	atgtttttta
8521	aactataaaa	tgacaataaa	actgactttt	gaaatgagtg	tttcggataa	gtgacttctg
8581	tcctgatctt	ataccataaa	taaagtactg	aagacgaaat	atgaagctct	tacccaaagg
8641	agttagctgt	atggaacaag	agtgaagctt	gaagatcagc	cacacagggc	acctcacact
8701	ttgttctcgt	ttatcttacg	atacagtaag	ggaaggcacc	atttagagcc	agcttgtgtt
8761	agttaaccac	tctcatactg	cccaactctt	gactgaactc	tggcactcaa	atacttggag
8821	tgagcttctt	tccaaggcca	cagaacagag	accaaccgaa	ttaccagctg	ttccatcat
8881	agctagttaa	ctttatctag	caacaatttc	cactccctgc	attggtttga	aaaaaaaaat
8941	gcaaagagac	agtatcaatg	tatgtaagtg	gattcactaa	taatacaacc	acactttaag
9001	tattaaaagt	gggtgagatg	gcttggct			

SEQ ID NO: 12 Mouse ARID2 Amino Acid Sequence (NP_780460.3)

1 manstgkapp derricklafi delrqfhhsr gspfkkipav ggkeldlhlgl ytrvttlggf
61 akvsekngwg eivreenfpr scsnaafalk qylyrlyleky ekvhhfgedd devppgnpkp
121 qlpigaipss ynyqqhsved ylrqsyglsm dfnspndynk lvsllslglp nevdfainvc
181 tlleneskhv mqlekdphkii tlllanagvf ddtlgsfssv fgeewrekt d rdtvfkwdi
241 vddnevrdli sdrnkahedt pgewiweslf hpprklgind iegqrvlqla vilrnlsfee

TABLE 1-continued

301	snvkliaanr	tclrflllsa	hshfislrl	gldtlgniaa	ellldpvdfr	tthlmfhtvt
361	kclmrdrfl	kmrgmeilgn	lckaedngvl	iceyvdqdsy	reichltlp	dvllvtstle
421	vlymltemgd	vactkiakve	ksidvlvclv	smdaqmfgpd	alaavklieh	pssshqvlse
481	irpqaieqvq	tqthiasgpa	sravvaqhaa	pppgiveids	ekfacqwlna	hfevnpdcsv
541	sraemyseyl	stcsklargg	iltstgfykc	lrcvfpnhtv	krvedstssg	qahihvigvk
601	rralplpiqm	yyqqqpistp	vvrvdavatl	sptpspagip	hgpqaagnhf	qtrpvtngss
661	nltarqmsfp	vqgihtvaqt	vsripnpstv	hthqqgnspv	tvignkapip	cevvkatviq
721	nsvpqtavpv	sivvggapaq	nsvgqnhsag	pqpvtvnsq	tllhhpsvmp	qpplhtvvp
781	qqvpsgtpt	viqqtvpqsr	mfgrvqsipa	ctstsvsgqq	littspqpmh	tssqtaags
841	qpqdtviiap	pgyvtttsasn	ivsatsvqnt	qvattgqvvti	agvpspqpsr	vgfqniapkp
901	lpsqqvpspv	vqqpiqqppq	paqqsvviws	qpaqqggaya	paihqiylan	paalpagqtv
961	qltqqpnitp	sspspvppt	nnqvptamss	sstlqsgqpp	ptvsqmlsvk	rqqqqqhspa
1021	apaqqvqvqv	qqqqqvqvqv	qpqqpsagvg	qpapnessli	kqlllpkrqp	stpggklilp
1081	apqippnnna	rapspqvvyq	vannqaagfg	vqqgtpaqql	lvqqnqvqlv	qsamppaggv
1141	qtvplisnlqi	lpqplisnsp	atiftggtsgn	qvtitvvpnt	sfaratvsqg	naaqliapag
1201	lsmgagqasa	glqvqtlpag	qsacttaplp	fkqdkiiicqk	eeeakeatgl	hvhherkievm
1261	enpserrgtt	ntengdtses	elqvgsellng	rkydsdslpp	snsqklqset	sqcsliisngp
1321	slelgengap	gkqnsepvdm	qdvkqdlkka	lvngicdfdk	gdgshlskni	pnhktshnvq
1381	ngelispvep	qtsgatqqdt	akgdqlervs	ngpvltlggs	pstssmqeap	svatpplsgt
1441	dlpngplass	lnsdvpqqrp	svvvsphsta	pviqghqvia	vphsgprvtp	salssdarst
1501	ngtaecktkv	rpaedndrdt	vpqipnkvgv	rivtisdpnn	agcsatmvav	pagadpstva
1561	kvaiesaaqg	kgqhhpytymq	svapqntmp	pspavqvqgg	psssqpsepvs	assqhadvpr
1621	kpgqntmcml	qscckkwfqt	sqvfyhaate	hggkdvyppg	clwegcepfq	rqrfsfithl
1681	qdkhcskdal	laglkqdepg	qvanqksstk	qptvggtgsa	praqkaiaash	psaalmalrr
1741	gsrnlvfrdf	tdekegpitk	hirltaalil	knigkysecg	rrllkrhenn	lsvlaisnme
1801	asstlakcly	elnftvqske	qekdsem1			

SEQ ID NO: 13 Human BRD7 cDNA Sequence Variant 1 (NM_001173984 2. CDS:
from 161 to 2119)

1	gagaggggca	tgcgcgcgc	cggcgcgcg	cgccccctg	cctcgcgcg	cgggggtctcg
61	cgggccccgc	tcccgcctc	cgtcgcgctg	gcccgaccg	gaagcgcg	cgcacggcct
121	gggcctggcg	cggggggcg	gcaccggggc	ccggtcgagc	atgggcaaga	agcacaagaa
181	gcacaagtgc	gacaaacacc	tctacagga	gtatgtagag	aagcccttga	agctggctct
241	caaagttaga	gggaacgaag	tcaccgaact	ctccacggcg	agctcgggcg	acgactccag
301	cctcttcgaa	gacaaaaacg	atcatgacaa	acacaaggac	agaaagcgga	aaaagagaaa
361	gaaaggagag	aagcagattc	caggggaaga	aaaggggaga	aaacggagaa	gagttaagga
421	ggataaaaa	aagcgagatc	gagaccgggt	ggagaatgag	gcagaaaaag	atctccagtg
481	tcacgcccc	gtgagattag	acttgccctc	tgagaagcct	ctcacaagct	ctttagccaa
541	acaagaagaa	gtagaacaga	cacccttcca	agaagccttg	aatcaactga	tgagacaatt
601	gcagagaaaa	ggtccaaagt	ctttcttttc	atttctctgt	actgatttta	ttgctcctgg
661	ctactccatg	atcattaaac	acccaatgga	ttttagtacc	atgaaagaaa	agatcaagaa
721	caatgactat	cagtcacatg	aagaactaaa	ggataaactc	aaactaatgt	gtactaatgc
781	catgatttac	aataaaccag	agaccattta	ttataaagct	gcaaaagaagc	tggtgcactc
841	aggaatgaaa	attcttagcc	aggaaagaat	tcagagcctg	aagcagagca	tagacttcat
901	ggctgacttg	cagaaaaatc	gaaagcgaaa	agatggaaca	gacacctcac	agagtgggga
961	ggacggagcg	tgctgctcga	gagagagaga	ggactctgga	gatgccgaag	cacacgacct
1021	caagagctcc	agcaaaagaa	ataaaaagaa	agacaaagat	atgcttgaag	ataagtttaa
1081	aagcaataat	ttagagagag	agcaggagca	gcttgaccgc	atcgtgaagg	aatctggagg
1141	aaagctgacc	aggcggcttg	tgaacagtc	gtgcgaattt	gaaagaagaa	aaccagatgg
1201	aacaacgacg	ttgggacttc	tccatcctgt	ggatcccat	gtaggagagc	caggctactg
1261	ccctgtgaga	ctgggaatga	caactggaag	acttcagctc	ggagtgaata	ctttgcaggg
1321	gttcaaaag	gataaaagga	acaaagtac	tccagtgtta	tatttgaatt	atgggcctta
1381	cagttcttat	gcaccgcatt	atgactccac	atttgcaaat	atcagcaagg	atgattctga
1441	tttaattctat	tcaacctatg	gggaagactc	tgatcttcca	agtgaattca	gcattccatga
1501	gtttttggcg	acgtgccaag	attatccgta	tgctatggca	gatagtttac	tggtatgttt
1561	aacaaaagga	gggcattcca	ggaccctaca	agagatggag	atgtcattgc	ctgaagatga
1621	aggccatcat	aggacacttg	acacagcaaa	agaaatggag	cagattacag	aagttagagcc
1681	accagggcgt	ttggactcca	gtactcaaga	caggctcata	gcgctgaaag	cagtaacaaa
1741	ttttggcggt	ccagtgtgaag	tttttgactc	tgaagaagct	gaaatattcc	agaagaaact
1801	tgatgagacc	accagattgc	tcagggaact	ccagggaagc	cagaatgaac	gtttgagcac
1861	cagacccccc	ccgaacatga	tctgtctctt	gggtccctca	tacagagaaa	tgcatcttgc
1921	tgaacaagtg	accaataatc	ttaaagaact	tgacagcaaa	gtaaactccag	gtgatatcgt
1981	aagcacgtat	ggagtccgaa	aagcaatggg	gatttccatt	ccttcccccg	tcatggaaaa
2041	caactttgtg	gatttgacag	aagacactga	agaacctaaa	aagacggatg	ttgctgagtg
2101	tggacctggg	ggaagttag	gctgcctggg	atttgattat	atattatgta	catacttttt
2161	cattcttaac	ttagaatgc	ttttcagaag	atattaaata	tttgtaaat	gtgtttttta
2221	ttaaactttg	gaacagcgaa	tttgatgtt	ccagaggttg	gacttgtatt	aggtataataa
2281	gctggacctg	ggactcgtga	ggaaggaatg	tgaaaaaaa	aaaaaaaaa	

SEQ ID NO: 14 Human BRD7 Amino Acid Sequence Isoform A (NP_001167455.1)

1	mgkxkxkxk	dkhlyeeyve	kplklvlkv	gnevtelstg	ssghdsslfe	dkndhdkhkd
61	rkrkrkrkge	kqipgeekgr	krvrkedkk	krdrrdrvene	aeakdlqchap	vrlldlppekp
121	ltsslakqee	vegtplqea	nqlmrqlqrk	dpsaffsfpv	tdfiapgyym	iikhpmdfst
181	mkekiknndy	qsleelkdnf	klmctnamiy	nkpetiyyka	aklllhsgmk	ilsqeriqsl
241	kqsldfmadl	qktrkqkdgt	dtsgsgedgg	cwqreredsg	daeahafksp	skenkdkdkd
301	mledkfkfnn	leregqqldr	ivkesggklt	rrlvnsqcef	errkpdgttt	lgllhpvdpi
361	vgepgypvpr	lgmttgrlqs	gvntlqgfke	dkrnkvtpvl	ylngypysy	aphydstfan
421	iskddsdliy	stygedsdlp	sdfsihefla	tcqdypvyma	dsllldvltkg	ghsrtlgeme
481	mslpdedegh	rtldtakeme	qiteveppgr	ldsscqrlli	alkavtnfgv	pvevfdseaa
541	eifqkklket	trllrelqea	qnerlstrpp	pnmicllgps	yremhlaeqv	tnnlkelaqq

TABLE 1-continued

601 vtpgdivsty gvrkamgisi spvmennfv dltedteepk ktdvaecpgg gs									
SEQ ID NO: 15 Human BRD7 cDNA Sequence Variant 2 (NM_013263.4. CDS: from 161 to 2116)									
1	gagaggggga	tcgcgcgcgc	cggcgcgcgc	cgccccctg	cctcgcgcgc	cggggtctcg			
61	cgggcccgcc	tcgcgcgcgc	cgctcgcgcg	gccccgaccg	gaagcggcgc	cgacgggcct			
121	gggcctggcg	cgggggggcg	gcaccggggc	cggtcggac	atgggcaaga	agcacaagaa			
181	gcacaagtgc	gacaaacacc	tctacgagga	gtatgtagag	aagccttoga	agctggctct			
241	caaagttaga	gggaacgaag	tcaccgaact	ctccacgggc	agctcggggc	acgactccag			
301	cctcttcgaa	gacaaaaacg	atcatgacaa	acacaaggac	agaaagcgga	aaaagagaaa			
361	gaaaggagag	aagcagattc	caggggaaga	aaaggggaga	aaacggagaa	gagttaagga			
421	ggataaaaaa	aagcgagatc	gagacggggt	ggagaatgag	gcagaaaaag	atctccagtg			
481	tcacgcccc	gtgagattag	acttgccctc	tgagaagcct	ctcacaagct	ctttagccaa			
541	acaagaagaa	gtagaacaga	cacccttcca	agaagctttg	aatcaactga	tgagacaatt			
601	gcagagaaaa	gatcccaagt	ctttcttttc	atttccctgt	actgatttta	ttgctcctgg			
661	ctactccatg	atcattaaac	acccaatgga	ttttagtagc	atgaaagaaa	agatcaagaa			
721	caatgactat	cagtccatag	aagaactaaa	ggataacttc	aaactaatgt	gtactaatgc			
781	catgatttac	aataaaccag	agaccattta	ttataaagct	gcaaaagaagc	tggtgcactc			
841	aggaatgaaa	attcttagcc	aggaagaat	tcagagcctg	aagcagagca	tagacttcat			
901	ggctgacttg	cagaaaaact	gaaagcagaa	agatggaaca	gacacctcac	agagtgggga			
961	ggacggaggc	tgctggcaga	gagagagaga	ggactctgga	gatgccgaag	cacacgcctt			
1021	caagagtc	agcaaaagaa	ataaaaaaaa	agacaaagat	atgcttgaag	ataagtttaa			
1081	aagcaataat	ttagagagag	agcaggagca	gcttgaccgc	atcgtgaagg	aatctggagg			
1141	aaagctgacc	aggcggcttg	tgaacagtc	gtgcgaactt	gaaagaagaa	aaccagatgg			
1201	aacaacgacg	ttgggacttc	tccatcctgt	ggatcccat	gtaggagagc	caggctactg			
1261	ccctgtgaga	ctgggaatga	caactggaag	acttcagttc	ggagtgaata	ctttgcaggg			
1321	gttcaaaag	gataaaaagga	acaaagtcac	tccagtgtta	tatttgaatt	atgggccccta			
1381	cagttcttat	gcaccgcatt	atgactccac	atttgcaaat	atcagcaagg	atgattctga			
1441	tttaattctat	tcaacctatg	gggaagactc	tgatcttcca	agtgatttca	gcacccatga			
1501	gtttttggcc	acgtgcccaag	attatccgta	tgctcatggca	gatagtttac	tggtatgttt			
1561	aacaaaagga	gggcatctcca	ggaccctaca	agagatggag	atgtcattgc	ctgaagatga			
1621	aggccatact	aggacacttg	acacagcaaa	agaaatggag	attacagaag	tagagccacc			
1681	aggcgctttg	gactccagta	ctcaagacag	gctcatagcg	ctgaaagcag	taacaaattt			
1741	tggcggtcca	gttggaagttt	ttgactctga	agaagctgaa	atattccaga	agaaacttga			
1801	tgagaccacc	agattgtcca	gggaactcca	ggaagcccag	aatgaacgtt	tgagcaccag			
1861	acccccctcg	aacatgatct	gtctcttggg	tcccccatc	agagaaatgc	atcttgctga			
1921	acaagtgaac	aataactctta	aagaacttgc	acagcaagta	actccagggt	atatcgtaag			
1981	cacgtatgga	gttcgaaaag	caatggggat	ttccattcct	tcccccgta	tggaaaacaa			
2041	cttgccggat	ttgacagaag	acactgaaga	acctaataag	acggatgttg	ctgagtgttg			
2101	acctgggtga	agttgaggt	gcctgggtat	tgattatata	ttatgtacat	acttttctat			
2161	tcttaactta	gaaatgcttt	tcagaagata	ttaaatattt	gtaaatgtg	ttttcaatta			
2221	aactttggaa	cagcgaaatt	ggatgttcca	gaggttgagc	ttgtattagg	taataaagct			
2281	ggacctggga	ctcgtgagga	aggaatgtga	aaaaaaaaaa	aaaaaaaa				
SEQ ID NO: 16 Human BRD7 Amino Acid Sequence Isoform B (NP_037395.2)									
1	mgkhhkhkhs	dkhlyeeyve	kplklvlkvg	gnevtelstg	ssghdsslfe	dkndhdkhkd			
61	rkckkrkkge	kqipgeekgr	krrrvkedkk	krdrdrvne	aekdlqchap	vrlldlppkcp			
121	ltsslakqee	veqtlplgeal	nqlmrqlqrk	dpsaffsfvp	tdfiapgyym	iikhpmdfst			
181	mkekiknndy	gsieelkdnf	klmctnamiy	nkpetiyyka	akllhsgmk	ilsgeriqsl			
241	kqsldfmadl	qtrkrqkdgt	dsqsgedgg	cwqreredsg	daeahafksp	skenkdkdkd			
301	mledkfknsn	lereqeqldr	ivkesggklt	rrlvnsqcef	errkpdgttt	lgllhpvdpi			
361	vgepgycpvr	lmgttglrqs	gvntlqgfke	dkrnkvtpvl	ylngypysy	aphydstfan			
421	iskddsdlly	stygedsdlp	sdfsihefla	tcqdyppvma	dsllldvltkg	ghsrtlqeme			
481	mslpedeght	rlcdtakeme	iteveppgrl	dsstqdrlia	lkavtnfvgp	vevfdseae			
541	ifqkkldeet	rlrlrelqeaq	nerlstrppp	nmicllgpsy	remhlaeqvt	nnklalaqqv			
601	tpgdivstyg	vrkamgisip	spvmennfv	ltdteepkk	tdvaecpgg	s			
SEQ ID NO: 17 Mouse BRD7 cDNA Sequence (NM_012047.2. CDS: from 238 to 2193)									
1	ggtttgccgg	cctctcgccc	tctcgccact	gggtgcgcgc	ttcggtgcgc	tcccgcgcgt			
61	ggtttttttt	ttttctcgty	agggacctcg	cgccgcgggg	cgctgcccgt	ccccctgcct			
121	cgcgccgcgg	gctctcgccg	gccccgctcc	cgccctccgc	tcgcctggcc	cggaaccgaa			
181	gcgccgcgcg	acggcctggg	cctggcgccg	ggggcggggt	ctggggcccg	gtcggacatg			
241	ggcaagaagc	acaagaagca	caagtccgac	cgccacttct	acgaggagta	cgtggagaag			
301	ccccgtaagc	tggtcctcaa	agtcgggggg	agcagaggtc	ccgagctctc	cacgggcagc			
361	tccgggcacg	actccagcct	cttcgaagac	agaagcgacc	acgacaaaca	caaggacaga			
421	aaacggaaaa	agaggagaag	agggcagaag	caggctccgg	gggaagagaa	ggggagaaaa			
481	cggagaagag	tcaaggagga	taaaaagaag	cgggatcgag	accgtgcaga	gaatgaggtg			
541	gacagagatc	tccagtgcca	tgctccctata	agattagact	tacctcctga	gaagcctctt			
601	acaagctcgt	tagccaaaca	agaagaagta	gaacagacac	cccttcagga	agctttgaat			
661	cagctcatga	gacaattgca	aagaaaagac	ccaagtgcct	tcttttcatt	tcctgtgacg			
721	gattttattg	cgcttggtta	ctccatgatt	attaacacac	caatggattt	tagtaccatg			
781	aaagaaaaag	tcaagaataa	cgactaccag	tccatagaag	aactaaagga	taacttcaag			
841	ctaattgtga	ctaattgcaat	gatttacaat	aagccagaga	ccatttatta	taagctgcca			
901	aagaagctgt	tgcaactcag	gatgaaaatt	ctcagtcagg	agagaattca	gagcctgaag			
961	cagagtatag	acttcatgtc	agacttgcat	aaaactcgga	agcagaaaga	acgaacagat			
1021	gcctgtcaga	gtggggagga	cagcggctgc	tggcagcgcg	agaggaagaa	ctctggagat			
1081	gctgaaacac	agggccttcag	aagccccgct	aaggacaata	aaaggaagaa	caaagatgtg			
1141	cttgaagaca	aattggagaag	cagcaactca	gaaaggagac	atgagcagat	tgagcgcggt			
1201	gtccaggagt	caggaggcaa	gctaaccacg	cggctggcaa	acagtcagtg	tgaatttgaa			

TABLE 1-continued

1261	agaagaaaaa	cagatgggac	aacaacactg	gggcttctcc	atcctgtgga	tccattgtg
1321	ggagagccag	gctactgccc	tgtgagattg	gggatgacaa	ctggaagact	gcagtctgga
1381	gtgaacactc	tgagggggtt	caaagaggat	aaaaggaaca	gagtaacccc	agtattatatac
1441	ttgaattatg	gacctacag	ttcttatgcc	ccacattatg	actctacatt	tgccaatatt
1501	agcaaatgat	attctgattt	aatctactca	acatatgggg	aagactctga	ccttccaaac
1561	aatttcagca	tctctgagtt	tttgccaca	tgccaagatt	accgctatgt	tatggcagat
1621	agtttactgg	atgttctaac	aaaaggagga	cattccagga	gcctgcagga	cttgacatg
1681	tcatctcctg	aagatgaagg	ccagaccaga	gcattggaca	cagcaaaaga	agcagagatt
1741	acacaaatag	agccaacagg	gcgtttggag	tccagcagtc	aggacaggct	cacagcactg
1801	caagctgtaa	caaccttttg	tgctccagct	gaagtctttg	actccgaaga	ggctgaggtg
1861	ttccagagga	agcttgatga	gacgacaaga	ttgctcaggg	agctccagga	ggcacagaat
1921	gagcgactga	gcactaggcc	tctcccaat	atgatctgtc	tctgggtcc	ttcttacaga
1981	gaaatgtacc	ttgctgaaca	agtgaccaat	aacctcaaa	aactcacaca	gcaagtgaat
2041	ccaggtgatg	ttgtaagcat	acacggagtg	cgaaaagcaa	tggggatttc	tggtccttcc
2101	cccactgtgg	gaaacagctt	cgtagatttg	acaggagagt	gtgaagaacc	taaggagacc
2161	agcactgctg	agtggtggcc	tgacgcgagc	tgaactagcc	tggtatttga	ttctattatg
2221	tacatagttt	ttcattctga	acttggaggt	gcttttcaga	agatattaac	tatttgtaaa
2281	ttgtgtttta	attaagcttt	gggacagttc	ctttcaatgt	tccaaagatt	ggctttgtat
2341	taggaataaa	agctgaacct	gggactgtga			

SEQ ID NO: 18 Mouse BRD7 Amino Acid Sequence (NP_036177.1)

1	mgkhhkhkhs	drhfyeeyve	kplklvlkv	gsevtelstg	ssghdsslfe	drsdhdkhkd
61	rkrkrkrkge	kqapgeekgr	krrrvkedkk	krdrdraene	vdrdlqchvp	irldlppkpk
121	ltsslakqee	vegtplqcal	nqlmrqlqrk	dpsaffsfpv	tdfiapgyssm	iikhpmdfst
181	mkekiknndy	qsieelkdnf	klmctnamiy	nkpetiyyka	aklllhsgmk	ilsqeriqsl
241	kqsidfmsdl	qtkrkqkert	dacqsgedsg	cwqreredsg	daetqafrsp	akdnkrkdkd
301	vledkwrssn	sereheqier	vvqesggklt	rrlansqcef	errkpdggtt	lgllhpvdpi
361	vgepgycpvr	lgmttgrlqs	gvntlqgfke	dkrrrvtpvl	ylnygpyssy	aphydstfan
421	iskddsdlly	stygedsdlp	nnfsisefla	tcqdyvyvma	dsllldvltkg	ghsrslqldld
481	msspedeqgt	raldtakeae	itqieptgrl	esssqdrlda	lqvattfgap	aevfdseeae
541	vfrklldett	rlrlrelqeaq	nerlstrppp	nmicllgpsy	remylaeqvt	nnlkeltqqv
601	tpgdvvsihg	vrkamgisvp	spivgnsfvd	ltgeceepke	tstaecgpda	s

SEQ ID NO: 19 Human PHF10 cDNA Sequence Variant 1 (NM_018288.3, CDS: from 80 to 1576)

1	ggcgggcgcg	gcagcgcgcg	cgcgggcgcg	gacaaggcgg	aggcgacggc	ggcgggcgcg
61	gcgcggggcg	cccgggctga	tgcgggcgcg	ggcgggggcc	ggggctgcgc	tgcccccgcg
121	gccgtgcgac	agcgaccocg	ccacccccgg	agcgacgtcc	ccgaaggatg	ataatgaaga
181	taattcaaat	gatgggaccc	agccatccaa	aaggaggcga	atgggctcag	gagatagttc
241	taggagttgt	gaaacttcaa	gtcaagatct	tggttttagt	tactatccag	cagaaaactt
301	gatagagtac	aaatggccac	ctgatgaaac	aggagaatac	tatatgcttc	aagaacaagt
361	cagtgaatat	ttgggtgtga	cctcctttaa	aaggaaatat	ccagatttag	agcgacgaga
421	tttgtctcac	aaggagaaac	tctacctgag	agagctaaat	gtcattactg	aaactcagtg
481	cactctaggc	ttaacagcat	tgcgacgtga	tgaagtgatt	gatttaatat	taaaagaata
541	tccagccaaa	catgctgagt	attctgttat	tctacaagaa	aaagaacgtc	aacgaattac
601	agaccattat	aaagagtatt	cccaaatgca	acaacagaat	actcagaaag	ttgaagccag
661	taaagtgcct	gagtatatta	agaaagctgc	caaaaaagca	gcagaattta	atagcaacct
721	aaacccggaa	cgcatggaag	aaagaagagc	ttattttgac	ttgcagacac	atgttatcca
781	ggtacctcaa	gggaagtaca	aagttttgcc	aacagagcga	acaaagggtca	gtccttacct
841	agtggtctct	atccccggac	agttccagga	atattataag	aggtagctac	cagatgagct
901	gcggtatctg	ccattaaaca	cagccctgta	tgagccccct	ctggatcctg	agctccctgc
961	tctagacagt	gagttgtgatt	cagatgatgg	cgaagatggg	cgaggtgatg	agaaacggaa
1021	aaataaaggc	acttcggaca	gctcctctgg	caatgtatct	gaaggggaaa	gccctcctga
1081	cagccagagc	gactctttcc	aggggaagaca	gaaatcaaaa	gacaaagctg	ccactccaag
1141	aaaagatggg	cccaaacggt	ctgtactgtc	caagtcagtt	cctgggtaca	agccaaagggt
1201	cattccaaat	gctatatgtg	gaatttgtct	gaagggttaag	gagtcacaac	agaaaggaaa
1261	ggctgaatca	cttatacact	gctcccaatg	tgagaatagt	ggccatcctt	cttgctctga
1321	tacgacaatg	gagctctgtt	ctatgattaa	gacctaccac	tgccagcgta	tggaatgtaa
1381	aacatgcatt	atatgtggac	aaccccccca	tgaagaagaa	atgatgttct	gtgatagtgt
1441	tgacagaggt	tatcatactt	tttgtgtggg	ccttgggtgct	attccatcag	gtcgctggat
1501	tttgtactgt	tgctcagcgg	cccccccaac	acccaggaaa	gtgggcagaa	gggggaaaaa
1561	cagcaaaagc	ggataaaata	gtttttgact	ctaatactgt	atatgcattt	aagtgggaata
1621	tttgtgtcca	tttaacaacat	tattttcatg	ccaataaaag	attttttttg	caaaaaaaa
1681	aaaaaaaaaa	aa				

SEQ ID NO: 20 Human PHF10 Amino Acid Sequence Isoform A (NP_060758.2)

1	maaaagpgaa	lsprpcdsdp	atpgagspkd	dnednsndgt	qpskrirmgs	gdssrscets
61	sqdlgtseyp	aelieykwp	pdetgeyml	geqvseylgv	tsfkrkypdl	errdlshkek
121	lylrelnvit	ecqctlglt	lrsdevidlm	ikeypakhae	ysvilqeker	qritdhykey
181	sqmqgqntk	veaskvpeyi	kkaakkaef	nsnlrnerme	errayfdlgt	hviqvpqgky
241	kvlptertkv	ssypvalipg	qfgeyykrys	pdelrylpln	talyeppldp	elpaldsdgd
301	sddgedgrgd	ekrknkgtst	sssgnvsege	sppdsqedsf	qgrqkskdka	atprkdgpkr
361	svlsksvpgy	kpkvipnaic	giclkgesn	kkkgaeslih	csqcensghp	scldmtmelv
421	smiktypwgc	mecktciaic	qphheemmf	cdmcdrgyht	fcvlglaips	grwicdcccqr
481	apptprkvgr	rgknskeg				

SEQ ID NO: 21 Human PHF10 cDNA Sequence Variant 2 (NM_133325.2, CDS: From 80 to 1570)

1	ggcgggcgcg	gcagcgcgcg	cgcgggcgcg	gacaaggcgg	aggcgacggc	ggcgggcgcg
61	gcgcggggcg	cccgggctga	tgcgggcgcg	ggcgggggcc	ggggctgcgc	tgcccccgcg

TABLE 1-continued

121	gcccgtgcgac	agcgaccacg	ccacccccgg	agcgagctcc	ccgaaggatg	ataatgaaga
181	taattcaaat	gatgggagcc	agccatccaa	aaggaggcga	atgggctcag	gagatagttc
241	taggagttgt	gaaacttcaa	gtcaagatct	tggttttagt	tactatccag	cagaaaactt
301	gatagagtac	aaatggccac	ctgatgaaac	aggagaatac	tatatgcttc	aagaacaagt
361	cagtgaatat	ttgggtgtga	cctcctttaa	aaggaaatat	ccagagcgac	gagatttgtc
421	tcacaaggag	aaactctacc	tgagagagct	aaatgtcatt	actgaaactc	agtgcactct
481	aggcttaaca	gcatttgcga	gtgatgaagt	gattgattta	atgataaaa	aatatccagc
541	caaacatgct	gagtattctg	ttattctaca	agaaaaagaa	cgtcaacgaa	ttacagacca
601	ttataaagag	tattcccaaa	tgcaacaaca	gaataactcag	aaagtgtgaag	ccagtaaagt
661	gcctgagtac	attgaagaag	ctgccaaaaa	agcagcagaa	tttaaatgca	acttaaacccg
721	ggaacgcatg	gaagaaagaa	gagcttattt	tgacttgcag	acacatgtta	tccaggtacc
781	tcaagggaag	tacaaagttt	tgccaacaga	gcgaacaaag	gtcagttctt	accagtggtc
841	tctcatcccc	ggacagttcc	aggaatatta	taagaggtac	tcaccagatg	agctgcggtg
901	tctgccatta	aacacagccc	tgtatgagcc	ccctctggat	cctgagctcc	ctgctctaga
961	cagtgtaggt	gatttcagatg	atggcgaaag	tggctcaggt	gatgagaaac	ggaaaaataa
1021	aggcacttcg	gcagactccc	ctggcaatgt	atctgaagg	gaaagcccc	ctgacagcca
1081	ggaggactct	ttccagggaa	gacagaaatc	aaaagacaaa	gctgccactc	caagaaaaga
1141	tgggtccaaa	cgttctgtac	tgccaagtc	agttcctggg	tacaagccaa	aggtcattcc
1201	aaatgctata	tttggaattt	gtctgaaagg	taaggagtc	aacaagaaag	gaaaggctga
1261	atcacttata	cactgtctcc	aatgtgagaa	tagtggccat	ccttctcgcc	tggatatgac
1321	aatggagctt	gtttctatga	ttaagaccta	cccatggcag	tgtatggaat	gtaaaaatg
1381	cattatagtg	ggacaacccc	accatgaaga	agaaatgatg	ttctgtgata	tgtgtgacag
1441	aggttatcat	actttttgtg	tgggccttgg	tgctattcca	tcaggctcgt	ggatttgtga
1501	ctgttgtcag	cgggcccccc	caacaccacg	gaaagtgggc	agaaggggga	aaaacagcaa
1561	agagggataa	aatagttttt	gactctaata	ctgtatatgc	atttaagtgg	aatattttgt
1621	gccattttaca	acattatttt	catgccaata	aaagattttt	tttgcaaaaa	aaaaaaaaaa
1681	aaaaaa					

SEQ ID NO: 22 Human PHF10 Amino Acid Sequence Isoform B (NP_579866.2)

1	maaaagpgaa	lsprpcdsdp	atpgaqspkd	dnednsndgt	qpskrrrmgs	gdssrscets
61	sqdlgfysyp	aenlieykwp	pdetgeyml	qeqvseylgv	tsfkrkyper	rdlshekelly
121	lrelnvtet	qctlgltalr	sdevidlmik	eypakhaeys	vilqekerqr	itdhykeysq
181	mqqqntqkve	askvpeyikk	aakkaaeafn	nlmrermeer	rayfdlqthv	iqvpggkykv
241	lptertkvss	ypvalipggg	qeyykrysp	elrlyplnta	lyeppldpel	paldsdgdsd
301	dgedgrgdek	rkknktsdss	sgnvsegesp	pdsqedsfqq	rqlskdkkaat	prkdgpkrsv
361	lskvsppgypk	kvipnaicgi	clkgkesnkk	gkaeslihc	qcensghpsc	ldmtmelvsm
421	iktypwqcm	ekctciicgq	hheemmfcd	mcdrgyhtfc	vglgaipsgr	widcccgpr
481	ptprkvgrg	knskeg				

SEQ ID NO: 23 Mouse PHF10 cDNA Sequence (NM_024250.4. CDS: from 67 to 1560)

1	cgccggcgccg	ccgctgggac	tagggcgaagg	cgccgacgac	gacggaggcg	cgccggcgctt
61	gggctgatgg	cagcgcccg	gcccggggcg	gcgctgtccc	cgccggcggtg	cgacagcgac
121	ccggcctccc	ccggagcgca	gtccccaagg	gatgataatg	aagacaactc	aatgatggg
181	accatccat	gtaaaaggag	gcgaatgggc	tcaggagaca	gctcaagaag	ttgtgagact
241	tcaagtcaag	atcttagctt	cagttactac	ccagcagaaa	acttaactga	atacaaatgg
301	ccacctgatg	aaacaggaga	atactatatg	cttcaggagc	aagtcagtga	atatctgggt
361	gtgacctcct	tcaagcgga	atatccagat	ttagagcgac	gagatttatc	tcacaaggag
421	aaactatacc	tgagagaatt	aaacgtcatc	acggaacac	agtgacact	gggtttaaca
481	gcatttgcga	gtgatgaagt	gattgactta	atgataaaag	aatatccagc	taaacacgct
541	gaatattcgg	ttatcctaca	agaaaaggaa	cgtcagagaa	ttacagatca	ttataaagag
601	tattctcaaa	tgcaacaaca	gagtactcag	aaagtccaag	ccagcaaaag	acctgagtac
661	attaagaaag	cagccaagaa	ggcagctgag	ttcaacagca	acttaaacccg	ggagcgcatg
721	gaagaaagaa	gagcctattt	tgacttacag	acacatgtta	tccaagtggc	tcaaggaaag
781	tacaaagtgt	tgccgacaga	ccgaacgaag	gtcagttcct	accagtggtc	tctcatcccg
841	ggacagttcc	aggagtatta	taagaggtac	tcaccagatg	agcttcggtg	cttgccatta
901	aacacagccc	tgtatgagcc	gccccctggc	ccagagctcc	cgccactaga	tagtgatgga
961	gactcagatg	atggcggaaga	tggcggaagg	gatgagaagc	ggaagaataa	aggcacttcg
1021	gacagctcct	caggcaatgt	gtctgaagg	gacagccccc	ctgacagcca	ggaggacacc
1081	ttccacggaa	gacagaaatc	aaaagacaaa	atggccactc	caagaaaaga	cggtcccaa
1141	cgttctgtac	tgtccaaatc	agccccctgg	tacaagccaa	aggtcattcc	aatgctcta
1201	tgtggaattt	gtctgaagg	taaggagtc	aacaagaaag	gaaaggctga	atcacttata
1261	cactgtctcc	agtgtgataa	cagtggccac	ccttcttctg	tggatatgac	catggagctt
1321	gtttctatga	ttatgaccta	cccatggcag	tgtatggaat	gtaaagcatg	cattatatgt
1381	ggacagcccc	accatgaaga	agaaatgatg	ttctgtgatg	tgtgtgacag	aggttatcat
1441	actttttgtg	tgggccttgg	tgctattcct	tcaggctcgt	ggatttgtga	ctgttgtcag
1501	cgagctcccc	caacaccacg	gaaagtgggc	agaaggggga	aaaacagcaa	agaggggtta
1561	aataggcttt	gacctcatg	tttgggatat	ttggtgcaa	tttatttaca	acactttcat
1621	ttttacgcca	ataaaaactt	tttcgaaatt	aacgatgacc	ttaaa	

SEQ ID NO: 24 Mouse PHF10 Amino Acid Sequence (NP_077212.3)

1	maaaagpgaal	spgrcdsdpa	spgaqspkdd	nednsndgth	pckrrrmgs	dssrscetss
61	qdlsfsyypa	enlieykwp	detgeymlq	eqvseylgvt	sfkrkydpde	rrdlshkel
121	ylrelnvtet	tqctlgltal	rsdevidlmi	keypakhaey	svilqekerq	ritdhykeys
181	mqqqstqkv	easkvpeyik	kaakkaaeafn	snlnrermee	rrayfdlqrh	viqvpqgkyk
241	vlptdrtkvs	syvalipgg	fqeyykrysp	delrylplnt	alyeppldpe	lpaldsdgds
301	ddgedgggde	krknktsds	ssgnvsegs	ppdsqedtfh	grqlskdkma	tpdkdgsks
361	vlksapgyk	pkvipnalcg	iclkgesnkk	kgkaeslihc	sqcdnsghps	cldmtemlvs
421	miktypwqcm	ekctciicgq	phheemmfcd	dvcdrgyhtf	cvglgaips	rwidcccgpr
481	pptprkvgr	gknskeg				

TABLE 1-continued

SEQ ID NO: 25 Human KDM6A cDNA Sequence

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1 atgaaatcct gggagtgct gctcgtacc gccgccgctg ccgccgccgc ttccggtgat
61 gaggaagaaga aaatggcggc gggaaaagcg agcggcgaga gcgaggaggc gtccccacgc
121 ctgacagccg aggagagggg ggcgctcggc ggactggaca gccgcctctt tgggttcgtg
181 agattttcatg aagatggcgc caggacgaag gccctactgg gcaaggctgt tcgctgctat
241 gaatctctaa tcttaaaagc tgaaggaaaa gtggagctcg atttcttttg tcaattaggt
301 cacttcaacc tcttattgga agattatcca aaagcattat ctgcatacca gaggtactac
361 agtttacagt ctgactactg gaagaatgct gcctttttat atggctcttg ttggtctac
421 ttccattata atgctatttca gtgggcaatt aaagcatttc aggagtgct ttatgttgat
481 cccagctttt gtcgagccaa ggaattcat ttacgacttg ggcttatgtt caaagtgaac
541 acagactatg agtctagttt aaagcatttt cagttagctt tgggtgactg taatccctgc
601 actttgtcca atgctgaatt tcaatttcac attgccact tatatgaaac ccagaggaaa
661 tatcattctg caaaagaagc ttatgaacaa cttttgcaga cagagaatct ttctgcacaa
721 gtaaaagcaa ctgtcttaca acagttaggt tggatgcac acactgtaga tctctggga
781 gataaagcaa ccaaggaaag ctatgctatt cagtatctcc aaaagtcctt ggaagcagat
841 cctaattctg gccagtcctg gtatttcttc ggaaggtgct attcaagtat tgggaaagt
901 caggatgctt ttatatctta caggcagctt actgataaat cagaagcaag tgcagatata
961 tgggtgtcaa taatgcaaga atatcagcag caaatcagc ccattggatgc ttacaggcc
1021 tatatttggt ctgtacaatt ggaccatggc catgctgcag cctggatgga cctaggcact
1081 ctctatgaat cctgcaacca gccctcaggat gccattaaat gctacttaaa tgcacataga
1141 agcaaaagtt gtatgaatac ctctgcactt gcagcacgaa ttaagtattt ccaggtcag
1201 ttgtgtaacc ttccacaagg tagtctacag aataaaacta aattacttcc tagtattgag
1261 gaggcgtgga gectaccaat tcccgcagag cttacctcca ggcagggtgc catgaacaca
1321 gcacagcaga atactcttga caattggagt ggtggacatg ctgtgtcaca tctccagta
1381 cagcaacaag ctcatctatg gtgttgaca ccacagaat tacagcattt ggaacagctc
1441 cgcgcaataa gaaataattt aaatccagca cagaaactga tctggaaca gctggaaagt
1501 cagtttgctt taatgcaaca acaccaatg agaccaacag gatttgaca ggtacgatgt
1561 actggaattc ctaatgggac aacagctgac tcatcactgc ctacaaactc agtctctggc
1621 cagcagccac agcttgctct gaccagatg cctagcgtct ctacagcctg agtccgtctt
1681 gccctgccct ggcaagcctt gcccttctct caggccatgt tccctgtagc
1741 acatcaagaa cgtgggaag tacagacact atttgatag gcaataatca tataacagga
1801 agtggaaagta atggaaacgt gccctacctg cagcgaacgc cactcactct acctcataac
1861 cgcacaaacc tgaccagcag cgcagaggag ccgtggaaaa accaactatc taactccact
1921 caggggcttc acaagggtca gagttcacat tcggcaggtc ctaatggtga acgacctctc
1981 tcttccactg ggcttcccca gcatctccag gcagctggct ctggatttca gaatcagaac
2041 ggacatccca cctgcctag caattcagta acacagggg ctgctctcaa tcaactctcc
2101 tctcacactg ctacctcagg tggacaacaa ggcattacct taaccaaga gagcaagcct
2161 tcaggaacaa tattgacggt gccctgaaaca agcaggcaca ctggagagac acctaacagc
2221 actgcccagtg tcgagtgact tcctaatcat gtccatcaga tgacggcaga tgcgtgttgc
2281 agtcctagcc atggagattc taagtcacca ggtttactaa gttcagacaa tctcagctc
2341 tctgccttgt tgatgggaaa agccaataac aatgtgggta ctggaacctg tgacaaagt
2401 aataaacatc accagctgtg tcatacaaa agtgataact ctgttgctct tccacctct
2461 tcagccattt caacagcaac accttctcca aaatccactg agcagacaa cacaacagat
2521 gttaccagcc ttaacagccc tcacagtggg ctacacacaa ttaatggaga agggatggaa
2581 gaatctcaga gccctcctga aacagatctg cttctggtta accacaaacc tagtccacag
2641 atcataccat caatgtctgt gtccatatac ccagctcag cagaagttct gaaggcatgc
2701 aggaatctag gtaaaaatgg cttatctaac agtagcattt tgttgataaa atgtccacct
2761 ccaagaccac catcttcacc ataccctccc ttgcccaggg acaagttgaa tccacctaca
2821 cctagatttt acttgaaaaa taaacgtgat gctttcttct ctccattaca tcaattttgt
2881 acaaatccga acaacctctg tacagtaata cgtggccttg ctggagctct taagttagac
2941 ctgggacttt tctctactaa aactttggcg gaagctaaca atgaacatat ggtagaagtg
3001 aggcacacag tgttgagccc agcagatgaa aactgggac ccactggaac aaagaaaatc
3061 tggcattgtg aaagtaatat atctcactat acaattgcta aatatgcaca gtaccaggcc
3121 tctcactctc agtcaatcatt gagagaagaa aatgaaaaaa gaagtcatac taagaccac
3181 tcagatagtg aatctacatc gtcagataat tctgggagga ggaggaaagg acctttaa
3241 accataaagt ttgggaccaa tattgacct tctgatgaca aaaagtggaa gttgcagcta
3301 catgagctct ctaaaccttc tgcttttgtg cgtgtcgtat cagcaggaaa tcttctaagc
3361 catgttggtc ataccatatt gggcatgaac acagttcaac tatacatgaa agttccaggg
3421 agcagaacac caggtcatca ggaataaac aacttctggt cagttaacat aaatattggc
3481 ccaggtgact gcgaatgggt tgttgttctt gaagggtact ggggtgttct gaatgactc
3541 tgtgaaaaaa ataatttgaa ttctctaag ggttcttggt gcccaatct tgaagattc
3601 tatgaagcaa atgttccagt gtatagttt attcagcgac ctggagattt ggtctggata
3661 aatgcaggca ctgttcattg ggttcaggct attggtggtt gcaacaacat tgcctggaa
3721 gttggtccac ttacagcctg ccagtataaa ttggcagtg aacggtacga atggaacaaa
3781 ttgcaaaagt tgaagtcatt agtacctatg gttcatcttt cctggaatat ggcacgaaat
3841 atcaagtgct cagatccaaa gctttctgaa atgattaagc attgtcttct aagaactctg
3901 aagcaatgtc agacattgag ggaagctctc actgctgcag gaaaagagat tatatggcat
3961 gggcggacaa aagaagaacc agctcattac tgtagcattc gtgaagtggg ggtttttgat
4021 ctgctttttg tcaactaatg agtaattca cgaagacct acatagtaca ttgccaagat
4081 tgtgcacgaa aaacaagcgg aaacttgaa aactttgttg tgcataaaca gtacaaaatg
4141 gaggacctga tgcaagtcta tgaccaattt acattagctc ctccattacc atccgctcca
4201 tcttga

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SEQ ID NO: 26 Human KDM6A Amino Acid Sequence

```

1 mkscgvs1at aaaaaaafgd eekkmaagka sgeseasps ltaeereal gldsr1fgfv
61 rfhedgartk allgkavrcy eslilkaegk vesdffclqg hfnl1lledyp kalsayqrry
121 slqsydwkna aflyglglvy fhynafqwai kafqevlyvd pscfcrakeih lrlglmfkvn
181 tdyesslkhf qlalvdncncp t1snaeiqfh iahlyetqrk yhsakeayeq llqtenlsaq
241 vkatv1lqqlg wmhtvd1llg dkatkesyai qylqkslead pnsqsgwyf1 grcyssigkv

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TABLE 1-continued

301	qdafisyrs	idkseasadt	wsigvlyqq	qnqpmadalqa	yicavqlhdg	haaawmdlgt
361	lyescnqpgd	aikcylnatr	skscntsals	aarikylqag	lcnlpqgslq	nktklpsie
421	eawslpipae	ltsrqgamnt	aqntsdsnws	gghavshppv	qqgahswclt	pqklqhleql
481	ranrnnlnpa	qklmleqles	qfvlmqghqm	rptgvaqvr	tgipngptad	sslpnsvsg
541	qppqlaltrv	psvsgpqr	acpgqplang	pfsaghpvcs	tsrtlgstdt	ilignnhitg
601	sgsgnvpvyl	qrnaltlphn	rtnltssae	pwknqlsnst	qglhkgqssh	sagpngerpl
661	sstgpgshlq	aagsgiqnqn	ghptlpsnsv	tqgaalnhls	shtatsgggq	gitltkeskp
721	sgniltvpet	srhtgetpns	tasveglphn	vhqmtadavc	spshgdsks	gllsdnpql
781	sallmgkann	nvgtgtcdkv	nrihpavhtk	tdnsavassps	saistatp	ksaqttttns
841	vtslnsphsg	lhtingegme	esqspmktdl	llvnhkpspq	ipsmsvsiy	psaevlkac
901	rnlgknglsn	ssilldkcpp	prppsspypp	lpkdklnp	psiylenkrd	affpplhqfc
961	tnpnnpvtvi	rglagalkid	lgfstktlv	eannehmvev	rtqllgpade	nwdptgtkki
1021	whcesnrsh	tiakyaqyga	ssfqeslree	nekrshhkd	sdsessdn	sgrrrkqpfk
1081	tikfgtnidl	sddkkwklql	heltklpafv	rvvsagnlls	hvghtilgm	tvqlymkvpg
1141	srtpgghenn	nfcsvninig	pgdcwfvvp	egywgvlndf	ceknnlnflm	gswwpnledl
1201	yeavnpvyrf	iqprgdlvwi	nagtvhwvga	igwcnniawn	vgpltacqyk	laveryewnk
1261	lqsvksivpm	vhlswnmarn	ikvsdpklfe	mikycllrtl	kqcqtlreal	iaagkeiwh
1321	grtkeepahy	csicevevfd	llfvtnesns	rkytivyhcqd	carktsgnle	nfvvleqykm
1381	edlmqvdyqf	tlapplpas	s			

SEQ ID NO: 27 Mouse KDM6A cDNA Sequence

```

1  atgaaatcct gcggagtgtc gccgcgcgcgc ccgcgcgcgc ccgcgccttc
61  ggtgatgagg aaaagaaat gcgcgcgcgc aaagcgagcg gcgagagcga ggaggcgctc
121  ccagcctga cagcgaggga gaggagcggc ctgcgcgcgc tggacagcgc ccttttcggg
181  ttctgtgagt ttcatgaaga tggcgccagg atgaaggccc tgctgggcaa ggctgttcgc
241  tgctacgaat ctctaactct aaaagctgaa gggaaagtgg agtctgattt cttttgtcaa
301  ttaggtcact tcaacctctt attggaagat tatccaaaag cattatctgc ataccagagg
361  tactacagtt tacagcttga ttaactggaag aatgctgcct ttttatatgg tctgtgttgc
421  gtctacttcc attacaatgc atttcagtgg gctattaaag catttcagga ggtgctttat
481  gtcgatccca gcttttgcgc agccaaggaa attcatttac gacttgggct tatgttcaaa
541  gtgaacacag actatgagtc tagtttaagc cattttcagt tagctttggt tgactgtaat
601  ccctgcactt tgtccaatgc tgaattcag tttcacattg cccacttata tgaaccagc
661  aggaagtatc attctgcaaa agaagcttat gagcaacttt tgacagacga aaacctttct
721  gcacaagtaa aagcgaaactt ttacaacaa ttaggctgga tgcatcacac tgtggaatct
781  ctgggagata agccaccaa ggaaagttaa gctattcagt atctccagaa gtccttggaa
841  gcagatccaa attctggcca gtcctggtat ttccttggaa ggtgctattc aagattggg
901  aaagtctagg atgcctttat atcttacagc caatctattg ataatcaga agcaagtgc
961  gatacatggt gttcaatagg tgtgctctat caacagcaaa atcagcctat ggatgctttg
1021  caagcttata ttgtgctgt acaattggac cagcgctatg ctgcagcccg gatggatcta
1081  ggcactctct atgtaactct caaccaactc caggatgcta tcaaatgcta tttaaatgca
1141  actagaagca aaaattgtag taatacctct ggacttgacg cacgaattaa gtatttacag
1201  gctcagttgt gtaaccttcc acaaggtagt ctacagaata aaactaaatt acctcctagt
1261  attgaggagc atgagtcgcc accaatcccc gcagagctta ccccgaggca gggtgccatg
1321  aacacagcac agcagaatac tcccagataa tggagtgggt gcaatgcacc acctccagta
1381  gaacaacaaa ctattcatg gtgtttgaca ccacagaaat tacagcactt ggaacagctc
1441  cgagcaaaaa gaaataattt aaatccagca cagaaactaa tgctggaaca gctggaagt
1501  cagtttgtct taatgcagca acaccaaatg agacaaacag gagtgcaca ggtacggcct
1561  actggaattc ttaattggcc aacagttgac tcatcaccgc ctacaaactc agtttctggc
1621  cagcagccac atgcttctct gaccagaatg cctagtgtct ctgagcttgg agtccacact
1681  gcctgcccta ggagactttt ggccaatgga ccttttctgc caggccatgt tccctgtagc
1741  acatcaagaa cactgggaag tacagacact gttttgatag gcaataatca tgtaacagga
1801  agtggaaagt atgaaacagt gccttacctg cagcgaaacg caccactctc acctcaaac
1861  cgcacaaacc tgaccagcag cacagaggag ccgtggaaaa accaactatc taactccact
1921  caggggcttc aaaaaggctc gagttcacat ttggcaggtc ccaatggtga acgacctcta
1981  tcttcactgt ggcttcccc ccatctccag gcagctggct ctggtattca gaatcagaat
2041  ggacatccca cctgcctag caattcagta acacaggggg ctgctctcaa tcaactctcc
2101  tctcacactg ctacctcagg tggacaacaa ggcattacct taaccaaga gagcaagcct
2161  tcaggaaaca cattgacggg gcctgaaaca agcaggcaaa ctggagagac acctaacagc
2221  actgccagtg ttgagggaact tcctaatacat gtccatcagg tgatggcaga tgctgtttgc
2281  agtccatgac atggagattc taagtacca ggtttactaa gttcagacaa tctcagctc
2341  tctgccttgt tgatgggaaa agctaataac aatgtgggct ctggaaacct tgacaaagtc
2401  aataacatcc acccaactgt ccatacaaa actgataatt ctgttgctc ttcacactct
2461  tcagccattt ccacagcaac accttctcct aagtcactg aacagacac cacaacagct
2521  gttaccagcc ttaccagccc tcacagtggg ctgcacacaa ttaattggaga aggaatggaa
2581  gaatctcaga gccccattaa aacagatctg cttctagtta gccacagacc tagtctcag
2641  atcataccat caatgtctgt gtccatatat cccagctcag cagaagtctc gaaagcttgc
2701  aggaatctag gtaaaacagg cctgtctaat agtagcttc ttgtggataa atgtccgct
2761  ccaagaccac catcctcacc ataccctccc ttgcccagg acaagttaa tccacctaca
2821  cctagtattt atttggaaaa taaacgtgat gctttcttct ctccattaca tcaattttgt
2881  acaaacccaa acaactcctg tacagtaata cgtggccttg ccggagctct taaattagac
2941  ttgggacttt tctctactaa aactttggtg gaagctaaca atgaacatat ggtagaagtg
3001  aggacacagt tgttacaacc agcagatgaa aattgggacc ctactggaac caagaaaatc
3061  ttgcactgtg aagatgatac atctcatact acaattgtca aatatgtcga gtaccaggcc
3121  tctcatttcc aagaatcatt gagagaagaa aatgagaaaa gaagtacca taaagaccac
3181  tcagacagtg aatctacatc atcagataat tctgggaaaa gaagaaaaag accctttaa
3241  accattaaat ttgggaccaa cattgacctg tccgatgaca aaaagtgaaa gttacagcta
3301  catgagctga ctaaaccttc tgccttcgtg agagttgtat ctgcaggaaa tcttttaagc
3361  cagttggctg atactatac gggcatgaac acagttcaac tatacatgaa agttccagga
3421  agcagaacac caggtcatca agaaaataac aactctgtt cagttaatat aaattatggc
3481  ccaggtgact gtgaatggtt tgttgttctc gaaggctact ggggtgtttt gaatgacttc

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TABLE 1-continued

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3541 tgtgaaaaa ataatttgaa ttctttaatg ggttcttggt ggcccaacct tgaagatcta
3601 tatgaagcaa atgttccagt gtatagggtt attcagcgac ctggagatct ggtctggata
3661 aatgctggca ctgttcattg ggttcaagct attggctggt gcaacaacat tgcttggat
3721 gttgggtccac ttacagcctg tcagtataag ttagcagcgg aacgttatga atggaacaag
3781 ttgcaaatg taaagtcatt agtaccatg gttcatcttt cctggaatat ggcacgaaat
3841 atcaaggttt cagatccaaa gctttttgaa atgattaagt attgtcttct gagaacgctg
3901 aagcaatgtc agacattgag ggaagctcta attgctgcag gaaaagagat catatggcac
3961 gggcgagcaa aagaagaacc agctcattat tgtagtattt gtgagtgga ggtttttgat
4021 ctgctctttg tcaactatga gagcaattct cgaaaaacct acatagtaca ttgccaaagt
4081 tgtgcacgaa aaacaatggg gaactctgga aattttgcgg tgctagaaca gtacaaaatg
4141 gaggatctga tgcaagtcta tgaccaattt acattagtaa gtgaaatcaa catgctcctc
4201 cattaccatc cgctcatctc tgatattggt ccatggacat taaacatgag accttttctg
4261 ctattcagaa agtaa

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SEQ ID NO: 28 Mouse KDM6A Amino Acid Sequence

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1 mkscgvslat aaaaaaaaf gdeekkmaag kasgeseeas psltaeerea lggldsrflg
61 fvrfhedgar mkallgkavr cyeslilkae gkvesdfcql ghfnlilled ypkalsayqr
121 yyslqgsdywk naaflyvglg vyfhnafqwk aikafqevly vdpfcrake ihlrlglmfk
181 vntdyesslk hfqlalvdcn pctlснаeiq fhiahyetq rkyhsakeay eqltqtenls
241 aqvkatilqq lgwmhhtvdl lgdkatkesy aiqylqksle adpnsqgswy flgrcyssig
301 kvqdafisyrs qsidkseasa dtwscsigvly qqnnqpmadal qayicavql d hghaaawmdl
361 gtlyescnqp qdaikcylna trskncsnts glaariqylq aqlcnlpqgs lqntklkllps
421 ieeawslpip aeltstrggam ntaqntsdn wsggnappvv eqqthswclt pqklqhleql
481 ranrnlnlpa qklmlleqls qfvlmqghqm rqtgvaqvrp tgilngptvd sslptnsvsg
541 qggqlpltrm psvsqpgvht acprqtlang pfsaghpvc tsrtlgstdt vlinnhvtg
601 sgsngnvpvl qrnaptlphn rtnltsstee pwknqlsnst qglhkgpssh lagpngrpl
661 sstgpgshlq aagsgiqnqn ghptlpsnsv tqgaalnhls shtatgggq gitltkeskp
721 sgntlcvpst srtgetpns tasveglphn vhmvmadavc spshgdsksk glssdnppql
781 sallmgkann nvpggtcdkv nnihptvhtk tdnsvassps saistatpsp ksteqtttns
841 vtslnshphg lhtingegme esqspiktdl llvshrpasp iipsmvsiy pssaevlkac
901 rnlkgnglsn ssilldkcpg prppspyppl lpkdklmptt psyilenkrd affpplhqc
961 tnpnnpvtvi rglagalkld lglfstktlv eannehmvev rtqllqpade nwdptgtkki
1021 whcesnrshd tiakyaqyqa ssgfcslree nekrrshkdh sdsstssdn sgkrrkpgfk
1081 tikfgtnldl sddkkwklql heltklpafv rvvsagnlls hvghtilgmn tvqlymkvpg
1141 srtphgqenn nfcsvninlg pgdcwfvvp egywgvlnfd ceknnlnflm gswwpnledl
1201 yeavnpvyrf iqrpqdlvwi nagtvhwvga igwcnniawn vglptacqyk laveryewnk
1261 lqnvksivpm vhlswmmarn ikvsdpklte mikyclrltl kqcqtlreal iaagkeiwh
1321 grtkeepahy csicevevfd llfvtnesns rktyivhcqd carktsgnle nfvvleqykm
1381 edlmqvdydf tlvsleinmlh hyhphpldiv pwtlnmrptl lfrk

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SEQ ID NO: 29 Human ARID1A cDNA Sequence Variant 1 (NM_006015.4. CDS:

From 374 to 7231)

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1 cagaaagcgg agagtcacag cggggccagg ccctggggag cggagcctcc accgcccccc
61 tcattcccag gcaagggtct ggggggaatg agccgggaga gccgggtccc gagcctacag
121 agccgggagc agctgagcgg ccggcgccctc ggccggccgcc gccgcctcct cctcctccgc
181 cgccggccagc ccggagcctg agccggcgggg gcgggggggga gaggagcgag cgcagcgagc
241 cagcggagcc ccgcggagcc cgcccgggcg ggtggggagg gcagccgggg ggaactgggg
301 ccggggcggg gtgggagggg gggagaagac gaagacaggg ccgggtctct ccgcggagca
361 gacagcgggg atcatggcgg cgcaggctgc ccccgccgcc gccagcagcc tgggcaaccc
421 gccgcggcgg ccgcctcctg agctgaagaa agccgagcag cagcagcggg aggagggcgg
481 gggcgaggcg gcggcgggcg cagcggcgga gcggggggaa atgaaggcag ccgcgggca
541 ggaagagcag gggcccgccg tggggcgccc gcagccgctg ggaaggagc tgcaggagcg
601 gggcgagagc aatgggggtg gcggcgggcg cggagccggc agcggcgggc ggccggcgcc
661 ggagccggag ctgaagaact cgaacgggaa cgcgggccct aggcggccc tgaaacaata
721 cctcagcagg ccgcggcgcg cgggcggtgg cggcagcagc gatgggtggt gggcgctccc
781 tcaactagcc gcggcgccct tgccgcccc agcctacggc ttggggcaac cctacggccg
841 gagccgctct gccgtcgccg ccgcggcgcc cgccgtcttc caccacaac atggcgagca
901 acaagcccat ggctccggcg cgtgcagag cgccggcgcc gggggcctgg agccctacgc
961 ggggccccag cagaactctc acgaccacgg cttccccaac caccagtaca actcctacta
1021 ccccaaccgc aggcctacc ccccgcccg ccggcgctac gcgctgagct ccccgagagg
1081 tggcaactcg ggctccggcg cggcgggggc tgccggctcc aagccgctc cctcctccag
1141 cgctccggcc tctcgtcgtt cttcgtcctt cgtcagcag cgtctcggg ccatgggggg
1201 aggcggcccc tcggcgggcg gcgggggaac tccccagccc accgceacc ccaccctcaa
1261 ccaactgctc acgtcgccca gctcgcccg gggctaccag ggctaccgg gggcgagcta
1321 cagtggcggg cccagggagc gggcgccgg caaggggccc gcggacatgg cctcgcagtg
1381 ttgggggggt gcggcgggcg cagctgcggc ggcggcgccc tcgggagggg cccaacaaag
1441 gagcccccag gcgcccatga gccccgggag cagcgcgggc ggggggcagc cgctcgcccg
1501 gaccctcag ccatccagtc caatggatca gatgggcaag atgagacct agccatagg
1561 cgggactaac ccatactgc agcaacaggg acctccgcca ggaccgcagc aaggacatgg
1621 gtaccacagg ggtcccagac cccgcagcgg tacccagatg ccatcaggg
1681 ccggggcgag agtgccatgg gcggcctctc ttatacacag cagattcctc cttatggaca
1741 acaaggcccc agcgggtatg gtcaacaggg ccagactcca tattacaacc agcaaaagtcc
1801 tcaccctcag cagccatagc caccctactc ccagcaacca ccgtcccaga cccctcagc
1861 ccaacctcgt tatcagcagc agccacagtc tcaaccacca cagctccagt cctctcagcc
1921 tccatactcc cagcagccat cccagcctcc acatcagcag tccccggctc cataccctcc
1981 ccagcagtcg agccacacagc agcaccacca gagccagccc cctactcag agccacaggg
2041 tcagtctcct taccagcagc agcaacctca gcagccagca cctcgcagc tctcccagca
2101 ggctgcgtat cctcagcccc agtctcagca gtcccagcaa actgctctat cccagcagcg
2161 cttccctcca ccgcaggagc tatctcaaga ttctttggg tctcaggcat cctcagcccc
2221 ctcaatgacc tccagtaagg gagggcaaga agatataaac ccgagccttc agtcaagacc

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TABLE 1-continued

2281	ctccagcttg	cctgatctat	ctggttcaat	agatgacctc	cccatgggga	cagaaggagc
2341	tctgagtcct	ggagttagca	catcagggat	ttccagcagc	caaggagagc	agagtaatcc
2401	agctcagttc	cctttctctc	ctcatacctc	ccctcacctg	cctggcatcc	gaggcccttc
2461	cccgctccct	gttggtctctc	ccgccagtgt	tgctcagttc	cgctcaggac	cactctcgcc
2521	tgctgcagtg	ccaggcaacc	agatgccacc	tcggccaccc	agtggccagt	cggacagcat
2581	catgcacctc	tcctgaacc	aatcaagcat	tgcccaagat	cgaggttata	tgccagaggaa
2641	ccccagatg	ccccagtaca	gttccccca	gcccggctca	gccttatctc	cgcgtcagcc
2701	ttccggagga	cagatacaca	caggcatggg	ctcctaccag	cagaactcca	tggggagcta
2761	tggtccccag	gggggtcagt	atggccacac	agggtggctac	ccaggcagc	caaatataa
2821	tgcttgccc	aatgcccaact	acccagtg	aggcatggcc	ggaggcataa	accccatggg
2881	tgccggaggt	caaatgcatg	gacagcctgg	catcccacct	tatggcacac	tcctccagg
2941	gaggatgagt	cacgcctcca	tgggcaaccg	gccttatggc	cccaacatgg	ccaatatgcc
3001	acctcaggtt	gggtcaggga	tggtgtcccc	accagggggc	atgaaccgga	aaacccaaga
3061	aactgccgtc	gcatgcatg	ttgctgcca	ctctatccaa	aacaggccgc	caggctaccc
3121	caatatgaat	caagggggca	tgatgggaac	tggaacctcct	tatggacaag	ggattaatag
3181	tatggctggc	acgatcaacc	ctcagggacc	cccatattcc	atgggtggaa	ccatggccaa
3241	caattctgca	gggatggcag	ccagcccaga	gatgatgggc	cttggggatg	taaagttaac
3301	tcagccacc	aaaatgaaca	acaaggcaga	tgggacaccc	aagacagaat	ccaaatccaa
3361	gaaatccagt	tcttctacta	caaccaatga	gaagatcacc	aagtgtgatg	agctgggtgg
3421	tgagcctgag	aggaagatgt	gggtggaccg	ttatctggcc	ttcactgagg	agaaggccat
3481	gggcatgaca	aatctgcctg	ctgtgggtag	gaaacctctg	gacctctatc	gcctctatgt
3541	gtctgtgaag	gagatcggtg	gattgactca	ggtcaacaag	aacaaaaaat	ggcgggaact
3601	tgcaaccaac	ctcaatgtgg	gcacatcaag	cagtgtctgc	agctccttga	aaaagcagta
3661	tatccagtgt	ctctatgcct	ttgaatgcaa	gattgaacgg	ggagaagacc	ctccccaga
3721	catctttgca	gctgctgatt	ccaagaagtc	ccagcccgaag	atccagcctc	cctctcctgc
3781	gggatcagga	tctatgcagg	ggcccagac	tcccagtcga	accagcagtt	ccatggcaga
3841	aggaggagac	ttaaagccac	caactccagc	atccacacca	cacagtcaga	tcccccatt
3901	gccaggcatg	agcaggagca	atccagtggg	gatccaggat	gcctttaatg	atggaagtga
3961	ctccacattc	cagaagccga	attccatgac	tccaaacctc	gggtatcagc	ccagtatgaa
4021	tacctctgac	atgatggggc	gcagtgtcta	tgagccaaat	aaggatcctt	atggcagcat
4081	gaggaaaagt	ccagggagtg	atcccttcat	gtcctcaggg	caggggccca	acggcgggat
4141	gggtgacccc	tacagtcgcg	ctgccggccc	tggtctagga	aatgtggcga	tgggaccacg
4201	acagcactat	ccctatggag	gtccttatga	cagagttagg	acggagcctg	gaatagggcc
4261	tgaggggaa	atgagcactg	ggggcccaca	gccgaatctc	atgccttcca	accagactc
4321	gggatgtat	tctcctagcc	gctaccccc	gcagcagcag	cagcagcagc	agcaacgaca
4381	tgatttctat	ggcaatcagt	tctccacca	aggcaccctc	tctggcagcc	ccttccccag
4441	ccagcagact	acaatgtatc	aacagcaaca	gcagaattac	aagcggccaa	tggatggcac
4501	atatggccct	cctgccaaagc	ggcacgaagg	ggagatgtac	agcgtgccat	acagcactgg
4561	gcagggggc	cctcagcagc	agcagttgcc	cccagcccag	ccccagcctg	ccagccagca
4621	acaagctgcc	cagccttccc	ctcagcaaga	tgtatacaac	cagtatggca	atgcctatcc
4681	tgccactgcc	acagctgcta	ctgagcgccg	accagcaggg	ggccccaga	accaatttcc
4741	attccagttt	ggccgagacc	gtgtctctgc	acccccctgg	accaatgcc	agcaaaacat
4801	gccaccacaa	atgatgggcg	gccccataca	ggcatcagct	gaggttgctc	agcaaggcac
4861	catgtggcag	gggcgtaatg	acatgacctc	taattatgcc	aacaggcaga	gcacgggctc
4921	tgccccccag	ggcccgccct	atcatggcgt	gaaccgaaca	gacgaaatgc	tgccacacaga
4981	tcagagggcg	aaccacgaag	gctcgtggcc	ttcccatggc	acacgccagc	ccccatagg
5041	tccctctgcc	cctgtgcccc	ccatgacaag	gccccctcca	tctaactacc	agccccacc
5101	aagcatgcag	aatcacattc	ctcaggtatc	cagccctgct	cccttgcccc	ggccaatgga
5161	gaaccgcacc	tctcctagca	agtctccatt	cctgcactct	gggatgaaaa	tgccagaaggc
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5281	gcgggatatc	accttcccac	ctggctctgt	tgaagccaca	cagcctgtgt	tgaagcagag
5341	gagggcgctc	acattggaac	cccgaggcca	tgccgggtga	tgagctccct	tgagctccct
5401	caagtctggt	ctcctggcag	agagcacatg	ggcattagat	accatcaaca	tcctgctgta
5461	tgatgacaac	agcatcatga	ccttcaacct	cagtcagctc	ccaggggtgc	tagagctcct
5521	tgtagaatat	ttcccgat	gcctgattga	gatctttggc	attttaagg	agtatgagg
5581	gggtgaccca	ggacagagaa	cgctactgga	tcctgggagg	ttcagcaagg	tgtctagtcc
5641	agctccacag	gaggggtggg	aagaagaaga	agaacttcta	ggctcctaac	tagaaggaga
5701	agaagaagag	gaagtgttg	aaaatgatga	ggagatagcc	tttccaggca	aggcaagacc
5761	agcttcagag	aatagttagg	agaagctgat	cagtaagttt	gacaagcttc	cagtaagat
5821	cgtacagaag	aatgatccat	ttgtgggtga	ctgctcagat	aagcttgggc	gtgtgcagga
5881	gtttgacagt	ggcctgctgc	actggcgat	tggtgggggg	gacaccactg	agcatatcca
5941	gaccacattc	gagagcaaga	cagagctgct	gccttcccgg	cctcacgcac	cctgcccacc
6001	agccccctcg	aagcatgtga	caacagcaga	gggtacacca	gggacaacag	accaggaggg
6061	gccccacact	gatggacctc	cagaaaaacg	gatcacagcc	actatggatg	acatgttgtc
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6181	cagcaagttt	ccatttggca	ttagccacgc	acagagccac	cggaaacatca	agatcctaga
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6301	tcttgccaag	cgtcgtctct	gtgtgtccaa	taccattcga	agcctgtcat	ttgtgccagg
6361	caatgacttt	gagatgtcca	aacaccccag	gctgctgctc	atcctgggca	agctgatcct
6421	gctgcaccac	aacggaagca	ggcaccacta	acttatgaaa	aggaggagga	ggaggaggga
6481	acaggaccac	gggtgagct	gcaacaaagt	ggagtgtgtg	tggtactgct	tgagatgct
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6601	cccagagagc	atctgctgga	cggactccta	cactggggcg	tttgcccttc	cccgcagag
6661	agctgaagcc	caggacccct	tttccaccct	gggccccaat	gcccgtcctt	cccgcagag
6721	actggctctt	gaaacccctc	gcaaaactcag	catccaggac	aacaatgtgg	acctgattct
6781	ggccacaccc	cccttcagcc	gcctggagaa	gttgtatagc	actatgggtc	gcttccctag
6841	tgaccgaaag	aacccgggtg	gccgggagat	ggctgtggta	ctgctggcca	acctggctca
6901	gggggacaga	ctggcagctc	gtgccattgc	agtgcagaag	ggcagtatcg	gcaacctcct
6961	gggttcccta	gaggacagcc	ttgcccgcac	acagttccag	cagagccagg	ccagcctcct
7021	ccacatgcag	aacccaccct	ttgagccaac	tagtgtggac	atgatgcggc	gggctgccc

TABLE 1-continued

7081	cgcgctgctt	gccttggcca	aggtggacga	gaaccactca	gagtttactc	tgtacgaatc
7141	acggctgttg	gacatctcgg	tatcacctgt	gatgaactca	ttggcttcac	aagtcatttg
7201	tgatgtactg	tttttgattg	gccagtcacg	acagccgtgg	gacacctccc	ccccccgtgt
7261	gtgtgtgcgt	gtgtggagaa	cttagaaact	gactgttgcc	ctttatttat	gcaaaaccac
7321	ctcagaatcc	agtttaccct	gtgctgtcca	gcttctccct	tgggaaaaag	tctctcctgt
7381	ttctctctcc	tccttccacc	tcctctccct	ccatcacctc	acgcctttct	gttctctgtc
7441	ctcaccttac	tcctctcagg	accctacccc	accctctttg	aaaagacaaa	gctctgccta
7501	catagaagac	tttttttatt	ttaaccaaag	ttactgttgt	ttacagttag	ttgggggaaa
7561	aaaaataaaa	taaaaatggc	tttcccagtc	cttgcatcaa	cgggatgcca	catttcataa
7621	ctgtttttta	tggttagact	aaaaaaaaaa	aatacaaaaa	aaattctga	aggacaaaaa
7681	aggtgactgc	tgaactgtgt	gtggtttatt	gttgtacatt	cacaaccttg	caggagccaa
7741	gaagttcgca	gttgtgaaca	gacctgttcc	actggagagg	cctgtgcagt	agagtgtaga
7801	ccctttcatg	tactgtactg	tacacctgat	actgtaaaca	tactgttaata	ataatgtctc
7861	acatggaac	agaaaacgct	gggtcagcag	caagctgtag	tttttaaaaa	tgttttttagt
7921	taaacgttga	ggagaaaaaa	aaaaaaggct	tttcccccac	agtatcatgt	gtgaacctac
7981	aacaccttga	ccctcttctc	tcctccttga	ttgtatgaat	aacctggaga	tcacctctta
8041	gaactggttt	taacctttag	ctgcagcggc	tacgtgccca	cgtgtgtata	tatatgacgt
8101	tgtacattgc	acataccctt	ggatccccac	agtttggtcc	tcctcccgag	taccctctta
8161	tagtatgacg	agttaacaag	ttggtgacct	gcacaaagcg	agacacagct	atttaatctc
8221	ttgccagata	tcgccccctc	tggtgcgagc	ctgtacaggt	ctctgtaaaa	agtccttgct
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8341	tcctttcaat	cgaggtgtga	aaaagtctta	ggttcagttg	aagtctctgat	gaagaaacac
8401	aattgagatt	ttttcagtg	taaaatctgc	atatttgtat	ttcaacaatg	tagctaaaaac
8461	ttgatgtaaa	ttctcctctt	ttttcctttt	ttggcttaat	gaatatcatt	tattcagtat
8521	gaaatcttta	tactatatgt	tcacagtggt	aagaataaat	gtacattaaa	tcttggttaag
8581	acttt					

SEQ ID NO: 30 Human ARID1A Amino Acid Sequence isoform A (NP_006006.3)

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121	pgggggsssd	gvgapphsaa	aalpppaygf	gqpygrpsa	vaaaaavfh	qhgqgqspg
181	laalqsgggg	glepyagpqq	nshdhtpnh	qynsyypnrs	aypppapaya	lssprggtpg
241	sgaaaaagsk	ppssssasas	sssssfagqr	fgamggggps	aagggtpqpt	atptlnqlit
301	pspsaryyqg	ypggdysggp	qdgagkgp	dmasqcgaa	aaaaaaaaas	ggaqqrshha
361	pmspgsgggg	gqplartpqp	sspmqmgkm	rpqpyggtnp	ysqqqgppsg	pqqghgypgq
421	pygqtpqry	pmtmqgrags	amgglsttq	ippyqggggs	gyqqqgqtpy	ynqqspqhpq
481	qppysqgpp	sqtphagpsy	qqqpqsqppq	lqssqppysq	qpsqpphqqs	papypsqgst
541	tqqhqsqpp	ysqpqagpsy	qqqpqgppap	stlsgaaayp	qpqsqqsgqt	aysqqrffpp
601	qelsqdsfqs	qassapsmtc	skgggedmnl	slqsrpslpl	dlsgsiddlp	mgtegalasp
661	vstsgissq	qeqsnpaqsp	fsptspplp	girgspspvp	gspasvaqsr	sgplspaaav
721	gnqmprrpps	gqedsimhps	mnqssiaqdr	gymqrnpqmp	qyspqpqgsa	lsprqpssgg
781	ihrgmsyqq	nsmsgsyppq	gqygpqggyp	rqpnynalpn	anypsqmag	ginpmsgagg
841	mhgqpglppy	teekamgmsh	asmgnrpygp	nmanmppqvg	sgmcppppgm	nrktqetava
901	mhvaansiqn	rppgypnmnq	ggmmgtgppy	qgginsmagm	inpqgppysm	ggtmannsag
961	maaspemmg	gdvkltpatk	mnnkadgtpk	teskskssss	stttnekik	lyelggper
1021	kmwvdrylat	teekamgmsh	lpavgrkpld	lyrlvsvke	igglgtqvmk	kkwrelatnl
1081	nvgtsssaas	slkkqyiqcl	yafeckierg	edpppdi faa	adskksqpk	qppspagsgs
1141	mqgpgtqgst	ssmaeggdl	kpptpastph	sqipplpgms	rnsnsvgiqda	findgsdstf
1201	krnemtppng	yqpsmntsdm	mgmsyepnk	dpygsmrkap	gsdpfmsgg	gpnngmgdpy
1261	sraagpglgn	vamgprghyp	yggpydrvt	epgigpegnm	stgapqpnlm	pnpdsgmys
1321	psryppqqqq	qqqqrhdsyg	nqfstqgtps	gsfpfsqgtt	myqqqqqnyk	rpmqdygpp
1381	akrhgemys	vpystgggpp	qqqgippaqp	qpasqggaag	pspqgdvynq	ygnaypatat
1441	aaterppagg	pqnqtptqtg	rdrvsappgt	naqqnmpqpm	mggpiqasae	vaqggtmwgg
1501	rndmtynyan	rqstgsapgg	payhgvnrt	emlhtdgran	hegswpshgc	rqppypgsap
1561	vppmtprpps	nyqpppsmqn	hipqvsspap	lprpmenrts	psksplhsg	mkmqkagppv
1621	pashiapapv	qppmirrdit	fppgsveatq	pvlkqrrrlt	mkdigtpeaw	rvmmslksgl
1681	laestwaldt	inillyddns	imtfnlslpl	gllellveyf	rrclieftgi	lkeyevgdpg
1741	qrtlldpgrf	skvsspapme	ggееееееllg	pkleeeeeee	vvendeef	sgkdkpasen
1801	seekliskfd	klpvkivqkn	dpfvvdcskd	lgrvqefdsq	llhwrigggd	ttehiqthfe
1861	sktellpsrp	hapcpapack	hvttaegtpg	ttdegqpppd	gppekritat	mddmlstrss
1921	tltdgakss	eaikesskfp	fgispaqshr	nikiledeph	skdetplctl	ldwqdsllakr
1981	cvcvntirs	lsfvpndfde	mskhpgllli	lgklillhkh	hperkqaplt	yekeeeqdgq
2041	vscknveww	dclmlrent	lvtlanisgg	ldlspypesi	clpvlldgllh	wavcpsaeag
2101	dpfstlpgna	vlspprlvle	tlslksiqdn	nvdllilatpp	fsrleklyst	mvrflsdrkn
2161	pvcremavvl	lanlaggds	aaaraiavkg	signllgfle	dslaataqfqq	sqasllhmqn
2221	ppfptsdvdm	mrraaralla	lakvdenhse	ftlyesrllld	isvslpmsnl	vsqvicdvlf
2281	ligqs					

SEQ ID NO: 31 Human ARID1A cDNA Sequence Variant 2 (NM_139135.2. CDS: from 374 to 6580)

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121	agccgggagc	agctgagccg	cgggcgcctc	ggccggcgcc	gccgcctcct	cctcctccgc
181	cgccgcacag	cgggagcctg	agccggcggg	gcggggggga	gaggagcgag	cgcagcgag
241	cagcggagcc	cgcgagggcc	cgccggggcg	gggtggggagg	gcagccgggg	ggactggggc
301	cggggcgggg	gtgggagggg	gggagaagac	gaagacaggg	ccgggtctct	cgcgggacga
361	gacagcgggg	atcatggcgg	cgcaggtcgc	ccccggcgcc	gccagcagcc	tgggcaaccc
421	gggcgagggg	gcggcgggcg	cagcggcgga	gcgcggggga	atgaaggcag	cgcggggcca
481	ggaaagcggg	ggccccggcg	tgggggcgcc	gcagccgctg	ggaaaggagc	tgacggagcg
541	ggccgagagc	aatggggggtg	gcggcgggcg	cggagccggc	agcggcgggc	ggccggcgcc

TABLE 1-continued

661	ggagccggag	ctgaagaact	cgaacgggaa	cgccggccct	aggcccgccc	tgaacaataa
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781	tcactcagcc	gcggcgccct	tgccgcccc	agcctacggc	ttcgggcaac	cctacggcgg
841	gagcccgctt	gcgctcgccg	cgcccgccgc	cgccgtcttc	caccaacaac	atggcggaca
901	acaaagccct	ggcctggcag	cgctgcagag	cgccggcgcc	gggggcttgg	agccctacgc
961	ggggcccccag	cagaactctc	acgaccacgg	cttccccaac	caccagtaca	actcctacta
1021	ccccaacccg	agcgctacc	ccccgccgc	cccgccctac	gcgctgagct	ccccgagagg
1081	tggcactccg	ggctccggcg	cgccggcgcc	tgccggctcc	aagccgctc	cctcctccag
1141	cgctccggcc	tcctcgctcg	cttcgctctt	cgctcagcag	cgcttcgggg	ccatgggggg
1201	aggcgggccc	tccggggccg	gcgggagaa	tccccagccc	accgccaccc	ccaccctcaa
1261	ccaactgtct	acgtcgccca	gctcgggccc	gggtaccag	ggctacccc	ggggcgacta
1321	cagtggcggg	ccccaggagc	ggggcgccgg	caaggggccc	gcggacatgg	cctcgcagtg
1381	ttggggggct	cgccggcgcc	cagctgcggc	ggcgcccgcc	tcgggagggg	cccaacaag
1441	gagccaccac	gcgcccata	gccccgggag	cagcgccggc	ggggggcagc	cgctcgcccg
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1621	gtaccacagg	cagccatcac	ggtcccagac	cccgcagcgg	taccgcagca	ccatgcaggg
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1861	ccaacctctg	tatcagcagc	agccacagtc	tcaaccacca	cagctccagt	cctctcagcc
1921	tcataactcc	cagcagccat	cccagcctcc	acatcagcag	tccccggctc	cataccctcc
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2761	tgggtccccc	gggggtcagt	atggccccc	aggtggccac	ccagggcagc	caaaactata
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5341	ctgcccacca	gcccccgga	agcatgtgac	aacagcagag	ggtaaccacg	ggacaacaga
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TABLE 1-continued

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6061	cccgcagaga	ccggtcttgg	aaaccctcag	caaactcagc	atccaggaca	acaatgtgga
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6481	gtacgaatca	cggctggttg	acatctcggg	atcacccgtt	atgaactcat	tggtttcaca
6541	agtcatttgg	aggtgactgt	ttttgattgg	ccagtcatga	cagccgtggg	acacctcccc
6601	cccccggtg	tgtgcgcgtg	tgccgagaa	ttagaaactg	actgttgccc	ttattttatg
6661	caaaaccacc	tcagaatcca	gtttaccctg	tgtgtgccag	cttctccctt	gggaaaaagt
6721	ctctcctggt	ctctctcctt	ccttccacct	ccctcccttc	catcacctca	cgcttttccg
6781	ttcctgttgc	tcaccttact	cccctcagga	ccctacccca	cctcttttga	aaagacaaa
6841	ctctgcctac	atagaagact	ttttttat	taaccaaagt	tactgttgtt	tacagtggat
6901	ttggggaaaa	aaaaaatggt	ttcccagctc	ttgcataaac	gggatgccac	gggatgccac
6961	atttcataac	tgtttttaat	ggtaaaaaaa	aaaaaaaata	atacaaaaaa	aaattctgaa
7021	ggacaaaaaa	gggtgactgt	gaactgtgtg	tggtttattg	ttgcacattc	acaactctgc
7081	aggagccaag	aagtttcgag	ttgtgaacag	acccgtgtca	ctggagaggg	ctgtgcagta
7141	gagtgtagac	cctttcatgt	actgtactgt	acacctgata	ccgtaaacat	actgtaataa
7201	taatgtctca	catggaaaca	gaaaacgctg	ggtcagcagc	aagctgtagt	ttttaaaaat
7261	gttttttagtt	aaacgttgag	gagaaaaaaa	aaaaaggctt	tcccccaaaa	gtatcatgtg
7321	tgaacctaca	acacctgac	ctctttctct	cctccttgat	tgtatgaata	accctgagat
7381	cacctcttag	aactggtttt	aacctttagc	tgcagcggct	acgctgccac	gtgtgtatat
7441	atatgacgtt	gtacattgca	catacccttg	gatccccaca	gtttggtcct	cctccagctg
7501	accctcttat	agtatgacga	gttaacaagt	tggtgacctg	cacaaagcga	gacacagcta
7561	tttaactctt	tgcacagatat	cgccctctct	gggtgcgatg	tgtacaggtc	tctgtaaaaa
7621	gtcctctgct	tctcagcagc	caatcaactt	atagttttat	tttttctggg	tttttctgtt
7681	gttttgtttt	ctttctaatc	gaggtgtgaa	aaagtcttag	gttcagttga	agttctgacg
7741	aagaacacac	actgagattt	tttcagtgat	aaaatctgca	tatttgtatt	tcaacaatgt
7801	agctaaaact	tgatgtaaat	ttctcctttt	tttctctttt	tggtctaagt	aatatcatct
7861	attcagtatg	aaatctttat	actatatgtt	ccacgtgtta	agaataaatg	tacattaat
7921	ctcggtaaga	cttt				

SEQ ID NO: 32 Human ARID1A Amino Acid Sequence isoform B (NP_624361.1)

1	maaqvaaaa	sslgnpptppp	pselkkaeqq	qreeaggeaa	aaaaaergem	kaaaggeseq
61	pavppppplg	kelqdaesn	gggggggags	gggpgaepld	knsgnagpr	palnnllep
121	pggggggssd	gvgaphhsaa	aalpppaygf	gqpygrpsa	vaaaaavfh	qhgqgqspg
181	laalqsgggg	glepyagpqq	nshdhgfnh	qynsyypnrs	aypppapaya	lssprggtpg
241	sgaaaaagsk	pppssasas	sssssfagqr	fgamggggps	aagggtppqt	atptlnqlt
301	spssaryyqg	ypggdysggp	qdgagkgp	dmascqwgaa	aaaaaaaas	ggaqqrshha
361	pmspgsgggg	gqplartpqp	sspmqmgkm	rpqpyggtnt	ysqqqgppsg	pqqghgypgq
421	pygqtpqpry	pmtcmgrags	amgglstytq	ippyggqggs	gygqgqgtpy	ynqqqphpqq
481	qpppysgqpp	sqtphaqpsy	qqqpqsqppq	lqssqppysq	qpsqpphqqs	papypsqgst
541	tqghpqsqpp	ysqpqaqspy	qqqpqqqpap	stlsqqaayp	qpqsqqsqqt	aysqqrfppt
601	qelsqdsfsg	qassapsmts	skgggedmnl	slqsrpslpl	dlsgsiddlp	mgtegalasp
661	vstsgisssg	geqsnpaqsp	fsphtsphlp	girgspspvp	gspsasvaqr	sgplspaavp
721	gnqmprrpps	gqdsdsimhps	mqgssiaqdr	gymqrnpqmp	qysspqqgsa	lsprpaggq
781	ihtgmgsyqg	nsmgsygpqg	gqygpqggyp	rqpnynalpn	anypsqmag	ginpmsgagg
841	mhgqpgippy	gtlppgrmsh	asmgnrpygp	nmanmppqvg	sgmcpptggm	nrtkqetava
901	mhvaansiqn	rppgyppnmq	ggmmtgppy	ggginsmagm	inpqqppysm	ggtmannsag
961	maaspeemgl	gdvkltpatk	mnnkadgtpk	teskskksss	stttnekikt	lyelggeper
1021	kmwvdrylaf	teekamgmtn	lpavgrkpld	lyrlyvsvke	iggltgqvnkn	kkwrelatnl
1081	nvgtsssaas	slkkqyiqcl	yafeckierg	edpppdifaa	adskksqpk	qppspagsgs
1141	mqgptqpgst	ssmaadsgdl	kpttpastph	sqipplpgms	rsnsvgiqda	findgsdrfq
1201	krnsmtpnpg	yqpsmtsdm	mgmsyepnk	dpygsmrkap	gsdpfmsgg	gpnngmgdpy
1261	sraagpglgn	vamgprqhpy	yggpydrvt	epgigpegm	stgapqpnlm	pnpdsgmys
1321	psryppqgqg	qqqqrhdsyg	nqfstqgtps	gspfpssqgt	myqqqqqvss	paplrprmen
1381	rtspksqpf	hsgmkmqkag	ppvpashiap	apvqppmirr	ditfppgsve	atqpvkqrr
1441	rltmkdigtg	eawrvmslk	sgllaestwa	ldtinillyd	dnsimtfnl	qlpglllellv
1501	eyfrrcliei	fgilkeyevg	dpgqrtlldp	grfskvsspa	pmegeeeeee	llgpkleeee
1561	eeevvndee	iafsgkdka	senseeklis	kfdklpvkiv	qkndpfvdc	sdklgrvgef
1621	dsgllhwrig	ggdttehiqt	hfesktellp	srphapcpa	prkhvttatg	tpgttdqegp
1681	ppdgppekri	ttatmdmlst	rstltedga	ksseaikeess	kfpfgispaq	shrnikiiled
1741	ephskdetpl	ctllwdqds	akrcvcvnt	irslsfvpgn	dfemskhpgl	llilgklill
1801	hhkhperkqa	pltyekeeeq	dqgvscnkve	wwwdclemlr	entlvtlani	sgqldlspyp
1861	esiclpvldg	llhwavcpsa	eaqdpfstlg	pnavlspqrl	vletlslklsl	qdnvdlila
1921	tppfsrlekl	yscmvrlfled	rknpcvcrema	vvllanlagg	dslaaraia	qkgsignllg
1981	fledslaatq	fqqsqasllh	mqnppfepts	vdmmrraara	llalakvden	hsfeftlyesr
2041	lldsivspml	nslvsgvied	vlfllggs			

TABLE 1-continued

SEQ ID NO: 33 Mouse ARID1A cDNA Sequence (NM_001080819.1. CDS: from 1 to 6852)

1	atggccgcgc	aggtcgcccc	cgccgcccgc	agcagcctgg	gcaaccgcgc	gccgcccgc
61	tcggagctga	agaaagccga	gcagcaacag	cgggaggagg	cgggggcgga	ggcgccgcgc
121	gcagcggccg	agcgcgggga	aatgaaggca	gccgcccggc	aggagagcga	gggccccgcc
181	gtggggccgc	cgagcgcgct	gggaaggag	ctgcaggacg	gggcccagag	caatgggggt
241	ggcgccgcgc	gcgagcgcgc	cagcggcgcc	gggcccgcgc	cgagcccgga	cctgaagaac
301	tcgaacggga	acgcccggcc	taggcccgc	ctgaacaata	acctcccgga	gccgcccggc
361	ggcgccgcgc	gcgcccggcag	cagcagcagc	gacgggggtg	gggcccctcc	tcaactcgcc
421	gccgcccgc	tgcccgcgcc	agcctacggc	ttcgggcaag	cctacggccg	gagcccgtct
481	gcccgcgcgc	ccgcccgggc	cgccgtcttc	caccaacaac	atggcggaca	acaaagccct
541	ggcctggcag	cgctgcagag	cgccggcgcc	ggggggttgg	agccctacgc	cgggcccag
601	cagaactcgc	acgaccacgg	cttcccacac	caccagtaca	actcctacca	ccccaacccg
661	agcgccctac	ccccgcctcc	ccaggcctac	gcgctgagct	ccccgagagg	tggcactccg
721	ggctccggcg	cgccggcgcc	cgccggctcc	aagccgcctc	cctcctccag	cgccctctgc
781	tcctcgctcg	cttcgctcct	cgcacagcag	cgcttcgggg	cctatggggg	aggcggcccc
841	tcagcggccg	gcgggggaac	tccccagccc	accgccaccc	ccacccccaa	ccactgctc
901	acgtcgccca	gctcgccccc	tggtaccag	ggctaccccc	ggggcgacta	cgccggcgcc
961	ccccaggacg	ggggcgccgg	caaaggcccg	gcggacatgg	cctcgagctg	ctggggggct
1021	ggcgccgcgc	cgccggcgcc	ggcagcggcc	gtctcgggag	gggcccacaa	aaggagccac
1081	cacgcgcccc	tgagcccccg	gagcagcggc	ggcggggggg	agccgctcgc	ccggacccct
1141	cagtcaccca	gtccaatgga	tcagatggga	aagatgagac	ctcagccgta	tgggtgggact
1201	aaccctact	cgcaacaaca	gggacctcct	tcaggaccgc	aacaaggaca	tgggtaccca
1261	gggcagccat	atgggtccca	gactccacag	cggtacccca	tgaccatgca	gggcccggct
1321	cagagtgcag	tgggcgagcc	ctcttatgca	cagcagattc	caccttatgg	ccagcaaggc
1381	cccagtgctg	atggccagca	gggcccagct	ccataactata	accagcaaa	tcctcatccc
1441	cagcagcagc	ccccctacgc	ccagcaacca	ccatcccaga	ccccctatgc	ccagccttcg
1501	tatcagcagc	acgcgcagac	tcagcaacca	cagcttcagt	cctctcagcc	tcctattccc
1561	cagcagccat	cccagcctcc	acatcagcag	tccccaaact	catatccctc	ccagcagctc
1621	accacacaac	agcatcccca	gagccagccc	ccctactcac	aaccacaggc	acagctctccc
1681	taccagcagc	agcaaccctc	gcagccagca	tcctcgtcgc	ttccccagca	ggctgcata
1741	cctcagcccc	agcctcagca	gtcccagcaa	actgcctatt	cccagcagcg	cttccctcca
1801	ccacaggagc	tttctcaaga	ttcatttggg	tctcaggcat	cctcagcccc	ctcaatgacc
1861	tcacagtaag	gagggccaaga	agatatgaac	ctgagtcttc	agtcaggccc	ctccagcttg
1921	cctgatctgt	ctggttcaat	cgatgatctc	cccatgggga	cagaaggagc	tctgagtcct
1981	ggcgtgagca	catcagggat	ttccagcagc	caaggagagc	agagcaatcc	agctcagctc
2041	ccctttcttc	ctcacacctc	ccctcacctc	cctggcatcc	gagggccgct	ccctccctcc
2101	gttggtcttc	ctgcccagct	cgccagctct	cgctcaggac	cactctcgcc	tgctgcagtg
2161	ccaggcaacc	agatgccacc	tcggccaccc	agtggccagt	cagacagcat	catgcacctc
2221	tcctatgaac	aatcaagcat	tgcccagaat	cgagggtata	tgagaggcaa	ccccagatg
2281	ccccagtaac	cttcccctca	gcctggctcg	gccttatccc	cacgtcagcc	gtctggaggga
2341	cagatgcact	cgggcgctgg	ctcctaccag	cagaactcca	tggggagcta	cgggccccag
2401	ggcagtgca	atggcccaca	aggaggctat	cctaggcagc	ctaaactata	tgcccttgcc
2461	aacgccaact	accccatagc	aggcatggcc	ggaagtatga	acctatggg	tgccggagggt
2521	cagatgcag	ggcagcctgg	aatcccacct	tacggcacac	tcctccaggg	gagaatggct
2581	catgcgtcta	tgggcaacag	gcctatggc	cctaataatg	ccaatatgcc	acctcaggtt
2641	gggtcaggga	tgtgtcctcc	accaggggga	atgaacagga	aaactcaaga	gtctgctgtt
2701	gccatgcag	ttgctgcca	ctctatccaa	aacaggccac	caggctaccc	aaatatgaat
2761	caagggggca	tgagcctccc	tggacctccc	tatggacagg	ggatcaatag	tatggctggc
2821	atgatcaacc	ctcaggggacc	cccatatcct	atgggtggaa	ccatggccaa	caatcagca
2881	gggatggcag	ccagcccaga	gatgatgggc	cttggggatg	ttaagttaac	tcccgccaca
2941	aaaatgaaca	acaaggcaga	tggaacaccc	aagacagaat	ccaaatctaa	gaaatccagt
3001	ttctctacca	ccaccaatga	gaagatcac	aaattgtatg	agttgggtgg	tgagcccag
3061	aggaagatgt	gggtggaccg	gtacctggcc	ttcacagagg	agaaggccat	gggcatgaca
3121	aatctgctct	ctgtggggag	gaagcctctg	gacctctatc	gcctctatgt	gtctgtgaag
3181	gagattgggt	ggttgactca	ggtcaacaag	aacaaaaaat	ggcgggaact	tgcaaccaac
3241	ctcaatgtgg	gtacatcaag	cagtgctgcc	agctcactga	aaaagcagta	tatccaatgt
3301	ctctatgcct	agcagtgcaa	gatcgagcgt	ggagaagacc	ctcccccgca	tatcttcgca
3361	gctgctgact	ccaagaagtc	ccaacccaag	atccagcccc	cctctcctgc	gggatcaggg
3421	tctatgcagg	ggccacaaac	tcctcagtea	accagcagtt	ctatggcaga	aggaggagac
3481	ctgaagccac	caactccagc	atccacacca	catagtcaaa	ttccccctt	accaggcatg
3541	agcaggagca	actcagtcgg	aatccaggat	gcctttcctg	atggaagtga	ccccacattc
3601	cagaagcgga	atcccatgac	tccaaacct	gggtaccagc	ccagtatgaa	tacctctgac
3661	atgatggggc	gcatgtccta	tgagccaaat	aaggatcctt	atggcagcat	gaggaagcgc
3721	ccagggaagt	atcccttcat	gtcctcaggg	caggggccca	atggcgggat	gggtgatccc
3781	tacagccgtg	ctgctggccc	tgggctggga	agtgtggcga	tgggaccacg	gcagcactat
3841	ccctatggag	gtccttacga	cagagtggag	acggagcctg	gaatcgggcc	tgaaggaaat
3901	atgggcactg	gagccctcca	gccaaatctc	atgccttcca	ccccagattc	ggggatgtat
3961	tctcctagcc	gctaccccc	gcagcagcag	cagcaacagc	agcaacaaca	tgattcctat
4021	ggcaatcaat	ttctctacca	aggcacccct	tccagcagcc	ccttccccag	ccagcagacc
4081	acaatgtatc	agcagcagca	gcagaattat	aagaggccaa	tgatgggcac	atatggcccc
4141	cctgccaaag	ggcatgaagg	ggagatgtac	agtgtgccgt	acagcgtctg	gcaaggccag
4201	cctcaacagc	atgagctgcc	tcagcctcag	tccagcctg	ccagccagcc	acaagctgcc
4261	cagccttccc	ctcagcagga	cgtgtacaac	cagtacagca	atgcctaccc	tgccctccgc
4321	accgctgcta	ctgactcgcc	accagcaggg	ggcccccaga	accaatttcc	attccagttt
4381	ggccgagagc	gagtcctctg	acctcctggg	tccagtcccc	agcagaacat	gccaccacaa
4441	atgatgggtg	gccccataca	ggcatcagct	gaggttgctc	agcagggcac	catgtggcag
4501	gggcgaatg	acatgacctc	caattatgcc	aacaggcaga	acacaggctc	tgccaccag
4561	ggccctgcgt	atcatggtgt	gaaccgaaca	gatgaaatgc	tcacacaga	tcagagggcc
4621	aaccatgaag	gccccatggc	ttcccatggc	acacgcccag	ctccgtatgg	tccttcagcc

TABLE 1-continued

4681	cctgttcccc	ccatgacaag	gccccctcca	tctaactacc	agccccacc	aagcatgccg
4741	aatcacattc	ctcaggtatc	cagccccgct	cccctccccc	ggcccatgga	gaaccgtact
4801	tctcctagca	agtctccatt	cctgcactct	gggatgaaaa	tgcaaaaggc	gggtccacog
4861	gtgcctgctt	cgcacatagc	gcctacccct	gtgcagccgc	ctatgattcg	gcgggatatc
4921	accttcccac	ctggctctgt	agaggccact	cagcctgtgt	tgaagcagag	aaggcggtc
4981	acaatgaaa	acattggaac	cccggaggca	tggcgggtaa	tgatgtocct	caagtcoggg
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5101	agcattatga	ccttcaacct	cagccagctc	ccaggcttgc	tagagctcct	tgtggaatat
5161	ttcctgtagt	gcctaattga	aatctttggc	attttaagg	agtatgaggt	aggggaccca
5221	ggacagagaa	cattataga	ccctgggaga	ttaccaagg	tgtatagtcc	agcccataca
5281	gaggaagaag	aggaagaaca	ccttgatcct	aaactggagg	aggaagagga	agaaggggtt
5341	ggaaatgatg	aggagatggc	ctttttgggc	aaggacaagc	catcttcaga	gaataatgag
5401	gagaagctag	tcagtaagtt	tgacaagctt	cggtaaaga	tcgtgcagag	gaatgaccca
5461	tttgtggtgg	actgctcaga	taagcttggg	cgctgacagg	agtttgacag	tggcctgcta
5521	cactggcgga	ttggtgggtg	ggataccact	gagcatatcc	agaccactt	tgagagcaag
5581	atagagctgc	tgccctccc	gccttatgtg	ccctgcccaa	cgccccctcg	gaacacactc
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5701	ccagagaaaa	ggatcacagc	caccatggat	gacatgttgt	ctaccgggtc	tagcacattg
5761	actgatagg	gggcaaaag	tgacagaggc	accaaggaaa	cgacgaagt	tccatttggc
5821	attagcccag	cacagagcca	ccggaacatc	aaaatttttag	aggatgaacc	ccatagtaag
5881	gatgagacc	cactgtgtac	ccttctggac	tggcaggatt	cccttgctaa	gcgctgtgtc
5941	tgtgtctcca	ataccatccg	gagcctgtcg	tttgtgccag	gcaacgaact	tgagatgtcc
6001	aaacaccacg	ggctgctgct	tatcctgggc	aagctgatcc	tgctgcacca	caagcaccca
6061	gagcggaagc	aggcaccact	aacttatgag	aaggaggagg	aacaggacca	aggggtgagc
6121	tgtgacaaa	ttgagtggtg	gtgggactgc	ttggagatgc	tccgagaaaa	cacgctggtc
6181	accctcgcca	acatctcggg	gcaattggac	ctatccccat	atcctgagag	catctgctcg
6241	cctgtctctg	acggactcct	acactgggca	gtttgcctt	cagctgaagc	ccaggacccc
6301	ttctcaaccc	tagccccc	tgccgtctct	tccccccaga	gatttggtct	ggaacccctc
6361	agcaaaactca	gcattccagg	caacaatgtg	gacctgatcc	tggccactcc	cccttttagc
6421	cgctcgagag	agttgtatag	taccatgggt	cgcttctcca	gtgaccgaaa	gaacccagtg
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6541	cgggccattg	cagtgcagaa	gggcagcatc	ggcaacctcc	tgggtttcct	ggaggacagc
6601	cttgctgcca	cacagttcca	gcagagccag	gcaagcctcc	tgcatatgca	gaatccaccc
6661	tttgaaacca	ctagttggga	catgatcgcg	cgggctgccc	gagcactgct	tgccctggcc
6721	aaggtggatg	agaaccactc	agagttcact	ctgtatgagt	cacgggtgtt	ggacatctcc
6781	gtgtcaccac	tgatgaactc	atttggttca	caagtcattt	gtgatgtact	gtttttgatt
6841	ggccagtcag	gcagcccggt	ggacacctcc	cctccccgtg	tggtgtgtag	tggtggaga
6901	acttagaana	tgactgttgc	cctttattta	tgcaaaaaca	cctcagaatc	cagtttacct
6961	tgtgctgtcc	agcttctccc	ttgggaaaag	ctctcctggt	ctctctctcc	ccccccctca
7021	ctccctcaca	ctcttctggt	ccccatctcc	acctgcttcc	ctcaggaccc	cacctatttt
7081	gaaaagacaa	agctctgcct	acatagaaga	cttttttatt	taacccaaag	ttactgttgt
7141	ttacagttag	tttggggaaa	aaaatggctt	tcccagtcct	tgcatcaacg	ggatgccaca
7201	tttcataact	gttttttaag	gttaaaaaaa	aaaaaaaaaa	aaggaaaaaa	aatacaaaaa
7261	aaccctgaag	gacaaaaggt	actgctgagc	tgtgtggttt	gtcgctgtcc	attcacaatc
7321	tcgcaggagc	cgagaagttc	gcagttgcga	gcagaccctg	ttcactggag	aggcctgtgc
7381	agtagagtg	gagccctttc	atgtactgta	ctgtacacct	gatactgtaa	acatactgta
7441	ataataatgt	ctcacatgga	aacgagagaa	gacgtcggtg	cagcagcaag	ctgtagtttt
7501	taaaaaatgt	tttagttaa	tgttgaggag	aaaaaaaaat	gctttccccc	caaaagtatcc
7561	tgtgtgaacc	tacaacgccc	tgacctcttt	ctctcctcct	tgattgtatg	aatagccctg
7621	agatcacctc	ttagacctgg	ttttaacctt	tagctgcagc	ggctgcgctg	ccacgtgtgt
7681	atatatatga	tgttgtacat	tgacataacc	cttgaatctc	cacagtttgg	tccctctccc
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7801	gctatttaat	ctcttgcagc	acattgcccc	tcttggtgca	gtgctctaca	ggtctctgta
7861	aaaagccctt	gtgctctcag	cagccaatca	acttacagtt	tatttttttc	tgggtttttg
7921	tttgtgtttg	tttctattct	aatcgaggtg	tgaaaaagtt	ctagggtcag	ttgaagttcc
7981	tgatgaagaa	acacaattga	gattttttca	gtgataaaat	ctgcatattt	gtatttcaac
8041	aatgtagcta	aaaacttgat	gtaaattcct	cctttttttt	cctttttttg	cttaaatgat
8101	atcatttatt	cagtatgaaa	tctttatact	atatgttcca	cgtgttaaga	ataaatgtac
8161	attaaatctt	ggttaa				

SEQ ID NO: 34 Mouse ARID1A Amino Acid Sequence (NP_001074288.1)

1	maagvaaaa	sslgnpptpp	selkkaeqqg	reeaggeaaa	aaargemka	aagqesegpa
61	vgppqplgke	lqdgaaesngg	gggggagsgg	gpgaepdlkn	sngnagprpa	lnnnlpeppg
121	ggggggssss	dvgvapphsa	aaalpppayg	fggaygrsps	avaaaaaavf	hqghggqsgp
181	glaalqsggg	ggl1epyagpq	qnshdhgfpn	hqynsyypnr	sayppppqay	alsprggtg
241	sgsaaaaags	kpppsasasa	sssssfacq	rfgamggggp	saagggtpqp	tatptlnqll
301	tspsargyqg	gypgdyggg	pqdgagkgp	admasqcwga	aaaaaaaaaa	vsggaqqrsh
361	hapmppsag	gggqplartp	qssspmdqmg	kmrqpvyggt	npysqqggpp	sgpqgghyyp
421	gqpygsqtqp	rypmtmggra	qsamgslysa	qqippygqgg	psayggqggt	pyynqgsphp
481	qgqppyaqgp	ysgtphagps	yqgqpqtqgp	qlqssqppys	qggsqpphqq	sptypsqsgs
541	ttqghpqsqp	pyeqpqaqsp	yqgqpqgpa	sslsqqaay	pqpqpqsgsq	taysqgrfpp
601	pqelsqdsfg	sqassapsmt	sskggqedmn	lslsqsrpsl	pdlsgsiddl	pmgtgalsp
661	gvstsgisss	qqeqsnpaqs	pfshtsphi	pgirgspsp	vgpasvags	rsqplspaav
721	pgnqmprrpp	sgqsdsimhp	smnqssiaqd	rgymqrnpqm	pqytpqpgs	alsprqpagg
781	qmhsqgvsyq	qnsmsysgqp	gsqygpqggy	prqpnynalp	nanypnagma	gsmnmpgagg
841	qmhsqgqgip	ysgtlppgrma	hasmgnrpyg	pnmamnpqp	qsgmcpqpgg	mnrktqesav
901	amhvaansiq	nrrppgynmn	qggmmgtgpp	ygqginsmag	minpqpypyp	mggtmannsa
961	gmaaspeemm	lgdvkltpat	kmnnkadgtp	kteskskkss	sstttnekit	klyelggepe
1021	rkmwvdryla	fteekamgmt	nlpavgrkpl	dlyrlysvsk	eigglqvnk	nkkwrelatn
1081	lnvgtssaaa	ssllkqyiqc	lyafeckier	gedpppdifa	aadsksqpk	iqppspagsg

TABLE 1-continued

1141	smqgpgtqps	tssmaeggd	lkpptpasrp	hsqipplpgm	srensvgiqd	afpdgsdptf
1201	qkrnsmtnpn	gyqpsmstsd	mmgrmsyepn	kdpysmrka	pgsdpfmsag	qgpnngmgdp
1261	ysraagpglg	svamgprqhy	pyggpydrvr	tepgigpegn	mgtgapqpnl	mpstpdsgmy
1321	spsrpppqgq	qqqgqhdsy	gnqfstqgtp	sssfpsqqt	tmyqqqqny	krpmdgtygp
1381	pakrhegemy	svpysagggq	pqqqqlpaag	sqpasqpqaa	qpspqdvyn	qysnaypasa
1441	taatdrpag	gpqnqpfqf	grdrvsappg	ssaqnmppq	mmggpiqasa	evaqqgtmwq
1501	grndmtynya	nrqntgsatg	gpayhgvnrt	demlhtdgra	nhegwpshg	trqppygpsa
1561	pvpmttrppp	snypqppsm	nhipqvsspa	plprpmenrt	spksfpflhs	gmkmqkagpp
1621	vpashiaptp	vqppmirrdi	tfppgsveat	qpvlkqrrrl	tmkdigtpea	wrvmslksq
1681	llaestwald	tinillyddn	simtfnlsq	pglllellvey	frrclieifg	ilkeyevgdp
1741	gqrtlldpgr	ftkvyspah	eeeeehldp	kleeeeeeegv	gndeemaflg	kdkpsenne
1801	eklvskfdkl	pvkivgrndp	Iwdcsdklg	rvqefcdsgll	hwrigggdt	ehiqthfes
1861	iellpsrpyv	pcptprkhl	ttvegtpgtt	eqegppdgl	pekritatmd	dmlstrsstl
1921	tdegaksaea	tkesskfpfg	ispaqshni	kiledephsk	detplctlld	wqdsalakrcv
1981	cvstntirsls	fvpqndfems	khpglllilg	klillhkhkp	erkqapltie	keeeqdgqvs
2041	cdkvwewdc	lemlrentlv	tlanisgqld	lspypesicl	pvlidgllhwa	vcpsaeaagd
2101	fstlgpnavl	spqrlvletl	sklsiqdnnv	dlilatppfs	rleklystmv	rflsdrknpv
2161	cremavilla	nlaqgdslaa	raiavqkgsi	gnllgfleds	laatqfqqsq	asllhmqnpp
2221	feptsdmmr	raarallala	kvdenhseft	lyesrllidis	vsplmnsllvs	qvicdvfli
2281	gqs					

SEQ ID NO: 35 Human ARID1B cDNA Sequence Variant (NM_017519.2, CDS:
from 1 to 6711)

1	atggcccata	acgcggggcg	cgcgcccgcc	gccggcaccc	acagcgccaa	gagcgggcg
61	tccgagggcg	ctctcaagga	gggtggaagc	gccgcgcgcg	tgctctctcc	ctctctctcc
121	tccgagggcg	cagcgggcg	atcctctctc	tcctcgtcgg	gcccggggtc	ggccatggag
181	acgggggtgc	tccccaaaca	caaatgaaa	accgttgggc	aagcccccg	cgcgcggccc
241	caccagcagc	accaccacca	ccaccatgcc	caccaccacc	accaccatgc	ccaccaccct
301	caccaccacc	acgcactaca	gcagcagcta	aaccagtctc	agcagcagca	gcagcagcag
361	caacagcagc	agcagcagca	gcagcaacag	caacatccca	tttccaaaca	caacagcttg
421	ggcgggcgcg	gcgcggcgcg	gcctcagccc	ggccccgaca	tgagcagacc	gcaacatgga
481	ggcgccaaag	acagtgtctg	ggcgggcgag	gccgaccccc	cgggcccgcc	gctgctgagc
541	aagccggggc	acgaggacga	cgcgcggccc	aagatggggg	agccggcggg	cgcccgctac
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661	ggcgggcgcc	cgcgggcgct	cccgagttt	aataattact	atggcagcgc	tgccccctcg
721	agcgggcgcc	ccggcgggcg	cgctgggcct	tgctttgatc	aacatggcg	acaacaaagc
781	cccggggatg	ggatgtgca	ctccgcctcc	gccgcgcgcg	ccggggcccc	cggcagcatg
841	gacccccctg	agaactccca	cgaagggtac	cccaacagcc	agtgcaccca	ttatccgggc
901	tacagccggc	ccggcgcggg	cgccggcgcc	ggcgggcggg	gcggaggagg	aggaggcagc
961	ggaggaggag	gaggaggagg	aggagcagga	gcaggaggag	caggagcggg	agctgtggcg
1021	gcggcgggcg	cgcgggcgcc	ggcagcagca	ggaggcgggc	gcggcgggcg	ctatgggggc
1081	tcgtccggcg	ggtacggggg	gctgagctcc	ccccggcagc	agggcgggcg	catgatgacg
1141	ggccccggcg	cgcgggggcg	cgcgagcctc	agcaaggcg	cccgcggtc	ggcgggggcg
1201	ggcttccagc	gcttcgcgg	ccagaaccag	caccggtcgg	gggccacccc	gacctcaat
1261	cagctgctca	cctcgccca	ccccatgatg	cggagctacg	gcggcagcta	ccccgagtag
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1381	ggggcgggcg	cgggcgggca	gcaggcgccg	gcgggcatgg	gcttggggca	ggacatgggc
1441	gccagctacg	ccgctgccag	cccgccctgg	gcggcccgcg	aacaaaggag	tcacccggcg
1501	atgagccccc	gcaccccgcg	accgaccatg	ggcagatccc	agggcagccc	aatggatcca
1561	atggtgatga	agagacctca	gttgtatggc	atgggcagta	accctcattc	tcagcctcag
1621	cagagcagtc	cgtaccacag	agggtccctat	ggccctccag	gcccacagcg	gtatccaatt
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1741	ccacctcagt	atggacagca	aggtgtgagt	ggttactgcc	agcaggggca	acagccatag
1801	tacagccagc	agccgcagcc	ccgcacctc	ccacccagcg	cgcagtatct	gccgtcccg
1861	tccagcagca	ggcagcagcc	gcagcaggac	atgtctcagg	aaggctatgg	aactagatct
1921	caacctctct	tgcccccgcg	aaaacctaac	catgaagact	tgaacttaat	acagcaagaa
1981	agaccatcaa	gtttaccaga	tctgtctggc	tccattgatg	acctccccac	gggaacggga
2041	gcaactttga	gcagcagcag	cagtgcatcc	gggtccacga	gcagccaaag	ggatcagagc
2101	aaccggcgcg	agtcgctctt	ccccccacat	gcgtccctcc	atctctccag	catcccgggg
2161	ggcccatctc	cctctcctgt	tggtctctct	gtaggaagca	accagctctg	atctggccca
2221	atctctcctg	caagtatccc	aggtagtcag	atgcctccgc	agccaccccg	gagccagtag
2281	gaatccaggt	cccatcccg	cttgagccag	tcaccaatgc	cacaggaaag	aggttttatg
2341	gcaggcacac	aaagaaaccc	tcagatggcc	cagtatggac	ctcaacagac	aggaccatcc
2401	atgtcgctcc	atctctctcc	tgggggccag	acgcagctcg	gaatcagtag	ctttcagcag
2461	agtaactcaa	gtgggactta	cgggtccacag	atgagccagt	atggaccaca	aggtaactac
2521	tccagacccc	cagcgtatag	tggggtgccc	agtgcagact	acagcgggcc	agggcccggt
2581	atggggtatc	gtgccacaac	ccagatgcat	ggacaaggcg	caagccagcc	atgtgctgct
2641	gtgcccctgg	gacgaatgcc	atcagctggg	atgcagaaca	gaccatttcc	tggaaatatg
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2761	ccaatgccaa	cgtgcagccg	taaggcacag	gaggcagccg	cagcagtgat	gcaggctctg
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2881	gcttccagct	ctccctacag	ccagcccatg	aacaacagct	ctagcctgat	gaacacgcag
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3061	gaagaaggca	ctccacagcc	cgagagcaag	tcaaaagagt	ccagctcctc	caccactact
3121	ggggagaaga	tcacgaaggt	gtacgagctg	gggaatgagc	cagagagaaa	gctctgggtc
3181	gaccgatacc	tcaccttcat	ggaagagaga	ggctctcctg	tctcaagctc	gcctgccgtg
3241	ggcaagaagc	ccttgagcct	gttccgactc	tacgtctgag	tcaaaagagt	cggggggttg
3301	gccaggtta	ataaaaacaa	gaagtggcgt	gagctggcaa	ccaaacctaa	cgttggcacc
3361	tcaagcagtg	cagcgagctc	cctgaaaaag	cagtatattc	agtaacctgt	tgcttctgag

TABLE 1-continued

3421	tgaagatg	aacgtgggga	ggagccccg	ccggaagtct	tcagcaccgg	ggacacccaa
3481	aagcagccca	agctccagcc	gcatctctct	gctaactcgg	gacctctgca	agggccacag
3541	acccccagct	caactggcag	caattccatg	gcagaggttc	caggtgacct	gaagccacct
3601	acccccagct	ccacccctca	cggccagatg	actccaatgc	aaggtggaag	aagcagtaca
3661	atcagtggtc	acgacccatt	ctcagatgtg	agtgattcat	ccttcccga	acggaactcc
3721	atgactccaa	acgcccccta	ccagcagggc	atgagcatgc	cagatgtgat	gggcaggatg
3781	ccctatgagc	ccaacaagga	cccctttggg	ggaatgagaa	aagtgcctgg	aagcagcgag
3841	ccctttatga	cgcaaggaca	gatgccccac	agcagcatgc	aggacatgta	caaccaaagt
3901	ccctccggag	caatgtctaa	cctgggcatg	gggcagcgcc	agcagtttcc	ctatggagcc
3961	agttacgacc	gaaggcatga	accttatggg	cagcagtatc	caggccaagg	ccctccctcg
4021	ggacagccgc	cgtatggagg	gcaccagccc	ggcctgtacc	cacagcagcc	gaattacaaa
4081	cgccatattg	acggcatgta	cgggccccca	gccaagcgcc	acgagggcga	catgtacaac
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4201	ccggaccgca	ggcccatcca	gggccagtac	ccgtatccct	acagcagggg	gaggatgcag
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4381	ccctaccaga	acaggcaggg	ccctggcggc	cctacacagg	cgccccctta	cccaggcatg
4441	aaccgcacag	acgatatgat	ggtacccgat	cagaggataa	atcatgagag	ccagtggcct
4501	tctcacgtca	gcacgctca	gccttatatg	tcgtccctcag	cctccatgca	gcccatacaca
4561	cgccccaccac	agccgtccta	ccagacgcca	ccgtcactgc	caaatcacat	ctccagggcg
4621	cccagcccgag	cgtccctcca	gcgctccctg	gagaaccgca	tgccctccaa	caagtctcct
4681	ttcttgccgt	ctatgaagat	gcagaaggtc	atgccccagg	tcccccacatc	ccaggtcacc
4741	gggcccaccac	cccaaccacc	cccaatcaga	agggagatca	cctttcctcc	tggtcagta
4801	gaagcatcac	aaccagtcct	gaaacaaagg	cgaaagatta	cctccaaaga	tatcgttact
4861	cctgagcggt	ggcggtgat	gatgcccctc	aaatcagggtc	ttttggctga	gagtagctgg
4921	gctttggaca	ctattaatat	tcttctgtat	gatgacagca	ctgttgctac	tttcaatctc
4981	ttccagttgt	ctggatttcc	cgaactttta	gtcaggtact	ttagaaaaatg	cctgattgac
5041	atttttggaa	ttcttatgga	atatgaagtg	ggagacccca	gccccaaagc	acttgatcac
5101	aacgcagcaa	ggaaggatga	cagccagtc	ttggcagacg	attctgggaa	agaggaggaa
5161	gatgctgaat	gtattgatga	cgacgaggaa	gacgaggagg	atgaggaggga	agacagcgag
5221	aagacagaaa	gcgatgaaaa	gagcagcatc	gctctgactg	ccccggacgc	cgcctgcagac
5281	ccaaaggaga	agcccaagca	agccagtaag	ttcgacaagc	tgccaataaa	gatagtcaaa
5341	aagaacaacc	tgtttgttgt	tgaccgatct	gacaagttgg	ggcggtgtgca	ggagttcaat
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5881	accctgggga	agctgattct	tcttcaccac	gagcatccag	agagaaagcg	agcaccgcag
5941	acctatgaga	aagagaggga	tgaggacaag	ggggtggcct	gcagcaaa	tgagtggtgg
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6061	cagctagact	tgctgcttta	cacggaaaagc	atctgcttgc	caattttgga	tggtctgtctg
6121	cactggatgg	tgtgcccgct	tcagagggca	caagatccct	ttccaaactgt	gggacccaac
6181	tcggtcctgt	cgctcagag	acttgtgctg	gagaccctct	gtaaaactcag	tatccaggac
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6301	acattagtta	ggtacgttgg	ggatcgcaaa	aaccagtcct	gtcgagaaat	gtccatggcg
6361	cttttatcga	accttgccca	aggggacgca	ctagcagcaa	gggccatagc	tgtgcagaaa
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6601	gaattccttt	tgacagaggg	ccggttgctg	gatatctcga	tatcagctgt	cctgaactct
6661	ctggttgcat	tgatcatctg	tgatgtactg	tttcagattg	ggcagttatg	acataagtga
6721	gaaggcaagc	atgtgtgagt	gaagattaga	gggtcacata	taactggctg	ttttctgttc
6781	ttgtttatcc	agcgtaggaa	gaaggaaaag	aaaatctttg	ctcctctgcc	ccattcacta
6841	tttaccattt	gggaattaaa	gaaataatta	atttgaacag	tcatgaaat	aatatttgc
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6961	gctgtctagt	gcattcaaa	gccaactttt	gttcttcaca	gatcttttta	atgtttcttc
7021	ccatgttgta	ttgcattttt	gggggaagca	aattgacttt	aaagaaaaaa	gttgtggcaa
7081	aagatgctaa	gatgcgaaaa	tttcaccaca	ctgagtcaaa	aaggtgaaaa	attatccatt
7141	tcctatgcgt	tttactctc	agagaatgaa	aaaaactgca	tcocatcacc	caaagttctg
7201	tgcaaataga	atttctacag	atcacgggat	aggggctcaa	ggaggtatgt	cggtcagtag
7261	tcaaaactat	gaaatgatac	tggtttctcc	acaggaaatat	ggttccatta	ggctgggagc
7321	aaaaacaatt	tttttaaga	ttgagaatac	atacctgaca	acgatccgga	aactgctcct
7381	caccactccc	gtcactcctg	ctgtcggcgt	ttgacctcc	acgtgacagt	tcttcacaa
7441	tcctttcatc	atttttttaa	tatttttttt	actgcctatg	ggctgtgatg	tatatagaag
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7621	tttattttta	ttttatatat	tttttcatta	gggccacatc	tcacaaaaaa	gaaagaaaaa
7681	atacaaaaaa	caaaaaacaa	aaaaaaagag	ggtaattgtac	aagtttctgt	atgtataaag
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7921	acattttgtat	ttcaagataa	tgtagtttaa	aaaaaaataa	aagaaaaaaa	cttgatgtaa
7981	attcctcctt	ttcctctggc	ttaatgaata	tcattttatc	agtataaaat	cttttatatg
8041	tcacacatgt	aagaataaat	gtacattaaa	tcttgtaaag	cactgtgatg	gggtgtcttg
8101	aatactgttc	tactttcctt	aaagtgggtt	cctagtaatc	aagttattta	caagaaatag
8161	gggaatgcag	cagtgatttc	acattataaa	accctacatt	tggaagagac	ctttaggggt

TABLE 1-continued

8221	tacctacttt	agagtgggga	gcaacagttt	gatttttctca	aattacttag	ctaattagtc
8281	tttcttcgaa	gcaattaact	ctaacgacat	tgaggatga	tcattttcag	tatttatggg
8341	aggtgggtgc	tgaccactt	gaggtgagat	ctcagaagct	taactggcct	gaaaatgtaa
8401	cattctgctt	tttactaact	ccatcttagt	ttaatcaaag	ttcaatctat	tcottgttct
8461	ttctgtgtgc	ctcagagcta	ttttgcattt	agtttactcc	accgtgtata	atatttatac
8521	tgtgcaatgt	taaaaaagaa	tctgttatat	tgtatgtggt	gtacatagtg	caaatgtatg
8581	atttctatct	cagggcatat	tatggctctc	atattccttc	ctacctgggtg	cacagtagct
8641	ttttaatact	agtcacttct	aatttaaact	ttctcttctc	gggtcattga	ctgttactgt
8701	gtaataatcg	atttctttga	aactgctgca	taattatgct	gttagtggac	ctctacctct
8761	tctcttccct	ctcccaatca	cagtatactc	agaatcccca	gccctcgca	tacattgtgt
8821	cggttcacat	tactcacagt	aatatatgga	agagtttagac	aagaacatgc	agttacagtc
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9001	aaatatatgt	tgaattaaag	acccatttat	aattctgctt	taaacacatc	tgcttgctaa
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9301	atgcgcaaca	cagtcctctag	agactaatcc	aggaagactt	tagctctcct	tccatttct
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9481	aggcttttca	gaaatttaca	ggatgccag	actttacatg	tgtaacaaaa	aaaaaaaaaa
9541	gataaaaaat	aaaggtgcaa	agaaagttaa	gtattttgga	atggtgctat	aaagttgaaa
9601	aaaaaaaaaa					

SEQ ID NO: 36 Human ARID1B Amino Acid Sequence isoform A (NP_059989.2)

1	mahnagaaaa	agthsaksgg	seaalkeggs	aaalssssss	saaaaaasss	sssgpgsame
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121	qqqqqqqqq	qhpisnnsl	ggagggapqp	gpdmeqpqhg	gakdsaaagg	adppgpplls
181	kpgdeddapp	kmgepaggry	ehpglgaigt	qpppvavpgg	gggpaavpef	nnnygsaapa
241	sgpgpggragp	cfdqhgqgqs	pgmghmhsas	aaaagapgsm	dplqnshegy	pnsqcnhypp
301	ysrpaggggg	gggggggggs	ggggggggag	aggagagava	aaaaaaaaaa	gggggggyyg
361	ssagyvllss	prqggggmmm	gpqgggaasl	skaaggsaag	gfqrfaqngq	hpsgatptln
421	qlltspgpm	rsygygyypey	sspsappppp	sqpqsqaaaa	gaaaggqgaa	agmglgkdmg
481	aqyaaaspaw	aaagqrshpa	mspgtpgptm	grsggspmdp	mvnkrpqlyg	mgsnphsqpp
541	qsspypgggy	gpppgqrypi	giqgrtsgam	agmqypqqqm	ppqygqqgvs	gyccqqqqpy
601	ysqpqpqphl	pqgaqlpsq	sqgryqpqgd	msqegygtts	qpplapqkpn	hedlnliqge
661	rpsslpdlsg	siddlptgte	atlsasvas	gstssqgdqs	npaqsfpfsh	asphlssipg
721	gpspspvgs	vgsnqersgp	ispasipgsq	mppqppgsqs	essshpalsq	spmpqergfm
781	agtgrrnpqma	qygpqgtggs	msphpspggq	mhaqissfqq	snssgygpgq	msqygpqgny
841	srppaysgvp	sasysgpgpg	mgisannqmh	gqgspqpcga	vplgrmpsag	mqnrrptpgnm
901	ssmtpspgpm	sqgggpgmgp	pmpvtvrkaq	aaaaavmqaa	ansaqrqgs	fpgmnqsglm
961	asspysgpm	rvsfstgdtk	appysmapam	vnsaaavgl	admmspgesk	lplplkadgk
1021	eegtpqpesk	skkssssttt	gekitkvyl	gneperklwv	dryltfmeer	gspvsslpav
1081	gkklpdlfrl	yvcvkeiggl	aqvknkknwr	elatnlvgt	sssaasslkk	qyiqylfafa
1141	ckiergeepg	pevfstgdtk	kqpklpqpsp	ansgslqgpg	tpqstgnsnm	aevpgdlkpp
1201	tpastphqgm	tpmqggrrst	isvhdpfsvd	sdssfpkrns	mtpnapyqqg	msmpdvmgrm
1261	pyepnkdpfg	gmkrkvpssse	pfmtqgqmpn	ssmqdmynqs	psgamenlgn	gqrqdfpyga
1321	sydrrehpyg	qypgqgpps	gqppygghqp	glypqgpnk	rhmdgmygpp	akrhgedmyn
1381	mqysqqqgm	ynqyggsyeg	pdrripqggy	pypysrermq	gpgqigthgi	ppqmmggplq
1441	ssseegqgqn	mwaarndmpy	pyqnrqgpgg	ptqappypgm	nrtddmmvpd	qrinhesqwp
1501	shvsgqgpm	ssasraapit	rppqpsyqtp	pslpnhisra	spasfqrsl	enrmepsksk
1561	flpsmkmgkv	mpsvtsqvt	gpppqpppir	reitfppgsv	easqpvlkqr	rkitskdiwt
1621	peawrvmmnl	ksgllaestw	aldtinilly	ddstvatfnl	sqslsgflell	veyfrkclid
1681	ifgillmeyev	gdqslkalldh	naarkddsqs	laddsgkeee	daecidddee	deedeedese
1741	ktesdekssi	altapdaaad	pkekpkqask	fdklpikivk	knnlfvvdrr	dklgrvqefn
1801	sgllhwqlgg	gdttehiqth	feskmeippr	rrppplssa	grkkeqegkg	dseeqqeksi
1861	iatiddvlsa	rgalpedan	pgpqtesskf	pfigqgaksh	rnkilledep	rsrdetpct
1921	iahwqdelak	rcicvsnivr	slsfvpgnda	emskhpglvl	ilgklillhh	ehperkrapp
1981	tyeekeddck	gvacskdeww	wdclevlrdn	tlvtlanisg	qldlsaytes	iclpildgll
2041	hwmvcpsaea	qdpfptvgpn	svlspqrlvl	etlcklsiqd	nnvdlilarp	pfsrqekfya
2101	tlvryvgdrk	npvcremsma	llsnlaqgda	laaraiaavqk	gsignlisfl	edgvtmagyq
2161	qsqhlmlhmq	pppleppsvd	mmcraakall	amarvdenrs	efllhegrll	disisavlms
2221	lvasvicdvl	fqigql				

SEQ ID NO: 37 Human ARID1B cDNA Sequence Variant 2 (NM_020732.3. CDS: from 1 to 6750)

1	atggcccata	acgcgggcgc	cgcgccgcgc	gccggcacc	acagcgccaa	gagcgggcgc
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121	tccgcgccgc	cagcgccgcg	atcctctctc	tcctcgctcg	gcccgggctc	ggccatggag
181	acggggctgc	tccccacca	caaactgaaa	accgttggcg	aagccccgcg	cgcgcgcgcc
241	caccagcagc	accaccacca	ccaccatgcc	caccaccacc	accaccatgc	ccaccacccc
301	caccaccacc	gcagcagcta	aaccagttcc	agcagcagca	gcagcagcag	gcagcagcag
361	caacagcagc	agcagcagca	gcagcaacag	caacatccca	tttccaacaa	caacagcttg
421	ggcgggcgcg	gcgcgggcgc	gcctcagccc	ggccccgaca	tgagcagcgc	gcaacatgga
481	ggcgccaaag	acagtgctgc	ggcgggccag	gccgaccccc	cgggcccgcc	gccgctgagc
541	aagccggcgc	acgaggaacg	cgcgcgcgcc	aagatggggg	agccggcgcg	cgcccgctac
601	gagcaccgcg	cgttgggcgc	cctgggcacg	cagcagccgc	cggtcgccgt	gcccgggggc
661	ggcgggcgcg	cggcgccgcg	cccgagttt	aataactact	atggcagcgc	tgccctcgcg
721	agcgggcgcc	ccggcgccgc	cgctgggcct	tgctttgatc	aacatggcgc	acaacaaagc

TABLE 1-continued

781	ccccggatgg	ggatgatgca	ctccgctcc	gccgcccgcg	ccggggcccc	cggcagcatg
841	gacccccctgc	agaactccca	cgaagggtac	cccaacagcc	agtgcacca	ttatccgggc
901	tacagccggc	ccggcgccgg	cggcgccggc	ggcgccggcg	gcggaggagg	aggaggcagc
961	ggaggaggag	gaggaggagg	aggagcagga	gcaggaggag	caggagccgg	agctgtggcg
1021	cgccggccgc	cgccggccgc	ggcagcagca	ggaggccggc	gcggcgccgg	ctatgggggc
1081	tcgtccgccc	ggtagccggg	gctgagctcc	ccccggcagc	aggcgccggg	catgatgatg
1141	ggccccgggg	gcggcgccgg	cgcgagcctc	agcaaggccg	ccgcggcctc	ggcgccgggg
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1261	cagctgctca	cctcgcccag	ccccatgatg	cggagctacg	gcggcagcta	ccccagtagc
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1501	atgagccccc	gcacccccc	accgacctag	ggcagatccc	agggcagccc	aatggatcca
1561	acggtagatga	agagacctca	gttgtatggc	atgggcagta	accctcattc	tcagcctcag
1621	cagagcagtc	cgtaccacag	aggttcctat	ggccctccag	gcccacagcg	gtatccaatt
1681	ggcatccagg	tcgggaactcc	cggggccatg	gccggaatgc	agtaacctca	gcagcaggac
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1801	gggtgtgagtg	gttactgccca	gcaggggcaa	cagccatatt	acagccagca	gccgcagccc
1861	ccgcacctcc	caccccgccc	gcagtatctg	ccgtcccagt	cccagcagag	gtaccagccg
1921	cagcaggaca	tgtctcagga	aggctatgga	actagatctc	aacctcctct	ggcccccgga
1981	aaacctaaac	atgaagactt	gaacttaata	cagcaagaaa	gacctcaag	tttaccagat
2041	ccgtctggct	gcattgatga	cctccccacg	ggaaacggag	caactttgag	ctcagcagtc
2101	agtgcacccg	ggcccccag	cagccaaggg	gatcagagca	acccggcgca	gtcgccttct
2161	tccccacatg	cgtccccctca	tctctccagc	atcccggggg	gccccatctc	ctctcctgtt
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2281	ggtagtcaga	tgctcccgca	gccacccggg	agccagtcag	aatccagttc	ccatcccgcc
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3361	aagtggcgtg	agctggcaac	caacctaaac	gttggcacct	caagcagtcg	agcgagctcc
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3781	cagcagggca	tgagcatgcc	cgatgtgatg	ggcaggatgc	cctatgagcc	caacaaggac
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TABLE 1-continued

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9541	gatgccacaga	ctttacatgt	gtaccaaaaa	aaaaaaaaaag	ataaaaaata	aaggtgcaaa
9601	gaaagtttag	tatttttgaa	tggtgctata	aagtgaaaaa	aaaaaaaaa	

SEQ ID NO: 38 Human ARID1B Amino Acid Sequence isoform B (NP_065783.3)

1	mahnagaaaa	agthsaksgg	seaalkeggs	aaalssssss	saaaaaasss	sssgpgsame
61	tglllpnhklk	tvgeapaapp	hqqhllhhha	hhhhhhahhl	hhhhhalqqq	ngfqgggggg
121	qqqqqqqqqq	qhpisnnns1	ggagggapqp	gpdmeqpqh	gakdsaaagg	adppppp1ls
181	kpgdeddapp	kmgepaggry	ehpglga1gt	qpppvavpgg	gggpaavpef	nnyygsaapa
241	sgggpggragp	cfdqhgggqs	pmgmmhsas	aaaagapggm	dplqnshegy	pnsqcnhypp
301	ysrpgagggg	gggggggggg	ggggggggag	aggagagava	aaaaaaaaaa	gggggggggg
361	ssagyv1ss	prqggggmmm	gpqgggaasl	skaaagaaag	gfqrfaqgnq	hpsgatpt1n
421	qltspspmm	rsyggssyep	sspsappppp	sqppsqaaaa	gaaagggqaa	agmg1gkdmg
481	aqyaaaspaw	aaaqqrshpa	mepgtppgpm	grsqgspmdp	mvmk1rpqlyg	mgsnphsqpp
541	qsspypgggy	gppgpq1rypi	giqgrcpgam	agmqypqqqd	sgdatwketf	wlmpqyggqg

TABLE 1-continued

601	gvsgycqcgqg	qpysqcpqp	phlppqayl	psqsgqryqp	qddmsqegyq	trsqpplapg
661	kpnhedlnli	qqrpslpld	lsgsiddlpt	gteatlssav	sasgstssqg	dqsnpaqspf
721	sphasphlss	ipggpspspv	gspvgsnqsr	sgpispasip	gsqmpqpqp	sqsessshpa
781	lsqspmpqer	yfmagtrqrp	qmagygpqqt	gpsmsphpsp	ggqmhagiss	fqqsnsstgy
841	gpqmsqygpq	gnysrppays	gvpsasygsp	gpgmgisann	qmhgqgpps	cgavplgrmp
901	sagmqnrpfp	gnmssmtpps	pgmsqgggpg	mgppmptvnr	kageaaaavm	qaaansaqsr
961	qgsfpgmnqs	glmassspys	qpmnssslm	ntqappysma	pamvnssaas	vgladmsspg
1021	esklplplka	dgkeegtppp	eskskkssss	tttgekitkv	yelgneperk	lwdryltfm
1081	eergspvssl	pavgkpldl	frlyvcvkei	gglagvknk	kwrelatnl	vgtsssaass
1141	llkqyiqylf	afeckierge	epppevfstg	dtkkqklqp	pspansgslq	gpqtpqstgs
1201	nsmaevpgdl	kpptpastph	gqmtpmgggr	stisvhdpf	sdvsdssfpk	rnsmtpnapy
1261	qgmsmpdvm	grmpyepnkd	pfggmrkvpg	ssepfmtqgq	mpnssmqdmy	nqspsgamsn
1321	lmggqrqgfp	ygasydrhrh	pygqyppgqg	ppsgqpyygg	hqpqlypqgp	nykrhmdgmy
1381	gppakrhegd	mynmqysqq	qemyngyggg	ysgpdrrpiq	gqypypysre	rmqgpgqigt
1441	hgippqmmgg	plqsssssegp	qgnmwaarnd	mpypypnrgg	pggptqappy	pgmnrtdm
1501	vpdgrinhes	qwpshvsqrq	pymssasmq	pitrrppqsy	qtppslpnhi	srapsasfsg
1561	rslenrmeps	kspflpsmk	qkvmtvpts	qvtgppppqp	pirceitfpp	gsveasqpvl
1621	kqrkrktskd	ivtpeawrv	melksgllae	stwaldtini	llyddstvat	fnlsqslsgf
1681	ellveyfrkc	liidifilme	yevgdpsqka	ldhnaarkdd	sqsladdsgk	eedeacidd
1741	deedeede	dsektesdek	ssialrapta	aadpkckpkq	askfdklpik	ivkknlnfvv
1801	drsdklgrqv	efnsgllhwq	lggddttehi	qthfeskmei	pprrrppppl	ssagrkkeqe
1861	gkgdseeqq	ksliatiddv	lsarpgalpe	danpgqtes	skfpfgiqqa	kshrnlikle
1921	deprsrdetp	lctiahwqds	lakrcicvsn	ivrslsfvpq	ndaemskhpg	lvllilgkll
1981	lhhehperkr	apqtyekee	edkgvacskd	ewwwdclevl	rdnclvtlan	isgqldlsay
2041	tesiclpild	gllhwmvcp	aeaqpfpptv	gpnsvlspqr	lvletckls	iqdnnvdlil
2101	atppfsrqek	fyatlrvyvg	drknppvcre	smallsnlaq	gdalaaraia	vqkgsignli
2161	sfledgvtma	qyqsgqnlm	hmqppllepp	svdmmcraak	allamarvde	nrsefllheg
2221	rlldisisav	lnslvasvic	dvlfqigql			

SEQ ID NO: 39 Human ARID1B cDNA Sequence Variant 3 (NM_001346813.1,

CDS: from 76 to 6945)

1	ggggggcgcg	gcgacggcgg	cggcgccgctg	aacagtgtgc	accaccaccc	cctgctcccc
61	cgtcacgaac	tcaacatggc	ccataacgcg	ggcgccgctg	ccgcccgccg	caccacacgc
121	gccaaagagc	gcggctccga	ggcgctctc	aaggagggtg	gaagcgccgc	cgcgctgtcc
181	tcctcctcct	cctcctccgc	ggcgcgagcg	gcggcatcct	cttcctcctc	gtcgggcccgc
241	ggctcgccca	tggagacggg	gctgctcccc	aaccacaaac	tgaaccacgt	tggcggaagcc
301	ccgcgcgcgc	gcgccaccca	gcagcaccac	caccaccacc	atgccacca	ccaccaccac
361	catgcccacc	acctccacca	ccaccacgca	ctacagcagc	agctaaacca	gttccagcag
421	cagcagcagc	agcagcaaca	gcagcagcag	cagcagcagc	aacagcaaca	tccatttcc
481	aacaacaaca	gcttgggcgg	cgcggcgccg	ggcgcgccct	agcccgccgc	cgacatggag
541	cagccgcaac	atggaggcgc	caaggacagt	gctcgggcgc	gccaggccga	cccccggggc
601	ccgcgcgtgc	tgagcaagcc	ggcgacgagc	gacgacgcgc	cgcccaagat	gggggagccg
661	gcggcgccgc	gctacgagca	ccggggcttg	ggcgccctgg	gcagcgagca	cgcccggttc
721	gccgtgcccg	ggggcgccgc	cggcccgccg	gccgtcccg	agtttaataa	ttactatggc
781	agcgctgccc	ctcgagcggg	cggcccccgc	ggcgcgctg	ggccttgctt	tgatcaacat
841	ggcgagcaac	aaagcccgcc	gatggggatg	atgcaactcg	cctcccgccg	cgcccgccgg
901	gcccccgcca	gcattggaacc	cctgcagaac	tcccacgaag	ggtaccacca	cagccagtcg
961	aaccattatc	cgggctacag	ccggcccgcc	gcggcgccgc	gcggcgccgc	cggcgccgga
1021	ggaggaggag	gcagcggagg	aggaggaggga	ggaggaggag	caggagcagg	aggagcaggga
1081	gcgggagctg	tggcgccg	ggcccgccgc	gcggcgccag	cagcaggagg	cggcgccg
1141	ggcggtatg	ggggctcgct	cgccgggtac	gggggtgtga	gctccccccg	gcagcagggc
1201	ggcgggcatg	tgatggggcc	cggggcgccg	ggggcgccga	gcctcagcaa	ggcgcccgcc
1261	ggctcgccgc	cggggggcct	ccagcgcttc	gccggccaga	accagcacc	gtcggggg
1321	accccgaccc	tcaatcagct	gctcacctcg	cccagcccca	tgatgaggag	ctacggcgcc
1381	agctacccgc	aatcacagag	ccccagcgcg	ccgcgcgcgc	cgccgtcgca	gcccagctcc
1441	caggcgccgc	cggcgggggc	ggcgcgccgc	ggccagcagg	cggcgccggg	catgggcttg
1501	ggcaaggaca	tgggcgcccc	gtacgcgcct	gccagcccg	cctggcgccg	cgcgcaacaa
1561	aggagtcacc	cgccgatgag	ccccggcacc	ccgggaccca	ccatggcgag	atccaggggc
1621	agcccaatgg	atccaatggt	gatgaagaga	cctcagttgt	atggcatggg	cagtaaccct
1681	cattctcagc	ctcagcagag	cagtcctgtac	ccaggagggt	cctatggccc	tccaggccca
1741	cagcggtatc	caattggcat	ccagggtcgg	actcccgggg	ccatggccgg	aatgcagtac
1801	ccccagcagc	agatgccacc	tcagtatgga	cagcaagggt	tgagtgggta	ctgccagcag
1861	ggccaacagc	catattacag	ccagcagccg	cagcccccgc	acctccacc	ccaggcgccg
1921	tatccgcctg	ccatgcccca	gcagaggtag	cagccgcagc	aggacatgtc	tcagggaagg
1981	tatggaaacta	gatctcaacc	tcctctggcc	cccggaaaac	ctaaccatga	agacttgaac
2041	taaatacagc	aagaaagacc	atcaagttta	ccagatctgt	ctgggtccat	tgatgacctc
2101	ccacgggaa	cggagcgaac	tttagctca	gcagtcagtg	catccgggtc	cacgagcagc
2161	caaggggatc	agagcaaccc	ggcgagctcg	cctttctccc	cacatgcgtc	ccctcatctc
2221	tccagcatcc	cggggggccc	atctccctct	cctgttggtt	ctcctgtagg	aagcaaccag
2281	tctcgatctg	gcgccaatct	tcctgcaagt	atcccaggta	gtcagatgcc	tccgagccca
2341	ccggggagcc	agtcagaatc	cagttcccat	ccgccttga	gccagtcacc	aatgccacag
2401	gaaagagggt	ttatggcagg	cacacaaaaga	aacctccaga	tggtcagta	tggacctcaa
2461	cagacaggag	catccatgct	gcctcatctc	tctcctgggg	gccagatgca	tgctggaacc
2521	agtagctttc	agcagagtaa	ctcaagtggg	acttacgggc	cacagatgag	ccagtatgga
2581	ccacaaggta	actactccag	acccccagcg	tacagtgggg	tgcccagtg	aagctacagc
2641	ggccccaggc	ccggtagggg	tatcagtgcc	aacaaccaga	tgcatggaca	agggccaaag
2701	cagccatgtg	gtgctgtgcc	cctgggacga	atgccatcag	ctgggatgca	gaacagacca
2761	tttctctgaa	atatgagcag	catgaccccc	agttctcctg	gcaggtctca	gcagggaggg
2821	ccagggaatg	ggccgccaat	gccaaactgt	aaccgtaagg	cacaggaggg	agccgcagca
2881	gtgatgcagg	ctgctgcgaa	ctcagcacia	agcaggcaag	gcagtttccc	cggcatgaac

TABLE 1-continued

2941	cagagtggac	tcatggtctc	cagctctccc	tacagccagc	ccatgaacaa	cagctctagc
3001	ctgatgaaca	cgcaggcgcc	gccctacagc	atggcgcccg	ccatggtgaa	cagctcggca
3061	gcattctgcg	gtcttgcaga	tatgatgtct	cctggtgaat	ccaaactgcc	cctgcctctc
3121	aaagcagacg	gcaaagaaga	aggcactcca	cagcccagag	gcaagtcaaa	ggatagctac
3181	agctctcagg	gtatttctca	gcccccaacc	ccaggcaacc	tgccagcccc	ttcccaatg
3241	tccccagctc	ctgctagcat	ctcctcattt	catggagatg	aaagtgatag	cattagcagc
3301	ccaggctggc	caaagactcc	atcaagccct	aagtccagct	cctccaccac	tactggggag
3361	aagatcacga	aggtgtacga	gctggggaat	gagccagaga	gaaagctctg	ggtcgaccga
3421	tacctcacct	tcacggaaga	gagaggtctc	cccgtctcaa	gtctgcctgc	cgtgggcaag
3481	aagccccctg	acctgttccg	actctacgtc	tgcttcaaa	agatcgggg	tttggcccag
3541	gttaataaaa	acaagaagtg	gcgtgagctg	gcaaccaacc	taaaactgtg	cacctcaagc
3601	agtgcacgca	gccccctgaa	aaagcagtat	atccagtagc	tggttgcctt	tgagtgcag
3661	atcgaacgtg	gggaggagcc	ccgcgcggaa	gtcttcagca	ccggggacac	caaaaagcag
3721	cccaagctcc	agccgcccac	tcctgctaac	tcgggacctc	tgcaaggccc	acagaccccc
3781	cagtcacact	gcagcaattc	catggcagag	gtcccagggt	acctgaagcc	acctacccca
3841	gcctccaccc	ctcacggcca	gatgactcca	atgcaagggt	gaagaagcag	tacaatcagt
3901	gtgcacgacc	cattctcaga	tgtgagtgat	tcctccttcc	cgaacgggaa	ctccatgact
3961	ccaaacgccc	cctaccagca	gggcatgagc	atgcccagtg	tgatgggag	gatgcccctt
4021	gagcccaaca	aggacccttt	tgggggaatg	agaaaagtgc	ctggaagcag	cagacccttt
4081	atgacgcaag	gacagatgcc	caacagcagc	atgcaggaca	tgtacaacca	aagccccccc
4141	ggagcaatgt	ctaactctgg	catggggcag	cgccagcagt	ttccctatgg	agccagttac
4201	gaccgaagcg	atgaacctta	tgggcagcag	tatccaggcc	aaggccctcc	ctcgggacag
4261	ccgcgctatg	gagggcacca	gcccggcctg	taccacagc	agccgaatta	caaacgccat
4321	atggacggca	tgatcggggc	cccagccaa	cgccacgagg	gcgacatgta	caacatgcag
4381	tacagcagcg	agcagcagga	gatgtacaac	cagtagggag	gtccctactc	ggggccggac
4441	cgacggccca	tcaggggcca	gtaccgctat	ccctacagca	gggagaggat	gcaggggccc
4501	gggcagatcc	agacacacgg	aatcccgctc	cagatgatgg	gcggcccgct	gcagtcgtcc
4561	tcagtgagg	ggcctcagca	gaatatgtgg	gcagcacgca	atgatatgcc	ttatccctac
4621	cagaacaggc	agggccctgg	cgcccttaca	caggcgcccc	cttaccagg	catgaaccgc
4681	acagacgata	tgatggtacc	cgatcagagg	acaaatcatg	agagccagtg	gccttctcac
4741	gtcagccagc	gtacgacctt	tatgtcgtcc	tcagccctca	tgagcccat	cacacgcccc
4801	ccacagccgt	cctaccagac	gccaccgtca	ctgccaaatc	acacctccag	ggcgcccgag
4861	ccagcgtcct	tcacgcgctc	cctggagaac	cgcatgtctc	caagcaagtc	tccttttctg
4921	ccgtctatga	agatgcagaa	ggtcatgccc	acggctccca	catcccaggt	caccgggcca
4981	ccacccccaa	cacccccaat	cagaaggagg	atcacctttc	ctcctggctc	agtagaagca
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5101	gcgtggcgct	tgatgatgtc	ccttaaatca	ggcttttgg	ctgagagtag	tgggctttg
5161	gacactacta	atattctccc	gcacgatgac	agcactgttg	ctactttcaa	tctctcccag
5221	tcgcttgagc	ttctcgaaac	cttagtcgag	tactctagaa	aacgcctgac	tgacattcct
5281	gggaattctta	tggaatatga	agtgaggagc	cccagccaaa	aagcacttga	tcacaacgca
5341	gcaaggaaag	acgacagcca	gtcctcgcca	gacgattctg	ggaaagagga	ggaagatgct
5401	gaatgtattg	atgacgacga	ggaagacgag	gaggacgagg	aggaagacag	cgagaagaca
5461	gaaagcgatg	aaagacgag	catcgctctg	actgccccgg	acgcgctg	agacccaaag
5521	gagaagccca	agcaagccag	taagtctgac	aagctgcca	tgaagacagt	caaaaagaac
5581	aacctgtttg	tcgttgaccg	atctgacaag	ttggggcggt	tgccaggagt	caatagtggc
5641	ctctcgactc	ggcagctcgg	cggggggtgac	accaccgagc	acattcagac	tcactttgag
5701	agcaagatgg	aaatcccccc	ccgcaggcgc	ccaccccccc	cctcaagctc	cgccaggcaga
5761	aagaaagagc	aagaaggcaa	aggcgactct	gaagagcagc	aagagaaaa	caccatagca
5821	accatcgatg	acgtcctctc	tgctcgccca	ggggcattgc	ctgaagacgc	aaaccctggg
5881	cccagaccgc	aaagcagtaa	gtttcccttt	ggtagccagc	aagccaaaag	tcaccggaa
5941	accaagctgc	tgaggagaca	gcccaggagc	cgagacgaga	ctcctctgtg	taccatcgcg
6001	cactggcagc	actcgctggc	taagcgatgc	atctgtgtgt	ccaatattgt	ccgtactgtg
6061	tcactcgtgc	ccggcaacga	cgccgaaacg	tcacaaatc	caggcccggt	gctgatcctg
6121	gggaagctga	ttcttcttca	ccacgagcat	ccagagagaa	agcgagcacc	gcagacctat
6181	gagaaagagg	aggttaggga	caagggggtg	gcctgcagca	aagatgagtg	gtggtagggac
6241	cgctcgagg	tcttgaggga	taacacgttg	gtcacgttgg	ccaacatttc	cgggcagcta
6301	gacttgtctg	ccttacacga	aagcatctgc	ttgccaattt	tggttggtgt	gctgcactgg
6361	atggtgtgcc	ttctgtcaga	ggcacaaagt	ccctttccaa	ctgtgggacc	caactcggcc
6421	ctgtcgccctc	agagacttgt	gctggagacc	ctctgtaaac	tcagtatcca	ggacaataat
6481	gtggacctga	tcttggccac	tcctccattt	agtcgtcagg	agaaattcta	tgctacatta
6541	gttaggtacg	ttggggaccg	caaaaaccca	gtctgcccag	aaatgtccat	ggcgctttta
6601	tcgaacctcg	cccaagggga	cgactagaca	gcaagggcca	tagctgtgca	gaaaggaaag
6661	actggaacac	cgacaagctt	cctagaggat	ggggccacga	tgggccagca	ccagcagagc
6721	cagcaccaac	tcatgcacat	gcagcccccg	cccctggaac	cacctagcgt	agacatgacg
6781	tgacggcgcg	ccaaggcttt	gctagccatg	gccagagtg	acgaaaaacg	ctcggaattc
6841	cctttgcacg	agggcgggct	gctggatacc	tcgatatac	ctgccccgaa	ctctctgggtc
6901	gcattctgtc	ctgtgtgac	actgtctcag	attgggcagt	tatgacataa	gtgagaagcc
6961	aagcatgtgt	gagcgaagac	tagagggcca	catacaactg	gcccgtttct	gttctcgttt
7021	atccagcgta	ggaagaagga	aaagaaaacc	tttgcCctc	tgccccattc	actattttacc
7081	aattgggaac	taagaaata	atataattga	acagttatga	aattaatatt	tgctgtcgtg
7141	gtgtacaagt	acatcctttg	gggtttttct	tttccctctt	tttaaaccaa	agttgtcgtc
7201	tag-gcattc	aaaggtcacc	ttttgttctt	cacagatctt	tttaattgtc	ttccccatgt
7261	cgtattgtac	ttttggggga	agcaaatgga	ctttaaagaa	aaaagttgtg	gcaaaagacg
7321	ctaagacgcg	aaaatttcac	cacactgagt	caaaaagggt	aaaaactatc	cacttctctat
7381	gcgttttact	cctcagagaa	tgaaaaaaac	tgcatcccat	cacccaaagt	tctgtgcaat
7441	agaaatttct	acagatcacg	gtataggggg	tcaaggaggt	atgtcggtca	gtagtcaaaa
7501	ccatgaaatg	acactggctc	ctccacagga	ataggttcc	actaggtcgt	gagcaaaaa
7561	aatgtttttt	aagattgaga	atacatacct	gacaacgac	cggaaactgc	tcctcaccac
7621	cccgctcagc	cctgctgtcg	gcgtttgacc	ttccacgcca	cagctcctca	caatcccttc
7681	catcattttt	taaatatttt	ttttactgcc	tatgggctgt	gatgtatata	gaagttgtac

TABLE 1-continued

7741	attaacata	ccctcatttc	ttctttttct	ttctttttcc	ttcttttagt	acaaagttcc
7801	agttttcttt	tcatgatgtg	gtaactacga	agtgatggta	gatttaaata	attttttatt
7861	tctactctat	atattttttc	actagggcca	tatctccaaa	aaaagaaaga	aaaaatacaa
7921	aaaacaaaa	caaaaaaaa	agagggtaat	gtacaagctt	ctgcacgtat	aaagccatgc
7981	tcgatttcag	gagagcagct	gatcacaatt	tgcttcatga	atcaagggtg	ggaaatgggt
8041	ata-atggat	tgatttagaa	aatggttacc	agtacagtca	aaaaagagaa	aacgaaaaaa
8101	atacaactaa	aaggaagaaa	cacaacttca	aagatttttc	agtgatgaga	atccacactt
8161	gtattttcaag	ataatgtagt	taaaaaaaa	aaaaaagaaa	aaaacttgat	gtaaatctct
8221	cctcttcctc	tggtctaatg	aataccattt	attcagtata	aaatctttat	atgttcacaca
8281	tggttaagaat	aaatgtatgt	taaatcttgt	taagcactgt	gatgggtgtt	cttgaatact
8341	gttctagttt	ccttaaaagt	gtttcctagt	aatcaagtta	tttacaagaa	ataggggaat
8401	gcagcagtg	attcacatta	taaaacccta	catttggaag	agacctttag	gggttaccta
8461	ctttagagt	gggagcaaca	gtttgatttt	ctcaaatat	ttagctaatt	agtctttctt
8521	tgaagcaatt	aactctaacg	acattgaggt	atgatcattt	tcagtattta	tgggaggtgg
8581	ctgctgaccc	acttgaggtg	agatctcaga	agcttaactg	gcctgaaaa	gtaacattct
8641	gccttttact	aactccatct	tagtttaatc	aaagttcaat	ctattccttg	ttctctctgt
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8761	atgttaaaaa	agaatctgtc	atattgtatg	tggtgtacat	agtgcacagt	gatgatctct
8821	atttcaggcg	acatttatgt	tctcacattc	cttctacct	gggtgcacagt	agctttttaa
8881	tactagtcc	ttctaattta	aaactttctt	tcctgggtca	ttgactgtta	ctgtgtaata
8941	atcgatttct	ttgaaactgc	tgcataatta	tgctgttagt	ggacctctac	ctctctctct
9001	ccctctccca	atcacagtat	actcagaatc	cccagccct	cgatacatt	gtgtcggttc
9061	acattactca	cagtaataata	tggaagagtt	agacaagaac	atgcagttac	agtcattgtg
9121	agacgtgact	ctccagtgct	acgaggaata	aaatcatctt	ttctgcaaac	agtcctctat
9181	ctgtcaactc	ctacagtact	gagtcacaca	gtcttcttac	ataacaatgc	aaccaaatat
9241	atgttgaatt	aaagacccat	ttataattct	gctttaaata	catctgcttg	ctaagaacag
9301	atttcagtg	tccaagcttc	aaatatggag	atttgttaaga	gggaattcaa	tattattcta
9361	atttctctct	tacagagtac	aaataaaagg	tgtatacaaa	ctccgaacat	atccagattt
9421	ccaatttcct	tgtaacycag	aagagtaaaa	taattaacaa	aagactgttg	ttatgggttg
9481	cattgtaacc	gatacgcaga	gtctgaccgt	tgggcaacaa	gtttttctat	cctgatgcgc
9541	aacacagctc	ctagagacta	atccaggaag	actttagcct	cctttccata	ttctcaccoc
9601	cgaatcaaga	tttacagaag	cccacgaaga	atttacagcc	tgcttgagat	catcttgctt
9661	ataaactgag	ttattgcttt	gtcctaaaaa	ttagtcgggt	tttttttttc	tatgaggtct
9721	ttcagaaatt	tacagagtgc	ccagacttta	catgtgtacc	aaaaaaaaaa	aaaagataaa
9781	aaataaaggt	gcaagaagag	tttagtattt	tggatgggtg	ctataaagtt	gaa

SEQ ID NO: 40 Human ARID1B Amino Acid Sequence isoform C

(NP_001333742.1)

1	mahnagaaaa	agthsaksgg	seaalkeggs	aaalssssss	saaaaaasss	sssgpggsame
61	tgllpnhklk	tvgeapaapp	hqghhhhhha	hhhhhhahhl	hhhhhalqqql	nqfqqqqqqq
121	qqqqqqqqqq	qhpisnnns1	ggagggapqp	gpdmeqpqhg	gakdsaaaggq	adpppgpplls
181	kpgdeddapp	kmgepaggry	ehpglgaigt	qpppvavpgg	gggpaavpef	nnyygsaapa
241	sgpgpggragp	cfqdhggqqs	pgmgmmhsas	aaaagapgsd	dplnshhegy	pnsqcnhypp
301	ysrpagagggg	gggggggggs	gggggggggag	aggagagava	aaaaaaaaaa	gggggggyyg
361	ssagyvllss	prqggggmm	gpggggaasl	skaaagsaag	gfqrfaqgnq	hpsgatptln
421	qltspspmm	rsyvgyppey	sspsappppp	sqpsqaaaa	gaaaggqaaa	agmglgkdmg
481	aqyaaaspaw	aaaqqrshpa	mspgtpgpc	grsqspsmdp	mvmkrrpqlg	mgspnhsqpp
541	qsspypggys	gpppgryp1	glqgrtpgam	agmqypqqqm	ppqyggqgvs	gyccqggqpy
601	ysqppqpphl	ppqaqylpsq	sqqryqpqqd	msqegygtts	qpplapgkpn	hedlnliqqe
661	rpsslpdlsg	siddlptgte	atlssavsas	gstssqddqs	npaqspfsph	asphlssipg
721	gspspvqsp	vgnqrsrgsp	ispasipgsq	mppqppgsqs	essshpalsq	spmpqergfm
781	agtqrnpqma	qygpqqtgss	msphpspggg	mhaqissfqq	snssgtypgq	msqygpqgny
841	srppaysgvp	ysaysgpgg	mgisannqmh	ggqpsqpcga	vp1grmpsaq	mqnrpfpgnm
901	ssmtpsppgm	sqgggpgmgp	pmpvtvrkaq	eaaaaavmqaa	ansaqsrqgs	fpgmqnsglm
961	asspsypqpm	nnssslnmtq	appysmapam	vnssaaavgl	admmsspgesk	lp1plkadgk
1021	eegtpqpesk	skdsysgggi	sqpptpgnlp	vpssmpsssa	sisffhgdss	dsisspgwpk
1081	tpsspkssss	tttgekitkv	yelgneperk	lwdrylrfm	eergspvssl	pavgkkpdl
1141	frlyvcvkei	gglaqvknkn	kwrelatnln	vgtsssaass	lkkqyiqylf	afeckierge
1201	epppevfstg	dtkkqpk1qp	papansgslq	gpqtpqscgs	nsmavpgdl	kpptpastph
1261	gqmtpmqggr	sstisvhdpl	sdvsdssfpk	rnsmtpnapy	qggmsmpdvm	grmpyepnkd
1321	pfggmrvkvp	ssepfmtqgg	mpnssmqdmy	nqspsgamsn	lmggqrqqfp	ygasydrre
1381	pygqypqpgg	ppsgpppygg	hqpglypqqp	nykrhmdgmy	gppakrhegd	mynmqyssqq
1441	qemynqyqqs	ysqpdrripq	qyypypysre	rmqppqgiqt	hgippqmmgg	plqssssseq
1501	qqnmwaarnd	mpypqnrqg	pggptqappy	pgmnrtdmm	vdpqrinhes	qwpshvsqrq
1561	pymssasmq	pitrrppqpy	qtpplpnhi	srapsasfq	rslenrmsps	kspflpsmk
1621	qkvmpvtvps	qvtgppppqp	pirreitfpp	gsveasqpvl	kqrrktskd	ivtpeawrv
1681	mslksllae	stwaldtini	llyddstvat	fnlsq1sgfl	ellveyfrkc	lidifgilme
1741	yevgdpsqka	ldhnaarkdd	sqsladdsgk	eeedaecidd	deedeedeee	dsektesdek
1801	ssialtapda	aadpkekpkq	askfdklpik	ivkkn1fvv	drsdklgrvq	efnsgllhwq
1861	lgggdttethi	qthfeskmei	prrrrppppl	ssagrkkkeq	gkgdseeqqe	ksiatiddv
1921	lsarpgalpe	danpgpqtgs	skfptgiqqa	kshrn1kle	deprsrdetp	lctiahwqds
1981	lakrcicvsn	ivrs1sfvpg	ndaemskhpg	lv1ilgk1li	lhhehperkr	apqtyeeked
2041	edkgvacsdk	ewwvdclevl	rdntlvltan	isgqldlsay	tesic1pild	gllhwmvcp
2101	aeaqpdpftv	gpnsvlspqr	lvletlckls	iqdnnvd1li	atppf1srqek	fyatlrvryv
2161	drknvpcrem	smallenlaq	gdalaaraia	vqkgsignli	stledgvtma	qyqqsghnlm
2221	hmqppllepp	svdmmcraak	allamarvde	nrsef1l1heg	r1ldis1sav	lnslvasvic
2281	dvlfqigql					

TABLE 1-continued

SEQ ID NO: 41 Mouse ARID1B cDNA Sequence (NM_001085355.1, CDS: from 22 to 6756)

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121 caccaccacc accatgcccc ccacctccac cactccacc accaccacgc actacagcag
181 cagctaaacc agttccagca gccgcagccg ccgcagccac agcagcagca gccgcgcgca
241 ccgcccgcagc agcagcatcc cactgccaac aacagcctgg gcggtgcggg cggcggcgcg
301 cctcagcccc gcccgacat ggagcagccg caacatggag gcgccaaagg cagtgtcgcg
361 ggcaatcagg ctgaccgcga gggccagcct ctgctgagca aaccggggcg caggagcagc
421 gcgcgcgcca agatggggga gccggcgggc agccgctatg agcaccggg cctgggcgcg
481 cagcagcagc ccgcgcgggt cgcctgccc gggggcgggc gcggcccagc ggccgtctcg
541 gagtttaata attactatgg cagcgtgccc cctgctagcg gcggcccggg cggcgcgctc
601 gggccttgct ttgatcaaca tggcggaaca caaagccccc ggatggggat gatgcactcc
661 gcctctgccc ccgcggggc cccagcagc atggaccccc tgcagaactc ccacgaaggg
721 taccacaaca gccagtacaa ccattatccg ggctacagcc gggccggcgc gggcgcgccc
781 gggcgcgggc gcggaggagg aggagggcagc ggaggagggt gaggaggagg aggagcagga
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901 gcggcgcgcg cagcagcagc agcagcagga ggaggcggtg gcggcggtta tgggagctcg
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1021 cccggggggc gcggggcgcc gagcctcagc aaggcgggcg ccggcgcggc ggcgcgcgcg
1081 gggggcttcc agcgttctgc cggccagaac cagcaccctg cgggggctac accgaccctc
1141 aaccagctgc tcacctcacc cagccccatg atgaggagct accggcgtag ctaccocgac
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1321 cagtacgcgc ctaccagccc ggccctggcg gccgcgcaac aaaggagtca ccggcgatg
1381 agccccggca ccccgggacc gacctggggc agatcccagg gcagcccgat ggacccaatg
1441 gtgatgaaga gacctcagtt gtatgggatg ggtactcacc cccactccca gccacagcag
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1561 atgcaggggc gggctccagg ggccctggga ggcttgagc acccgagca gcagatgcca
1621 ccgagtcagc gacagcaagc tgtgagtggt tactgccagc aaggccagca gccatactac
1681 aaccagcagc cgcagccctc gcacctcccg cccagggcac agtaacctga gccgcgcgcg
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1981 cagagcaaac cagcgcagtc tcctttctcc ccacatgcat cactcactc ctccagcagc
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2281 ccattccatgt cgcctcacc accccctggg ggcagatgc atcctgggat cagtaacttt
2341 cagcagagta actcaagtgg cagctacggc ccacagatga gccagtatgg accccaaggc
2401 aactactcca gaaccccaac atatagcggg gtaccagtg caagctacag cggcccaggg
2461 cccggtatgg gcattcaatgc caacaaccag atgcatggac aagggccagc ccagccatgt
2521 ggtgctatgc cctggggagc aatgccttca gctgggatgc agaacagacc atttcctgga
2581 accatgagca ggcgtacccc cagttctcct ggcagtgctc aacaggaggg gccagggaatg
2641 ggccccacca tgcccactgt gaaccggaag gccagggaag ctgccgcagc tgtgatgcag
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3661 cctcatggag agatgactcc catgcaaaagc ggaagaagca gtacagtcag tgtgatgac
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4141 atgtacgggc ctccagccaa gcggcagcag ggagacatgt acaacatgca gtatggcagc
4201 cagcagcagc agatgtataa ccagtatgga ggtcctact ctggcccgga cagaaggccc
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4561 cgcagccctt acatgtcatc ttggccctcc atgcagccca tcacgcgcc acctcagtca
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TABLE 1-continued

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4801	cctccaccaa	tcagaaggga	gattaccttt	cctcctggct	ccgtagaagc	atcacagcca
4861	atcctgaaac	aaaggcgaaa	gattacctca	aaagatattg	ttactccoga	ggcgtggcgt
4921	gtgatgatgt	cccttaaatc	gggtctgttg	gctgagagca	cgtgggctct	ggacaccacc
4981	aatattctcc	tctatgatga	cagcaccgtc	gccaccttca	atctttocca	gctgtctgga
5041	ttcctggaac	tattagtaga	gtactttcga	aaatgcctaa	ttgacatttt	cggaaattctt
5101	atggaatatg	aagtgggtga	ccccagccaa	aaggctcttg	atcaccggtc	agggagaagaa
5161	gatgacagcc	agtccctgga	agatgattct	gggaagggaag	acgatgatgc	tgagtgtctt
5221	gtggaagagg	aggagaggag	agaggaggag	gaggaagaca	gtgaaaagat	agagtcagag
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5341	aagcaggcca	gtaagtttga	caagctgccc	ataaagattg	tcaaaaagaa	caagctgttt
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6061	gtcttgccgg	ataacacctt	ggtcacgttg	gcgaacattt	ccgggagcgt	agacttgtct
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6181	ccgtccgcag	aggctcagga	cccccttccc	actgtggggc	ccaactcagt	cctgtcgcgc
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6361	gttggggatc	gcaaaaaatcc	agtctgtcga	gaaatgtcca	tggcgccttt	atcgaaacctt
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6721	atctgcgatg	tactgtttca	gattgggcag	ttatgacatc	cgtgaaggca	cacatgtgtg
6781	agtgaacatt	agagggtcac	atataactgg	ctgttttctg	ttctcgttta	tccagtgtaa
6841	gaagaaggaa	aagaaaaaatc	tttgctcctc	tgccccgttt	actatttacc	aattgggaat
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6961	cctctggcgg	ttttctgttt	cttttttttt	taaccaaagt	tgcgctctag	tgcattcaaa
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7261	ccaaagctct	gtgcaataga	aaactctaga	gatatagtgt	aggggctgaa	ggaggtatgg
7321	cagcagtagt	caggggtcaat	gatactgctt	tctccaccgg	aaagtgggtta	cgttaggcct
7381	cgagcaaaaa	acagcgctct	cagataggtg	caaaaaatcca	ctcctagcag	ccaacagcag
7441	gatcgcttcc	tcaccacgac	cgccatgtct	gctgtggctc	agcctccacg	ggacaaagct
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7561	tatagaagtt	tgacatttaa	cataccctca	tttttttctt	cttttctttt	tttctttttt
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7681	actacgaagt	gatggtagat	ttaaataatt	ttttattttt	attttatata	ttttttcatt
7741	aggacctatc	ctccaaaaaa	caagaaaaag	aaacaaaaaa	tacaaaaaat	aaaaacaaac
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8161	aagaataaat	gtacattaaa	tcttgttacg	cactgtgatg	gggtgtcttg	aatgctgttc
8221	tagtttgctt	agcatgggtg	ccatagtaac	caagttattt	acaggaaata	gggaagatgt
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8461	caacccactg	tgacatgggg	ccctgcgaagc	taactggcct	gaacaccaga	ccttctgcct
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9241	caagtggcgt	tatagcagcc	aggcccagaa	cagcctcgca	gcacacacta	acaccaagcg
9301	gggtgtctgag	ctctcctagg	aagccttgtg	cctgcccctc	ctccattcac	ccagatccga
9361	ccctgggaag	cccacgaag	agtcaccctt	tgtctcacat	ttcctgacga	taccgagttg
9421	ctgctctgtc	ctaaaaatat	tagttctttt	ccagggcttt	cagaaatttg	caggatgccc

TABLE 1-continued

9481	atactctaaa	tgtgtaccaa	aaagagagag	aaataaaggt	gcgaagaaag	tttagtattt
9541	tggaaatgggt	cgataaaatg	gaatctgttg	gtttttaatg	taacataaga	tactattggc
9601	tggcactggc	taaaaaaaat	atctaagtg	tggagttgga	tgacacatca	acttttactt
9661	agctattcaa	agagtactta	tgttttccaa	gttaaaacag	acttggtttt	gacaggggcc
9721	gtgggtgggtc	ttatacaatg	ccagctccta	actgcagctt	ctgagaactg	gatatcggtt
9781	gcccctgagag	ctgcccgctc	ccaactatgt	gctgctgctg	cctgtgtgct	tcagcccaca
9841	aggatgtgga	gactggatag	acaacccctt	gcttcctgct	gggtgtgtgt	gagttccttg
9901	cagtcacagtc	aagtgccag	agctaccagc	ctacgtccct	catgcatcca	agagaaatga
9961	tcttgactat	catgatcaaa	acagctgtag	taataattct	agtaaatatt	tctgatgact
10021	ctgtgtaaatc	tcctacaaca	ggacactatt	cattaacttg	acagagacat	gtgggcatgt
10081	ggtcctgctt	tagtttaaca	gacaagtcaa	ccagttctca	ttacttagga	agagtggagg
10141	tatgtctggt	acaatcccaa	tgtgtgtgct	gcccttatcc	aaagacagtc	cgggggccct
10201	gtctgctgga	actgatgttc	gctccctctt	gggcttccca	ctgggatgtg	aaaagataac
10261	caatggctcc	caggttccca	gtgcccccca	aaccagtaat	cagggtctgg	actacagAAC
10321	ccgcaaaatc	atacacaggc	tgtttcaaa	ccagtactct	ctttatactc	ctgcttcctc
10381	cagcccccat	ttcacacccc	acccaaatca	caaggctctc	tgaagtctca	gaactccaaa
10441	ttaacgttgg	gatttaacat	gtgaatgctg	aggagaaaat	tgggagttgg	tgggagatca
10501	ccaaattgtc	aaaactatga	aactcatctg	tcttcccaaa	tctgacctca	gggacttggg
10561	gggttctact	tggtctctgc	cacagtattt	tctggggaac	caaaggcctc	gggaatagag
10621	aaacaggttg	ccgatatatc	tggaggtcta	agccatactg	accagtttgt	cttgagtgtt
10681	ttctttgtga	gectggaact	gtcccgggac	ccctttcttt	taaacatggt	tcaggacttt
10741	aaaaaaaagc	actgtatttt	ttttatgtaa	gccaaagatg	cctccctagc	agagatagcg
10801	ttgaactgtc	tctagtctct	tagcctgaga	gacttaaatc	gtttaaacttc	agtgtccttg
10861	tccactctgt	tgaactgcta	aggattctat	tgaatgtgtt	ctttgagggt	tgggaggagt
10921	tgtctgggtg	tgaagtcctg	catccctttg	cctgggatgt	gtatatattt	cctttgctctg
10981	gctgtgtatc	gttcttcagt	gtaagtacac	ccacactctg	tattcctttg	cctgtcctcc
11041	gcccccccat	acacacacat	cctgcatagt	ttttaaataa	ggcctgagag	actgtttcta
11101	ttctctgtca	tagctgtgga	cttttaacag	ttagggcgaa	tggcctgtca	cttgctggg
11161	ttcccgctcag	gggtgatcca	tggaaactcct	cagtggaaac	gaatttagga	cagaagatcc
11221	caccttcctt	ccaggcctgg	ggagaatcag	actgtgagat	aaacatgat	gctgcccatt
11281	cccactgccc	caccttgctt	ttaaaataaa	gtgcctccta	acgtc	

SEQ ID NO: 42 Mouse ARID1B Amino Acid Sequence (NP_001078824.1)

1	metgllphhk	lkavgeapaa	pphqghhhhh	ahhhhhhhah	hlhlhlhhha	lqqqlnqfqg
61	pqpqpqqqqq	ppppppqqhp	tannslggag	ggapqpqpd	eqpqhggakd	svagnqadpq
121	qqpllskpgd	eddappkmg	pagstyehtp	lgaqqqpapv	avpgggggpa	avsefnnyyq
181	saapasggpg	gragpcfdqh	ggqgspmgm	mhsasaaaga	pssmdplqns	hegyppnsqyn
241	hypgyrprga	gggggggggg	ggsggggggg	gaggagaaaa	aaagagavaa	aaaaaaaaaa
301	aagggggggg	gssssgygvl	ssprqggggm	mmgpggggaa	slskaaagaa	aaaggfqrfa
361	gqngqhpqat	ptlnqltsp	spmmrsyggg	ypdyssssap	pppsqpqsga	aaagaaggqq
421	aaagmgllgkd	lgaqyaaasp	awaaaqqrsh	pamspgtppg	tmgrsqgspm	dpmvmkrpql
481	ygmgtphsqg	pqqsspyppg	sygppgaqry	plmgqgrapg	alggllqypqg	qmpqpyggqa
541	vsgycqggqg	pyynqgqps	hlppqaqlq	paaagsggry	qpqgdmseqg	ygtrsqppla
601	pgkenhedln	liqgerpsl	pdlsgsiddl	ptgteatlss	avsasgstss	qgdqenpaqs
661	pfsphasphl	sslpggpsps	pvgspvgsnq	srsqspispas	ipgsqmpppp	pqsqsesssh
721	palsqspmpg	ergfmggtqr	npqmsqyggp	qtgpsmsphp	spggqmhpqi	snfqgsnssg
781	tygppqmsqyg	pqgnysrtpt	ysgvpsasys	pgpgmgina	nnqmhgggpa	qpcgampigr
841	mpsagmqnpr	fpgtmssvtp	sspgmsqggg	pgmgppmptv	nrkaqeaataa	vmqaaansaq
901	srqgsfpgmn	qsglvasssp	yqsmnnns	lmstqagpys	mtptmvsnst	asmgladmms
961	psekslsvpl	kadgkeegvs	qpeskskdsy	gsqgisqppt	pgnlpvpspm	spssasissf
1021	hgdesdsiss	pgwpktppsp	ksssssttge	kitkvyelgn	eperklwvdr	yltfmeergs
1081	pvsplpavqg	kpldlfrlyv	cvkeigglaq	vnknkkwrel	acnlngvtss	saasslkkqy
1141	iqylfafeck	tergepppe	vfstgdskkq	pklqppspan	sgslqgpqtp	qstgnsmae
1201	vpqdlkpptp	astphgqmtp	mqsgsrstvs	vhdpfsdvds	saypkrrsmt	pnapyqggmg
1261	mpdmngpmqss	epnkldfsgm	rkvpgsssepf	mtggqvpsng	mqdmyngqps	gamsnlmgqg
1321	rqgfpygtsy	drtheayggq	ypgggppptg	ppygghqpgl	ypqqpnykrh	mdgmygppak
1381	rhegdmynmq	ygsqqqemyn	qyggysgpd	rrpiqqgyyp	pynrermqgp	gqmqqhgpip
1441	qmmggpmqss	sseegpdkmw	atrndmpypy	qsrgqpggpa	qappypgmnr	tddmvpeqrg
1501	inhesqwpsh	vsqrqpmys	sasmqpitrp	pqssyqtpps	lpnhisraps	pasfqrsls
1561	rmspsksphl	ptnmkmqvm	tvptsqvtgp	ppqpppirre	itfppgsvea	sqpilkqrk
1621	itskdiptve	awrvmslks	gllaestwal	dtinillydd	stvatfnlsq	lsqflllell
1681	yfrkclidif	gilmeveygd	psqkaldhrs	gkdddsqsl	ddsgkeddda	eclvveeeee
1741	eeeeedseki	esegksspal	aapdasvdpk	etpkqaskfd	klpikivkkn	klfvvdrrsd
1801	lgrvqefssy	llhwqlgggd	ttehiqthfe	skmeiprrrr	ppplststgk	kkelegkgsd
1861	eeqpeksiaa	tiddvlsarp	galpedtnpg	pqcdsgkfpf	giqqakshrn	irllledeprs
1921	rdetpctia	hwqdsalakr	icvsnivrs	sfvpgndaem	skhpgllvlil	gklillhheh
1981	perkraptqy	ekseedkgv	acskdewwd	clevlrndtl	vtlanisqgl	dlsaytesic
2041	lpildglilhw	mvcpseaaqd	pfptvgpns	lspqrlvlet	lcklsiqdnn	vdllilatppf
2101	srqektyatl	vryvgdrknp	vcremsmall	snlaqgdtda	araiavqkgs	ignlisfied
2161	gvtmaqqyqs	qhnllhmqqp	pleppsvdmm	craakallam	arvdencsef	llhegrllidi
2221	sisavlnslv	asvicdvlfq	igql			

SEQ ID NO: 43 Human CRB1 cDNA Sequence Variant 1 (NM_201253.2, CDS: from 210 to 4430)

1	cctcccgctgt	aagtgtgct	aagaagcaca	aactgcactt	tgaatctaag	cccctgtatt
61	ttccgtgaag	gagctgcaag	taggggtggg	cagagatggc	acctgggggc	tctgaggcac
121	ccgctcctct	ctgagacaga	cagggatcag	gagccggact	gggaccagac	caccagcaac
181	acaccagagg	atgtttctcta	aataagacca	tggcacttaa	gaacattaac	tacctttcta
241	ttctctacct	cagtttctca	ctgcttatct	acataaaaaa	ttccttttgc	aataaaaaaa
301	acaccaggtg	cctctcaaat	tcttgccaaa	acaattctac	atgcaaaagt	ttttcaaaag

TABLE 1-continued

361	acaatgattg	ttcttgttca	gacacagcca	ataatttggga	caaagactgt	gacaacatga
421	aagacccttg	cttccccaat	ccctgtcaag	gaagtgccac	ttgtgtgaac	accccaggag
481	aaaggagctt	tttgtgcaaa	tgctcctctg	ggtacagtgg	gacaatctgt	gaaactacca
541	ttggttcctg	tggaagaac	tcctgccaac	atggaggtat	ttgocatcag	gacctatatt
601	atcctgtctg	catctgccct	gctggatag	ctggaagatt	ctgtgagata	gatcacgatg
661	agtggtcttc	cagcccttgc	caaaatgggg	ccgtgtgcca	ggatggaatt	gatggttact
721	cctgccctctg	tgtcccagga	tatcaaggca	gacactgcga	cttggaagtg	gatgaatgtg
781	cttcagatcc	ctgcaagaac	gaggctacat	gcctcaatga	aataggaaga	tatacttgta
841	tctgtcccca	caattattct	ggtgtaaaat	gtgaattgga	aattgacgaa	tgttgggtccc
901	agccttggtt	aaatgctgca	acttgctcagg	atgctctggg	ggcctatttc	tgcgactgtg
961	cccttggtt	cctgggggat	caactgtgaac	tcaacactga	tgagtgtgcc	agtcaacctt
1021	gtctccatgg	agggctgtgt	gtggatggag	aaaacagata	tagctgtaac	tgcacgggtg
1081	gtggattcac	agggaacac	tgtagacact	tgatgccctt	ttgttggtca	aaaccttgct
1141	acaataatgc	tacatgtgag	gacagtgttg	acaattacac	ttgtcactgc	tggcctggat
1201	acacaggtgc	ccagtgtgag	atcgacctca	atgaatgcaa	tagtaacccc	tgccagtcca
1261	atggggaaatg	cttgccctgg	tcctcagaga	aacaatatgg	acgcatacct	ggactgcctt
1321	cttctttcag	ctaccatgaa	gcctcaggtt	atgtctgtat	ctgtcagcct	ggattcacag
1381	gaatccactg	cgaagaagac	gtcaatgaat	gttcttcaaa	cccttgccaa	aatggttgta
1441	cttgttgaaa	aatgctggg	aattatactt	gccattgccc	atttgataac	ctttctagaa
1501	ctttttatgg	aggaaggac	tggtctgata	ttctcctggg	ctgtacccat	cagcaatgtc
1561	taaaataatg	aacatgcctc	cctcacttcc	aagatggcca	gcattggattc	agctgcctgt
1621	gtccatctgg	ctacaccggg	tcctgtgtg	aaatcgcaac	cacactttca	tttagggcgg
1681	atggcctcct	gtgggtcaaa	agtggctcag	tgacaaccaa	gggctcagtt	tgtacatag
1741	ccctcagggt	tcagactgtt	cagccaatgg	ctcttctact	tttccgaagc	aacagggatg
1801	tggtttgtaa	cttggtgagctg	ctaaagtggct	acattcactt	atcaattcag	gtcaataatc
1861	agtcaaaagt	gcttctgttc	atttccocaca	acaccagcga	tggagagtgg	catttctgtg
1921	aggtaatatt	tgcagaggct	gtgaccttca	ccttaatcga	cgactcctgt	aaggagaaat
1981	gcctgcgcaa	agctcctact	ccacttgaaa	gtgatcaatc	aatatgtgct	tttcagaact
2041	cctttttggg	tggtttacca	gtgggaatga	ccagcaatgg	tggtgctctg	cttaacttct
2101	ataatatgcc	atccacacct	tcgtttgtag	gctgtctcca	agacattaaa	attgattgga
2161	atcacattac	cctggagaac	atctcgtctg	gctcatcatt	aaatgtcaag	gcaggctgtg
2221	tgagaaagga	ttggtgtgaa	agccaacctt	gtcaaagcag	aggacgctgc	atcaacttgt
2281	ggcgcaggtt	ccagtgtgac	tgccacaggg	cctatgaagg	ccccaaactgt	ctgagagagt
2341	atgtggcagg	cagattttggc	caggatgact	ccactggtta	tgctatcttt	actcttgatg
2401	agagctatgg	agacaccatc	agcctctcca	tggttgtccg	aacgcttcaa	ccatcaggct
2461	tactcttagc	tttggaaaac	agcacttata	aatatatccg	tgtctgggta	gagcggcgga
2521	gactagcaat	gctgactcca	aactccccc	aattagtagt	aaaatttgtt	cttaatgatg
2581	gaaatgtcca	cttgatatct	ttgaaaatca	agccatataa	aattgaactg	tatcagttct
2641	cacaaaacct	aggatttatt	tctgcttcta	cgtggaaaat	cgaaaaggga	gatgtcatct
2701	acattgtgtg	cctgactgac	aagcaagaga	ctgaacttaa	tggtggattc	ttcaaaaggt
2761	gtatccaaga	tgtaaagacta	aacaacccaa	atctggaatt	ctttccaaat	ccaacaaaca
2821	atgcactctc	caatccagtt	cttgtcaatg	taacccaaag	ctgtgctgga	gacaacagct
2881	gcaagtccaa	ccctctgtcac	aatggagggtg	tttgccattc	ccgggtgggtg	gacttctcct
2941	gttcctgttc	tgccctcaca	agtgggaaag	cctgtgagga	ggttcagtg	tggtgattca
3001	gcccgtgttc	tcacggagcc	cagtggcagc	cggtgcttca	aggatttgaa	tgtattgcaa
3061	atgctgtttt	taatggacaa	agcggtcaaa	tattattcag	aagcaatggg	aatattacca
3121	gagaactcac	caatatcaca	tttggtttca	gaacaaggga	tgcaaatgta	ataatattgc
3181	atgcagaaaa	agagcctgaa	tttcttaaca	ttagcattca	agattccaga	ttattcttct
3241	aattgcaaa	tggaacacgc	ttttatatgc	taagtctgac	aagtttgacg	tcagtgaatg
3301	atggcacatg	gcacgaagt	accctttcca	tgacagaccc	actgtcccag	acctccaggt
3361	ggcaaatgga	agtggaacac	gaaacacctt	ttgtgaccag	cacaattgct	actggaagcc
3421	tcactttttt	gaaggataat	acagatattt	atgtgggaga	cagagctatt	gacaatataa
3481	agggcctgca	aggggtgtcta	agtacaatag	aaatcggagg	catttatctc	tcttactttg
3541	aaaatgttca	tggtttcatt	aataaacctc	aggaagagca	atttctcaaa	atctctacca
3601	attcagtggt	cactggctgt	ttgcagttaa	atgtctgcaa	ctccaaaccc	tgtttgatg
3661	gaggaaactg	tgaagacatc	tatagctctt	atcattgtctc	ctgtcccttg	ggatgggtcag
3721	ggaaacactg	tgaactcaac	atcgatgaat	gctttttcaa	ccctgtatc	catggcaact
3781	gctctgacag	agttgcagcc	taccactgca	catgtgagcc	tgataacact	ggtgtgaact
3841	gtgaagtggga	tatagacaac	tgccagagtc	accagtgtgc	aaatggagcc	acctgcattta
3901	gtcatactaa	tggtctattct	tgccctctgtt	ttggaaaattt	tacaggaaaa	ttttgcagac
3961	agagcagatt	acctcaaca	gtctgtggga	atgagaagac	aaatctcact	tgctacaatg
4021	gaggcaactg	cacagagttc	cagactgaat	taaaatgtat	gtgccggcca	ggttttactg
4081	gagaatgggt	tgaaggagac	attgatgagt	gtgcctctga	tcctgtgtgc	aatggagggtc
4141	tgtgcagagg	cttactcaac	aaattccagt	gcctctgtga	tggttgcctt	gctggcgagc
4201	gctgcgaggt	ggacttggca	gatgacttga	tctccgacat	tttccaccct	attggctcag
4261	tgactgtcgc	ctgtgtactg	atcctcttgc	tgcccatgtt	tgcttctgtt	gtcacctcca
4321	acaaaagggc	aactcaggga	acctacagcc	ccagccgtca	ggagaaggag	ggctcccag
4381	tggaatgtg	gaactctgatg	ccacccctg	caatggagag	actgatttag	gagcattgtg
4441	tcctctcgag	atggggatcc	acacactgtg	aatgtgatga	ctgtacttca	ggtatctctg
4501	acatacttga	caatgttaat	ctgcaactgg	gattacactg	gaactacagg	aatgattcct
4561	ttgaccaact	taaaaacttt	cacagtgggt	ccgctcgaca	ccactgtttt	attatattat
4621	atcagccaat	tgcaaaaaaa	gtctgtgcca	gtaatttcag	ccttataaatt	agcaaaaaaca
4681	tcttcacag	aataaagtct	tctgtggctt	tagtggccat	cactgaaact	ctttcctctt
4741	ttcaacctgg	gaacaaattt	tagttttcat	tttaggttcc	tgtaactttct	gtagtttctg
4801	tgtaaaactgc	catatgttta	catggaaact	acaggaaaaa	attggctaca	tttctcactt
4861	ctcctatcat	gtgggtcaag	ttattgttgt	ataccagcga	tgggattgat	acttttgtcc
4921	ttcattcatg	gattcagaga	aagctctggg	aacgacttat	ggcccaaaaa	agtgacccaa
4981	tggcaacaaa	taaaaattga	aatgcaaaaa	aaaaaaaaaa	aaaa	

TABLE 1-continued

SEQ ID NO: 44 Human CRB1 Amino Acid Sequence Isoform A (NP_957705.1)

1	malkninyll	ifylsfelli	yiknsfcnk	ntrclsnscq	nntckdfsk	dndcsesdta
61	nnldkdednm	kdpfcfnpcq	gsatcvntpg	ersflckcpp	gyegticett	igscgknscc
121	hggichqdpi	ypvcicpagy	agrfeidhd	ecasspcqng	avcqdgidgy	scfcvpgyqg
181	rhcdelevdec	asdpckneat	clneigrytc	icphnysgvn	celeidecws	qpclngatcq
241	dalgayfcdc	apgfldghe	lntdecasqp	clhggclcvd	enryscnctg	sgftgthcet
301	lmpclwskpc	hnnatcedsv	dnytchcwpg	ytgaqceidl	necnsnpcqs	ngcevelsse
361	kqygritglp	ssfsyheasg	yvcicqpgft	gihceedvne	cssnppcngg	tcenlpngny
421	chcpfdnlr	tlyggrdcsd	illgcthqcc	lmgctciphf	qdgqhgfscl	cpsgygtgsl
481	eiattlsteg	dglflwksqs	vttkgsvcni	alrfqtvqpm	alllfrsntd	vfvkllellsg
541	yihlsiqvnn	qskvllfish	ntsdgewhfv	evifaeavtl	tliddsckek	ciakaptple
601	sdqsaicafn	sflgglpvgm	tsngvallnf	ynmpstpsfv	gclqdkidw	nhitleniss
661	gsslnvkagc	vrkdwcscqp	cqsrgrcinl	wlsyqcdchr	pyegpnclre	yvagrfgqdd
721	stgyviftld	esygdtisls	mfvrtlqpsg	lllalensty	qyrvwlerg	rlamltpnsp
781	klvkvfvlnd	gnvhlisliki	kpykielyqs	sqnlgfisas	twkiekgdvi	yiggldpkqe
841	telnggffkg	cigdvrlnnq	nleffpnptn	naslnpvlvn	vtgcagadns	cksnpcnngg
901	vchsrwddfs	cscpaltsqk	aceevqwcg	spcphgaqcq	pvlqgfecia	navfngqsgg
961	ilfrsngnit	reltnitfgf	rtrdanvil	haekepefln	isigdsrlff	qlqsgnstym
1021	lsltslqsvn	dgtwhevltls	mtdpisqtsr	wqmevdnetp	fvtstiatgs	lnflkdntdi
1081	yvgdraidni	kglggclsti	eiggiylsyf	envhgfinkp	qeeqflkist	nsvvtgclql
1141	nvcsnppclh	ggncediys	yhcscplgws	gkhcelnide	cfnpcihgn	csdrvaayhc
1201	tcepgygtgn	cevidhncqs	hqcangatci	shtngyscl	fgnftgkfc	qsrpsetvcg
1261	neknltcyn	ggncetefgt	lkcmcrpgft	gewcekdide	casdpcvngg	lcqdlilnkfq
1321	clcdvafage	rcevdldadd	isdifttigs	vtvalllill	laivasvvt	nkratqgtys
1381	psrqekegsr	vemwnlmppp	amerli			

SEQ ID NO: 45 Human CRB1 cDNA Sequence Variant 2 (NM_001193640.1, CDS; from 210 to 4094)

1	cctccccgtg	aagtgatgct	aagaagcaca	aactgcattt	tgaatctaag	tccctgtatt
61	ttctgtgaag	gagctgtaag	taggggtgga	cagagatggc	acctgggggt	tctgaggcac
121	ccgctcctct	ctgagacaga	cagggatcag	gagccggact	gggaccagac	caccagcaac
181	acaccagagg	atgttctcta	aataagacca	tggcaactta	gaacattaac	taccttctca
241	tcttctacct	cagtttctca	ctgcttatct	acataaaaaa	tctcttttgc	aataaaaaaa
301	acaccagggt	ccctctcaaat	tcttgccaaa	acaattctac	atgcaaaagt	ttttcaaaag
361	acaatgatgt	ttcttgttca	gacacagcca	ataatttggg	caaagactgt	gacaacatga
421	aagacccttg	cttctccaat	ccctgtcaag	gaagtgccac	ttgtgtgaac	accccaggag
481	aaaggagctt	ttctgtcaaa	tgctcctcgt	ggtacagtgg	gacaatctgt	gaaactacca
541	ttggttctct	tggaagaagc	tcctgccaac	atggaggtat	ttgccatcag	gacctatttt
601	atcctgtctg	catctgccct	gctggatatg	ctgggaagatt	ctgtgagata	gatcacgatg
661	agtggtcttc	cagcccttgc	caaaatgggg	ccgtgtgcca	ggatggaaat	gatggttact
721	cctgctctct	tgctccagga	tatcaaggca	gacactgcga	cttggaagtg	gatgaatgtg
781	cttcagatcg	ctgcaagaac	gaggtacat	gcctcaatga	aatagggaag	tatacttgta
841	tctgtcccca	caattattct	ggatacacag	gtgccacgtg	tgagatcgac	ctcaatgaat
901	gcaatagtaa	ccctgcgcag	tcctaatggg	aatgtgtgga	gctgtcctca	gagaacaat
961	atggagcgat	cactggactg	ccttcttctt	tcagctacca	tgaagcctca	ggttatgtct
1021	gtatctgtca	gcctgtgattc	acaggaatcc	actgcgaaga	agacgtcaat	gaatgttctt
1081	caaacccctg	ccaaatgggt	ggtacttgtg	agaacttgcc	tgggaattat	acttgccatt
1141	gcccatttga	taacctttct	agaacttttt	atggaggaag	ggactgttct	gatattctcc
1201	tgggtctgac	ccatcagcaa	tgcttaaaac	atggaacatg	catccctcac	ttccaagatg
1261	gccagcatgg	attcagctgc	ctgtgtccat	ctggctacac	cgggtccctg	tgtgaatcgc
1321	caaccacact	ttcattttag	ggcgatggct	tcctgtgggt	caaaagtggc	tcagtgacaa
1381	ccaagggctc	agtttctaac	atagccctca	gggttcagac	tggttcagca	atggctcttc
1441	tacttttccg	aagcaacagg	gatgtgtttg	tgaagctgga	gctgtcaagt	ggctacattc
1501	acttatcaat	tcaggtcaat	aatcagtcac	aggtgcttct	gttcatttcc	cacaacacca
1561	gcgatggaga	gtggaggttc	gtggaggtac	tatttgcaga	ggctgtgacc	cttaccttaa
1621	tcgacgactc	ctgtaaggag	aaatgcactc	cgaaagctcc	tactccactt	gaaagtgtac
1681	aatcaaatat	tgctttttag	aaactccttt	tgggtgtgtt	accagtgagg	atgaccagca
1741	atgggtgttc	ctctgttaac	ttccataata	tgccatccac	acctctgttt	tgaggctgtc
1801	tccaagacat	taaaattgat	tggaaacaca	ttaccctgga	gaacatctcg	tctggctcat
1861	cattaaatgt	caaggcagcg	tgtgtgagaa	aggattgggt	tgaagcccaa	ccttgtcaaa
1921	gcagaggagc	ctgcatacaac	ttgtggctga	gttaccagtg	tgactgccac	aggccctatg
1981	aaggccccaa	ctgtctgaga	gagtatgtgg	caggcagatt	tggccaggat	gactccactg
2041	gttatgtcat	ctttactctt	gatgagagct	atggagacac	catcagcctc	tccatgtttg
2101	tcggaacgct	tcacatcaac	ggcttacttc	tagctttgga	aaacagcact	tatcaatata
2161	tccgtgtctg	gctagagcgc	ggcagactag	caatgctgac	tcctaaactct	cccaaatag
2221	tagtaaaatt	tggttctaat	gatggaaatg	tccacttgat	atctttgaaa	atcaagccat
2281	ataaaattga	actgtatcag	tcttcacaaa	acctaggatt	tatttctgct	tctacgtgga
2341	aaatcgaaaa	gggagatgtc	atctacattg	gtggcctacc	tgacaagcaa	gagactgaac
2401	ttaattgggtg	attcttcaaa	ggctgtatcc	aagatgtaag	actaaacaac	caaaatctgg
2461	aattctttcc	aaatccaaca	aacaatgcat	ctctcaatcc	agttctgtgc	aatgcaaccc
2521	aaggctgtgc	tgagacaaac	agctgcaagt	ccaacccctg	tcacaatgga	ggtgtttgcc
2581	attcccgggtg	ggatgacttc	tcctgttctt	gtcctgccct	cacaagtggg	aaagcctgtg
2641	aggaggttgc	gtgggtggga	ttcagccctg	gtcctcacgg	agcccgatgc	cagccgggtgc
2701	tccaaggatt	tgaatgtatt	gcaaatgctg	tttttaattg	acaaagcggg	caaatattat
2761	tcagaagcaa	tgggaatatt	accagagaac	tcaccaatat	cacatttggt	ttcagaacaa
2821	gggatgcaaa	tgaataaata	ttgcatgcat	aaaaagagcc	tgaatttctt	aatattagca
2881	ttcaagattc	cagattattc	tttcaattgc	aaagtggcaa	cagcttttat	atgctaagcc
2941	tgacaagttt	gcagtcagtg	aatgatggca	catggcacga	agtgaccctt	tccatgacag
3001	accactgtc	ccagactctc	aggtggcaaa	tggaaagtga	caacgaacaa	ccttttgtga
3061	ccagacaaat	tgctactgga	agcctcaact	ttttgaagga	taatacagat	atttatgtgg

TABLE 1-continued

3121	gagacagagc	tattgacaat	ataaagggcc	tgaagggtg	tctaagtaca	atagaaatcg
3181	gaggcattta	tctctcttac	ttgaaaaatg	ttcatggttt	cattaataaa	cctcaggaag
3241	agcaatttct	caaaaatctct	accaattcag	tggtcactgg	ctgtttgcag	ttaaatgtct
3301	gcaactccaa	ccctctgttg	catggaggaa	accgtgaaga	catctatagc	tcttatcatt
3361	gctcccgtcc	cttgggtagg	tcagggaaac	actgtgaact	caacatcgat	gaatgctttt
3421	caaaccctcg	tatccatggc	aactgctctg	acagagtgtg	agcctaccac	tgcacatgtg
3481	agcctggata	cactgggtgtg	aactgtgaag	tggatataga	caactgccag	agtcaccagt
3541	gtgcaaatgg	agccacctgc	attagtcata	ctaattggcta	ttcttgcttc	tgttttgtaa
3601	attttacagg	aaaattttgc	agacagagca	gattaccctc	aacagttctgt	gggaatgaga
3661	agacaaatct	cacttctctac	aatggaggca	actgcacaga	gttcagact	gaattaaat
3721	gtatgtgccg	gccaggtttt	actggagaat	gggttgaaaa	ggacattgat	gagtgtgcct
3781	ctgatccgtg	tgtcaatgga	ggctctgtgc	aggacttact	caacaaattc	cagtgcctct
3841	gtgatgttgc	ctttgctggc	gagcgtgcg	agggtgactt	ggcagatgac	ttgatctccg
3901	acattttcac	cactattggc	tcagtgtctg	tcgcttgggt	actgatcttc	ttgcccggcca
3961	tcgttgcttc	tgttgtcacc	tccaacaaaa	gggcaactca	gggaacctac	agccccagcc
4021	gtcaggagaa	ggagggtccc	cgagtggaaa	tgtggaactt	gatgccaccc	cctgcaatgg
4081	agagactgat	ttaggagcat	tgtgtccctt	cgagatgggg	atccacacac	tgtgaatgtg
4141	atgactgtac	ttcaggtatc	tctgacatac	ctgacaatgt	taactctgca	ctgggattac
4201	actgggaacta	cagggaatgat	tcctttgacc	accttaaaaa	ctttcacagt	ggttccgctc
4261	gacaccattg	ttttattata	ttataccagc	caattgcaaa	aaaagtctgt	gccagtaatt
4321	tcagcccttat	aattagcaaa	aacatcttcc	agagaataaa	gtcttctgtg	gcttttagtg
4381	ctatcactga	aactctttcc	tcttttcaac	ctgggaacaa	attttagttt	tcatttttag
4441	ttctctgtact	ttctgtagtt	tctgtgtaaa	ctgcatatg	tttcatgga	aactacagga
4501	aaaaattggc	tacattttct	actttctcta	tcatgtggtc	aaagttattg	ttgtatacca
4561	gcgatgggat	gtatactttt	gtccttcatt	catggattca	gagaaagctc	tgggaatgac
4621	ttatgggtcca	aaaaagtgc	ccaatggcaa	caaataaaaa	ttgaaatgca	aaaaaaaaaa
4681	aaaaaaaa					

SEQ ID NO: 46 Human CRB1 Amino Acid Sequence Isoform B (NP_0011X0569.1)

1	malnkniy11	ifylsfsl1i	yiknsfenkn	ntrelnsnscq	nnstckdfsk	dnndscsdta
61	nnldkddnm	kdpfcfnpcq	gsatcvntpg	ersflckcpp	gygsticett	igscgknscq
121	hggichqdpi	ypvcicpagy	agrfeidhd	ecasspcqng	avcqdgidgy	scfcvpgyqg
181	rhcldlevdec	asdpckneat	clneigrtyc	icphnysgyt	gaqceidlne	cnsnpcqng
241	ecvelssekq	ygritglpss	fsyheasgyv	cicqpgftgi	hceadvnecs	snpcqnggtc
301	enlpngnytch	cpfdnlrtf	yggrdcstil	lgcthqgcln	ngtciphfqd	qghgfsclcp
361	sgytgslcei	attlsfegdg	flwvksqsvt	tkgsvenial	rftqvqpmal	lfrsnrdvf
421	vkllslsgyi	hlslqvnngs	kvllfishnt	sdgwhfvev	ifaeavtltl	iddsckekci
481	akaptplead	qslcafqnaf	lgglpvgmts	ngvallnfyn	mpstpsfvgc	lqdikidwnh
541	itlenissgs	slnvkagcvr	kdwcscqpcq	srgrcinlwl	syqcdchrpy	egpnclreyv
601	agrfgqddst	gyviflides	ygdtslsmf	vrtlqpgll	lalenstyqy	irvwlergrl
661	amltpnspkl	vvkfvlnldgn	vhllslkikp	ykielyqssq	nlgfisastw	kiekgdviyi
721	ggldpkqete	lnggffkgei	qdvrlnnqnl	effpnptnna	slnpvlvnt	ggcagdnscck
781	snpchnggv	hswddffscs	cpaltsgkac	eevqwcgfs	cpghagcqv	lqgfeciana
841	vfnqsgsqil	frnngnitre	ltnitfgfrr	rdanvilha	ekepeflnis	iqdsrlffql
901	qsgnsfymls	ltslqsvndg	twhevtlsm	dpisqtsrwq	mevndnetpfv	tstiatgsln
961	flkdntdiyv	gdraidnikg	lqglstiei	ggilylsyfen	vhgfinkpge	eqflkistns
1021	vvtgclqlnv	cnsnplhgg	ncedyssyh	cscplwsgk	hcelnidecf	snpcihgnscs
1081	drvaayhtc	epgytgvnce	vdidncqshq	cangatacish	tngysclcf	nftgkfcrrqs
1141	ripstvcgne	ktnlctyngg	ncrefqtelk	cmcrpgftge	wckdkideca	sdpvcvnglc
1201	qdllnkfgcl	cdvafagerc	evdladdlis	difttigsvt	vallllilla	ivasvvtstnk
1261	ratqgtytsp	rqekegrve	mwnlmpppam	erli		

SEQ ID NO: 47 Human CRB1 cDNA Sequence Variant 3 (NM_001257965.1, CDS: from 340 to 4488)

1	atgtgcgcg	acgcgcgttt	acgcattgctc	cttaagtctc	ccgtactccc	tcggagagccc
61	tagctacaac	ccgaatccgt	tactccgggt	tttcgcagtg	gctcgggtggc	ctaccccgat
121	cgaaacctag	tctggaactg	aacctacaat	atctctgagg	gaggacacat	ctatgactag
181	cagtggcatg	tgctcaggaa	agattccttt	tgcaataaaa	acaaccacag	gtgcctctca
241	aattcttgcc	aaaacaattc	tacatgcaaa	gatttttcaa	aagacaatga	ttgttcttgt
301	tcagacacag	ccaataattt	ggacaaagac	tgtgacaaca	tgaagagccc	ttgcttctcc
361	aatccctgtc	aaggaaagtg	cacttggtgtg	aacaccccag	gagaaaggag	ctttctgtgc
421	aatgtctctc	ctgggtacag	tgggacaatc	tgtgaaacta	ccattgggtc	ctgtggcaag
481	aactcctgcc	aacatggagg	tatttgccat	caggacccta	tttatcctgt	ctgcatctgc
541	cctgctggat	atgtcggagg	attctgtgag	atagatcacg	atgagtgtgc	ttccagccct
601	tgccaaaatg	gggcccgtgtg	ccaggatgga	atcgatggtt	actcctgctt	ctgtgtccca
661	ggatatcaag	gcagacactg	cgacttgga	gtggacgaat	gtgcttcaga	tccctgcaag
721	aacgaggctg	gtgacctcaa	tgaatatagg	agatatactt	gtatctgtcc	ccacaattat
781	tctggtgtaa	actgtgaatt	ggaaattgac	gaatgttggt	ccagccttg	tttaaatggg
841	gcaactgtgc	aggacgctct	gggggcctat	ttctgcgacc	gtgccctctg	attcctgggg
901	gatcactgtg	gaactcaaac	tgatgagtg	gccagtcaac	cttgtctcca	tggagggtctg
961	tgtgtggatg	gagaaaaacg	atatagctgt	aactgcacgg	gtagtggatt	cacagggaca
1021	cactgtgaga	ccttgatgcc	tctttgttgg	tcaaaacctt	gtcacaataa	tgctacatgt
1081	gaggacagtg	ttgacaatta	cacttgctac	tgctggcctg	gatacacagg	tgcccaggtg
1141	gagatcgacc	tcaatgaatg	caatagtaac	ccctgccagt	ccaatgggga	atgtgtggag
1201	ctgtcctcag	agaacaata	tggacgcac	actggactgc	cttcttcttt	cagctaccat
1261	gaagcctcag	gttatgtctg	tatctgtcac	cctggattca	caggaaatcca	ctgcgaagaa
1321	gacgtcaatg	aattgtcttc	aaacccttgc	caaaatgggtg	gtacttgtga	gaacttgcc
1381	gggaattata	cttgccattg	cccatattgat	aacctttcta	gaacttttta	tggagggaagg
1441	gactgtctcg	atattctcct	gggtgtgacc	catcagcaat	gtctaaataa	tggaaacatgc
1501	atccctcact	tccaagatgg	ccagcatgga	ttcagctgcc	tgtgtccatc	tggctacacc

TABLE 1-continued

1561	gggtccctgt	gtgaaatcgc	aaccacactt	tcattttagg	gcgatggctt	cctgtgggtc
1621	aaaagtggct	cagtgcacac	caagggtcga	gtttgtaaca	tagccctcag	gtttcagact
1681	gttcagccaa	tggtctcttc	acttttccga	agcaacaggg	atgtgtttgt	gaagctggag
1741	ctgctaagtg	gtcacattca	cttatcaatt	caggtcaata	atcagcaaaa	ggtgcttctg
1801	ttcatttccc	acaacaccag	cgatggagag	tggcatttcg	tggaggtaat	atttgcagag
1861	gctgtgaccc	ttaccttaat	cgacgactcc	tgtaaggaga	aatgcacgcg	gaaagctcct
1921	actccacttg	aaagtgtatc	atcaatatgt	gcttttcaga	actccttttt	gggtgggtta
1981	ccagtgggaa	tgaccagcaa	tggtgttgct	ctgcttaact	tctataatat	gccatccaca
2041	ccttcgtttg	taggtgtgct	ccaagacatt	aaaattgatt	ggaatcacat	tacctggag
2101	aacatctcgt	ctggctcact	attaaatgtc	aaggcaggct	gtgtgagaaa	ggattggtgt
2161	gaaagccaac	cttgtcaaa	cagaggacgc	tgcatcaact	tgtggctgag	ttaccagtgt
2221	gactgccaca	ggccctatga	aggccccaac	tgctctgagag	agtatgtggc	aggcagattt
2281	ggccaggatg	actccactgg	ttatgtcatc	tttactcttg	atgagagcta	tggagacacc
2341	atcagcctct	ccatgtttgt	ccgaacgctt	caaccatcag	gcttacttct	agctttggaa
2401	aacagcactt	atcaatatat	ccgtgtctgg	ctagagcgcg	gcagactagc	aatgctgact
2461	ccaaactctc	ccaaatttagt	agtaaaattt	gttcttaatt	atggaaatgt	ccacttgata
2521	tctttgaaaa	tcaagccata	taaaattgaa	ctgtatcagt	cttcacaaaa	cctaggattt
2581	atttctgctt	ctacgtggaa	aatcgaaaag	ggagatgtca	tctacatttg	tggcctacct
2641	gacaaagca	agaccgaact	taattggtgga	ttcttcaaa	gctgtatcca	agatgtaaga
2701	ctaacaaccc	aaaatctgga	attctttcca	aatccaacaa	acaatgcata	tctcaatcca
2761	gttcttgtea	atgtaaccca	aggctgtgct	ggagacaaca	gctgcaagag	gcagaccaat
2821	gtgggaaggg	cactcactga	gttgggatcc	agaggaccta	agtaccaagt	ttcactgttt
2881	cgcttctgtg	taggatcttg	ggcaactgga	aacaccttct	ttttatcatc	tataaaacca
2941	ggatcccaac	cctgtcacia	tggagggtgt	tgccattccc	ggtgggatga	cttctcctgt
3001	ttcctgtcct	cctctcacia	tgggaaaagc	tgtgaggagg	ttcagtgggt	tggattcagc
3061	ccgtgtcctc	acggagccca	gtgccagccg	gtgcttcaag	gatttgaatg	tattgcaaat
3121	gctgttttta	atggacaaa	cggtcaaaata	ttatttcagaa	gcaatgggaa	tattaccaga
3181	gaactcacca	atatcacatt	tggtttcaga	acaagggatg	caaatgtaat	aatattgcat
3241	gcagaaaaag	agcctgaatt	tcttaatat	agcattcaag	attccagatt	attctttcaa
3301	ttgcaaaagt	gcaacagctt	ttatatgcta	agtctgacaa	gtttgcagtc	agtgaatgat
3361	ggcacatggc	acgaagtgtac	cctttccatg	acagaccac	tgccccagac	ctccagggtg
3421	caaatgggag	tggacaacga	aacacctttt	gtgaccagca	caattgtctac	tgggaagctc
3481	aactttttga	aggataatac	agatatattt	gtggggagaca	gagctattga	caatataaag
3541	ggcctgcaag	ggtgtctaa	tacaatagaa	atcggaggca	tttatctctc	ttactttgaa
3601	aatgttcacg	gtttcattaa	taaacctcag	gaagagcaat	ttctcaaaat	ctctaccaat
3661	tcagtggtea	ctggctgttt	gcagttaaat	gtctgcaact	ccaacccctg	tttgcattgga
3721	ggaaactgtg	aagacatcta	tagctcttat	cattgctcct	gtcccttggg	atggtoaggg
3781	aaacactgtg	aactcaacat	cgatgaatgc	ttttcaaaac	cctgtatcca	tggcaactgc
3841	tctgacagag	ttgcagccta	ccactgcaca	tgtgagcctg	gatacactgg	tgtgaactgt
3901	gaagtggata	tagacaaact	ccagagtcac	cagtgtgcaa	acggagccac	ctgcattagt
3961	cataactaat	gctattcttg	cctctgtttt	ggaaatttta	caggaaaaat	ttgcagacag
4021	agcagattac	cctcaacagt	ctgtgggaat	gagaagacaa	atctcacttg	ctacaatgga
4081	ggcaactgca	acttgatcca	gactgaatta	aaatgtatgt	gccggccagg	ttttactgga
4141	gaatggtgtg	aaaaggacat	tgatgagtgt	gcctctgata	cgtgtgtcaa	tggaggctct
4201	tgcagagact	tactcaacaa	attccagtgc	ctctgtgatg	ttgcctttgc	tggcgagcgc
4261	tgcgaggtgg	acttggcaga	tgaactgtat	tccgacattt	tcaccactat	tggctcagtg
4321	actgtcgctt	tgttactgat	cctcttgctg	gccattgttg	cttctgttgt	cacctccaac
4381	aaaagggcaa	ctcaggggac	ctacagcccc	agccgtcagg	agaaggaggg	ctcccagagt
4441	gaaatgtgga	acttgatgcc	accccttgca	atggagagac	tgatttagga	gcatttgtgc
4501	ccttcagagt	ggggatccac	acactgtgaa	tgtgatgact	gtacttcagg	tatctctgac
4561	atacctgaca	atgttaactc	gcaactggga	ttacactgga	actacaggaa	tgattccttt
4621	gaccacctga	aaaactttca	cagtggttcc	gctcgacacc	atgtgtttat	tatatattat
4681	cagccaattg	caaaaaaagt	ctgtgccagt	aatttcagcc	ttataattag	caaaaaatc
4741	ttccagagaa	taaagtcctc	tgtggcttta	gtggctatca	ctgaaactct	ttcctctttt
4801	caacctggga	acaaatttta	gttttcattt	taggtttctg	tactttctgt	agtttctgtg
4861	taaactgcca	tatgtttaca	tggaaactac	aggaaaaaat	tggctacatt	tctcacttct
4921	cctatcatgt	ggtcaaaagt	attgttgtat	accagcgatg	ggatgtatac	ttttgtcctt
4981	cattcatgga	ttcagagaaa	gctctgggaa	tgacttatgg	tccaaaaaag	tgacccaatg
5041	gcaacaaata	aaaattgaaa	tgcacaaaaa	aaaaaaaaaa	aa	

SEQ ID NO: 48 Human CRB1 Amino Acid Sequence Isoform C (NP_001244894.1)

1	mkdpcfsnnc	qgsatcvntp	gersflckcp	pgysgticet	tigscgknsc	qhgghichqdp
61	iyppvcicpag	yagrfeicidh	decasspcqn	gavcqdgidg	yscfvpgyq	grhcdlevde
121	casdpcknea	tclineigrvt	cicphnysgv	nceleidecw	sqpcnlngatc	qdalgaytcd
181	capgflgdhc	elntdecasq	pclhgglcvd	genrysncnt	sgsftgtthce	tlmplcwskp
241	chnnatceds	vdnytchcwp	gytgaqceid	lneconsnpq	sngecvlss	ekqygritgl
301	pssfsyheas	gyvicqpgf	tgiheeavn	ecssnpcng	gtcenlpngy	tchcpfdnls
361	rtfyggdrds	dillgcthqg	clnngtciph	fqdgqghfcs	lcpsgygtgl	ceiatltste
421	gdglfwksg	svttkgsven	ialrfqtvqp	malllfrsnr	dsvfkllells	gyihlsiqvn
481	nqskvllfis	smfvrtlqps	vevifaeavt	ltliddscke	kciakaptpl	esdgsicafq
541	nsflgglpvg	mtsngvalln	fynmpetpsf	vgclgdkid	wnhitlenis	sgsslnvkag
601	cvrkdwesq	pqsrgrcin	lwlsyqdcch	rpypgnclr	eyvagrfgqd	dstgyviftl
661	desygdtsl	smfvrtlqps	glallalenst	yqyirvwler	grlamltpps	pklvkvfvl
721	dgnvhlislk	ikpykielyq	ssqnlgfisa	stwkiekgdv	iyigglpdkq	etelnnggffk
781	gciqdvrlnn	qnletfpnpt	nnaalnplvl	nvtggcagdn	skrqtnvgr	altelgsrgp
841	kyqvsllfrf	vgsatwntf	flssikpgsn	pchnngvchs	rwdldscscp	altsgkacee
901	vqwcgfcpcp	hgaqccpvlq	gfecianavf	ngqsgqilfr	sngnitrelt	nitfgfrtrd
961	anvilhaek	epeflnisiq	dsrlffqlqs	gnsfymslt	slqsvndgtw	hevtlsmtdp
1021	lsqtsrwqme	vdnetpfvts	tlatgslnfl	kdntdiyvgd	raidnikglq	gclstieigq
1081	ylsyfenvh	gfinkpqqeq	flkistnsvv	tgclqlnvcn	snpcnlhggnc	ediyssyhcs

TABLE 1-continued

1141	cpilgswgkch	elnidecfns	pcihgncsdr	vaayhtceep	gytgvncevd	idncqshqca
1201	ngatcishtn	gysclcfgnf	tgkfcqrsl	pstvcgnek	nltcynggnc	tefqtelkcm
1261	crpgftgewc	ekdidecasd	pcvngglcqd	llnkfqclcd	vafagercev	dladdltsdi
1321	fttigsvtva	llllllllaiv	asvvtnskra	tqgtyspsrq	ekegsrvemw	nlpmpamer
1381	li					

SEQ ID NO: 49 Human CRB1 cDNA Sequence Variant 4 (NM_001257966.1., CDS: from 210 to 2822)

1	cctccccgtgt	aagtgtatgct	aagaagcaca	aactgcattt	tgaatctaag	tccctgtatt
61	ttctgtgaag	gagctgtaag	taggggtggga	cagagatggc	acctgggggt	tctgaggcac
21	ccgctcctct	ctgagacaga	cagggatcag	gagccggact	gggaccagac	caccagcaac
181	acaccagagg	atgtttctcta	aataagacca	tggcacttaa	gaacattaac	tacctttctca
241	tctttacact	cagttttctca	ctgcttatct	acataaaaaa	ttccttttgc	aataaaaaaca
301	acaccaggtg	ccctcacaat	tcttgccaaa	acaattctac	acgcaaaagat	ttttcaaaaag
361	acaatgattg	ttcttgttca	gacacagcca	ataatttggg	caaagactgt	gacaacatga
421	aagacccttg	cttctccaat	ccctgtcaag	gaagtggcac	ttgtgtgaac	accccaggag
481	aaaggagctt	tctgtgcaaa	tgtcctcctg	ggtacagtgg	gacaatctgt	gaaactacca
541	ttggttccctg	tggcaagaac	tcctgccaac	atggaggtat	ttgccatcag	gacctatttt
601	atcctgtctg	catctgcctc	gctggatag	ctggaagatt	ctgtgagata	gatcacgatg
661	agtggtcttc	cagcccttgc	caaaatgggg	ccgtgtgcca	ggatggaatt	gatggttact
721	cctgctctctg	tgtcccagga	tatcaaggca	gacactgcga	cttggaagtg	gatgaatgtg
781	cttcagatcc	ctgcaagaac	gaggctacat	gcctcaatga	aataggaaga	tatacttgta
841	tctgtcccca	caattattct	ggtgtaaaat	gtgaattgga	aattgacgaa	tgttggtccc
901	agccttgttt	aaatggtgca	acttgtcagg	atgctctggg	ggcctatttc	tgcgactgtg
961	ccctctggat	ctcgggggat	cactgtgaac	tcaacactga	tgagtgtgcc	agtcaacctt
1021	gtctccatgg	agggctgtgt	gtggatggag	aaaacagata	tagctgtaac	tgcacgggta
1081	gtggattcac	agggacacac	tgtgagacct	tgatgcctct	ttgttggtca	aaaccttgtc
1141	acaataatgc	tacatgtgag	gacagtgttg	acaattacac	ttgtcactgc	tggcctggat
1201	acacaggtgc	ccagtgtgag	atcgacctca	atgaatgcaa	tagtaacccc	tgccagtcca
1261	atggggaaatg	tgtggagctg	tcctcagaga	aacaatatgg	acgcatcact	ggactgcctt
1321	cttctttcag	ctaccatgaa	gcctcaggtt	atgtctgtat	ccgtcagcct	ggattccacag
1381	gaatccactg	cgaagaagac	gtcaatgaat	gttcttcaaa	cccttgccaa	aatggtggta
1441	cttgtgagaa	cttgccctggg	aattataact	gccattgccc	atttgataac	ctttctagaa
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1561	taaataatgg	aacatgcatc	cctcacttcc	aagatggcca	gcattggattc	agctgcctgt
1621	gtccatctgg	ctacaccggg	tcctctgtgtg	aaatcgcaac	cacactttca	tttgaggggc
1681	atggcttcc	gtgggtcaca	agtggctcag	tgacaaccaa	gggcccagtt	tgtaacatag
1741	ccctcaggtt	tcagactgtt	cagccaatgg	ctctcttact	tttccgaagc	aacagggatg
1801	tgtttgtgaa	gctggagctg	ctaagtggct	acattcactt	accaattcag	gtcaataatc
1861	agtcaaaagg	gcttctgttc	atttcccaca	acaccagcga	tggagagtgg	catttctgtg
1921	aggtaatatt	tgcagaggct	gtgaccttca	ccttaatcga	cgaactcctgt	aaggagaaat
1981	gcacgcgcaa	agctcctact	ccacttgaaa	gtgatcaatc	aatatgtgct	tttcagaact
2041	cttttttggg	tggtttacc	gtgggaatga	ccagcaatgg	tgttgctctg	cttaacttct
2101	ataatatgcc	atccacacct	tcgtttgtag	gctgtctcca	agacattaaa	attgatggga
2161	atcacattac	cctgggagac	atctcgtctg	gctcatcatt	aaatgtcaag	gcaggctgtg
2221	ttgagaaagga	ttgggtgtgaa	agccaacctt	gtcaaaagcag	aggacgtctg	atcaacttgt
2281	ggctgagtta	ccagtgtgac	tgccacaggc	cctacgaagg	ccccaaactgt	ctgagaggaa
2341	aatttttcag	acagagcaga	ttacctcaa	cagtctgtgg	gaatgagaag	acaaatctca
2401	cttgctacaa	tggaggcaac	tgcacagagt	tcagactga	attaaaatgt	atgtgccggc
2461	caggttttac	tggagaatgg	tgtgaaaagg	acattgatga	gtgtgcctct	gatccgtgtg
2521	tcaatggagg	tctgtgccag	gacttactca	acaaaattcca	gtgcctctgt	gatgttgctt
2581	ttgctggcga	gcgctgcgag	gtggacttgg	cagatgactt	gatctccgac	attttaccac
2641	ctattggctc	agtgaactgtc	gccttggtac	tgatcctctt	gctggccatt	gttgcttctg
2701	ttgtcacctc	caacaaaagg	gcaactcagg	gaacctacag	ccccagccgt	caggagaagg
2761	agggctcccc	agtggaatga	tggaaactga	tggcaacccc	tgaatggag	agactgattt
2821	aggagcattg	tgtccctctg	agatggggat	ccacacactg	tgaatgtgat	gactgtactt
2881	caggatctct	tgacatacct	gacaaatgta	atctgcaact	gggattacac	tggaaactaca
2941	ggaatgattc	ctttgaccac	cttaaaaaat	ttcacagtgg	ttccgctcga	caccattgtt
3001	ttattatatt	atatcagcca	attgcaaaaa	aagtctgtgc	cagtaatttc	agccttataa
3061	ttagcaaaaa	catcttccag	agaataaagt	cttctgtggc	tttagtggct	atcactgaaa
3121	ctcttttctc	ttttcaacct	gggaacaaat	tttagtttct	attttaggtt	tctgtacttt
3181	ctgtagtttc	tgtgtaaaat	gcatatgtt	tacatggaaa	ctacaggaaa	aaattggcta
3241	cattttctac	ttctcctatc	atgtgggtcaa	agttattgtt	gtataccagc	gatgggatgt
3301	atacttttgt	cttctcattca	tggattcaga	gaaagctctg	ggaaactgct	atgggtccaaa
3361	aaagtgacct	aatggcaaca	aataaaaaat	gaaatgcaaa	aaaaaaaaaa	aaaaaa

SEQ ID NO: 50 Human CRD1 Amino Acid Sequence Isoform D (NP_001244895.1)

1	malnkinyll	ifylsfelli	yiknsfcnk	ntclnsncq	nstckdfsk	dndscsdta
61	nnldkdcndm	kdpfcfnpcq	gsatcvnrrp	ersflckcpp	gysgticett	igscgknsqc
121	hggichqdpi	ypvcicpapy	agrfeidhd	ecasspcqng	avcqdgidgy	scfcvpgyqg
181	rhcdlevdec	asdpckneat	clneigrytc	icphnysgvn	celeidecws	qpclngatcq
241	dalgayfcdc	apgfldghce	lntdecasqp	clhggclcvd	enryscnctg	sgftgtthcet
301	lmpicwskpc	hnnatcedsv	dnythcwpg	ytgaqceld	necnspcqs	ngcevelsse
361	kqygritglp	ssfsyheasg	yvcicqpgft	gihceedvne	cssnpcqngg	tcnlpgnytt
421	chcpfdnlr	tfggrdcdd	illgcthqcc	lnngtciphf	qdgqhgfscl	cpsgytslsc
481	eiattlslfg	dyflwvks	vttkgsvcni	alrfqtvpqm	alllfrsnrd	vfvklellsg
541	yihlsiqvnn	qskvlflfish	ntsdgewhfv	evifaeavtl	tliddsckek	ciakaptple
601	sdqsicafqn	sflglplpvm	tengvallnf	ynmpscpsfv	gclqdkidw	nhitleniss
661	gsslnvkagc	vrkdwcscqp	cqsrgrcinl	wlsyqdcchr	pyegpnclrg	kfcgrsrlps
721	tvcegncktnl	tcyngngctc	fqtelkcmcr	pgftgewcek	didecasdpc	vngglcqdll

TABLE 1-continued

781 nkfqclcdva fagercevd1 addlisdift tigsvtvall lillilaivas vvtsnkratq
841 gtypsrqrqk egsrvemwnl mpppamerli

SEQ ID NO: 51 Mouse CRB1 cDNA Sequence (NM_133239.2, CDS: from 167 to 4384)

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1  gaagtgcctt  ctgattctct  gtctgtggag  gagccctggg  aggggtggga  cagagatggc
61  atcctggctc  tctgaggcac  ctgctcttct  ctgaaccaca  caggagtcaa  gagccaaaca
121  gggatagctt  cagcagcact  tcagagggtg  ttctctaagt  aagaacatga  agctcaagag
181  aactgcctac  cttctcttcc  tgtacctcag  ctctcactcg  ctcatctgca  taaagaattc
241  attttgcaat  aaaaacaata  ccaggtgcct  ttcaggctct  tgccaaaaca  attctacgcg
301  caagcatttt  ccacaagaca  acaattgttg  cttagacaca  gccataaatt  tggacaaga
361  ctgtgaagat  ctgaaagacc  cttgtctctc  gagtccctgc  caaggaaattg  ccacttgtgt
421  gaaaatccca  ggggaaggga  acttcccgtg  tcagtgtcct  cctgggtaca  gcgggctgaa
481  ctgtgaaact  gccaccaatt  cctgtggagg  gaacctctgc  caacatggag  gcacctgccg
541  taaagacctt  gagcacccctg  tctgtatctg  ccctcctgga  tatgtctgga  ggttctgtga
601  gactgatcac  aatgagtgtg  cttctagccc  ttgccacaat  ggggctatgt  gccaggatgg
661  aatcaatggc  tactcctgct  tctgtgtgcc  tggataccaa  ggcaggcatt  gtgacttggg
721  agtggatgaa  tgtgttctct  atccctgcaa  gaatgaggct  gtgtgcctca  atgagatagg
781  aagatacaat  tgtgtctgcc  ctcaagagtt  ttctggcggtg  aactgtgagt  tggaaattga
841  tgaatgcaga  tcccagcctt  gtctccacgg  tggcacatgt  caggacgctc  cagggggcta
901  ctctctgtac  tgtgcaacct  gattccttgg  agagcactgt  gaactcagcg  ttaatgaatg
961  tgaagtcag  ccggtgtctcc  atggagggtt  atgtgtggat  ggaagaaaca  gttaccactg
1021  tgactgcaca  ggtagtggat  tcacagggat  gcactgtgag  tccttgattc  ctcttctgtg
1081  gtcaaaacct  tgtccacaag  acgcgacatg  tgaagatact  gttgacagct  atatttgtca
1141  ctgcccgcct  ggatacacag  gtgccctgtg  tgagacagac  ataatgaat  gcagttagcaa
1201  cccctgccaa  ttttgggggg  aatgtgtcga  gctgtctca  gagggcttat  atggaaacac
1261  tgetggcctg  ccttctctct  tcagctatgt  tggagcctcg  ggctatgtgt  gtatctgtca
1321  gcctgggatt  acaggaaatt  actgtgaaga  agacgttgat  gaatgtttac  tgcaccttgg
1381  cctaaatggt  ggtacttgtg  agaacctgcc  tgggaattat  gcctgtcact  gtccctttga
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1501  tcataccag  tgtctgaaca  atggaaaatg  tatccctcat  tcccaaatg  gccagcatgg
1561  attcaactgc  cagtgtcttt  ctggctatgc  ggggcccctg  tgtgaaactg  tcaccacact
1621  ttcaattggg  agcaatggct  tcctatgggt  cacaagtggc  tcccatcacg  gcataaggcc
1681  agaatgtaac  atatccttga  ggtttcacac  tgttcaacca  aacgcacttc  tcctatccg
1741  aggcaacaag  gacgtgtcta  tgaagctgga  gttgtgtaat  ggttgtgttc  acttatcaat
1801  tgaagtcttg  aatcagttaa  aggtgctcct  gtctatttct  cacaacacca  gtgatggaga
1861  atggcatttc  gtggaggtaa  caatcgacga  aactccaacc  ctggccttag  ttggcgctc
1921  ctgcaaggag  aagtgcacca  ccaagtcttc  tgttccagtt  gagaatcatc  aatcaatatg
1981  tgettttgac  gactcttttt  tgggtggcct  accaatggg  acagccaaca  acagtgtgtc
2041  tgtgtctaac  atcttaaatg  tgcggtccac  accttctttt  gtaggctgtc  tccaagacat
2101  tagatttgat  ttgaatcaca  ttaactctga  gaacgtttca  tctggcctgt  catcaaatgt
2161  taaagcagcg  tgctgggaa  aggactggtg  tgaagtcaa  cctgtcaca  acagaggacg
2221  ctgcataaac  atgttcgagg  gttatcagtg  tgaatgtgac  aggcctata  caggctccaa
2281  ctgcctgaaa  gagtatgtag  cgggaagatt  tggccaagat  gactccacag  gatatcgggc
2341  ctttagtgtt  aatgataaatt  atggacagaa  cttcagctct  tcaatgtttg  tccgaacacg
2401  tcaacctctg  gccttacttc  tggctttgga  aaatagtact  taccagtatg  tcagtgtctg
2461  gctagagcac  ggcagcctag  cactgcagac  tccaggctct  cccaagtcca  tggtaaaactt
2521  tttctcagtg  gatggaaaatg  ttcaactaat  atctttgaga  atcaaaacca  atgaaaattga
2581  actgtatcag  tcttcacaaa  acctaggatt  catttctgtt  cctacatgga  caattcgaag
2641  aggagacgct  atcttcattg  gtggcttaac  tgacagagag  aagactgaag  ttatgtgtgg
2701  cttcttcaaa  ggtgtgtgtc  aagatgtcag  attaaacagc  cagactctgg  aattctttcc
2761  caattcaaca  aacatgcat  acgatgaccc  aattctgtc  aatgtgactc  aaggctgtcc
2821  cggagacaac  acatgtaagt  ccaacctctg  tcataatgga  ggtgtctgcc  actcctctgtg
2881  ggatgacttc  tctgtctccc  gccctacaaa  cacagcgggg  agagcctgcg  agcaagtcca
2941  gtgggtgcac  atgtgaccaa  gtccctccac  tgcagagtgc  cagctgtctc  ctcaagggtt
3001  tgaatgtatc  gcaaacgctg  ttttcagcgg  attaagcaga  gaaatactct  tcagaagcaa
3061  tgggaacatt  accagagaac  tcaccaaat  cacatttgct  ttcagaacac  atgatacaaa
3121  tgtgatgata  ttgcctgcag  aaaaagaacc  agagtttctt  aatatagca  tccaagatgc
3181  cagattatct  tttcaattgc  gaagtggcaa  cagcttttat  acgctgcacc  tgatgggttc
3241  ccaattgggt  aatgatggca  catggcacca  agtgacttct  tccatgatag  acccagtggt
3301  ccagacctcc  cgttggtgaa  tggagggtga  cgaccagaca  cctttgtga  taagtgaagt
3361  tgctactgga  agcctgaact  ttttgaagga  caatacagac  atctatgtgg  tgaccaatc
3421  tgttgacaa  ccgaaaggcc  tgcagggtcg  tctgagcaca  atagagattg  gaggcatata
3481  tctttcttac  tttgaaattc  tacatgggtt  ccctggtaag  cctcaggaag  agcaattctc
3541  caaagtctct  acaaatatgg  tacttactgg  ctgtttgcca  tcaaatgcct  gccactccag
3601  cccctgtttg  catggaggaa  actgtgaaga  cagctacagt  tcttatcggt  gtgctgtctc
3661  ctggggatgg  tcagggacac  actgtgaaat  caacattgat  gagtgtttt  ctagccctcg
3721  tatccatggc  aactgtctct  atggagtgc  agcctaccac  tgcagggtgt  agcctggata
3781  caccggtgtg  aactgtgagg  tggatgtaga  caattgcaag  agtcatcagt  gtgcaaatgg
3841  ggccacctgt  gttcgtgaag  ctcatggcta  ctctgtctc  tgcttggaa  attttacggg
3901  gagattttgc  agacacagca  gattaccctc  aacagtctgt  gggaatgaga  agagaaactt
3961  cacttgctac  aatggaggca  gctgtccat  gttccaggag  gactggcaat  gtatgtgctg
4021  gccaggttct  actggaaggt  ggtgtgaaga  ggacatcaac  gagtgtgct  ccgctccctg
4081  catcaatgga  ggaactgtga  gggacttggt  caacaggttc  ctatgcatct  gtgatgtggc
4141  cttcgtgggc  gagcgtgtg  agctggacct  ggctgatgac  aggtcctctg  gcaatttcc
4201  cgctgttggc  tccggaactt  tggccctgtt  cttcatctc  ttgttgcgt  ggggtgcttc
4261  tcttattgcc  tccaaacaaa  gggcgactca  aggaacctac  agccccagcg  gtccaggagaa
4321  ggctggccct  cgagtggaaa  tgtggatcag  gatgccgcc  ccggcactgg  aaaggctcat
4381  ctaggagact  gctgctcttc  tcaggacaga  gaagaacatg  atgataccg  ggtcgtgctc
4441  gagtgaagat  ggctttacat  cactagagat  acatacagct  gggactgtgg  gaaggacctt

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TABLE 1-continued

4501	cctgtggagt	cactgagtag	ttatgtcatc	cattcacaga	agagtgtccc	tgtgtttgcc
4561	tgtcagcctc	agaattagca	aaacatctag	cagacagaga	acacagtatt	tcagaagaac
4621	tccagaggct	gccccctaaa	ctctttactg	gttgatccac	ataaaatgct	tagtagccaa
4681	gtgccattaa	ttatacagag	cc			

SEQ ID NO: 52 Mouse CRB1 Amino Acid Sequence (NP_573502.2)

1	mkllkrtayll	flylssslili	ciknsfcnk	ntrcslsgpcq	nnstckhfpq	dnnccldtan
61	nldkdcedlk	dpcfsppcqq	iatcvkipge	gnflcqcpqg	ysglncetat	nscggnlcqh
121	ggterckdpeh	pvcicppgya	grfcetdhne	casspchnga	mcqdggingys	cfcvpgyqgr
181	hcdlevdecv	sdqckneavc	lneigrytcv	cpqetsgvnc	eleidecrsq	pclhgatecd
241	apggyscdca	pgflgehcel	svnecesqpc	lhggclcvdgr	nsyhcdctgs	gftgmhcesl
301	iplcwsfkpc	ndatcedtvd	syichcrpgy	tgalcetdin	ecssnpcqfw	gecvlseseg
361	lygntaglp	sfsyvgasgy	vcicqpgftg	ihceedvdec	llhpcclnggt	cenlpgnyac
421	hcptddtst	fyggencsei	llgcthhqcl	nngkciphfq	ngqhgftcqc	lsgyagplce
481	tvttlsfsgn	gflwvtsgh	tgigpecnis	lrfhtvqpna	lllirgnkdv	smklllncg
541	vhlsievwnq	lkvllsishn	tsdgewhfe	vtiaetltia	lvggscckec	ttkssvpven
601	hqsicalqds	flggplmgta	nnsvsvlniy	nvpstpsfvg	clqdirfdln	hitlenvssg
661	lssnvkagcl	gkdwcscqpc	qnrgcrinlw	qgyqcecdp	ytgsnclkey	vagrfgqdds
721	tgaaafsvnd	nyqgnfslsm	fvrtrqplgl	llalenstyq	yvsvwlhegs	lalqtpgspk
781	fmvnfflsdg	nvhllslrik	pneielyqss	qnlgfisvpt	wtirrgdvif	igglpdrekt
841	evyggfkgc	vqdvrlnsqt	leffpnstnn	ayddpilvnt	tqcgcpdntc	knpchnngv
901	chslwddfs	scptntagra	ceqvwcqls	pcpptaecql	lpqgfecian	avfsglsrei
961	lfrsnngnitr	eltnitfaf	thdtnvmilh	aekepeflni	siqdarlffq	lrsngsfytl
1021	hlmgsqlvnd	gtwhqvtfsm	idpvaqtsrw	qmevndqtpf	visevatgsl	nflkdntdiy
1081	vgdqvndpk	glggclstie	iggiylsyfe	nlhgffpgkp	eeqflkvstn	mltqclpsn
1141	achsspcilh	gncedsysy	rcacslgws	thceinidec	fsspcihgnc	sdgvaayhcr
1201	cepygtgvnc	evdvndcksh	qcanatcvc	eahgysclcf	gnftgrfrch	srtpstvcgn
1261	ekrnftcyng	gscsmfqedw	qcmcwpgftg	ewceedinec	asdpcinggl	crdlvnrlfc
1321	icdvafager	celldladrl	lgiftavgs	tlalfillll	agvasliasn	kratqgtysp
1381	sgqekagprv	emwirmpppa	lerli			

SEQ ID NO: 53 Human BRG1 cDNA Sequence Variant 1 (NM_001128849.1, CDS: from 75 to 5114)

1	ggcgggggag	gcgccgggaa	gtcgacggcg	ccggcgggctc	ctgcaggagg	ccactgtctg
61	cagctcccg	gaagatgtcc	actccagacc	caccctctggg	cggaaactcc	cggccaggctc
121	cttccccggg	ccctggccct	tccccctggag	ccatgctggg	ccctagcccc	ggctccctcgc
181	cggtctccgc	ccacagcatg	atggggccca	gcccagggcc	gcccctagca	ggacacccca
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301	tggagtcctc	gcgatgaga	ggcatgtcgg	acgaccggcg	ctacaaccag	atgaaaggaa
361	tggggatg	gtcagggggc	catgctggga	tggggccccc	gcccagcccc	atggaccagc
421	actcccaagg	ttaccctcgc	ccccctgggtg	gctctgagca	tgctctagct	ccagttccag
481	ccagtgggcc	gtctccgggg	ccccagatgt	ctccccgggc	aggaggtgcc	ccgctgggatg
541	gtgctgaccc	gcagggcctt	gggcagcaga	accggggccc	aacccattt	aaccagaacc
601	agctgcacca	gctcagagct	cagatcatgg	cctacaagat	gctggccagg	gggcagcccc
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781	ctggccccgg	cccggtgtcc	ggccccggc	ctccaaatta	cagcaggcct	catggtatgg
841	gagggcccaa	catgcctccc	ccaggacct	cggtcggtgc	ccccgggatg	ccaggccagc
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1021	ctgcccgtcc	acccgcccgc	tgcgccgtga	tgccaccgca	gaaccagctc	ccccggcagc
1081	cgccccagcc	cccgcccatg	gtgccactgc	accagaagca	gagccgcctc	acccccctcc
1141	agaagcccg	gggcccgcag	cctgtggaga	tcctgcagga	gcgcgagtag	aggctgcagg
1201	ctcgcacatc	acaccgaatt	caggaaactg	aaaaccttcc	cgggtccctg	gcccgggatt
1261	tgcgaaacca	agcgaccatt	gagctcaagg	ccctcaggct	gctgaaattc	cagagggcagc
1321	tgcgccagga	ggtggtggtg	tgcatgcgga	gggacacagc	gctggagaca	gccctcaatg
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1441	tggagaagca	gcagaagatc	gagcaggagc	gcaagcgccg	gcagaagcac	caggaaatcc
1501	tcaatagcat	tctccagcat	gccaaggatt	tcaaggaaata	tcacagatcc	gtcacaggca
1561	aaatccagaa	gctgaccaag	gcagtgccca	cgtaccatgc	caacacggag	cgggagcaga
1621	agaaagagaa	cgagcggtatc	gagaaggagc	gcattgcggag	gctcatggct	gaagatgagg
1681	aggggtaccg	caagctcatc	gaccagaaga	aggacaagcg	cctggcctac	ctcttgacgc
1741	agacagacga	gcacgtggct	aacctcacgg	agctggtg	gcagcacaa	gctgcccagg
1801	tcgccaaagg	gaaaaagaa	aaaaagaaa	agaagaaggg	agaaaatgca	gaaggacaga
1861	cgctcgccat	tgggccggat	ggcgagcctc	tggaacagac	cagccagatg	agcgacctcc
1921	cggtgaaggt	gatccacgtg	gagagtggga	agatcctcac	aggcacagat	gcccccaagg
1981	ccgggcagct	ggaggccttg	ctcgagatga	acccggggta	tgaagtactg	ccgaggtctg
2041	atagtgaaga	aagtggctca	gaagaagagg	aagaggagga	ggaggaaagg	cagccgcagg
2101	cagcacagcc	tccccacctg	cccgtggagg	agaagaagaa	gattccagat	ccagacagcg
2161	atgacgtctc	tgacgtggac	gcgcgggcaca	tcattgagaa	tgccaaagca	gatgtcgatg
2221	atgaatatgg	cgtgtcccag	gcccttgca	gtggcctgca	gtcctactat	gccgtggccc
2281	atgctgtcac	tgagagatg	gacaagcagt	cagcgcttat	ggctcaatgg	gtcctcaaac
2341	agtaccagat	caaaagtgtt	gagtggtg	tgctccctga	caacaacaac	ctgaacggca
2401	tcctggccga	cgagatgggc	ctggggaaga	ccatccagac	catcgcgctc	atcacgtacc
2461	tcatggagca	caaacgcctc	aatgggcctt	tcctcatcat	cgtgcctctc	tcaacgctgt
2521	ccaactgggc	gtacaggttt	gacaagtggg	ccccctccgt	gggtaagggtg	tcttacaagg
2581	gatccccagc	agcaagacgg	gcctttgtcc	cccagctccg	gagtgggagg	ttcaacgtct
2641	tgctgacgac	gtacgagtag	atcatcaaa	acaagcacat	cctcgccaag	atccgttgga
2701	agtacatgat	tgtggagcaa	ggtcaccgca	tgaagaacca	ccactgcaag	ctgacgcagg
2761	tgctcaaac	gcactatgtg	gcaccccgcc	gcctgctgct	gacgggcaca	ccgctgcaga

TABLE 1-continued

2821	acaagcttcc	cgagctcttg	gcgctgctca	acttctgct	gcccaccatc	ttcaagagct
2881	gcagcacctt	cgagcagtg	tttaacgcac	cctttgccat	gaccggggaa	aaggtggacc
2941	tgaatgagga	ggaaccatt	ctcatcatcc	ggcgtctcca	caaagtgtg	cggcccttct
3001	tgctccgacg	actcaagaag	gaagtccagg	cccagttgcc	cgaaaagggtg	gagtaacgtca
3061	tcaagtgcga	catgtctgag	ctgcagcgag	tgctctaccg	ccacatgcag	gccaagggcg
3121	tgctgctgac	tgatggctcc	gagaaggaca	agaagggcaa	aggcggcacc	aagaccctga
3181	tgaacacccat	catgcagctg	cggaagatct	gcaaccaccc	ctacatgttc	cagcacatcg
3241	aggagtcctt	ttccgagcac	ttggggttca	ctggcgccat	tgccaagggtg	ctggacctgt
3301	accgagcctc	gggtaaat	gagcttcttg	atagaattct	tcccaaacctc	cgagcaacca
3361	accacaaagt	ggctgctgtt	tgccaaatga	cctccctcat	gaccatcatg	gaagattact
3421	ttgctgatcg	cggttttaa	tacctcaggc	ttgatggaac	cacgaaggcg	gaggaccggg
3481	gcctgctgct	gaaaaccttc	aacgagcccg	gctctgagta	cttcatcttc	ctgctcagca
3541	cccgggctgg	ggggctcgcc	ctgaacctcc	agtccggcaga	cactgtgatc	atTTTTgaca
3601	gcgactggaa	tcctcaccag	gacctgcaag	cgaggagccg	agcccaccgc	atcgggcagc
3661	agaacgaggt	gcgtgtgctc	cgctctgca	ccgtcaacag	cgtaggaggag	aagatcctag
3721	ctgcagccaa	gtacaagctc	aacgtggacc	agaagggtgat	ccaggccggc	atgttcgacc
3781	agaagtctct	cagccatgag	cggcgcgcct	tcctgcaggc	catcctggag	cacgaggagc
3841	aggatgagag	cagacactgc	agcacgggca	gcggcagtg	cagcttcgcc	cacactgcc
3901	ctccgcagc	ggcgctcaac	cccgaacttg	aggagccacc	ttctaaaggag	gaagacgagg
3961	tgcccgcaga	cgagaccgtc	aaccagatga	tcgcccggca	cgaggaggag	ttgatctgt
4021	tcctgctgcat	ggacctggac	cgagggcgcg	aggaggcccg	caaccccaag	cggaagccgc
4081	gcctcatgga	ggaggacgag	ctccctcgt	ggatcatcaa	ggacgacgcg	gaggtggagc
4141	ggctgacctg	tgaggaggag	gaggagaaga	tgctcgccg	tggtcccg	caccgcaagg
4201	aggtggacta	cagcgactca	ctgacggaga	agcagtggt	caagaaaatt	acaggaaaag
4261	atatccatga	cacagccagc	agtgtggcac	gtgggtaca	attccagcgt	ggccttcagt
4321	tctgcacacg	tgctcaaa	gcatcgagg	agggcacgct	ggaggagatc	gaagaggagg
4381	tcgggcagaa	gaaatcatca	cggaagcgca	agcgagacag	cgacgcccgc	tcctccaccc
4441	cgaccaccag	cccgccagc	cgacacaagg	acgacgagag	caagaagcag	aagaagccgc
4501	ggcgcccgcc	tgccgagaaa	ctctccccta	acccacccaa	cctcaccaag	aagatgaaga
4561	agattgtgga	tgccgtgatc	aagtacaagg	acagcagcag	tgagctcag	ctcagcgagg
4621	ttctcatcca	gctgcctcgc	cgaaaggagc	tgcccagata	ctacagctc	atccgcaagc
4681	ccgtggactt	caagaagata	aaggagcgca	ttcgcaacca	caagtaccgc	agcctcaacg
4741	acctagagaa	ggacgtcatg	ctcctgtgct	agaacgcaca	gaccttcaac	ctggagggtt
4801	ccctgatcta	tgaagactcc	atcgtcttgc	agtcggtctt	caccagcgtg	cggcagaaaa
4861	tcgagaagga	ggatgacagt	gaaggcgagg	agagtgaagga	ggagggaagag	ggcgaggagg
4921	aaggctccga	atccgaatct	cggtccgtca	aagtgaagat	caagcttggt	cggaaggaga
4981	aggcacagga	ccggctgaag	ggcgcccgcc	ggcgcccgag	ccgagggtcc	cgagccaagc
5041	cggtcgtgag	tgacgatgac	agtgaggagg	aacaagagga	ggaccgctca	ggaagtggca
5101	gcgaagaaga	ctgagccccc	acattccagt	ctcgaccccc	agccctcctg	tccagagctg
5161	agatggcata	ggccttagca	gtaacgggta	gcagcagatg	tagcttcaga	cttgtagtaa
5221	aactgtataa	acaaaagaat	cttccatatt	tatacagcag	agaagctgta	ggactgtttg
5281	tgactggccc	tgccctggca	tcagttagcat	ctgtaacagc	attaactgtc	ttaaagagag
5341	agagagagaa	tcocgaattg	gggaacacac	gatacctgtt	ttctttttcc	gttgctggca
5401	gtactgttgc	gccgcagttt	ggagtcaactg	tagttaagt	tgatgcatg	tgctcaccg
5461	tccactcctc	ctactgtatt	ttattggaca	ggtcagactc	gcccggggcc	cggcgagggt
5521	atgtcagtg	cactgtagt	caaacagtaa	taaattaaac	caacaacaaa	acgcacagcc
5581	aaaaaaaa					

SEQ ID NO: 54 Human BRG1 Amino Acid Sequence Isoform A (NP_001122321.1,
CDS: from 75 to 5114)

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1  mstpdplpgg tprpgpspgp gspgamlgp spgspgsah smmgpspgpp saghpiptgg
61  pggyqdmnh qmhkpmesm h ekmsddpry nqmkgmgrs gghagmgypp spmdqhsqgy
121  psp1ggseha sspvpasggs sgpmsspgp gapldgadp alqgqnrqpt pfnqnlhlql
181  raqlmaykml argqpldhl qnavqgkrpm pgmqqqmptl ppsvsatgp gpgpgpgpgp
241  gpgpappnys rphgmgsppnm ppppsgvpv gmpgqppggp pkpwpegpma naaaptstpq
301  klippqptgr pspappavpp aaspvmpqgt qspgqaqpa pmvplhqkqs ritpiqkprg
361  ldpveilqer eyrlqariah rigelenlpg slagdlrtka tielkalrll nfrqlrlqev
421  vvcmmrdtal etalnakayk rskrqslrea riteklekqg kieqerkrrg khqeylnsil
481  qhakdfkeyh rsvtgkiqkl tkavatyhan tereqkkene riekermrrl maedeegyryk
541  lidqkdkkrl ayllqqtdey vanltelvrq hkaaqvakek kkkkkkkkae naegqtpaig
601  pdgepldets qmsdlpvkvi hvesgkiltg tdapkgaggle awlenmpgye vaprsdsees
661  gseeeeeeee eeqpqaagpp tlpveekkki pdpdsddvse vdarhiiena kqdvddeygv
721  sqalarglqs yyavahavte rvdksalmv ngvlkqyqik glewlvslvn nlnlglade
781  mglgkttiqt alitylme h ringpfliiv plstlsnway efdkwapsv kvsykgspaa
841  rrafvpqlrs gkfnvlltty eyiikdkhil akirwkymiv deghrmknhh cklqtvlnth
901  yvaprrillt gtplqnklpe lwallnflp tlfkscstfe qwnapgamt gekvdlneee
961  tiliirrlhk vlrrpflrrl kkeveaqlpe kveyvikcdm salqrlyrh mqakgvlltd
1021 gsekdkkgkg gktklmmtim qlrkicnhpy mfhieesfs ehlgftggiv qgldlyrasg
1081 kfelldrilp klratnhkvl lfcqmtsmt imedyfayrg fkyllrdgtt kaedrgmllk
1141 tfnepgseyf ifllstragg lglnlqsadt viifdsdwnp hqldlaqdra hrigqqnevr
1201 vlrlctvnsv eekilaaky klnvdqkviq agmtdqkss herraflqai leheeqdesr
1261 hcstgsgsas fahtapppag vnpdleppl keedevpdde tvnqmiarhe eefdlfmrmd
1321 ldrrearean prkrprlme delpswiik daeverltce eeeekmfgrg shrkrdevds
1381 dsltekqwlk kitgkdihdt assvarglqf qrglqfctra skaieegtle eieeervrqlk
1441 ssckrkrdsd agsstpttst rerdkddesk kqkkrgrppa eklspnppnl tkkmkkivda
1501 vikykdsssg rqlsevfqi prskelpy elirkpvdfk kikerirnhk yrslnldlek
1561 vmllcqnaqt fnlegslie dsivlqsvft svrqkieked dsegeeseee eegeesges
1621 esrvkvkik lgrkekaqdr lkgrrrrprsr gsrakpvvss ddseeegeed rsgsgseed

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TABLE 1-continued

SEQ ID NO: 55 Human BRG1 cDNA Sequence Variant 2 (NM_001128844.1, CDS:
from 361 to 5304)

1	ggagaggccg	ccgcggtgct	gagggggagg	ggagccggcg	agcgcgcgcg	cagcgggggc
61	gcgggtggcg	cgcgtgtgtg	tgaagggggg	gcggtggccg	aggcggggcg	gcgcgcgcgc
121	gaggcttccc	ctcgtttggc	ggcggcggcg	gcttctttgt	ttcgtgaaga	gaagcgagac
181	gcccattctg	ccccggggcc	cgcgcggagg	ggcgggggag	gcgccgggaa	gtcgacggcg
241	ccggcggctc	ctcgtctctg	cccttttgcc	caggctagag	tgcaagtggg	cggatcatgt
301	tcactgcagc	ctcaacctcc	tggactcagc	aggaggccac	tgtctgcagc	tcccgtagag
361	atgtccactc	cagaccacc	cctggggcga	actcctcggc	caggtccttc	cccggggcct
421	ggcccttccc	ctggaggcca	gctgggccct	agccggggtc	cctcgccggg	ctccgcccac
481	agcatgatgg	ggcccagccc	agggcccggc	tcagcaggac	accccatccc	caccaggggg
541	cctggagggt	accctcagga	caacatgcac	cagatgcaca	agcccatgga	gtccatgcac
601	gagaaggcga	tgtcggacga	ccgcgctac	aaccagatga	aaggaatggg	gatgcggcca
661	gggggcatg	ctgggatggg	gccccggccc	agcccatggg	accagcactc	ccaagggtac
721	ccctcgcccc	tgggtggctc	cagcatgccc	tctagtccag	ttccagccag	tggcccgctg
781	tcggggcccc	agatgtcttc	cgggccaggga	ggtgccccgc	tggatggtgc	tgacccccag
841	gccttggggc	agcagaaccg	gggcccaccc	ccatttaacc	agaaccagct	gcaccagctc
901	agagctcaga	tcattggccta	caagatgctg	gccagggggc	agccccctcc	cgaccacctg
961	cagatggcgg	tgaggggcaa	cgggcggatg	cccgggatgc	agcagcagat	gccaacgcta
1021	cctccacctc	cgggtgtccg	aacaggaccc	ggccttggcc	ccggccctgg	ccccggcccc
1081	ggtccccggc	cggcacctcc	aaattacagc	aggcctcatg	gtatggggag	gccccacatg
1141	cctccccagc	gaccctcggg	cgtgcccccc	gggatgccag	gccagcctcc	tggaggggct
1201	cccaagccct	ggcctgaagg	acccatggcg	aatgtgctg	ccccacagag	caccctctag
1261	aagctgattc	cccccgagcc	aacggggcgc	ccttcccccg	cgccccctgc	cgccccaccc
1321	gccgcctcgc	ccgtgatgcc	accgcagacc	cagtcccccg	ggcagccggc	ccagcccgcg
1381	cccatggtgc	actgcaccca	gaagcagagc	cgcacacccc	ccatccagaa	gcgcgggggc
1441	ctcgacctg	tggagatcct	gcaggagcgc	gagtagaggc	tcagggctcg	catcgcacac
1501	cgaattcagg	aacttgaata	ccttccccgg	tccttgcccg	gggatttgcg	aaccaaagcg
1561	accattgagc	tcaaggccct	caggctgctg	aacttcacga	ggcagctgcg	ccaggaggtg
1621	gtggtgtgca	tgcggaggga	cacagcgctg	gagacagccc	tcaatgctaa	ggcctacaag
1681	cgacgcaagg	gcccagtcct	cgcgcaggcc	cgcacacctg	agaagctgga	gaagcagcag
1741	aagatcgagc	aggagcgcaa	gcgccggcag	aagcaccagg	aatacctcaa	tagcattctc
1801	cagatgcaca	aggatttcaa	ggaatatcac	agatccgtca	caggcaaaat	ccagaagctg
1861	accaaggcag	tcacggagct	ccatgccaac	acggagcggg	agcagaagaa	agagaacgag
1921	cggatcgaga	aggagcgcat	gcggaggctc	atggctgaag	atgaggaggg	gtaccgcaag
1981	ctcatcgacc	agaagaagga	caagcgctg	gcctacctct	tcagcagagc	agacaggtac
2041	gtggctaac	ggcccttctc	ggtgcggcag	cacaaggctg	cccaggtcgc	caaggagaaa
2101	aagaagaaaa	agaaaaagaa	gaaggcagaa	aatgcagaa	gacagacgcc	tgccattggg
2161	ccggatggcg	agcctctgga	cagagaccag	cagatgagcg	acctcccggt	gaaggtgatc
2221	cacgtggaga	tgcggaaagt	cctcacaggc	acagatgccc	ccaaagcccg	cgagctggag
2281	gcctggctcg	agatgaaccc	ggggtatgaa	gtagctccga	ggtctgatag	tgaagaaagt
2341	ggctcagaag	aagagggaag	ggaggaggag	gaagagcagc	cgcaggcagc	acagcctccc
2401	accctgccc	tcaggagagc	gaagaagatt	ccagatccag	acagcgatga	cgctcttgag
2461	gtggacgcgc	ggcacatcat	tgagaatgcc	aagcaagatg	tcgatgatga	atatggcggt
2521	tcccaggccc	tgcacgtgg	cctgcagtc	tactatgccc	tggcccatgc	tgctactgag
2581	agagtggaga	tcagtcagc	gcttatggtc	aatgggtgct	tcaaacagta	ccagatcaaa
2641	ggtttggagt	ggctggtgtc	cctgtacaac	aacaacctga	acggcatcct	ggccgacgag
2701	atgggctctg	ggaagaccat	ccagaccatc	gcgctcatca	cgtacctcat	ggagcacaac
2761	cgcatcaatg	ggcccttctc	catcatcggt	cctctctcaa	cgctgtccaa	ctgggcgtac
2821	gagtttgaca	agtgggcccc	ctccgtgggt	aaggtgtctt	acaagggatc	cccagcagca
2881	agacgggctc	ttgtccccca	gctccggagt	gggaagttca	acgtcttgct	gacgacgtac
2941	gagtaacatc	tcaaaagcaa	gcacatctcc	gccaaagatc	gttggaagta	catgattgtg
3001	gacgaaggct	accgcatgaa	gaaccaccac	tgcaagctga	cgcagggtgc	caacacgcac
3061	tatgtggcgc	ccgcgcgcct	gctgctgacg	ggcacaccgc	tcgagaacaa	gcttcccag
3121	ctctggggcg	tcctcaactt	cctgctgccc	accatcttca	agagctgcag	caccttcgag
3181	cagtggttta	acgcaccctt	tgccatgacc	gggaaaaagg	tggacctgaa	tgaggaggaa
3241	accattctca	tcctccggcg	tctccacaaa	gtgctggcgc	ccttcttgct	ccgacgactc
3301	aagaaggaa	tcaggggcca	gttgcccgaa	aaggtggagt	acgtcatcaa	gtgcgacatg
3361	tctgcgctgc	agcaggtgct	ctaccgccac	atgcaggcca	aggcgctgct	gctgactgat
3421	ggctccgaga	aggacaagaa	gggcaaaagg	ggcaccagaa	ccctgatgaa	caccatcatg
3481	cagctgcgga	agatctgcaa	ccaccctac	acgttccagc	acatcgagga	gtccttttcc
3541	gagcacttgg	ggttcaactg	cggcattgtc	caagggttgg	acctgtaccg	agcctcgggt
3601	aaatttgagc	ttctttgacg	aattcttccc	aaactccgag	caaccaacca	caaagtgtgt
3661	ctgttctgct	aaatgacctc	cctcatgacc	atcatggaa	attactttgc	gtcccgggc
3721	tttaaatacc	tcaggcttga	tggaaaccag	aaggcggagg	accggggcat	gctgctgaaa
3781	accttcaacg	agcccggtgc	tgagtacttc	atcttctctg	tcagcaccgc	ggctgggggg
3841	ctcggcctga	accctcagtc	ggcagacact	gtgatcatct	ttgacagcga	ctggaaacct
3901	caccaggacc	tgcaagcgca	ggaccgagcc	caccgcacgc	ggcagcagaa	cagaggtgct
3961	gtgctccgcc	tctgcaccgt	caacagcgtg	gaggagaaga	tctagctgct	agccaaagtac
4021	aagctcaacg	tggaacagaa	ggtgatccag	gccggcatgt	tcgacacgaa	gtcctccagc
4081	catgagcggc	gcgccttctc	gcaggccatc	ctggagcagc	aggagcagga	tgagagcaga
4141	cactgcagca	ggggcagcgg	cagtgccagc	ttcgcccaca	ctgccctccc	gccagcgggc
4201	gtcaaccccc	acctcggagg	gccacctcta	aaggaggaa	acgaggtgct	cgacgacgag
4261	accgtcaacc	agatgatcgc	ccggcacgag	gaggagtttg	atctgttcat	gcgcatggac
4321	ctggaccgca	ggcgcgagg	ggcccgcaac	cccaagcgga	agccgcgctc	catggaggag
4381	gacgagctcc	cctcgtggat	catcaaggac	gacgcggagg	tggagcggct	gacctgtgag
4441	gaggaggagg	agaagatggt	cggccgtggc	tcccgcaccc	gcaaggaggt	ggactacagc
4501	gactcactga	cggagaagca	gtggctcaag	gccatcgagg	agggcacgct	ggaggagatc
4561	gaagaggagg	tcgggcagaa	gaaatcatca	cggaaagcga	agcgagacag	cgacgccggc
4621	tcctccaccc	cgaccaccag	caccgcgacg	cgcgacaagg	acgacgagag	caagaagcag

TABLE 1-continued

4681	aagaagcgcg	ggcggccgccc	tgccgagaaa	ctctccccta	acccacccaa	cctcaccaag
4741	aagatgaaga	agatttgtga	tgccgtgatc	aagtacaagg	acagcagcag	tggacgtcag
4801	ctcagcgagg	tcttcatcca	gctgcctcgc	cgaaggagc	tgcccagta	ctacagagtc
4861	atccgcaagc	ccgtggactt	caagaagata	aaggagcgca	ttcgcaacca	caagtaccgc
4921	agcctcaacg	acctagagaa	ggacgtcatg	ctcctgtgcc	agaacgcaca	gaccttcaac
4981	ctggagggct	cctgatcta	tgaagactcc	atcgtcttgc	agtcggtctt	caccagcgtg
5041	cggcagaaaa	tcgagaagga	ggatgacagt	gaaggcgagg	agagtgaagga	ggagggaagag
5101	ggcgaggagg	aaggctccga	atccgaatct	cggctccgtca	aagtgaagat	caagcttggc
5161	cggaaaggaga	aggcacagga	ccggctgaag	ggcggccggc	ggcggccgag	ccgagggtcc
5221	cgagcccaagc	cggctcgtgag	tgacgatgac	agtgaaggag	aacaagagga	ggaccgctca
5281	ggaagtggca	gcgaagaaga	ctgagccccc	acattccagt	ctcgacccc	agccccctcg
5341	tccagagctg	agatggcata	ggccttagca	gtaacgggta	gcagcagatg	tagtttcaga
5401	cttggagtaa	aactgtataa	acaaaagaat	cttccatatt	tatacagcag	agaagctgta
5461	ggactgtttg	tgaactggcc	tgtcctggca	tcagtagcat	ctgtaacagc	attaactgtc
5521	ttaaagagag	agagagagaa	ttccgaattg	gggaacacac	gatacctgtt	tttcttttcg
5581	gttgctggca	gtactgttgc	gccgcagttt	ggagtcaactg	tagttaagt	tggatgctgc
5641	tgcgtcaccg	tccactcctc	ctactgtatt	ttattggaca	ggtcagactc	gccggggggc
5701	cggcgagggt	atgtcagtg	cactggatgt	caaacagtaa	taaatataac	caacaacaaa
5761	acgcacagcc	aaaaaaaaa				

SEQ ID NO: 56 Human BRG1 Amino Acid Sequence Isoform B (NP_001122316.1)

1	mstpdpllg	tprrpgp	gspgamlgp	spgspgsah	smmgpsgpp	saghipptqg
61	pggyppqdmh	qmhkpmesm	ekgmsddpry	nqmkgmgrs	gghagmgppp	spmdqhsqgy
121	psplggseha	ssvpasggs	sgpqmssgpg	gapldgadpq	algqgnrgpt	pfnqnqlhql
181	raqimaykml	argqplpdhl	qmavqgkrpm	pgmqgqmpptl	pppsvstgpp	gpgpgp
241	gpgpappnys	rphmgggnm	pppgpsgvp	gmpgqppgpp	pkpwpegpma	naaaptstpq
301	klippqgtgr	pspappavpp	aaspvmpqt	qspgqpaqpa	pmvplhqkqs	ritpiqkprg
361	ldpveilqer	eyrlqariah	riqelenlpg	slagdlrtka	tielkalrl	nfqrqlrgev
421	vvcmmrdtal	etalnakayk	rskrqslrea	riteklekq	kiegerkrrq	khqeylnsil
481	qhakdfkeyh	rsvtgiqkl	tkavatyhan	tereqqkene	riekermrrl	maedeeygrk
541	lidqkkdkrl	ayllqgtdey	vanltelvrq	hkaaqvakek	kkkkkkkkkae	naeggtpaig
601	pdgepldets	qmsdlpvkvi	hvesgkiltg	tdapkaggle	awlemnpgye	vaprsdsees
661	gseeeeeeee	eeqpqaagpp	tlpveekkki	pdpdsddvse	vdarhiiena	kqdvddeygv
721	sqalarglqs	yyavahavte	rvdkqsalmv	ngvlkqyqik	glewlvslyn	nlnlilade
781	mglgktigt	alitylmehk	ringpfliiv	plstlsnway	etdkwapsvv	kvsykgspaa
841	rrafvpqlrs	gkfnvlltty	eyiikdkhil	akirwkymiv	deghrmknhh	ckltqvlnt
901	yvaprrlllt	gtplqnlkpe	lwallnflp	tifkscstfe	qwnapfam	gekvdnee
961	tiliirrlhk	vlpfllrrl	kkeveaqlpe	kveyvikcdm	salqrvlyrh	mqakgvlltd
1021	gsekdkkkg	gtktlmncim	qlrkicnhpy	mfqhiesfs	ehlgftggiv	qgldlyrasg
1081	kfelldrilp	klratnkhvl	lfcqmtsmt	imedyfayrg	fkylrlrdgt	kaedrgmllk
1141	tfnepgseyf	ifillstragg	lglnlqsadt	viifdsdwnp	hqdlqaqdra	hrigqgnevr
1201	vrlrctvns	eekilaaaky	klndvdkviq	agmfdqkss	herraflqai	leheeqdesr
1261	hctsgsgsls	fyatavppag	vnpdleppl	keedevpde	tvnqmiarhe	eefdlfrmd
1321	ldrrreearn	pkkrprrlme	delpswiikd	daeverltce	eeekmfgrg	srhrkevdy
1381	dsltekwqlk	aeegtlelei	eeevrqkss	rkrkrdsdag	sstpttstrs	rdkddeskkq
1441	kkrgrrppak	lspnppnlk	kmkivdavi	kykdsqgrq	lsevfqlps	rkelpyyel
1501	irkpvdffki	kerirnhkyr	slndlekdv	llcqnagtf	legsllyeds	ivlqsvftsv
1561	rqlikedds	eegeeseees	geeegseses	rsvkvkiklg	rkekaqdrk	ggrrrpsrgs
1621	rakpvvsddd	seegeedrs	gsgseed			

SEQ ID NO: 57 Human BRG1 cDNA Sequence Variant 3 (NM_003072.3, CDS: from 285 to 5228)

1	ggagagggcg	cgcgggtgct	gagggggagg	ggagccggcg	agcgcgcgcg	cagcgggggg
61	gcggggcgcg	cgcgtgtgtg	tgaagggggg	gcgggtggcg	aggcggggcg	gcgcgcgcgc
121	gaggtctccc	ctcctttggc	ggcggcgcg	gcttccttgt	ttcgtgaaga	gaagcgagac
181	gccattctcg	ccccggggcc	cgcgcggagg	ggcgggggag	gcgcggggaa	gtcgacggcg
241	ccggcggtgc	ctgcaggagg	ccactgtctg	cagctcccg	gaagatgtcc	actccagacc
301	cacccctggg	cggcaactcc	cggccaggtc	cttccccggg	ccctggccct	tcccctggag
361	ccatgctggg	ccctagcccg	ggctccctcg	cgggctccgc	ccacagcatg	atggggccca
421	gccagggggc	gcctcagca	ggacacccca	tccccaccca	ggggcctgga	gggtaccctc
481	aggacaacat	gcaccagatg	cacaagccca	tggagtccat	gcatagaga	ggcatgtcgg
541	acgaccgcgc	ctacaaccag	atgaaaggaa	tggggatgag	gtcagggggc	catgctggga
601	tggggggccc	gcccagcccc	atggaccagc	actcccaagg	ttaccctcgc	cccctgggtg
661	gctctgagca	tgccctatgc	ccagttccag	ccagtggccc	gccttcgggg	ccccagatgt
721	cttcggggcc	aggaggtg	ccgctggatg	gtgctgaccc	ccaggccttg	ggcgagcaga
781	accggggccc	aacccccatt	aaccagaacc	agctgcacca	gctcagagct	cagatcatgg
841	cctacaagat	gctggccagg	gggcagcccc	tccccagcca	cctgcagatg	gcgggtgcagg
901	gcaagcgccc	gatgcccggg	atgcagcagc	agatgccaac	gctacctcca	ccctcggtgt
961	ccgcaacagg	accggccct	ggccctggcc	ctggcccccg	cccggtctcc	ggccccggac
1021	ctccaaatta	cagcaggcct	catggtatgg	gagggcccaa	catgcctccc	ccaggacctg
1081	cggcgctg	ccccgggatg	ccaggccagc	ctcctggagg	gcctcccaag	ccctggcctg
1141	aaggaccat	ggcgaatgct	gctgccccca	cgagcaccac	tcagaagctg	attcccccg
1201	agccaacggg	cgcgccttcc	cccgccggcc	ctgcgctccc	accgcgcgc	tcggcctgga
1261	tgccaccgca	gacccagtc	cccgggcagc	cggcccgacc	cgcgcccatg	gtgccactgc
1321	accagaagca	gagccgcac	acccccatcc	agaagccggc	gggcctcgac	cctgtggaga
1381	tcctgcagga	gcgcagatg	aggctgcagg	ctgcctcgc	acaccgaatt	cagggaactg
1441	aaaaccttcc	cgggtccctg	gccggggatt	tgccaaccaa	agcgaccatt	gagctcaagg
1501	ccctcaggct	gctgaacttc	cagaggcagc	tgcccgagga	gggtggtggtg	tgcatgcgga
1561	gggacacagc	gctggagaca	gccctcaatg	ctaaggccta	caagcgagc	aagcgccagt
1621	ccctgcgcga	ggccccgatc	actgagaagc	tggagaagca	gcagaagatc	gagcaggagc

TABLE 1-continued

1681	gcaagcgccg	gcagaagcac	caggaataacc	tcaatagcat	tctccagcat	gccaaggatt
1741	tcaaggaata	tcacagatcc	gtcacaggca	aaatccagaa	gctgaccaag	gcagtggcca
1801	cgtaccatgc	caacacggag	cgggagcaga	agaagagaa	cgagcggatc	gagaaggagc
1861	gcctgcggag	gctcatggct	gaagatgagg	aggggtatccg	caagctcatc	gaccagaaga
1921	aggacaagcg	cctggcctac	ctcttgacgc	agacagacga	gtacgtggct	aacctcacgg
1981	agctgggtgcg	gcagcacaa	gctgcccagg	tcgccaagga	gaaaaagaag	aaaaagaaaa
2041	agaagaagcg	agaaaaatgca	gaaggacaga	cgccctgccat	tgggcccggat	ggcgcgcctc
2101	tggacgagac	cagccagatg	agcgcacctc	cggtgaaggt	gatccacgtg	gagagtggga
2161	agatcctcac	aggcacagat	gcccccaaa	ccgggcagct	ggaggcctgg	ctcgagatga
2221	acccggggta	tgaagtatgt	ccgaggtctg	atagtgaaga	aagtggctca	gaagaagagg
2281	aagaggagga	ggaggaagag	cagccgcagg	cagcacagcc	tcccacctg	cccgtggagg
2341	agaagaagaa	gattccagat	ccagacagcg	atgacgtctc	tgagggtggac	gcgcggcaca
2401	tcatttgaaa	tgccaagcaa	gatgtcgatg	atgaatatgg	cggtgccag	gccccttgca
2461	gtggcctgca	gtcctactat	gcccgtggccc	atgctgtcac	tgagagagt	gacaagcagt
2521	cagcgcttat	ggtcaatggt	gtcctcaaac	agtaccagat	caaaagtttg	gagtggtggg
2581	tgtccctgta	caacaacaac	ctgaacggca	tcctggccga	cgagatgggc	ctggggaaga
2641	ccatccagac	catcgcgctc	atcacgtacc	tcattggagca	caaacgcata	aatggggcct
2701	tcctccatcat	cgtgcctctc	tcaacgctgt	ccaactgggc	gtacgagttt	gacaagtggg
2761	ccccctccgt	gtgtgaagtg	tcttacaagg	gatccccagc	agcaagacgg	gccccttgcc
2821	cccagctccg	gagtggaag	ttcaacgtct	tgctgacgac	gtacgagtac	atcatcaaa
2881	acaagcacat	ctccgccaa	atccgttgga	agtacatgat	tgtggacgaa	ggtcaccgca
2941	tgaagaacca	ccactccaag	ctgacgcagg	tgctcaaac	gcactatgtg	gcaccccgcc
3001	gcctgctgct	gacgggcaca	ccgctgcaga	acaagcttcc	cgagctctgg	gcgctgctca
3061	acttccctgct	gcccaccatc	ttcaagagct	gcagcacctt	cgagcagttg	tttaaccgac
3121	cctttgccc	gacccgggaa	aaggtggacc	tgaatgagga	ggaaaccatt	ctcatcatcc
3181	ggcgtctcca	caaagtgcgt	cggcccttct	tgctccgacg	actcaagaag	gaagtgcagg
3241	cccagttgccc	cgaaaagggt	gagtagctca	tcaagtgcga	catgtcccg	ctgcagcgag
3301	tgctctacgc	ccacatgcag	gccaaggcg	tgctgctgac	tgatggctcc	gagaaggaca
3361	agaagggcaa	aggcggcacc	aagaccctga	tgaacacccat	catgcagctg	cggaagatct
3421	gcaaccaccc	ctacatgttc	cagcacatcg	aggagtccct	ttccgagcac	tgggggttca
3481	ctggcggcat	tgtccaagg	ctggacctgt	accgagcctc	gggtaaattt	gagcttcttg
3541	atagaattct	tcccaaaact	cgagcaacca	accacaaagt	gctgctgttc	tgccaaatga
3601	cctccctcat	gaccatcatg	gaagattact	ttgcgtatcg	cggctttaa	tacctcaggc
3661	ttgatggaac	gacggaagg	gaggaccggg	gcattgctgt	gaaacacctc	aacgagcccg
3721	gctctgagta	cttcatcttc	ctgctcagca	cccgggctgg	ggggctcggc	ctgaacctcc
3781	agtccgcaga	caactgtgat	atttttgaca	gcgactggaa	tcctcaccag	gacctgcaag
3841	cgagggaccg	agcccaaccg	atcgggcagc	agaacagagt	gcgtgcgctc	cgctctgca
3901	ccgtcaacag	cgtggaggag	aagatcctag	ctgcagccaa	gtacaagctc	aacgtggacc
3961	agaagggtgat	ccaggccggc	atgttcgacc	agaagtctc	cagccatgag	cggcgcgcct
4021	tcctgcagcg	catcctggag	cacgaggagc	aggatgagag	cagacactgc	agcacgggca
4081	gcggcagtg	cagcttcgcc	cacactgccc	ctccgccagc	gggcgtcaac	cccgacttgg
4141	aggagccacc	tctaaaggag	gaagacgagg	tgcccgcagc	cgagaccgtc	aaccagatga
4201	tcgcccgcca	cgagggagg	tttgatctgt	tcattgcgat	ggacctggac	cgcaggcgcg
4261	aggaggcccg	caaccccaag	cggaaagccg	gcctcatgga	ggaggacgag	ctccctctcg
4321	ggatcatcaa	ggacgacg	gaggtggagc	ggctgacctg	tgaggaggag	gaggagaaga
4381	tgctcgccg	tgctcccg	caccgcaagg	agggtggacta	cagcgactca	ctgacggaga
4441	agcagtggtg	caaggccatc	gaggagggca	cgctggaggga	gatcgaagag	gaggtccggc
4501	agaagaatc	atcacggaag	cgcaagcgag	acagcgacgc	cggctcctcc	accccgacca
4561	ccagcacccg	cagccgcgac	aaggacgacg	agagcaagaa	gcagaagaag	cgcgggcggc
4621	cgctgcgcga	gaaactctcc	cctaaccac	ccaacctcac	caagaagacg	aagaagattg
4681	tggatgccgt	gatcaagtac	aaggacagca	gcagtggacg	tcagctcagc	gaggtcttca
4741	tcagctgcc	ctccgcaaa	gagctgccc	agtactacga	gctcatccg	aagcccgtgg
4801	acttcaagaa	gataaaggag	cgcattcgca	accacaagta	ccgcagcctc	aacgacctag
4861	agaaggacgt	catgctcctg	tgccagaacg	cacagacctt	caacctggag	ggctccctga
4921	tctatgaaga	ctcctcgctc	ttgcagtcgg	tcttcaccag	cgctgcggcag	aaaatcgaga
4981	aggagatga	cagtgaaggc	gaggagagtg	aggaggagga	agagggcgag	gaggaaagct
5041	ccgaatccga	atctcgggtc	gtcaaaagtga	agatcaagct	tggccggaa	gagaaggcac
5101	aggaccggct	gaaggcgccg	cggcgccggc	cgagccgagg	gtcccagacc	aagccggctc
5161	tgagtgcaga	tgacagtgag	gaggaacaag	aggaggaccg	ctcaggaaat	ggcagcgaa
5221	aagactgagc	cccagacatt	cagctctgac	cccagacccc	tcgttcacga	gctgagatgg
5281	cataggcctt	agcagtaacg	ggtagcagca	gatgtagttt	cagacttga	gtaaaactgt
5341	ataaacaata	gaatcttcca	tattatata	gcagagaagc	tgtaggactg	tttgtgactg
5401	gcccctgtct	ggcatcagta	gcattctgta	cagcattaac	tgtctttaa	agagagagag
5461	agaattccga	attgggggaac	acacgatacc	tggttttctt	ttccgttgct	ggcagtagtg
5521	ttgcgcgcga	gtttggagtc	actgtagtta	agtgtggatg	catgtgcgtc	accgtccact
5581	cctcctactg	tactttattg	gacaggtcag	actcgcggcg	ggcccggcga	gggtatgtca
5641	gtgtcactgg	atgtcaaaac	gtaataaaact	aaaccaacaa	caaaacgcac	agccaaaaaa
5701	aaa					

SEQ ID NO: 58 Human BRG1 cDNA Sequence Variant 4 (NM_001128845.1, CDS:
from 1 to 4854)

1	atgtccactc	cagaccacc	cctggggcgga	actcctcggc	caggtccttc	cccggggccct
61	ggcccttccc	ctggagccat	gctggggccct	agcccgggtc	cctcgccggg	ctccgcccac
121	agcatgatgg	ggcccagccc	agggccgccc	tcagcaggac	accccatccc	caccagggg
181	cctggagggt	accctcagga	caacatgcac	cagatgcaca	agcccattga	gtccatgcac
241	gagaagggca	tgccggaacg	cccgcgctac	aaccagatga	aaggaatggg	gatgcgggtca
301	gggggcatg	ctgggatggg	gcccccgccc	agcccatgg	accagcactc	ccaagggtac
361	ccctcgcccc	tgggtggctc	tgagcatgcc	tctagtccag	tcccagccag	tggcccgctc
421	tcggggcccc	agatgtcttc	cgggccaggga	ggtgccccgc	tggatgggtc	tgacccccag
481	gccttggggc	agcagaaccg	gggcccaccc	ccatttaacc	agaaccagct	gcaccagctc

TABLE 1-continued

541	agagctcaga	tcatggccta	caagatgctg	gccagggggc	agcccccctc	cgaccaacctg
601	cagatggcgg	tcaggggcaa	gcggccgatg	cccgggatgc	agcagcagat	gccaacgcta
661	cctccaccct	cgggtgtccg	aacaggaccc	ggccctggcc	ctggccctgg	ccccggcccg
721	ggccccggcc	cgccacctcc	aaattacagc	aggcctcatg	gtatgggagg	gccccacatg
781	cctccccagc	gacctctggg	cgtgcccccc	gggatgccag	gccagcctcc	tggagggcct
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901	aagctgattc	ccccgcagcc	aacgggcgcg	cctccccccg	cgccccctgc	cgtcccaccc
961	gccgcctcgc	ccgtgatgcc	accgcagacc	cagtcccccg	ggcagccggc	ccagcccgcg
1021	cccatgggtg	cactgcacca	gaagcagagc	cgcatacccc	ccatccagaa	gccgcggggc
1081	ctcgaccctg	tggagatcct	gcaggagcgc	gagtagacgg	tgaggctcgc	catcgcacac
1141	cgaattcagg	aacttgaaaa	ccttccccgg	tccttgcccg	gggatttgcg	aaccaaagcg
1201	accattgagc	tcaggccctc	caggctgctg	aacttccaga	ggcagctcgc	ccaggagggtg
1261	gtgggtgtgc	tgccggaggga	cacagcgctg	gagacagccc	tcattgctaa	ggcctacaag
1321	cgcagcaagc	gccagtcctc	gcgcgaggcc	cgcatactcg	agaagctgga	gaagcagcag
1381	aagatcgagc	aggagcgcaa	gcgcggcgag	aagcaccagg	aatacctcaa	tagcatcttc
1441	cagcatgccca	aggatttcaa	ggaatatcac	agatccgtca	caggcaaaa	ccagaagctg
1501	accaaggcag	tggccacgta	ccatgccaac	acggagcggg	agcagaagaa	agagaacgag
1561	cggatcgaga	aggagcgcat	gcggaggctc	atggctgaag	atgaggaggg	gtaccgcaag
1621	ctcatcgacc	agaagaagga	caagcgccctg	gcctacctct	tgacgcagac	agacgagtac
1681	gtggctaacc	tcacggagct	ggtgcggcag	cacaaggctg	cccaggtcgc	caaggagaaa
1741	aagaagaaaa	agaaaaagaa	gaaggcgaaa	aatgcagaag	gacagacgcc	tgcatttggg
1801	ccgtagggcg	agcctctgga	cgagaccagc	cagatgagcg	accccccggt	gaagggtgatc
1861	cacgtggaga	gtgggaagat	cctcacaggc	acagatgccc	ccaaagcccg	gcagctggag
1921	gcctggctcg	agatgaaccc	ggggatgtaa	gtagctccga	ggctctgatag	tgaagaaagt
1981	ggctcagaag	aagaggagaa	ggaggaggag	gaagagcagc	cgcaggcagc	acagcctccc
2041	accctgcccc	tggaggagaa	gaagaagatt	ccagatccag	acagcgatga	cgtctctgag
2101	tgggacgcgc	ggcacatcat	tgagaatgcc	aagcaagatg	tcgatgatga	atatggcggtg
2161	tccaggcccc	ttgcacgtgg	cctgcagtcg	tactatgccg	tgcccatatg	tgtcactgag
2221	agagtggaca	agcagtcagc	gcttatggtc	aatgggtgtc	tcaaacagta	ccagatcaaa
2281	ggtttggagt	ggctggtgtc	cctgtacaac	aacaacctga	acggcatcct	ggccgacgag
2341	atgggctggg	ggaagaccat	ccagaccatc	gcgctcatca	cgtaacctcat	ggagacaaa
2401	cgcataaatg	ggcccttccct	catcatcgtg	cctctctcaa	cgtgtccaa	ctgggcgtac
2461	gagtttgaca	agtgggcccc	ctccgtggtg	aaggtgtctt	acaagggatc	cccagcagca
2521	agacggggcct	ttgtccccc	gctccggagt	gggaagtcca	acgtcttgct	gacgactac
2581	gagtacatca	tcaaagacaa	gcacatcctc	gccaaagatc	ggtggaagta	catgattgtg
2641	gacgaaggtc	accgcgatga	gaaccaccac	tgcaagctga	cgcagggtgt	caacacgcac
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2761	ctctggggcg	tgctcaactt	cctgctgccc	accatcttca	agagctgcag	caccttcgag
2821	cagtgggtta	acgcaccctt	tgccatgacc	ggggaaaaag	tggacctgaa	tgaggaggaa
2881	accattctca	tcataccggc	gttgcggcgc	gtgctgcggc	ccttcttgct	ccgacgactc
2941	aagaagggaag	tcgaggccca	gttgccccga	aaggtggagt	acgtcatcaa	gtgcgacatg
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3061	ggctccgaga	aggacaagaa	gggcaaaagg	ggcaccacga	ccctgatgaa	caccatcatg
3121	cagctgcgga	agatctgcaa	ccaccctac	atgttccagc	acatcgagga	gtccttttcc
3181	gagcacttgg	ggttcactgg	cggcattgtc	caagggtcgg	acctgtaccg	agcctcgggt
3241	aaatttgagc	ttctctgatg	aattcttccc	aaactccgag	caaccaccca	caaatgtctg
3301	ctgtttctgcc	aaatgacctc	cctcatgacc	atcatggaag	attactttgc	gtatcgcggc
3361	tttaaatacc	tcaggcttga	tggaaaccag	aaggccggag	acccggggcat	gctgctgaaa
3421	accttcaagc	agcccggtct	tgagtacttc	atcttctctg	tcagcacccg	ggctgggggg
3481	ctcggctcga	acctccagtc	ggcagacact	gtgatcattt	ttgacagcga	ctggaatcct
3541	caccaggacc	tccaagcgca	ggaccgagcc	caccgcacgc	ggcagcagaa	cgagggtcgt
3601	gtgctccgct	gtgcacccgt	caacagcgctg	gaggagaaga	tcctagctgc	agcccaagtac
3661	aagctcaacg	tggaccagaa	ggtgatccag	gccggcatgt	tcgaccagaa	gtcctccagc
3721	catgagcggc	gjccttctct	gcaggccatc	ctggagcacg	aggagcagga	tgaggaggaa
3781	gacgaggtgc	ccgacagcga	gaccgtcaac	cagatgatcg	cccggcacga	ggaggagtct
3841	gatctgttca	tgccgatgga	cctggaccgc	aggcgcgagg	aggcccgcga	ccccaaagcg
3901	aagccgcgcc	tcattggagga	ggacgagctc	ccctcgtgga	tcataagga	cgacgcggag
3961	gtggagcggc	tgacctgtga	ggaggaggag	gagaagatgt	tcggccggtg	ctcccgcac
4021	cgcaaggagg	tggactacag	cgactcactg	acggagaagc	agtggtctaa	gacctgaag
4081	gccatcgagg	agggcacgct	ggaggagatc	gaagaggagg	tcggcgagaa	gaaatcatca
4141	cggaagcgca	agcgagacag	cgacgcgggc	tcctccaccc	cgaccaccag	caccgcgagc
4201	cgcgacaagg	acgacgagag	caagaagcag	aagaagcgcg	ggcgcccgcc	tggcgagaaa
4261	ctctccccta	acccacccaa	cctcaccaag	aagatgaaga	agattgtgga	tggcgtgatc
4321	aagtacaagg	acagcgcagc	tggacgtcag	ctcagcgagg	tccttcatca	gctgcctcgc
4381	cgaaaggagc	tgcccagata	ctacgagctc	atccgcaagc	ccgtggactt	caagaagata
4441	aaggagcgca	ttcgcaacca	caagtaccgc	agcctcaacg	acctagagaa	ggacgtcatg
4501	ctcctgtgcc	agacgcacac	gaccttcaac	ctggagggtc	ccctgatcta	tgaagactcc
4561	atcgtcttgc	agtcggctct	caccagcgtg	cggcagaaaa	tcgagaagga	ggatgacagt
4621	gaaggcgagg	agagtggaga	ggaggaagag	ggcgaggagg	aaggctccga	atccgaatct
4681	cggtccgctca	caagcttggc	cggaaggaga	aggcacagga	ccggctgaag	ccggctgaag
4741	ggcggccggc	ggcggccgag	ccgagggtcc	cgagccaaagc	cggctcgtgag	tgacgatgac
4801	agtgaggagg	aacaagagga	ggaccgctca	ggaagtggca	gcgaagaaga	ctgagccccg
4861	acattccagt	cccgaccctcg	agccccctgt	tcagagctcg	agatggcata	ggccttagca
4921	gtaacgggta	gcagcagatg	tagtttcaga	cttgaggata	aactgtataa	acaaaagaat
4981	cttccatatt	tatacagcag	agaagctgta	ggactgtttg	tgactggccc	tgtcctggca
5041	tcagttagct	ctgtaacagc	attaaactgtc	ttaaagagag	agagagagaa	ttccgaattg
5101	gggaacacac	gatacctgtt	tttcttttcc	gttctgtgga	gtactgttgc	gccgcagttt
5161	ggagtcactg	tagttaagtg	tggatgcatg	tgcgtcaccg	tcactccttc	ctactgtatt
5221	ttatttgaca	ggtcagactc	gcggggggcc	cgccgagggt	acgtcagttg	cactggatgt
5281	caaacagtaa	taaatataac	caacaacaaa	acgcacagcc	aaaaaaaa	

TABLE 1-continued

SEQ ID NO: 59 Human BRG1 Amino Acid Sequence Isoform C (NP_001122317.1)

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1  mstpdpplpg tprpgpspgp gspsgamlgp spgspgsah smmgpspgpp saghpiptqg
61  pggypqdnmh qmhkpmesmh ekqmsddprry nqmkgmgmrs gghagmgppp spmdqhsqgy
121  psplggseha sspvpasggs sgpmssggp gapldgadpq algqqnrgpt pfnqnqlhql
181  raqimaykml argqplpdhl qmavqgkrrp pgmqqqmptl ppsvsvsatp gpgpgpgpgp
241  gpgpappnys rphgmggpnm ppgpsgvpv gmpgqppggp pkpwpegpma naaaptstpq
301  klippqgtgr pspappavpp aaspvmppqt qspgqpqpa pmvplhqqs ritpiqkprg
361  ldpveilqer eyrlqariah riqelenlpg slagdlrcka tielkalrll nfrqqlrgev
421  vvcmmrdtal etalnakyk rskrqslrea riteklekqg kieqerkrxg khqeylnsil
481  qhakdfkeyh rsvtgiqkl tkavatyhan tereqkkene riekermrrl maedeegyryk
541  lidqkdkkrl aylqgtdey vanltelvrq hkaaqvakek kkkkkkkkae naegqtpaig
601  pdgepldets qmsdlpvkvi hvesgkiltg tdapkaqgle awlemppgye vaprsdsees
661  gseeeeeeee eeqpqaagpp tlpveekkki pdpdsddvse vdarhiiena kqdvddeygv
721  sqalarglqs yyavahavte rvdqksalmv ngvlkqyqik glewlvslyn nlnlilade
781  mglgkttiqli alitylmehk ringpfliiv plstlsnway efdkwapsvv kvsykgspaa
841  rrafvpqlrs gkfnvlltty eyiikdkhil akirwkymiv deghrmknhh ckltpvlnth
901  yvaprrlllt gtlplqnklpe lwallnflfp tlfkscstfe qwnapfamt gekvdlneee
961  tiliirrlhk vlrflllrrl kkeveaqle kveyvikcdm salqrlylrh mqakgvlltd
1021  gsekdkkkgk gttklmmtim qlrkicnhy mfhieesfs ehlgftggiv qgldlyrasg
1081  kfeldrirlp klratnhkvl lfcqmtsamt imedyfayrg fkyrlrdgtt kaedrgmllk
1141  tfnepgseyf ifillstragg lglnlqsadt viifdsdwnp hqdlqaqdra hrigqgnevr
1201  vlrlctvnsv eekilaaaky klnvdqkviq agmfdqkss herafllgai leheeqdeee
1261  devpddetvn qmiarheeeef dlfrmdldr rreearnpkr kprlmeedel pswiikddae
1321  verltceeee ekmfgrgsrh rkevdsdsl tekqwlktlk aieegtleei eeervqkss
1381  rkrkrdsdag sstpttstrs rdkddeskkk kkrgrppaek lspnppnltk kmkkivdavi
1441  kykdsessgrq lsevfliqlps rkelpyyel irkpvdffki kerirnhkyr slndlekdv
1501  llcnaqgtfn legsllyeds ivlqsvftsv rkiekedds egeeseeeee geeegseses
1561  rsvkvkiklg rkekaqdrllk ggrrrpsrgs rakpvvsddd seeeqeedrs gsgseed

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SEQ ID NO: 60 Human BRG1 cDNA Sequence Variant 5 (NM_001128846.1, CDS: From 1 to 4851)

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1  atgtccactc cagaccacc cctggggcga actcctcggc caggtccttc cccgggacct
61  ggcccttccc cctggggcct agcccgggtc cctgcgcggg ctccgcccac
121  agcatgatgg ggcacagccc agggccgccc tcagcaggac accccatccc caccagggg
181  cctggagggg acctcagga caacatgcac cagatgcaca agcccatgga gtccatgcat
241  gagaaggcca tgcgcgaca cccgcgctac aaccagatga aaggaaatgg gatgcggcca
301  gggggcccat ctgggatggg gcccccgccc agcccatgga accagcactc ccaaggttac
361  cccctgcccc tgggtggctc tgagcatgcc tctagtcagg ttcagccagc tggcccgctc
421  tcggggcccc agatgtcttc cgggccaggga ggtgccccgc tggatggtgc tgacccccc
481  gccttggggc agcagaaccg gggcccaacc ccatttaacc agaaccagct gcaccagctc
541  agagctcaga tcattggccta caagatgtcg gccagggggc agccctctcc cgaccacctg
601  cagatggcgg tgcgcgcaa cggcgcgatg cccgggatgc agcagcagat gccaacgcta
661  cctccaccct cgggtgtccg aacaggaccc ggccctggcc ctggccctgg ccccgcccg
721  ggtcccgccc cggcacctcc aaattacagc aggcctcatg gtatgggagg gcccaacatg
781  cctcccccag tgcgcgccc cgtgcccccc gggatgccag gccagcctcc tggagggccc
841  cccaagccct ggctgaagg acccatggcg aatgtctgtg ccccccagag caccctcag
901  aagctgattc ccccgagccc aacggggccc ccttcccccg cgcctcccgc cgtccccccc
961  gccgcctccc cctgtgatgc accgcagacc cagtcctccc ggagcccgcc ccagcccgcc
1021  cctgatgtgc cactgcacca gaagcagagc cgcacacccc ccatccagaa gccgcggggc
1081  ctgcacccct tggagatcct gcaggagcgc gagtacaggg tgcaggctcg catgcacac
1141  cgaattcagg aatttcgaaa ccttccccgg tccctggccc gggatgtgag aaccaaagcg
1201  accattgagc tcaaggccct caggctgctg aacttcagaa ggagctgag ccaggaggtg
1261  gtggtgtgca tgcggaggga cacagcgtcg gagacagccc tcaatgctaa ggcctacaag
1321  cgcagcagag gccagtcctc cgcgcaggcc cgcacacccc agaagctgga gaagcagcag
1381  aagatcgagc agagagcga ggcggggcag aagcaccagg aatacctcaa tagcattctc
1441  cagcatgcca aggatattca ggaatatcac agatccgtca caggcaaaat ccagaagctg
1501  accaagcagc tggccacgta ccatgccaac acggagcggg agcagaagaa agagaagcag
1561  cggatcgaga agagagcgc ggcggagctc atggctgaag atgaggaggg gtaccgcaag
1621  ctactgacc agaagaagga caagcggccc gcctacctct tgcagcagac agacaggtac
1681  gtggctaac ctcgcgagct ggtgcggcag cacaaggctg cccaggtcgc caaggagaaa
1741  aagaagaaaa agaaaaagaa gaagcagaaa aatgcagaag gacagacgcc tgccattggg
1801  ccggatgggc agcctctgga cgagaccagc cagatgagcg acctcccggt gaaggtgatc
1861  cagctggaga cgtgggaagt cctcacaggc acagatgccc ccaaagccgg gcagctggag
1921  gcctggctcg agatgaaccc ggggatgaa gtactccga ggtctgatag tgaagaaagt
1981  ggctcagaag aagaggaaga ggaggaggag gaagagcagc cgcaggcagc acagcctccc
2041  accctgccc tggaggagaa gaagaagatt ccagatccag acagcagatg cgtctctgag
2101  gtggacgcgc gccacatcat tgagaatgcc aagcaagatg tcgatgatga atatggcgtg
2161  tcccaggccc ttgcacgtcg cctgcagtc tactacgcgc tggcccatgc tgcactgag
2221  agagtggaca agcagtcagc gcttatggtc aatgggtgct tcaaacagta ccagatcaaa
2281  ggtttggagt ggctggtgct cctgtacaac aacaacctga acggcatcct ggccgacgag
2341  atgggctctg ggaagacctc ccagaccatc gcgctcatca cgtacctcat ggagcacaaa
2401  cgcacatcat ggcctctcct catcatcggt cctctctcaa cgtgtccaa ctggcgctac
2461  gagtttgaca agtgggcccc ctccgtggtg aaggtgtctt acaagggatc cccagcagca
2521  agacgggccc tcgtccccc gctccggagt ggggaagttca acgtcttgct gacgacgtac
2581  gactacatca tcaaaagaaa gcacatcctc gccaaagatc gttggaagta catgatgtg
2641  gacgaaggtc accgcatgaa gaaccaccac tgcaagctga cgcaggtgct caaacgcac
2701  tatgtggcac cccgcgcct gctgtgacg ggcacaccgc tgcagaacaa gcttcccag
2761  ctctggcgcg tgcctaacct cctgctgccc accatcttca agagctgcag cactctgag
2821  cagtggttta acgcaccctt tgccatgacc ggggaaaagg tggacctgaa tgaggaggaa

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TABLE 1-continued

2881	accatttctca	tcateccggcg	tctccacaaa	gtgctgcggc	ccttcttgct	cgcagcactc
2941	aagaaggaag	tcgaggccca	gtgccccga	aaggtggagt	acgtcaccaa	gtgcgacatg
3001	tctgcgtgc	agcgagtgc	ctaccgccac	atgcaggcca	agggcgtgct	gctgactgat
3061	ggctccgaga	aggacaagaa	gggcaaggc	ggcaccaga	ccctgatgaa	caccatcatg
3121	cagctgcgga	agatctgcaa	ccaccctac	atgttccagc	acatcgagga	gtccttttcc
3181	gagcacttg	ggttcactgg	cggcattgtc	caagggctgg	acctgtaccg	agcctcgggt
3241	aaatttgagc	ttcttgatag	aattcttccc	aaactccgag	caaccaacca	caaagtgcgt
3301	ccgttctgcc	aaatgacctc	cctcatgacc	atcatggaag	attactttgc	gtatcgcggc
3361	tttaaatacc	tcaggcttga	tggaaaccag	aaggcggagg	accggggcat	gctgctgaaa
3421	accttcaacg	agcccggtc	tgagcacttc	atctccctgc	tcagcaccgc	ggctgggggg
3481	ctcggcctga	acctccagtc	ggcagacact	gtgatcattt	ttgacagcga	ctggaatcct
3541	caccaggacc	tgcaggcgca	ggaccgagcc	caccgcatcg	ggcagcagaa	cgaggtgcgt
3601	gtgctccgct	tctgcaccgt	caacagcggt	gaggagaaga	tcctagctgc	agccaagtac
3661	aagctcaacg	tggaccagaa	ggtgatccag	gccggcatgt	tcgaccagaa	gtcctccagc
3721	catgagcggc	ggccttccct	gcaggccatc	ctggagcacg	aggagcagga	tgaggaggaa
3781	gacgaggtgc	ccgacgacga	gaccgtcaac	cagatgatcg	cccggcacga	ggaggagttt
3841	gatctgttca	tgcgcatgga	cctggaccgc	aggcgcgagg	aggcccgcga	ccccaaagcg
3901	aagccgcgcc	tcattggagga	ggacgagctc	ccctcgtgga	tcataagga	cgacgcggag
3961	gtggagcgcc	tgacctgtga	ggaggaggag	gagaagatgt	tcggccgtgg	ctcccgcac
4021	cgcaaggagg	tggactacag	cgactcactg	acggagaagc	agtggcccaa	gacctgaag
4081	gccatcgagg	agggcacgct	ggaggagatc	gaagaggagg	tcggcgagaa	gaatatcatc
4141	cggaaagcga	agcgagacag	cgacgcgggc	tcctccaccc	cgaccaccag	caccgcgacg
4201	cgcgacaagg	acgacgagag	caagaagcag	aagaagcgcg	ggcggccgcc	tgccgagaaa
4261	ctctccccta	acccacccaa	cctcaccaag	aagatgaaga	agattgtgga	tgccgtgatc
4321	aagtacaagg	acagcagtg	acgtcagctc	agcgaggtct	tcactccagc	gcccctcgca
4381	aaggagctgc	ccgagtaact	cgagctcatc	cgcaagcccg	tggacttcaa	gaagataaag
4441	gagcgattc	gcaaccacaa	gtaccgcagc	ctcaacgacc	tagagaagga	cgatcatgct
4501	ctgtgccaga	acgcacagac	cttcaacctg	gagggtccc	tgatctatga	agactccatc
4561	gtcttgcagt	cggctctcac	cagcgtgcgg	cagaaaatcg	agaaggagga	tgacagtga
4621	ggcgaggaga	gtgaggagga	ggaagagggc	gaggaggaa	gctccgaatc	cgaatctcgg
4681	tcctgcaaa	tgaagatcaa	gcttggccgg	aaggagaagg	cacaggaccg	gctgaagggc
4741	ggcggcgccg	ggccgagccg	agggtcccga	gccaagccgg	tcgtgagtga	cgatgacagt
4801	gaggaggaac	aagaggagga	ccgctcagga	agtggcagcg	aagaagactg	agccccgaca
4861	ttccagctc	gaccccgagc	ccctcggtcc	agagctgaga	tgccataggg	cttagcagta
4921	acgggtagca	gcagatgtag	tttcagactt	ggagtaaaac	tgtataaaca	aaagaatctt
4981	ccatatttat	acagcagaga	agctgttaga	ctgttttgta	ctggccctgt	cctggcatca
5041	gcagcatctg	taacagcatt	aactgtctta	aagagagaga	gagagaattc	cgaattgggg
5101	aacacacgat	acctgttttt	ctttccggtt	gctggcagta	ctgttgccgc	gcagtttgga
5161	gtcactgtag	ttaagtgtgg	atgcattgtc	gtcaccgtcc	actcctccta	ctgtatttta
5221	ttggacaggt	cgactcggcc	gggggcccgg	cgagggtatg	tcagttgtac	tggatgtcaa
5281	acagtaataa	attaacacaa	caacaaaacg	cacagccaaa	aaaaaa	

SEQ ID NO: 61 Human BRG1 Amino Acid Sequence Isoform D (NP_001122318.1)

1	mstpdplpgg	tprrpgspgp	gspgamlgp	spgspgsh	smmgpspgpp	saghiptqg
61	pggyqdnm	qmhmksmhm	ekgmsddpry	nqmkgmms	gghagmgppp	spmdqhsqgy
121	psplggseha	sspvpasggs	sgpqmssgpg	gapldgadpq	algqqrngpt	pfnqqlhlql
181	raqimaykml	argqplpdhl	qmvqgkkrpm	pgmqgmptl	pppsvsatgp	gpgpgpgpgp
241	gpgpappnys	rphgmggpnm	pppgpsgvp	gmpgqppgpp	pkpwpepgma	naaaptstpq
301	klippqgtgr	pspappavpp	aasvmpqgt	qspgqpaqpa	pmvplhqkqs	ritpiqkprg
361	ldpveilqer	eyrlqariah	riqelenlpg	slagdlrtka	tielkalrl	nfrqqlrgev
421	vvcrrrdtal	etalnakayk	rskrqslrea	riteklekq	kiegerkrrq	khqeylnsil
481	qhakdfkeyh	rsvtgkiqkl	tkavatyhan	tereqqkene	riekermrrl	maedeeygry
541	lidqkkdkrl	ayllqgtdey	vanltelvrq	hkaaavakek	kkkkkkkkae	naeqtpaig
601	pdgepldets	qmssldpvkvi	hvesgkiltg	tdapkaggle	awlemnpgye	vaprsdsees
661	gseeeeeeee	eeqpqaagpp	tlpveekkki	pdpsddvse	vdarhiiena	kqdvddeygv
721	sqalarglqs	yyavahavte	rvdkqsalmv	ngvlkqvqik	glewlvslyn	nnlmgilade
781	mglgktiqt	alitylmehk	ringpfliiv	plstlsnway	efdkwapsvv	kvsykgspaa
841	rrafvpglrs	gkfntvlrtty	eyiikdkhil	akirwkymiv	deghrmknhh	ckltqvlnc
901	yvaprrlllt	gtpnlqnkpe	lwallnflp	tifkscstfe	qwnapfam	gekvdlnnee
961	tiliirrlhk	vlrpflrrl	kkeveaqlpe	kveyvikcdm	salqrlyrh	mqakgvlltd
1021	gsekdkkkg	gktklmmtim	qlrkicnhpy	mfqhieesfc	ehlgftggiv	ggldlyrasg
1081	kfelldrllp	klratnhkvl	ltcqmstlmt	imedytayrg	fkylrldgtt	kaedrgmllk
1141	tfnepgseyf	ifillstragg	lglnlqsadt	viifdsdwnp	hqdlqagdra	hrigqqnevr
1201	vlrlctvnsv	eeekilaaky	klndvdkvii	agmfdqkss	herraflgai	leheeqdeee
1261	devpddetvn	qmlarheee	dlfmrmdldr	rreearnpkr	kprlmeedel	pswiikddae
1321	verltceeee	ekmfgrgrsrh	rkevdydsdl	tekqwlktlk	aieegtleei	eeevrqkkss
1381	rkrkrdsdag	sspttststrs	rdkddeskkq	kkrgprpaek	lspnppnltk	kmkkivdavi
1441	kykdsqsrql	sevfqlpsr	kelpyeyeli	rkpvdffkik	erirnhkyrs	lndlekdvml
1501	lcqnaqtnl	egsliyeds	vlqsvftsvr	qkiekeddse	geeseeeeeg	eeegsesesr
1561	svkviklgr	kekaqdrllg	grrrpsrgsr	akpvvsddds	eeeqeedrsg	sgseed

SEQ ID NO: 62 Human BRG1 cDNA Sequence Variant 6 (NM_001128847.1, CDS: from 1 to 4845)

1	atgtccactc	cagaccaccc	cctggggcga	actcctcggc	caggtccttc	cccgggccct
61	ggcccttccc	ctggagccat	gctgggccc	agcccggggc	cctcgccggg	ctccgcccac
121	agcatgatgc	ggccccagccc	agggccgccc	tcagcaggag	accccatccc	caccagggg
181	cctggagggt	acctcagga	caacatgcac	cagatgcaca	agcccatgga	gtccatgcac
241	gagaaggcca	tgtcggacga	cccgcgctac	aaccagatga	aagggaatgg	gatgcgggtca
301	gggggcatg	ctgggaggg	gccccgccc	agcccatgg	accagcactc	ccaaggttac
361	ccctcgcccc	tgggtggctc	tgagcatgcc	tctagtcacg	tccagaccag	tggcccgctc

TABLE 1-continued

421	tccggggcccc	agatgtcttc	cgggccagga	ggtgccccgc	tggatggtgc	tgacccccag
481	gccttggggc	agcagaaccg	gggcccacc	ccatttaacc	agaaccagct	gcaccagctc
541	agagctcaga	tcattggccta	caagatgtg	gccagggggc	agccccctcc	cgaccaccctg
601	cagatggcgg	tgacgggcaa	gcggccgatg	cccgggatgc	agcagcagat	gccaacgcta
661	cctccacctc	cgggtgtccg	aacaggacc	ggccctggcc	ctggccctgg	ccccggcccc
721	ggtcccgccc	cggcacctcc	aaattacagc	aggcctcatg	gtatgggagg	gccccacatg
781	cctccccag	gacctccggg	cgtgcccccc	gggatgccag	gccagcctcc	tggagggcct
841	cccaagccct	ggcctgaagg	acccatggcg	aatgctgctg	ccccacagag	cacccctcag
901	aagctgatcc	ccccgcagcc	aacggggccg	ccttcccccg	cgccccctgc	cgteccaccc
961	gcccgcctcg	ccgtgatgcc	accgcagacc	cagtcccccg	ggcagccggc	ccagcccgcg
1021	cccatggtgc	cactgcacca	gaagcagagc	cgcatcacc	ccatccagaa	gccgcggggc
1081	ctcgaccctg	tggagatcct	gcaggagcgc	gagtacaggg	tgacggctcg	catcgcacac
1141	cgaattcagg	aacttggaata	ccttccccgg	tccttgcccg	gggatttgcg	aaccaaagcg
1201	accattgagc	tcaaggccct	caggctgctg	aacttccaga	ggcagctgcg	ccaggagggtg
1261	gtggtgtgca	tgccggaggga	cacagcgctg	gagacagccc	tcaatgctaa	ggcctacaag
1321	cgacgcaagc	ggcagtcctc	cgccgaggcc	cgcatcactg	agaagctgga	gaagcagcag
1381	aagatcgagc	aggagcgcaa	gcgccggcag	aagcaccagg	aatacctcaa	tagcattctc
1441	cagatgcca	aggatttcaa	ggaatatcac	agatccgtca	caggcaaaat	ccagaagctg
1501	accaaggcag	tgcccatgta	ccatgccaac	acggagcggg	agcagaagaa	agagaacgag
1561	cggatcgaga	aggagcgcat	gcggaggctc	atggctgaag	atgaggaggg	gtaccgcaag
1621	ctcatcgacc	agaagaagga	caagcgcccg	gcctacctct	tgacgcagac	agacgagtac
1681	gtggctaac	tcacggagct	ggtgcccag	cacaaggctg	cccaggctcg	caaggagaaa
1741	aagaagaaaa	agaaaaagaa	gaaggcagaa	aatgcagaag	gacagacgcc	tgccattggg
1801	ccgatggggc	agcctctgga	cgagaccagc	cagatgagcg	acctccccgt	gaaggatgatc
1861	cacgtgga	gtgggaagat	cctcacaggc	acagatgcc	ccaaagcccg	gcagctggag
1921	gcctggctcg	agatgaaccc	ggggtatgaa	gtagctccga	ggtctgatag	tgaagaaagt
1981	ggctcagaag	aagaggaaga	ggaggaggag	gaagagcagc	cgcaggcagc	acagcctccc
2041	accctgccc	tgaggagaa	gaagaagact	ccagatccag	acagcgatga	cgtctctgag
2101	gtggacgcgc	ggcacatcat	tgagaatgcc	aagcaagatg	tcgatgatga	atatggcgtg
2161	tcccaggccc	ttgcacgtgg	cctgcagtc	tactatgccg	tggcccatgc	tgctactgag
2221	agagtggaca	agcagtcagc	gcttatggtc	aatgggtgct	tcaaacagta	ccagatcaaa
2281	ggtttggagt	ggctggtgtc	cctgtacaac	aacaacctga	acggcatcct	ggccgacgag
2341	atgggctcgg	ggaagaccat	ccagaccatc	gcgctcatca	cgtacctcat	ggagcacaaa
2401	cgcatcaatg	ggcccttctc	catcatcgct	cctctctcaa	cgctgtccaa	ctggcgctac
2461	gagtttgaca	agtgggcccc	ctccgtggtg	aagggtgtct	acaagggatc	cccagcagca
2521	agacgggccc	ttgtccccca	gctccggagt	gggaagttca	acgtcttgct	gacgacgtac
2581	gagtacatca	tcaaaagaaa	gcacatcctc	gccaaagatc	gttggaaagta	catgattgtg
2641	gacgaaggct	accgcgatga	gaaccaccac	tgcaagctga	cgcagggtgc	caacacgcac
2701	tatgtggcac	ccgcgcgcct	gcccgtgacg	ggcacaccgc	tgcaagaaca	gcttccccag
2761	ctctgggcgc	gtctcaactt	cctgctgccc	accatcttca	agagctgcag	caccttcgag
2821	cagtgggtta	acgcaccctt	tgccatgacc	ggggaagagg	tggacctgaa	tgaggaggaa
2881	accattctca	tcacccggcg	tctccacaaa	gtgctggcgc	ccttctctgt	ccgacgactc
2941	aagaaggag	tgccagccca	gttgcccgaa	aagggtggag	acgtcatcaa	gtgcgacatg
3001	tctgcgctgc	agcagtgctc	ctaccgccac	atgcaggcca	agggcggtgc	gctgactgat
3061	ggctccgaga	aggacaagaa	gggcaaaagg	ggcaccaaga	ccctgatgaa	caccatcatg
3121	cagctgcgga	gcactcgcaa	ccaccctcac	atgttccagc	acatcgagga	gtccttttcc
3181	gagcaactgg	ggttcaactg	cggcattgtc	caagggtctg	acctgtaccg	agcctcgggt
3241	aaatttgagc	ttctttgatag	aattcttccc	aaactccgag	caaccaacca	caaagtgtct
3301	ctgttctgcc	aaatgacctc	cctcatgacc	atcatggaa	attactttgc	gtatcgcggc
3361	tttaaatacc	tcaggcttga	tggaaaccag	aaggcggagg	accggggcat	gctgctgaaa
3421	accttcaacg	agcccggtgc	tgagtacttc	atcttctctc	tcagcacccg	ggctgggggg
3481	ctcgccgctg	actaacagtc	ggcagacact	gtgatcattt	ttgacagcga	ctggaaacct
3541	caccaggacc	tgcaagcgca	ggaccgagcc	caccgcacgc	ggcagcagaa	cgagggtgct
3601	gtgctccgcc	tctgcaccgt	caacagcgtg	gaggagaaga	tcctagctgc	agccaaagtac
3661	aagctcaacg	tggaacagaa	ggtgatccag	gccggcatgt	tcgacacagaa	gtcctccagc
3721	catgagcgcc	gcgccttctc	gcaggccatc	ctggagcagc	aggagcagga	tgaggaggaa
3781	gacgaggtgc	ccgacgacga	gaccgtcaac	cagatgatcg	cccgccacga	ggaggagttt
3841	gatctgttca	tgccgatgga	cctggaccgc	aggcgcgagg	aggcccgcaa	ccccaaagcg
3901	aagccgcgcc	tcattggagga	ggacgagctc	ccctcgtgga	tcattcaagga	cgacgcggag
3961	gtggagcgcc	tgacctgtga	ggaggaggag	gagaagatgt	tcggccgtgg	ctcccgccac
4021	cgcaaggagg	tggactacag	cgactcacgc	acggagaagc	agtggctcaa	ggccatcgag
4081	gagggcacgc	tgaggagatg	cgaaggaggag	gtccggcaga	agaaatcatc	accgaaagcg
4141	aagcgagaca	gagacgcggg	ctcctccacc	ccgaccacca	gcaccgcgag	ccgcgacaag
4201	gacgacgaga	gcaagaagca	gaagaagcgc	ggggcgccgc	ctcccgagaa	actctccctc
4261	aaccaccaca	acctcaccaa	gaagatgaag	aagattgtgg	atgccgtgat	caagtacaa
4321	gacagcagca	gtggacgtca	gctcagcgag	gtcttccacc	agctgccctc	gcgaaaggag
4381	ctgcccaggt	actacgagct	catccgcaag	cccggtgact	tcagaagatg	aaaggagcgc
4441	attcgcaacc	acaagtaccg	cagcctcaac	gacctagaga	aggacgtcat	gctcctgtgc
4501	cagaacgcac	agaccttcaa	cctggagggg	tccttgatct	atgaagactc	catcgtcttg
4561	cagtcggctc	tcaccagcgt	gcggcagaaa	atcgagaagg	aggatgacga	tgaaggcgag
4621	gagagtgaag	aggaggaaga	ggcgaggagg	gaaggctccg	aatccgaatc	tcggtccgtc
4681	aaagtgaaga	tcaagcttgg	ccggaaggag	aaggcacagg	accggctgaa	ggggcgcccg
4741	cgccggccga	gccgagggtc	ccgagccaa	ccggtcgtga	gtgacgatga	cagtgaggag
4801	gaacaagagg	aggaccgctc	aggaaagtgg	agcgaagaag	actgagcccc	gacattccag
4861	tctcgacccc	gagccctcgc	ttccagagct	gagatggcat	aggccttagc	agtaaacggg
4921	agcagcagat	tgagtttcag	acttgagta	aaactgtata	aacaaagaa	tcttccat
4981	ttatacagca	gagaagctgt	aggactgttt	gtgactggcc	ctgtcctggc	atcagtagca
5041	tctgtaacag	cattaaactgt	cttaaagaga	gagagagaga	attccgaatt	ggggaacaca
5101	cgatacctgt	ttttcttttc	cgttgctggc	agtactgttg	cgccgagctt	tggagtcact
5161	gtagttaagt	gtggatgcat	gtgcgtcacc	gtccactcct	cctactgtat	ttatttgac

TABLE 1-continued

5221	aggtcagact	cgccgggggc	cgggcgaggg	tatgtcagtg	tcactggatg	tcaaacagta
5281	ataaattaaa	ccaacaacaa	aacgcacagc	caaaaaaaaa		

SEQ ID NO: 63 Human BRG1 Amino Acid Sequence Isoform E (NP_001122319.1)

1	mstpdplpgg	trprgpgpgp	gspgpgamlg	spgpgpgsah	smmgpgpgpp	saghpptqg
61	pggypqdnmh	qmhkpmesm	ekgmddpdy	nqmkgmgrs	gghagmgppp	spmdqhsqgy
121	psplggseha	ssvpasggs	sgpmsggpg	gapldgadp	algqqrngpt	pfnqqlhlql
181	raqimaykml	argqplpdhl	qmvavgkkrp	pgmqqqmptl	pppsvsatgp	gpgpgpgpgp
241	gpgpappnys	rphgmggpnm	pppgpsgvpp	gmpgqpppgp	pkpwpegpma	naaaptstpq
301	klippqptgr	psfpavpvp	aaspvmppqt	qspgqpaqpa	pmvplhqkqs	ritpiqkprg
361	ldpveilqer	eyrlqariah	rigelenlpg	slagdlrtka	tielkalrll	nfrqlrlqev
421	vvmrrrdtal	etalnakayk	rskrqslrea	riteklekq	kieqerkrq	khqeylnsil
481	qhakdfkeyh	rsvtgkikl	tkavatyhan	tereqqkene	riekermrl	maedeegyrr
541	lidqkkdkrl	ayllqqtdey	vanltelvrq	hkaaqvakek	kkkkkkkkkae	naeggtpaig
601	pdgepldets	qmedlpvkvi	hvesgkiltg	tdapkaggle	awlemnpgye	vaprsdsees
661	gseseeeeee	eeqpqaagpp	tlpveekkki	pdpsddvse	vdarhiiena	kqdvddeygv
721	sqalarglqs	yyavahavte	rvdkgsalmv	ngvlkqyqik	glewlvslyn	nnlngilade
781	mglgkltqti	alitylmehk	ringpfliiv	plstlsnway	efdkwapsvv	kvsykgspaa
841	rratvpqlrs	qkfnvlltty	eyiikdkhil	akitwkymiv	deghrmknhh	ckltqvlnth
901	yvaprrlllt	gtplqnklpe	lwallnfllp	tifkscstfe	qwnapfamf	gekvdlneee
961	tiliirrlhk	vlrpfllrrl	kkeveaqlpe	kveyvikodm	salqrvlyrh	mqakgvlltd
1021	gsekdkkqkg	gkttlmtim	qlrkicnhpy	mtqhieesfs	ehlgftggiv	qgldlyrasg
1081	kfelldrilp	klratnhkvl	lfcqmtslmt	imedyfayrg	fkylrldgtt	kaedrgmllk
1141	tfnepgseyf	ifllstragg	lglnlqsadt	viifdsdwnp	hqdlqaqdra	hrigqqnevr
1201	vlrlctvnsv	eehilaaaky	klndvdkvii	agmfdqkss	herraflqai	leheeqdeee
1261	devpddetvn	qmlarheee	dlfmrmdldr	rreearnpkr	kprlmeedel	pswiikddae
1321	verltceeee	ekmfgrgrsrh	rkevdydsdl	tekqwlkaie	egtleieeee	vrqksssrkr
1381	krdsdagssst	pttstrsrdrk	ddeskkqkkr	grppaelksp	nppnltkkkm	kivdaviyky
1441	dsssgqrqlse	vfiqlpsrke	lpeyyelirk	pvdffkkiker	irnhkyrsln	dlekdvmllc
1501	qnaqtfnlqg	sliyedsviv	qsvftsvrqr	iekeddsge	eseeeeegge	egsesesrsv
1561	kvikiklgrke	kaqdrilkggr	rrpsrgsrak	pvsdaddsee	eqeedrsgsg	seed

SEQ ID NO: 64 Human BRG1 cDNA Sequence Variant 7 (NM_001128848.1, CDS: from 1 to 4842)

1	atgtccactc	cagaccacc	cctggggcgga	actcctcggc	caggtccttc	cccggggccct
61	ggccctctcc	ctggagccat	gctggggccct	agcccggggc	cctcgccggg	ctccgcccac
121	agcatgatgg	ggcccagccc	agggccgccc	tcagcaggac	accocatccc	caccagggg
181	cctggagggg	accctcagga	caacatgcac	cagatgcaca	agcccatgga	gtccatgcat
241	gagaagggga	tgctggagca	cccgcgtac	aaccagatga	aaggaatggg	gatcggtca
301	ggggggccatg	ctgggatggg	gcccccgccc	agcccatggg	accagcactc	ccaaggttac
361	ccctcgcccc	tggttggtc	tgagcatgcc	tctagtcagg	ttccagccag	tggcccgctc
421	tcggggcccc	agatgtcttc	cgggccagga	ggtgccccgc	tggatggtgc	tgacccccac
481	gccttggggc	agcagaaccg	gggcccacc	ccatttaacc	agaaccagct	gcaccagctc
541	agagctcaga	tcattggccta	caagatgctg	gccagggggc	agccctctcc	cgaccacctg
601	cagatggcgg	tgccagggca	gcggccgatg	cccgggatgc	agcagcagat	gccaacgcta
661	cctccaccct	cgggtgtccg	aacaggacc	ggccctggcc	ctggccctgg	ccccggccc
721	ggtcccgccc	cgccaccccc	aaattacagc	aggcctcatg	gcattggagg	gcccaacatg
781	cctccccccg	gaccctcggg	cgtgcccccc	gggatgccc	gccagcctcc	tggaggggct
841	cccaagccct	ctcgtgaagg	accatggcg	aattgctgct	ccccccagag	gaccctcag
901	aagctgattc	ccccgcagcc	aacgggccgc	cctccccccc	cgccccctgc	cgctccacc
961	gcgcctcgc	ccgtgatgcc	accgcagacc	cagtccccc	ggcagccggc	ccagcccgcg
1021	cccatggctg	cactgcacca	gaagcagagc	cgcacacccc	ccatccagaa	cgcgggggg
1081	ctcgaccctg	tgagatcct	gcaggagcgc	gagtacaggc	tgagggtcgc	catcgacac
1141	cgaattcagg	aaattgaaaa	ccttccccgg	tccctggccg	gggatttgcc	aaccaaagcg
1201	accattgagc	tcaaggccct	caggctgctg	aaattccaga	ggcagctgcg	ccaggaggtg
1261	gtggtgtgca	tgccgaggga	cacagcgctg	gagacagccc	tcaatgctaa	ggcctacaa
1321	cgcagcaagc	gcccagtcct	gcgcgaggcc	cgcacactg	agaagctgga	gaagcagcag
1381	aagatcgagc	aggagcgcaa	gcgcggcgag	aagcaccagg	aatacctcaa	cagcatctc
1441	cagcatgcca	aggatttcaa	ggaatatcac	agatccgtca	caggcaaaat	ccagaagctg
1501	accaaggcag	tgccaccgca	ccatgccaac	acggagcggg	agcagaagaa	agagaacag
1561	cggatcgaga	aggagcgcat	gcggaggctc	atggctgaag	atgaggagg	gtaccgcaag
1621	ctcatcgacc	agaagaagga	caagcgccg	gcctacctcc	tgacgagac	agacgagtac
1681	gtggctaacc	tcacggagct	ggtgcggcag	cacaaggctg	cccaggctgc	caaggagaaa
1741	aagaagaaaa	agaaaagaaa	gaaggcagaa	aacgcagaa	gacagacgcc	tgccattggg
1801	ccgatggcgc	agcctctgga	cgagaccagc	cagatgagcg	acctcccggt	gaagggtgac
1861	cacgtggaga	gtgggaagat	cctcacaggc	acagatgccc	ccaaagcccg	gcagctggag
1921	gcctggctcg	agatgaaccc	ggggtatgaa	gtagctccga	ggctctgata	tgaagaaagt
1981	ggctcagaag	aagaggga	ggaggaggag	gaagagcagc	cgagggcagc	acagcctccc
2041	acctgcccc	tgaggagaa	gaagaagatt	ccagatccag	acagcgatga	cgtctctgag
2101	gtggacgcgc	ggcacatcat	tgagaatgcc	aagcaagatg	tcgatgatga	atatgctgtg
2161	tcccaggccc	ttgcacgtgg	cctgcagctc	tactatgccg	tgcccatgac	tgctactgag
2221	agagtggaca	agcagtcagc	gcttatggtc	aatgggtgtc	tcaaacagta	ccagatcaaa
2281	gggttgagtc	ggcgtgtgtc	cctgtacaac	aacaacctga	acggcatcct	ggccgacgag
2341	atgggctcgg	ggaagaccat	ccagaccatc	gcgctcatca	cgtacctcat	ggagcaca
2401	cgcacatgat	ggcccttctc	catcatcgtg	cctctctcaa	cgtctgtcaa	ctgggctgac
2461	gagtttgaca	aggtggcccc	ctccgtgggt	aaggtgtctt	acaaggatc	cccagcagca
2521	agacgggcct	ttgtccccca	gctccggagt	gggaagtcca	acgtcttgct	gacgacgtac
2581	gagtagatca	tcaaagacaa	gcacatctc	gccaaagatc	gttggaagta	catgattgtg
2641	gacgaaggta	accgatgaa	gaaccaccac	tgcaagctga	cgcaggtgct	caacacgac
2701	tatgtggcac	cccgcgcct	gctgctgagc	ggcacaccgc	tgcaagaaca	gcttcccgag

TABLE 1-continued

2761	ctctggg	tgctcaactc	cctgctgccc	accatcttca	agagctgcag	caccttcgag
2821	cagtggttta	acgcaccctt	tgccatgacc	ggggaaaagg	tggacctgaa	tgaggaggaa
2881	accattctca	tcacccggcg	tctccacaaa	gtgctgcggc	ccttcttgct	ccgacgactc
2941	aagaagggaag	tcgaggccca	gttgcccga	aaggtggagt	acgtcatcaa	gtgcgacatg
3001	tctgcgtgc	agcgagtgtc	ctaccgccac	atgcaggcca	agggcgtgct	gctgactgat
3061	ggctccgaga	aggacaagaa	gggcaaaagg	ggcaccaga	cctgatgaa	caccatcatg
3121	cagctgcgga	agatctgcaa	ccaccctac	atgttccagc	acatcgagga	gtccttttcc
3181	gagcacttgg	ggttcactgg	cggcattgtc	caagggtctg	acctgtaccg	agcctcgggt
3241	aaattttgagc	ttcttgacag	aattcttccc	aaactccgag	caaccaacca	caaagtgtcg
3301	ctgttctgcc	aatgacacctc	cctcatgacc	atcatggaag	attactttgc	gtatcgcggc
3361	tttaaatacc	tcaggcttga	tggaaccacg	aaggcggagg	accggggcat	gctgctgaaa
3421	accttcaacg	agcccggctc	tgagtacttc	atcttctctg	tcagcacccg	ggctgggggg
3481	ctcggcctga	acctccagtc	ggcagacact	gcatcatttt	tcgacagcga	ctggaatcct
3541	caccaggacc	tgcaagcgca	ggaccgagcc	caccgcatcg	ggcagcagaa	cgaggtgcgt
3601	gtgtccgccc	tctgcaccgt	caacagcgtg	gaggagaaga	tcttagctgc	agccaaagtac
3661	aagctcaacg	tggaaccagaa	ggtgatccag	gccggcatgt	tcgaccagaa	gtccctccag
3721	catgagcggc	gcgccttctc	gcaggccatc	ctggagcacg	aggagcagga	tgaggaggaa
3781	gacgaggtgc	ccgacgacga	gaccttcaac	cagatgatcg	cccgccacga	ggaggagttt
3841	gatctgttca	tgcgcatgga	cctggaccgc	aggcgcgagg	aggcccgcaa	ccccaaagcg
3901	aagccgcgccc	tcatggagga	ggacgagctc	ccctcgtgga	tcatcaagga	cgacgcggag
3961	gtggagcggc	tgacctgtga	ggaggaggag	gagaagatgt	tcggccgtgg	ctcccggcac
4021	cgcaaggagg	tggaactacag	cgactcactg	acggagaagc	agtggctcaa	ggccatcgag
4081	gagggcagcg	tgaggagatg	cgaagaggag	gtccggcaga	agaaatcatc	acggaagcgc
4141	aagcgagaca	ggcagcccg	ctcctccacc	ccgaccacca	gcacccgag	ccgcgacaag
4201	gacgacgaga	gcaagaaagca	gaagaagcgc	ggcgcccgcc	ctcccgagaa	actctccctc
4261	aaccaccacca	acctcaccaa	gaagatgaag	aagattgttg	atgccgtgat	caagtacaag
4321	gacagcagtg	gacgtcagct	cagcgaggtc	ttcatccagc	tgccctcccg	aaaggagctg
4381	cccagtgact	acgagctcat	ccgcaagccc	gtggacttca	agaagataaa	ggagcgcat
4441	cgcaaccaca	agtaccgcag	cctcaacgac	ctagagaagg	acgtcatgct	cctgtgccag
4501	aacgcacaga	ccttcaacct	ggagggtctc	ctgatctatg	aagactccat	cgtcttgcag
4561	tcggtcttca	ccagcgtgcg	gcagaaaatc	gagaaggagg	atgacagtga	aggcgaggag
4621	agtgaggagg	aggaagaggg	cgaggaggaa	ggctccgaat	ccgaatctcg	gtccgtcaaa
4681	gtgaagatca	agcttggccg	gaaggagaag	gcacaggacc	ggctgaaggg	ggcccgccgg
4741	cgcccgagcg	gaggggtccc	agccaagccg	gtcgtgagtg	acgatgacag	tgaggaggaa
4801	caagaggagg	accgctcagg	aagtggcagc	gaagaagact	gagcccgac	attccagctc
4861	cgaccccgag	ccctcgttcc	cagagctgag	atggcatagg	ccttagcagt	aacgggtagc
4921	agcagatgta	gtttcagact	tgagtaaaaa	ctgtataaac	aaaagaatct	tccatattta
4981	tacagcagag	aagctgtagg	actgtttgtg	actggccctg	tcccggcatc	agtagcatct
5041	gtaacacagc	taactgtctt	aaagagagag	agagagaatt	ccgaattggg	gaacacacga
5101	tactgttttt	ttctttccgt	tgctggcagc	actgttgcgc	cgcaatttgg	agtcactgta
5161	gttaagtgtg	gatgcattgt	cgtaaccgtc	cactcctcct	actgtatttt	attggacagg
5221	tcagactcgc	cgggggcccg	gcgaggggtat	gtcagcgtca	ctggatgtca	aacagtaata
5281	aattaaacca	acaacaaaaa	gcacagccaa	aaaaaaa		

SEQ ID NO: 65 Human BRG1 Amino Acid Sequence Isoform F (NP_001122320.1)

1	mstpdpplpg	tprrpgpspgp	gpspgamlgp	spgspgsah	smmgpspgpp	saghpipptgg
61	pggyppqdnmh	qmhkpmesmh	ekgmsddpry	nqmkgmgmrs	gghagmgppp	spmdqhsqgy
121	psplggseha	ssvpvasggs	sgpqmssgpg	gapldgadpq	algqqnrgpt	pfnqnqlhql
181	raqimaykml	arggplpdhl	qmavqgkrpm	pgmqggmptl	pppsvstatp	gpgpgpgpgp
241	gpgpappnys	rphmgggpnm	pppgpsgvpp	gmpgqppggp	pkpwpeppma	naaapcstpq
301	klippqptgr	pspappavpp	aasvmpptq	qspgqpapqa	pmvplhqkqs	ritpiqkprg
361	ldpveilgrt	eyrlgariah	rigelenlpg	slagdrlrka	tielkalrl	nfqrlqrgev
421	vvcmmrdtal	etalnakayk	rskrqslrea	riteklekq	kierkerkrrq	khqeylnsil
481	qhakdfkeyh	rsvtgiqkl	tkavatyhan	tereqqkene	riekermrrl	maedeegyrk
541	lidqkkdkrl	ayllqgtdey	vanltelvrq	hkaaavakek	kkkkkkkkkae	naegqtpaig
601	pdgepldets	qmsdlpvkvi	hvesgkiltg	tdapkgaggle	awlemnpgye	vaprsdsees
661	gseeeeeeee	eeqpqaagpp	tlpveekkki	pdpdsddvse	vdarhiiena	kqdvdddegvy
721	sqalarqlqs	yyavahavte	rvdkqsalmv	ngvlkqyqik	glewlvslvn	nlnlgilade
781	mglgkltigt	alitylmehk	ringpfliiv	plstlsnway	efdkwapsvv	kvsykgsppa
841	rrafvpqlrs	gkfnvlltty	eyiikdkhil	akirwkymiv	deghrmknhh	ckltqvlnth
901	yvaprrlllt	gtplgnklpe	lwallnfllp	tifkscstfe	qwfnapfam	gekvdlnnee
961	tiliirrlhk	vlrpflrlrl	kkeveaqlpe	kveyvikcdn	salqrvlyrh	mqakgvlltd
1021	gsekdkkkgk	grktlmtim	qlrkicnhpy	mfighieesfs	ehlgftggiv	ggldlyrasg
1081	kfelldrlkl	klratnhkvl	lfcqmtsllm	imedyfayrg	fkylrlrdgtt	kaedrgmllk
1141	tfnepgseyf	ifillstragg	lglnlgsadt	viifdsdwnp	hqdlqagdra	hrigqqnevr
1201	vrlrlctvns	eekilaaaky	klndvdqkviq	agmfdqksss	herraflqai	leheeqdeee
1261	devpddeqtn	yyavahavee	dlfmrmdldr	rreearnprk	kprlmeedel	pswiikddae
1321	verltceeee	ekmfgrgsrh	rkevdsdsl	tekqwlkale	egtleieeee	vrqkksrkr
1381	krdsdagsst	pttstrsrdk	ddeskkqkkr	grppaeklsp	nppnltkmk	kivdaviky
1441	dssgrqlsev	peyyelirp	vdfkkikeri	rnkhysrlnd	lekdvmlleq	
1501	naqtfnlleg	liyedshivlq	svftsvrqi	ekeddsegee	seeeeegeee	gseserssvk
1561	vkiklgrkek	aqdrklkgrr	rpsrgsrakp	vvsdddseee	qeedrsgsgs	eed

SEQ ID NO: 66 Mouse BRG1 cDNA Sequence Variant 1 (NM_001174078.1, CDS: from 261 to 5114)

1	ggcaagtggga	gcgggtagac	agggaggcgg	ggggcgcgcg	cgggcgcgctg	cggtgggggg
61	gggtggcctg	gcgaagccca	gcggcgcgcg	gcgcgaggct	ttccactcgc	cttgccagcg
121	gcggagacgg	cttctttgtt	tcctgaggag	aagcgagacg	cccactctgt	ccccagcccc
181	tcgtggaggg	ttggggcgcg	cgccaggaag	gttacggcgc	cgttaacctc	aggagaccag
241	tgctgttagc	tccagtaaa	atgtctactc	cagaccacc	cttgggtggg	actcctcgcc

TABLE 1-continued

301	ctggctcttc	ccagagccct	ggctcttcac	ctggtgcaat	gctgggtcct	agccctggcc
361	cctcaccagg	ttctgcccac	agcatgatgg	ggccaagccc	aggacctcct	tcagcaggac
421	atcccatgcc	caccagggg	cctggagggt	acccccagga	caacatgcat	cagatgcaca
481	agcctatgga	gtccatgcac	gagaaggcca	tgccatgatga	ccacagatac	aaccagatga
541	aagggtatgg	catgcggtca	ggggcccaca	caggcatggc	acctccacct	agtcccatgg
601	accagcattc	tcaaggttac	ccctcaccoc	tcggcggtcc	tgaacatgcc	tccagtcctg
661	tcccagccag	tggtccatct	tcaggccccc	agatgtcctc	tggtccaggga	ggggcccac
721	tagatggttc	tgatccccag	gccttgggac	agcaaaacag	aggcccaacc	ccatttaacc
781	agaaccagct	gcatacaact	agagctcaga	taattggcta	caagatgttg	gccaggggcc
841	agccattgcc	cgaccacctg	cagatggccg	tgcaaggcaa	gcggccgatg	cctggaatgc
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961	ccggcccttg	ccctggccct	ggcccaggac	cagccctccc	aaattacagt	agaccccatg
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1261	ggcagccagc	ccagcctgct	ccattggtgc	cactgcacca	gaagcagagc	cgaatcacc
1321	ccatccagaa	gccccaggcc	cttgaccctg	tgagatcct	acaagagcgg	gagtacaggg
1381	ttcagggtcg	aatcgacac	agaattcagg	aacttgaaa	cctccctggg	tccctggctg
1441	gggaccttcg	aaccaaagca	accatcgaac	tcaaggccct	taggttgctg	aacttcaga
1501	ggcagctgcg	ccaggaggtg	gcggtgtgca	tgccgaagaga	cacagccctg	gagacagccc
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1681	agtacctcaa	cagcattctg	cagcatgcc	aggacttcag	ggagtatcac	agatcagta
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1801	agcagaagaa	agaaaatgag	cgcattgaga	aggagcgaat	gcggaggctt	atggctgaag
1861	atgaggaggg	ctaccgcaaa	ctcattgacc	agaagaagga	caagcgctg	gcctaccttc
1921	tgacagcagc	agatgagtat	gtggccaaac	tcacagagct	gggtcgccag	cacaagctg
1981	cccagggttc	caaggagaag	aagaagaaaa	agaaaaagaa	gaaggcagaa	aatgctgaag
2041	gacagacacc	tgctattgga	ccagatggtg	agcctctgga	tgagaccagc	cagatgagtg
2101	acctccctgt	gaaggtgatc	cacgtggaga	gtggcaagat	cctcactggc	acagatgcc
2161	caaaagccgg	gcagctggaa	gcctggcttg	aaatgaaccc	agggtatgaa	gtagccccc
2221	ggtcagacag	tgaagaaagt	ggctctgaag	aggaggagga	ggaggagga	gaggagcagc
2281	ctcagcccg	acagccccc	acaactgctg	tggaagaaaa	gaagaagatt	ccagaccag
2341	acagcgatga	tgctctctgag	gtggacgccc	gacacattat	tgaaacgccc	aagcaagatg
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2641	catacctcat	ggagcacaa	cgcatacaac	ggcctttcct	catcatcggt	cctctctcga
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2761	acaagggtcc	tcagctgca	aggcgagctt	ttgccccaca	gcttcgcagt	gggaagtcca
2821	acgtcttact	gcccaccctat	gaatatatca	tcaagacaa	gcataccta	gccaagatcc
2881	gctggaagta	catgatttgt	gatgaaggcc	accgatgaa	aaaccaccac	tgcaagttga
2941	cgcaggtcct	taacacacac	tacgtggccc	ctcggcgcc	gctttctaca	ggcacaccac
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3121	tggaacctga	tgaagaggag	actatcctca	ttattcgtcg	cctacacaaa	gttctcgggc
3181	ccttcctgct	gcggcggttc	aagaagggaag	ttgaagccca	gctccctgag	aaggtagagt
3241	atgtcatcaa	atgcgacatg	tcagccctgc	agcgtgtgct	gtaccgtcac	atgcaggcca
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3361	cactgatgta	cactattatg	caactgcgta	agatctgcaa	ccacccccac	atgttcacgc
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3661	accggggcat	gctgttgaaa	acctttaatg	aacctggctc	tgagcatttc	attttctcgc
3721	tcagtaccgg	tgctgggggg	ctgggcccga	atctgcagtc	agctgacact	gtgatcatct
3781	ttgacagtga	ctggaatccc	caccaggacc	tgcaagcaca	ggatcgagcc	catcgcattg
3841	gacagcagaa	tgagggtcgt	gttcttcgcc	tgtgcacgg	caacagtgtg	gaagagaaga
3901	tactggctgc	tgccaaatac	aaactcaatg	tggaacagaa	ggatgacag	gcaggcatgt
3961	tcgaccagaa	gtcgtccagc	catgagaggc	gtgccttcc	gcaggccatc	ctggagcagc
4021	aggagcagga	tgaggaggaa	gatgaggtgc	ctgatgatga	gacctcaac	cagatgattg
4081	ccgggcacga	agaagagttt	gacctcttca	tgccatgga	cttggaacgc	cggcgtgaag
4141	aagcccgcaa	ccccaaagg	aagccacgcc	tgatggaaga	ggatgagctc	ccatcctgga
4201	tcatcaagga	tgatgccgag	gtggagcggc	tgacatgtga	agagggaag	gagaagatgt
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4321	agtggctcaa	gaccttgaag	gctatcgagg	aggcgacgct	ggaggagatc	gaaggaggag
4381	tccggcgaga	gaaatcttca	cgtaagcgta	agcgagacag	cagggcgggc	tcctccaccc
4441	cgacaccagc	gacccacagc	cgtgacaagg	atgaggagag	caagaagcag	aagaaacgtg
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4561	agatcgtgga	tgctgtgatc	aagtacaaag	acagcagcag	tggaagctcag	ctcagcgagg
4621	tgttcaccca	tgctccctct	cgcaaggagc	ttcctgagta	ctatgagctc	atccgaaagc
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4741	acctggagaa	ggatgtgatg	ctgctgtgcc	agaacgctca	gacgttcaac	ctcagggggt
4801	ccctgatcta	tgaggactcc	atcgtcctgc	agtctgtctt	caccagcgta	cggcagaaga
4861	ttgagaagga	ggacgacagt	gaaggcgagg	aaagcgagga	ggaggaggag	ggcgaggagg
4921	aaggctccga	gtctgagtc	cgctccgtca	aggtgaagat	caagctgggc	cgcaaggaga
4981	aggcccaaga	ccgactcaag	ggggcccgcc	ggcgcccaag	ccggggatcc	cgggccaagc
5041	cggttgtgag	tgacgatgac	agtgaggagg	agcaggagga	ggaccgctca	ggaagtggca

TABLE 1-continued

5101	gtgaggaaga	ctgaaccaga	cattcctgag	tcctgacccc	gaggcgctcg	tcccagccaa
5161	gatggagtag	cccttagcag	tgatgggtag	caccagatgt	agtttcgaac	ttggagaact
5221	gtacacatgc	aatcttcac	atcttttaggc	agagaagtat	aggcctgtct	gtcggccctg
5281	gcctggcctc	gagtccttac	cagcattaac	tgtctagaga	ggggacctcc	tgggagcacc
5341	atccacctcc	ccaggcccca	gtcactgtag	ctcagtggtg	gcctgcgcgt	gccggccgct
5401	ccttgtaactg	taccttactg	gacagggcca	gctctccagg	aggctcacag	gcccagcggg
5461	tatgtcagtg	tcactggagt	cagacagtaa	taaatataag	caatgacaag	ccaccactgg
5521	ctccctggag	tccttgctgt	cagcagtggt	tcgggggcca	cagagaagaa	agaaagacct
5581	ttaggaactg	gggtctaactt	atgggcaaa	tacttgctct	gccaggtgta	tgggttttgc
5641	attcccatca	cccacacacc	ctaacaacgc	caagtccagt	agcttcaagt	tagagcctcc
5701	acctcaatgt	gtacgtggaa	agcaatcaaa	gatgatgcct	agcatccacc	tctggccctc
5761	atgtgcagat	gtacacacac	tgaattacat	acacgggaca	cacacatcca	cacggaggca
5821	gtccatgact	tgcaactggg	agatggtag	ataggcgaaa	gtgccacagg	cacagggcca
5881	ggctaattta	gtcctgcagt	cctgtgctct	taagatgaag	gcacaaagag	gaaccccagg
5941	cgctccaact	agcatgccag	gcagtgcaca	gacctgctt	caaatgaatc	agagcccaca
6001	tcctagattg	ccctcttacc	cgatgcgatg	cccatgcctt	cacatgatga	tgcgtatata
6061	tacatacata	cgtaaaaata	ttctttttta	aattatagac	atttttgtgt	gaatgttttg
6121	cctgaatgtg	tgtgtgtgtg	tgtgtgtgtg	tgtgtgtgtg	tgtgtgtgtg	tatcaagtac
6181	attctatag	ctcacacagg	tcaaggagg	gcattggatc	tggaaactga	gtcacatgag
6241	gctgtgagca	actgcgtggg	ttcctgggcc	tttgcaacag	cagtttagtac	tcttcaccac
6301	tgagccattt	ctccaatctc	aaaaagaagc	attcttttaa	atgaagactg	aaataaataa
6361	gtaggacttg	cccttg				

SEQ ID NO: 67 Mouse BRG1 Amino Acid Sequence Isoform A (NP_001167549.1)

1	mstpdplpg	tprrpgp	gspgamlgp	spgspgsah	smmgpspgp	saghpmpqg
61	pggyppqdmh	qmhkpmesm	ekgmpddpry	nqmkgmgrs	gahtgmapp	spmdqhsqgy
121	psplggseha	ssvpasggs	sgpqmssgpg	gapldgsdpq	algqgnrgpt	pfnqnqlhql
181	raqimaykml	argcpdpdl	qmavqgkrpm	pgmqgqmpdl	pppsvsatgp	gpgpgpqp
241	gpgpappnys	rphmgggnm	pppgpsgvp	gmpgqppgpg	pkpwpegpma	naaaptstpq
301	klippqptgr	pspappavp	aaspvmppt	qspgqpaqpa	plvplhqkqs	ritpiqkprg
361	ldpveilger	eyrlqariah	riqeielpg	slagdlrtka	tielkalrl	nfqrlrgev
421	vvcmmrdtal	etalnakayk	tskrqslrea	riteklekq	kieqerkrq	khqeylnsil
481	qhakdfreyh	rsvtgklqkl	tkavatyhan	tereqqkene	riekermrrl	maedeeyrk
541	lidqkdkrl	kyllqgtdey	vanltelvrq	hkaaqvakek	kkkkkkkkkae	naeggtpaig
601	pdgepldets	qmsdlpvkvi	hvesgkiltg	tdapkgaggle	awlemmpgye	vaprsdsees
661	gseeeeeeee	eeqpqpaqpp	tlpveekkki	pdpdsddvse	vdarhiiena	kqdvddeygv
721	sqalarglqs	yyavahavte	rvdkqsalmv	ngvlkqyqik	glewlvsllyn	nlnlgilade
781	mglgktigt	alitylmehk	ringpfliiv	plstlsnway	efdkwapsvv	kvsykgsppa
841	rralvpqlrs	gkfnvlltty	eyiikdkhil	akirwkymiv	deghrmknhh	ckltqvlnt
901	yvaprrillt	gtpqlnkllpe	lwallnflp	tifkscstfe	qwnapfam	gekvdlnee
961	tiliirrlhk	vlpfllrrl	kkeveaqlpe	kveyvikcdm	salqrlyrh	mqaqgvlltd
1021	gsekdkkkg	gtktlmtim	qlrkicnhpy	mfqhiesfs	ehlgftggiv	qgldlyrasg
1081	kfelldrlp	klratnhkvl	lfcqmtsmt	imedyfayrg	fkylrlldgt	kaedrgmllk
1141	tfnepgseyf	ifillstragg	lglnlqsadt	viifdsdwnp	hqdlgaqdra	hrigqgnevr
1201	vrlrctvns	eekilaaaky	klndvdqkviq	agmfdqkss	herraflqai	leheeqdeee
1261	devpddetv	qmiaheeeef	dlfmrmdldr	rreearnprk	kprlmeedel	pswiikddae
1321	verltceeee	ekmfgrgsrh	rkevdydsdl	tekqwlktlk	aieegtleei	eeevrkkss
1381	rkrkrdseag	sstpttstrs	rdkdeeskkq	kkrgprpaek	lspnppnltk	kmkkivdavi
1441	kykdsssggr	lsevfliqlps	rkelpyyel	irkpvdffki	kerirnhkyr	slndlekdv
1501	llcqnagtfn	legsllyeds	ivlqsvftsv	rqlikedds	eegeeeeee	geegseses
1561	rsvkvkiklg	rkekaqdrk	grrrrprgs	rakpvvsdd	seeqeeds	gsgsseed

SEQ ID NO: 68 Mouse BRG1 cDNA Sequence Variant 2 (NM_011417.3, CDS:

from 261 to 5105)

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121	gcggagacgg	cttctttgtt	tcctgaggag	aagcgagacg	cccactctgt	ccccgacccc
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241	tgctctgtagc	tcagtaaa	atgtctactc	cagaccacc	cttgggtggg	actcctcggc
301	ctggcccttc	cccaggccct	ggtccttcac	ctggtgcaat	gctgggtcct	agccctggcc
361	cctcaccagg	ttctgccac	agcatgatgg	ggccaagccc	aggacctcct	tcagcaggac
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541	aagggatgtc	gtcctgggtca	ggggccca	caggcatggc	acctccacct	agtcccatgg
601	accagcattc	tcaaggttac	ccctcacc	tcggcggtc	tgaacatgcc	tcagtcctg
661	ccccagccag	tgcccatct	tcaggccccc	agatgtctc	tgggccagga	ggggccccc
721	tagatgggtc	tgatgccag	gccttgggac	agcaaaacag	aggcccaacc	ccatttaacc
781	agaaccagct	gcatacaact	agagctcaga	taattggccta	caagatgttg	gccaggggcc
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901	agcaacagat	gccaaaccta	cctccacct	cagtgtccgc	cacaggacc	ggacctggac
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1141	ccccacaag	caccacacag	aagctgattc	ctccgcaacc	aacaggccgt	ccttcacctg
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1261	ggcagccagc	ccagcctgct	ccattgggtc	cactgcacca	gaagcagagc	cgaatcacc
1321	ccatccagaa	gccccaggc	cttgacctg	tgagatcct	acaagagcgg	gagtagaggg
1381	ttcaggtctg	aatcgcacac	agaattcagg	aacttgaaaa	cctcctgggg	tccttggtg
1441	gggacctcg	aaccacaagca	accatcgaa	tcaaggccct	taggttgctg	aacttcaga
1501	ggcagctgcg	ccaggagggt	gtggtgtgca	tgcaagaga	cacagcccc	gagacagccc

TABLE 1-continued

1561	tcaatgccaa	ggcctacaag	cgcagcaaac	gtcagtcact	acgggaggcc	cgcacactg
1621	agaagtgtga	gaagcagcag	aagattgaac	aggagcgcaa	gcgccgccag	aagcaccagg
1681	agtacctcaa	cagcattctg	cagcatgccca	aggacttcag	ggagtatcac	agatcagtea
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1981	cccagggtgc	caaggagaag	aagaagaaaa	agaaaaagaa	gaaggcagaa	aatgctgaag
2041	gacagacacc	tgctatttga	ccagatgggtg	agcctctgga	tgagaccagc	cagatgagtg
2101	acctccctgt	gaagggtgatc	cacgtggaga	gtggcaagat	cctcactggc	acagatgccc
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2221	ggtcagacag	tgaagaaagt	ggctctgaag	aggaggagga	ggaggaggaa	gaggagcagc
2281	ctcagccccc	acagccccc	acactgcctg	tggagaaaa	gaagaagatt	ccagaccag
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2761	acaagggtcc	tccagctgca	aggcgagctt	ttgtcccaca	gcttcgcagt	gggaagtcca
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3361	cactgatgaa	cactattatg	caactgcgta	agatctgcaa	ccacccttac	atgttccagc
3421	acatcgagga	gtccttttct	gagcacttgg	gggtcacccg	cggcatcgctg	caaggattgg
3481	acctttaccg	tgctcagggt	aaatttgaac	ttcttgatag	aattctaccc	aaactccgtg
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4561	atgctgtgat	caagtacaaa	gacagcagca	gtggacgtca	gctcagcgag	gtgttcatcc
4621	agctcccttc	tcgcaaggag	cttctctgag	actatgagct	catccgaaa	cctgtggact
4681	tcaagaagat	gaaggaaagc	atccgaaaac	acaagtaccg	cagcctcaat	gaactggaga
4741	aggatgtgat	gctgctgtgc	cagaacgctc	agacgttcaa	cctcgagggt	tccttgatct
4801	atgaggactc	catcgctcctg	cagtctgtct	tcaccagcgt	acggcgagaag	attgagaagg
4861	aggacgacag	tgaaggcgag	gaaagcgagg	aggaggagga	ggcgagggag	gaaggctccg
4921	agtctgagtc	ccgctccgct	aaggtgaaga	tcaagctggg	ccgcaaggag	aaggcccgag
4981	accgactcaa	ggggggccgc	cggcggccaa	gccggggatc	ccgggccaag	ccggttgtga
5041	gtgacgatga	cagtgaaggag	gagcaggagg	aggaccgctc	aggaagtggc	agtgaaggaa
5101	actgaaccag	acattcctga	gtcctgaccc	cgaggcgctc	gtcccagcca	agatggagta
5161	gcccttagca	gtgatgggtg	gcaccagatg	tagtttcgaa	cttgaggaa	tgtacacatg
5221	caatcttcca	cattttttag	cagagaagta	taggctgtgc	tgtcggccct	ggcctggcct
5281	cgagtctcta	ccagcattaa	ctgtctagag	aggggacctc	ctgggagcac	catccacctc
5341	cccaggcccc	agtcactgta	gctcagtgga	tgcagtcgag	tgccggccgc	tccttgta
5401	gtatcttact	ggacagggcc	agctctccag	gaggctcaca	ggcccagcgg	gtatgtcagt
5461	gtcactggag	tcagacagta	ataaattaaa	gcaatgacaa	gccaccactg	gctccctgga
5521	ctccttgctg	tcagcagtg	ctccggggcc	acagagaaga	aagaaagact	tttaggaact
5581	gggtctaa	tatgggcaaa	gtacttgct	tgccaggtgt	atgggttttg	cattcccatc
5641	accacacac	cctaacaag	ccaagtcatg	gagcttcaag	ttagagcctc	cacctcaatg
5701	tgtacgtgga	aagtaaccaa	agatgatgcc	tagcatccac	ctctggccct	catgtgcaga
5761	tgtacacaca	ctgaattaca	tacacgggac	acacacatcc	acacggaggc	agtcctatgac
5821	ttgcaactgg	gagatgggtac	cataggcgaa	agtgccacag	gcacagggcc	aggctaattt
5881	agtcctgcag	tcctgtgctc	ttaagatgaa	ggcacaaga	ggaacccag	gcgctccaac
5941	tagcatgcca	ggcagtgaca	agacctgct	tcaaatgaat	cagagccac	atcagtat
6001	gccctcttac	ccgatgcgat	gccccatgcc	tcacatatga	atgtgtatat	atacatat

TABLE 1-continued

6061	acgtaaaata	attctttttt	aaattataga	cattttttgtg	tgaatgtttt	gcctgaatgt
6121	gtgtgtgtgt	gtgtgtgtgt	gtgtgtgtgt	gtgtgtgtgt	gtatcaagta	cattcctaga
6181	gcctacagag	gtcaaggagg	ggcattggat	ctggaactgg	agtcacatga	ggctgtgagc
6241	aactgtgtgg	gttctctggc	ctttgcaaca	gcagtttaga	ctcttcacca	ctgagccatt
6301	tctccaatct	caaaaagaag	cattctttta	aatgaagact	gaaataaata	agtaggactt
6361	gccttgg					

SEQ ID NO: 69 Mouse BRG1 Amino Acid Sequence Isoform B (NP_035547.2)

1	mstpdpplpg	tprrgppspg	gspgamlgp	spgspgsah	smmgppspgpp	saghpmpgtg
61	pggypqdnmh	qmhkpmesm	ekgmpddpry	nqmkmgmrs	gahtgmapp	spmdqhsqgy
121	psplggseha	ssvpasggs	sgpqmssggp	gapldgsdpq	algqnnrgpt	pfnqnqlhql
181	raqimaykml	argqplpdl	qnavqgkrpm	pgmqqqmptl	pppsvsatgp	gpgpgpgpgp
241	gpgpappnys	rphgmgsppm	pppgpsgvpv	gmpgqppggp	pkpwpegpma	naaptstpq
301	klippqgtgr	pspappavpp	aaapvmppqt	qspgqpaqpa	plvplhqkqs	ritpiqkprg
361	ldpveillqr	eyrlqariah	riqelenlpg	slagdlrtka	tielkalrl	nfrqrlrgev
421	vvcmmrdtal	etalnakayk	rskrqslrea	riteklekqg	kiegerkrrq	khqeylnsil
481	qhakdfreyh	rsvtqklqkl	tkavatyhan	tereqqkene	riekermrrl	maedeegyryk
541	lidqkdkkrl	ayllqgtdey	vanltelvrq	hkaaqvakek	kkkkkkkkkae	naeggtpaig
601	pgpepldets	qmsdlpvkvi	hvesgkiltg	tdapkaggle	awlemnpgye	vaprdses
661	gseeeeeeee	eeqpqpaqpp	tlpveekkki	pdpdsddvse	vdarhiiena	kqdvdeygv
721	sqalarglqs	yyavahavte	rvdkgsalmv	ngvlkqyqik	glewlvslvn	nnlmgilade
781	mglgttiqti	alitylmehk	ringpfliiv	plstlsnway	efdkwapsvv	kvsykgspaa
841	rrafvpqlrs	gkfnvlltty	eyiikdkhil	akirwkymiv	deghrmknhh	ckltqvlnth
901	yvaprrlllt	gtplqnklpe	lwallnflfp	tifkscstfe	qwnapfam	gekvdlnsee
961	tiliirllrhk	vlrpfllrrl	kkeveaqle	kveyvikcdm	salgrvlyrh	mqakgvlltd
1021	gsekdkkkgk	gtktlmttim	qlrkicnhpy	mfqhieesfs	ehlgftggiv	qgldlyrasg
1081	kfelldrllp	klratnhkvl	lfcqmtsllm	imediyfayrg	fkylrldgtt	kaedrgmllk
1141	tfnepgseyf	ifllstragg	lglnlqsadt	viifdsdwnp	hqldlgaqdra	hrigqgnevr
1201	vrlrctvnsv	eekilaaaky	klndvdkviq	agmfdqkss	heraflgai	leheeqdeee
1261	devpddetvn	qmiaheeeef	dlfmrmdldr	rreearnpkr	kprlmeedel	pswiikddae
1321	verltceeee	ekmfgrgsrh	rkevdsdsl	tekqwlkaie	egtleieeee	vrqkksrkr
1381	krdeagssst	pttstsrdrk	deeskkgkkr	grppaeklsp	nppnltkmk	kivdaviky
1441	dsssggrqlse	vfiqlpsrke	lpeyyelirk	pvdffkkiker	irnhkyrsln	dlekdvmlkc
1501	qnagtfnleg	sllyedsivl	qsvftsvrqk	iekeddsese	eseeeeeege	egsesesrsv
1561	kvkiklgrke	kaqdrilkggr	rrpsrgsrak	pvsdddsee	eqeedrsgsg	seed

SEQ ID NO: 70 Mouse BRG1 cDNA Sequence Variant 3 (NM_001174079.1, CDS: from 261 to 5102)

1	ggcaagtga	gcggttagac	agggaggcgg	gggcgcgcgg	cgggcgcgtg	cgggtggggg
61	gggtggcctg	gcgaagccca	gcgggcgcgc	gcgcgaggct	ttccactcgt	cttgccagcg
121	gcggagacgg	cttctttgtt	tcctgaggag	aagcgagacg	cccactctgt	ccccgacccc
181	tcgtggaggg	ttggggggcg	cgccaggaag	gttacggcgc	cgttacctcc	aggagaccag
241	tgctctagc	ttccagtaaa	atgtctactc	cagaccaccc	cttgggtggg	actcctcggc
301	ctggtccttc	cccaggccct	ggtccttcac	ctggtgcaat	gctgggtcct	agccctggcc
361	cctcaccagg	ttctgcccac	agcatgatgg	ggccaagccc	aggacctcct	tcagcaggac
421	atcccatgcc	caccaggggg	cctggagggt	acccccagga	caacatgcat	cagatgcaca
481	agcctatgga	gtccatgcac	gagaaggcca	tgctgatga	cccacgatac	aaccagatga
541	aagggatggg	catgcggtca	ggggccccc	caggcatggc	acctccacct	agtcacctgg
601	accagattcc	gcacagttac	ccctcaccce	tcggcggtct	tgaaacatgc	tccagtcctg
661	tcccagccag	tgcccatctc	tcaggccccc	agatgtctct	tgggccagga	ggggccccc
721	tagatgggtc	tgatccccag	gccttggggc	agcaaaacag	aggcccaacc	ccatttaacc
781	agaacacag	gcacacattc	agagctcaga	taattggccta	caagatgttg	gccagggggc
841	agccattgcc	cgaccacctg	cagatggccg	tgcaaggcaa	gcggccgatg	cctggaatgc
901	agcaacagat	gccaacacta	cctccacctc	cagtgtccgc	cacaggaccc	ggacctggac
961	ccggccctgg	ccctggccct	ggcccaggac	cagccctccc	aaattacagt	agaccccatg
1021	gtatgggagg	gccccacatg	cctccccacg	gacctcagg	tgtgcccccc	gggatggcctg
1081	gtcagcgccg	tggaggggct	cccaagccat	ggcctgaagg	acccatggcc	aatgtctgtg
1141	ccccacacag	caccacacag	aagctgattc	ctccgcaacc	aacaggccgt	ccttcactctg
1201	cacctcctgc	tgtcccgcct	gctgcctcac	ctgtaatgcc	accacaaaca	cagtcctccag
1261	ggcagccagc	ccagcctgct	ccattggtgc	cactgcacca	gaagcagagc	cgaatcacc
1321	ccatccagaa	gcaccgaggc	cttgaccctg	tggagatcct	acaagagcgg	gagtagaggg
1381	ttcaggctcg	aatcgccac	agaattcagg	aacttgaaaa	cctccctggg	tccctggctg
1441	gggacacctg	acccaagaca	accatcgaa	tcaaggccct	taggtgtgtg	aacttcagaa
1501	ggcagctcgg	ccaggaggtg	gtgggtgca	tgcgaaagaa	cacagccctg	gagacagccc
1561	tcaatgccaa	ggcctacaag	cgcagcaaac	gtcagtcact	acgggaggcc	cgcacactctg
1621	agaagttgga	gaagcagcag	aagattgaac	aggagcgcaa	gcggccgccag	aagcaccagg
1681	agtacctcaa	cagcatctctg	cagcatgcca	aggacttcag	ggagatcac	agatcagtc
1741	caggcaaa	ccagaaactc	accaaggctg	tggccacctc	ccatgccaac	actgagcggg
1801	agcagaagaa	agaaaatgag	cgcattgaga	aggagcgaa	gcggaggctt	atggctgaag
1861	atgaggaggg	ctaccgcaaa	ctcattgacc	agaagaagga	caagcgcctg	gcctaccttc
1921	tgacagcagc	agatgagtat	gtggccaacc	tcacagagct	ggtgcccagc	cacaaagctg
1981	cccaggttgc	caaggagaag	aagaagaaaa	agaaaaagaa	gaaggcagaa	aatgctgaag
2041	gacagacac	tgctattgga	ccagatgggt	agcctctgga	tgagaccagc	cagatgagtg
2101	acctccctgt	gaaggtgatc	cacgtggaga	gtggcaagat	cctcactggc	acagatgccc
2161	caaaagccgg	gcagctggaa	gcctggcttg	aaatgaacca	agggtatgaa	gtagccccc
2221	ggtagacagc	tgaagaaagt	ggctctgaag	aggaggagga	ggaggagga	gaggagcagc
2281	ctcagcccg	acagccccc	acactgcctg	tggaaagaaa	gaagaagatt	ccagaccag
2341	acagcgatga	tgtctctgag	gtggacgccc	gacacattat	tgagaacgcc	aagcaagatg
2401	tggacgatga	gtacagtggt	tcccaggccc	ttgctcgtgg	cctgcagctc	tactatgctg
2461	tggcccatgc	agtcacagag	agagttagata	agcagtcgcg	ccccatggct	aacggtgtcc

TABLE 1-continued

2521	tcaaacagta	ccagatcaag	ggtttggagt	ggctggcgtc	cctgtacaac	aacaacctga
2581	atggcatcct	ggctgatgag	atggggctgg	ggaagaccat	ccagaccatc	gcgctcatca
2641	cataacctcat	ggagcacaag	cgcatacaac	ggcctttcct	catcatcgtg	cctctctcga
2701	cactgtcaaa	ctgggcgtat	gaatttgaca	agtgggcccc	ctctgtgggtg	aaggtttctt
2761	acaagggctc	tccagctgca	aggcgagctt	ttgtcccaca	gctccgcagt	gggaagttca
2821	acgtcttact	gaccacotat	gaatatatca	tcaaagacaa	gcatactcta	gccaaagatcc
2881	gctggaagta	catgatttgt	gatgaaggcc	accgcacgaa	aaaccaccac	tgcaagttga
2941	cgcaggctct	taacacacac	taegtggccc	ctcggcgccct	gcttcttaca	ggcacaccac
3001	tgcagaacaa	gctaccggag	ctctggggccc	tgtttaactt	cctgctcccc	actatcttca
3061	agagctgcag	cacctctgaa	cagtgggtca	atgcaccctt	tgccatgact	ggagaaaagg
3121	tggacctgaa	tgaagaggag	actatcctca	ttattcgtcg	cctacacaaa	gttctgcggc
3181	ccttctctgct	gcggcggtc	aagaaggag	ttgaagccca	gctccctgag	aaggtagagt
3241	atgtcatcaa	atgcgacatg	tcagccctgc	agcgtgtgct	gtaccgtcac	atgcaggcca
3301	aaggtgtgct	gctgactgac	ggctccgaga	aggacaagaa	gggcaaaagt	ggcaccaaga
3361	cactgatgaa	cactattatg	caactgcgta	agatctgcaa	ccacccttac	atgttccagc
3421	acatcgagga	gtccttttct	gagcacttgg	gggtcacccg	cggcatcgtg	caaggattgg
3481	acctttaccg	tgctcagggt	aaatttgaac	ttcttgatag	aattctaccc	aaactccgtg
3541	caacgaacaa	taaagtgtct	ctcttttgcc	aaatgacctc	cctcatgacc	atcatggag
3601	actacttttg	ataccgtggc	ttcaaatacc	tcaggcttga	tggaaccaca	aaagcaagag
3661	accggggcat	gctgttgaaa	acctttaatg	aacctggctc	tgagtatttc	attttcctgc
3721	tcagtaccgg	tgctgggggg	ctgggctcga	atctgcagtc	agctgacact	gtgatcatct
3781	ttgacagtga	ctggaatccc	caccaggacc	tgcaagcaca	ggatcgagcc	catcgcattg
3841	gacagcagaa	tgaggtgcgt	gttcttcgcc	tgtgcacggt	caacagtgtg	gaagagaaga
3901	tactggctgc	tgccaaatac	aaactcaatg	tggatcagaa	ggtgatccag	cgaggcatgt
3961	tcgaccagaa	gtcgtccagc	catgagaggc	gtgccttctc	gcaggccatc	ctggagcacg
4021	aggagcagga	tgaggaggaa	gatgaggtgc	ctgatgatga	gaccgtcaac	cagatgattg
4081	cccggcacga	agaagagttt	gacctcttca	tgcgcatgga	cttggaaccg	cggcgtgaag
4141	aagcccgcaa	ccccaaagcg	aagccacgcc	tgatggaaga	ggatgagctc	ccatctgga
4201	tcatacaagga	tgatgccgag	gtggagcggc	tgacatgtga	agagggaagag	gagaagatgt
4261	tcggccgtgg	ttctcgccac	cgaaggagg	tagactacag	cgactcactg	acagagaagc
4321	agtggctcaa	ggctatcgag	gagggcacgc	tggaggagat	cgaagaggag	gtccggcaga
4381	agaaatcttc	acgtaagcgt	aagcgagaca	gcgaggccgg	ctcctccacc	ccgaccacca
4441	gcaccccgag	ccgtgacaa	gatgaggaga	gcaagaagca	gaagaaacgt	ggggggccac
4501	ctgctgagaa	gtcgtcccca	aaccacaccta	acctcaccac	gaagatgaag	aagatcgtgg
4561	atgctgtgat	caagtacaaa	gacagcagtg	gacgtcagct	cagcgagggtg	ttcatccagc
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4861	acgacagtga	aggcagtgaa	agcagaggaa	aggaggagg	cagaggagaa	ggctccgagt
4921	ctgagtcctc	ctcgtcaag	gtgaagatca	agctgggccc	caaggagaag	gccagggacc
4981	gactcaagg	ggggccgcgg	cggccaagcc	ggggatcccc	ggccaagccg	gttgtgagtg
5041	acgatgacag	tggagggagg	caggaggagg	accgctcagg	aagtggcagt	gagggaagact
5101	gaaccagaca	tccttgagtc	ctgaccccca	ggcgtctgct	ccagccaaga	tggagttagcc
5161	cttagcagtg	atgggttagca	ccagatgtag	tttcgaactt	ggagaactgt	acacatgcaa
5221	ttctccacac	ttttaggcag	agaagtatag	gcctgtctgt	cggccctggc	ctggcctcga
5281	gtctctacca	gcattaaactg	tctagagagg	ggacctcctg	ggagcaccat	ccacctcccc
5341	aggccccagt	cactgtagct	cagtggatgc	atgcgcgtgc	cggccgctcc	ttgtactgta
5401	tcttactgga	cagggccagc	tctccaggag	gctcacagcc	ccagcgggta	tgtcagtgct
5461	actggagtca	gacagtaata	aattaaagca	atgacaagcc	accactggct	ccctggactc
5521	cttgcctgta	gcagtggtc	cggggccaca	gagaagaaag	aaagactttt	aggaaactggg
5581	tctaaactat	tgtgcaagta	cttgcttgc	caggtgtatg	ggttttgcat	tcccataccc
5641	cacacacct	aaacaagcca	agtcagtgag	cttcaagtta	gagcctccac	ctcaatgtgt
5701	acgtggaag	caatcaaaga	tgatgcctag	catccacctc	tggccctcat	gtgcagatgt
5761	acacacactg	aactacatc	acgggacaca	cacatccaca	cggaggcagt	ccatgacttg
5821	cactggggag	atggtaccat	aggcgaaagt	gccacaggca	caggggccagg	ctaatttagt
5881	cctgcagctc	tgtgctctta	agatgaaggg	acaaagagga	accccaggcg	ctccaaactag
5941	catgccaggg	agtgacaaga	ccctgcttca	aatgaatcag	agccacatt	cagtattgcc
6001	ctcttaccgg	atgcatgccc	catgcccctc	catatgaatg	tgtatatata	catacatacg
6061	taaaataaat	ctttttttaa	ttatagacat	ttttgtgtga	atgttttgcc	tgaatgtgtg
6121	tgtgtgtgtg	tgtgtgtgtg	tgtgtgtgtg	tgtgtgtgtg	tcaagtacat	tcttagagcc
6181	tacagaggtc	aagggagggc	atttgatctg	gaactggagt	cacatgaggc	tgtgagcaac
6241	tgtgtgggtt	cctgggccc	tgcaacagca	gttagtactc	ttcaccactg	agccatttct
6301	ccaatctcaa	aaagaagcat	tcttttaaat	gaagactgaa	ataaataagt	aggacttgcc
6361	ttgg					

SEQ ID NO: 71 Mouse BRG1 Amino Acid Sequence Isoform C (NP_001167550.1)

1 mstpdpplpg tprpgrpspg gspsgamlgp spgspgsah smmgpspgpp saghpmptgg
 61 pggypqdnmh qmhkpmesm ekmpddprr nqmkgmgrs gahtgmapp spmdqhsqgy
 121 pspglgseha sspvpasggs spgmssgpg gapldgsdpq alqgnrgpt pfnqnlhlql
 181 raqimaykml argqplpdlh qmavqgkrpm pgmqgqmtl ppsvsvatgp spgpgpgpgp
 241 gpgpappnys rphmgggpnm pppgspgvpp gmpgqppggp pkpwppegpma naaaptstpg
 301 klippqptgr pspappavpp aaspvmppt qspgqpaqpa plvplhqkqs ritpiqkprg
 361 ldpveilqer eyrlqariah rigelenlpg slagdlrtka tielkalrl1 nfqrqlrgev
 421 vvmrrdtal etalnakayk rskrqslrea riteklekqg kieqerkrq khqeylnsil
 481 qhakdfreyh rsvtqklqkl tkavatyhan tereqkkene riekermr1 maedeeyrk
 541 lidqkkdkrl ayllqgtdey vanltelvrq hkaaqvakek kkkkkkkkae naegqtpaig
 601 pdgepldets qmsdlpvkvi hvesgkiltg tdpakaggle awlemnpgye vaprsdsees
 661 gsaeaeaeae eepqpqapp tlpveekki pdpsddvse vdarhiiena kqdvddeygv
 721 sqalarglqs yyavahavte rvdksqalmv ngvlkqyqik glewlvsllyn nlnlgilade

TABLE 1-continued

781	mglgkkti	alitylmehk	ringpfliiv	plstlsnway	efdkwapsvv	kvsykgspaa
841	rrafvpqlrs	gkfnvlltty	eyiikdkhil	akirwkymiv	deghrmknhh	ckltqvlnth
901	yvaprrlllt	gtplqnklpe	lwallnflrp	tifkscstfe	qwnapfamt	gekvdlnnee
961	tiliirrlhk	vlrpfllrrl	kkeveaqlpe	kveyvikcdm	salqrvlyrh	mqakgvlltd
1021	gsekdkkkgk	gktklmnmim	qlrkicnhy	mfqhieesfs	ehlgftggiv	ggldlyrasg
1081	kfelldrllp	klratnhkvl	lfcqmtslmt	imedyfayrg	fkylrldgtt	kaedrgmllk
1141	tfnepgseyf	ifillstragg	lglnlqsadt	viifdsdwnp	hqdlqaqdra	hrigqgnevr
1201	vlrlctvnsv	eekilaaaky	klndvdkviq	agmfdqksss	herraflqai	leheeqdeee
1261	devpddetvn	qmiarheeeef	dlfmrmdldr	rreearnpkr	kprlmeedel	pewiikddae
1321	verltceeee	ekmfggrsrh	rkevdydsdl	tekqwlkaie	egtleeeeee	vrqkksrkr
1381	krdseagssst	pttstrsrdrk	deeskkgkkr	grppaeklsp	ppnltkkmk	kivdaviky
1441	dssgrqlsev	fiqlpsrkel	peyyelirkp	vdffkkikeri	rnhykyslnd	lekdvmlloq
1501	naqtfnlqgs	liyedsivlq	svftsvrqi	ekeddsegee	seeeeeeeee	gsesesrsvk
1561	vkiklgrkek	aqdrllkgrr	rprrgrakp	vvdddseeee	qeedrsgsgs	eed
SEQ ID NO: 72 Human BRM cDNA Sequence Variant 1 (NM_003070.4, CDS: from 223 to 4995)						
1	gcgtcttc	gcgcccgc	aggagggc	gggtggac	tgggcggag	ccgagtttag
61	gaagaggag	ggacggctgt	catcaatgaa	gtcatattca	taactagtc	ctctctccct
121	ctgtttctgt	actctgggtg	actcagagag	ggaagagatt	cagccagcac	actcctcgcg
181	agcaagcatt	actctactga	ctggcagaga	caggagaggt	agatgtccac	gcccacagac
241	cctgggtcga	tgcccacccc	agggccttcg	ccggggcctg	ggccttcccc	tggggccaatt
301	cttgggccta	gtccaggacc	aggaccatcc	ccaggttccg	tccacagcat	gatggggcca
361	agtccctggac	ctccaaagtgt	ctcccatcct	atgcccagca	tggggteccac	agacttccca
421	caggaaggca	tgcatcaaat	gcataagccc	atcgatggta	tacatgacaa	ggggattgta
481	gaagacatcc	attgtggatc	catgaagggc	actggatgac	gaccacotca	ccaggcgatg
541	ggccctcccc	agagtccaat	ggatcaacac	agccaagggt	atatgtcacc	acacccatct
601	ccattaggag	ccccagagca	cgtctccagc	cctatgtctg	gaggaggccc	aactccacot
661	cagatgccac	caagccagcc	gggggcccct	atcccagggt	atccgcaggc	catgagccag
721	cccaacagag	gtccctcacc	tttcagtcct	gtccagctgc	atcagcttcg	agctcagatt
781	ttagcttata	aaatgctggc	ccgaggccag	ccccctcccc	aaacgctgca	gcttgcagtc
841	caggggaaaa	ggacgttgcc	tggcttgacg	caacaacagc	agcagcaaca	gcagcagcag
901	cagcagcagc	agcagcagca	gcagcagcaa	cagcagccgc	agcagcagcc	gcccacaacca
961	cagacgcagc	aacaacagca	gccggccctt	gttaactaca	acagaccatc	tggcccgggg
1021	ccggagctga	gcggcccag	ccccccgag	aagctgccc	tggccgcgccc	cgccggccgg
1081	ccctcgcccc	cgccccccgc	agcccgccag	ccgcccgcgg	ccgcagtgcc	cgggccctca
1141	gtgcccagc	agcagccggg	gcagccctcg	cccgtcctcc	agctgcagca	gaagcagagc
1201	cgcatcagcc	ccatccagaa	accgcaaggc	ctggaccccc	tggaaattct	gcaagagcgg
1261	gaatacagac	ttcaggcccc	catagctcat	aggatacaag	aactggaaaa	tctgctgggc
1321	tccttgccac	cagatttaag	aaccaaagca	accgtggaac	taaaagcact	tcggttactc
1381	aatttccagc	gtcagctgag	acaggaggtg	gtggcctgca	tgcgcaggga	cacgaccctg
1441	gagacggctc	tcactctcaa	agcatacaaa	cgagcagagc	gccagactct	gagagaagct
1501	cgcatgaccc	agaactctgga	gaagcagcag	aagattgagc	aggagaggaa	acgcgctcag
1561	aaacaccagg	aatacctgaa	cagcattttg	caacatgcaa	aagattttaa	ggaatatcat
1621	cggtctgtgg	ccggaagat	ccagaagctc	tccaaagcag	tggcaacttg	gcattgccaac
1681	actgaaagag	agcagaagaa	ggagacagag	cggattgaaa	aggagagaat	gcggcgactg
1741	atggctgaag	atgaggaggg	ttatagaaaa	ctgattgac	aaaagaaaaga	caggcgttta
1801	gcttaccttt	tcagcagac	cgatgagtat	gtagccaatc	tgaccaatct	ggtttgggag
1861	cacaagcaag	cccagcgagc	caaagagaag	aagaagagga	ggaggaggaa	gaagaaggct
1921	gaggagaatg	cagaggggtg	ggagtctgcc	ctgggaccgg	atggagagcc	catagatgag
1981	agcagccaga	tgagtgaact	ccctgtcaaa	gtgaccacac	cagaaaaccgg	caaggttctg
2041	ttcggaccag	agcaagtcag	ctggagccct	ggctggaaat	gaatcctggt	gaatcctggt
2101	tatgaagtgg	ccctagatc	tgacagtga	gagagtgaat	ctgattatga	ggaagaggat
2161	gaggaagaag	agtccagtag	gcaggaaacc	gaagagaaaa	tactcctgga	tccaaatagc
2221	gaagaagttt	ctgagaagga	tgctaagcag	atcattgaga	cagctaagca	agacgtggat
2281	gatgaataca	gcattcgagta	cagtgcacag	ggctcccagt	cctaactacac	cgtggctcat
2341	gccatctcgg	agaggggtga	gaaacagctc	gccctcctaa	ttaatgggac	cctaagacat
2401	taccagcttc	agggcctgga	atggatgggt	tccctgtata	ataacaact	gaacgggaatc
2461	ttagccgatg	aaacggggct	tggaaagacc	atacagacca	ttgcactcat	cacttatctg
2521	atggagcaca	aaagactcaa	tggcccctat	ctcatcattg	ttcccctttc	gactctatct
2581	aactggacat	atgaatttga	caaattgggt	ccttctgtgg	tgaagatttc	ttacaagggt
2641	actcctgcca	tcgctcgctc	ccttgtcccc	cagctacgga	gtggcaaat	caatgtcttc
2701	ttgactactt	atgagtatat	tataaaagac	aagcacattc	ttgcaagat	tcgggtggaa
2761	tacatgatag	tgagcgaagg	ccaccgaatg	aagaatcacc	actgcaagct	gactcaggtc
2821	ttgaacactc	actatgtggc	ccccagaagg	atcctcttga	ctgggacccc	gctgcagaat
2881	aagctccctg	aactctgggc	cctcctcaac	ttcctcctcc	caacaatttt	taagagctgc
2941	agcacatttg	aaacatgggt	caatgctcca	tttgccatga	ctgggtgaaa	gggtgactta
3001	aatgaagaag	aaactatatt	gatcatcagg	cgtctacata	aggtgttaag	accattttta
3061	ctaaggagac	tgaagaaaga	agttgaatcc	cagcttcccc	aaaaagtggg	atatgtgatc
3121	aagtgtgaca	gtcagctctc	gcagaagatt	ctgtatgcgc	atatgcaagc	caaggggatc
3181	cttctcacag	atgggtctga	gaaagataag	aaggggaaag	gaggtgctaa	gacacttatg
3241	aacactatta	tgacgttgag	aaaaatctgc	aaccaccocat	atatgtttca	gcacattgag
3301	gaatcctttg	ctgaacacct	aggctattca	aatgggggtc	tcaatggggc	tgaactgtat
3361	cgggcctcag	ggaagtgtga	gctgcttgat	cgtattctgc	caaaattgag	agcgactaat
3421	caccagagctc	tgtctttctg	ccagatgaca	tccctcatga	ccatcatgga	ggattatttt
3481	gcttttcgga	actctcttta	cctacgcctt	gatggcacca	ccaagtctga	agatcgtgct
3541	gcttttgctga	agaaattcaa	tgaacctgga	tccagattat	tcattttctt	gctgagcaca
3601	agagctgggtg	gcttggggtt	aaatcttcag	gcagctgata	cagtggtcat	ctttgacagc
3661	gactggaatc	ctcatcagga	tctgcaggcc	caagaccgag	ctcacccgat	cgggagcagc
3721	aacgaggtcc	gggtactgag	gctctgtacc	gtgaacagcg	tggaggaaaa	gatcctcgcg

TABLE 1-continued

3781	gccgcacaaat	acaagctgaa	cgtggatcag	aaagtgatcc	aggcgggcat	gtttgaccaa
3841	aagtcttcaa	gccacgagcg	gagggcattc	ctgcaggcca	tcttgagca	tgaggaggaa
3901	aatgaggaag	aagatgaagt	accggacgat	gagactctga	accacatgat	tgctcgacga
3961	gaagaagaat	ttgacctttt	tatgcggatg	gacatggacc	ggcggaggga	agatgcccg
4021	aaccggaac	ggaagccccg	tttaatggag	gaggatgagc	tgccctcctg	gatcattaag
4081	gatgacgctg	aagtagaaag	gctcacctgt	gaagaagagg	aggagaaaat	atttgggagg
4141	gggtcccgcc	agcgccgtga	cgtggactac	agtgcgccc	tcacggagaa	gcagtggcta
4201	agggccatcg	aagacggcaa	ttggaggaa	atggaagagg	aagtacggct	taagaagcga
4261	aaaagacgaa	gaaatgtgga	taaagatcct	gcaaaagaag	atgtggaaaa	agctaagaag
4321	agaagaggcc	gccctcccg	tgagaaactg	tcaccaaatc	cccccaact	gacaaagcag
4381	atgaacgcta	tcacgatac	tgtgataaac	tacaaagata	ggtgtaacgt	ggagaaggtg
4441	cccagtaatt	ctcagttgga	aatagaagga	aacagttcag	ggcgacagct	cagtgaagtc
4501	ttcatttcagt	taccttcaag	gaaagaatta	ccagaatact	atgaattaat	taggaagcca
4561	gtggatttca	aaaaataaaa	ggaaaggatt	cgtaatacata	agtaccggag	cctaggcgac
4621	ctggagaag	atgtcatgct	tctctgtcac	aacgctcaga	cgttcaacct	ggagggatcc
4681	cagatctatg	aagactccat	cgtcttacag	tcagtgttta	agagtgcctg	gcagaaaatt
4741	gccaaagagg	aagagagtga	ggatgaaagc	aatgaagagg	aggaagagga	agatgaagaa
4801	gagtcagagt	ccgaggcaaa	atcagtcag	gtgaaaatta	agctcaataa	aaaagatgac
4861	aaaggccggg	acaaaggcaa	aggcaagaaa	aggccaaatc	gaggaaaagc	caaacctgta
4921	gtgagcgatt	ttgacagcga	tgaggagcag	gatgaacgtg	aacagtcaga	aggaagtggg
4981	acggatgatg	agtgatcagt	atggaccttt	ttccttggtg	gaactgaatt	ccttcctccc
5041	ctgtctcatt	tctaccaggt	gagttcattt	gtcatatagg	cactgggttg	ttctatatc
5101	atcatcgctt	ataaactagc	tttaggatag	tgccagacaa	acatatgata	tcattggtga
5161	aaaaacacac	acatacacaa	atatttgtaa	catattgtga	ccaaatgggc	ctcaagatt
5221	cagattgaaa	caaacacaaa	gcttttgatg	gaaaatatgt	gggtggatag	tatatattca
5281	tgggtgggtc	taatttggta	acggtttgat	tgtgcctggt	tttatcacct	gttcagatga
5341	gaagattttt	gtcttttcta	gcactgataa	ccaggagaag	ccattaaaag	ccactgggtta
5401	ttttattttt	ctccaggcaa	ttttcgaggt	ttttatttgt	tcggtattgt	ttttttacac
5461	tgtggtacat	ataagcaact	ttaatagggt	ataaatgtac	agtagttaga	tttcacctgc
5521	atatacattt	ttccatttta	tgctctatga	tctgaacaaa	agctttttga	attgtataag
5581	atttatgtct	actgtaaac	ttgcttaatt	ttttgtctct	tgatttaaaa	aaaagttttg
5641	ttgaaagcgc	tattgaatat	tgcaatctat	atagtgtatt	ggatggcttc	ttttgtcacc
5701	ccgatctcct	atgttaccac	tgtgtatcgt	ctccttctcc	ctaaagtgtg	cttaattctt
5761	gctttctttg	cacaatgtct	ttggttgcaa	gtcataagcc	tgaggcaaat	aaaattccag
5821	taatttcgaa	gaatgtgggt	ttggcgcttt	cctaataaag	aaataattta	gcttgacaaa
5881	aaaaaaaaaa	aa				

SEQ ID NO: 73 Human BRM Amino Acid Sequence Isoform A (NP_003061.3)

1	msrptdpgam	phpgpspgpg	pspgpilggs	pgpgpspgsv	hsmmgpspgp	psvshpmpmtm
61	gstdfpgegm	hgmhkpidgi	hdkgivedih	cgsmkgtgmr	pphpgmgppp	spmdqhsqgy
121	msphpsplga	pehvsspmgs	ggptppqmpg	sqpgalipgd	pqamsqpnrg	pspfspvqlh
181	qlraqilayk	mlarggplpe	tlqlavqgkr	tlpglqqqqq	qqqqqqqqqq	qqqqqqqqqp
241	qgppqpgtqg	qgdpalvny	rspgpgpels	gpstpqklpv	papggrrpsa	ppaaaqppaa
301	avpgpsvpqp	apggpspvlq	lqqkqsrisp	iqkpqgldpv	eilqereyrl	qariahrige
361	lenlpgslpp	dltkatvel	kairllnfqr	qlrqevvacm	rrdtltetal	nskeykrskr
421	qtlrearnte	klekqgkieq	erkrrqkhqe	ylnsilqhak	dfkeyhrsra	gkiqklskav
481	atwhantere	qkketeriek	ermrllmaed	eegyrklidq	kkdrllayll	qqtdeyvanl
541	tnlvwehkqa	qaakekkkrr	rrkkkaeena	eggesalpgd	gepidessqm	sdipkvvtht
601	etgkvlfgpe	apkasqlidaw	lemmpgyeva	prsdseeds	dyeedeese	ssrgeteeki
661	lldpnseevs	ekdakqiet	akqdvdeys	mqsarsqs	yytvahaise	rveksallli
721	ngtlkhyqlq	glewmvslyn	nnlngilade	mglgkltgti	alilylmehk	rlngpyliiv
781	plstlsnwtg	apkwapsv	kisykgtpam	rrslvpqlrs	gkfnvlltty	eyiikdkhil
841	akirwkymiv	deghrmknhh	ckltqvlnth	yvaprrillt	gtplqnlkpe	lwallnflfp
901	tifkscstfe	qwnapfam	gervdlneee	tiliirrlhk	vlrpfllrrl	kkevesqlpe
961	kveyvikcdm	salqkilyrh	mqakgilltd	gsekdkkgkg	gaktlmntim	qlrkicnhpy
1021	mfqhiesefa	ehlgysngvi	ngaelyrasg	kfelldrillp	klratnhrvl	lfcqmrslmt
1081	imedyafrn	flylrlldgt	ksedraallk	kfnepgsqyf	ifllstragg	lglnlqaadt
1141	vvifdsdwnp	hqdlqagdra	hrigqgnevr	vlrlctvns	eeekilaaak	klndvqkviq
1201	agmfdqkss	herraflgai	leheeneeee	devpddetln	qmiaarreeef	dlfmrmdmdr
1261	rredarnpkr	kprlmeedel	pswiikddae	verltceeee	ekifgrgsrq	rrdvdsdal
1321	tekqwlraie	dgnleemeee	vrllkrkrrr	nvdkdaked	vekakrrrg	ppaeklsnp
1381	pkltkqmnai	idtvinykdr	cnvekvpsns	gleiegnssg	qrlsevfqi	psrkelpyey
1441	elirkpvdvk	kikerirnhk	yrsldglekd	vmlchnaqt	fnlegsqiye	dsivqlsvfk
1501	sargkiake	esedesneee	eeedeesees	eaksvkvkik	lnkkddkgrd	kgkgkkrpnr
1561	gkakpvvsdf	dsdeeqdere	qseegstdd			

SEQ ID NO: 74 Human BRM cDNA Sequence Variant 2 (NM_139045.3, CDS: from 223 to 4941)

1	gcgtcttccg	gcgcccggcg	aggaggcgag	gggtgggacgc	tgggcggagc	ccgagtttag
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121	ctgtttctgt	actctgggtg	actcagagag	ggaagagatt	cagccagcac	actcccccg
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301	cttgggccta	gtccaggacc	aggaccatcc	ccagggtccg	tcacagcat	gatggggcca
361	agtctctggc	ctccaagtgt	ctcccatcct	atgccgacga	tgggggtccac	agacttccca
421	caggaaaggc	tgcatcaaat	gcataagccc	atcgatggta	tacatgacaa	ggggattgta
481	gaagacatcc	attgtggatc	catgaaggcg	actggtatgc	gaccacctca	ccaggcatg
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601	ccattaggag	ccccagagca	cgtctccagc	cctatgtctg	gaggaggccc	aactccacct
661	cagatgccac	caagccagcc	gggggcccct	atcccagggt	atccgcaggc	catgagccag

TABLE 1-continued

721	cccaacagag	gtccctcacc	tttcagtcct	gtccagctgc	atcagcttcg	agctcagatt
781	ttagcttata	aaatgctggc	ccgagggcag	cccctccccg	aaacgctgca	gcttgcagtc
841	caggggaaaa	ggacgttgcc	tggtttgcag	caacaacagc	agcagcaaca	gcagcagcag
901	cagcagcagc	agcagcagca	gcagcagcaa	cagcagccgc	agcagcagcc	gcgcgaacca
961	cagcagcagc	aacaacagca	gcgggcccctt	gttaactaca	acagaccatc	tgggccgggg
1021	ccggagctga	ggggcccgag	caccccgag	aagctgcccg	tgcccgcgcc	cgccggccgg
1081	ccctcgcccc	cgccccccgc	agccgcgcag	ccgcccgcgg	ccgcaagtgc	cgggccctca
1141	gtgcgcgagc	cgccccccgg	gcagccctcg	cccgtcctcc	agctgcagca	gaagcagagc
1201	cgcatcagcc	ccatccagaa	accgcaaggc	ctggaccocc	tggaattct	gcaagagcgg
1261	gaatacagac	ttcaggcccc	catagctcat	aggatacaag	aactggaaaa	tctgctggc
1321	tctttgccac	cagatttaag	aaaccaagca	accgtggaac	taaaagcact	tcggttacc
1381	aatttccagc	gtcagctgag	acaggaggtg	gtggccctgca	tgccgaggga	cacgaccctg
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1561	aaacaccagg	aatccctgaa	cagtattttg	caacatgcaa	aagattttta	ggaatatcat
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1921	gaggagaatg	cagaggggtg	ggagtctgcc	ctgggaccgg	atggagagcc	catagatgag
1981	agcagccaga	tgagtgacct	ccctgtcaaa	gtgactcaca	cagaaaccgg	caaggttctg
2041	ttcggaccag	aagcaccocaa	agcaagtca	ctggacgcct	ggctggaaat	gaatcctgg
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2281	gatgaatata	gcattgcagt	cagtgcacgg	ggctcccagt	cctactacac	cgtggctcat
2341	gccatctcgg	agaggggtga	gaaacagctc	gccctcctaa	ttaattggga	cctaagcat
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2821	ttgaacactc	actatgtggc	ccccagaagg	atcctcttga	ctgggacccc	gctgcagaat
2881	aagctccctg	agctctgggc	cctcctcaac	ttcctcctcc	caacaatttt	taagagctgc
2941	agcacatttg	aacaatgggt	caatgctcca	tttgccatga	ctgggtgaa	gggtgactta
3001	aatgaagaag	aaactatatt	gatcatcagg	cgtctacata	aggtgttaag	accattttta
3061	ctaaggagag	tgaagaagaa	agttgaatcc	cagcttcccc	aaaaagtgga	atatgtgatc
3121	aagtgtgaca	tgctcagctc	gcagaagatt	ctgtatcgcc	atagcaagc	caaggggac
3181	cttctcacag	atgggtctga	gaaagataag	aaggggaaag	gaggtgtctaa	gacacttatg
3241	aacactatta	tgctttttcg	aaaaatctgc	aaccacccat	atagtttca	gcacattgag
3301	gaatcctttg	ctgaacacct	aggctattca	aatgggttca	tcaatggggc	tgaactgtat
3361	cgggccctcag	ggaagtttga	gctgcttgat	cgtatcctgc	caaaattgag	agcgactaat
3421	caccgagtg	tgctttttcg	ccagatgaca	tctctcatga	ccatcatgga	ggattatttt
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3841	aagtcttcaa	gccacgagcg	gagggcattc	ccgcaggcca	tcttgagaca	tgaggaggaa
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3961	gaagaagaat	ttgacctttt	tatgcggatg	gacatggacc	ggcgaggagg	agatgcggcg
4021	aaccgcaaac	ggaagccccg	tttaattggag	gaggtatgag	tgccccctcg	gatcattaa
4081	gatgacgctg	aagttagaaag	gctcacctgt	gaagaagagg	aggagaaaa	atttgggagg
4141	gggtcccgcc	agcgccgtga	cgtggactac	agtgcagccc	tcacggagaa	gcagtggtca
4201	agggccatcg	aagacggcaa	tttgaggagaa	atggaagagg	aagtacggct	taagaagcga
4261	aaaagacgaa	gaaatgtgga	taaagatcct	gcaaaagaag	atgtggaaaa	agctaagaag
4321	agaagaggcc	gccctcccg	tgagaaactg	tcaccaaatc	cccccaact	gacaaagcag
4381	acgaacgcta	tcacgatac	tgtgataaac	tacaaagata	gttcaggggc	acagctcagt
4441	gaagtcttca	ttcagttacc	ttcaaggaaa	gaattaccag	aatactatga	attaattagg
4501	aagccagtg	atttcaaaaa	aataaaggaa	aggattctga	atcataagta	ccggagccta
4561	ggcgacctgg	agaaggtatg	catgcttctc	tgtcacacgc	ctcagacgtt	caacctggag
4621	ggatcccgca	tctatgaaga	ctccatcgct	ttacagtcag	tgtttaagag	tgccccggcg
4681	aaaattgcca	agaggaagaa	gagtgaggat	gaaagcaatg	aagaggagga	agaggaaagt
4741	gaagaagagt	cagagtccga	ggcaaaatca	gtcaagggtga	aaattaaagt	caataaaaaa
4801	gatgacaaag	gccgggacaa	agggaaaagg	aagaaaaggc	caaatcgagg	aaaagccaaa
4861	cctgtagtga	gcgatttcga	cagcgatgag	gagcaggatg	aacgtgaaca	gtcagaagga
4921	agtgggacgg	atgatgagtg	atcagtatgg	acctttttcc	ttggtagaac	tgaattcctt
4981	cctccctctg	ctcattttca	cccagtgagt	tcattttgtca	tataggcact	gggttggttc
5041	tatatcatca	tcgtctataa	actagcttta	ggatagtgcc	agacaacat	atgatatac
5101	gggtgtaaaa	acacacacat	acacaaatat	ttgcaacata	ttgtgaccaa	atgggcctca
5161	aagattcaga	ttgaaacaaa	caaaaagctt	ttgatggaaa	atatgtgggt	ggatagtata
5221	ttctatggg	tggtgtctaat	ttggtaacgg	tttgattgtg	cctgggttta	tcacctgttc
5281	agatgagaag	atttttctct	ttttagcac	tgataaccag	gagaagccat	taaaagccac
5341	tggttatttt	atttttctc	aggcaatttt	cagaggtttt	atgtgttcgg	tattgctttt
5401	ttacactgtg	gtacatatata	gcaactttaa	taggtgataa	atgtacagta	gtcagatttc
5461	acctgcatat	acattttttc	attttatgct	ctatgatctg	aacaaaagct	ttttgaactg

TABLE 1-continued

5521	tataagattt	atgtctactg	taaacattgc	ttaatttttt	tgctcttgat	ttaaaaaaaa
5581	gctttgttga	aagcgctatt	gaatattgca	atctatatag	tgtattggat	ggcttctttt
5641	gtcaccctga	tctcctatgt	taccaatgtg	tatcgtctcc	tctccctaa	agtgacttta
5701	atcttttgc	tctttgcaca	atgtctttgg	ttgcaagtca	taagcctgag	gcaataaaaa
5761	ttccagtaat	ttcgaagaat	gtggtgttgg	tgctttccta	ataaagaaat	aatttagctt
5821	gacaaaaaaa	aaaaaaaa				

SEQ ID NO: 75 Human BRM Amino Acid Sequence Isoform B (NP_620614.2)

1	mstptdpgam	phpgpspgps	pspgpilgps	pgpgpspgsv	hsmmgpspgp	psvshpmpm
61	gstdfpgeg	hgmhkpigdi	hdkgivedih	cgsmkgtgmr	pphpgmgspp	spmdqhsqgy
121	msphpsplga	pehvspsmg	ggptppqmp	sqpgalipgd	pqamsqpnrg	pspfspvqlh
181	qlraqilayk	mlargqplpe	tlqlavqgkr	tlpglqqqqq	qqqqqqqqqq	qqqqqqqqpq
241	qgppqpgtqg	qqqpalvny	rpqpgpels	gpstpqklpv	papggpspa	ppaaagppaa
301	avpgpsvpqg	apqpspvlq	lqqkqsrsp	iqkpqglpv	eilqereyrl	qariahriq
361	lenlpgslpp	dlrtkatvel	kalrlnlfr	qlrqevvacm	rrdtltetal	nskaykrskr
421	qtlrearmte	klekqgkieq	erkrrqkhqe	ylnsilqhak	dfkeyhrsva	gkiqklskav
481	atwhantere	qketeriek	ermrlmaed	eegyrklidq	kkdrllayll	qqtdeyvanl
541	tnlvwehkga	qaakekkkrr	rrkkkaeena	eggesalpgd	gepidessgm	sdlpvkvtht
601	etgkvlfgpe	apklasqldaw	lemmpgyeva	prsdseesds	dyeedeeree	ssrgeteeki
661	lldpnseevs	ekdakqiet	akqdvdeys	mqysargsgs	yytvahaise	rvekqsalli
721	ngtlkhyqlq	glewmvslyn	nnlmgilade	mglgkttiqt	alitylmehk	rlngpyliiv
781	plstlsnwtg	etdkwapsv	kisykgtpam	rrslvpqlrs	gkfnvlltty	eyiikdkhil
841	akirwkymiv	deghrmknhh	ckltqvlnth	yvaprrillt	gtplqnlkpe	lwallnflfp
901	tifkscstfe	qwnapfam	gervdlnnee	tiliirrlhk	vlrpfllrrl	kkevesqlpe
961	kveyvikcdm	salqkilyrh	mqakgilltd	gsekdkkgkg	gaktlmntim	qlrkicnhpy
1021	mfqhieesfa	ehlgysngvi	ngaelyrasg	kfelldrillp	klratnhrvl	lfcqmtsmt
1081	imedyafrn	flylrlldgt	ksedraallk	kfnepgsqyf	ifllstragg	lglnlqaadt
1141	vvifdsdwnp	hqdlqagdra	hrigqgnevr	vlrlctvnsv	eeakilaaaky	klndvqkviq
1201	agmfdqkss	herraflgai	leheeneeee	devpddetln	qmiaarreeef	dlfmrmdmdr
1261	rredarnpkr	kprlmeedel	pswiikddae	verltceeee	ekifgrgsrq	rrdvdsdal
1321	tekqwlraie	dgnleemeee	vrllkrkrrr	nvdkdaked	vekakrrrr	ppaelksnp
1381	pkltkqmnai	idtvinykds	sgrqlsevf	qlpsrkelp	yyelirkpvd	fkikierirn
1441	hkyrsldgle	kdvmlchna	qtnflegsqi	yedsivlqsv	fksarqkiak	eeesedesne
1501	eeeeeeeee	esavskvkv	iklnkkddkg	rdkgkgkkkrp	nrgkakpvvs	dfdsdeeqde
1561	reqsegsgtd	de				

SEQ ID NO: 76 Human BRM cDNA Sequence Variant 3 (NM_001289396.1, CDS: from 210 to 4982)

1	tcagaagaaa	gccccgagat	cacagagacc	cggcgagatc	acagagaccc	ggcctgaagg
61	aacgtggaaa	gaccaatgta	cctgttttga	cgggttgcc	ggagcaagaa	gttcagttg
121	gggagaattt	tcagaagata	aagtcggaga	ttgtggaaag	acttgacttg	cagcattact
181	ctactgactg	gcagagacag	gagaggtaga	tgctccacgc	cacagaccct	gggtgcgatg
241	cccaccaggg	gccttcggcg	gggctggggc	cttccctgg	gccaatctt	gggctagtc
301	caggaccagg	accatcccca	ggttcogtcc	acagcatgat	ggggccaagt	cctggacctc
361	caagtgtctc	ccatcctatg	ccgacgatgg	gggtccacaga	cttccacacg	gaaggcatgc
421	atcaaatgca	taagcccatc	gatggtatac	atgacaaggg	gattgtagaa	gacatccatt
481	gtggatccat	gaagggcact	ggtatgagac	cacctcacc	aggcatgggc	cctccccaga
541	gtccaatgga	tcaacacagc	caaggttata	tgctaccaca	cccatctcca	ttaggagccc
601	cagagcacgt	cccccgccct	atgtctggag	gaggcccaac	tccacctcag	atgccacca
661	gccagccggg	ggcctctatc	ccaggtgatc	cgcaggccat	gagccagccc	aacagaggtc
721	cctcaccttt	cagtcctgtc	cagctgcac	agcttcgagc	tcagatttta	gcttataaaa
781	tgctggcccg	aggccagccc	ctccccgaaa	cgtgcagct	tgagtcacag	gggaaaagga
841	cgttgctctg	cttgacagaa	caacagcagc	agcaacagca	gcagcagcag	cagcagcagc
901	agcagcagca	gcagcaacag	cagccgcagc	agcagccgcc	gcaaccacag	acgcagcaac
961	aacagcagcg	ggcctctgtt	aactacaaca	gacctctgg	ccggggcgcg	gagctgagcg
1021	gcccgagcac	cccgagaaag	ctgccggtgc	ccgcgcccg	cggccggccc	tcgcccgcg
1081	cccccgagc	cgccgagccg	cccgccggcg	cagtgcccg	gcctcagtg	ccgcagccgg
1141	ccccgggggc	gcctcgcccc	gtcctccagc	tgacagagaa	gcagagccgc	atcagcccca
1201	tccagaaacc	gcaaggcctg	gaccccgctg	aaattctgca	agagcgggaa	tacagacttc
1261	aggcccgcat	agctcatagg	atacaagaac	tggaaaatcc	gcctggctct	ttgccaccag
1321	atttaagaac	caaaagcaac	gtggaactaa	aagcaactcg	gttactcaat	ttccagcgtc
1381	agctgagaca	ggaggtggtg	gcctgcacgc	gcagggacac	gacctgggag	acggctctca
1441	actccaaagc	atacaaacgg	agcaagcgcc	agactctgag	agaagctcgc	atgaccgaga
1501	agctggagaa	gcagcagaa	attgagcagg	agaggaaacg	ccgtcagaaa	caccagggaat
1561	acctgaacag	tattttgcaa	catgcaaaa	attttaagga	atatcatcgg	tctgtggccg
1621	gaaagatcca	gaagctctcc	aaagcagtg	caacttgcca	tgccaacact	gaaagagagc
1681	agaagaagga	gcagagcggg	attgaaaagg	agagaatgag	gcgactgatg	gctgaagatg
1741	aggagggtta	tagaaaaactg	attgatcaaa	agaaagacag	gcgttttagct	taccttttgc
1801	agcagaccga	tgagtatgta	gccaatctga	ccaatctggt	ttgggagcac	aagcaagccc
1861	aggagcccaa	agaggaagaa	aagaggagga	ggagggaaga	gaaggctgag	gagaatgcag
1921	agggtgggga	gtctgcccgt	ggaccggatg	gagagcccat	agatgagagc	agccagatga
1981	gtgacctccc	tgtcaaatgt	actcacacag	aaaccggcaa	ggttctgttc	ggaccagaag
2041	caccacaagc	agctcagctg	gacgcctggc	tggaaatgaa	tcctggttat	gaagttgccc
2101	ctagatctga	cagtgaaag	agtgattctg	attatgagga	agaggatgag	gaagaagagt
2161	ccagtaggca	ggaaccggaa	gagaaaaatc	tcctggatcc	aaatagcgaa	gaagtttctg
2221	agaaggtatg	taagcagatc	attgagacag	ctaagcaaga	cgtggacgat	gaatcacaga
2281	tgacgtacag	tgccaggggc	tcccagtcct	actacaccgt	ggctcatgcc	atctcggaga
2341	gggtggagaa	acagctctgcc	ctcctaatta	atgggaccct	aaagcattac	cagctccagg
2401	gcctgggaatg	gatggtttcc	ctgtataata	acaacttgaa	cggaaatcta	gccgatgaaa
2461	tggggcttgg	aaagaccata	cagaccattg	cactcatcac	ttatctgatg	gagcacaaaa

TABLE 1-continued

2521	gactcaatgg	cccttatctc	atcattgttc	ccctttcgac	tctatctaac	tggacatacg
2581	aatttgacaa	atgggctcct	tctgtggtga	agattttcta	caaggggtact	cctgccatgc
2641	gtcgctccct	tgtcccccag	ctacggagtg	gcaaattcaa	tgtcctcttg	actacttatg
2701	agtatattat	aaaagacaag	cacattcttg	caaagattcg	gtggaaatac	atgatagtgg
2761	acgaaggcca	ccgaatgaag	aatcaccact	gcaagctgac	tcaggctctg	aacactcact
2821	atgtggcccc	cagaaggatc	ctcttgactg	ggaccccgc	gcagaataag	ctccctgaac
2881	tctgggcccc	cctcaacttc	ctcctcccaa	caatttttaa	gagctgcagc	acatttgaac
2941	aatgggtcaa	tgtccattt	gccatgactg	gtgaaagggt	ggacttaaat	gaagaagaaa
3001	ctatattgat	catcaggcgt	ctacataagg	tgttaagacc	atttttacta	aggagactga
3061	agaaagaagt	tgaattcccag	cttcccga	aaagtgaata	tgtgatcaag	tgtgacatgt
3121	cagctctgca	gaagattctg	tatcgccata	tgcaagccaa	ggggatcctt	ctcacagatg
3181	gttctgagaa	agataagaag	gggaaaggag	gtgctaagac	acttatgaac	actattatgc
3241	agttgagaaa	aatctgcaac	cacctata	tgtttcagca	cattgaggaa	tcctttgctg
3301	aacacctagg	ctattcaaat	ggggtcatca	atggggctga	actgtatcgg	gcctcaggga
3361	agtttgagct	gcttgatcgt	attctgcca	aattgagagc	gactaatcac	cgagtgctgc
3421	ttttctgcca	gatgaacatc	ctcatgacca	tcattggagga	ttattttgct	tttcggaact
3481	tcctttacct	acgccttgat	ggcaccacca	agtctgaaga	tcgtgctgct	ttgctgaaga
3541	aattcaatga	acctggatcc	cagtatttca	ttttcttgct	gagcacaa	gctgggtgcc
3601	tgggcttaaa	ttctcaggca	gctgatacag	tgttcactct	tgacagcgac	tggaaatcctc
3661	atcaggatct	gcaggcccaa	gaccgagctc	accgcacatc	gcagcagaac	gaggtccggg
3721	tactgaggct	ctgtaccgtg	aacagcgtgg	aggaaaagat	cctcgcggcc	gcaaaataca
3781	agctgaacgt	ggatcacgaa	gtgatccagg	cgggcatggt	tgacaaaaag	tcctcaagcc
3841	acgagcggag	ggcattcctg	caggccatct	tggagcatga	ggaggaaaat	gaggaagaag
3901	atgaagtacc	ggacgatgag	actctgaacc	aaatgattgc	tcgacgagaa	gaagaatttg
3961	acctttttat	gcggatggac	atggaccggc	ggagggaaga	tgcccgaac	ccgaaccgga
4021	agccccgttt	aatggaggag	gatgagctgc	cctcccggt	cattaaggat	gacgctgaag
4081	tagaaaggct	cacctgtgaa	gaagaggagg	agaaaaatatt	tgggaggggg	tcgccccagc
4141	gccgtgacgt	ggactacagt	gacgccccta	cggagaagca	gtggctaagg	gccatcgaag
4201	acggcaattt	ggaggaaatg	gaagagggaag	tacggcttaa	gaagcgaaaa	agacgaagaa
4261	atgtggataa	agatcctgca	aaagaagatg	tggaaaaagc	taagaagaga	agagggccgc
4321	ctcccgtgga	gaaactgtca	ccaaatcccc	ccaaactgac	aaagcagatg	aacgctatca
4381	tcgatactgt	gataaactac	aaagataggt	gtaacgtgga	gaaggtgccc	agtaattctc
4441	agttggaat	agaaggaatc	agttcagggc	gacagctcag	tgaagtcttc	attcagttac
4501	cttcaaggaa	agaattacca	gaatactatg	aattaattag	gaagccagtg	gatttcaaaa
4561	aaataaagg	aaggattcgt	aatcataagt	accggagcct	aggcgacctg	gagaaggatg
4621	tcattgctct	ctgtcacca	gctcagacgt	tcaacctgga	gggatccacg	atctatgaag
4681	actccatcgt	cttaccgtca	gtgtttaaga	gtgcccggca	gaaaattgcc	aaagagggaag
4741	agagtggagg	tgaagcaat	gaagaggagg	aagagggaaga	tgaagaagag	tcagagtcgc
4801	agggcaaatc	agtcaagggt	aaaattaaagc	tcaataaaaa	agatgacaaa	ggccgggaca
4861	aaagggaagg	caagaaaagg	ccaaatcgag	gaaaagccaa	acctgtagtg	agcctatttg
4921	acagcgatga	ggagcaggat	gaacgtgaac	agtcagaagg	aagtgggacg	gatgatgagt
4981	gatcagcatg	gacctttttc	cttggtagaa	ctgaattcct	tcctcccctg	tctcatttct
5041	accagtgag	ttcatttgct	atataggcac	tgggttggtt	ctatctatc	atcgtctata
5101	aactagcttt	aggatagtgc	cagacaaaca	tatgatatca	tgggtgtaaaa	aacacacaca
5161	tacacaata	tttgtaacat	atttgtacca	aatgggcctc	aaagattcag	attgaacaaa
5221	acaaaaagct	tttgtgggaa	aatatgtggg	tggatagtat	atttctatgg	gtgggtctaa
5281	tttggttaacg	gtttgatatt	gcctgggttt	atcacctggt	cagatgagaa	gattttttgtc
5341	ttttgtagca	ctgataacca	ggagaagcca	ttaaaagcca	ctgggttattt	tatttttcat
5401	caggcaattt	tcgaggtttt	tatttggtcg	gtattgtttt	tttacctgt	ggtacatata
5461	agcaacttta	ataggtgata	aatgtacagt	agttagattt	cacctgcata	tacatttttc
5521	cattttatgc	tctatgatct	gaacaaaagc	tttttgaatt	gtataagatt	tatgtctact
5581	gtaaacattg	cttaattttt	ttgctcttga	tttaaaaaaa	agttttgttg	aaagcgtat
5641	tgaatattgc	aatctatata	gtgtattgga	tggcttcttt	tgtcacccctg	atctctatag
5701	ttaccaatgt	gtatcgtctc	cttctcccta	aagtgtactt	aatctttgct	ttctttgcac
5761	aatgtctttg	gttgcaagtc	ataagcctga	ggcaaataaa	attccagtaa	tttcgaagaa
5821	tgtggtgttg	gtgctttcct	aataaagaaa	taatttagct	tgacaaaaaa	aaaaaaaaa

SEQ ID NO: 77 Human BRM cDNA Sequence Variant 4 (NM_001289397.1, CDS:
from 223 to 4767)

1	gcgtcttccg	gcgcccgcgg	aggaggcgag	gggtgggacgc	tgggcggagc	ccgagtttag
61	gaagaggagg	ggacggctgt	catcaatgaa	gtcatattca	taatctagtc	ctctctccct
121	ctgtttctgt	actctgggtg	actcagagag	ggaagagatt	cagccagcac	actcctcgcg
181	agcaagcatc	actctactga	ctggcagaga	caggagaggt	agatgtccac	gcccacagac
241	cctgggtcga	tgcctcaccc	agggccttcg	ccggggcctg	ggccttcccc	tgggccaatt
301	cttgggccta	gtccaggacc	aggaccatcc	ccagggtccg	tcacagcat	gatggggcca
361	agtccctggc	ctccaagtgt	ctcccatcct	atgccgacga	tggggctccac	agacttccca
421	cagggaaggc	tgacccaat	gcataagccc	atcgatggta	tacatgacaa	ggggatgta
481	gaagacatcc	attgtggatc	catgaagggc	actggtatgc	gaccacctca	ccaggcagtg
541	ggccctcccc	agagtccaat	ggatcaacac	agccaagggt	acatgtcacc	acacccatct
601	ccattaggag	ccccagacga	cgtctccagc	cctatgtctg	gaggaggccc	aactgcacct
661	cagatgccac	caagccagcc	gggggcccctc	accccagggtg	atccgcaggc	catgagccag
721	cccaacagag	gtccctcacc	tttcagtcct	gtccagctgc	atcagcttcg	agctcagatt
781	ttagcttata	aaatgtggc	ccgaggccag	cccctccccg	aaacgctgca	gcttgacgtc
841	caggggaaaa	ggacgttgcc	tggcttgacg	caacaacagc	agcagcaaca	gcagcagcag
901	cagcagcagc	agcagcagca	gcagcagcaa	cagcagccgc	agcagcagcc	gccgcaacca
961	cagacgcagc	aacaacagca	gcggccctt	gttaactaca	acagaccatc	tggcccgggg
1021	ccgagctgga	gcggcccgag	caccccgag	aagctgcggg	tgcccgcgcc	cggcgcccg
1081	ccctgcggcg	cgcccccgcc	agccgcgcag	ccgcccgcgg	ccgcagtgcc	cgggccccta
1141	gtgcgcgagc	cgggcccggg	gcagccctcg	ccgctcctcc	agctgcagca	gaagcagagc
1201	cgcacgcagc	ccatccagaa	accgcaaggc	ctggaccccg	tggaaactct	gcaagagcgg

TABLE 1-continued

1261	gaatacacagac	ttcagggcccc	catagctcat	aggatacaag	aactggaaaa	tctgcctggc
1321	tctttggccac	cagattttaag	aaccaaaagca	accgtggaac	taaaagcact	tcggttactc
1381	aattttccagc	gtcagctgag	acaggagggtg	gtggcctgca	tgcgcaggga	cacgaccctg
1441	gagacggctc	tcaactccaa	agcatacaaa	cggagcaagc	gccagactct	gagagaagct
1501	cgcatacgcc	agaagctgga	gaagcagcag	aagattgagc	aggagaggaa	acgccgtcag
1561	aaacaccagg	aatacctgaa	cagtattttg	caacatgcaa	aagattttta	ggaatatcat
1621	cggctctgtg	ccggaaagat	ccagaagctc	tccaaagcag	tggcaacttg	gcatgccaac
1681	actgaaagag	agcagaagaa	ggagacagag	cggattgaaa	aggagagaat	gcggcgactg
1741	atggctgaag	atgaggagg	ttatagaaaa	ctgattgatc	aaaagaaa	caggcgttta
1801	gcttaccttt	tgcagcagac	cgatgagtat	gtagccaatc	tgaccaatct	ggtttgggag
1861	cacaagcaag	cccaggcagc	caaagagaag	aagaagagga	ggaggaggaa	gaagaaggct
1921	gaggagaatg	cagagggttg	ggagtctgcc	ctgggacccg	atggagagcc	catagatgag
1981	agcagccaga	tgagtgcact	ccctgtcaaa	gtgactcaca	cagaaaccgg	caagggtctg
2041	ttcggaccag	aagcacccaa	agcaagtcag	ctggacgcct	ggctggaaat	gaatcctggg
2101	tatgaagtgg	ccctcagatc	tgacagtga	gagagtgtat	ctgattatga	ggaagaggat
2161	gaggaagaag	agtcocagtag	gcaggaaaac	gaagagaaaa	tactcctgga	tccaaatagc
2221	gaagaagttt	ctgagaagga	tgctaagcag	atcattgaga	cagctaagca	agacgtggat
2281	gatgaataca	gcatacgagt	cagtgcacag	ggctcccagt	cctactacac	cgtggctcat
2341	gccatctcgg	agagggtgga	gaaacagtct	gccctcctaa	ttaatgggac	cctaagcat
2401	taccagctcc	agggcctgga	atggatgggt	tccctgtata	ataacaactt	gaacggaatc
2461	ttagccgatg	aaatggggct	tgaaaagacc	atacagacca	ttgcactcat	cacttatctg
2521	atggagcaca	aaagactcaa	tggccctat	ctcatcattg	ttccctttc	gacttatctg
2581	aactggacat	atgaatttga	caaatgggct	ccttctgtgg	tgaagatttc	ttacaagggt
2641	actcctgcca	tgctgcgctc	ccttgtcccc	cagctacgga	gtggcaaat	caatgtcctc
2701	ttgactactt	atgagtatat	tataaaagac	aagcacattc	ttgcaagat	tcggtggaaa
2761	tacatgatag	tgacgaaggg	ccaccgaatg	aagaatcacc	actgcaagct	gactcaggtg
2821	gacttaaatg	aagaagaagac	tatatgtatc	atcaggcgct	tacataaggt	gttaagacca
2881	ttttacttaa	ggagactgaa	gaaagaagtt	gaatccagc	ttcccgaaaa	agtggaaat
2941	gtgatcaagt	gtgacatgtc	agctctgcag	aagattctgt	atcgccatat	gcaagccaag
3001	gggactcttc	tcacagatgg	ttctgagaaa	gataagaagg	ggaaaaggag	tgctaagaca
3061	cttatgaaca	ccattatgca	gttgagaaaa	atctgcaacc	accatatat	gtttcagcac
3121	attgaggaat	cctttgtctg	acacctaggc	tattcaaatg	gggtcatcaa	tggggctgaa
3181	ctgtatcggg	cctcaggggaa	gtttgagctg	cttgatcgta	ttctgccaaa	attgagagcg
3241	actaatcacc	gagtgctgct	ttctgcccag	atgacatctc	tcagaccat	catggaggat
3301	tattttgctt	ttcggaaact	cctttaccta	cgcttctgat	gcaccaccaa	gtctgaagat
3361	cgtgctgctt	tgctgaagaa	attcaatgaa	cctggatccc	agtatttcat	tttcttctgt
3421	agcacaaag	ctgggtggcct	gggtttaa	cttcaggcag	ctgatacagt	ggtcatcttt
3481	gacagcgact	ggaatcctca	tcaggatctg	caggcccaag	accgagctca	ccgcatcggg
3541	cagcagaacg	aggtccgggt	actgaggctc	tgtaccgtga	acagcgtgga	ggaaaaagtc
3601	ctcgccggcc	caaaaatacaa	gctgaacgtg	gatcagaaa	tgatccagcc	gggcatgttt
3661	gaccaaaagt	cttcaagcca	cgagcggagg	gcattctctg	aggccatctt	ggagcatgag
3721	gaggaagagt	aggaagaaga	tgaagtacgg	gacgatgaga	ctctgaacca	aatgattgct
3781	cgacgagaga	gaggtcttga	cctttttatg	cggatggaca	tggaccggcg	gagggagagat
3841	gcccggaacc	cgaaacggaa	gccccgttta	atggaggagg	atgagctgcc	ctcctggatc
3901	attaaggatg	acgctgaagt	agaaaaggctc	acctgtgaag	aagaggaggga	gaaaatattt
3961	gggagggggt	ccgcgcagcg	ccgtgacgtg	gactacagtg	accgccctac	ggagaagcag
4021	tggctaaggg	ccatcgaaga	cggcaatttg	gaggaaatgg	aagagggaat	accgcttaag
4081	aagcgaaaaa	gacgaagaaa	tgtggataaa	gatcctgcaa	aagaagatgt	ggaaaaagct
4141	aagaagagaa	gagggcggcc	tcccgctgag	aaactgtcac	caaatcccc	caaaactgaca
4201	aagcagatga	acgctatcat	cgatactgtg	ataaactaca	aagatagttc	aggcgacacg
4261	ctcagtgaag	tcttcatcca	gttaccttca	aggaagaagat	taccagaata	ctatgaatta
4321	attaggaagc	cagttggattt	caaaaaata	aaggaaaagga	ttcgtaatca	taagtaccgg
4381	agcctaggcg	acctggagaa	ggatgtcatg	cttctctgtc	acaacgctca	gacgttcaac
4441	ctggaggggat	cccagatcta	tgaagactcc	atcgtcttac	agtcagtgtt	taagagtgcc
4501	cggcagaaaa	ttgctcaagaa	ggaagagagt	gaggatgaaa	gcaatgaaga	ggaggagagag
4561	gaagatgaag	aagagtcaga	gtccgaggca	aatcagtcga	agggtgaaat	taagctcaat
4621	aaaaaagatg	acaaaggccg	ggacaaaagg	aaaggcaaga	aaaggccaaa	tcgagggaaa
4681	gccaaacctg	tagtgagcga	ttttgacagc	gatgaggagc	aggatgaacg	tgaacagtcga
4741	gaagggaagt	ggacggatga	tgagtgatca	gtatggacct	ttttccttgg	tagaactgaa
4801	ttccttctct	ccctgtctca	tttctaccca	gtgagttcat	ttgtcatata	ggcactgggt
4861	tgtttctata	tcattcatcgt	ctataaacta	gctttaggat	agtgccagac	aaacatatga
4921	tatcatgggt	taaaaaacac	acacatacac	aatatttgt	aacatattgt	gaccaaattg
4981	gcctcaaa	ttcagattga	aacaaacaaa	aagcttttga	tggaaaatat	gtgggtggat
5041	agtatatttc	tatgggtggg	tctaatttgg	taacggtttg	attgtgcctg	gttttatcac
5101	ctgttcagat	gagaagattt	ttgtcttttg	tagcactgat	aaccaggaga	agccattaaa
5161	agccactggg	tattttattt	ttcatcaggg	aattttcgag	gtttttattt	gttcggtatt
5221	gtttttttac	actgtgggtac	atataagcaa	ctttaatagg	tgataaatgt	acagtattga
5281	gatttcacct	gcataatcat	ttttccattt	tatgctctat	gatctgaaca	aaagcttttt
5341	gaattgtata	agatttatgt	ctactgtaaa	cattgcttaa	tttttttggc	cttgatttaa
5401	aaaaaagttt	ttgtgaaagc	gctattgaat	attgcaatct	atatagtga	ttggatggct
5461	tcctttgtca	ccctgatctc	ctatgttacc	aatgtgtatc	gtctccttct	ccctaaagtg
5521	tacttaattct	ttgctttctt	tgcacaatgt	ctttgggttc	aagtcataag	cctgaggcaa
5581	ataaaattcc	agtaatttcg	aagaatgtgg	tgttgggtgc	ttcctaataa	agaaataatt
5641	tagcttgaca	aaaaaaaaaa	aaaa			

SEQ ID NO: 78 Human BRM Amino Acid Sequence Isoform C (NP_001276326.1)

1 mstptdpgam phpgpbgpg pspgpilggs ppgpbgpsv hsmmgpspgp psvshpmpm
61 gstdfpqegm hqmhkpidgi hdkgivedih csgmkgtgmr pphpgmmpg spmdqhsqgy
121 mspbpsplga pehvspmsg ggtpppqmp sgpgalipgd pqamsgpnrg pspfpvqlh
181 qlraqilayk mlargqplpe tlqlavqgkr tlpqlqqqqq qqqqqqqqqq qqqqqqqqqq

TABLE 1-continued

241	qppppqptqqt	qqqpalvny	rpssgpgpels	gpstpqklpv	papggrrpspa	ppaaaqppaa
301	avpgpsvvpq	apggpsvvlq	lqqkqsrisp	iqkpqgldpv	eilqereyrl	qariahriqe
361	lenlpgslpp	dlttkatvel	kalrlilnfr	qlrgevvacm	rrdtlletal	nskeykrskr
421	qtlrearmte	klekqgkieq	erkrrqkhqe	ylnsilqhak	dfkeyhrsva	gkiqklskav
481	atwhantere	qkketeriek	ermrrlmaed	eegyrklidq	kkdrriayll	qqtdeyvanl
541	tnlvvvhkqa	qaakekkkrr	rrkkkaeena	eggesalgp	gepidessqm	sdlpvkvtht
601	etgkvlfqpe	apkasqldaw	lemmpgyeva	prsdseesds	dyeedeedee	ssrqeteeki
661	lldpnseevs	ekdakqiet	akqdvdeys	mqysargsgs	yytvahaise	rvekqsalli
721	ngtlkhyqlq	glewmvslyn	nnlmgilade	mglgttiqti	alitylmehk	ringpyliiv
781	plstlsnwt	efdkwapsv	kisykgtpam	rrslvpqlrs	gkfnvltty	eyiikdkhil
841	akirwkymiv	deghrmknhh	ckltqvdlne	eetiliirrl	hkvlrpfillr	rlkkevesql
901	pekveyvikc	dmsalqkily	rhmqaqgill	tdgsekdkkg	kkgaktlmnt	imqlrkicnh
961	pymfqihees	faehlgysng	vingaelyra	sgkfellidri	Ipklratnhr	vllfcqmts1
1021	mtimedyfaf	rnflyrlldg	ttksedraal	lkkfnepgsq	yfifllstra	gglglnlqaa
1081	dtvvifdsdw	nphqdlqagd	rahrigqgne	vrvlrlctvn	sveekilaaa	kyklnvdqkv
1141	iqagmfdqgs	ssheraflq	aileheene	eedevpddet	lnqmiarree	efdlfmrmdm
1201	drredarnp	krkprlmeed	elpswiikdd	aeverltcee	eeekifgrgs	rqrdrvdysd
1261	altekwqlra	iedgnleeme	eevrlkkrkr	rrnvdkdpak	edvekakkrr	grppaekisp
1321	npplltkymn	efdkwapsv	kisykgtpam	rrslvpqlrs	gkfnvltty	eyiikdkhil
1381	rnhykyslqd	lekdvmlch	naqtfnllegs	qiyedsivlq	svfksarqki	akeeesedes
1441	neeeeeedee	eseseaksvk	vkiklnkkdd	kgrdkgkgkk	rpnrgkakpv	vsdfdsdeeq
1501	dereqsegsg	tdde				

SEQ ID NO: 79 Human BRM cDNA Sequence Variant 5 (NM_001289398.1, CDS:
from 203 to 949)

1	cttgagagagg	cggaggtgga	aacgatgcgc	aggagttggc	ttggggcttt	ttgtttgcgt
61	gtccctgttt	acctattcat	aatcatggat	cccctctgct	ttgtgatact	gtgaaccacg
121	cataacagca	attctttaca	ccaccgggtt	gagaagaagg	cgcttgaggg	tgactttctg
181	gacgtccgct	cacgcagtaa	agatgtggtt	ggccatcgaa	gacggcaatt	tggaggaaat
241	ggaagaggaa	gtacggctta	agaagcgaaa	aagacgaaga	aatgtggata	aagatcctgc
301	aaaagaagat	gtggaaaaag	ctaagaagag	aagaggccgc	ccctccgctg	agaaactgtc
361	accaaattccc	cccaaactga	caaagcagat	gaacgctatc	atcgatactg	tgataaaacta
421	caaagatagt	tcaggggcgac	agctcagtg	agtcttcatt	cagttacctt	caaggaaaaga
481	attaccagaa	tactatgaat	taattaggaa	gccagtggtt	ttcaaaaaaa	taaaaggaaag
541	gattcgtaat	cataagttacc	ggagcctagg	cgacctggag	aaggatgtca	tgcttctctg
601	tcacaacgct	cagacgttca	acctggaggg	atccccagatc	tatgaagact	ccatcgtctt
661	acagtcaagt	tttaagagtg	cccggcagaa	aattgccaaa	gaggaaagaga	gtgaggatga
721	aagcaatgaa	gaggaggaag	aggaagatga	agaagagtca	gagtcgagg	caaaatcagt
781	caagggtgaa	attaagctca	ataaaaaaga	tgacaaaagg	cgggacaaaag	ggaaaaggcaa
841	gaaaaggcca	aatcgaggaa	aagccaaaac	tgtagtgagc	gattttgaca	cgatgaggaa
901	gcaggatgaa	cgtgaacagt	cagaaggaag	tgggacggat	gatgagtgat	cagtatggac
961	ctttttctct	ggtagaactg	aattccttcc	tccctctgct	cattttctacc	cagttaggtt
1021	atttgtcata	tactcatgag	gttgtttcta	tatcatcatc	gtctataaac	tagcttttag
1081	atagtgcacg	acaaacatat	gatatacatg	tgtaaaaaac	acacacatac	acaaatattt
1141	gtaacatat	gtgacaaaat	gggcctcaaa	gattcagatt	gaaacaaaaca	aaaagctttt
1201	gatgggaaaat	tttgagggtg	atagatatatt	tctatgggtg	ggctctaatt	ggtaacggtt
1261	tgattgtgcc	tggttttatc	acctgttcag	atgagaagat	ttttgtcttt	tgtagcactg
1321	ataaccagga	gaagccatta	aaagccactg	gttattttat	ttttcatcag	gcaattttcg
1381	aggtttttat	tttggtcggt	ttgttttttt	acactgtggt	acataataag	aactttaata
1441	ggtgataaat	gtacagtagt	tagattttac	ctgcatatac	atttttccat	tttatgctct
1501	atgatctgaa	caaaaagctt	ttgaattgta	taagatttat	gtctactgta	aacattgctt
1561	aatatttttt	cttcgtgatt	aaaaaaaagt	ttgttgtaa	gcgctattga	atattgcaat
1621	ctatatagtg	tattggatgg	cttcttttgt	cacctgatc	tcctatgtta	ccaatgtgta
1681	tcgtctcctt	ctccctaaag	tgtacttaat	ctttgctttc	tttgccacaat	gtcctttggt
1741	gcaagtcata	agcctgaggg	aaataaaaat	ccagtaattt	cgaagaatgt	ggtgtgtggt
1801	ctttcctaat	aaagaaataa	tttagcttga	caaaaaaaa	aaaaaa	

SEQ ID NO: 80 Human BRM Amino Acid Sequence Isoform D (NP_001276327.1)

1	mwlaiiedgnl	eemeeevrlk	krkrnrnvdk	dpakedveka	kkrrgrppae	klsppnpklt
61	kqmnaiidtv	inykdssgrq	lsevfqlps	rkelppeyyl	irkpvdffki	kerirnhkyr
121	slgdlkdvm	llchnaqtfn	legsqiyeds	ivlqsvfksa	rqkiakeees	edesneeeee
181	edeeseesea	kskvkvikln	kkddkgrdkg	kgkkirpnrgk	akpvvsdfds	deeqdereqs
241	egsgtdde					

SEQ ID NO: 81 Human BRM cDNA Sequence Variant 6 (NM_001289399.1, CDS:
from 106 to 936)

1	attcacttca	ttaaatctag	aggcagttga	gcatggggagc	cgtctgtatg	ttgaattagg
61	gctcgcactc	ttgcgcaaca	cgccaccagt	cggaaactgg	ggctgatgaa	gagactagca
121	gctcgctgct	ttgctggctt	gttaatttta	tccccactaa	ctgtgatttc	tgatagccgg
181	cctgctgata	gttgtaaggg	catcgaagac	ggcaatttgg	aggaatagg	agaggaaagta
241	cggcttaaga	agcgaaaaag	acgaagaaat	gtggataaag	atcctgcaaa	agaagatgtg
301	gaaaaagcta	agaagagaag	aggccgcctc	cccgcgtgaga	aactgtcacc	aaatcccccc
361	aaactgacaa	agcagatgaa	cgctatcatc	gatactgtga	taaaactaca	agatagttca
421	ggcgacagc	tcagtgaagt	cttcattcag	ttaccttcaa	ggaaagaatt	accagaatac
481	tatgaattaa	ttaggaagcc	agtggtattc	aaaaaaaata	aggaaaggat	tcgtaaatcat
541	aagtaccgga	gcctagggcga	cctggagaag	gatgtcatgc	ttctctgtca	caacgctcag
601	acgttcaacc	tgaggggatc	ccagatctat	gaagactcca	tcgtcttaca	gtcagtgttt
661	aagagtgtccc	ggcagaaaaa	tgccaaagag	gaagagagtg	aggatgaaag	caatgaagag
721	gaggaaagag	agagtcaaga	ccgaggccaa	aactcagtaa	ggtgaaaatt	
781	aagctcaata	aaaaagatga	caaaggccgg	gacaaaggga	aaggcaagaa	aaggccaaat

TABLE 1-continued

841	cgagggaaaag	ccaaacctgt	agtggagcgat	tttgacagcg	atgaggagca	ggatgaacgt
901	gaacagtcag	aaggaagtgg	gacggatgat	gagtgatcag	tacggacctt	tttccttggt
961	agaactgaat	tccttcctcc	cctgtctcat	ttctaccag	tgagttcatt	tgatcatatg
1021	gcactgggtt	gtttctatat	catcatcgto	tataaactag	ctttaggata	gtgccagaca
1081	aacacatgat	atcatggtgt	aaaaaacaca	cacatacaca	aacatttgta	acatatgttg
1141	accaaatggg	cctcaaaagt	tcagattgaa	acaacaaaa	agcttttgat	ggaaaaatag
1201	tgggtggata	gtaattttct	atgggtgggt	ctaatttggt	aacggtttga	ttgtgctcgg
1261	ttttatcacc	tgttcagatg	agaagatgtt	tgtcttttgt	agcactgata	accaggagaa
1321	gccattaaaa	gccactgggt	attttatgtt	tcatacaggca	attttcgagg	tttttatgtg
1381	ttcgggtattg	ttttttttaca	ctgtgggtaca	tataagcaac	tttaaatagg	gataaatgta
1441	cagtagttag	atttcacctg	catatacatt	tttccatttt	atgctctatg	atctgaacaa
1501	aagctttttg	aattgtataa	gatttatgtc	tactgtaaac	attgcttaat	ttttttgtct
1561	ttgatattaa	aaaaagtttt	gttgaaagcg	ctattgata	ttgcaatcta	tatagtgtat
1621	tggatggcct	cttttgtcac	cctgatctcc	tatgttacca	acgtgtatcg	tctccttctc
1681	cctaaagtgt	acttaattct	tgctttcttt	gcacaatgtc	tttggttgca	agtcataaag
1741	ctgaggcaaa	taaaattcca	gtaatttcga	agaatgtgg	gttggtgctt	tcctaataaa
1801	gaaataattt	agcttgacaa	aaaaaaaaaa	aaa		

SEQ ID NO: 82 Human BRM Amino Acid Sequence Isoform E (NP_001276328.1)

1 mkrlaarcfa gllilslptv isdsrpadsg kaiedgnlee meeevrlkkr krrrnvdkgp
 61 akedvekakk rrrgrppaekl spnppkltkq mnaidvtin ykdssgrqls evfiqlpsrk
 121 elpeyyelir kpvdfklike rirnkhkyrs lgdlekdvmll chnaqtfnl egsqiyedsiv
 181 lqsvfkasrqi kiakeeesees esneeeeee eseseseaks kvkviklnkk ddkgrdkgkg
 241 kkrpnrgkak pvvsvdfdsde eqdereqseg sgtdde

SEQ ID NO: 83 Human BRM cDNA Sequence Variant 7 (NM_001289400.1, CDS:

from 521 to 1357)

1 acttcattaa atctagaggc agttgagcat gggagccgctc tgtatgttga attagggctc
 61 gcactcctgc gcaacacgctc accagtcgga aactgggggtc tcgcttctgt gatttatctc
 121 attattgtgc tggtaaaagg tttggaaggg aattctttttt gggggtagta ctttagcatt
 181 gtgtagcaag ttttggggtt ttttctgtgt gtgaccccc agccccagc gctgagtttg
 241 agtcagttga gccagtttag taaataattt tttaaaataa aagaacagtt taaatctcc
 301 atgaacaatt ttacttacat gcaggagtaa tcctactcta ctctttacgt gcgaaaagea
 361 ttgggaagtg ttgtctgaat tgatttccat tagaaaaaga cccttagaaa tcacagaaca
 421 taaagcactg catatggatg tgtttggggt ctttggggag gaggggaagt gttttgtagc
 481 tctctgcatt cctgcataaa acctagtttt gaggggaata atgctgatga agagactagc
 541 agctcgctgc ttgtctggct tgttaatttt atccccacta actgtgattt ctgatagccg
 601 gcctgctgat agtggttaagg ccatacgaaga cggcaatttg gaggaagtgg aagaggaagt
 661 acggcttaag aagcgaaaaa gacgaagaaa tgtggataaa gaccttgcaa aagaagatgt
 721 ggaaaaagct ggtgtcggaa gaggcgcgcc tcccgctgag aaactgtcac caaatcccc
 781 caaactgaca aagcagatga acgctatcat cgatactgtg ataaactaca aagatagttc
 841 agggcgacag ccagtggaag tcttcattca gttacctca aggaaagaat taccagaata
 901 ctatgaatta attaggaagc cagtggtatt caaaaaata aaggaaagga ttcgtaatca
 961 taagtaccgg agcctaggcg acctggagaa ggatgtcatg cttctctgtc acaacgctca
 1021 gacgttcaac ctggagggat cccagatcta tgaagactcc atcgctctac agtcagtgtt
 1081 caagagtgcg ccgcagaaaa ttgccaaaaa ggaagagagt gaggatgaaa gcaatgaaga
 1141 ggaggaagag gaagtgaag aagagtcaga gtccgaggca aaatcagtca aggtgaaaat
 1201 taagctcaat aaaaaagatg acaaaaggcg ggacaaaagg aaaggcaaga aaaggccaaa
 1261 tcgaggaaaa gccaaacctg tagtgagcga ttttgacagc gatgaggagc aggatgaaacg
 1321 tgaacagtca gaaggaagtg ggacggatga tgagtgtatc gtatggacct ttttctctgg
 1381 tagaactgaa ttcttctctc cctgtgtctc tttctaccca gtgagttcat ttgtcatata
 1441 ggcactgggt tgtttctata tcatactcgt ctataaacta gctttaggat agtgccagac
 1501 aaacatatga tatcatggtg taaaaaacac acacatacac aaatatttgt aacatattgt
 1561 gaccaaaatg gctcctaaaga ttcagattga aacaaacaaa aagcttttga tggaaaatat
 1621 gtgggtggat agtatatttc tatgggtggg tctaatttgg taacggtttg attgtgctcg
 1681 gttttatcac ctgttcagat gagaagattt ttgtcttttg tagcactgat aaccaggaga
 1741 agccattaaa agccactggt tattttattt tccatcagcg aattttcgag gtttttattt
 1801 gttcggtatt gtttttttac actgtggtac atataagcaa ctttaaatagg tgataaatgt
 1861 acagtagtta gatttcacct gcatatacat ttttccattt tatgctctat gatctgaaca
 1921 aaagcttttt gaattgtata agatttatgt ctactgtaaa cattgcttaa tttttttgct
 1981 cttgatttaa aaaaaagttt tgttgaaagc gctattgaat attgcaatct atatagtga
 2041 ttgatggctt tcttttgtca ccctgatctc ctatgttacc aatgtgtatc gtctcctctc
 2101 ccctaaagtg tacttaattct ttgctttctt tgcacaatgt ctttggttgc aagtcataag
 2161 cctgaggcaa ataaaattcc agtaatttcg aagaatgtgg tgttggtgct ttcctaataa
 2221 agaaataatt tagcttgaca aaaaaaaaaa aaaa

SEQ ID NO: 84 Human BRM Amino Acid Sequence Isoform F (NP_001276329.1)

1 mlmkrlaarc fagllilslpl tvisdsrpad sgkaiedgnl eemeevrlk krrrrnvdkg
 61 dpakedveka krrgrppae klspnppklt kqmnaidvt inykdsgrq lsevfqlps
 121 rkelpyyel irkpvdfkki kerirnhkyr slgdlekdvm llchnaqtfn legsqiyeds
 181 ivlqsvfkas rqiakaees edesneeees edeeseesea ksvkvikln kdkdkgrdkg
 241 kgkkrpnrgk akpvsdfds deeqdereqs egsgtdde

SEQ ID NO: 85 Mouse BRM cDNA Sequence Variant 1 (NM_011416.2, CDS: from 111 to 4862)

1 ctgcgccccct ctgtttctgt acttgggtg actcagagag ggaagattca gccagcacac
 61 tgctcggcag caagtgtcac tctgctaact ggcagagcca ggagacctag atgtccacac
 121 ccacagaccc agcagcaatg ccccatcctg ggcctcctcc ggggctctga ccctctcctg
 181 gaccaattct ggggctagt ccaggaccag gaccatcccc aggtttctgtg cacagcatga
 241 tgggtcctag tcccggaact cccagcgtct cacatcctct gtcaacgatg ggctctgcag

TABLE 1-continued

301	acttcccaca	ggaaggcatg	caccaattac	ataagcccat	ggatgggata	catgacaaag
361	ggattgtaga	agatgtccac	tgtggatcca	tgaagggcac	cagcatgcgc	ccccacacc
421	caggaatggg	ccctccacag	agcccatagg	accagcacag	ccaaggttat	atgtcaccac
481	atccgtctcc	tctgggagcc	ccggagcacg	totctagccc	tatatctgga	ggaggcccaa
541	ccccacccca	gatgccaccg	agccagccag	gggcactcat	cccaggagat	ccgcaggcca
601	tgaaccagcc	taacagaggt	ccctcgctt	tcagtcctgt	gcagctgcat	cagcttcgag
661	ctcagatttt	agcttacaaa	atggtggcca	ggggccagcc	ttccccgaa	actctgcagc
721	tggcagtcga	gggaaaaagg	accttgctcg	gcctgcagca	gcagcagcag	caacaaacac
781	aacagcagca	gcagcagcag	cagcagcagc	agcaacagca	gcaacaacag	cagccccagc
841	agcctcagca	gcagcagcag	gcacagccccc	agcagcagca	gcaacagcag	cagcagccag
901	ctcttgttag	ctataatcga	ccatctggcc	ccgggcagga	gctgctactg	agtggccaga
961	gcgctccgca	gaagctgtca	gcaccagcac	caagcggcgg	accttcaccg	gcaccccagg
1021	ccgcgctcca	gtcccagccc	acagcggcgc	ccgggcccctc	cgtgcagcag	ccgcccccag
1081	ggcagccgctc	tccggtccta	cagctgcaac	agaagcagag	ccgcatcagc	cccatccaga
1141	aaccgcgaag	cctggaccgg	gtggagatcc	tgcaggaaacg	agagtacaga	cttcaagctc
1201	gcctcgctca	tgtcgccaaa	gaactggaaa	gtctgcctgg	ttccttgcca	ccagatttac
1261	gcaccaaagc	aaccgtggaa	ctgaaagcac	ttcgcttact	caacttccaa	cgtcagctga
1321	gacaggaggt	gggtggcctgc	atgcggaggg	acaccacccct	ggagacggcc	ctcaactcca
1381	aagcatataa	gcaggagcag	cgccagaccc	tgctgtaggc	acgcatgaca	gagaaactgg
1441	agaagcagca	gaagatagaa	caggagagga	aacgccggca	gaaacaccag	gaatacctga
1501	acagtatttt	gcaacatgca	aaagatttta	aggaatatca	ccggtctgtg	gccgggaaga
1561	tccagaagct	gtcccagcca	gtggcgactt	ggcatgctaa	cacagaaagg	gagcagaaga
1621	aggagacgga	gcggatcgag	aaggagagaa	tgccggaggct	gatggccgaa	gatgaagagg
1681	gctacaggaa	gcttattgac	caaaagaaag	acagacgtct	cgctaccta	ttgcagcaga
1741	ccgatgagta	tgtcgccaat	ctgaccaaac	tggtgtggga	gcacaagcag	gcccaagcag
1801	ccaaagagaa	gaagaagagg	aggaggagga	agaagaaggc	tgaagagaat	gcagagggag
1861	gggaacctgc	cctgggacca	gatggagagc	caatagatga	aagcagccag	atgagtgaac
1921	tgctcgccaa	agtgcacac	acagaaactg	gcaaggtcct	ctttggacca	gaagcaccca
1981	aagcaagtca	gctggatgcc	tggtgggaga	tgaatcctgg	ttacgaagt	gcacccagat
2041	ctgacagtga	agagagtga	tcggactacg	aggaggagga	tgaagaagaa	gagtcagata
2101	ggcaggaac	cgaggagaag	atactgctgg	atcccaacag	tgaagaagt	tccgaaaagg
2161	atgcccaagca	gatcattgag	actgcgaagc	aggacgtgga	cgacgaatac	agcatgcagt
2221	acagtgcag	aggctctcag	tcctactaca	cggtggctca	cgctatctct	gagaggggtg
2281	agaagcagtc	tgccctctct	attaacggca	ccctaaagca	ttaccagctc	cagggcctgg
2341	aatggatggt	ttccctgtat	aataacaatc	tgaacggaat	cttagctgat	gaaatggggc
2401	taggcaagac	catccagacc	attgcactca	tcacgtatct	gatggagcac	aaaaggctca
2461	atggtcctca	ctccatcatc	gtccccctct	cgactctgtc	taactggaca	tatgaatttg
2521	acaaatgggc	tccttctgtg	gtgaaaattt	cttacaaggg	tacccctgcc	atgcgacgct
2581	ccctcgcttc	ccagctacgg	agtggaacat	tcaatgtccc	cctgactact	tacgagtaca
2641	ttataaaaag	gacacacatt	cttgcaaaag	ttcgggtggaa	gtacatgac	gtggcgaag
2701	gccaccggat	gaagaatcac	cactgcaagc	taaccacagg	cctgaacaca	cactatgtgg
2761	ccccaggggc	gacccctctg	actgggaccc	cactgcagaa	taagcttccg	gaactctggg
2821	ccctccctca	gtccctctct	cctacaatct	tcaagagtgg	cagcacattt	gagcagtggt
2881	ttaatgctcc	atttgccatg	accggtgaaa	gggtggacct	gaacgaagaa	gaaacgattt
2941	tgatcatcag	gcgtctacac	aaggtgctga	gacccctttt	actgaggagg	ctgaagaag
3001	aggttgagtc	tcagcttccg	gaaaagggtg	agtatgtgat	caagtgtagc	atgtcagctc
3061	tgacagaagt	tctgtaccgt	cacatgcaag	ccaaggggat	cctcctcacg	gacgggtctg
3121	agaaagataa	gaaggggaaa	ggaggtgcca	agacacttat	gaacaccatc	atgcagctga
3181	gaaaaaatatg	caaccaccca	tatatgttcc	agcacattga	ggaaatcctt	gctgaacacc
3241	tgggtctattc	gaatggggtc	atcaatgggg	ctgagctgta	tcgggcctcg	ggaaagtgtg
3301	agctgctcga	tcgcattctg	cccaaatgga	gagcgactaa	ccaccgcgtg	ctgctttctg
3361	gccagatgac	gtcactcatg	accattatgg	aggattactt	tgcttttcgg	aactctctgt
3421	acctgcgcct	tgacggcacc	accaagtctg	aagatcgtgc	tgctttgcta	aagaaattca
3481	atgaaccttg	gtcccagtat	ttcattttct	tgctgagcac	aagagcaggg	ggcctggggt
3541	taaatcttca	ggcgcagac	acgggtggta	tatttgacag	cgactggaa	cctcaccagg
3601	atctgcaggc	ccaagaccga	gctcaccgca	ttggccaaca	aaacagaggtc	cgggtgctga
3661	ggctttgcac	cgtcaacagt	gtggaggaaa	agattctcgc	ggctgccaa	tacaagctga
3721	acgtggatca	gaaggttatc	caagcaggca	tggttgacca	gaagtcattc	agccacgagc
3781	ggagggccct	cctgcaggcc	attctggagc	acgaggagga	gaatgaggaa	gaagatgagg
3841	taccagacga	cgagaccctg	aaccagatga	ttgctcgccg	ggagggaaga	tttgatcttt
3901	ttatgcgcat	ggacatggac	cggcgagggg	aggatgcccg	gaaccggaag	cgcaaacccc
3961	gcttgatgga	ggaagatgag	ctgccctcct	ggattatcaa	ggatgacgcc	gaagtggaaa
4021	ggctcacctg	tgaagaagag	gaggagaaga	tatttgggag	gggctctcgc	cagcgccggg
4081	atgtggacta	gtcaccagac	agcaatggct	ctcaccgaga	caggggccatc	gaagacggca
4141	atttggaaga	aatggaagag	gaggtacggc	ttaaagaagag	aaaaagacga	agaaatgtgg
4201	ataaagaccc	cgtgaaggaa	gatgtggaaa	aagcgaagaa	aagaagaggc	cgccctccgg
4261	ctgagaaggt	gtcaccaaat	cccccaaaac	taacgaagca	gatgaacgcc	atcattgata
4321	ctgtgataaa	ctacaagac	agttcagggc	gacagctcag	tgaagtcttc	attcagttac
4381	cttccaggaa	agacttacca	gaataactatg	aattaattag	gaagccagtg	gatttcaaaa
4441	agataaagga	gggaatccgt	aaccataagt	atcggagcct	gggagaccgtg	gagaaagacg
4501	tcattgctct	ctgtcacac	gcacagacat	tcaacttgga	aggatcccag	atctacgaag
4561	actccattgt	cctacagtca	gtgtttaaga	gtgctcggca	gaaaattggc	aaagaagaag
4621	agagtgaagg	gaagaagag	gaagaagag	aagaagatga	tgaagaggag	tcggagtcag
4681	aggcgaatat	tgtgaagggt	aaaatcaagc	tgaataaaaa	ggaagagaaa	ggccgggaca
4741	cagggaaggg	caagaagcgg	ccaaaccgag	gcaaaagcaa	acccgtcgtg	agcgattttg
4801	acagtgaaga	ggaaacagaa	gagaaacgaa	agtcagaagc	aagtggaaact	gataacagat
4861	gaccatccctg	gacgtgagct	tcccgcgggtg	gcagaaccga	atgctttcct	ccccctctcc
4921	ttctccccc	gtgagttcac	ttgccattcg	ggcacactgg	gttattttct	cgtctctcatt
4981	gtcatctaga	actagcttta	gggtagtgcc	agacaaacat	atgatcat	ggtgtaaaaa
5041	aagaacacac	tgctgtcaga	cacactacac	acacacacac	acacacacac	acacacacac

TABLE 1-continued

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5101 acacatatatt gtaacatat gtagccaaat gggcctcaaa gattcaaaga ttaaaaaaaa
5161 aaagccttttg atggaaaaga tgggggtgga tagtatattt ctacaggtgg gtcagggttg
5221 gtagcagttt gatgtgctgg gttctgtcat ctgttctgat gagaagattt ttatcttctg
5281 cagtgtctgat ggcggggagg aaccattcaa agccactggg tattttgttt ttcacaggc
5341 gattttcaag attttcattt gtttcagtat tgggtggttt ctcttctctc tttttacac
5401 tgtggtacat ataagcaact tgactagtga caaatgtaca gtagttagat atcacctaca
5461 tatacatttt tccattttat gctctatgat ctgaagaaca aaaaaaaaag ctttttgact
5521 tgtataagat ttatgtctac tgtaaacatt gcggaatttt tttttgttct tgttttattg
5581 acaatgctat tgagtattac agtgtctaga ataccctgga tggcttctct tgtccaccgg
5641 atctcccggt ttaccaatgt gtatggctct cttctcccgga aagtgtactt aatctttgct
5701 ttctttgcac aatgtctttg gttgcaagtc ataagcctga ggcaataaaa attccagtaa
5761 tttccaagaa tgtggtggtg gtacttttct aataaaccca taacgtacct tgaaaaaaaa
5821 aaaaaaaaaa a

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SEQ ID NO: 86 Mouse BRM Amino Acid Sequence Isoform A (NP_035546.2)

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1 mstptdpaam phpgpspgpg pspgpilggs pgpgpspgsv hsmmgpspgp psvshplstm
61 gsadfpqegm hqhlkpmddi hdkgivedvh csgmkgtsmr pphpgmgppq spmdqhsqgy
121 mspbpsplga pehvsspisg ggtpppqmpg sqpgalipgd pqamnpnrg pspfpvqlh
181 qlraqilayk mlargqplpe tlqlavqgkr tlpqmqqqqq qqqqqqqqqq qqqqqqqqqq
241 qpqqpqqqag aqpqqqqqqq qpqalvsynr psgpgqelll sgqsapqkls apapsgrsp
301 apqaavqpta tavpgpsvqg papgqpsvpl qlqkqksris piqkpggldp veilqereyr
361 lqariahriq eleslpgslp pdlrtkatve lkallrlnfq rqlcgevvac mrrdtlleta
421 lnskaykrsk rqltrearmt eklekqgkie qerkrqkhq eylnsilgha kdfkeyhrsv
401 agkikqlska vatwhanter eqkketerie kermrrlmae deegyrklid gkkdrllayl
541 lqqtdeyvan ltnlvvehkq aqaakekkkr rrrkkkaeen aeggepalgp dgepidessq
601 msdlpvkvth tetgkvlfpg eapkasqlda wlemnpgyev aprsdseese sdyeedeese
661 essrqeteek illdpnseev sekdakqiee takqdvdddey smqysargsq syytvahaais
721 erveqksall ingtlkhyql qglewmvsly nnnlmgilad emglgkttgt ialitylmeh
781 krlngpylil vpstlsnwt yefdkwapsv vkisykgtpa mrrslvpqlr sgkfnvlltt
841 yeylikdkhi lakirwkymi vdeghrmknh hckltqvlnt hyvaprrill tgtplqnklp
901 elwallnflf ptifkscstf eqwfnafpam tgervdlnee etiliirrlh kvlrpfillrr
961 lkkevesqlp ekveyvikcd msalqkilyr hmqaqgillt dgeekdkkkg ggaaktlmnti
1021 mqlrkicnph ymfqhieesf aehlgysngv ingaelyras gkfelldril pklatnrhrv
1081 llfqcmstslm timedyafar nflylrlldgt tkseadraall kkfnepgsqy fifllstrag
1141 glglnlqaad tvvifdsdwn phqdlqagdr ahrigqqnev rvrlrlctvns weekilaak
1201 yklndvdqkvi qagmfdqkss sherrafllq ileheeneee edevpddetl nqmiarreee
1261 fdlfmrmdmd rrrredarnpk rkprlmeede lpswiikdda everlcteee eekifgrgsr
1321 qrrdvdsyda lrekwqlrai edgnleemee evrlkrkrkr rnvdkdpvke dvekakkrrg
1381 rppaeklspp ppklrtqgma iidtvinykd ssgrqlsevf iqlpsrkdip eyyelirkpv
1441 dfkkikerir nhkyrsldgl ekdvmllchn aqtfnlsgsq iyedsivlqs vfksarqkia
1501 keeeseesn eeeeeedeee seseaksvkv kiklnkkeek grdtgkqkkr pnrqkpkpvv
1561 sdfdsdeeqe eneqseasgt dne

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SEQ ID NO: 87 Mouse BRM cDNA Sequence Variant 2 (NM_026003.2, CDS: from 301 to 1011)

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1 ttcacctcat taaatctaga ggcggttcag catgggagcc gtctgtatgt tgaattaggg
61 ctgcctctct tgcgcaacac gtcaccagtc ggaactggg ggtttgcttc tgtgatttat
121 ttcattattg tgcttgtaaa agctgatgaa gagactagca gctcgtctgt ttgcccgtt
181 gtttaatttta tccccactaa ctgtgatctc cgatagccgg cctgctgata tgggtaagt
241 cggctggctc tggtttaaa caagcgtttg caggccatcg aagacggcaa ttggaagaa
301 atggaagagg aggtacggct taagaagaga aaaagacgaa gaaatgtgga taaagacccc
361 gtgaaggaag atgtggaaaa agcgaagaaa agaagaggcc gccctccggc tgagaagttg
421 tcaccaaatc ccccaaaact aacgaagcag atgaacgcca tcattgatac tgtgataaac
481 tacaagaca gttcaggcg acagctcagt gaagtcttca ttcagttacc ttccaggaaa
541 gacttaccag aatactatga attaattagg aagccagtgg atttcaaaaa gataaaggag
601 cgaatccgta atcataagta tcggagcctg ggagacctgg agaaagacgt catgcttctc
661 tgtcacaacg cacagacatt caacttgga ggatccaga tctacgaaga ctccattgtc
721 ctacagtcag tgtttaagag tgctcggcag aaaattgcca aagaagaaga gagttaggaa
781 gaaagcaatg aagaagagga agaagatgat gaagaggagt cggagtcaga ggcgaaatct
841 gtgaagtgta aatcaagct gaataaaaag gaagagaaag gccgggacac agggaaagggc
901 aagaagcggc caaaccgagg caaagccaaa ccgctcgtga gcgattttga cagtgcagag
961 gaacaggaag agaacgaaca gtcagaagca agtggaactg ataacgagt accatcctgg
1021 acgtgagctt ccgcgggtgg cagaaccgaa tgctttcttc cccctctctc tctcccag
1081 tgagttcact tgccattcgg gcacactggg ttatttcttc gtccctcatt tcatctagaa
1141 ctagctttag ggtagtgcga gacaaacata tgatatcatg ggtgaaaaaa agaaacacat
1201 gcgtgcagac aactacaca cacacacaca cacacacaca cacatatttg
1261 taacatatgg tgacaaaatg ggcctcaaa attcaaatg taaaaacaaa aagcttttga
1321 tggaaaaagt gtgggtggat agtatatttc tacaggtggg tcaggttttg tagcagtttg
1381 attgctggg tttctgctc ttgtctgatg agaagatttt tatctctcgc agtgctgatg
1441 gccgggagga accattcaaa gccactgggt attttgtttt tcatcaggcg attttcaaga
1501 ttttcatttg ttccagtatt gttggttttc tcttttctct tttttacact gtggtacata
1561 taagcaactt gactagtgc aaatgtacag tagttagata tcacctacat atacattttt
1621 ccattttatg ctctatgatc tgaagaacaa aaaaaaagc tttttgactt gtaagaatt
1681 tatgtctact gtaaacattg cgaatttttt ttttgttctt gttttattga caatgctatt

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TABLE 1-continued

1741	gagtattaca	gtgtctagaa	taccctggat	ggcttctctt	gtccacccga	tctcccgtgt
1801	taccaatgtg	tatggctctc	ttctcccgaa	agtgtactta	atctttgctt	tctttgcaca
1861	atgtcttttg	ttgcaagtca	taagcctgag	gcaataaaaa	ttccagtaat	ttccaagaat
1921	gtggtgttgg	tactttctca	ataaacccgat	aacgtacctt	gaaa	

SEQ ID NO: 88 Mouse BRM Amino Acid Sequence Isoform B (NP_080279.1)

1	meeevrlkkr	krnrndkdp	vkedvekakk	rrgrppaekl	spnppkltkq	mnaidvtin
61	ykdsgrqls	evfiqlpsrk	dlpeyyelir	kpvdffkike	rnrnhkyrs1	gdlekdvml1
121	chnaqtfnle	gsqiyedsiv	lqsvfksarq	kiakeeese	esneeeedd	eeeseseaks
181	vkviklknk	eeigrdrtgk	kkprnrgkak	pvvsdfdsde	eqeeneqsea	sgtdne

SEQ ID NO: 89 Mouse BRM cDNA Sequence Variant 3 (NM_001347439.1, CDS: from 180 to 1010)

1	acacacacac	acacacacac	acgcaggctg	aagtatgctt	aactctttta	acttggtgtg
61	ggcttttttag	caccatattg	gttcttttct	gacgtccgga	cccgaaagag	tgacagtgtg
121	ctttaaggaa	agaggttacct	caccaaactt	ccctgtagtt	gtgcctcacc	atttagctga
181	tgaagagact	agcagctcgc	tgttttgcgc	gcttggtaat	tttatcccca	ctaactgtga
241	tttccgatag	ccggcctgct	gatatgtgta	aggccatcga	agacggcaat	ttggaagaaa
301	tggaagagga	ggtacggctt	aagaagagaa	aaagacgaag	aaatgtggat	aaagaccccg
361	tgaagggaaga	tgtggaaaaa	gcgaagaaaa	gaagaggccg	ccctccggct	gagaagttgt
421	caccaaattcc	cccaaacta	acgaagcaga	tgaacgccat	cattgatact	gtgataaaact
481	acaaagacag	ttcaggggcg	cagctcagtg	aagtcttcat	tcagttacct	tccaggaaag
541	acttaccaga	atactatgaa	ttaattagga	agccagtgga	tttcaaaaag	ataaaggagc
601	gaatccgtaa	tcataagtat	cggagcctgg	gagacctgga	gaaagacgtc	atgctttctct
661	gtcacacacg	acagacattc	aacttggaag	gatcccgat	ctacgaagac	tccattgtcc
721	tacagtcagt	gtttaagagt	gctcggcaga	aaattgccaa	agaagaagag	agtgagggaag
781	aaagcaatga	agaagaggaa	gaagatgatg	aagaggagtc	ggagtcagag	gcgaaatctg
841	tgaaggtgaa	aaccaagctg	aataaaaaag	aagagaaagg	ccgggacaca	gggaagggca
901	agaagcggcc	aaaccgaggg	aaagccaaac	ccgtcgtgag	cgattttgac	agtgacgagg
961	aacaggaaga	gaacgaacag	tcagaagcaa	gtggaaactga	taacgagtg	ccatcctgga
1021	cgtgagcttc	cccggtgtgc	agaaccgaat	gctttcttcc	ccctctcctt	cctcccgagt
1081	gagttcactt	gccattcggg	cacactgggt	tattctctcg	tcctcattgt	catctagaac
1141	tagcttttag	gtagtccag	acaaacatat	gatatcatgg	tgtaaaaaaa	gaaacacatg
1201	cgtgcagaca	cactacacac	acacacacac	acacacacac	acacacacac	acatatgtgt
1261	aacatatgtt	gaccaaatgg	gcctcaaaga	ttcaaagatt	aaaaacaaaa	agctttttag
1321	ggaaaagatg	tgggtggata	gtatatctct	acaggtgggt	caggttttgt	agcagtttga
1381	tgtgtgtggg	tctgtcatct	gttctgatga	gaagattttt	atcttctgca	gtgctgatgg
1441	ccggaggagaa	ccattcaaag	ccactgggta	ttttgttttt	catcaggcga	ttttcaagat
1501	tttcatttgt	ttcagttatt	ttggttttct	cttttctctt	ttttacactg	tggtacatat
1561	aagcaacttg	actagtgaca	aatgtacagt	agttagatat	cacctacata	tacatttttc
1621	cattttatgc	tctatgatct	gaagaacaaa	aaaaaaagct	ttttgacttg	tataagattt
1681	atgtctactg	taaacattgc	ggaattttct	tttgttcttg	ttttattgac	aatgctattg
1741	agtattacag	tgctatgaat	accctggatg	gcttcccttg	tccaccgat	ctcccggtgt
1801	accaatgtgt	atggtctcct	tctccgaaaa	gtgtacttaa	tctttgcttt	ctttgcacaa
1861	tgtcttttgt	tgcaagtcat	aagcctgagg	caaataaaat	tccagtaatt	tccaagaatg
1921	tgggtgttgt	actttctcaa	taaacccgata	acgtaccttg	aaaaaaaaaa	aaaaaaaaaa

SEQ ID NO: 90 Mouse BRM Amino Acid Sequence Isoform C (NP_001334368.1)

1	mkrlaarcfa	gllilpltlv	idsrpadsg	kaiedgnlee	meeevrlkkr	krnrndkdp
61	vkedvekakk	rrgrppaekl	spnppkltkq	mnaidvtin	ykdsgrqls	evfiqlpsrk
121	dlpeyyelir	kpvdffkike	rnrnhkyrs1	gdlekdvml1	chnaqtfnle	gsqiyedsiv
181	lqsvfksarq	kiakeeese	esneeeedd	eeeseseaks	vkviklknk	eeigrdrtgk
241	kkprnrgkak	pvvsdfdsdc	eqeeneqsea	sgtdne		

SEQ ID NO: 91 Human EGFR cDNA Sequence Variant 1 (NM_005228.4, CDS: from 258 to 3890)

1	gtccgggagc	ccccgggagc	agcggggcgc	cagcagcctc	cgccccccgc	acgggtgtgag
61	cgccccgacg	ggccgaggcg	gcccagagtc	cgagctagcc	ccggcgggcg	cgcccgccca
121	gaccggagca	caggccacct	cgtcggcgct	cgcccgagtc	ccgcctcgcg	cgccaacgcc
181	acaaccaccg	cgcacggccc	cctgactccg	tccagttatt	atcgggagag	ccggagcgag
241	ctcttcgggg	agcagcgatg	cgacctcccg	ggacggccgg	ggcagcgctc	ctggcgctgc
301	tggctgcgct	ctgcccggcg	agtcgggctc	tggaggaaaa	gaaagtttgc	caaggcacga
361	gtaacaagct	cacgcagttg	ggcacttttg	aagatcactt	tctcagcctc	cagaggatgt
421	tcaataactg	tgaggtgggt	cttggaattt	tggaaattac	ctatgtgcag	aggaattatg
481	atctttctct	cttaaaagacc	atccaggagg	tggctgggta	tgtcctcatt	gccctcaaca
541	cagtgaggcg	aattcctttg	gaaaacctgc	agatcatcag	aggaaatatg	tactacgaaa
601	attctatgc	cttagcagtc	ttatctaact	atgatgcaaa	taaaaccgga	ctgaaggagc
661	tgcccatgag	aaatttacag	gaaatcctgc	atggcgccgt	gcgggttcagc	aacaacctcg
721	ccctgtgcaa	cgtggagagc	atccagtggt	gggacatagt	cagcagtgac	tttctcagca
781	acatgtcgag	cttagctcag	aaccacctgg	gcagctgcca	aaagtgtgat	ccaagctgtc
841	ccaatgggag	ctgctggggt	gcaggagagg	agaactgcca	gaaactgacc	aaaatcatct
901	gtgcccagca	gtgctccggg	cgctgccgtg	gcaagtcccc	cagtgactgc	tgccacaacc
961	agtgtgctgc	aggctgcaca	ggcccccggg	agagcgactg	cccggtctgc	cgcaaatccc
1021	gagacgaagc	cacgtgcaag	gacacctgcc	ccccactcat	gctctacaac	cccaccacgt
1081	accagatgga	tgtgaacccc	gagggcaaat	acagcttttg	tgccacctgc	gtgaagaagt
1141	gtccccgtaa	ttatgtgggt	acagatcacg	gctcgtgcgt	ccgagcctgt	ggggccgaca
1201	gctatgagat	ggaggaaagc	ggcgtccgca	agtgtaaaga	gtgcgaaggg	ccttgccgca
1261	aagtgtgtaa	cgaatatagg	attggcgaat	ttaaagactc	actctccata	aatgctacga
1321	atattaaaca	cttcaaaaac	tgcaacctca	tcagttggca	tctccacatc	ctgcccgtgg
1381	catttagggg	tgactccttc	acacatactc	ctcctctgga	tccacaggaa	ctggatattc

TABLE 1-continued

1441	tgaaaacgct	aaaggaaatc	acagggtttt	tgctgattca	ggcttggcct	gaaaacagga
1501	cggacctcca	tgcccttgag	aacctagaaa	tcatacgcgg	caggaccaag	caacatggtc
1561	agttttctct	tcagtcgtc	agcctgaaca	taacatcctt	gggattacgc	tcctctaaagg
1621	agataagtga	tgagatgtg	acaatttcag	gaaacaaaaa	tttgtgctat	gcaaatacaa
1681	taaaactggaa	aaaactgttt	gggacctccg	gtcagaaaaac	caaaattata	agcaacagag
1741	gtgaaaacag	ctgcaaggcc	acaggccagg	tctgccatgc	cttgtgtctcc	cccaggggct
1801	gctggggccc	ggagcccgag	gactgcgtct	cttgccggaa	tgcaagccga	ggcagggaat
1861	gcgtggacaa	gtgcaacctt	ctggagggtg	agccaaggga	gttgtgtggag	aactctgagt
1921	gcatacagtg	ccaccagag	tgccctgcctc	aggccatgaa	catcacctgc	acaggacggg
1981	gaccagacaa	aatctccgat	tgtgccact	acattgacgg	ccccactgc	gtcaagacct
2041	gcccggcagg	agtcatggga	gaaaacaaca	ccctggctctg	gaagtacgca	gacgcgggcc
2101	atgtgtgcca	cctgtgccac	ccaaactgca	cctacggatg	cactggggcca	ggtcttgaag
2161	gctgtccaac	gaatgggcct	aagatccgt	ccatcgccac	tggtatgggt	ggggccctcc
2221	tcttgctgct	ggtgggtggc	ctggggatcg	gcctcttcat	gcgaaggcgc	cacatcgctc
2281	ggaagcgcac	gctgcggagg	ctgctgcagg	agagggagct	tgtggagcct	cttacacca
2341	gtggagaagc	tcaccaacaa	gctctcttga	ggatcttgaa	ggaaactgaa	ttcaaaaaga
2401	tcaaagtgtc	gggctccggt	gcgttcggca	cggtgtataa	gggactctgg	atcccagaag
2461	gtgagaagtg	taaaattccc	gtcgctatca	aggaattaag	agaagcaaca	tctccgaag
2521	ccaacaagga	aatcctcgat	gaagcctacg	tgatggccag	cgtggacaac	ccccacgtgt
2581	gccgcctgct	gggcatctgc	ctcacctcca	ccgtgcagct	catcacgcag	ctcatgccct
2641	tcggctgcct	cctggactat	gtccgggaac	acaaagacaa	tattggctcc	cagtacctgc
2701	tcaactgggt	tgtgcagatc	gcaaaggcca	tgaaactactt	ggaggaccgt	cgcttgggtc
2761	accgcgacct	ggcagccagg	aacgtactgg	tgaaaacacc	gcagcatgtc	aagatcacag
2821	attttgggct	ggccaaactg	ctgggtgcgg	aagagaaaga	ataccatgca	gaaggaggca
2881	aagtgcctat	tcagggtgatg	gcattggaat	caattttaca	cagaatctat	accaccaga
2941	gtgatgtctg	gagctacggg	gtgactgttt	gggagttgat	gacctttgga	tccaagccat
3001	atgacggaat	ccctgccagc	gagatctcct	ccatcctgga	gaaaggagaa	cgctccctcc
3061	agccaccat	atgtaccatc	gatgtctaca	tgatcatggt	caagtgtctg	atgatagacg
3121	cagatagtcg	cccaaagtgc	cgtgagttga	tcacogaatt	ctccaaatg	gcccgagacc
3181	cccagcgcta	ccttgtcatt	cagggggatg	aaagaatgca	tttgccaaat	cctacagact
3241	ccaactctca	ccgtgcctctg	atggatgaag	aagacatgga	cgagctgggt	gatgcgcagc
3301	agtaacctcat	cccacagcag	ggcttcttca	gcagccctcc	cacgtcacgg	actccctcc
3361	tgagctctct	gagtgcaacc	agcaacaatt	ccaccgtggc	ttgcatgtat	agaaatgggc
3421	tgaagagctg	tccatccaag	gaagacagct	tcctgcagcg	atcacgtcca	gacccacag
3481	gcgccttgac	tgaggacagc	atagacgaca	ccttccctcc	agtgccctgaa	tacataaacc
3541	agtcgcttcc	caaaaggccc	gctggctctg	tgcaaatcc	tgtctatcac	aatcagcctc
3601	tgaaccccg	gcccagcaga	gaccacact	accaggaccc	ccacagcact	gcagtgggca
3661	accocgagta	tctcaacact	gtccagccca	cctgtgtcaa	cagcacattc	gacagccctg
3721	cccactgggc	ccagaaggcc	agccacccaa	ttagcctgga	caaccctgac	taccagcagg
3781	acttcttctc	caagggaagcc	aagccaaatg	gcactcttaa	gggctccaca	gctgaaaatg
3841	cagaataacct	aagggtcgcg	ccacaaagca	gtgaatttat	tgagcatga	ccacggaggga
3901	tagtatgagc	cctaaaaatc	cagactcttt	cgatacccg	gaccaagcca	cagcaggtcc
3961	tccatcccaa	gcacatgcc	cgcattagct	cttagaccca	cagactgggt	ttgcaacgct
4021	tacaccgact	agccaggaag	tacttccacc	tcgggcacat	tttgggaagt	tgcatctctt
4081	tgtcttcaaa	ctgtgaaagca	tttacagaaa	cgcatccagc	aagaatattg	tcctcttgag
4141	cagaaattta	tctttcaaa	aggtatat	gaaaaaaa	aaaagtatat	gtgaggattt
4201	ttattgattg	gggatcttgg	agtttttcat	tgctcgctatt	gatttttact	tcaatgggct
4261	cttccaacaa	ggaagaagct	tgctggtagc	acttgctacc	ctgagttcat	ccaggcccaa
4321	ctgtgagcaa	ggagcacaag	ccacaagtct	tcagaggagc	gcttgattcc	agtggttctg
4381	cttcaaggct	tccactgcaa	aacactaaag	atccaagaag	gccttcatgg	ccccagcagg
4441	ccggatcggt	actgtatcaa	gtcatggcag	gtacagtagg	ataagccact	ctgtcccttc
4501	ctgggcaaa	aaggaacgga	ggggatggaa	ttcttcttta	gacttacttt	tgtaaaaatg
4561	tcccacgggt	acttactccc	cactgatgga	ccagtggctt	ccagtcatga	gcgttagact
4621	gacttgtttg	tcttccattc	cattgttttg	aaactcagta	tgctgcccct	gtcttgtgtg
4681	catgaaatca	gcaagagagg	atgacacatc	aaataataac	tcggattcca	gcccacattg
4741	gattcatcag	catttggacc	aatagcccac	agctgagaat	gtggaatacc	taaggatagc
4801	accgcttttg	ttctgcgaaa	aacgtatctc	ctaatttgag	gcccagatga	aatgcacag
4861	gtcctttggg	gcatacatca	gaagactaca	aaaatgaagc	tgctctgaaa	tctccttag
4921	ccatcacccc	aaccccccaa	aattagtttg	tgctacttat	ggaagatagt	tttctccttt
4981	tacttcactt	caaaagcttt	ttactcaaa	agtatatgtt	ccctccaggt	cagctgcccc
5041	caaaccctcc	ccttacgctt	tgtcacacaa	aaagtgtctc	tgcttgagt	catctattca
5101	agcacttaca	gctctggcca	caacaggcca	ttttacaggt	gcgaatgaca	gtagcattat
5161	gagtagtggt	gaattcaggt	agtaaatatg	aaactagggt	ttgaaattga	taatgctttc
5221	acaacatttg	cagatgtttt	agaaggaaaa	aagtctcttc	ctaaaataat	tctctacaa
5281	ttggaagatt	ggaagattca	gctagttagg	agccacctt	ttttccta	ctgtgtgtgc
5341	cctgtaacct	gactgggtta	cagcagctct	ttgtaaacag	tggttttaac	tctctatgct
5401	aatatccacc	ccatcaagta	tatcaaggaa	gaaatgggtc	agaaaatatt	ttcagcctac
5461	agttatgttc	agtcacacac	acatacaaaa	tggtcctttt	gcttttaaa	taatttttga
5521	ctcccgagct	agtcagagcc	cctacagcat	tggttaagaaa	gtatttgatt	ttgtctcaa
5581	tgaaaataaa	actaatctca	tttccactct	attatgctct	caaatacccc	taagcatcta
5641	tactagcctg	gtatgggtat	gaaagataca	aagataaata	aaacatagtc	cctgattcta
5701	agaaattcac	aattttagcaa	aggaaatgga	ctcatagatg	ctaaccctaa	aacaacgtga
5761	caaatgccag	acaggaccaca	tcagccaggc	actgtgagag	cacagagcag	ggaggttggg
5821	tctcgctcga	ggagacctgg	aaggaggccc	tcacaggagg	atgaccaggt	ctcagtcagc
5881	ggggagggtg	aaagtgcagg	tgcatcaggg	gcaccctgac	cgaggaaaca	gctgccagag
5941	gcctccactg	ctaaagtcca	cataaggctg	aggtcagtca	ccctaaca	cctgctccct
6001	ctaagccagg	ggatgagctt	ggagcatccc	acaagttccc	taaaagtgtc	agccccagg
6061	gggattttga	gctatcatct	ctgcacatgc	ttagttagaa	gactacacaa	cattttctaag
6121	aatctgagat	tttatattgt	cagttaacca	cttctattat	tcattccact	caggacatgc
6181	agaaatattt	cagtcagaac	tggaagacag	aaggacctac	attctgctgt	cacttatgtg

TABLE 1-continued

6241	tcaagaagca	gatgatcgat	gaggcaggtc	agttgtaagt	gagtcacatt	gtagcattaa
6301	attctagtat	ttttgtagtt	tgaaacagta	acttaataaa	agagcaaaaag	ctaaaaaaaa
6361	aaaaaaaa					

SEQ ID NO: 92 Human EGFR Amino Acid Sequence Isoform A (NP_005219.2)

1	mrpsgtagaa	llallaaalcp	asraleekkv	cqgtsnklq	lgtfedhfls	lqrmfnncev
61	vlgnleityv	qrnydlslfk	tiqevagyvl	ialntverip	lenlqiirgn	myyensyala
121	vlсныdankt	glkelpmrnl	qeilhgavrf	snnpalcnve	siqwrdivss	dflsnmsmdf
181	qnhlgsccqkc	dpscpngscw	gagencqkl	tliicacqcs	grcrkgspsd	cchnqcaagc
241	tgpresdclv	crkfrdeatc	kdtecpplmly	npttyqmdvn	pegkysfgat	cvkkcprnyv
301	vtdhgscvra	cgadsyemee	dgvrkckkce	gpcrkvcngi	gigefkdsls	inatnikhfk
361	nctsisgdllh	ilpvafrgds	fthtppldpq	eldilktvke	itgflliqaw	penrtdlhaf
421	enleiirgrt	kqhggfslav	vslnitslgl	rsakeisdgd	viisgnknlc	yantinwkl
481	fgtsgqtkti	isnrgensck	atgqvchalc	spegcwgppep	rdcvscrnvs	rgrecvdkcn
541	llegeprefv	enseciqlchp	ecelpqamnit	ctgrgpdnci	qcahyidgph	cvktcpagvm
601	genntlvwky	adaghvhclh	hpnctygs	pglegcptng	pkipsiatgm	vgalllllvv
661	algiglfmr	rhlvrkrtlr	rlqerelve	pltpsgaepn	qallrilket	efkkikvlgs
721	gafgtvykgl	wipegekvi	pvaikelrea	tspkankeil	deayvmavsd	nphvcrllgi
781	cltsrvqlit	qvlmpfclld	yvrehkdni	sqyllnwcvq	iakgmnyled	rrlvhrdlaa
841	rnvlvktpqh	vktdfqlak	llgaeekyeh	aegkvvpikw	malesilhri	ythqsdvwsy
901	gvtvvelmtf	gskpydgipa	seissilekg	erlpqpict	idvymimvkc	wmidadsrpk
961	freliiefsk	mardepqylv	iqgdermhlp	sptdsnfya	lmdeedmdvd	vdadeyilipq
1021	qgffsspsst	rtplsslsa	tsnnstvac	drnglqscpi	kedsflqrys	sdptgalted
1081	siddtflpvp	eylnqsvpkr	pagsvqnpvy	hngplnpaps	rdphyqdpbs	tavgnpeyln
1141	tvqptcvnst	fdspahwaqk	gshqisldnp	dyqqdfffpe	akpngifkgs	taenaeylrv
1201	apqssefiga					

SEQ ID NO 93 Human HGFR CPNA Sequence Variant 2 (NM_201282.1, CDS:
from 247 to 2133)

1	ccccggcgca	gcccggcgcc	agcagcctcc	gcccccgca	cggtgtgagc	gcccagcgcg
61	gcccaggcgg	ccggagtcce	gagctagccc	cgccggcgcc	cgccgcccag	accggacgac
121	aggccaccctc	gtcggcgctcc	gcccaggtcc	ccgcctcgcc	gccaacgccca	caaccaccgc
181	gcacggcccc	ctgactccgt	ccagtattga	tcgggagagc	cggagcgagc	ctctcgggga
241	gcagcgatgc	gaccctccgg	gacggccggg	gcagcgctcc	tgccgctgct	ggctgcgctc
301	tgcccgcgca	gtcgggtctc	ggaggaaaag	aaagtttgcc	aaggcacgag	taacaagctc
361	acgcagttgg	gcacttttga	agatcatttt	ctcagcctcc	agaggatgtt	caataactgt
421	gaggtggctc	ttgggaattt	ggaaattacc	tatgtgcaga	ggaattatga	ctttctcttc
481	ttaaagacca	tccaggaggt	ggctgggtat	gtcctcattg	ccctcaacac	agtgagcgca
541	attcctttgg	aaaacctgca	gatcatcaga	ggaaatatgt	actacgaaaa	ttcctatgcc
601	ttagcagctc	tatctaaact	tgatgcaaat	aaaaccggac	tgaaggagct	gcccagaga
661	aatttacagg	aaatcctgca	tgccgcccgt	cggttcagca	acaaccctgc	cctgtgcaac
721	gtggagagca	tccagtgggc	ggacatagtc	agcagtgact	ttctcagcaa	catgtcgatg
781	gacttcagca	gaccctccgg	cagctgccaa	aagtgtgatc	caagctgtcc	caatgggagc
841	tgctgggggtg	caggagagga	gaactgccag	aaactgacca	aaatcatctg	tgcccagcag
901	tgctccggggc	gctgcgctgg	caagtccccc	agtgactgct	gccacaacca	gtgtgctgca
961	ggctgacacg	gcccccgggg	gagcgactgc	ctggctgtgc	gcaaatcccg	agacgaagcc
1021	acgtgcaagg	acacctgccc	ccactcatg	ctctacaacc	ccaccacgta	ccagatggat
1081	gtgaaccccc	aggggaaata	cagcttttgt	gccacctgcg	tgaagaagtg	cccccgtaat
1141	tatgtgggtg	cagatcacgg	ctcgtgcgtc	cgagcctgtg	gggcccagac	ctatgagatg
1201	gaggaagacg	gcgtccgcaa	gtgtaagaag	tgcaaggggc	cttgccgcaa	agtggtgaac
1261	ggaataggtg	ttgggtgaat	taaagactca	ctctccataa	atgctacgaa	tattaaacac
1321	ttcaaaaaat	gtccctccat	cagtgggcat	ctccacatcc	tgccggtggc	atttaggggt
1381	gactccttca	cacatactcc	tcctctggat	ccacaggaa	tgatattctt	gaaaaccgta
1441	aaggaaatca	cagggttttt	gctgattcag	gcttggcctg	aaaaacaggac	ggacctccat
1501	gcctttgaga	acctcaaat	catagcgggc	aggaccaagc	aaatggtgca	gtttctctct
1561	gcagtcgtca	gcctgaacat	aaatcctctg	ggattacgct	ccctcaagga	gataagtgat
1621	ggagatgtga	taatttcagg	aaacaaaaat	ttgtgctatg	caaatacaat	aaactggaaa
1681	aaactgtttg	ggacctccgg	tcagaaaaacc	aaaattataa	gcaacagagg	tgaaaaacgc
1741	tgcaaggcca	caggccaggt	ctgccatgcc	ttgtgctccc	ccgagggctg	ctggggcccg
1801	gagcccgagg	actgcgtctc	ttgccggaat	gtcagccgag	gcaggggaatg	cgtgggcaag
1861	tgcaaccttc	tgagggtgga	gccaagggag	tttgtggaga	actctgagtg	catacagtcg
1921	caccagagtg	gctgcctca	ggccatgaac	atcacctgca	caggacgggg	accagacaa
1981	tgatccagtg	gtgcccacta	cattgacggc	ccccactgcg	tcaagacctg	cccggcagga
2041	gtcatggggg	aaacaaacac	cctggtctgg	aagtacgcag	acccgggcca	tggtgcccac
2101	ctgtgccatc	caaatgcac	ctacgggtcc	taataaatct	tcactgtctg	actttagctc
2161	cccactaaaa	ctgcatttcc	ttcttacaat	ttcaatttct	ccctttgcct	caataaaagt
2221	cctgacacta	ttcatttga				

SEQ ID NO: 94 Human EGFR Amino Acid Sequence Isoform B (NP_958439.1)

1	mrpsgtagaa	llallaaalcp	asraleekkv	cqgtsnklq	lgtfedhfls	lqrmfnncev
61	vlgnleityv	qrnydlslfk	tiqevagyvl	ialntverip	lenlqiirgn	myyensyala
121	vlсныdankt	glkelpmrnl	qeilhgavrf	snnpalcnve	siqwrdivss	dflsnmsmdf
181	qnhlgsccqkc	dpscpngscw	gagencqkl	tliicacqcs	grcrkgspsd	cchnqcaagc
241	tgpresdclv	crkfrdeatc	kdtecpplmly	npttyqmdvn	pegkysfgat	cvkkcprnyv
301	vtdhgscvra	cgadsyemee	dgvrkckkce	gpcrkvcngi	gigefkdsls	inatnikhfk
361	nctsisgdllh	ilpvafrgds	fthtppldpq	eldilktvke	itgflliqaw	penrtdlhaf
421	enleiirgrt	kqhggfslav	vslnitslgl	rsakeisdgd	viisgnknlc	yantinwkl
481	fgtsgqtkti	isnrgensck	atgqvchalc	spegcwgppep	rdcvscrnvs	rgrecvdkcn
541	llegeprefv	enseciqlchp	ecelpqamnit	ctgrgpdnci	qcahyidgph	cvktcpagvm
601	genntlvwky	adaghvhclh	hpnctygs			

TABLE 1-continued

SEQ ID NO: 95 Human EGFR cDNA Sequence Variant 3 (NM_201283.1, CDS:
from 247 to 1464)

```

1  ccccggcgca ggcggcgccg agcagcctcc gccccccgca cgggtgtgagc gcccgacgcg
61  gccgagggcg ccggagtgcc gagctagccc cggcgggccgc cgcgcgccag accggacgac
121  agggccacctc gtccggcgcc gcccagagtc ccgcctcgcc gccaacgccca caaccaccgc
181  gcacgggcccc ctgactccgt ccagtattga tcgggagagc cggagcgagc tcttcgggga
241  gcagcgatgc gaccctccgg gacggccggg gcagcgctcc tggcgctgct ggctgcgctc
301  tgcccggcga gtccggcgct ggaggaaaag aaagtttgcc aaggcacgag taacaagctc
361  acgcagttgg gcacttttga agatcatttt ctacgcctcc agaggatgtt caataactgt
421  gaggtggtcc ttgggaattt ggaaattacc tatgtgcaga ggaattatga tcttctcttc
481  ttaagacca tccaggaggt ggctggttat gtcctcattg ccctcaacac agtggagcga
541  attcctttgg aaacctgca gatcatcaga ggaatatgt actacgaaaa ttcctatgcc
601  ttagcagttc tatctaacta tgatgcaaat aaaaccggac tgaaggagct gcccatgaga
661  aatttacagg aaatcctgca tggcgccgtg cggtttcagca acaacctgct cctgtgcaac
721  gtggagagca tccagtgccg ggacatagtc agcagtgact ttctcagcaa catgtcgatg
781  gacttcagca accacctggg cagctgcca aagtgtgac caagctgtcc caatgggagc
841  tgctgggggtg caggagagga gaactgccag aaactgacca aaatcatctg tgcccagcag
901  tgctccgggc gctgcgctgg caagtcccc agtgactgct gccacaacca gtgtgctgca
961  ggctgcacag gcccccgga gacgactgc ctggtctgcc gcaaatccg agacgaagcc
1021  acgtgcaagg acacctgccc cccactcatg ctctacaacc ccaccacgta ccagatggat
1081  gtgaaccccc agggcaaat aagcttttgt gccacctgct tgaagaagtg tcccgtaat
1141  tatgtggtga cagatcacgg ctctgctgct cgagcctgtg gggccgacag ctatgagatg
1201  gaggaagacg gcgtccgcaa gtgtaagaag tgcgaagggc cttgcccga aagtgtgtaac
1261  ggaataggta ttggtgaatt taaagactca ctctccataa atgtacgaa tattaacac
1321  ttcaaaaact gcacctccat cagtggcgat ctccacatcc tgcgggtggc atttaggggt
1381  gactccttca cacatactcc tctctggat ccacaggaac tggatattct gaaaaccgta
1441  aaggaaatca caggtttgag ctgaattatc acatgaatat aaatgggaaa tcatgtgttt
1501  agagagagaa cttttcgaca ttttctgt tcccttggaa taaaaacatt tcttctgaaa
1561  ttttaccggt aaaaaaaaa aaaaaaaa aaaaa

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SEQ ID NO: 96 Human EGFR Amino Acid Sequence Isoform C (NP_958440.1)

```

1  mrpsgtagaa llallaalp asraleekkv cqgtsnklq lgtfedhfls lqrmfnncev
61  vlgneleityv qnydlsflk tiqevagyvl ialntverip lenlqiirgn myyensyala
121  vlsnydantk glkelpmrnl qeilhgavrf snnpalcnve siqwrdivss dflsnmsmdf
181  qnhlgscqkc dpncpncscw gageencqkl tkiicagqcs grergkspsd cchnqcaagc
241  tgpresdclv crkfrdeatc kdtcpplmly npttyqmdvn pegkysfgat cvkkcpnrny
301  vtdhgsdvra cgadsyemee dgvrkckkce gpcrkvcngi gigeftkdsis inatnikhfk
361  nctsisgdlh ilpvafrgds fthtppldpq eldiltkvke itgls

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SEQ ID NP: 97 Human EGFR cDNA Sequence Variant 4 (NM_201284.1, CDS:
from 247 to 2364)

```

1  ccccggcgca ggcggcgccg agcagcctcc gccccccgca cgggtgtgagc gcccgacgcg
61  gccgagggcg ccggagtgcc gagctagccc cggcgggccgc cgcgcgccag accggacgac
121  agggccacctc gtccggcgcc gcccagagtc ccgcctcgcc gccaacgccca caaccaccgc
181  gcacgggcccc ctgactccgt ccagtattga tcgggagagc cggagcgagc tcttcgggga
241  gcagcgatgc gaccctccgg gacggccggg gcagcgctcc tggcgctgct ggctgcgctc
301  tgcccggcga gtccggcgct ggaggaaaag aaagtttgcc aaggcacgag taacaagctc
361  acgcagttgg gcacttttga agatcatttt ctacgcctcc agaggatgtt caataactgt
421  gaggtggtcc ttgggaattt ggaaattacc tatgtgcaga ggaattatga tcttctcttc
481  ttaagacca tccaggaggt ggctggttat gtcctcattg ccctcaacac agtggagcga
541  attcctttgg aaacctgca gatcatcaga ggaatatgt actacgaaaa ttcctatgcc
601  ttagcagttc tatctaacta tgatgcaaat aaaaccggac tgaaggagct gcccatgaga
661  aatttacagg aaatcctgca tggcgccgtg cggtttcagca acaacctgct cctgtgcaac
721  gtggagagca tccagtgccg ggacatagtc agcagtgact ttctcagcaa catgtcgatg
781  gacttcagca accacctggg cagctgcca aagtgtgac caagctgtcc caatgggagc
841  tgctgggggtg caggagagga gaactgccag aaactgacca aaatcatctg tgcccagcag
901  tgctccgggc gctgcgctgg caagtcccc agtgactgct gccacaacca gtgtgctgca
961  ggctgcacag gcccccgga gacgactgc ctggtctgcc gcaaatccg agacgaagcc
1021  acgtgcaagg acacctgccc cccactcatg ctctacaacc ccaccacgta ccagatggat
1081  gtgaaccccc agggcaaat aagcttttgt gccacctgct tgaagaagtg tcccgtaat
1141  tatgtggtga cagatcacgg ctctgctgct cgagcctgtg gggccgacag ctatgagacg
1201  gaggaagacg gcgtccgcaa gtgtaagaag tgcgaagggc cttgcccga aagtgtgtaac
1261  ggaataggta ttggtgaatt taaagactca ctctccataa atgtacgaa tattaacac
1321  ttcaaaaact gcacctccat cagtggcgat ctccacatcc tgcgggtggc atttaggggt
1381  gactccttca cacatactcc tctctggat ccacaggaac tggatattct gaaaaccgta
1441  aaggaaatca cagggttttt gctgattcag gcttggcctg aaaaaggagc ggacctccat
1501  gcctttgaga acctagaaat catacgccgc aggaccaagc aacatgggta gttttctctt
1561  gcagtcgtca gctgaacat aacatccttg ggattacgct cctcaagga gataagtgat
1621  ggagatgtga taatttcagg aaacaaaaat ttgtgctatg caaacacaat aaactggaaa
1681  aaactgtttg ggacctccgg tcagaaaacc aaaattataa gcaacagagg tgaaaaacgc
1741  tgcaaggcca caggccaggt ctgccatgcc ttgtgctccc ccgaggggtg ctggggcccg
1801  gagcccgagg actgctgctc ttgcccgaat gtcagccgag gcaggggaat cgtggacaag
1861  tgcaaccttc tggaggggtg gccaaaggag tttgtggaga actctgagtg catacagtg
1921  caccacaggt gctgctctca ggcatgaac atcacctgca caggacgggg accagacaa
1981  tgtatccagt gtgcccacta cattgacggc cccactgctg tcaagacctg ccgggcagga
2041  gtcatgggag aaaaacacac cctgtctggt aagtacgcag acgcccggca tgtgtgccac
2101  ctgtgccatc caaactgcac ctacgggcca ggaatgaga gtctcaaac catgttatto
2161  tgctttttta aactaatc atgtaataca agtaatgatg gcagcgtgtc ccaccagagc
2221  gggagcccgag ctgctcagga gtcatgctta ggatggatcc cttctcttct gccgtcagag

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TABLE 1-continued

2281	tttcagctgg	gttgggggtg	atgcagccac	ctccatgect	ggccttctgc	atctgtgatc
2341	atcacggcct	cctcctgcc	ctgagcctca	tgccttcacg	tgtctgttcc	ccccgctttt
2401	cctttctgcc	acccctgcac	gtgggcccgc	aggttcccaa	gagtatocct	cccatctcct
2461	tccttccact	ccctttgcc	gtgcctctca	ccccaactag	tagctaacca	tcacccccag
2521	gactgacctc	ttctctctcg	ctgccagatg	attgttcaaa	gcacagaatt	tgtcagaaac
2581	ctgcagggac	tcctatgctg	cagccttctc	cgtaattagc	atggcccccag	tcctatgcttc
2641	tagccttggt	tccttctgcc	cctctgtttg	aaattctaga	gccagctgtg	ggacaattat
2701	ctgtgtcaaa	agccagatgt	gaaaacatct	caataacaaa	ctggctgctt	tgttcaatgc
2761	tagaacaacg	cctgtcacag	agtagaaact	caaaaatatt	tgtctagtgga	atgaacaaat
2821	gaataaatgc	ataataaata	attaaccacc	aatccaacat	ccaga	

SEQ ID NO: 98 Human EGFR Amino Acid Sequence Isoform D (NP_95844.1)

1 mrpsgtagaa llallaalcp asraleekkv cqgtsnkltg lgtfedhfls lqrmfnnceev
61 vlgmleityv qrnnydlfslk tiqevagyvl ialntverip lenlqiirgn myyensyala
121 vlsnydankt glkelpmrnl qeilhgavrf snnpalcneve siqwrdivss dflsnmsmdf
181 qnhlgscqkc dpcpcpncsw gageencqkl tkiiicagqcs gcrgrkspsd cchnqcaagc
241 tgpresdcclv crkfrdeatc kdtcpplmly npttyqmdvn pegkyatgat cvkckprnyv
301 vtdhgscvra cgadsyemee dgvrkckkce gpcrkvcngi gigeftkdsls inatnikhfk
361 nctsisgdlh ilpvafrgds fthtppldpq eldilkvtke itgflliqaw penrtldhaf
421 enleiirgrt kqhggfslav vslnitslgl rslkeisdgd viisgnknlc yantinwklk
481 fgtsgqtkki isnrngensk atgqvchalc spegcwpgpep rdcvscrnvs rgrecvdkcn
541 llegeprefv enseciqchp eclpqamnit ctgrgpdnci qcayhidgph cvktcpagvm
601 genntlvwky adaghvchlc hpnctygpqn eslamlfcl fklsscnqsn dgsvshqsgs
661 paaqescclgw ipellpsefq lgwggcshlh awpasvilt assch

SEQ ID NO: 99 Human EGFR cDNA Sequence Variant 5 (NM_001346897.1, CDS:

from 258 to 3533)

1 gtccggggcag cccccggcgc agcgcgggcg cagcagcctc cgcggggcgc acgggtgtgag
61 cgcccgagcgc ggccgaggcg gccggagctc cgagctagcc ccggcggcgc ccggccgcca
121 gaccggacga caggccacct cgtcggcgtc cgcggcgagtc ccgcgcctgc cgccaacgccc
181 acaaccacccg cgacagggccc cctgactccg tccagtattg atcgggagag ccggagcgag
241 ctcttcgggg agcagcagtg cgaccctccg ggacggccgg ggcagcgctc ctggcgctgc
301 tggctgcgct ctgcccggcg agtcggcgctc tggaggaaaa gaaagtgttc caaggcacga
361 gtaacaaagc dpcgcaggtg ggcacttttg aagatcattt tctcagcctc cagaggatgt
421 tcaataaactg tgaggtggctc ctgggaattt tggaaattac ctatgtgcag aggaattatg
481 atcttttctt cttaaagacc atccaggagg tggctgtgta tgtcctcatt gccctcaaca
541 cagtggagcg aattcctttg gaaaacctgc agatcatcag aggaatatg tactacaaa
601 attcctatgc cttagcagtc ttatctaact atgatgcaaa taaaaccgga ctgaaggagc
661 tgcccatgag aaatttacag ggccaaaagt gtgatccaag ctgtcccaat gggagctgct
721 ggggtgcagc agaggagaaac tgaccagaac catctgtgccc cagcagtgct
781 ccggggcgctg ccgtggcgaag tccccagtg actgctgcca caaccagtgct gctgcaggct
841 gcacaggccc ccgggagagc gactgcctgg tctgcccga attcggagac gaagccacgt
901 gcaaggacac cctcccccca ctcatgctct acaaccccac cagctaccag atggatgtga
961 accccgaggg caataacagc ttgtgtgcca cctgcgtgaa gaagtgtccc cgtaattatg
1021 tgggtgacaga tcacggctcg tgcgtccgag cctgtggggc cgacagctat gagatggagg
1081 aagacggcgt ccgcaagtgt aagaagtgcg aagggccttg ccgcaaatg tgtaacggaa
1141 taggtatttg tgaatttaaa gactcactct ccataaatgc tacgaatatt aaacacttca
1201 aaaactgcac ctccatcagt ggcgatctcc acatcctgcc ggtggcattt aggggtgact
1261 ccttcacaca tactcctcct ctggatccac aggaactgga tattctgaaa accgtaaaagg
1321 aaatcacagg gtttttgcgt attcaggctt ggcctgaaaa caggacggac ctccatgcct
1381 ttgagaacct agaaatcata cgcggcagga ccaagcaaca tggatcagtt tctcttgacg
1441 tctcagcctg gaacataaca tccttgggat tacgctccct caaggagata agtgatggag
1501 atgtgatgat ttacaggaaac aaaaatttgt gctatgcaaa tacaataaac tggaaaaaac
1561 tgtttgggac ctccggctcg aaaacccaaa ttataagcaa cagaggtgaa aacagctgca
1621 agggccacag ccaggctctgc catgccttgt gctccccga gggctgtgag gggccggagc
1681 ccagggagctg cgtctcttgc cggaatgtca gccgaggcag ggaatgcgtg gacaagtga
1741 accttctgga ggggtgagcca agggagtttg tggagaactc tgagtgcata cagtgcacc
1801 cagagtgcct gctccaggcc atgaacatca cctgcacagg acggggacca gacaactgta
1861 tccagtgtgc ccaactacatt gacggccccc actgcgtcaa gacctgccc gaggagtc
1921 tgggagaaaa caacaccctg gtctggaagt acgcagacgc cggccatgtg tgccacctgt
1981 gccatccaaa ctgcacccatc ggatgcactg ggccaggtct tgaaggctgt ccaacgaatg
2041 ggcctaagat ccctgccatc gccactggga tgggtggggc cctcctcttg ctgctgggtg
2101 tggccctggg gatcggcctc ttcatgcaaa ggcgccacat cgttcggaag cgcacgctgc
2161 ggaggctgct gcaggagagg gagcttgtgg agcctcttac acccagtgga gaagctccca
2221 accaagctct ctgaggatc ttgaaggaaa ctgaattcaa aaagatcaaa gtgctgggct
2281 ccggtgcgtt cggcagcggg tataagggac tctggatccc agaaggtgag aaagttaaaa
2341 ttcccgtcgc tgcacaaggaa ttaagagaa caacatctcc gaaagccaac aaggaaatcc
2401 tcatggaagc ctacgtgatg gccagcgtgg acaaccccca cgtgtgccgc ctgctgggca
2461 tctgctcacc ctccaccgtg cagctcatca cgcagctcat gcccttcggc tgccctcctg
2521 actatgtccc ggaaacacaaa gacaattatt gctcccagta cctgctcaac tgggtgtgctg
2581 agatcgcaaa gggcatgaac tacttgagg accgtgcgtt ggtgcaccgc gacctggcag
2641 ccaggaacct actgggtgaa acaccgcagc atgtcaagat cacagatttt gggctggcca
2701 aactgctggg tgcgggaagg atgcagaagg aggcacaagt ccatcaagt
2761 ggatggcatt ggaatcaatt ttacacagaa tctataccca ccagagtgat gtctggagct
2821 accgggtgac tgtttgggag ttgatgacct ttggatccaa gccatagac ggaatccctg
2881 ccagcgagat ctctccatc ctggagaaa gagaacgcct ccctcagcca cccatagta
2941 ccacgatgat ctacatgatc atggtcaagt gctggatgat agacgcagat agtcgcccac
3001 agttccgtga gttgatcatc gaattctcca aaatggccc agacccccag cgtaccttg
3061 tcattcagg ggatgaaga atgcatttgc caagtccac agactccac ttctaccgtg
3121 cctgatgga tgaagaagac atggacgagc tgggtggatgc cgacgagtac ctcatccac

TABLE 1-continued

3181	agcagggtctt	cttcagcagc	ccctccacgt	cacggactcc	cctcctgagc	tctctgagtg
3241	caaccagcaa	caattccacc	gtggcttgca	ttgatagaaa	tgggctgcaa	agctgtccca
3301	tcaaggagaa	cagcttcttg	cagcgataca	gctcagaccc	cacaggcgcc	ttgactgagg
3361	acagcataga	cgacaccttc	ctcccagtcg	ctgggtgagt	gcttgtctgg	aaacagtcct
3421	gctcctcaac	ctcctcgacc	cactcagcag	cagccagtct	ccagtgtcca	agccagggtgc
3481	tcctctccagc	atctccagag	ggggaacag	tggcagattt	gcagacacag	tgaagggcgt
3541	aaggagcaga	taaacacatg	accgagcctg	cacaagctct	ttgttgtgtc	tggttgtttg
3601	ctgtacctct	gttgtaagaa	tgaatctgca	aaatttctag	cttatgaagc	aaatcacgga
3661	catacacatc	tgtgtgtgtg	agtgttcatg	atgtgtgtac	atctgtgtat	gtgtgtgtgt
3721	gtatgtgtgt	gtttgtgaca	gatttgatcc	ctgttctctc	tgctggctct	atcttgacct
3781	gtgaaacgta	tatttaacta	atataatatt	agttaatatt	aataaatttt	aagctttatc
3841	cagaaaaaaa	aaaaaaaaa				

SEQ ID NO: 100 Human EGFR Amino Acid Sequence Isoform E (NP_001333826.1)

1 mrpsgtagaa llallaalcp asraleekkv cqgtsnklq lgtfedhfls lqrmfnncev
61 vlgneleityv qnydlsflk tiqevagyvl ialntverip lenlqiirgn myyensyala
121 vlsnydankt glkelpmrnl ggqkcdpsep ngscwgagee ncqkltkiic aqqcsgrcrq
181 kspdcchngc caagctgpre sdclvcrkfr deatckdccc plmlynptty qmdvnpesky
241 sfgatcvkkc prnyvvt dhg scvracgads yemeedgvrk ckkecgpcrk vcnigigef
301 kdsleinatn ikhfkntcsi sgdlhilpva frgdsfthtp pldpqeldil ktvkeitgfl
361 liqawpenrt dlhafenlei irgrtkqhgg fslavvslni tsiglrlslke isdgdviisg
421 nknlcyanli nwkklfgtsg qtkkiisnrg enskatgqv chalcspcgc wpeprdcvs
481 crnvsrgrec vdkcnllge prefvnscc iqchpeclpq amnitctgrg pncicqcahy
541 idgphvcvkc pagvmgennt lvwkyadagh vchlcnpnct ygctgpgleg cptngpkips
601 iatgmvgall lllvvalgig lfmrrrhivr krtlrllqe relvepltps geapnqallr
661 ilketefkki kvlgsgafgt vykgwipieg ekvkipvaik elreatspka nkeildeayv
721 masvndphvc rllgieltst vqlitqlmpf gclldyvreh kdnigsqyll nwcqvkiakgm
781 nyledrrlvh rdlaarnvlv ktpqhvkitd fglakllgae ekeyhaeggk vpikwmales
841 ilhriythsq dwsvgyvtvw elmtfgskpy dgipaseiss ilekgerlpq ppictidvym
901 imvkcwmida dsrpkrfeli iefskmardp qrylviggede rmhlpsptds nfyralmdce
961 dmdvdvdade yllpqggffs spstertpll ssleatsnns tvacidrngl qscpikedsf
1021 lqrysdpdptg altdedsiddt flpvpgevlw wkqscsstss thsaaaslqc psqvlppasp
1081 egetvadlqt q

SEQ ID NO: 101 Human EGFR cDNA Sequence Variant 6 (NM_001346898.1, CDS:
from 258 to 3668)

1 gtccggggcag cccccggcgc agcgcggcgc cagcagcctc cccccccgc acggtgtgag
61 cgcccgagcgc ggccgaggcg gcccgagctc cgagctagcc ccggcgggcg ccgcccgcga
121 gaccggacga caggccacct cgtcggcgtc cgcccgagtc ccgcgcctgc cgccaaacgc
181 acaaccacgc cgacaggccc cctgactcgc tccagtattg atccggagag ccggagcgag
241 ctcttcgggg agcagcagat cgaccctccg ggacggccgg ggcagcgctc ctggcgctgc
301 tggtgtcgct ctgcccggcg agtcggcgtc tggaggaaaa gaaagtgttc caaggcacga
361 gtaacaagct cagcgacttg gccacttttg aagatcattt tctcagcctc cagaggatgt
421 tcaataaactg tgaggtggct cttgggaatt tggaaattac ctatgtgcag aggaattatg
481 atctttcctt cttaaagacc atccaggagg tggctgggta tgcctcatt gccctcaaca
541 cagtggagcg aattcctttg gaaaacctgc agatcatcag aggaatatg tactacgaaa
601 attcctatgc cttagcagtc ttatctaact atgatgcaaa taaaaccgga ctgaaggagc
661 tgccccatgag aaatttacag gaaacctgc atggcgccgt gcggttcagc aacaacctgc
721 cctgtgtcaa cgtggagagc atccagtggc gggacatagt cagcagtgac ttcctcagca
781 acatgtcgat ggaactccag aaccacctgg gcagctgcca aaagtgtgat ccaagctgtc
841 ccaatgggag ctgctggggt gcaggagagg agaactgcca gaaactgacc aaatcatct
901 gtgccacga cgtgtccggg cgtcgcctgc gcaagtcgcc cagtactgc tgccacaacc
961 agtgtgtctc aggtctgaca ggccccggg agagcgactg cctggctctc cgcaaatcc
1021 gagacgaagc cagctgcaag gacacctgcc ccccaactcat gccctacaac cccaccacgt
1081 accagatgga tgtgaacccc gagggcaaat acagctttgg tgccacctgc gtgaagaagt
1141 gtccccgtaa ttatgtgggt acagatcacg gctcgtcgct ccgagcctgt gggggcgaca
1201 gctatgagat ggaggagagc ggcgtccgca agtgtaagaa gtgcgaaggg ccttgccgca
1261 aagtggtgaa cgtgaatagg attggtgaat ttaagactc actctccata aatgctacga
1321 atattaaaca cttcaaaaac tgcacctcca tcagtggcga tctccacatc ctgccggtgg
1381 catttagggg tgactccttc acacatactc ctctcttgga tccacaggaa ctggatattc
1441 tgaaaaccgt aaaggaaatc acagggtttt tgctgattca ggcttggcct gaaaacagga
1501 cggacctcca tgcttttagg aacctagaaa tcatacgcgg caggaccaag caaatggtc
1561 agttttctct tgcagtcgtc agcctgaaca taacatcctt gggattacgc tccctcaagg
1621 agataagtag tggagatgtg ataatttcag gaaacaaaaa tttgtgctat gcaaatacaa
1681 taaactggaa aaaaactgtt gggacctccg gtcagaaaaa caaaattata agcaacagag
1741 gtgaaaacag ctgcaaggcc acaggccagg tctgccatgc cttgtgtctc cccgagggct
1801 gctggggccc cggagccagg gactgcgtct cttgccgga tgcagccga ggcagggaat
1861 gcgtggacaa gtgcaacctt ctggagggtg agccaaggga gtttgtggag aactctgagt
1921 gcatacagtg ccaccagagc tgcctgctc agggccatgaa catcactgc acaggacggg
1981 gaccagacaa cttgatccag tgtgccact acattgacgg cccccactgc gtcaagacct
2041 gcccggcagg agtcatggga gaaaacaaca cctgggtctg gaagtacgca gacgcggcc
2101 atgtgtgcca cctgtgccat ccaaaactgca cctacggatg cactgggcca ggtcttgaag
2161 gctgtccaag gaatggccct aagatccgt ccatcgccac tgggatgggt gggggcctcc
2221 tcttgcgtct ggtggtggcc ctggggatcg gcctctcat gcgaaggcgc cacatcgttc
2281 ggaagcgcac gctgcggagg ctgctgcagg agaggagct tgtggagcct cttacaccca
2341 gtggagaagc tcccaaccaa gctctcttga ggatcttga ggaactgaa tccaaaaga
2401 tcaaatgtct gggctccggt gcgttcggca cggtgtataa gggactctgg atcccagaag
2461 gtgagaaggt taaaattccc gtcgctatca aggaattaag agaagcaaca tctccgaaag
2521 ccaacaagga aatcctcgat gaagcctacg tgatggccag cgtggacaac cccacgtgt
2581 gccgcctgct gggcatctgc ctcacctcca ccgtgcagct catcacgcag ctcatgcctc

TABLE 1-continued

2641	tccgctgctc	cctggactat	gtccgggaac	acaaagacaa	tattggctcc	cagtacctgc
2701	tcaactgggt	tgtgcagatc	gcaaagggca	tgaactactt	ggaggaccgt	cgcttggtgc
2761	accgcgacct	ggcagccagg	aacgtactgg	tgaaaacacc	gcagcatgtc	aagatcacag
2821	attttgggct	ggccaaactg	ctgggtgcgg	aagagaaaga	atccatgca	gaaggaggca
2881	aagtgcctat	caagtggatg	gcattggaat	caattttaca	cagaatctat	acccaccaga
2941	gtgatgtctg	gagctacggg	gcgactgttt	gggagttgat	gacctttgga	tccaagccat
3001	atgacggaat	ccctgccagc	gagatctcct	ccatcctgga	gaaaggagaa	cgccctccctc
3061	agccacccat	atgtaccatc	gatgtctaca	tgatcatggt	caagtgcctg	atgatagacg
3121	cagatagtcg	cccaaagtcc	cgtaggttga	tcacgaatt	ctccaaaatg	gcccagagacc
3181	cccagcgcta	ccttgtcatt	cagggggatg	aaagaatgca	tttgccaagt	cctacagact
3241	ccaacttcta	ccgtgccctg	atggatgaag	aagacatgga	cgacgtggtg	gatgccgacg
3301	agtaacctcat	cccacagcag	ggctttctca	gcagccctcc	cacgtcacgg	actccctcc
3361	tgagctctct	gagtgcaacc	agcaacaatt	ccaccgtggc	ttgcattgat	agaaatgggc
3421	tgcaaaagctg	tcccatcaag	gaagacagct	tcttgacgcg	atacagctca	gacccacacg
3481	gcgccttgac	tgaggacagc	atagacgaca	ccttccctcc	agtgctctgt	gagtggttg
3541	tctgggaaca	ctctgctccc	tcaacctcct	cgaccactc	agcagcagcc	agtcctcagt
3601	gtccaagcca	ggtgctccct	ccagcatctc	cagaggggga	aacagtggca	gatttgcaga
3661	cacagtgaag	ggcgtgaagga	gcagataaac	acatgaccca	gctgcacaaa	gctctttgtt
3721	gtgtctgggt	gtttgtctgt	cctctgttgt	aagaatgaat	ctgcaaaatt	tctagcttat
3781	gaagcaaatc	acggacatac	acatctgtgt	gtgtgagtg	tcatgatgtg	tgtacatctg
3841	tgtatgtgtg	tgtgtgtatg	tgtgtgtttg	tgacagatgt	gatccctgtt	ctctctgctg
3901	gctctacctt	gacctgtgaa	acgtatatatt	aactaattaa	atattagtta	atattaataa
3961	attttaagct	ttatccagaa	aaaaaaaaaa	aaaa		

SEQ ID NO: 102 Human EGFR Amino Acid Sequence Isoform F (NP_001333827.1)

1	mrpsgtagaa	llalllaalcp	asraleekkv	cqgtcnklitq	lgtfedhfls	lqrmfnncev
61	vlgnleityv	qrrydlslfk	tiqevagyvl	ialntverip	lenlqiirgn	myyensyala
121	vlshndankt	glkclpmrnl	qeilhgvrf	snnpalcnve	siqwrdivss	dflsnmsmdf
181	qnhlgscqkc	dpccpnsgcw	gagencqkl	tikiacqqs	grcrgkpspd	cchnqcaagc
241	tgpresdclv	crkfrdeatc	kdtcpplmly	npttyqmdvn	pegkysfgat	cvkkcprnyv
301	vtdhgscvra	cgadsyemee	dgvrkckkce	gpcrkvcngi	gigefkdsls	inatnikhfk
361	nctsisgdll	ilpvafrgds	fthtppldpq	eldilktvke	itgflliqaw	penrtdlhaf
421	enleirgrt	kqhgqtslav	vslnitslgl	rsлкеisddg	viisgnknlc	yantinwkkll
481	fgtsgqtkki	isrnrgensck	atgqvchalc	spegcwgpep	rdcvscrnvs	rgrecvdckn
541	llegeprefv	enseciqchp	eclpqamnit	ctgrgpdnci	qcahyidgph	cvktcpagvm
601	genntlvwky	adaghvchlc	hpnctygtcg	pglegcptng	pkipsiatgm	vgallllllv
661	alvgglfmr	rhivrkrtlr	rlqerelvc	pltpsgcapn	qallrilket	efkkikvlgs
721	gafgtvykgl	wipegekvti	pvaikelrea	tspkankeil	deayvmavsd	nphvcrllgi
781	cltstvtqlit	qlmpfgclld	yvrehkdng	sqyllnwcvq	iakgmnyled	rrlvhrdlaa
841	rnvlvktpph	vkltddfglak	llgaeekeyh	aegkvvpikw	malesilhri	ythqsdvwsy
901	gvtvvelmtf	gskpydgipa	seissilekg	erlpqppict	idvymimvkc	wmidadsrpk
961	freliiefsk	marpdqrylv	iqgdermhlp	sptdsnfyra	lmdeedmdv	vdadeylipq
1021	qgffsspsps	rtpllsllsa	tsnnstvac	drnlgscpi	kedsflqrys	sdptgalted
1081	siddtflvpv	gewlvwkqsc	sstssthsaa	aslqcpssqvl	ppaspegetv	adlqtq

SEQ ID NO: 103 Human EGFR cDNA Sequence Variant 7 (NM_001346899.1, CDS: from 258 to 3755)

1	gtccgggcag	ccccgggcgc	agcgcggcgc	cagcagcctc	cgccccccgc	acgggtgtgag
61	cgcccgacgc	ggccgaggcg	gccggagtc	cgagctagcc	ccggcgggcg	cgcccgccca
121	gaccggacga	caggccacct	cgccggcgtc	cgcccgagtc	ccgcctcgc	cgccaacgcc
181	acaaccaccg	cgcacggccc	cctgactccg	tccagtattg	atcgggagag	ccggagcgag
241	ctcttcgggg	agcagcgatg	cgaccctccg	ggcggcgccg	ggcagcgctc	ctggcgctgc
301	tggctgcgct	ctgcccggcg	agtcgggctc	tggaggaaaa	gaaagtgtgc	caaggcacga
361	gtaacaagct	cacgcagttg	ggcacttttg	aagatcattt	tctcagcctc	cagaggatgt
421	tcaataaact	tgaggtggtc	cttgggaatt	tggaaattac	ctatgtgcag	aggaattatg
481	atctttcctt	cttaagagcc	atccaggagg	tggctggtta	tgtcctcatt	gccctcaaca
541	cagtggagcg	aattcctttg	gaaaacctgc	agatcatcag	aggaaatatg	tactacgaaa
601	attcctatgc	cttagcagtc	ttatctaact	atgatgcaaa	taaaaccgga	ctgaaggagc
661	tgcccatgag	aaatttacag	ggccaaaagt	gtgatccaag	ctgtcccaat	gggagctgct
721	gggggtgcag	agaggagaac	tgccagaaac	tgacaaaaat	catctgtgcc	cagcagtgct
781	ccggggcgctg	cgtctggcaag	tccccagtg	actgctgcca	caaccagtg	gctgcaggct
841	gcacaggccc	ccgggagagc	gactgcctgg	tctgccgcaa	attccgagac	gaagccacgt
901	gcaaggacac	ctgcccccca	ctcatgctct	acaacccccc	cacgtaccag	atggatgtga
961	accccgaggg	caaatcacgc	tttggtgcca	cctgcgtgaa	gaagtgtccc	cgtaattatg
1021	tgggtgacaga	tcacggctcg	tgcgtccgag	cctgtggggc	cgacagctat	gagatggagg
1081	aagacggcgt	ccgcaagtgt	aagaagtgcg	aaggggcctg	ccgcaagtgt	tgtaacggaa
1141	taggtatttg	tgaatttaaa	gactcactct	ccataaatgc	tacgaatat	aaacacttca
1201	aaaactgcac	ctccatcagt	ggcgatctcc	acatcctgcc	ggtggcattt	aggggtgact
1261	ccttcacaca	tactcctcct	ctggatccac	aggaactgga	tattctgaaa	accgtaaaag
1321	aaatcacagg	gttttctgctg	atccaggtct	ggcctgaaaa	caggacggac	ctccatgctc
1381	ttgagaacct	agaaatcata	cgccgcaggga	ccaagcaaca	tgtcagttt	tctcttgacg
1441	tcgtcagcct	gaacataaca	tccttgggat	tacgctccct	caaggagata	agtgatggag
1501	atgtgataat	ttacaggaac	aaaaatttgt	gctatgcaaa	tacaataaac	tggaaaaaac
1561	tggttgggac	ctccggtcag	aaaacaaaaa	ttataagcaa	cagaggtgaa	aacagctgca
1621	agggccacag	ccaggtctgc	catgccttgt	gctcccccca	gggctgctgg	ggcccgagag
1681	ccagggaactg	cgtctcttgc	cggaatgtca	gccgaggcag	gggaatgcgtg	gacaagtgtca
1741	accttctgga	gggtgagcca	agggagtttg	tggagaactc	tgagtgcata	cagtgccacc
1801	cagagtgcct	gcctcaggcc	atgaacatca	cctgcacagg	acggggacca	gacaactgtg
1861	tccagtgctc	ccactacatt	gacggcccc	actgcgtcaa	gacctgccc	gcaggagtca
1921	tgggagaaaa	caacacctg	gtctggaagt	acgcagacgc	cgcccatgtg	tgccacctgt

TABLE 1-continued

1981	gccatccaaa	ctgcacctac	ggatgcactg	ggccaggctc	tgaagcgtgt	ccaacgaatg
2041	ggcctaagat	cccgtccatc	gccactggga	tggtgggggc	cctcctcttg	ctgctgggtg
2101	tggccctggg	gatcggcctc	ttcatgcgaa	ggcgccacat	cgttcgggaag	cgcacgctgc
2161	ggaggctgct	gcaggagagg	gagcttggtg	agcctcttac	accagtgga	gaagctccca
2221	accaagctct	cttgaggatc	ttgaaggaaa	ctgaattcaa	aaagatcaaa	gtgctgggct
2281	ccggtgctgt	cggcacgggtg	tataagggac	tctggatccc	agaagggtgag	aaagttaaaa
2341	ttcccgtcgc	tatcaaggaa	ttaagagaag	caacatctcc	gaaagccaac	aaggaaatcc
2401	tcgatgaagc	ctacgtgatg	gccagcgtgg	acaaccccc	cgtgtgccgc	ctgctgggca
2461	tctgcctcac	ctccacccgtg	cagctcatca	cgcagctcat	gcccttcggc	tgccctcctgg
2521	actatgtccg	ggaacacaaa	gacaatattg	gctcccagta	cctgctcaac	tggtgcgtgc
2581	agatcgcaaa	gggcatgaac	tacttgaggg	accgtcgctt	ggtgcaccgc	gacctggcag
2641	ccaggaacgt	actggtgaaa	acaccgcagc	atgtcaagat	cacagatttt	gggctggcca
2701	aactgctggg	tgcggaagag	aaagaatacc	atgcagaagg	aggcaaaagt	cctatcaagt
2761	ggatggcatt	ggaatcaatt	ttacacagaa	tctataccca	ccagagtgat	gtctggagct
2821	acgggggtgac	tgtttggggag	ttgatgacct	ttggatccaa	gccatgatgac	ggaatccctg
2881	ccagcgagat	ctcctccatc	ctggagaaa	gagaacgcct	ccctcagcca	cccatatgta
2941	ccatcgatgt	ctacatgatc	atggtcaagt	gctggatgat	agacgcagat	agtcgcccc
3001	agttccgtga	gttgatcatc	gaattctcca	aaatggcccg	agacccccag	cgtacacctg
3061	tctattcagg	ggatgacaaa	atgcatttgc	caagtcctac	agactccaac	ttctaccgtg
3121	ccctgatgga	tgaagaagac	atggacgacg	tggtggatgc	cgacgagtag	ctcatcccac
3181	agcagggtct	cttcagcagc	ccctccacgt	cacggactcc	cctcctgagc	tctctgagtg
3241	caaccagcaa	caattccacc	gtggcttgca	ttgatagaaa	tgggctgcaa	agctgtccca
3301	tcaaggaaga	cagcttcttg	cagcgataca	gctcagaccc	cacaggcgcc	ttgactgagg
3361	acagcataga	cgcacacctc	ctcccagtg	ctgaatacat	aaaccagtc	gttccccaaa
3421	ggcccgctgg	ctctgtgcag	aatcctgtct	atcacatca	gcctctgaac	cccgcgccca
3481	gcagagaccc	acactaccag	gacccccaca	gcactgcagt	gggcaacccc	gagtatctca
3541	acactgtcca	gccccacctg	gtcaacagca	cattcgacag	ccctgcccac	tgggcccaga
3601	aaggcagcca	ccaaattagc	ctggacaacc	ctgactacca	gcaggacttc	tttcccaagg
3661	aagccaagcc	aaatggcatc	tttaagggtc	ccacagctga	aaatgcagaa	tacctaaagg
3721	tcgcgccaca	aagcagtgaa	tttatgggag	catgaccacg	gaggatagta	tgagccctaa
3781	aaatccagac	tctttcgata	cccaggacca	agccacagca	ggctcctcc	cccaacagcc
3841	atgcccgcac	tagctcttag	accacagac	tggttttgca	acgtttacac	cgactagcca
3901	ggaagtactt	ccacctcggg	cacattttgg	gaagttgcat	tcctttgtct	tcaaaactgtg
3961	aagcatttac	agtaacgcac	ccagcaagaa	tattgtccct	ttgagcagaa	atztatcttt
4021	caaagaggta	tatttgaaaa	aaaaaaaaag	tatatgtgag	gattttttatt	gattggggat
4081	cttggagttt	ttcattgtcg	ctattgattt	ttacttcaat	gggctcttcc	aacaagggaag
4141	aagcttgctg	ctagcacttg	ctaccctgag	ttcatccagg	cccaactgtg	agcaaggagc
4201	acaagccaca	agtcttccag	aggatgcttg	attccagtg	ttctgcttca	aggcttccac
4261	tgcaaaacac	taaatgatca	agaaggcctt	catggcccca	gcaggccgga	tcggtagctg
4321	atcaagtcat	ggcaggtaca	gtaggataag	ccactctgtc	ccttctcggg	caaaagaagaa
4381	acggaggggg	tggaattctt	ccttagactt	actttttaa	aaatgtcccc	acggtagctta
4441	ctccccactg	atggaccagt	ggtttccagt	catgagcgtt	agactgactt	gtttgtcttc
4501	catcctatg	ttttgaaact	cagtatgctg	ccctgtctt	gctgtcatga	aatcagcaag
4561	agaggatgac	acatcaaata	ataactcggg	ttccagccca	cattggattc	atcagcattt
4621	ggaccaatag	cccacagctg	agaatgtgga	atacctaagg	atagcacccg	ttttgttctc
4681	gcaaaaaact	agatccta	ttgaggctca	gatgaaatgc	atcaggtcct	tgggggcata
4741	gatcagaaga	ctacaaaaat	gaagctgctc	tgaatctcc	tttagccatc	accccaaccc
4801	cccaaaatta	gtttgtgtta	cttatggag	atagttttct	ccttttactt	cacttcaaaa
4861	gcttttctac	caaagagtat	atgttccctc	caggtcagct	gcccccaaac	ccctcctta
4921	cgctttgtca	cacaaaaagt	gtctctgcct	tgagtcactc	attcaagcac	ttacagctct
4981	ggccacaaca	gggcatttta	caggtgcgaa	tgacagttag	attatgagta	gtgtggaatt
5041	acgtagttaa	atatgaaact	agggtttgaa	attgataatg	ctttcacac	atttgcagat
5101	gttttagaag	gaaaaaagtt	ccttcctaaa	ataatttctc	tacaattgga	agattggaag
5161	attcagctag	ttaggagccc	accttttttc	ctaactctgt	tgtgcctgtg	aaactgactg
5221	gttaacagca	gtcctttgta	aacagtgttt	taaactctcc	tagtcaatat	ccacccatc
5281	caatttatca	aggaagaat	ggttcagaaa	atattttcag	cctacagtta	tgttcagtc
5341	cacacacata	caaaatgttc	cttttgcttt	taaagtaatt	tttgactccc	agatcagtc
5401	gagccctcac	agctattgta	agaaagtatt	tgatttttgt	ctcaatgaaa	ataaaactat
5461	attcatttcc	actctattat	gctctcaaat	accctaagc	atctatacta	gcctggtagt
5521	ggatgaaag	atacaaatg	aaataaaaca	tagtccctga	ttctaagaaa	ttcacaaatt
5581	agcaaaaggaa	atggactcat	agatgcta	cttaaaacaa	cgtgacaaat	gccagacagg
5641	accatcagc	caggcactgt	gagagcacag	agcaggagg	ttgggtcctg	cctgaggaga
5701	cctggaagg	aggcctcaca	ggaggatgac	caggtctcag	tcagcgggga	ggtggaaggt
5761	gcagggtgac	cagggcacac	ctgaccgagg	aaacagctgc	cagaggcctc	cactgtcaaa
5821	gtccacataa	ggctgaggtc	agtcacccta	aacaacctgc	tcctctaa	ccaggggtag
5881	agcttgaggc	atccacaa	ttccctaaaa	gttgacagccc	ccagggggat	tttgagctat
5941	catctctgca	catgcttagt	gagaagacta	cacaacattt	ctaagaatct	gagattttat
6001	attgtcagtt	aaccactttc	attatttcatt	cacctcagga	catgcagaaa	tatttcagtc
6061	agaactggga	aacagaagga	cctacattct	gctgtcactt	atgtgtcaag	aagcagatga
6121	tcgatgaggc	aggtcaggtg	taagttagtc	acattgtagc	attaaattct	agtatttttg
6181	tagtttgaaa	cagtaactta	ataaaagagc	aaaagctaaa	aaaaaaaaaa	aaaa

SEQ ID NO: 104 Human EGQFR Amino Acid Sequence Isoform G (NP_001333828.1)

1 mrpsetgaaq llallaaalp asraleekkv cqgtsnklitq lgtfedhfls lqrmfnncev
 61 vlgnleityv qnydlsflk tiqevagyvl ialntverip lenlqiirgn myyensyala
 121 vlsnydankt glkelpmrnl gqgkcdpscp ngscwagee ncqkltkiic aqgcgrgrg
 181 kspedchohnq caagctgpre sdclvrkfr deatckdtp plmlynptty qmdvnpky
 241 sfgatcvkkc prnyvvdthg scvracgads yemeedgvrk ckckegpckr vengigigef
 301 kdslninatn ikhfkncetsi sgdlhilpva frgdsfthtp pldpqldil ktkvktgfl
 361 liqawpenrt dlhafenlei irgrtkqhgg fslavvslni tsiglrlske isdgdiviisg

TABLE 1-continued

421	nknlcyanti	nwkklfqtsg	qtkkiisnrg	ensckatgqv	chalcspegc	wgpeprdcvs
481	crnvsgrec	vdkenllege	prefvensec	iqchpeclpq	amnitctgrg	pdnciqcahy
541	idgphvcvtc	pagvmgennt	lvwkyadagh	vchlchpnct	ygctgpgleg	cpnngpkips
601	iatgmvgall	lllvvalgig	lfmrrrhivr	krtlrlllqe	relvepltps	geapngallr
661	ilketefkki	kvlgsgafgt	vykglwipeg	ekvkipvaik	elreatspka	nkeildeayv
721	masvdnphvc	rligicttst	vqlitqlmpf	gclddyvreh	kdnigsqyll	nwcvciaqgm
781	nyledrrlvh	rdlaarnvlv	ktpqhvkitd	fglakllgae	ekeyhaeggk	vpikwmales
841	ilhriythqs	dvwsygvttv	elmtfgskpy	dgipaseiss	ilekgerlpq	ppictidvym
901	imvkcwmida	dsrpkfrelt	iefskmardp	qrylvigqde	rmlhlpptds	nfyralmdce
961	dmddvvdade	ylipqggffs	spstsrtpll	sslsatnns	tvacidrnl	qscpikedsf
1021	lqrysdsptg	altdesiddt	flpvpeyinq	svpkrrpagsv	qnpvyhnpql	npapserdphy
1081	qdphtstavn	peylntvqpt	cvnstfdspa	hwaqkqshqi	sldnpydqgd	ffpkcapkng
1141	ifkgstaena	eylrvaqpss	efiga			

SEQ ID NO: 105 Human EGFR cDNA Sequence Variant 8 (NM_001346900.1, CDS: from 214 to 3687)

1	ccttttgaat	gagctctaaa	acagttcttc	actggacttc	agaacaagag	ggagctctgg
61	gctgctggct	gggtgtgcat	ttgctgtggg	tcccccccg	caggcgacct	ctccgcgctg
121	agaaggttat	ccggataaacc	aatttgccaa	ggcacgagta	acaagctcac	gcagttggcg
181	acttttgaag	atcatttcct	cagcctccag	aggatgttca	ataactgtga	ggtggtcctt
241	gggaatttgg	aaattaccta	tgtgcagagg	aattatgatc	tttccctctt	aaagaccatc
301	caggaggtgg	ctggttatgt	cctcattgcc	ctcaacacag	tggagcgaat	tcctttggaa
361	aacctgcaga	tcacagagg	aaatatgtac	tacgaaaatt	cctatgcctt	agcagtcctt
421	tctaactatg	atgcaataaa	aaccggactg	aaggagctgc	ccatgagaaa	tttacaggaa
481	atcctgcatg	gcgcccgtgc	gttcagcaac	aacctgcctc	tgtgcaacgt	ggagagcatc
541	cagtggcggg	acatagtcag	cagtgaactt	ctcagcaaca	tgtcgatgga	cttcacagaa
601	cacctgggca	gctgcacaaa	gtgtgatcca	agctgtccca	atgggagctg	ctgggggtga
661	ggagaggaga	actgcacaaa	actgacccaa	accatctgtg	cccagcagtg	ctccggcgcg
721	tggcgtggga	agtccccag	tgactgctgc	cacaaccagt	gtgctgcagg	ctgcacaggc
781	cccggggaga	gcgactgcct	ggctgtccgc	aaattccgag	acgaagccac	gtgcaggagc
841	acctgcccc	cactcatgct	ctacaacccc	accacgtacc	agatggatgt	gaaccccag
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961	gatacggctt	cgtgcgtccg	agcctgtggg	gccgacagct	atgagatgga	gggaagcggc
1021	gtccgcaagt	gtgaagagtg	cgaaggccct	tgccgcaaa	tgtgtaacgg	aatagggtatt
1081	ggtgaattta	aagactcact	ctccataaat	gctacgaata	ttaaacactt	caaaaactgc
1141	acctccatca	gtggcgatct	ccacatctcg	ccggtggcat	ttagggggtga	ctccttcaca
1201	catactcctc	ctctggatcc	acaggaactg	gatattctga	aaaccgtaaa	ggaaatcaca
1261	gggtttttgc	tgattcaggc	ttggcctgaa	aacaggacgg	acctccatgc	ctttgagaac
1321	ctagaaatca	tacgcggcag	gaccaagcaa	catggtcagt	tttctcttgc	agtcgtcagc
1381	ctgaacataa	catccttggg	attacgctcc	ctcaaggaga	taagtgtatg	agatgtgata
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1621	tgcgtctctt	gccggaatgt	cagccgaggc	agggaatgct	tggacaagtg	caacctcttg
1681	gagggtgagc	caaggagatt	tgtggagaac	tctgagtcca	tacagtcca	cccagagtgc
1741	ctgcctcagg	ccatgaacat	cacctgcaca	ggacggggac	cagacaactg	tatccagtgt
1801	gcccaactaca	ttgacggccc	ccactgcctc	aagacctgcc	cggcaggagt	catgggagaa
1861	aacaaccccc	tggtctggaa	gtacgcagac	gccggccatg	tgtgccacct	gtgccatcca
1921	aactgcacat	aggaatgcac	tgggccagggt	cttgaaggct	gtccaaacga	tgggcctaa
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2041	gggatcgccc	tcttcatgct	aaggcgccac	atcgctcgga	agcgcacgct	gcggaggctg
2101	ctgcaggaga	gggagcttgc	ggagcctctt	acacccagtg	gagaagctcc	caaccaagct
2161	ctcttgaggga	tcttgaaggga	aactgaattc	aaaagatcca	aagtgtctgg	ctccggtgcg
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2341	gcctacgtga	tggccagcgt	ggacaacccc	cacgtgtgcc	gcctgctggg	catctgcctc
2401	acctccaccg	tgcagctcat	cacgcagctc	atgcccttcg	gctgcctcct	ggactatgtc
2461	cgggaacaca	aagacaatat	tggctcccag	tacctgctca	actgggtgtg	gcagatcgca
2521	aagggtcatga	accacttgga	ggaccgtcgc	ttggtgcacc	gcgacctggc	agccaggaa
2581	gtactggtga	aaacaccgca	gcagtgtcaag	atcacagatt	ttgggtgggc	caaactgctg
2641	ggtgcggaag	agaaagaata	ccatgcagaa	ggaggcgaag	tgccatcaa	tgggatggca
2701	ttggaatcaa	ttttacacag	aatctatacc	caccagagtg	atgtctggag	ctacggggtg
2761	actgtttggg	agtttgatgac	ctttggatcc	aagccatag	acggaatccc	tgccagcgag
2821	atctcctcca	tctctggaga	aggagaaacgc	ctccctcagc	cacccatag	tacctcgat
2881	gtctacatga	tcattggtcaa	gtgctggatg	atagacgcag	atagtcgccc	aaagtccgct
2941	gagttgatca	tgaattcttc	caaaatggcc	cgagaccccc	agcgctacct	tgctattcag
3001	ggggatgaaa	gaatgcattt	gccaaagtct	acagactcca	acttctaccg	tgccctgatg
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3121	ttcttcagca	gcccctccac	gtcacggact	cccctcctga	gctctctgag	tgcaaccagc
3181	aacaaattcca	ccgtggcttg	cattgataga	aatgggctgc	aaagctgtcc	catcaaggaa
3241	gacagctctt	tcagcagata	cagctcagac	cccacaggcg	ccttgactga	ggacagcata
3301	gacgacacct	tctctccagt	gcctgaatac	ataaaccagt	ccgttcccaa	aaggcccgc
3361	ggctctgtgc	agaaatctgt	ctatcacaat	cagcctctga	accccgccgc	cagcagagac
3421	ccacactacc	aggaccccc	cagcactgca	gtgggcaacc	ccgagtatct	caacactgtc
3481	cagccacacct	gtgtcaacag	cacattcgac	agccctgccc	actgggcccc	gaaaggcagc
3541	caccaaatta	gcctggacaa	ccctgactac	cagcaggagt	ttcttcccaa	ggaagccaa
3601	ccaaatggca	tctttaaggg	ctccacagct	gaaaatgcag	aatacctaag	ggtcgcgcca
3661	caaagcagtg	aatttatttg	agcatgacca	cggaggatag	tatgagccct	aaaaatccag
3721	actcttctga	taccaggac	caagccacag	caggtcctcc	atcccaacag	ccatgcgcgc
3781	attagctctt	agaccacag	actggttttg	caacgtttac	accgactagc	caggaaagtac

TABLE 1-continued

3841	ttccacctcg	ggcacatctt	gggaagttgc	attcctttgt	cttcaaaactg	tgaagcattt
3901	acagaaacgc	atccagcaag	aatattgtcc	ctttgagcag	aaatttatct	ttcaaagagg
3961	tatatattgaa	aaaaaaaaaa	agtatatgtg	aggattttta	ttgattgggg	atcttggagt
4021	ttttcatcgt	cgctattgat	ttttacttca	atgggctctt	ccaacaagga	agaagcttgc
4081	tggtagcact	tgctaccctg	agttcatcca	ggcccaactg	tgagcaagga	gcacaagcca
4141	caagtcttcc	agaggatgct	tgattccagt	ggttctgctt	caaggcttcc	actgcaaaac
4201	actaaagatc	caagaaggcc	ttcatggccc	cagcaggccg	gacgggtact	gtatcaagtc
4261	atggcaggta	cagtaggata	agccactctg	tcccttctcg	ggcaaagaag	aaacggaggg
4321	gatggaattc	ttccttagac	ttacttttgt	aaaaatgtcc	ccacggtact	tactccccac
4381	tgatggacca	gtgggtttcc	gtcatgagcg	ttagactgac	ttgtttgtct	tccattccat
4441	tgttttgaaa	ctcagtagtc	tgccctgtgc	ttgctgtcat	gaaatcagca	agagaggatg
4501	acacatcaaa	taataactcg	gattccagcc	cacattggat	tcatacagat	ttggaccaat
4561	agcccacagc	tgagaattgt	gaatacctaa	ggatagcacc	gcttttggtc	tcgcaaaaac
4621	gtatctccta	atttgaggct	cagatgaaat	gcatacagtc	ctttggggca	tagatcagaa
4681	gactacaaaa	atgaagctgc	tctgaaatct	cctttagcca	tcaccccaac	cccccaaaat
4741	tagttttgtg	tacttttggg	agatagtttt	ctccttttac	ttcacttcaa	aaagcttttta
4801	ctcaaaagag	atatgttccc	tccagggtcag	ctgcccccaa	acccccctct	tacgctttgt
4861	cacacaaaaa	gtgtctctgc	cttgagtcac	ctattcaagc	acttacagct	ctggccacaa
4921	cagggcattt	tacaggtgcg	aatgacagta	gcattatgag	tagtgtggaa	ttcaggtagt
4981	aaatatgaaa	ctagggtttg	aaattgataa	tgctttcaca	acatttgcag	atgttttaga
5041	agggaaaaag	ttccttcccta	aaacaatttc	tctacaattg	gaagattgga	agattcacgt
5101	agttaggact	tccttttttt	tcctaactcg	tggtgtgccct	gtaacctgac	tggttaacag
5161	cagtcctttg	taaacagtgt	tttaaaactct	cctagtcaat	atccacccca	tccaatttat
5221	caagggaaga	atgggtccaga	aaatatcttc	agcctacagt	tatgttcagt	cacacacaca
5281	tacaaaaatg	tcctttttgt	tttaaagtaa	tttttgactc	ccagatcagt	cagagccctc
5341	acagcattgt	taagaaagta	tttgattttt	gtctcaatga	aaataaaact	atatcatttt
5401	ccactctatt	atgctctcaa	ataccctcaa	gcactctaac	tagccttgta	tgggtatgaa
5461	agatacaaa	ataataaaaa	catagtcctc	gattctaaga	aattcacaat	ttagcaaaag
5521	aaatggactc	atagatgcta	accttaaaac	aacgtgacaa	atgccagaca	ggaccatca
5581	gccaggcact	gtgagagcac	agagcaggga	ggttgggtcc	tgccctgagga	gacctggaag
5641	ggaggcctca	cagcaggatg	accagggtct	agtcagcggg	gaggtggaaa	gtgcagggtc
5701	atcaggggca	ccctgaccga	ggaaacagct	gccagaggcc	tccactgcta	aagtcacat
5761	aaggctgagg	tcagtcaccc	taaacacact	gctccctcta	agccagggga	tgagcttgga
5821	gcattcccaca	agttccctaa	aagttgcagc	cccaggggg	attttgagct	atcatctctg
5881	cacatgctta	gtgagaagac	tacacaacat	ttctaagaat	ctgagatttt	atatgttcag
5941	ttaaccactt	tcattattca	ttcacctcag	gacatgcaga	aatatttcag	tcagaaactg
6001	gaaacagaag	gacctacatt	ctgctgtcac	ttatgcgtca	agaagcagat	gacgatgag
6061	gcaggtcagt	tgtaagttag	tcacattgta	gcattaaatt	ctagtatttt	tgtagtttga
6121	aacagtaact	taataaaaga	gcaaaagcta	aaaaaaaaaa	aaaaaa	

SEQ ID NO: 106 Human EGFR Amino Acid Sequence Isoform H (NP_001333829.1)

1 mfnnccewlg nleityvqrn ydlsflkti q evagyvlial ntveriplen lqiirgnmyy
61 ensyalavls nydankttgk elpmrnlgei lhgavrfsnn palcnvesiq wrdivssdfl
121 snmmdfqnq igscqkcdps cpngscwag eencqkltki icaqqcsgrc rgkspsdch
181 nqcaagctgp resdcivcrk frdeatckdt cpllmlynpt tyqmdvnpeg kysfgatcvk
241 keprnyvvt d hsgcvracga dseyeedgv rkckkceggp rkvenigig efkdslsina
301 tnkhfknc t siggdhilt vafrgdsfrh tppldpqeld ilktvkeitg fliliqawpen
361 rtdlhafenl eiirgrtkqh gqfslavvsl nitslglrls keisdgdvii sgnknlcyan
421 tinwkklift sgqtkliisn rgensckatg qvchalcspe gcwgppeprdc vscrnvsrgr
481 ecvdkcnlle geprefvens eciqchpecl pqamnitctg rgpdnciqca hyidgphcvk
541 tcpagvmgen ntlvwkyada ghvchlchpn ctygctgpgl egcptngpki psiatgmvg
601 lllllvvalg igtfmrnrhi vrkrtlrll qerelvept psgeapnqal lrilketefk
661 kikvlgsgef gtyvkgylwip egekvikpva ikelreatsp kankeildea yvmasvndph
721 vcrlligiclt stvqlitqlm pfgclldyvr ehkdnigsqy llnwcvqiak gmnyledrll
781 vhrdlaarnv lvtktphvki tdfglakllg aeekeyhaeg gkvpikwmal esilhriyth
841 qsdvwsygv t vwelmftgsk pydgipasei ssilekgerl ppppictidv ymimvkcwmi
901 dadsrpkfre liiefskmar dpqrylviqg dermhlpstps dsnfyralm d eedmdvvd
961 deyliqqgf fsspstsrtp llsslsatsn nstvacidrn glqscpiked sflqrysdp
1021 tgaltedsid dtflpvpeyi nqsvpkp pag svqnpvyhnq plnpapsrdp hyqdpbstav
1081 gnpeylncvq ptcvnsctfs pahwaqkqsh qisldndpyq qdffpkeakp ngifkgstae
1141 naeylrvaq ssefiga

SEQ ID NO: 107 Human EGFR cDNA Sequence Variant 9 (NM_001346941.1, CDS: from 258 to 3089)

1 gtccgggagc ccccgggcgc agcggggcgc cagcagcctc cgccccccgc acgggtgtgag
61 cgcccgagcgc ggcggaggcg gccggagctc cgagctagcc cggcgggcgc ccgcccgcga
121 gaccggagca caggccaccc cgtcggcgct cgcccgagtc ccgcctcgc cgcccaagcc
181 acaaccaccc cgacaggccc cctgactccg tccagtattg atcgggagag ccggagcgag
241 ctcttcgggg agcagcgatg cgaccctccg ggacggcccg ggcagcgctc ctggcgctgc
301 tggctcgctg ctgcggcgcg agtcgggctc tggaggaaaa gaaaggtaat tatgtgggta
361 cagatcacgg ctgctgctgc cgagcctgtg gggccgacag ctatgagatg gaggaagacg
421 gcgtccgcaa gtgtaagaag tgcgaaggcg cttgccgcaa agtgtgtaac ggaataggta
481 ttgttgaaat tgaactcaaa atgctacgaa tactaaacac tcaaaaaact
541 gcacctccat cagtggcgat ctccacatcc tgcgggtggc atttaggggt gactcctcca
601 cacatactcc tctctgtgat ccacaggaac tggatattct gaaaaccgta aaggaatca
661 cagggttttt gcttgattcag aaaacaggac ggacctccat gcctttgaga
721 acctagaaat catacggcgc aggaccaagc aacatggtca gttttctctt gcagtcgtca
781 gcctgaacat aacatccttg ggattacgct cctcgaagga gataagtgat ggagatgtga
841 taatttcagg aaacaaaaat ttgtgctatg caatacaaat aaactggaaa aaactgtttg
901 ggacctccgg tcagaaaacc aaaattataa gcaacagagg tgaacacagc tgcaaggcca

TABLE 1-continued

961	caggccaggt	ctgccatgcc	ttgtgctccc	cagagggctg	ctggggcccg	gagcccaggg
1021	actgcgtctc	ttgccggaat	gtcagccgag	gcaggggaatg	cgtgggacaag	tgcaaccttc
1081	tggagggtga	gccaaaggag	tttgtggaga	actctgagtg	catacagtg	caccagaggt
1141	gcctgctca	ggccatgaac	atcacctgca	caggacgggg	accagacaac	tgatccaggt
1201	gtgcccacta	cattgacggc	ccccactg	tcaagacctg	cccgccagga	gtcatgggag
1261	aaaacaacac	cctggctccg	aagtacgcag	acgcccggcca	tgtgtgccac	ctgtgccacc
1321	caaaactgcac	ctacggatgc	actgggcccag	gtcttgaagg	ctgtccaacg	aatgggcta
1381	agatcccgctc	catcgccact	gggatgggtg	gggcccctct	cttgcctgct	gtgggtggcc
1441	tggggatcgg	cctcttcatg	cgaaggcgcc	acatcgctcg	gaagcgcacg	ctgcggaggc
1501	tgctgcagga	gagggagctt	gtggagcctc	ttacaccacg	tgagaagct	cccaaccaag
1561	ctctcttgag	gatcttgaag	gaaactgaat	tcaaaaagat	caaagtgtgt	ggctccgggtg
1621	cgttcggcac	gggtgataag	ggactctgga	tcccagaagg	tgagaaagt	aaaattcccg
1681	tcgctatcaa	ggaattaaga	gaagcaacat	ctccgaaagc	caacaaggaa	atcctcgatg
1741	aagcctacgt	gatggccagc	gtggacaacc	cccacgtgtg	ccgcctgctg	ggcatctgcc
1801	tcacctccac	cgtgcagctc	atcacgcagc	tcacgcccct	cggctgcctc	ctggactatg
1861	tccgggaaca	caaagacaat	atgggctccc	agtagctgct	caactgggtg	gtgcagatcg
1921	caaaaggcat	gaactacttg	gaggaccgtc	gcttgggtgca	ccgcgacctg	gcagccagga
1981	acgtactggt	gaaaaacacg	cagcatgtca	agatcacaga	ttttgggtg	gccaaactgc
2041	tgggtgcgga	agagaagaa	taccatgcag	aaggaggcaa	agtgcccatc	aagtggatgg
2101	cattggaatc	aattttacac	agaatctata	cccaccagag	tgatgtctgg	agctacgggg
2161	tgactgtttg	ggagtgtatg	acctttggat	ccaagccata	tgacggaatc	cctgccagcg
2221	agatctcttc	cttgcaggag	aaaggagaac	gcctccctca	gccaccata	tgtaccatcg
2281	atgtctacat	gatcatggtc	aagtgtgga	tgatagacgc	agatagtgc	ccaaagtctc
2341	gtgagttgat	catcgaaatc	tccaaaatgg	cccagagacc	ccagcgtac	cttgtcattc
2401	agggggatga	aagaaatgc	ttgccaaagt	ctacagactc	caactctac	cgtgccctga
2461	tggatgaaga	agacatggac	gacgtgggtg	atgccgacga	gtacctcatc	ccacagcagg
2521	gcttcttcag	cagcccctcc	acgtcacgga	ctcccctcct	gagctctctg	agtgcaacca
2581	gcaacaattc	caccgtggct	tgcatgtata	gaaatgggt	gcaaagctgt	cccatcaagg
2641	aagacagctt	cttgcagcga	tacagctcag	accccacagg	cgccttgact	gaggacagca
2701	tagacgacac	cttccctcca	gtgcctgaat	acataaacca	gtccgttccc	aaaaggcccg
2761	ctggctctgt	gcagaatcct	gtctatcaca	atcagcctct	gaaccccgcg	cccagcagag
2821	accacacata	ccaggacccc	cacagcactg	cagtgggcaa	ccccagatg	ctcaacactg
2881	tccagcccac	ctgtgtcaac	agcacattcg	acagccctgc	ccactggggc	cagaaaggca
2941	gccaccaa	tagcctggac	aacctgact	accagcagga	cttcttccc	aagggaagcca
3001	agccaaatgg	catctttaag	ggctccacag	ctgaaaatgc	agaataccta	agggctcgcg
3061	cacaaagcag	tgaatttatt	ggagcatgac	cacggaggat	agtatgagcc	ctaaaaatcc
3121	agactctttc	gatacccagg	accaagccac	agcaggtcct	ccatcccaac	agccatggcc
3181	gcatttagctc	ttagacccac	agactgggtt	tgcaacgttt	acaccgacta	gccagggaagt
3241	acttccacct	cgggcacatt	ttgggaagtt	gcattctctt	gtcttcaaac	tgtgaagcat
3301	ttacagaaac	gcattccagca	agaatattgt	ccctttgagc	agaaatttat	ctttcaaaaga
3361	ggatattttg	aaaaaaaaaa	aaagtatatg	tgaggatttt	tattgattgg	ggatcttgga
3421	gtttttcatt	gtcgctattg	atttttactt	caatgggctc	ttccaacaag	gaagaagctt
3481	gctggtagca	cttgcatacc	tgagttcatc	caggcccac	tgtgagcaag	gagcacaaagc
3541	cacaagtcag	ccagaggatg	cttgattcca	gtggttctgc	ttcaaggctt	ccactgcaaa
3601	acactaaaga	tccaagaagg	ccttcatggc	cccagcaggc	cggatcggtg	ctgtatcaag
3661	tcatggcagg	tacagttaga	taagccactc	tgtcccttcc	tgggcaaaaga	agaaacggag
3721	gggatggaat	tcttctcttag	acttactttt	gtaaaaatgt	ccccacggta	cttactcccc
3781	actgatggag	cagtgggttc	cagtcattgag	cggttagactg	acttgtttgt	cttccattcc
3841	attgttttga	aactcagtat	gctgcccctg	tcttgcctgc	atgaaatcag	caagagaggga
3901	tgacacatca	aataataaact	cggattccag	cccacattgg	attcatcagc	atttggacca
3961	atagcccaca	gctgagaatg	tggaatacct	aaggatagca	ccgcttttgt	tctcgcaaaa
4021	acgtatctcc	taatttgagg	ctcagatgaa	atgcacagg	tctttcgggg	catagatcag
4081	aagactacaa	aaatgaagct	gctctgaaat	ctccttttagc	catcacccca	accccccaaa
4141	attagtttgt	gttacttatg	gaagatagtt	ttctcctttt	acttcaactc	aaaagctttt
4201	tactcaaaaga	gtatatgttc	cctccaggtc	agctgcccc	aaacccctc	cttacgctt
4261	gtcacacaaa	aagtgtctct	gccttgagtc	atctattcaa	gcacttacag	ctctggccac
4321	aacagggcac	tttacagggtg	cgaatgacag	tagcattatg	agtagtgtgg	aattcaggta
4381	gtaaatatga	aactagggtt	tgaaattgat	aatgctttca	caacatttgc	agatgtttta
4441	gaaggaaaaa	agctccttcc	taaaaataat	tctctacaat	tgggaagattg	gaagattcag
4501	ctagtttagg	gcccaccttt	tttctaate	tgtgtgtgccc	ctgtaacctg	actggttaac
4561	agcagtcctt	tgtaaacagt	gttttaaaact	ctcctagtca	atatccaccc	catccaattt
4621	atcaagggaag	aaatggttca	gaaaaatatt	tcagccctaca	gttatgttca	gtcacacaca
4681	catacaaaat	gttcttttgg	cttttaaaagt	aatttttgac	tcccagatca	gtcagagccc
4741	ctacagcaact	gttaagaaag	tatttgattt	ttgtctcaat	gaaaataaaa	ctatattcat
4801	ttccactcta	ttatgctctc	aaatacccct	aagcatctat	actagcctgg	tatgggtatg
4861	aaagatacaa	agataataaa	aacatagtc	ctgattctaa	gaaattcaca	atttagcaaa
4921	ggaaatggac	tcatatagtc	taaccttaaa	acaacgtgac	aaatgccaga	caggacccat
4981	cagccaggca	ctagtggagc	acagagcagg	gaggttgggt	cctgcctgag	gagacctgga
5041	agggaggcct	cacaggagga	tgaccaggtc	tcagtcagcg	gggaggtgga	aagtgcagggt
5101	gcacacgggg	caccctgacc	gaggaaacag	ctgccagagg	cctccactgc	taagttccac
5161	ataaggctga	ggtcagtcac	cctaacaac	ctgctccctc	taagccagggt	gatgagcttg
5221	gagcatccca	caagttccct	aaaagttgca	gccccagggt	ggatttcag	ctatcatctc
5281	tgcacatgct	tagtgagaag	actacacaac	atctcagaat	ttatatgttc	

TABLE 1-continued

5341	agttaaccac	tttcattatt	cattcacctc	aggacatgca	gaaatatattc	agtcagaact
5401	gggaacacaga	aggacctaca	ttctgctgtc	acttatgtgt	caagaagcag	atgatcgatg
5461	aggcaggtca	gttgtaagt	agtcacattg	tagcattaaa	ttctagtatt	ttttagtatt
5521	gaaacagtaa	cttaataaaa	gagcaaaagc	ta		

SEQ ID NO: 108 Human EGFR Amino Acid Sequence Isoform I (NP_001333870.1)

1	mrpsgtagaa	llalllaalcp	asraleekkg	nyvvtldhgsc	vracgadsye	meedgvrkck
61	kcepgckrvc	ngigigefkd	slsinatnik	hfkncstisg	dlhilpvafr	gdsfthtppl
121	dpqeldilk	vkeitgflli	qawpenrtdl	hafenleir	grtkqhgqfs	lavvslnits
181	lglsrlkeis	dgdvliisgnk	nlcyantinw	klifgtsgqk	tkiisnrgen	skatgqvch
241	alcspegcwg	peprdcvscr	nvsrgrecvd	kcnllegepr	efvenseciq	chpeclpqam
301	nitctgrgpd	nciqcahyid	gphcvktcpa	gvmgenntlv	wkyadaghvc	hlchpncityg
361	ctgpglegcp	tngpkipsia	tgmvgallll	lvvalgiglf	mrrrhivrkr	tlrrllgere
421	lvepltpse	apnqallril	kerefkikiv	lsgafgtvy	kglwipegek	vkipvaikel
481	reatspkank	eildeayvma	svdnphvcr	lgicltstvt	litqlmpfgc	lldyvhrehkd
541	nigsqyllnw	cvgiakgmy	ledrrlvhrd	laarnlvkt	pqhvktdfg	lakllgaeek
601	eyhaegkvkp	ikwmalesil	hriythqsdv	wsygvvswel	mtfgskpydg	ipaseissil
661	ekgerlpqpp	ictidvymim	vkcmidads	rpkfrelie	fskmardpqr	ylvigqderm
721	hlpsptdsnf	yralmdeedm	ddvvdadeyl	ipqggfsssp	stsrtpllss	lsatnnstsv
781	acidrnglqs	cpikedsfllq	ryssdptgal	tedsidtdfl	pvpeyinqsv	pkrpagsvqn
841	pyvhnqplnp	apsrdphyqd	phstavgnp	ylntvqptcv	nstfdespahw	aqkqshqisl
901	dnpdyqqdff	pkeakpngif	kgstaenaey	lrvaqgssef	iga	

SEQ ID NO: 109 Mouse EGFR cDNA Sequence Variant 1 (NM_207655.2, CDS: from 281 to 3913)

1	ctccccagc	cccgaccga	gctaactaga	cgtctgggca	gccccagcgc	aacgcgcagc
61	agcctccctc	ctcttcttcc	cgcactgtgc	gctcctcctg	ggctagggcg	tctggatcga
121	gtccccgagg	ctaccgcctc	ccagacagac	gacaggtcac	ctggacgcga	gcctgtgtcc
181	gggtctcgtc	gttgccggcg	cagtcactgg	gcacaaccgt	gggactccgt	ctgtctcgga
241	ttaatcccg	agagccagag	ccaacctctc	ccggtcagag	atgcgacct	cagggaccgc
301	gagaaccaca	ctgctgggtg	tgctgaccgc	gctctgcgcc	gcaggtgggg	cggtggagga
361	aaagaaagtc	tgccaaggca	caagtaacag	gctcacccaa	ctgggcactt	ttgaagacca
421	cttctctgagc	ctgcagagga	tgtacaacaa	ctgtgaagtg	gtccttggga	acttggaaat
481	tacctaagt	caaaaggaat	acgaccttcc	cttcttaaa	accatccagg	aggtggccgg
541	ctatgtcctc	attgcctcca	acaccgtgga	gagaatccct	ttggagaacc	tgcagatcat
601	caggggaaat	gctctttatg	aaaacaccta	tgccttagcc	atcccgtcca	actatgggac
661	aaacagaact	gggcttaggg	aactgcccac	gcggaactta	caggaaatcc	tgattggtgc
721	tgtgcgattc	agcaacaacc	ccatcctctg	caatatggat	actatccagt	ggagggacat
781	cgtccaaaac	gtctttatga	gcaacatgct	aattggactta	cagagccatc	cgagcagttg
841	cccaaatgtg	gatccaaagt	gtcccaatgg	aagctgctgg	ggaggaggag	aggagaaactg
901	ccagaaattg	accaaaatca	tctgtgccca	gcaatgttcc	catcgctgtc	gtggcaggtc
961	cccagtgac	tgtgtccaca	accaatgtgc	tgcgggggtg	acagggcccc	gagagagtga
1021	ctgtctggct	gtccaaaagt	tccaagatga	ggccacatgc	aaagacacct	gccaccact
1081	catgctgtac	aaacccacca	cctaccagat	ggatgtcaac	cctgaaggga	agtacagctt
1141	tgtgtgccac	tgtgtgaaga	agtgcccccg	aaactacgtg	gcgacagatc	atgggtcatg
1201	tgtccagctc	tgtgggctcg	actactacga	agtggaaaga	gatggcatcc	gcaagtgtaa
1261	aaaatgtgat	gggcccctgtc	gcaaaagttg	taatggcata	ggcattggtg	aaactaaaga
1321	cacactctcc	ataaatgtcta	caaacatcaa	acacttcaaa	tactgcactg	ccatcagcgg
1381	ggaccttcac	atcccccgag	tggcctttaa	gggggattct	ttcacgcga	ctcctcctct
1441	agacccacga	gaactagaaa	ttctaaaaac	cgtaaaggaa	ataacaggct	ttttgtgat
1501	tcaggcttgg	cctgataact	ggactgacct	ccatgcttcc	gagaacctag	aaataaatag
1561	tggcagacaa	aagcaacatg	gtcagtttcc	tttggcggtc	gttggcttga	acatcacatc
1621	actggggctg	cgttcctcca	aggagatcag	tgatggggat	gtgatcattt	ctggaaaccg
1681	aaatttggc	tacgcaaaac	caataaaactg	gaaaaaaact	ttcgggacac	ccaatcagaa
1741	aaacaaaact	atgaaacaac	gagctgagaa	agactgcaag	gcggtgaacc	acgtgtgcaa
1801	tcctttatgc	tcctcggaag	gctgtggggg	ccctgagccc	agggactgtg	tctcctgcca
1861	gaatgtgagc	agaggcaggg	agtgctgga	gaaatgcaac	atcctggagg	gggaaccaag
1921	ggagtgtgtg	gaaaattctg	aatgcatcca	gtgccatcca	gaatgtctgc	ccagggccat
1981	gaacatcacc	tgtacaggca	ggggaccaga	caactgcatc	cagtgtgccc	actacattga
2041	tggcccacac	tgtgtcaaga	cctgcccagc	tggcatcatg	ggagagaaca	acactctggt
2101	ctggaagtac	gcagatgccca	ataatgtctg	ccacctatgc	cacgccaact	gtacctatgg
2161	atgtgctggg	ccaggctctc	aaggatgtga	agtggtggcca	tctgggcca	agataccatc
2221	tattgccact	gggatgtgtg	gtggcctcct	cttcatagtg	gtggtggccc	ttgggattgg
2281	cctattcatg	cgaagacgtc	acattgttgc	aaagcgtaca	ctacgcgcc	tgcttcaaga
2341	gagagagctc	gtggaaacct	tcacacccag	cggagaagct	ccaaaccaag	cccacttgag
2401	gatattaaag	gaaacagaat	tcaaaaagat	caaagttctg	gggtcgggag	cattttggcac
2461	agtgataaag	ggtctctgga	tccagaagg	tgagaaagta	aaaatcccg	tggccatcaa
2521	ggagttaaga	gaagccacat	ctccaaaagc	caacaaagaa	atccttgacg	aagcctatgt
2581	gatggctagt	gtggacaacc	ctcatgtatg	ccgcctcctg	ggcatctgtc	tgacctccac
2641	tgtccagctc	attacacagc	tcattgccta	cggttgcttc	ctggactacg	tccagaaaca
2701	caaggacaac	attggctccc	agtaacctct	caactggtgt	gtgcagattg	caaagggcac
2761	gaactacctg	gaagatcggc	gtttggtgca	ccgtgacttg	gcagccagga	atgtactggt
2821	gaagacacca	cagcatgca	agatcacaga	ttttgggtcg	gccaaactgc	ttgggtgctga
2881	agagaaagaa	tatcatgccg	aggggggcaa	agtgccatc	aagtggatgg	ctttggaatc
2941	aattttacac	cgaatttaca	cacaccaaag	tgatgtctgg	agctatggtg	tcactgtgtg
3001	ggaactgatg	acctttgggt	ccaagcctta	tgatggatc	ccagcaagt	acatctcatc
3061	catcctagag	aaaggagagc	gccttcacac	gccacctatc	tgaccatc	atgtctacat
3121	gatcatggct	aagtgtctga	tgatagatgc	tgatagccgc	ccaaagtctc	gagagtgtat
3181	tcttgaattc	tccaaaatgg	cccagagacc	acagcgctac	cttgttatcc	agggggatga
3241	aagaatgcac	ttgccaagcc	ctacagactc	caacttttac	cgagccctga	tggatgaaga

TABLE 1-continued

3301	ggacatggag	gatgtagttg	atgctgatga	gtatcttatt	ccacagcaag	gcttcttcaa
3361	cagcccgctc	acgtcgagga	ctcccctctt	gagttctctg	agtgaacta	gcaacaattc
3421	cactgtggct	tgattaata	gaaatgggag	ctgccgtgtc	aaagaagacg	ccttcttgca
3481	gcggtagacg	tccgacccca	caggtgctgt	aacagaggac	aacatagatg	acgcattcct
3541	ccccgtacct	gaatatgtaa	accaatctgt	tcccagagg	ccagcaggct	ctgtgcagaa
3601	ccctgtctat	cacaatcagc	ccctgcattc	agctcctgga	agagacctgc	attatcaaaa
3661	tccccacagc	aatgcagttg	gcaaccctga	gtatctcaac	actgccacgc	ctacctgtct
3721	cagcagtgag	tttaacagcc	ctgcactctg	gatccagaaa	ggcagtcacc	aaatgagcct
3781	agacaaccct	gactaccagc	aggacttctt	ccccaaaggaa	accaagccaa	atggcatatt
3841	taagggcccc	acagctgaaa	atgcagagta	cctacgggtg	gcacctccaa	gcagtgagtt
3901	tattggagca	tgacaagaag	gggcatcata	ccagctataa	aatgtctgga	ctttctagaa
3961	tcccaggacc	aactatggca	gcacctccac	ttctggtagc	catgccccag	ctgtgtcaaa
4021	tgtcactcag	actggcttta	aagcataact	ctgatgggct	ttgtcactga	gccaagaagt
4081	gggcctctct	cctgatgcac	tttggaagt	tgaaggtaca	tcaattgatc	ttcgaactgt
4141	gaagattcca	caaaaaaggt	atccatcgag	aacattgtcc	attggaacag	aagtttgctt
4201	catggtagag	tacatcatgg	aaaaaaacag	acatatggag	cttatattta	gggaactttg
4261	ggattcttgt	ctttattgat	ttgattgatg	cactcttgta	gtctggtaga	cagagtggcc
4321	tgagagcaac	tgaccagaca	gttggttcca	ccagctctgc	atcaagacac	ttccgtggca
4381	agacaactaa	acagtataaga	agtcctagga	tgccctgagc	aggccacact	tgtacagcat
4441	taaacatagg	cagatacaat	aggataagcc	actttgttac	ttactggggc	tgggagaaga
4501	ggaatgacgg	ggtagaattt	tccctcagac	gtacttttta	tataaatatg	tccctggcac
4561	ctaacacgcg	ctagtcttacc	agtgttttct	attagacttc	cttctatgtt	ttctgtttca
4621	ttgttttgag	ttgtaaatat	gtgttctctg	cttcatttca	tgaagttaac	aaacaaacaa
4681	aaaaccagct	atttaagtatt	atcaagaagc	aaccatgatt	ccacatttga	acccattcaa
4741	accatcagta	tttgtaccac	aagcctttaa	ctaagaaggga	gtaaccatgc	aaaaatccat
4801	agaggaattt	aacccaaaat	tttagtctca	gcattgtgtc	tgctgagggt	tgtatatgag
4861	actacgaaag	tgaactactc	ttcaaatcca	ctttgccttc	actcctctat	accctaaatc
4921	tagtgtaaac	cagacatgga	ggataacttt	ttttttta	tttaaaagt	tttattagat
4981	atgtttttct	tccctggttaa	ctgcagccaa	acatcagtta	agagccattt	ttgataaaca
5011	ctatcacaa	gatctcggga	tccatccttt	ccgatttacc	aagtgtatga	tagacgtgaa
5101	ctcataaaca	ctaccataaa	gacaaaacaa	tgagtgcagc	acaagacatc	agccaggcac
5161	cagagcacag	agcaggactg	ggcaatctgt	tggagatatt	tagaaaagtt	acaaaggaaa
5221	caagattgtc	cactaccttg	tgagatctag	cagtcataaa	taccagggaa	atggaaaagt
5281	tgtttcttta	cagcaccagg	tcttcgatct	tcctaagct	gtgacctttt	aatacagttt
5341	gccatgttgt	ggtgaccccc	aaccataaaa	ttatttttgt	tgctacttca	taactgtaaa
5401	tttgctactc	ttacagacca	caatgtaaat	atctgatatt	ctatctgata	tgcaggctat
5461	ctgacagagg	tcgcaaaccc	caggttgaga	gccactgcct	tcaaggcttt	aatcaagaga
5521	gtagttagct	gagggcttta	ctggttaagc	aggggcaagt	ccaactcaat	catcctcaca
5581	tactggctgc	tccctcagcc	ctgagaatga	ggcttgagc	atcctctggt	ttcctaaccg
5641	ttatccatcc	ctgactctca	tctctgaaaa	tagatgtcat	ccatgaaat	aaggagttag
5701	aatattaagc	agcatttata	gagctcaaaa	ttccatgtca	tcaccaggaa	gtgccatgtt
5761	gatcacagag	aacacagagg	agacatatag	acaggggttt	gctcaaaat	gggatataga
5821	atgagcctgt	tcaggagcgg	taatccgtga	gagagaaccg	ttgcaagcca	
5881	ctctaactgt	agcaatgaaa	ccctagtagt	ttgtactttt	gaaatacttt	cttataacaa
5941	aataaagtag	caaaaaaact	gttcaaaaaa	aaaaaaaaaa	aaa	

SEQ ID NO: 110 Mouse EGFR Amino Acid Sequence Isoform A (NP_997538.1)

1	mrpsgtartt	llvlltalca	aggaleekkv	cqgtsnrltq	lgtfedhfls	lqrmynncev
61	vlgnleityv	qnydisflk	tiqevagyvl	ialntverip	lenlqiirgn	alyentyala
121	ilsnygtnrt	glrelpmrnl	qeiligavrf	snnpilmnd	tiqwrdivqn	vfmnsmsmdl
181	qshpscpkc	dpsscngscw	gggeencqkl	tikiacqcs	hrcgrspsd	cchnqcaagc
241	tgpsndclv	cqkfqdeatc	kdtcpplmly	npttyqmdvn	pegkysfgat	cvkckprnyv
301	vtahgscvra	cgpdyveeve	dgirkckkcd	gpcrkvcngi	gigefkdtls	inatnikhfk
361	yctaisgdll	ilpvafkgds	frtptpldpr	eleilkvtke	itgflliqaw	pdnwtdlhaf
421	enleirgrt	kqhggqslav	vglntslgl	rsлкеisdgd	viisgnrnlc	yantlnwkl
481	fgtpnqtkki	mnrraekdck	avnhvcnplc	ssegcwpgpep	rdcvscqnvs	rgrecvekcn
541	ilegeprefv	enseciqchp	ecelpqamnit	ctgrgpdnci	qcahyidgph	cvktcpagim
601	genntlwwky	adannvchlc	hanctygcag	pqlggcevw	sgpkipsiat	givggllfiv
661	vvalgiglfm	rrrhivrkr	lrlllgerel	vepltpsgca	pnqahlrlk	etefkklkvl
721	gsgafgtvyk	glwpegekv	kipvaikelr	eatspkanke	ildeayvmas	vdnphvcrl
781	gicltstvg	itqlmpygcl	ldyvrehkdn	igsqyllnwc	vqiakgmnyl	edrrlvhrdl
841	aarnvlvkt	qhvkitdfgl	akllgaecke	yaeggkvpi	kwmalelilh	riythqsdvw
901	sygvtvwelm	tfsgskpydgi	pasdissile	kgerlpqppi	ctidvymimv	kcmwidadsr
961	pkfrelirgt	skmaridpqr	lvigqdermh	lpsptdsnfy	ralmdeedme	divvdadeyli
1021	pqqgffnsp	tsrtpllsel	satsnnstva	cinrngscr	kedafllqrys	sdptgavted
1081	niddaflpvp	eyvnqsvpkr	pagsvqnpvy	hnqplhpap	rdlhyqnphs	navgnpeyln
1141	tagptclssg	fnpalwlgk	gshqmsldnp	dyqqdfpke	tkpngifkpg	taenaeylrv
1201	appssefiga					

SEQ ID NO: 111 Mouse EGFR cDNA Sequence Variant 2 (NM_007912.4, CDS:
from 281 to 22478)

1	ctccccagct	cccgaccoga	gctaactaga	cgtctgggca	gccccagcgc	aacgcgcagc
61	agcccccttc	ctctttcttc	cgcactgtgc	gctccccctg	ggctaggggc	ctctgtcgca
121	gtcccgagg	ctaccgcctc	ccagacagac	gacagggtcac	ctggacgcga	gcctgtgtcc
181	gggtctctgc	gttgcggggc	cagccactgg	gcacaaccgt	gggactccgt	ctgtctcgga
241	ttaatcccg	agagccagag	ccaacctctc	ccggtcagag	atcgaccctc	cagggaaccg
301	gagaaccaca	ccgctgggtg	tgctgaccgc	gctctgcgcc	gcaggtgggg	cggtggagga
361	aaagaaagtc	tgccaaggca	caagtaacag	gctcacccaa	ctgggcactt	ttgaagacca
421	cttctcgagc	ctgcagagga	tgtacaacaa	ctgtgaagtg	gtccttggga	acttggaaat
481	tacctatgtg	caaaggaatt	acgacctttc	cttcttaaag	accatccagg	aggtggccgg

TABLE 1-continued

541	ctatgtcctc	attgccctca	acaccgtgga	gagaatccct	ttggagaacc	tgcagatcat
601	caggggaaat	gtcttttatg	aaaacaccta	tgccttagcc	atcctgtcca	actatgggac
661	aaacagaaat	gggtcttagg	aactgcccat	gcggaactta	caggaaatcc	tgtatgggtgc
721	tgtgcatgac	agcaacaacc	ccatcctctg	caatatggat	actatccagt	ggagggagac
781	cgtccaaaac	gtcttttatg	gcaacatgtc	aatggactta	cagagccatc	cgagcagttg
841	ccccaaatgt	gatccaagct	gtcccaatgg	aagctgctgg	ggaggaggag	aggagaactg
901	ccagaaatgt	acaaaaatca	tctgtgccca	gcaatgttcc	catcgctgtc	gtggcaggct
961	cccagtgac	tgctgccaca	accaaagtgc	tgcggggtgt	acagggcccc	gagagagtga
1021	ctgtctggtc	tgccaaaagt	tccaagatga	ggccacatgc	aaagacacct	gcccaccact
1081	catgctgtac	aacccccacca	cctatcagat	ggatgtcaac	cctgaaggga	agtacagctt
1141	tggtgccacc	tgtgtgaaga	agtgcctccg	aaactacgtg	gtgacagatc	atggctcatg
1201	tgtccgagcc	tgtgggcctg	actactacga	agtggagaaa	gatggcatcc	gcaagtgtaa
1261	aaaatgtgat	gggcccctgtc	gcaaaagtgtg	taatggcata	ggcattgggtg	aatttaaaga
1321	cacactctcc	ataaatgtcta	caaacatcaa	acatttcaaa	tactgcactg	ccatcagcgg
1381	ggaccttcac	atcctgccag	tggcctttaa	gggggattct	ttcacgcga	ctcctcctct
1441	agaccacaga	gaactagaaa	ttctaaaaac	cgtaaaggaa	ataacaggct	ttttgtgat
1501	tcagggttgg	cctgataact	ggactgacct	ccatgcttcc	gagaacctag	aaataatacg
1561	tggcagaaca	aagcaacatg	gtcagtttcc	tttggcggtc	gttggcctga	acatacacatc
1621	actggggctg	cgttccctca	aggagatcag	tgatggggat	gtgatcattt	ctggaaaccg
1681	aaatttggtc	tacgcaaaaa	caataaactg	gaaaaaactc	ttcgggacac	ccaatcagaa
1741	aacaaaaatc	atgaacaaca	gagctgagaa	agactgcaag	gccgtgaacc	acgtctgcaa
1801	tctcttatgc	tgtgtgaaga	gctgctgggg	ccctgagccc	agggactgtg	tctcttgcca
1861	gaatgtgagc	agaggcaggg	agtgcgtgga	gaaatgcaac	atcctggagg	gggaaccaag
1921	ggagtgtgtg	gaaaattctg	aatgcatcca	gtgccatcca	gaatgtctgc	cccaggccat
1981	gaacatcacc	tgtacaggca	ggggaccaga	caactgcac	cagtgtgcc	actacattga
2041	tggccacac	tgtgtcaaga	cctgccagc	tggcatcatg	ggagagaaca	acactctggt
2101	ctggaagtat	gcagatgcca	ataatgtctg	ccacctatgc	cacgccaact	gtacctatgg
2161	atgtgctggg	ccaggtcttc	aaggatgtga	agtgtggcca	tctgggtacg	ttcaatggca
2221	gtggatctta	aagacctttt	ggatctaaga	ccagaagcca	tctctgactc	ccctctcacc
2281	tccagtttc	ttccaaatcc	tctgggccag	ccagaggtct	cagattctgc	cctcttgccc
2341	tgtgccacc	ttgttgacca	ctggacagca	tatgtgatgg	ctactgctag	tggcagcttc
2401	acaagagggt	aacactacgg	actagccatt	cttctatgt	atctgtttct	gcaataacag
2461	ccgctttact	taagtctcag	cacttcttag	tctctctttt	tctctcagt	agcccaaggg
2521	gtcatgtcac	aaacatgggt	tgaagggcta	ctttgtcaaa	tgaaaaggct	tatcttgggg
2581	ggcatttttt	tctttttttt	ttttcttgaa	acacattgcc	cagcaaaagg	aataaatttc
2641	tctcatcatt	ttgtttctga	taaattctta	ctattgat		

SEQ ID NO: 112 Mouse EGFR Amino Acid Sequence Isoform B (NP_031938.1)

1	mrpsgtartt	llvlltalca	aggaleekkv	cqgtsnrltq	lgtfedhfls	lqrmynncev
61	vlgnleityv	qnydisflk	tiqevagyvl	ialntverip	lenlqiirgn	alyentyala
121	ilsnygtnrt	glrelpmrnl	qeiligavrf	snnpilcnmd	tiqwrdivqn	vfmsnmsmdl
181	qshpspcpkc	dpccpngscw	gggeencqkl	kliicagqcs	hrccgrspcd	chnqcaagc
241	tgpresdciv	cqkfdeatc	kdtcpplmly	npptyqmdvn	pegkysfgat	cvkkcprnyv
301	vtahgscvra	cgpdyveeve	dgirkckkcd	gpcrkvcngi	gigefkdtls	inatnikhfk
361	yctaisgdll	ilpvafkgds	fttrtppldpr	eleilkktvke	itgflliqaw	pdnwtldhaf
421	enleirgrt	kqhgqfslav	vglntslgl	rsлкеisdgd	viisgnrnlc	yantinwkl
481	fgtpnqtkki	mnnaekdck	avnhvcnplc	ssegcwgpep	rdcvscqnvs	rgrecvekcn
541	ilegeprefv	ensecigchp	ecelpgamnit	ctgrgpdnci	qcahyidgph	cvktcpagim
601	genntlvwky	adannvchlc	hanctygcag	pqlggcevwp	sgyvqwqwil	ktfwi

* Included in Table 1 are RNA nucleic acid molecules (e.g., thymines replaced with uridines), nucleic acid molecules encoding orthologs of the encoded proteins, as well as DNA or RNA nucleic acid sequences comprising a nucleic acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or more identity across their full length with the nucleic acid sequence of any SEQ ID NO or biomarker described in Table 1 (see below for example), or a portion thereof. Such nucleic acid molecules can have a function of the full-length nucleic acid as described further herein.

* Included in Table 1 are orthologs of the proteins, as well as polypeptide molecules comprising an amino acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with an amino acid sequence of any SEQ ID NO or biomarker described in Table 1 (see below for example), or a portion thereof. Such polypeptides can have a function of the full-length polypeptide as described further herein.

* Included in Table 1 are one or more subunits of a SWI/SNF complex like BAF or PBAF, and mutations within the one or more subunits. In some embodiments, the biomarkers are a class of mutations encompassing the one or more subunits of a SWI/SNF complex, such as the class of synonymous and/or non-synonymous mutations of ARID2 and/or PBRM1, or the class of loss-of-function mutations for biomarkers shown in Tables 4-5. In other embodiment, the biomarkers are particular mutations of one or more subunits of a SWI/SNF complex, such as particular mutations described in the Tables and Examples (e.g., Tables 4-5). Thus, included in Table 1 is, for example, PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and/or EGFR, including any cDNA or polypeptide of PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR. Similarly, included in Table 1 is, for example, PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR nucleic acid and/or amino acid sequences encoding or representing PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR having reduced or eliminated function (e.g., truncating mutations causing encoding of incomplete protein of PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR). Many of these mutations were found in subjects having cancer and who were insensitive to immune checkpoint therapies. It is further determined that EGFR as a biomarker of immune checkpoint efficacy acts in opposite fashion to the other biomarkers described in Table 1 such that EGFR is mutated more frequently (e.g., hotspot mutations) in non-responders or less efficacious responders to immune checkpoint therapy rather than more frequently in subjects who respond to immune checkpoint therapy.

II. Subjects

In one embodiment, the subject for whom predicted likelihood of efficacy of an immune checkpoint therapy is determined, is a mammal (e.g., mouse, rat, primate, non-human mammal, domestic animal, such as a dog, cat, cow, horse, and the like), and is preferably a human.

In another embodiment of the methods of the present invention, the subject has not undergone treatment, such as chemotherapy, radiation therapy, targeted therapy, and/or immune checkpoint therapy. In still another embodiment, the subject has undergone treatment, such as chemotherapy, radiation therapy, targeted therapy, and/or immune checkpoint therapy.

In certain embodiments, the subject has had surgery to remove cancerous or precancerous tissue. In other embodiments, the cancerous tissue has not been removed, e.g., the cancerous tissue may be located in an inoperable region of the body, such as in a tissue that is essential for life, or in a region where a surgical procedure would cause considerable risk of harm to the patient.

The methods of the present invention can be used to determine the responsiveness to anti-immune checkpoint therapies of a cancer. In one embodiment, the cancer is one for which an immune checkpoint therapy (e.g., anti-PD-1 blocking antibody, anti-PD-L1 blocking antibody, CTLA-4 blocking antibody, and the like) is FDA-approved for treatment, such as those described in the Examples. In one embodiment, the cancers are solid tumors, such as lung cancer such as non-small cell lung cancer, bladder cancer, melanoma such as metastatic melanoma, and/or renal cell carcinoma. In another embodiment, the cancer is an epithelial cancer such as, but not limited to, brain cancer (e.g., glioblastomas) bladder cancer, breast cancer, cervical cancer, colon cancer, gynecologic cancers, renal cancer, laryngeal cancer, lung cancer, oral cancer, head and neck cancer, ovarian cancer, pancreatic cancer, prostate cancer, or skin cancer. In still other embodiments, the cancer is breast cancer, prostate cancer, lung cancer, or colon cancer. In still other embodiments, the epithelial cancer is non-small-cell lung cancer, nonpapillary renal cell carcinoma, cervical carcinoma, ovarian carcinoma (e.g., serous ovarian carcinoma), or breast carcinoma. The epithelial cancers may be characterized in various other ways including, but not limited to, serous, endometrioid, mucinous, clear cell, brenner, or undifferentiated. In yet other embodiments, the cancer is a mesenchymal cancer, such as sarcoma.

III. Sample Collection, Preparation and Separation

In some embodiments, biomarker amount and/or activity measurement(s) in a sample from a subject is compared to a predetermined control (standard) sample. The sample from the subject is typically from a diseased tissue, such as cancer cells or tissues. The control sample can be from the same subject or from a different subject. The control sample is typically a normal, non-diseased sample. However, in some embodiments, such as for staging of disease or for evaluating the efficacy of treatment, the control sample can be from a diseased tissue. The control sample can be a combination of samples from several different subjects. In some embodiments, the biomarker amount and/or activity measurement(s) from a subject is compared to a pre-determined level. This pre-determined level is typically obtained from normal samples. As described herein, a "pre-determined" biomarker amount and/or activity measurement(s) may be a biomarker amount and/or activity measurement(s)

used to, by way of example only, evaluate a subject that may be selected for treatment, evaluate a response to an immune checkpoint therapy, and/or evaluate a response to a combination immune checkpoint therapy. A pre-determined biomarker amount and/or activity measurement(s) may be determined in populations of patients with or without cancer. The pre-determined biomarker amount and/or activity measurement(s) can be a single number, equally applicable to every patient, or the pre-determined biomarker amount and/or activity measurement(s) can vary according to specific subpopulations of patients. Age, weight, height, and other factors of a subject may affect the pre-determined biomarker amount and/or activity measurement(s) of the individual. Furthermore, the pre-determined biomarker amount and/or activity can be determined for each subject individually. In one embodiment, the amounts determined and/or compared in a method described herein are based on absolute measurements.

In another embodiment, the amounts determined and/or compared in a method described herein are based on relative measurements, such as ratios (e.g., biomarker copy numbers, level, and/or activity before a treatment vs. after a treatment, such biomarker measurements relative to a spiked or man-made control, such biomarker measurements relative to the expression of a housekeeping gene, and the like). For example, the relative analysis can be based on the ratio of pre-treatment biomarker measurement as compared to post-treatment biomarker measurement. Pre-treatment biomarker measurement can be made at any time prior to initiation of anti-cancer therapy. Post-treatment biomarker measurement can be made at any time after initiation of anti-cancer therapy. In some embodiments, post-treatment biomarker measurements are made 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 weeks or more after initiation of anti-cancer therapy, and even longer toward indefinitely for continued monitoring. Treatment can comprise anti-cancer therapy, such as a therapeutic regimen comprising an anti-PD1 monoclonal antibody (e.g., nivolumab) alone or in combination with other anti-cancer agents, such as anti-PD-L1/PD-L2 antibodies, anti-VEGF agents (e.g., bevacizumab), agents described in the Examples, Figures, and Tables, or anti-PBRM1 (or anti-ARID2, anti-BRD7, anti-PHF10, or anti-KDM6A) agents.

The pre-determined biomarker amount and/or activity measurement(s) can be any suitable standard. For example, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from the same or a different human for whom a patient selection is being assessed. In one embodiment, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from a previous assessment of the same patient. In such a manner, the progress of the selection of the patient can be monitored over time. In addition, the control can be obtained from an assessment of another human or multiple humans, e.g., selected groups of humans, if the subject is a human. In such a manner, the extent of the selection of the human for whom selection is being assessed can be compared to suitable other humans, e.g., other humans who are in a similar situation to the human of interest, such as those suffering from similar or the same condition(s) and/or of the same ethnic group.

In some embodiments of the present invention the change of biomarker amount and/or activity measurement(s) from the pre-determined level is about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, or 5.0 fold or greater, or any range in between, inclusive. Such cutoff values apply equally when the measurement is based on

relative changes, such as based on the ratio of pre-treatment biomarker measurement as compared to post-treatment biomarker measurement.

Biological samples can be collected from a variety of sources from a patient including a body fluid sample, cell sample, or a tissue sample comprising nucleic acids and/or proteins. "Body fluids" refer to fluids that are excreted or secreted from the body as well as fluids that are normally not (e.g., amniotic fluid, aqueous humor, bile, blood and blood plasma, cerebrospinal fluid, cerumen and earwax, cowper's fluid or pre-ejaculatory fluid, chyle, chyme, stool, female ejaculate, interstitial fluid, intracellular fluid, lymph, menses, breast milk, mucus, pleural fluid, pus, saliva, sebum, semen, serum, sweat, synovial fluid, tears, urine, vaginal lubrication, vitreous humor, vomit). In a preferred embodiment, the subject and/or control sample is selected from the group consisting of cells, cell lines, histological slides, paraffin embedded tissues, biopsies, whole blood, nipple aspirate, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, and bone marrow. In one embodiment, the sample is serum, plasma, or urine. In another embodiment, the sample is serum.

The samples can be collected from individuals repeatedly over a longitudinal period of time (e.g., once or more on the order of days, weeks, months, annually, biannually, etc.). Obtaining numerous samples from an individual over a period of time can be used to verify results from earlier detections and/or to identify an alteration in biological pattern as a result of, for example, disease progression, drug treatment, etc. For example, subject samples can be taken and monitored every month, every two months, or combinations of one, two, or three month intervals according to the present invention. In addition, the biomarker amount and/or activity measurements of the subject obtained over time can be conveniently compared with each other, as well as with those of normal controls during the monitoring period, thereby providing the subject's own values, as an internal, or personal, control for long-term monitoring.

Sample preparation and separation can involve any of the procedures, depending on the type of sample collected and/or analysis of biomarker measurement(s). Such procedures include, by way of example only, concentration, dilution, adjustment of pH, removal of high abundance polypeptides (e.g., albumin, gamma globulin, and transferrin, etc.), addition of preservatives and calibrants, addition of protease inhibitors, addition of denaturants, desalting of samples, concentration of sample proteins, extraction and purification of lipids.

The sample preparation can also isolate molecules that are bound in non-covalent complexes to other protein (e.g., carrier proteins). This process may isolate those molecules bound to a specific carrier protein (e.g., albumin), or use a more general process, such as the release of bound molecules from all carrier proteins via protein denaturation, for example using an acid, followed by removal of the carrier proteins.

Removal of undesired proteins (e.g., high abundance, uninformative, or undetectable proteins) from a sample can be achieved using high affinity reagents, high molecular weight filters, ultracentrifugation and/or electrodialysis. High affinity reagents include antibodies or other reagents (e.g., aptamers) that selectively bind to high abundance proteins. Sample preparation could also include ion exchange chromatography, metal ion affinity chromatography, gel filtration, hydrophobic chromatography, chromatofocusing, adsorption chromatography, isoelectric focusing and related techniques. Molecular weight filters include

membranes that separate molecules on the basis of size and molecular weight. Such filters may further employ reverse osmosis, nanofiltration, ultrafiltration and microfiltration.

Ultracentrifugation is a method for removing undesired polypeptides from a sample. Ultracentrifugation is the centrifugation of a sample at about 15,000-60,000 rpm while monitoring with an optical system the sedimentation (or lack thereof) of particles. Electrodialysis is a procedure which uses an electromembrane or semipermeable membrane in a process in which ions are transported through semi-permeable membranes from one solution to another under the influence of a potential gradient. Since the membranes used in electrodialysis may have the ability to selectively transport ions having positive or negative charge, reject ions of the opposite charge, or to allow species to migrate through a semipermeable membrane based on size and charge, it renders electrodialysis useful for concentration, removal, or separation of electrolytes.

Separation and purification in the present invention may include any procedure known in the art, such as capillary electrophoresis (e.g., in capillary or on-chip) or chromatography (e.g., in capillary, column or on a chip). Electrophoresis is a method which can be used to separate ionic molecules under the influence of an electric field. Electrophoresis can be conducted in a gel, capillary, or in a microchannel on a chip. Examples of gels used for electrophoresis include starch, acrylamide, polyethylene oxides, agarose, or combinations thereof. A gel can be modified by its cross-linking, addition of detergents, or denaturants, immobilization of enzymes or antibodies (affinity electrophoresis) or substrates (zymography) and incorporation of a pH gradient. Examples of capillaries used for electrophoresis include capillaries that interface with an electrospray.

Capillary electrophoresis (CE) is preferred for separating complex hydrophilic molecules and highly charged solutes. CE technology can also be implemented on microfluidic chips. Depending on the types of capillary and buffers used, CE can be further segmented into separation techniques such as capillary zone electrophoresis (CZE), capillary isoelectric focusing (CIEF), capillary isotachopheresis (cITP) and capillary electrochromatography (CEC). An embodiment to couple CE techniques to electrospray ionization involves the use of volatile solutions, for example, aqueous mixtures containing a volatile acid and/or base and an organic such as an alcohol or acetonitrile.

Capillary isotachopheresis (cITP) is a technique in which the analytes move through the capillary at a constant speed but are nevertheless separated by their respective mobilities. Capillary zone electrophoresis (CZE), also known as free-solution CE (FSCE), is based on differences in the electrophoretic mobility of the species, determined by the charge on the molecule, and the frictional resistance the molecule encounters during migration which is often directly proportional to the size of the molecule. Capillary isoelectric focusing (CIEF) allows weakly-ionizable amphoteric molecules, to be separated by electrophoresis in a pH gradient. CEC is a hybrid technique between traditional high performance liquid chromatography (HPLC) and CE.

Separation and purification techniques used in the present invention include any chromatography procedures known in the art. Chromatography can be based on the differential adsorption and elution of certain analytes or partitioning of analytes between mobile and stationary phases. Different examples of chromatography include, but not limited to,

liquid chromatography (LC), gas chromatography (GC), high performance liquid chromatography (HPLC), etc.

IV. Biomarker Nucleic Acids and Polypeptides

One aspect of the present invention pertains to the use of isolated nucleic acid molecules that correspond to biomarker nucleic acids that encode a biomarker polypeptide or a portion of such a polypeptide. As used herein, the term “nucleic acid molecule” is intended to include DNA molecules (e.g., cDNA or genomic DNA) and RNA molecules (e.g., mRNA) and analogs of the DNA or RNA generated using nucleotide analogs. The nucleic acid molecule can be single-stranded or double-stranded, but preferably is double-stranded DNA.

An “isolated” nucleic acid molecule is one which is separated from other nucleic acid molecules which are present in the natural source of the nucleic acid molecule. Preferably, an “isolated” nucleic acid molecule is free of sequences (preferably protein-encoding sequences) which naturally flank the nucleic acid (i.e., sequences located at the 5' and 3' ends of the nucleic acid) in the genomic DNA of the organism from which the nucleic acid is derived. For example, in various embodiments, the isolated nucleic acid molecule can contain less than about 5 kB, 4 kB, 3 kB, 2 kB, 1 kB, 0.5 kB or 0.1 kB of nucleotide sequences which naturally flank the nucleic acid molecule in genomic DNA of the cell from which the nucleic acid is derived. Moreover, an “isolated” nucleic acid molecule, such as a cDNA molecule, can be substantially free of other cellular material or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized.

A biomarker nucleic acid molecule of the present invention can be isolated using standard molecular biology techniques and the sequence information in the database records described herein. Using all or a portion of such nucleic acid sequences, nucleic acid molecules of the present invention can be isolated using standard hybridization and cloning techniques (e.g., as described in Sambrook et al., ed., *Molecular Cloning: A Laboratory Manual*, 2nd ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989).

A nucleic acid molecule of the present invention can be amplified using cDNA, mRNA, or genomic DNA as a template and appropriate oligonucleotide primers according to standard PCR amplification techniques. The nucleic acid molecules so amplified can be cloned into an appropriate vector and characterized by DNA sequence analysis. Furthermore, oligonucleotides corresponding to all or a portion of a nucleic acid molecule of the present invention can be prepared by standard synthetic techniques, e.g., using an automated DNA synthesizer.

Moreover, a nucleic acid molecule of the present invention can comprise only a portion of a nucleic acid sequence, wherein the full length nucleic acid sequence comprises a marker of the present invention or which encodes a polypeptide corresponding to a marker of the present invention. Such nucleic acid molecules can be used, for example, as a probe or primer. The probe/primer typically is used as one or more substantially purified oligonucleotides. The oligonucleotide typically comprises a region of nucleotide sequence that hybridizes under stringent conditions to at least about 7, preferably about 15, more preferably about 25, 50, 75, 100, 125, 150, 175, 200, 250, 300, 350, or 400 or more consecutive nucleotides of a biomarker nucleic acid sequence. Probes based on the sequence of a biomarker

nucleic acid molecule can be used to detect transcripts or genomic sequences corresponding to one or more markers of the present invention. The probe comprises a label group attached thereto, e.g., a radioisotope, a fluorescent compound, an enzyme, or an enzyme co-factor.

A biomarker nucleic acid molecules that differ, due to degeneracy of the genetic code, from the nucleotide sequence of nucleic acid molecules encoding a protein which corresponds to the biomarker, and thus encode the same protein, are also contemplated.

In addition, it will be appreciated by those skilled in the art that DNA sequence polymorphisms that lead to changes in the amino acid sequence can exist within a population (e.g., the human population). Such genetic polymorphisms can exist among individuals within a population due to natural allelic variation. An allele is one of a group of genes which occur alternatively at a given genetic locus. In addition, it will be appreciated that DNA polymorphisms that affect RNA expression levels can also exist that may affect the overall expression level of that gene (e.g., by affecting regulation or degradation).

The term “allele,” which is used interchangeably herein with “allelic variant,” refers to alternative forms of a gene or portions thereof. Alleles occupy the same locus or position on homologous chromosomes. When a subject has two identical alleles of a gene, the subject is said to be homozygous for the gene or allele. When a subject has two different alleles of a gene, the subject is said to be heterozygous for the gene or allele. For example, biomarker alleles can differ from each other in a single nucleotide, or several nucleotides, and can include substitutions, deletions, and insertions of nucleotides. An allele of a gene can also be a form of a gene containing one or more mutations.

The term “allelic variant of a polymorphic region of gene” or “allelic variant”, used interchangeably herein, refers to an alternative form of a gene having one of several possible nucleotide sequences found in that region of the gene in the population. As used herein, allelic variant is meant to encompass functional allelic variants, non-functional allelic variants, SNPs, mutations and polymorphisms.

The term “single nucleotide polymorphism” (SNP) refers to a polymorphic site occupied by a single nucleotide, which is the site of variation between allelic sequences. The site is usually preceded by and followed by highly conserved sequences of the allele (e.g., sequences that vary in less than 1/100 or 1/1000 members of a population). A SNP usually arises due to substitution of one nucleotide for another at the polymorphic site. SNPs can also arise from a deletion of a nucleotide or an insertion of a nucleotide relative to a reference allele. Typically the polymorphic site is occupied by a base other than the reference base. For example, where the reference allele contains the base “T” (thymidine) at the polymorphic site, the altered allele can contain a “C” (cytidine), “G” (guanine), or “A” (adenine) at the polymorphic site. SNP's may occur in protein-coding nucleic acid sequences, in which case they may give rise to a defective or otherwise variant protein, or genetic disease. Such a SNP may alter the coding sequence of the gene and therefore specify another amino acid (a “missense” SNP) or a SNP may introduce a stop codon (a “nonsense” SNP). When a SNP does not alter the amino acid sequence of a protein, the SNP is called “silent.” SNP's may also occur in noncoding regions of the nucleotide sequence. This may result in defective protein expression, e.g., as a result of alternative splicing, or it may have no effect on the function of the protein.

As used herein, the terms “gene” and “recombinant gene” refer to nucleic acid molecules comprising an open reading frame encoding a polypeptide corresponding to a marker of the present invention. Such natural allelic variations can typically result in 1-5% variance in the nucleotide sequence of a given gene. Alternative alleles can be identified by sequencing the gene of interest in a number of different individuals. This can be readily carried out by using hybridization probes to identify the same genetic locus in a variety of individuals. Any and all such nucleotide variations and resulting amino acid polymorphisms or variations that are the result of natural allelic variation and that do not alter the functional activity are intended to be within the scope of the present invention.

In another embodiment, a biomarker nucleic acid molecule is at least 7, 15, 20, 25, 30, 40, 60, 80, 100, 150, 200, 250, 300, 350, 400, 450, 550, 650, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, 2000, 2200, 2400, 2600, 2800, 3000, 3500, 4000, 4500, or more nucleotides in length and hybridizes under stringent conditions to a nucleic acid molecule corresponding to a marker of the present invention or to a nucleic acid molecule encoding a protein corresponding to a marker of the present invention. As used herein, the term “hybridizes under stringent conditions” is intended to describe conditions for hybridization and washing under which nucleotide sequences at least 60% (65%, 70%, 75%, 80%, preferably 85%) identical to each other typically remain hybridized to each other. Such stringent conditions are known to those skilled in the art and can be found in sections 6.3.1-6.3.6 of *Current Protocols in Molecular Biology*, John Wiley & Sons, N.Y. (1989). A preferred, non-limiting example of stringent hybridization conditions are hybridization in 6× sodium chloride/sodium citrate (SSC) at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 50-65° C.

In addition to naturally-occurring allelic variants of a nucleic acid molecule of the present invention that can exist in the population, the skilled artisan will further appreciate that sequence changes can be introduced by mutation thereby leading to changes in the amino acid sequence of the encoded protein, without altering the biological activity of the protein encoded thereby. For example, one can make nucleotide substitutions leading to amino acid substitutions at “non-essential” amino acid residues. A “non-essential” amino acid residue is a residue that can be altered from the wild-type sequence without altering the biological activity, whereas an “essential” amino acid residue is required for biological activity. For example, amino acid residues that are not conserved or only semi-conserved among homologs of various species may be non-essential for activity and thus would be likely targets for alteration. Alternatively, amino acid residues that are conserved among the homologs of various species (e.g., murine and human) may be essential for activity and thus would not be likely targets for alteration.

Accordingly, another aspect of the present invention pertains to nucleic acid molecules encoding a polypeptide of the present invention that contain changes in amino acid residues that are not essential for activity. Such polypeptides differ in amino acid sequence from the naturally-occurring proteins which correspond to the markers of the present invention, yet retain biological activity. In one embodiment, a biomarker protein has an amino acid sequence that is at least about 40% identical, 50%, 60%, 70%, 75%, 80%, 83%, 85%, 87.5%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%,

98%, 99% or identical to the amino acid sequence of a biomarker protein described herein.

An isolated nucleic acid molecule encoding a variant protein can be created by introducing one or more nucleotide substitutions, additions or deletions into the nucleotide sequence of nucleic acids of the present invention, such that one or more amino acid residue substitutions, additions, or deletions are introduced into the encoded protein. Mutations can be introduced by standard techniques, such as site-directed mutagenesis and PCR-mediated mutagenesis. Preferably, conservative amino acid substitutions are made at one or more predicted non-essential amino acid residues. A “conservative amino acid substitution” is one in which the amino acid residue is replaced with an amino acid residue having a similar side chain. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine), non-polar side chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). Alternatively, mutations can be introduced randomly along all or part of the coding sequence, such as by saturation mutagenesis, and the resulting mutants can be screened for biological activity to identify mutants that retain activity. Following mutagenesis, the encoded protein can be expressed recombinantly and the activity of the protein can be determined.

In some embodiments, the present invention further contemplates the use of anti-biomarker antisense nucleic acid molecules, i.e., molecules which are complementary to a sense nucleic acid of the present invention, e.g., complementary to the coding strand of a double-stranded cDNA molecule corresponding to a marker of the present invention or complementary to an mRNA sequence corresponding to a marker of the present invention. Accordingly, an antisense nucleic acid molecule of the present invention can hydrogen bond to (i.e. anneal with) a sense nucleic acid of the present invention. The antisense nucleic acid can be complementary to an entire coding strand, or to only a portion thereof, e.g., all or part of the protein coding region (or open reading frame). An antisense nucleic acid molecule can also be antisense to all or part of a non-coding region of the coding strand of a nucleotide sequence encoding a polypeptide of the present invention. The non-coding regions (“5' and 3' untranslated regions”) are the 5' and 3' sequences which flank the coding region and are not translated into amino acids.

An antisense oligonucleotide can be, for example, about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 or more nucleotides in length. An antisense nucleic acid can be constructed using chemical synthesis and enzymatic ligation reactions using procedures known in the art. For example, an antisense nucleic acid (e.g., an antisense oligonucleotide) can be chemically synthesized using naturally occurring nucleotides or variously modified nucleotides designed to increase the biological stability of the molecules or to increase the physical stability of the duplex formed between the antisense and sense nucleic acids, e.g., phosphorothioate derivatives and acridine substituted nucleotides can be used. Examples of modified nucleotides which can be used to generate the antisense nucleic acid include 5-fluorouracil, 5-bromouracil, 5-chlorouracil, 5-iodouracil, hypoxanthine, xanthine, 4-acetylcytosine, 5-(carboxyhydroxymethyl) ura-

cil, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluracil, dihydrouracil, beta-D-galactosylqueosine, inosine, N6-isopentenyladenine, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-methylguanine, 3-methylcytosine, 5-methylcytosine, N6-adenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5'-methoxycarboxymethyluracil, 5-methoxyuracil, 2-methylthio-N6-isopentenyladenine, uracil-5-oxyacetic acid (v), wybutoxosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, uracil-5-oxyacetic acid methyl ester, uracil-5-oxy acetic acid (v), 5-methyl-2-thiouracil, 3-(3-amino-3-N-2-carboxypropyl) uracil, (acp3)w, and 2,6-diaminopurine. Alternatively, the antisense nucleic acid can be produced biologically using an expression vector into which a nucleic acid has been sub-cloned in an antisense orientation (i.e., RNA transcribed from the inserted nucleic acid will be of an antisense orientation to a target nucleic acid of interest, described further in the following subsection).

The antisense nucleic acid molecules of the present invention are typically administered to a subject or generated in situ such that they hybridize with or bind to cellular mRNA and/or genomic DNA encoding a polypeptide corresponding to a selected marker of the present invention to thereby inhibit expression of the marker, e.g., by inhibiting transcription and/or translation. The hybridization can be by conventional nucleotide complementarity to form a stable duplex, or, for example, in the case of an antisense nucleic acid molecule which binds to DNA duplexes, through specific interactions in the major groove of the double helix. Examples of a route of administration of antisense nucleic acid molecules of the present invention includes direct injection at a tissue site or infusion of the antisense nucleic acid into a blood- or bone marrow-associated body fluid. Alternatively, antisense nucleic acid molecules can be modified to target selected cells and then administered systemically. For example, for systemic administration, antisense molecules can be modified such that they specifically bind to receptors or antigens expressed on a selected cell surface, e.g., by linking the antisense nucleic acid molecules to peptides or antibodies which bind to cell surface receptors or antigens. The antisense nucleic acid molecules can also be delivered to cells using the vectors described herein. To achieve sufficient intracellular concentrations of the antisense molecules, vector constructs in which the antisense nucleic acid molecule is placed under the control of a strong pol II or pol III promoter are preferred.

An antisense nucleic acid molecule of the present invention can be an α -anomeric nucleic acid molecule. An α -anomeric nucleic acid molecule forms specific double-stranded hybrids with complementary RNA in which, contrary to the usual α -units, the strands run parallel to each other (Gaultier et al. (1987) *Nucleic Acids Res.* 15:6625-6641). The antisense nucleic acid molecule can also comprise a 2'-O-methylribonucleotide (Inoue et al. (1987) *Nucleic Acids Res.* 15:6131-6148) or a chimeric RNA-DNA analogue (Inoue et al. (1987) *FEBS Lett.* 215:327-330).

The present invention also encompasses ribozymes. Ribozymes are catalytic RNA molecules with ribonuclease activity which are capable of cleaving a single-stranded nucleic acid, such as an mRNA, to which they have a complementary region. Thus, ribozymes (e.g., hammerhead ribozymes as described in Haselhoff and Gerlach (1988) *Nature* 334:585-591) can be used to catalytically cleave mRNA transcripts to thereby inhibit translation of the pro-

tein encoded by the mRNA. A ribozyme having specificity for a nucleic acid molecule encoding a polypeptide corresponding to a marker of the present invention can be designed based upon the nucleotide sequence of a cDNA corresponding to the marker. For example, a derivative of a *Tetrahymena* L-19 IVS RNA can be constructed in which the nucleotide sequence of the active site is complementary to the nucleotide sequence to be cleaved (see Cech et al. U.S. Pat. No. 4,987,071; and Cech et al. U.S. Pat. No. 5,116,742). Alternatively, an mRNA encoding a polypeptide of the present invention can be used to select a catalytic RNA having a specific ribonuclease activity from a pool of RNA molecules (see, e.g., Bartel and Szostak (1993) *Science* 261:1411-1418).

The present invention also encompasses nucleic acid molecules which form triple helical structures. For example, expression of a biomarker protein can be inhibited by targeting nucleotide sequences complementary to the regulatory region of the gene encoding the polypeptide (e.g., the promoter and/or enhancer) to form triple helical structures that prevent transcription of the gene in target cells. See generally Helene (1991) *Anticancer Drug Des.* 6(6):569-84; Helene (1992) *Ann. N. Y. Acad. Sci.* 660:27-36; and Maher (1992) *Bioassays* 14(12):807-15.

In various embodiments, the nucleic acid molecules of the present invention can be modified at the base moiety, sugar moiety or phosphate backbone to improve, e.g., the stability, hybridization, or solubility of the molecule. For example, the deoxyribose phosphate backbone of the nucleic acid molecules can be modified to generate peptide nucleic acid molecules (see Hyrup et al. (1996) *Bioorganic & Medicinal Chemistry* 4(1): 5-23). As used herein, the terms "peptide nucleic acids" or "PNAs" refer to nucleic acid mimics, e.g., DNA mimics, in which the deoxyribose phosphate backbone is replaced by a pseudopeptide backbone and only the four natural nucleobases are retained. The neutral backbone of PNAs has been shown to allow for specific hybridization to DNA and RNA under conditions of low ionic strength. The synthesis of PNA oligomers can be performed using standard solid phase peptide synthesis protocols as described in Hyrup et al. (1996), supra; Perry-O'Keefe et al. (1996) *Proc. Natl. Acad. Sci. USA* 93:14670-675.

PNAs can be used in therapeutic and diagnostic applications. For example, PNAs can be used as antisense or antigene agents for sequence-specific modulation of gene expression by, e.g., inducing transcription or translation arrest or inhibiting replication. PNAs can also be used, e.g., in the analysis of single base pair mutations in a gene by, e.g., PNA directed PCR clamping; as artificial restriction enzymes when used in combination with other enzymes, e.g., S1 nucleases (Hyrup (1996), supra; or as probes or primers for DNA sequence and hybridization (Hyrup (1996), supra; Perry-O'Keefe et al. (1996) *Proc. Natl. Acad. Sci. USA* 93:14670-14675).

In another embodiment, PNAs can be modified, e.g., to enhance their stability or cellular uptake, by attaching lipophilic or other helper groups to PNA, by the formation of PNA-DNA chimeras, or by the use of liposomes or other techniques of drug delivery known in the art. For example, PNA-DNA chimeras can be generated which can combine the advantageous properties of PNA and DNA. Such chimeras allow DNA recognition enzymes, e.g., RNASE H and DNA polymerases, to interact with the DNA portion while the PNA portion would provide high binding affinity and specificity. PNA-DNA chimeras can be linked using linkers of appropriate lengths selected in terms of base stacking, number of bonds between the nucleobases, and orientation

(Hyrup (1996), supra). The synthesis of PNA-DNA chimeras can be performed as described in Hyrup (1996), supra, and Finn et al. (1996) *Nucleic Acids Res.* 24(17):3357-3363. For example, a DNA chain can be synthesized on a solid support using standard phosphoramidite coupling chemistry and modified nucleoside analogs. Compounds such as 5'-(4-methoxytrityl)amino-5'-deoxy-thymidine phosphoramidite can be used as a link between the PNA and the 5' end of DNA (Mag et al. (1989) *Nucleic Acids Res.* 17:5973-5988). PNA monomers are then coupled in a step-wise manner to produce a chimeric molecule with a 5' PNA segment and a 3' DNA segment (Finn et al. (1996) *Nucleic Acids Res.* 24:3357-3363). Alternatively, chimeric molecules can be synthesized with a 5' DNA segment and a 3' PNA segment (Peterser et al. (1975) *Bioorganic Med. Chem. Lett.* 5:1119-11124).

In other embodiments, the oligonucleotide can include other appended groups such as peptides (e.g., for targeting host cell receptors in vivo), or agents facilitating transport across the cell membrane (see, e.g., Letsinger et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:6553-6556; Lemaître et al. (1987) *Proc. Natl. Acad. Sci. USA* 84:648-652; PCT Publication No. WO 88/09810) or the blood-brain barrier (see, e.g., PCT Publication No. WO 89/10134). In addition, oligonucleotides can be modified with hybridization-triggered cleavage agents (see, e.g., Krol et al. (1988) *Bio Techniques* 6:958-976) or intercalating agents (see, e.g., Zon (1988) *Pharm. Res.* 5:539-549). To this end, the oligonucleotide can be conjugated to another molecule, e.g., a peptide, hybridization triggered cross-linking agent, transport agent, hybridization-triggered cleavage agent, etc.

Another aspect of the present invention pertains to the use of biomarker proteins and biologically active portions thereof. In one embodiment, the native polypeptide corresponding to a marker can be isolated from cells or tissue sources by an appropriate purification scheme using standard protein purification techniques. In another embodiment, polypeptides corresponding to a marker of the present invention are produced by recombinant DNA techniques. Alternative to recombinant expression, a polypeptide corresponding to a marker of the present invention can be synthesized chemically using standard peptide synthesis techniques.

An "isolated" or "purified" protein or biologically active portion thereof is substantially free of cellular material or other contaminating proteins from the cell or tissue source from which the protein is derived, or substantially free of chemical precursors or other chemicals when chemically synthesized. The language "substantially free of cellular material" includes preparations of protein in which the protein is separated from cellular components of the cells from which it is isolated or recombinantly produced. Thus, protein that is substantially free of cellular material includes preparations of protein having less than about 30%, 20%, 10%, or 5% (by dry weight) of heterologous protein (also referred to herein as a "contaminating protein"). When the protein or biologically active portion thereof is recombinantly produced, it is also preferably substantially free of culture medium, i.e., culture medium represents less than about 20%, 10%, or 5% of the volume of the protein preparation. When the protein is produced by chemical synthesis, it is preferably substantially free of chemical precursors or other chemicals, i.e., it is separated from chemical precursors or other chemicals which are involved in the synthesis of the protein. Accordingly such preparations of the protein have less than about 30%, 20%, 10%, 5%

(by dry weight) of chemical precursors or compounds other than the polypeptide of interest.

Biologically active portions of a biomarker polypeptide include polypeptides comprising amino acid sequences sufficiently identical to or derived from a biomarker protein amino acid sequence described herein, but which includes fewer amino acids than the full length protein, and exhibit at least one activity of the corresponding full-length protein. Typically, biologically active portions comprise a domain or motif with at least one activity of the corresponding protein. A biologically active portion of a protein of the present invention can be a polypeptide which is, for example, 10, 25, 50, 100 or more amino acids in length. Moreover, other biologically active portions, in which other regions of the protein are deleted, can be prepared by recombinant techniques and evaluated for one or more of the functional activities of the native form of a polypeptide of the present invention.

Preferred polypeptides have an amino acid sequence of a biomarker protein encoded by a nucleic acid molecule described herein. Other useful proteins are substantially identical (e.g., at least about 40%, preferably 50%, 60%, 70%, 75%, 80%, 83%, 85%, 88%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to one of these sequences and retain the functional activity of the protein of the corresponding naturally-occurring protein yet differ in amino acid sequence due to natural allelic variation or mutagenesis.

To determine the percent identity of two amino acid sequences or of two nucleic acids, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in the sequence of a first amino acid or nucleic acid sequence for optimal alignment with a second amino or nucleic acid sequence). The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences (i.e., % identity=# of identical positions/total # of positions (e.g., overlapping positions)×100). In one embodiment the two sequences are the same length.

The determination of percent identity between two sequences can be accomplished using a mathematical algorithm. A preferred, non-limiting example of a mathematical algorithm utilized for the comparison of two sequences is the algorithm of Karlin and Altschul (1990) *Proc. Natl. Acad. Sci. USA* 87:2264-2268, modified as in Karlin and Altschul (1993) *Proc. Natl. Acad. Sci. USA* 90:5873-5877. Such an algorithm is incorporated into the NBLAST and XBLAST programs of Altschul, et al. (1990) *J. Mol. Biol.* 215:403-410. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12 to obtain nucleotide sequences homologous to a nucleic acid molecules of the present invention. BLAST protein searches can be performed with the XBLAST program, score=50, wordlength=3 to obtain amino acid sequences homologous to a protein molecules of the present invention. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402. Alternatively, PSI-Blast can be used to perform an iterated search which detects distant relationships between molecules. When utilizing BLAST, Gapped BLAST, and PSI-Blast programs, the default parameters of the respective programs (e.g.,

XBLAST and NBLAST) can be used. See <http://www.ncbi.nlm.nih.gov>. Another preferred, non-limiting example of a mathematical algorithm utilized for the comparison of sequences is the algorithm of Myers and Miller, (1988) *Comput Appl Biosci*, 4:11-7. Such an algorithm is incorporated into the ALIGN program (version 2.0) which is part of the GCG sequence alignment software package. When utilizing the ALIGN program for comparing amino acid sequences, a PAM120 weight residue table, a gap length penalty of 12, and a gap penalty of 4 can be used. Yet another useful algorithm for identifying regions of local sequence similarity and alignment is the FASTA algorithm as described in Pearson and Lipman (1988) *Proc. Natl. Acad. Sci. USA* 85:2444-2448. When using the FASTA algorithm for comparing nucleotide or amino acid sequences, a PAM120 weight residue table can, for example, be used with a k-tuple value of 2.

The percent identity between two sequences can be determined using techniques similar to those described above, with or without allowing gaps. In calculating percent identity, only exact matches are counted.

The present invention also provides chimeric or fusion proteins corresponding to a biomarker protein. As used herein, a "chimeric protein" or "fusion protein" comprises all or part (preferably a biologically active part) of a polypeptide corresponding to a marker of the present invention operably linked to a heterologous polypeptide (i.e., a polypeptide other than the polypeptide corresponding to the marker). Within the fusion protein, the term "operably linked" is intended to indicate that the polypeptide of the present invention and the heterologous polypeptide are fused in-frame to each other. The heterologous polypeptide can be fused to the amino-terminus or the carboxyl-terminus of the polypeptide of the present invention.

One useful fusion protein is a GST fusion protein in which a polypeptide corresponding to a marker of the present invention is fused to the carboxyl terminus of GST sequences. Such fusion proteins can facilitate the purification of a recombinant polypeptide of the present invention.

In another embodiment, the fusion protein contains a heterologous signal sequence, immunoglobulin fusion protein, toxin, or other useful protein sequence. Chimeric and fusion proteins of the present invention can be produced by standard recombinant DNA techniques. In another embodiment, the fusion gene can be synthesized by conventional techniques including automated DNA synthesizers. Alternatively, PCR amplification of gene fragments can be carried out using anchor primers which give rise to complementary overhangs between two consecutive gene fragments which can subsequently be annealed and re-amplified to generate a chimeric gene sequence (see, e.g., Ausubel et al., supra). Moreover, many expression vectors are commercially available that already encode a fusion moiety (e.g., a GST polypeptide). A nucleic acid encoding a polypeptide of the present invention can be cloned into such an expression vector such that the fusion moiety is linked in-frame to the polypeptide of the present invention.

A signal sequence can be used to facilitate secretion and isolation of the secreted protein or other proteins of interest. Signal sequences are typically characterized by a core of hydrophobic amino acids which are generally cleaved from the mature protein during secretion in one or more cleavage events. Such signal peptides contain processing sites that allow cleavage of the signal sequence from the mature proteins as they pass through the secretory pathway. Thus, the present invention pertains to the described polypeptides having a signal sequence, as well as to polypeptides from

which the signal sequence has been proteolytically cleaved (i.e., the cleavage products). In one embodiment, a nucleic acid sequence encoding a signal sequence can be operably linked in an expression vector to a protein of interest, such as a protein which is ordinarily not secreted or is otherwise difficult to isolate. The signal sequence directs secretion of the protein, such as from a eukaryotic host into which the expression vector is transformed, and the signal sequence is subsequently or concurrently cleaved. The protein can then be readily purified from the extracellular medium by art recognized methods. Alternatively, the signal sequence can be linked to the protein of interest using a sequence which facilitates purification, such as with a GST domain.

The present invention also pertains to variants of the biomarker polypeptides described herein. Such variants have an altered amino acid sequence which can function as either agonists (mimetics) or as antagonists. Variants can be generated by mutagenesis, e.g., discrete point mutation or truncation. An agonist can retain substantially the same, or a subset, of the biological activities of the naturally occurring form of the protein. An antagonist of a protein can inhibit one or more of the activities of the naturally occurring form of the protein by, for example, competitively binding to a downstream or upstream member of a cellular signaling cascade which includes the protein of interest. Thus, specific biological effects can be elicited by treatment with a variant of limited function. Treatment of a subject with a variant having a subset of the biological activities of the naturally occurring form of the protein can have fewer side effects in a subject relative to treatment with the naturally occurring form of the protein.

Variants of a biomarker protein which function as either agonists (mimetics) or as antagonists can be identified by screening combinatorial libraries of mutants, e.g., truncation mutants, of the protein of the present invention for agonist or antagonist activity. In one embodiment, a variegated library of variants is generated by combinatorial mutagenesis at the nucleic acid level and is encoded by a variegated gene library. A variegated library of variants can be produced by, for example, enzymatically ligating a mixture of synthetic oligonucleotides into gene sequences such that a degenerate set of potential protein sequences is expressible as individual polypeptides, or alternatively, as a set of larger fusion proteins (e.g., for phage display). There are a variety of methods which can be used to produce libraries of potential variants of the polypeptides of the present invention from a degenerate oligonucleotide sequence. Methods for synthesizing degenerate oligonucleotides are known in the art (see, e.g., Narang (1983) *Tetrahedron* 39:3; Itakura et al. (1984) *Annu. Rev. Biochem.* 53:323; Itakura et al. (1984) *Science* 198:1056; Ike et al. (1983) *Nucleic Acid Res.* 11:477).

In addition, libraries of fragments of the coding sequence of a polypeptide corresponding to a marker of the present invention can be used to generate a variegated population of polypeptides for screening and subsequent selection of variants. For example, a library of coding sequence fragments can be generated by treating a double stranded PCR fragment of the coding sequence of interest with a nuclease under conditions wherein nicking occurs only about once per molecule, denaturing the double stranded DNA, renaturing the DNA to form double stranded DNA which can include sense/antisense pairs from different nicked products, removing single stranded portions from reformed duplexes by treatment with S1 nuclease, and ligating the resulting fragment library into an expression vector. By this method,

an expression library can be derived which encodes amino terminal and internal fragments of various sizes of the protein of interest.

Several techniques are known in the art for screening gene products of combinatorial libraries made by point mutations or truncation, and for screening cDNA libraries for gene products having a selected property. The most widely used techniques, which are amenable to high throughput analysis, for screening large gene libraries typically include cloning the gene library into replicable expression vectors, transforming appropriate cells with the resulting library of vectors, and expressing the combinatorial genes under conditions in which detection of a desired activity facilitates isolation of the vector encoding the gene whose product was detected. Recursive ensemble mutagenesis (REM), a technique which enhances the frequency of functional mutants in the libraries, can be used in combination with the screening assays to identify variants of a protein of the present invention (Arkin and Yourvan (1992) *Proc. Natl. Acad. Sci. USA* 89:7811-7815; Delgrave et al. 91993) *Protein Engineering* 6(3):327-331).

The production and use of biomarker nucleic acid and/or biomarker polypeptide molecules described herein can be facilitated by using standard recombinant techniques. In some embodiments, such techniques use vectors, preferably expression vectors, containing a nucleic acid encoding a biomarker polypeptide or a portion of such a polypeptide. As used herein, the term "vector" refers to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a "plasmid", which refers to a circular double stranded DNA loop into which additional DNA segments can be ligated. Another type of vector is a viral vector, wherein additional DNA segments can be ligated into the viral genome. Certain vectors are capable of autonomous replication in a host cell into which they are introduced (e.g., bacterial vectors having a bacterial origin of replication and episomal mammalian vectors). Other vectors (e.g., non-episomal mammalian vectors) are integrated into the genome of a host cell upon introduction into the host cell, and thereby are replicated along with the host genome. Moreover, certain vectors, namely expression vectors, are capable of directing the expression of genes to which they are operably linked. In general, expression vectors of utility in recombinant DNA techniques are often in the form of plasmids (vectors). However, the present invention is intended to include such other forms of expression vectors, such as viral vectors (e.g., replication defective retroviruses, adenoviruses and adeno-associated viruses), which serve equivalent functions.

The recombinant expression vectors of the present invention comprise a nucleic acid of the present invention in a form suitable for expression of the nucleic acid in a host cell. This means that the recombinant expression vectors include one or more regulatory sequences, selected on the basis of the host cells to be used for expression, which is operably linked to the nucleic acid sequence to be expressed. Within a recombinant expression vector, "operably linked" is intended to mean that the nucleotide sequence of interest is linked to the regulatory sequence(s) in a manner which allows for expression of the nucleotide sequence (e.g., in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell). The term "regulatory sequence" is intended to include promoters, enhancers and other expression control elements (e.g., polyadenylation signals). Such regulatory sequences are described, for example, in Goeddel, *Methods in Enzymology: Gene Expression Technology* vol. 185, Academic Press,

San Diego, CA (1991). Regulatory sequences include those which direct constitutive expression of a nucleotide sequence in many types of host cell and those which direct expression of the nucleotide sequence only in certain host cells (e.g., tissue-specific regulatory sequences). It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed, the level of expression of protein desired, and the like. The expression vectors of the present invention can be introduced into host cells to thereby produce proteins or peptides, including fusion proteins or peptides, encoded by nucleic acids as described herein.

The recombinant expression vectors for use in the present invention can be designed for expression of a polypeptide corresponding to a marker of the present invention in prokaryotic (e.g., *E. coli*) or eukaryotic cells (e.g., insect cells {using baculovirus expression vectors}, yeast cells or mammalian cells). Suitable host cells are discussed further in Goeddel, supra. Alternatively, the recombinant expression vector can be transcribed and translated in vitro, for example using T7 promoter regulatory sequences and T7 polymerase.

Expression of proteins in prokaryotes is most often carried out in *E. coli* with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion proteins. Fusion vectors add a number of amino acids to a protein encoded therein, usually to the amino terminus of the recombinant protein. Such fusion vectors typically serve three purposes: 1) to increase expression of recombinant protein; 2) to increase the solubility of the recombinant protein; and 3) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification. Often, in fusion expression vectors, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such enzymes, and their cognate recognition sequences, include Factor Xa, thrombin and enterokinase. Typical fusion expression vectors include pGEX (Pharmacia Biotech Inc; Smith and Johnson, 1988, *Gene* 67:31-40), pMAL (New England Biolabs, Beverly, MA) and pRIT5 (Pharmacia, Piscataway, NJ) which fuse glutathione S-transferase (GST), maltose E binding protein, or protein A, respectively, to the target recombinant protein.

Examples of suitable inducible non-fusion *E. coli* expression vectors include pTrc (Amann et al. (1988) *Gene* 69:301-315) and pET 11d (Studier et al., p. 60-89, In *Gene Expression Technology: Methods in Enzymology* vol. 185, Academic Press, San Diego, CA, 1991). Target biomarker nucleic acid expression from the pTrc vector relies on host RNA polymerase transcription from a hybrid trp-lac fusion promoter. Target biomarker nucleic acid expression from the pET 11d vector relies on transcription from a T7 gn10-lac fusion promoter mediated by a co-expressed viral RNA polymerase (T7 gn1). This viral polymerase is supplied by host strains BL21 (DE3) or HMS174(DE3) from a resident prophage harboring a T7 gn1 gene under the transcriptional control of the lacUV 5 promoter.

One strategy to maximize recombinant protein expression in *E. coli* is to express the protein in a host bacterium with an impaired capacity to proteolytically cleave the recombinant protein (Gottesman, p. 119-128, In *Gene Expression Technology: Methods in Enzymology* vol. 185, Academic Press, San Diego, CA, 1990. Another strategy is to alter the nucleic acid sequence of the nucleic acid to be inserted into an expression vector so that the individual codons for each amino acid are those preferentially utilized in *E. coli* (Wada et al., (1992) *Nucleic Acids Res.* 20:2111-2118). Such altera-

tion of nucleic acid sequences of the present invention can be carried out by standard DNA synthesis techniques.

In another embodiment, the expression vector is a yeast expression vector. Examples of vectors for expression in yeast *S. cerevisiae* include pYepSec1 (Baldari et al. (1987) *EMBO J.* 6:229-234), pMFa (Kurjan and Herskowitz (1982) *Cell* 30:933-943), pJRY88 (Schultz et al. (1987) *Gene* 54:113-123), pYES2 (Invitrogen Corporation, San Diego, CA), and pPicZ (Invitrogen Corp, San Diego, CA).

Alternatively, the expression vector is a baculovirus expression vector. Baculovirus vectors available for expression of proteins in cultured insect cells (e.g., Sf 9 cells) include the pAc series (Smith et al. (1983) *Mol. Cell Biol.* 3:2156-2165) and the pVL series (Lucklow and Summers (1989) *Virology* 170:31-39).

In yet another embodiment, a nucleic acid of the present invention is expressed in mammalian cells using a mammalian expression vector. Examples of mammalian expression vectors include pCDM8 (Seed (1987) *Nature* 329:840) and pMT2PC (Kaufman et al. (1987) *EMBO J.* 6:187-195). When used in mammalian cells, the expression vector's control functions are often provided by viral regulatory elements. For example, commonly used promoters are derived from polyoma, Adenovirus 2, cytomegalovirus and Simian Virus 40. For other suitable expression systems for both prokaryotic and eukaryotic cells see chapters 16 and 17 of Sambrook et al., supra.

In another embodiment, the recombinant mammalian expression vector is capable of directing expression of the nucleic acid preferentially in a particular cell type (e.g., tissue-specific regulatory elements are used to express the nucleic acid). Tissue-specific regulatory elements are known in the art. Non-limiting examples of suitable tissue-specific promoters include the albumin promoter (liver-specific; Pinkert et al. (1987) *Genes Dev.* 1:268-277), lymphoid-specific promoters (Calame and Eaton (1988) *Adv. Immunol.* 43:235-275), in particular promoters of T cell receptors (Winoto and Baltimore (1989) *EMBO J.* 8:729-733) and immunoglobulins (Banerji et al. (1983) *Cell* 33:729-740; Queen and Baltimore (1983) *Cell* 33:741-748), neuron-specific promoters (e.g., the neurofilament promoter; Byrne and Ruddle (1989) *Proc. Natl. Acad. Sci. USA* 86:5473-5477), pancreas-specific promoters (Edlund et al. (1985) *Science* 230:912-916), and mammary gland-specific promoters (e.g., milk whey promoter; U.S. Pat. No. 4,873,316 and European Application Publication No. 264,166). Developmentally-regulated promoters are also encompassed, for example the murine hox promoters (Kessel and Gruss (1990) *Science* 249:374-379) and the α -fetoprotein promoter (Camper and Tilghman (1989) *Genes Dev.* 3:537-546).

The present invention further provides a recombinant expression vector comprising a DNA molecule cloned into the expression vector in an antisense orientation. That is, the DNA molecule is operably linked to a regulatory sequence in a manner which allows for expression (by transcription of the DNA molecule) of an RNA molecule which is antisense to the mRNA encoding a polypeptide of the present invention. Regulatory sequences operably linked to a nucleic acid cloned in the antisense orientation can be chosen which direct the continuous expression of the antisense RNA molecule in a variety of cell types, for instance viral promoters and/or enhancers, or regulatory sequences can be chosen which direct constitutive, tissue-specific or cell type specific expression of antisense RNA. The antisense expression vector can be in the form of a recombinant plasmid, phagemid, or attenuated virus in which antisense nucleic

acids are produced under the control of a high efficiency regulatory region, the activity of which can be determined by the cell type into which the vector is introduced. For a discussion of the regulation of gene expression using antisense genes (see Weintraub et al. (1986) *Trends in Genetics*, Vol. 1(1)).

Another aspect of the present invention pertains to host cells into which a recombinant expression vector of the present invention has been introduced. The terms "host cell" and "recombinant host cell" are used interchangeably herein. It is understood that such terms refer not only to the particular subject cell but to the progeny or potential progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term as used herein.

A host cell can be any prokaryotic (e.g., *E. coli*) or eukaryotic cell (e.g., insect cells, yeast or mammalian cells).

Vector DNA can be introduced into prokaryotic or eukaryotic cells via conventional transformation or transfection techniques. As used herein, the terms "transformation" and "transfection" are intended to refer to a variety of art-recognized techniques for introducing foreign nucleic acid into a host cell, including calcium phosphate or calcium chloride co-precipitation, DEAE-dextran-mediated transfection, lipofection, or electroporation. Suitable methods for transforming or transfecting host cells can be found in Sambrook, et al. (supra), and other laboratory manuals.

For stable transfection of mammalian cells, it is known that, depending upon the expression vector and transfection technique used, only a small fraction of cells may integrate the foreign DNA into their genome. In order to identify and select these integrants, a gene that encodes a selectable marker (e.g., for resistance to antibiotics) is generally introduced into the host cells along with the gene of interest. Preferred selectable markers include those which confer resistance to drugs, such as G418, hygromycin and methotrexate. Cells stably transfected with the introduced nucleic acid can be identified by drug selection (e.g., cells that have incorporated the selectable marker gene will survive, while the other cells die).

V. Analyzing Biomarker Nucleic Acids and Polypeptides

Biomarker nucleic acids and/or biomarker polypeptides can be analyzed according to the methods described herein and techniques known to the skilled artisan to identify such genetic or expression alterations useful for the present invention including, but not limited to, 1) an alteration in the level of a biomarker transcript or polypeptide, 2) a deletion or addition of one or more nucleotides from a biomarker gene, 4) a substitution of one or more nucleotides of a biomarker gene, 5) aberrant modification of a biomarker gene, such as an expression regulatory region, and the like. a. Methods for Detection of Copy Number

Methods of evaluating the copy number of a biomarker nucleic acid are well known to those of skill in the art. The presence or absence of chromosomal gain or loss can be evaluated simply by a determination of copy number of the regions or markers identified herein.

In one embodiment, a biological sample is tested for the presence of copy number changes in genomic loci containing the genomic marker. A copy number of at least 3, 4, 5, 6, 7, 8, 9, or 10 is predictive of poorer outcome of anti-immune checkpoint treatment.

Methods of evaluating the copy number of a biomarker locus include, but are not limited to, hybridization-based assays. Hybridization-based assays include, but are not limited to, traditional "direct probe" methods, such as Southern blots, in situ hybridization (e.g., FISH and FISH plus SKY) methods, and "comparative probe" methods, such as comparative genomic hybridization (CGH), e.g., cDNA-based or oligonucleotide-based CGH. The methods can be used in a wide variety of formats including, but not limited to, substrate (e.g. membrane or glass) bound methods or array-based approaches.

In one embodiment, evaluating the biomarker gene copy number in a sample involves a Southern Blot. In a Southern Blot, the genomic DNA (typically fragmented and separated on an electrophoretic gel) is hybridized to a probe specific for the target region. Comparison of the intensity of the hybridization signal from the probe for the target region with control probe signal from analysis of normal genomic DNA (e.g., a non-amplified portion of the same or related cell, tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid. Alternatively, a Northern blot may be utilized for evaluating the copy number of encoding nucleic acid in a sample. In a Northern blot, mRNA is hybridized to a probe specific for the target region. Comparison of the intensity of the hybridization signal from the probe for the target region with control probe signal from analysis of normal RNA (e.g., a non-amplified portion of the same or related cell, tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid. Alternatively, other methods well known in the art to detect RNA can be used, such that higher or lower expression relative to an appropriate control (e.g., a non-amplified portion of the same or related cell tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid.

An alternative means for determining genomic copy number is in situ hybridization (e.g., Angerer (1987) *Meth. Enzymol* 152: 649). Generally, in situ hybridization comprises the following steps: (1) fixation of tissue or biological structure to be analyzed; (2) prehybridization treatment of the biological structure to increase accessibility of target DNA, and to reduce nonspecific binding; (3) hybridization of the mixture of nucleic acids to the nucleic acid in the biological structure or tissue; (4) post-hybridization washes to remove nucleic acid fragments not bound in the hybridization and (5) detection of the hybridized nucleic acid fragments. The reagent used in each of these steps and the conditions for use vary depending on the particular application. In a typical in situ hybridization assay, cells are fixed to a solid support, typically a glass slide. If a nucleic acid is to be probed, the cells are typically denatured with heat or alkali. The cells are then contacted with a hybridization solution at a moderate temperature to permit annealing of labeled probes specific to the nucleic acid sequence encoding the protein. The targets (e.g., cells) are then typically washed at a predetermined stringency or at an increasing stringency until an appropriate signal to noise ratio is obtained. The probes are typically labeled, e.g., with radioisotopes or fluorescent reporters. In one embodiment, probes are sufficiently long so as to specifically hybridize with the target nucleic acid(s) under stringent conditions. Probes generally range in length from about 200 bases to about 1000 bases. In some applications it is necessary to block the hybridization capacity of repetitive sequences. Thus, in some embodiments, tRNA, human genomic DNA, or Cot-I DNA is used to block non-specific hybridization.

An alternative means for determining genomic copy number is comparative genomic hybridization. In general, genomic DNA is isolated from normal reference cells, as well as from test cells (e.g., tumor cells) and amplified, if necessary. The two nucleic acids are differentially labeled and then hybridized in situ to metaphase chromosomes of a reference cell. The repetitive sequences in both the reference and test DNAs are either removed or their hybridization capacity is reduced by some means, for example by prehybridization with appropriate blocking nucleic acids and/or including such blocking nucleic acid sequences for said repetitive sequences during said hybridization. The bound, labeled DNA sequences are then rendered in a visualizable form, if necessary. Chromosomal regions in the test cells which are at increased or decreased copy number can be identified by detecting regions where the ratio of signal from the two DNAs is altered. For example, those regions that have decreased in copy number in the test cells will show relatively lower signal from the test DNA than the reference compared to other regions of the genome. Regions that have been increased in copy number in the test cells will show relatively higher signal from the test DNA. Where there are chromosomal deletions or multiplications, differences in the ratio of the signals from the two labels will be detected and the ratio will provide a measure of the copy number. In another embodiment of CGH, array CGH (aCGH), the immobilized chromosome element is replaced with a collection of solid support bound target nucleic acids on an array, allowing for a large or complete percentage of the genome to be represented in the collection of solid support bound targets. Target nucleic acids may comprise cDNAs, genomic DNAs, oligonucleotides (e.g., to detect single nucleotide polymorphisms) and the like. Array-based CGH may also be performed with single-color labeling (as opposed to labeling the control and the possible tumor sample with two different dyes and mixing them prior to hybridization, which will yield a ratio due to competitive hybridization of probes on the arrays). In single color CGH, the control is labeled and hybridized to one array and absolute signals are read, and the possible tumor sample is labeled and hybridized to a second array (with identical content) and absolute signals are read. Copy number difference is calculated based on absolute signals from the two arrays. Methods of preparing immobilized chromosomes or arrays and performing comparative genomic hybridization are well known in the art (see, e.g., U.S. Pat. Nos. 6,335,167; 6,197,501; 5,830,645; and 5,665,549 and Albertson (1984) *EMBO J.* 3: 1227-1234; Pinkel (1988) *Proc. Natl. Acad. Sci. USA* 85: 9138-9142; EPO Pub. No. 430,402; *Methods in Molecular Biology*, Vol. 33: In situ Hybridization Protocols, Choo, ed., Humana Press, Totowa, N.J. (1994), etc.). In another embodiment, the hybridization protocol of Pinkel, et al. (1998) *Nature Genetics* 20: 207-211, or of Kallioniemi (1992) *Proc. Natl Acad Sci USA* 89:5321-5325 (1992) is used.

In still another embodiment, amplification-based assays can be used to measure copy number. In such amplification-based assays, the nucleic acid sequences act as a template in an amplification reaction (e.g., Polymerase Chain Reaction (PCR)). In a quantitative amplification, the amount of amplification product will be proportional to the amount of template in the original sample. Comparison to appropriate controls, e.g. healthy tissue, provides a measure of the copy number.

Methods of "quantitative" amplification are well known to those of skill in the art. For example, quantitative PCR involves simultaneously co-amplifying a known quantity of

a control sequence using the same primers. This provides an internal standard that may be used to calibrate the PCR reaction. Detailed protocols for quantitative PCR are provided in Innis, et al. (1990) *PCR Protocols, A Guide to Methods and Applications*, Academic Press, Inc. N.Y.). Measurement of DNA copy number at microsatellite loci using quantitative PCR analysis is described in Ginzinger, et al. (2000) *Cancer Research* 60:5405-5409. The known nucleic acid sequence for the genes is sufficient to enable one of skill in the art to routinely select primers to amplify any portion of the gene. Fluorogenic quantitative PCR may also be used in the methods of the present invention. In fluorogenic quantitative PCR, quantitation is based on amount of fluorescence signals, e.g., TaqMan and SYBR green.

Other suitable amplification methods include, but are not limited to, ligase chain reaction (LCR) (see Wu and Wallace (1989) *Genomics* 4: 560, Landegren, et al. (1988) *Science* 241:1077, and Barringer et al. (1990) *Gene* 89: 117), transcription amplification (Kwoh, et al. (1989) *Proc. Natl. Acad. Sci. USA* 86: 1173), self-sustained sequence replication (Guatelli, et al. (1990) *Proc. Nat. Acad. Sci. USA* 87: 1874), dot PCR, and linker adapter PCR, etc.

Loss of heterozygosity (LOH) and major copy proportion (MCP) mapping (Wang, Z. C., et al. (2004) *Cancer Res* 64(1):64-71; Seymour, A. B., et al. (1994) *Cancer Res* 54, 2761-4; Hahn, S. A., et al. (1995) *Cancer Res* 55, 4670-5; Kimura, M., et al. (1996) *Genes Chromosomes Cancer* 17, 88-93; Li et al., (2008) *MBC Bioinform.* 9, 204-219) may also be used to identify regions of amplification or deletion.

b. Methods for Detection of Biomarker Nucleic Acid Expression

Biomarker expression may be assessed by any of a wide variety of well known methods for detecting expression of a transcribed molecule or protein. Non-limiting examples of such methods include immunological methods for detection of secreted, cell-surface, cytoplasmic, or nuclear proteins, protein purification methods, protein function or activity assays, nucleic acid hybridization methods, nucleic acid reverse transcription methods, and nucleic acid amplification methods.

In preferred embodiments, activity of a particular gene is characterized by a measure of gene transcript (e.g. mRNA), by a measure of the quantity of translated protein, or by a measure of gene product activity. Marker expression can be monitored in a variety of ways, including by detecting mRNA levels, protein levels, or protein activity, any of which can be measured using standard techniques. Detection can involve quantification of the level of gene expression (e.g., genomic DNA, cDNA, mRNA, protein, or enzyme activity), or, alternatively, can be a qualitative assessment of the level of gene expression, in particular in comparison with a control level. The type of level being detected will be clear from the context.

In another embodiment, detecting or determining expression levels of a biomarker and functionally similar homologs thereof, including a fragment or genetic alteration thereof (e.g., in regulatory or promoter regions thereof) comprises detecting or determining RNA levels for the marker of interest. In one embodiment, one or more cells from the subject to be tested are obtained and RNA is isolated from the cells. In a preferred embodiment, a sample of breast tissue cells is obtained from the subject.

In one embodiment, RNA is obtained from a single cell. For example, a cell can be isolated from a tissue sample by laser capture microdissection (LCM). Using this technique, a cell can be isolated from a tissue section, including a

stained tissue section, thereby assuring that the desired cell is isolated (see, e.g., Bonner et al. (1997) *Science* 278:1481; Emmert-Buck et al. (1996) *Science* 274:998; Fend et al. (1999) *Am. J. Path.* 154:61 and Murakami et al. (2000) *Kidney Int.* 58:1346). For example, Murakami et al., supra, describe isolation of a cell from a previously immunostained tissue section.

It is also possible to obtain cells from a subject and culture the cells in vitro, such as to obtain a larger population of cells from which RNA can be extracted. Methods for establishing cultures of non-transformed cells, i.e., primary cell cultures, are known in the art.

When isolating RNA from tissue samples or cells from individuals, it may be important to prevent any further changes in gene expression after the tissue or cells has been removed from the subject. Changes in expression levels are known to change rapidly following perturbations, e.g., heat shock or activation with lipopolysaccharide (LPS) or other reagents. In addition, the RNA in the tissue and cells may quickly become degraded. Accordingly, in a preferred embodiment, the tissue or cells obtained from a subject is snap frozen as soon as possible.

RNA can be extracted from the tissue sample by a variety of methods, e.g., the guanidium thiocyanate lysis followed by CsCl centrifugation (Chirgwin et al. (1979) *Biochemistry* 18:5294-5299). RNA from single cells can be obtained as described in methods for preparing cDNA libraries from single cells, such as those described in Dulac, C. (1998) *Curr. Top. Dev. Biol.* 36:245 and Jena et al. (1996) *J. Immunol. Methods* 190:199. Care to avoid RNA degradation must be taken, e.g., by inclusion of RNasin.

The RNA sample can then be enriched in particular species. In one embodiment, poly(A)+RNA is isolated from the RNA sample. In general, such purification takes advantage of the poly-A tails on mRNA. In particular and as noted above, poly-T oligonucleotides may be immobilized within on a solid support to serve as affinity ligands for mRNA. Kits for this purpose are commercially available, e.g., the MessageMaker kit (Life Technologies, Grand Island, NY).

In a preferred embodiment, the RNA population is enriched in marker sequences. Enrichment can be undertaken, e.g., by primer-specific cDNA synthesis, or multiple rounds of linear amplification based on cDNA synthesis and template-directed in vitro transcription (see, e.g., Wang et al. (1989) *PNAS* 86, 9717; Dulac et al., supra, and Jena et al., supra).

The population of RNA, enriched or not in particular species or sequences, can further be amplified. As defined herein, an "amplification process" is designed to strengthen, increase, or augment a molecule within the RNA. For example, where RNA is mRNA, an amplification process such as RT-PCR can be utilized to amplify the mRNA, such that a signal is detectable or detection is enhanced. Such an amplification process is beneficial particularly when the biological, tissue, or tumor sample is of a small size or volume.

Various amplification and detection methods can be used. For example, it is within the scope of the present invention to reverse transcribe mRNA into cDNA followed by polymerase chain reaction (RT-PCR); or, to use a single enzyme for both steps as described in U.S. Pat. No. 5,322,770, or reverse transcribe mRNA into cDNA followed by symmetric gap ligase chain reaction (RT-AGLCR) as described by R. L. Marshall, et al., *PCR Methods and Applications* 4: 80-84 (1994). Real time PCR may also be used.

Other known amplification methods which can be utilized herein include but are not limited to the so-called "NASBA"

or "3SR" technique described in *PNAS USA* 87: 1874-1878 (1990) and also described in *Nature* 350 (No. 6313): 91-92 (1991); Q-beta amplification as described in published European Patent Application (EPA) No. 4544610; strand displacement amplification (as described in G. T. Walker et al., *Clin. Chem.* 42: 9-13 (1996) and European Patent Application No. 684315; target mediated amplification, as described by PCT Publication WO9322461; PCR; ligase chain reaction (LCR) (see, e.g., Wu and Wallace, *Genomics* 4, 560 (1989), Landegren et al., *Science* 241, 1077 (1988)); self-sustained sequence replication (SSR) (see, e.g., Guatelli et al., *Proc. Nat. Acad. Sci. USA*, 87, 1874 (1990)); and transcription amplification (see, e.g., Kwoh et al., *Proc. Natl. Acad. Sci. USA* 86, 1173 (1989)).

Many techniques are known in the state of the art for determining absolute and relative levels of gene expression, commonly used techniques suitable for use in the present invention include Northern analysis, RNase protection assays (RPA), microarrays and PCR-based techniques, such as quantitative PCR and differential display PCR. For example, Northern blotting involves running a preparation of RNA on a denaturing agarose gel, and transferring it to a suitable support, such as activated cellulose, nitrocellulose or glass or nylon membranes. Radiolabeled cDNA or RNA is then hybridized to the preparation, washed and analyzed by autoradiography.

In situ hybridization visualization may also be employed, wherein a radioactively labeled antisense RNA probe is hybridized with a thin section of a biopsy sample, washed, cleaved with RNase and exposed to a sensitive emulsion for autoradiography. The samples may be stained with hematoxylin to demonstrate the histological composition of the sample, and dark field imaging with a suitable light filter shows the developed emulsion. Non-radioactive labels such as digoxigenin may also be used.

Alternatively, mRNA expression can be detected on a DNA array, chip or a microarray. Labeled nucleic acids of a test sample obtained from a subject may be hybridized to a solid surface comprising biomarker DNA. Positive hybridization signal is obtained with the sample containing biomarker transcripts. Methods of preparing DNA arrays and their use are well known in the art (see, e.g., U.S. Pat. Nos. 6,618,679; 6,379,897; 6,664,377; 6,451,536; 548,257; U.S. 20030157485 and Schena et al. (1995) *Science* 20, 467-470; Gerhold et al. (1999) *Trends In Biochem. Sci.* 24, 168-173; and Lennon et al. (2000) *Drug Discovery Today* 5, 59-65, which are herein incorporated by reference in their entirety). Serial Analysis of Gene Expression (SAGE) can also be performed (See for example U.S. Patent Application 20030215858).

To monitor mRNA levels, for example, mRNA is extracted from the biological sample to be tested, reverse transcribed, and fluorescently-labeled cDNA probes are generated. The microarrays capable of hybridizing to marker cDNA are then probed with the labeled cDNA probes, the slides scanned and fluorescence intensity measured. This intensity correlates with the hybridization intensity and expression levels.

Types of probes that can be used in the methods described herein include cDNA, riboprobes, synthetic oligonucleotides and genomic probes. The type of probe used will generally be dictated by the particular situation, such as riboprobes for in situ hybridization, and cDNA for Northern blotting, for example. In one embodiment, the probe is directed to nucleotide regions unique to the RNA. The probes may be as short as is required to differentially recognize marker mRNA transcripts, and may be as short as,

for example, 15 bases; however, probes of at least 17, 18, 19 or 20 or more bases can be used. In one embodiment, the primers and probes hybridize specifically under stringent conditions to a DNA fragment having the nucleotide sequence corresponding to the marker. As herein used, the term "stringent conditions" means hybridization will occur only if there is at least 95% identity in nucleotide sequences. In another embodiment, hybridization under "stringent conditions" occurs when there is at least 97% identity between the sequences.

The form of labeling of the probes may be any that is appropriate, such as the use of radioisotopes, for example, ³²P and ¹⁵S. Labeling with radioisotopes may be achieved, whether the probe is synthesized chemically or biologically, by the use of suitably labeled bases.

In one embodiment, the biological sample contains polypeptide molecules from the test subject. Alternatively, the biological sample can contain mRNA molecules from the test subject or genomic DNA molecules from the test subject.

In another embodiment, the methods further involve obtaining a control biological sample from a control subject, contacting the control sample with a compound or agent capable of detecting marker polypeptide, mRNA, genomic DNA, or fragments thereof, such that the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof, is detected in the biological sample, and comparing the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof, in the control sample with the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof in the test sample.

c. Methods for Detection of Biomarker Protein Expression

The activity or level of a biomarker protein can be detected and/or quantified by detecting or quantifying the expressed polypeptide. The polypeptide can be detected and quantified by any of a number of means well known to those of skill in the art. Aberrant levels of polypeptide expression of the polypeptides encoded by a biomarker nucleic acid and functionally similar homologs thereof, including a fragment or genetic alteration thereof (e.g., in regulatory or promoter regions thereof) are associated with the likelihood of response of a cancer to an immune checkpoint therapy. Any method known in the art for detecting polypeptides can be used. Such methods include, but are not limited to, immunodiffusion, immunoelectrophoresis, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, Western blotting, binder-ligand assays, immunohistochemical techniques, agglutination, complement assays, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, and the like (e.g., *Basic and Clinical Immunology*, Sites and Terr, eds., Appleton and Lange, Norwalk, Conn. pp 217-262, 1991 which is incorporated by reference). Preferred are binder-ligand immunoassay methods including reacting antibodies with an epitope or epitopes and competitively displacing a labeled polypeptide or derivative thereof.

For example, ELISA and RIA procedures may be conducted such that a desired biomarker protein standard is labeled (with a radioisotope such as ¹²⁵I or ³⁵S, or an assayable enzyme, such as horseradish peroxidase or alkaline phosphatase), and, together with the unlabelled sample, brought into contact with the corresponding antibody, whereon a second antibody is used to bind the first, and radioactivity or the immobilized enzyme assayed (competitive assay). Alternatively, the biomarker protein in the sample is allowed to react with the corresponding immobi-

lized antibody, radioisotope- or enzyme-labeled anti-biomarker protein antibody is allowed to react with the system, and radioactivity or the enzyme assayed (ELISA-sandwich assay). Other conventional methods may also be employed as suitable.

The above techniques may be conducted essentially as a “one-step” or “two-step” assay. A “one-step” assay involves contacting antigen with immobilized antibody and, without washing, contacting the mixture with labeled antibody. A “two-step” assay involves washing before contacting, the mixture with labeled antibody. Other conventional methods may also be employed as suitable.

In one embodiment, a method for measuring biomarker protein levels comprises the steps of: contacting a biological specimen with an antibody or variant (e.g., fragment) thereof which selectively binds the biomarker protein, and detecting whether said antibody or variant thereof is bound to said sample and thereby measuring the levels of the biomarker protein.

Enzymatic and radiolabeling of biomarker protein and/or the antibodies may be effected by conventional means. Such means will generally include covalent linking of the enzyme to the antigen or the antibody in question, such as by glutaraldehyde, specifically so as not to adversely affect the activity of the enzyme, by which is meant that the enzyme must still be capable of interacting with its substrate, although it is not necessary for all of the enzyme to be active, provided that enough remains active to permit the assay to be effected. Indeed, some techniques for binding enzyme are non-specific (such as using formaldehyde), and will only yield a proportion of active enzyme.

It is usually desirable to immobilize one component of the assay system on a support, thereby allowing other components of the system to be brought into contact with the component and readily removed without laborious and time-consuming labor. It is possible for a second phase to be immobilized away from the first, but one phase is usually sufficient.

It is possible to immobilize the enzyme itself on a support, but if solid-phase enzyme is required, then this is generally best achieved by binding to antibody and affixing the antibody to a support, models and systems for which are well-known in the art. Simple polyethylene may provide a suitable support.

Enzymes employable for labeling are not particularly limited, but may be selected from the members of the oxidase group, for example. These catalyze production of hydrogen peroxide by reaction with their substrates, and glucose oxidase is often used for its good stability, ease of availability and cheapness, as well as the ready availability of its substrate (glucose). Activity of the oxidase may be assayed by measuring the concentration of hydrogen peroxide formed after reaction of the enzyme-labeled antibody with the substrate under controlled conditions well-known in the art.

Other techniques may be used to detect biomarker protein according to a practitioner's preference based upon the present disclosure. One such technique is Western blotting (Towbin et al., *Proc. Nat. Acad. Sci.* 76:4350 (1979)), wherein a suitably treated sample is run on an SDS-PAGE gel before being transferred to a solid support, such as a nitrocellulose filter. Anti-biomarker protein antibodies (unlabeled) are then brought into contact with the support and assayed by a secondary immunological reagent, such as labeled protein A or anti-immunoglobulin (suitable labels including ^{125}I , horseradish peroxidase and alkaline phosphatase). Chromatographic detection may also be used.

Immunohistochemistry may be used to detect expression of biomarker protein, e.g., in a biopsy sample. A suitable antibody is brought into contact with, for example, a thin layer of cells, washed, and then contacted with a second, labeled antibody. Labeling may be by fluorescent markers, enzymes, such as peroxidase, avidin, or radiolabelling. The assay is scored visually, using microscopy.

Anti-biomarker protein antibodies, such as intrabodies, may also be used for imaging purposes, for example, to detect the presence of biomarker protein in cells and tissues of a subject. Suitable labels include radioisotopes, iodine (^{125}I , ^{121}I), carbon (^{14}C), sulphur (^{35}S), tritium (^3H), indium (^{112}In), and technetium ($^{99\text{m}}\text{Tc}$), fluorescent labels, such as fluorescein and rhodamine, and biotin.

For in vivo imaging purposes, antibodies are not detectable, as such, from outside the body, and so must be labeled, or otherwise modified, to permit detection. Markers for this purpose may be any that do not substantially interfere with the antibody binding, but which allow external detection. Suitable markers may include those that may be detected by X-radiography, NMR or MRI. For X-radiographic techniques, suitable markers include any radioisotope that emits detectable radiation but that is not overtly harmful to the subject, such as barium or cesium, for example. Suitable markers for NMR and MRI generally include those with a detectable characteristic spin, such as deuterium, which may be incorporated into the antibody by suitable labeling of nutrients for the relevant hybridoma, for example.

The size of the subject, and the imaging system used, will determine the quantity of imaging moiety needed to produce diagnostic images. In the case of a radioisotope moiety, for a human subject, the quantity of radioactivity injected will normally range from about 5 to 20 millicuries of technetium-99. The labeled antibody or antibody fragment will then preferentially accumulate at the location of cells which contain biomarker protein. The labeled antibody or antibody fragment can then be detected using known techniques.

Antibodies that may be used to detect biomarker protein include any antibody, whether natural or synthetic, full length or a fragment thereof, monoclonal or polyclonal, that binds sufficiently strongly and specifically to the biomarker protein to be detected. An antibody may have a K_d of at most about 10^{-6}M , 10^{-7}M , 10^{-8}M , 10^{-9}M , 10^{-10}M , 10^{-11}M , 10^{-12}M . The phrase “specifically binds” refers to binding of, for example, an antibody to an epitope or antigen or antigenic determinant in such a manner that binding can be displaced or competed with a second preparation of identical or similar epitope, antigen or antigenic determinant. An antibody may bind preferentially to the biomarker protein relative to other proteins, such as related proteins.

Antibodies are commercially available or may be prepared according to methods known in the art.

Antibodies and derivatives thereof that may be used encompass polyclonal or monoclonal antibodies, chimeric, human, humanized, primatized (CDR-grafted), veneered or single-chain antibodies as well as functional fragments, i.e., biomarker protein binding fragments, of antibodies. For example, antibody fragments capable of binding to a biomarker protein or portions thereof, including, but not limited to, Fv, Fab, Fab' and F(ab')₂ fragments can be used. Such fragments can be produced by enzymatic cleavage or by recombinant techniques. For example, papain or pepsin cleavage can generate Fab or F(ab')₂ fragments, respectively. Other proteases with the requisite substrate specificity can also be used to generate Fab or F(ab')₂ fragments. Antibodies can also be produced in a variety of truncated forms using antibody genes in which one or more stop

codons have been introduced upstream of the natural stop site. For example, a chimeric gene encoding a F(ab')₂ heavy chain portion can be designed to include DNA sequences encoding the CH₁ domain and hinge region of the heavy chain.

Synthetic and engineered antibodies are described in, e.g., Cabilly et al., U.S. Pat. No. 4,816,567; Cabilly et al., European Patent No. 0,125,023 B1; Boss et al., U.S. Pat. No. 4,816,397; Boss et al., European Patent No. 0,120,694 B1; Neuberger, M. S. et al., WO 86/01533; Neuberger, M. S. et al., European Patent No. 0,194,276 B1; Winter, U.S. Pat. No. 5,225,539; Winter, European Patent No. 0,239,400 B1; Queen et al., European Patent No. 0,451,216 B1; and Padlan, E. A. et al., EP 0519596 A1. See also, Newman, R. et al., *BioTechnology*, 10: 1455-1460 (1992), regarding primatized antibody, and Ladner et al., U.S. Pat. No. 4,946,778 and Bird, R. E. et al., *Science*, 242: 423-426 (1988) regarding single-chain antibodies. Antibodies produced from a library, e.g., phage display library, may also be used.

In some embodiments, agents that specifically bind to a biomarker protein other than antibodies are used, such as peptides. Peptides that specifically bind to a biomarker protein can be identified by any means known in the art. For example, specific peptide binders of a biomarker protein can be screened for using peptide phage display libraries.

d. Methods for Detection of Biomarker Structural Alterations

The following illustrative methods can be used to identify the presence of a structural alteration in a biomarker nucleic acid and/or biomarker polypeptide molecule in order to, for example, identify PBRM1 (or ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, EGFR, and the like) proteins that having mutations such as described herein.

In certain embodiments, detection of the alteration involves the use of a probe/primer in a polymerase chain reaction (PCR) (see, e.g., U.S. Pat. Nos. 4,683,195 and 4,683,202), such as anchor PCR or RACE PCR, or, alternatively, in a ligation chain reaction (LCR) (see, e.g., Landegran et al. (1988) *Science* 241:1077-1080; and Nakazawa et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:360-364), the latter of which can be particularly useful for detecting point mutations in a biomarker nucleic acid such as a biomarker gene (see Abravaya et al. (1995) *Nucleic Acids Res.* 23:675-682). This method can include the steps of collecting a sample of cells from a subject, isolating nucleic acid (e.g., genomic, mRNA or both) from the cells of the sample, contacting the nucleic acid sample with one or more primers which specifically hybridize to a biomarker gene under conditions such that hybridization and amplification of the biomarker gene (if present) occurs, and detecting the presence or absence of an amplification product, or detecting the size of the amplification product and comparing the length to a control sample. It is anticipated that PCR and/or LCR may be desirable to use as a preliminary amplification step in conjunction with any of the techniques used for detecting mutations described herein.

Alternative amplification methods include: self sustained sequence replication (Guatelli, J. C. et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:1874-1878), transcriptional amplification system (Kwoh, D. Y. et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:1173-1177), Q-Beta Replicase (Lizardi, P. M. et al. (1988) *Bio-Technology* 6:1197), or any other nucleic acid amplification method, followed by the detection of the amplified molecules using techniques well known to those of skill in the art. These detection schemes are especially

useful for the detection of nucleic acid molecules if such molecules are present in very low numbers.

In an alternative embodiment, mutations in a biomarker nucleic acid from a sample cell can be identified by alterations in restriction enzyme cleavage patterns. For example, sample and control DNA is isolated, amplified (optionally), digested with one or more restriction endonucleases, and fragment length sizes are determined by gel electrophoresis and compared. Differences in fragment length sizes between sample and control DNA indicates mutations in the sample DNA. Moreover, the use of sequence specific ribozymes (see, for example, U.S. Pat. No. 5,498,531) can be used to score for the presence of specific mutations by development or loss of a ribozyme cleavage site.

In other embodiments, genetic mutations in biomarker nucleic acid can be identified by hybridizing a sample and control nucleic acids, e.g., DNA or RNA, to high density arrays containing hundreds or thousands of oligonucleotide probes (Cronin, M. T. et al. (1996) *Hum. Mutat.* 7:244-255; Kozal, M. J. et al. (1996) *Nat. Med.* 2:753-759). For example, biomarker genetic mutations can be identified in two dimensional arrays containing light-generated DNA probes as described in Cronin et al. (1996) *supra*. Briefly, a first hybridization array of probes can be used to scan through long stretches of DNA in a sample and control to identify base changes between the sequences by making linear arrays of sequential, overlapping probes. This step allows the identification of point mutations. This step is followed by a second hybridization array that allows the characterization of specific mutations by using smaller, specialized probe arrays complementary to all variants or mutations detected. Each mutation array is composed of parallel probe sets, one complementary to the wild-type gene and the other complementary to the mutant gene. Such biomarker genetic mutations can be identified in a variety of contexts, including, for example, germline and somatic mutations.

In yet another embodiment, any of a variety of sequencing reactions known in the art can be used to directly sequence a biomarker gene and detect mutations by comparing the sequence of the sample biomarker with the corresponding wild-type (control) sequence. Examples of sequencing reactions include those based on techniques developed by Maxam and Gilbert (1977) *Proc. Natl. Acad. Sci. USA* 74:560 or Sanger (1977) *Proc. Natl. Acad. Sci. USA* 74:5463. It is also contemplated that any of a variety of automated sequencing procedures can be utilized when performing the diagnostic assays (Naeve (1995) *Biotechniques* 19:448-53), including sequencing by mass spectrometry (see, e.g., PCT International Publication No. WO 94/16101; Cohen et al. (1996) *Adv. Chromatogr.* 36:127-162; and Griffin et al. (1993) *Appl. Biochem. Biotechnol.* 38:147-159).

Other methods for detecting mutations in a biomarker gene include methods in which protection from cleavage agents is used to detect mismatched bases in RNA/RNA or RNA/DNA heteroduplexes (Myers et al. (1985) *Science* 230:1242). In general, the art technique of "mismatch cleavage" starts by providing heteroduplexes formed by hybridizing (labeled) RNA or DNA containing the wild-type biomarker sequence with potentially mutant RNA or DNA obtained from a tissue sample. The double-stranded duplexes are treated with an agent which cleaves single-stranded regions of the duplex such as which will exist due to base pair mismatches between the control and sample strands. For instance, RNA/DNA duplexes can be treated with RNase and DNA/DNA hybrids treated with SI nuclease to enzymatically digest the mismatched regions. In other

embodiments, either DNA/DNA or RNA/DNA duplexes can be treated with hydroxylamine or osmium tetroxide and with piperidine in order to digest mismatched regions. After digestion of the mismatched regions, the resulting material is then separated by size on denaturing polyacrylamide gels to determine the site of mutation. See, for example, Cotton et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:4397 and Saleeba et al. (1992) *Methods Enzymol.* 217:286-295. In a preferred embodiment, the control DNA or RNA can be labeled for detection.

In still another embodiment, the mismatch cleavage reaction employs one or more proteins that recognize mismatched base pairs in double-stranded DNA (so called "DNA mismatch repair" enzymes) in defined systems for detecting and mapping point mutations in biomarker cDNAs obtained from samples of cells. For example, the mutY enzyme of *E. coli* cleaves A at G/A mismatches and the thymidine DNA glycosylase from HeLa cells cleaves T at G/T mismatches (Hsu et al. (1994) *Carcinogenesis* 15:1657-1662). According to an exemplary embodiment, a probe based on a biomarker sequence, e.g., a wild-type biomarker treated with a DNA mismatch repair enzyme, and the cleavage products, if any, can be detected from electrophoresis protocols or the like (e.g., U.S. Pat. No. 5,459,039.)

In other embodiments, alterations in electrophoretic mobility can be used to identify mutations in biomarker genes. For example, single strand conformation polymorphism (SSCP) may be used to detect differences in electrophoretic mobility between mutant and wild type nucleic acids (Orita et al. (1989) *Proc Natl. Acad. Sci USA* 86:2766; see also Cotton (1993) *Mutat. Res.* 285:125-144 and Hayashi (1992) *Genet. Anal. Tech. Appl.* 9:73-79). Single-stranded DNA fragments of sample and control biomarker nucleic acids will be denatured and allowed to renature. The secondary structure of single-stranded nucleic acids varies according to sequence, the resulting alteration in electrophoretic mobility enables the detection of even a single base change. The DNA fragments may be labeled or detected with labeled probes. The sensitivity of the assay may be enhanced by using RNA (rather than DNA), in which the secondary structure is more sensitive to a change in sequence. In a preferred embodiment, the subject method utilizes heteroduplex analysis to separate double stranded heteroduplex molecules on the basis of changes in electrophoretic mobility (Keen et al. (1991) *Trends Genet.* 7:5).

In yet another embodiment the movement of mutant or wild-type fragments in polyacrylamide gels containing a gradient of denaturant is assayed using denaturing gradient gel electrophoresis (DGGE) (Myers et al. (1985) *Nature* 313:495). When DGGE is used as the method of analysis, DNA will be modified to ensure that it does not completely denature, for example by adding a GC clamp of approximately 40 bp of high-melting GC-rich DNA by PCR. In a further embodiment, a temperature gradient is used in place of a denaturing gradient to identify differences in the mobility of control and sample DNA (Rosenbaum and Reissner (1987) *Biophys. Chem.* 265:12753).

Examples of other techniques for detecting point mutations include, but are not limited to, selective oligonucleotide hybridization, selective amplification, or selective primer extension. For example, oligonucleotide primers may be prepared in which the known mutation is placed centrally and then hybridized to target DNA under conditions which permit hybridization only if a perfect match is found (Saiki et al. (1986) *Nature* 324:163; Saiki et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:6230). Such allele specific oligonucleotides are hybridized to PCR amplified target DNA or a

number of different mutations when the oligonucleotides are attached to the hybridizing membrane and hybridized with labeled target DNA.

Alternatively, allele specific amplification technology which depends on selective PCR amplification may be used in conjunction with the instant invention. Oligonucleotides used as primers for specific amplification may carry the mutation of interest in the center of the molecule (so that amplification depends on differential hybridization) (Gibbs et al. (1989) *Nucleic Acids Res.* 17:2437-2448) or at the extreme 3' end of one primer where, under appropriate conditions, mismatch can prevent, or reduce polymerase extension (Prossner (1993) *Tibtech* 11:238). In addition it may be desirable to introduce a novel restriction site in the region of the mutation to create cleavage-based detection (Gasparini et al. (1992) *Mol. Cell Probes* 6:1). It is anticipated that in certain embodiments amplification may also be performed using Taq ligase for amplification (Barany (1991) *Proc. Natl. Acad. Sci USA* 88:189). In such cases, ligation will occur only if there is a perfect match at the 3' end of the 5' sequence making it possible to detect the presence of a known mutation at a specific site by looking for the presence or absence of amplification.

3. Anti-Cancer Therapies

The efficacy of immune checkpoint therapy is predicted according to biomarker amount and/or activity associated with a cancer in a subject according to the methods described herein. In one embodiment, such immune checkpoint therapy or combinations of therapies (e.g., anti-PD-1 antibodies) can be administered once a subject is indicated as being a likely responder to immune checkpoint therapy. In another embodiment, such immune checkpoint therapy can be avoided once a subject is indicated as not being a likely responder to immune checkpoint therapy and an alternative treatment regimen, such as targeted and/or untargeted anti-cancer therapies can be administered. Combination therapies are also contemplated and can comprise, for example, one or more chemotherapeutic agents and radiation, one or more chemotherapeutic agents and immunotherapy, or one or more chemotherapeutic agents, radiation and chemotherapy, each combination of which can be with immune checkpoint therapy.

The term "targeted therapy" refers to administration of agents that selectively interact with a chosen biomolecule to thereby treat cancer. For example, anti-PBRM1 agents (or anti-ARID2 agents, anti-BRD7 agents, anti-PHF10 agents, anti-KDM6A agents, etc.), such as therapeutic monoclonal blocking antibodies, which are well-known in the art and described above, can be used to target tumor microenvironments and cells expressing unwanted PBRM1 (or ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, EGFR, and the like). Similarly, nivolumab (Opdivo®) is a human IgG4 anti-PD-1 monoclonal antibody that blocks PD-1 activity (see, for example, Wang et al. (2014) *Cancer Immunol. Res.* 2:846-856; Johnson et al. (2015) *Ther. Adv. Med. Oncol.* 7:97-106; and Sundar et al. (2015) *Ther. Adv. Med. Oncol.* 7:85-96).

Immunotherapy is one form of targeted therapy that may comprise, for example, the use of cancer vaccines and/or sensitized antigen presenting cells. For example, an oncolytic virus is a virus that is able to infect and lyse cancer cells, while leaving normal cells unharmed, making them potentially useful in cancer therapy. Replication of oncolytic viruses both facilitates tumor cell destruction and also produces dose amplification at the tumor site. They may also act as vectors for anticancer genes, allowing them to be specifically delivered to the tumor site. The immunotherapy

can involve passive immunity for short-term protection of a host, achieved by the administration of pre-formed antibody directed against a cancer antigen or disease antigen (e.g., administration of a monoclonal antibody, optionally linked to a chemotherapeutic agent or toxin, to a tumor antigen). Immunotherapy can also focus on using the cytotoxic lymphocyte-recognized epitopes of cancer cell lines. Alternatively, antisense polynucleotides, ribozymes, RNA interference molecules, triple helix polynucleotides and the like, can be used to selectively modulate biomolecules that are linked to the initiation, progression, and/or pathology of a tumor or cancer.

The term "untargeted therapy" refers to administration of agents that do not selectively interact with a chosen biomolecule yet treat cancer. Representative examples of untargeted therapies include, without limitation, chemotherapy, gene therapy, and radiation therapy.

In one embodiment, chemotherapy is used. Chemotherapy includes the administration of a chemotherapeutic agent. Such a chemotherapeutic agent may be, but is not limited to, those selected from among the following groups of compounds: platinum compounds, cytotoxic antibiotics, antimetabolites, anti-mitotic agents, alkylating agents, arsenic compounds, DNA topoisomerase inhibitors, taxanes, nucleoside analogues, plant alkaloids, and toxins; and synthetic derivatives thereof. Exemplary compounds include, but are not limited to, alkylating agents: cisplatin, treosulfan, and trofosfamide; plant alkaloids: vinblastine, paclitaxel, docetaxol; DNA topoisomerase inhibitors: teniposide, crinatonol, and mitomycin; anti-folates: methotrexate, mycophenolic acid, and hydroxyurea; pyrimidine analogs: 5-fluorouracil, doxifluridine, and cytosine arabinoside; purine analogs: mercaptopurine and thioguanine; DNA antimetabolites: 2'-deoxy-5-fluorouridine, aphidicolin glycinolate, and pyrazoloimidazole; and antimitotic agents: halichondrin, colchicine, and rhizoxin. Compositions comprising one or more chemotherapeutic agents (e.g., FLAG, CHOP) may also be used. FLAG comprises fludarabine, cytosine arabinoside (Ara-C) and G-CSF. CHOP comprises cyclophosphamide, vincristine, doxorubicin, and prednisone. In another embodiment, PARP (e.g., PARP-1 and/or PARP-2) inhibitors are used and such inhibitors are well known in the art (e.g., Olaparib, ABT-888, BSI-201, BGP-15 (N-Gene Research Laboratories, Inc.); INO-1001 (Inotek Pharmaceuticals Inc.); PJ34 (Soriano et al., 2001; Pacher et al., 2002b); 3-aminobenzamide (Trevigen); 4-amino-1,8-naphthalimide; (Trevigen); 6(5H)-phenanthridinone (Trevigen); benzamide (U.S. Pat. Re. 36,397); and NU1025 (Bowman et al.). The mechanism of action is generally related to the ability of PARP inhibitors to bind PARP and decrease its activity. PARP catalyzes the conversion of .beta.-nicotinamide adenine dinucleotide (NAD+) into nicotinamide and poly-ADP-ribose (PAR). Both poly (ADP-ribose) and PARP have been linked to regulation of transcription, cell proliferation, genomic stability, and carcinogenesis (Bouchard V. J. et al. *Experimental Hematology*, Volume 31, Number 6, June 2003, pp. 446-454(9); Herceg Z.; Wang Z.-Q. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis*, Volume 477, Number 1, 2 Jun. 2001, pp. 97-110(14)). Poly(ADP-ribose) polymerase 1 (PARP1) is a key molecule in the repair of DNA single-strand breaks (SSBs) (de Murcia J. et al. 1997. *Proc Natl Acad Sci USA* 94:7303-7307; Schreiber V, Dantzer F, Ame J C, de Murcia G (2006) *Nat Rev Mol Cell Biol* 7:517-528; Wang Z Q, et al. (1997) *Genes Dev* 11:2347-2358). Knockout of SSB repair by inhibition of PARP1 function induces DNA double-strand breaks (DSBs) that can trigger synthetic

lethality in cancer cells with defective homology-directed DSB repair (Bryant H E, et al. (2005) *Nature* 434:913-917; Farmer H, et al. (2005) *Nature* 434:917-921). The foregoing examples of chemotherapeutic agents are illustrative, and are not intended to be limiting.

In another embodiment, radiation therapy is used. The radiation used in radiation therapy can be ionizing radiation. Radiation therapy can also be gamma rays, X-rays, or proton beams. Examples of radiation therapy include, but are not limited to, external-beam radiation therapy, interstitial implantation of radioisotopes (1-125, palladium, iridium), radioisotopes such as strontium-89, thoracic radiation therapy, intraperitoneal P-32 radiation therapy, and/or total abdominal and pelvic radiation therapy. For a general overview of radiation therapy, see Hellman, Chapter 16: Principles of Cancer Management: Radiation Therapy, 6th edition, 2001, DeVita et al., eds., J. B. Lippencott Company, Philadelphia. The radiation therapy can be administered as external beam radiation or teletherapy wherein the radiation is directed from a remote source. The radiation treatment can also be administered as internal therapy or brachytherapy wherein a radioactive source is placed inside the body close to cancer cells or a tumor mass. Also encompassed is the use of photodynamic therapy comprising the administration of photosensitizers, such as hematoporphyrin and its derivatives, Vertoporphin (BPD-MA), phthalocyanine, photosensitizer Pc4, demethoxy-hypocrellin A; and 2BA-2-DMHA.

In another embodiment, hormone therapy is used. Hormonal therapeutic treatments can comprise, for example, hormonal agonists, hormonal antagonists (e.g., flutamide, bicalutamide, tamoxifen, raloxifene, leuprolide acetate (LUPRON), LH-RH antagonists), inhibitors of hormone biosynthesis and processing, and steroids (e.g., dexamethasone, retinoids, deltoids, betamethasone, cortisol, cortisone, prednisone, dehydrotestosterone, glucocorticoids, mineralocorticoids, estrogen, testosterone, progestins), vitamin A derivatives (e.g., all-trans retinoic acid (ATRA)); vitamin D3 analogs; antigestagens (e.g., mifepristone, onapristone), or antiandrogens (e.g., cyproterone acetate).

In another embodiment, hyperthermia, a procedure in which body tissue is exposed to high temperatures (up to 106° F.) is used. Heat may help shrink tumors by damaging cells or depriving them of substances they need to live. Hyperthermia therapy can be local, regional, and whole-body hyperthermia, using external and internal heating devices. Hyperthermia is almost always used with other forms of therapy (e.g., radiation therapy, chemotherapy, and biological therapy) to try to increase their effectiveness. Local hyperthermia refers to heat that is applied to a very small area, such as a tumor. The area may be heated externally with high-frequency waves aimed at a tumor from a device outside the body. To achieve internal heating, one of several types of sterile probes may be used, including thin, heated wires or hollow tubes filled with warm water; implanted microwave antennae; and radiofrequency electrodes. In regional hyperthermia, an organ or a limb is heated. Magnets and devices that produce high energy are placed over the region to be heated. In another approach, called perfusion, some of the patient's blood is removed, heated, and then pumped (perfused) into the region that is to be heated internally. Whole-body heating is used to treat metastatic cancer that has spread throughout the body. It can be accomplished using warm-water blankets, hot wax, inductive coils (like those in electric blankets), or thermal chambers (similar to large incubators). Hyperthermia does not cause any marked increase in radiation side effects or complications. Heat applied directly to the skin, however,

can cause discomfort or even significant local pain in about half the patients treated. It can also cause blisters, which generally heal rapidly.

In still another embodiment, photodynamic therapy (also called PDT, photoradiation therapy, phototherapy, or photochemotherapy) is used for the treatment of some types of cancer. It is based on the discovery that certain chemicals known as photosensitizing agents can kill one-celled organisms when the organisms are exposed to a particular type of light. PDT destroys cancer cells through the use of a fixed-frequency laser light in combination with a photosensitizing agent. In PDT, the photosensitizing agent is injected into the bloodstream and absorbed by cells all over the body. The agent remains in cancer cells for a longer time than it does in normal cells. When the treated cancer cells are exposed to laser light, the photosensitizing agent absorbs the light and produces an active form of oxygen that destroys the treated cancer cells. Light exposure must be timed carefully so that it occurs when most of the photosensitizing agent has left healthy cells but is still present in the cancer cells. The laser light used in PDT can be directed through a fiber-optic (a very thin glass strand). The fiber-optic is placed close to the cancer to deliver the proper amount of light. The fiber-optic can be directed through a bronchoscope into the lungs for the treatment of lung cancer or through an endoscope into the esophagus for the treatment of esophageal cancer. An advantage of PDT is that it causes minimal damage to healthy tissue. However, because the laser light currently in use cannot pass through more than about 3 centimeters of tissue (a little more than one and an eighth inch), PDT is mainly used to treat tumors on or just under the skin or on the lining of internal organs. Photodynamic therapy makes the skin and eyes sensitive to light for 6 weeks or more after treatment. Patients are advised to avoid direct sunlight and bright indoor light for at least 6 weeks. If patients must go outdoors, they need to wear protective clothing, including sunglasses. Other temporary side effects of PDT are related to the treatment of specific areas and can include coughing, trouble swallowing, abdominal pain, and painful breathing or shortness of breath. In December 1995, the U.S. Food and Drug Administration (FDA) approved a photosensitizing agent called porfimer sodium, or Photofrin®, to relieve symptoms of esophageal cancer that is causing an obstruction and for esophageal cancer that cannot be satisfactorily treated with lasers alone. In January 1998, the FDA approved porfimer sodium for the treatment of early nonsmall cell lung cancer in patients for whom the usual treatments for lung cancer are not appropriate. The National Cancer Institute and other institutions are supporting clinical trials (research studies) to evaluate the use of photodynamic therapy for several types of cancer, including cancers of the bladder, brain, larynx, and oral cavity.

In yet another embodiment, laser therapy is used to harness high-intensity light to destroy cancer cells. This technique is often used to relieve symptoms of cancer such as bleeding or obstruction, especially when the cancer cannot be cured by other treatments. It may also be used to treat cancer by shrinking or destroying tumors. The term "laser" stands for light amplification by stimulated emission of radiation. Ordinary light, such as that from a light bulb, has many wavelengths and spreads in all directions. Laser light, on the other hand, has a specific wavelength and is focused in a narrow beam. This type of high-intensity light contains a lot of energy. Lasers are very powerful and may be used to cut through steel or to shape diamonds. Lasers also can be used for very precise surgical work, such as repairing a damaged retina in the eye or cutting through

tissue (in place of a scalpel). Although there are several different kinds of lasers, only three kinds have gained wide use in medicine: Carbon dioxide (CO₂) laser—This type of laser can remove thin layers from the skin's surface without penetrating the deeper layers. This technique is particularly useful in treating tumors that have not spread deep into the skin and certain precancerous conditions. As an alternative to traditional scalpel surgery, the CO₂ laser is also able to cut the skin. The laser is used in this way to remove skin cancers. Neodymium:yttrium-aluminum-garnet (Nd:YAG) laser—Light from this laser can penetrate deeper into tissue than light from the other types of lasers, and it can cause blood to clot quickly. It can be carried through optical fibers to less accessible parts of the body. This type of laser is sometimes used to treat throat cancers. Argon laser—This laser can pass through only superficial layers of tissue and is therefore useful in dermatology and in eye surgery. It also is used with light-sensitive dyes to treat tumors in a procedure known as photodynamic therapy (PDT). Lasers have several advantages over standard surgical tools, including: Lasers are more precise than scalpels. Tissue near an incision is protected, since there is little contact with surrounding skin or other tissue. The heat produced by lasers sterilizes the surgery site, thus reducing the risk of infection. Less operating time may be needed because the precision of the laser allows for a smaller incision. Healing time is often shortened; since laser heat seals blood vessels, there is less bleeding, swelling, or scarring. Laser surgery may be less complicated. For example, with fiber optics, laser light can be directed to parts of the body without making a large incision. More procedures may be done on an outpatient basis. Lasers can be used in two ways to treat cancer: by shrinking or destroying a tumor with heat, or by activating a chemical—known as a photosensitizing agent—that destroys cancer cells. In PDT, a photosensitizing agent is retained in cancer cells and can be stimulated by light to cause a reaction that kills cancer cells. CO₂ and Nd:YAG lasers are used to shrink or destroy tumors. They may be used with endoscopes, tubes that allow physicians to see into certain areas of the body, such as the bladder. The light from some lasers can be transmitted through a flexible endoscope fitted with fiber optics. This allows physicians to see and work in parts of the body that could not otherwise be reached except by surgery and therefore allows very precise aiming of the laser beam. Lasers also may be used with low-power microscopes, giving the doctor a clear view of the site being treated. Used with other instruments, laser systems can produce a cutting area as small as 200 microns in diameter—less than the width of a very fine thread. Lasers are used to treat many types of cancer. Laser surgery is a standard treatment for certain stages of glottis (vocal cord), cervical, skin, lung, vaginal, vulvar, and penile cancers. In addition to its use to destroy the cancer, laser surgery is also used to help relieve symptoms caused by cancer (palliative care). For example, lasers may be used to shrink or destroy a tumor that is blocking a patient's trachea (windpipe), making it easier to breathe. It is also sometimes used for palliation in colorectal and anal cancer. Laser-induced interstitial thermotherapy (LITT) is one of the most recent developments in laser therapy. LITT uses the same idea as a cancer treatment called hyperthermia; that heat may help shrink tumors by damaging cells or depriving them of substances they need to live. In this treatment, lasers are directed to interstitial areas (areas between organs) in the body. The laser light then raises the temperature of the tumor, which damages or destroys cancer cells.

The duration and/or dose of treatment with anti-immune checkpoint therapies may vary according to the particular anti-immune checkpoint agent or combination thereof. An appropriate treatment time for a particular cancer therapeutic agent will be appreciated by the skilled artisan. The present invention contemplates the continued assessment of optimal treatment schedules for each cancer therapeutic agent, where the phenotype of the cancer of the subject as determined by the methods of the present invention is a factor in determining optimal treatment doses and schedules.

Any means for the introduction of a polynucleotide into mammals, human or non-human, or cells thereof may be adapted to the practice of this invention for the delivery of the various constructs of the present invention into the intended recipient. In one embodiment of the present invention, the DNA constructs are delivered to cells by transfection, i.e., by delivery of "naked" DNA or in a complex with a colloidal dispersion system. A colloidal system includes macromolecule complexes, nanocapsules, microspheres, beads, and lipid-based systems including oil-in-water emulsions, micelles, mixed micelles, and liposomes. The preferred colloidal system of this invention is a lipid-complexed or liposome-formulated DNA. In the former approach, prior to formulation of DNA, e.g., with lipid, a plasmid containing a transgene bearing the desired DNA constructs may first be experimentally optimized for expression (e.g., inclusion of an intron in the 5' untranslated region and elimination of unnecessary sequences (Felgner, et al., *Ann NY Acad Sci* 126-139, 1995). Formulation of DNA, e.g. with various lipid or liposome materials, may then be effected using known methods and materials and delivered to the recipient mammal. See, e.g., Canonico et al, *Am J Respir Cell Mol Biol* 10:24-29, 1994; Tsan et al, *Am J Physiol* 268; Alton et al., *Nat Genet.* 5:135-142, 1993 and U.S. Pat. No. 5,679,647 by Carson et al.

The targeting of liposomes can be classified based on anatomical and mechanistic factors. Anatomical classification is based on the level of selectivity, for example, organ-specific, cell-specific, and organelle-specific. Mechanistic targeting can be distinguished based upon whether it is passive or active. Passive targeting utilizes the natural tendency of liposomes to distribute to cells of the reticulo-endothelial system (RES) in organs, which contain sinusoidal capillaries. Active targeting, on the other hand, involves alteration of the liposome by coupling the liposome to a specific ligand such as a monoclonal antibody, sugar, glycolipid, or protein, or by changing the composition or size of the liposome in order to achieve targeting to organs and cell types other than the naturally occurring sites of localization.

The surface of the targeted delivery system may be modified in a variety of ways. In the case of a liposomal targeted delivery system, lipid groups can be incorporated into the lipid bilayer of the liposome in order to maintain the targeting ligand in stable association with the liposomal bilayer. Various linking groups can be used for joining the lipid chains to the targeting ligand. Naked DNA or DNA associated with a delivery vehicle, e.g., liposomes, can be administered to several sites in a subject (see below).

Nucleic acids can be delivered in any desired vector. These include viral or non-viral vectors, including adenovirus vectors, adeno-associated virus vectors, retrovirus vectors, lentivirus vectors, and plasmid vectors. Exemplary types of viruses include HSV (herpes simplex virus), AAV (adeno associated virus), HIV (human immunodeficiency virus), BIV (bovine immunodeficiency virus), and MLV (murine leukemia virus). Nucleic acids can be administered in any desired format that provides sufficiently efficient

delivery levels, including in virus particles, in liposomes, in nanoparticles, and complexed to polymers.

The nucleic acids encoding a protein or nucleic acid of interest may be in a plasmid or viral vector, or other vector as is known in the art. Such vectors are well known and any can be selected for a particular application. In one embodiment of the present invention, the gene delivery vehicle comprises a promoter and a demethylase coding sequence. Preferred promoters are tissue-specific promoters and promoters which are activated by cellular proliferation, such as the thymidine kinase and thymidylate synthase promoters. Other preferred promoters include promoters which are activatable by infection with a virus, such as the α - and β -interferon promoters, and promoters which are activatable by a hormone, such as estrogen. Other promoters which can be used include the Moloney virus LTR, the CMV promoter, and the mouse albumin promoter. A promoter may be constitutive or inducible.

In another embodiment, naked polynucleotide molecules are used as gene delivery vehicles, as described in WO 90/11092 and U.S. Pat. No. 5,580,859. Such gene delivery vehicles can be either growth factor DNA or RNA and, in certain embodiments, are linked to killed adenovirus. Curiel et al., *Hum. Gene. Ther.* 3:147-154, 1992. Other vehicles which can optionally be used include DNA-ligand (Wu et al., *J. Biol. Chem.* 264:16985-16987, 1989), lipid-DNA combinations (Felgner et al., *Proc. Natl. Acad. Sci. USA* 84:7413-7417, 1989), liposomes (Wang et al., *Proc. Natl. Acad. Sci.* 84:7851-7855, 1987) and microprojectiles (Williams et al., *Proc. Natl. Acad. Sci.* 88:2726-2730, 1991).

A gene delivery vehicle can optionally comprise viral sequences such as a viral origin of replication or packaging signal. These viral sequences can be selected from viruses such as astrovirus, coronavirus, orthomyxovirus, papovavirus, paramyxovirus, parvovirus, picornavirus, poxvirus, retrovirus, togavirus or adenovirus. In a preferred embodiment, the growth factor gene delivery vehicle is a recombinant retroviral vector. Recombinant retroviruses and various uses thereof have been described in numerous references including, for example, Mann et al., *Cell* 33:153, 1983, Cane and Mulligan, *Proc. Nat'l. Acad. Sci. USA* 81:6349, 1984, Miller et al., *Human Gene Therapy* 1:5-14, 1990, U.S. Pat. Nos. 4,405,712, 4,861,719, and 4,980,289, and PCT Application Nos. WO 89/02,468, WO 89/05,349, and WO 90/02,806. Numerous retroviral gene delivery vehicles can be utilized in the present invention, including for example those described in EP 0,415,731; WO 90/07936; WO 94/03622; WO 93/25698; WO 93/25234; U.S. Pat. No. 5,219,740; WO 9311230; WO 9310218; Vile and Hart, *Cancer Res.* 53:3860-3864, 1993; Vile and Hart, *Cancer Res.* 53:962-967, 1993; Ram et al., *Cancer Res.* 53:83-88, 1993; Takamiya et al., *J. Neurosci. Res.* 33:493-503, 1992; Baba et al., *J. Neurosurg.* 79:729-735, 1993 (U.S. Pat. No. 4,777,127, GB 2,200,651, EP 0,345,242 and WO91/02805).

Other viral vector systems that can be used to deliver a polynucleotide of the present invention have been derived from herpes virus, e.g., Herpes Simplex Virus (U.S. Pat. No. 5,631,236 by Woo et al., issued May 20, 1997 and WO 00/08191 by Neurovex), vaccinia virus (Ridgeway (1988) Ridgeway, "Mammalian expression vectors," In: Rodriguez R L, Denhardt D T, ed. *Vectors: A survey of molecular cloning vectors and their uses.* Stoneham: Butterworth; Baichwal and Sugden (1986) "Vectors for gene transfer derived from animal DNA viruses: Transient and stable expression of transferred genes," In: Kucherlapati R, ed. *Gene transfer.* New York: Plenum Press; Coupar et al. (1988) *Gene*, 68:1-10), and several RNA viruses. Preferred viruses

include an alphavirus, a poxvirus, an arena virus, a vaccinia virus, a polio virus, and the like. They offer several attractive features for various mammalian cells (Friedmann (1989) Science, 244:1275-1281; Ridgeway, 1988, supra; Baichwal and Sugden, 1986, supra; Coupar et al., 1988; Horwich et al. (1990) J. Virol., 64:642-650).

In other embodiments, target DNA in the genome can be manipulated using well-known methods in the art. For example, the target DNA in the genome can be manipulated by deletion, insertion, and/or mutation are retroviral insertion, artificial chromosome techniques, gene insertion, random insertion with tissue specific promoters, gene targeting, transposable elements and/or any other method for introducing foreign DNA or producing modified DNA/modified nuclear DNA. Other modification techniques include deleting DNA sequences from a genome and/or altering nuclear DNA sequences. Nuclear DNA sequences, for example, may be altered by site-directed mutagenesis.

In other embodiments, recombinant biomarker polypeptides, and fragments thereof, can be administered to subjects. In some embodiments, fusion proteins can be constructed and administered which have enhanced biological properties. In addition, the biomarker polypeptides, and fragment thereof, can be modified according to well-known pharmacological methods in the art (e.g., pegylation, glycosylation, oligomerization, etc.) in order to further enhance desirable biological activities, such as increased bioavailability and decreased proteolytic degradation.

4. Clinical Efficacy

Clinical efficacy can be measured by any method known in the art. For example, the response to a therapy, such as anti-immune checkpoint therapies, relates to any response of the cancer, e.g., a tumor, to the therapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant chemotherapy. Tumor response may be assessed in a neoadjuvant or adjuvant situation where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or palpation and the cellularity of a tumor can be estimated histologically and compared to the cellularity of a tumor biopsy taken before initiation of treatment. Response may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or cellularity or using a semi-quantitative scoring system such as residual cancer burden (Symmans et al., *J. Clin. Oncol.* (2007) 25:4414-4422) or Miller-Payne score (Ogston et al., (2003) *Breast* (Edinburgh, Scotland) 12:320-327) in a qualitative fashion like "pathological complete response" (pCR), "clinical complete remission" (cCR), "clinical partial remission" (cPR), "clinical stable disease" (cSD), "clinical progressive disease" (cPD) or other qualitative criteria. Assessment of tumor response may be performed early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed.

In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6

months out from the end of therapy. The shorthand for this formula is $CBR = CR + PR + SD$ over 6 months. In some embodiments, the CBR for a particular anti-immune checkpoint therapeutic regimen is at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more.

Additional criteria for evaluating the response to anti-immune checkpoint therapies are related to "survival," which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); "recurrence-free survival" (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.

For example, in order to determine appropriate threshold values, a particular anti-immune checkpoint therapeutic regimen can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any immune checkpoint therapy. The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following immune checkpoint therapy for whom biomarker measurement values are known. In certain embodiments, the same doses of anti-immune checkpoint agents are administered to each subject. In related embodiments, the doses administered are standard doses known in the art for anti-immune checkpoint agents. The period of time for which subjects are monitored can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of an immune checkpoint therapy can be determined using methods such as those described in the Examples section.

5. Further Uses and Methods of the Present Invention

The methods described herein can be used in a variety of diagnostic, prognostic, and therapeutic applications. In any method described herein, such as a diagnostic method, prognostic method, therapeutic method, or combination thereof, all steps of the method can be performed by a single actor or, alternatively, by more than one actor. For example, diagnosis can be performed directly by the actor providing therapeutic treatment. Alternatively, a person providing a therapeutic agent can request that a diagnostic assay be performed. The diagnostician and/or the therapeutic interventionist can interpret the diagnostic assay results to determine a therapeutic strategy. Similarly, such alternative processes can apply to other assays, such as prognostic assays. The compositions described herein can also be used in a variety of diagnostic, prognostic, and therapeutic applications regarding biomarkers described herein, such as those listed in Table 1. Moreover, any method of diagnosis, prognosis, prevention, and the like described herein can be applied to a therapy or test agent of interest, such as immune checkpoint therapies, EGFR therapies, anti-cancer therapies, and the like.

a. Screening Methods

One aspect of the present invention relates to screening assays, including non-cell based assays. In one embodiment, the assays provide a method for identifying whether a cancer is likely to respond to immune checkpoint therapy and/or whether an agent can inhibit the growth of or kill a cancer cell that is unlikely to respond to immune checkpoint therapy.

In one embodiment, the present invention relates to assays for screening test agents which bind to, or modulate the biological activity of, at least one biomarker listed in Table 1. In one embodiment, a method for identifying such an agent entails determining the ability of the agent to modulate, e.g. inhibit, the at least one biomarker listed in Table 1.

In one embodiment, an assay is a cell-free or cell-based assay, comprising contacting at least one biomarker listed in Table 1, with a test agent, and determining the ability of the test agent to modulate (e.g. inhibit) the enzymatic activity of the biomarker, such as by measuring direct binding of substrates or by measuring indirect parameters as described below.

For example, in a direct binding assay, biomarker protein (or their respective target polypeptides or molecules) can be coupled with a radioisotope or enzymatic label such that binding can be determined by detecting the labeled protein or molecule in a complex. For example, the targets can be labeled with ^{125}I , ^{35}S , ^{14}C , or ^3H , either directly or indirectly, and the radioisotope detected by direct counting of radioemission or by scintillation counting. Alternatively, the targets can be enzymatically labeled with, for example, horseradish peroxidase, alkaline phosphatase, or luciferase, and the enzymatic label detected by determination of conversion of an appropriate substrate to product. Determining the interaction between biomarker and substrate can also be accomplished using standard binding or enzymatic analysis assays. In one or more embodiments of the above described assay methods, it may be desirable to immobilize polypeptides or molecules to facilitate separation of complexed from uncomplexed forms of one or both of the proteins or molecules, as well as to accommodate automation of the assay.

Binding of a test agent to a target can be accomplished in any vessel suitable for containing the reactants. Non-limiting examples of such vessels include microtiter plates, test tubes, and micro-centrifuge tubes. Immobilized forms of the antibodies of the present invention can also include antibodies bound to a solid phase like a porous, microporous (with an average pore diameter less than about one micron) or macroporous (with an average pore diameter of more than about 10 microns) material, such as a membrane, cellulose, nitrocellulose, or glass fibers; a bead, such as that made of agarose or polyacrylamide or latex; or a surface of a dish, plate, or well, such as one made of polystyrene.

In an alternative embodiment, determining the ability of the agent to modulate the interaction between the biomarker and a substrate or a biomarker and its natural binding partner can be accomplished by determining the ability of the test agent to modulate the activity of a polypeptide or other product that functions downstream or upstream of its position within the signaling pathway (e.g., feedback loops). Such feedback loops are well-known in the art (see, for example, Chen and Guillemin (2009) *Int. J. Tryptophan Res.* 2:1-19).

The present invention further pertains to novel agents identified by the above-described screening assays. Accordingly, it is within the scope of this invention to further use an agent identified as described herein in an appropriate

animal model. For example, an agent identified as described herein can be used in an animal model to determine the efficacy, toxicity, or side effects of treatment with such an agent. Alternatively, an antibody identified as described herein can be used in an animal model to determine the mechanism of action of such an agent.

b. Predictive Medicine

The present invention also pertains to the field of predictive medicine in which diagnostic assays, prognostic assays, and monitoring clinical trials are used for prognostic (predictive) purposes to thereby treat an individual prophylactically. Accordingly, one aspect of the present invention relates to diagnostic assays for determining the amount and/or activity level of a biomarker listed in Table 1 in the context of a biological sample (e.g., blood, serum, cells, or tissue) to thereby determine whether an individual afflicted with a cancer is likely to respond to immune checkpoint therapy, whether in an original or recurrent cancer. Such assays can be used for prognostic or predictive purpose to thereby prophylactically treat an individual prior to the onset or after recurrence of a disorder characterized by or associated with biomarker polypeptide, nucleic acid expression or activity. The skilled artisan will appreciate that any method can use one or more (e.g., combinations) of biomarkers listed in Table 1.

Another aspect of the present invention pertains to monitoring the influence of agents (e.g., drugs, compounds, and small nucleic acid-based molecules) on the expression or activity of a biomarker listed in Table 1. These and other agents are described in further detail in the following sections.

The skilled artisan will also appreciate that, in certain embodiments, the methods of the present invention implement a computer program and computer system. For example, a computer program can be used to perform the algorithms described herein. A computer system can also store and manipulate data generated by the methods of the present invention which comprises a plurality of biomarker signal changes/profiles which can be used by a computer system in implementing the methods of this invention. In certain embodiments, a computer system receives biomarker expression data; (ii) stores the data; and (iii) compares the data in any number of ways described herein (e.g., analysis relative to appropriate controls) to determine the state of informative biomarkers from cancerous or pre-cancerous tissue. In other embodiments, a computer system (i) compares the determined expression biomarker level to a threshold value; and (ii) outputs an indication of whether said biomarker level is significantly modulated (e.g., above or below) the threshold value, or a phenotype based on said indication.

In certain embodiments, such computer systems are also considered part of the present invention. Numerous types of computer systems can be used to implement the analytic methods of this invention according to knowledge possessed by a skilled artisan in the bioinformatics and/or computer arts. Several software components can be loaded into memory during operation of such a computer system. The software components can comprise both software components that are standard in the art and components that are special to the present invention (e.g., dCHIP software described in Lin et al. (2004) *Bioinformatics* 20, 1233-1240; radial basis machine learning algorithms (RBM) known in the art).

The methods of the present invention can also be programmed or modeled in mathematical software packages that allow symbolic entry of equations and high-level speci-

fication of processing, including specific algorithms to be used, thereby freeing a user of the need to procedurally program individual equations and algorithms. Such packages include, e.g., Matlab from Mathworks (Natick, Mass.), Mathematica from Wolfram Research (Champaign, Ill.) or S-Plus from MathSoft (Seattle, Wash.).

In certain embodiments, the computer comprises a database for storage of biomarker data. Such stored profiles can be accessed and used to perform comparisons of interest at a later point in time. For example, biomarker expression profiles of a sample derived from the non-cancerous tissue of a subject and/or profiles generated from population-based distributions of informative loci of interest in relevant populations of the same species can be stored and later compared to that of a sample derived from the cancerous tissue of the subject or tissue suspected of being cancerous of the subject.

In addition to the exemplary program structures and computer systems described herein, other, alternative program structures and computer systems will be readily apparent to the skilled artisan. Such alternative systems, which do not depart from the above described computer system and programs structures either in spirit or in scope, are therefore intended to be comprehended within the accompanying claims.

c. Diagnostic Assays

The present invention provides, in part, methods, systems, and code for accurately classifying whether a biological sample is associated with a cancer that is likely to respond to immune checkpoint therapy. In some embodiments, the present invention is useful for classifying a sample (e.g., from a subject) as associated with or at risk for responding to or not responding to immune checkpoint therapy using a statistical algorithm and/or empirical data (e.g., the amount or activity of a biomarker listed in Table 1).

An exemplary method for detecting the amount or activity of a biomarker listed in Table 1, and thus useful for classifying whether a sample is likely or unlikely to respond to immune checkpoint therapy involves obtaining a biological sample from a test subject and contacting the biological sample with an agent, such as a protein-binding agent like an antibody or antigen-binding fragment thereof, or a nucleic acid-binding agent like an oligonucleotide, capable of detecting the amount or activity of the biomarker in the biological sample. In some embodiments, at least one antibody or antigen-binding fragment thereof is used, wherein two, three, four, five, six, seven, eight, nine, ten, or more such antibodies or antibody fragments can be used in combination (e.g., in sandwich ELISAs) or in serial. In certain instances, the statistical algorithm is a single learning statistical classifier system. For example, a single learning statistical classifier system can be used to classify a sample as a based upon a prediction or probability value and the presence or level of the biomarker. The use of a single learning statistical classifier system typically classifies the sample as, for example, a likely immune checkpoint therapy responder or progressor sample with a sensitivity, specificity, positive predictive value, negative predictive value, and/or overall accuracy of at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

Other suitable statistical algorithms are well known to those of skill in the art. For example, learning statistical classifier systems include a machine learning algorithmic technique capable of adapting to complex data sets (e.g., panel of markers of interest) and making decisions based upon such data sets. In some embodiments, a single learning

statistical classifier system such as a classification tree (e.g., random forest) is used. In other embodiments, a combination of 2, 3, 4, 5, 6, 7, 8, 9, 10, or more learning statistical classifier systems are used, preferably in tandem. Examples of learning statistical classifier systems include, but are not limited to, those using inductive learning (e.g., decision/classification trees such as random forests, classification and regression trees (C&RT), boosted trees, etc.), Probably Approximately Correct (PAC) learning, connectionist learning (e.g., neural networks (NN), artificial neural networks (ANN), neuro fuzzy networks (NFN), network structures, perceptrons such as multi-layer perceptrons, multi-layer feed-forward networks, applications of neural networks, Bayesian learning in belief networks, etc.), reinforcement learning (e.g., passive learning in a known environment such as naive learning, adaptive dynamic learning, and temporal difference learning, passive learning in an unknown environment, active learning in an unknown environment, learning action-value functions, applications of reinforcement learning, etc.), and genetic algorithms and evolutionary programming. Other learning statistical classifier systems include support vector machines (e.g., Kernel methods), multivariate adaptive regression splines (MARS), Levenberg-Marquardt algorithms, Gauss-Newton algorithms, mixtures of Gaussians, gradient descent algorithms, and learning vector quantization (LVQ). In certain embodiments, the method of the present invention further comprises sending the sample classification results to a clinician, e.g., an oncologist.

In another embodiment, the diagnosis of a subject is followed by administering to the individual a therapeutically effective amount of a defined treatment based upon the diagnosis.

In one embodiment, the methods further involve obtaining a control biological sample (e.g., biological sample from a subject who does not have a cancer or whose cancer is susceptible to immune checkpoint therapy), a biological sample from the subject during remission, or a biological sample from the subject during treatment for developing a cancer progressing despite immune checkpoint therapy.

d. Prognostic Assays

The diagnostic methods described herein can furthermore be utilized to identify subjects having or at risk of developing a cancer that is likely or unlikely to be responsive to immune checkpoint therapy. The assays described herein, such as the preceding diagnostic assays or the following assays, can be utilized to identify a subject having or at risk of developing a disorder associated with a misregulation of the amount or activity of at least one biomarker described in Table 1, such as in cancer. Alternatively, the prognostic assays can be utilized to identify a subject having or at risk for developing a disorder associated with a misregulation of the at least one biomarker described in Table 1, such as in cancer. Furthermore, the prognostic assays described herein can be used to determine whether a subject can be administered an agent (e.g., an agonist, antagonist, peptidomimetic, polypeptide, peptide, nucleic acid, small molecule, or other drug candidate) to treat a disease or disorder associated with the aberrant biomarker expression or activity.

e. Treatment Methods

The compositions described herein (including dual binding antibodies and derivatives and conjugates thereof) can be used in a variety of in vitro and in vivo therapeutic applications using the formulations and/or combinations described herein. In one embodiment, anti-immune checkpoint agents can be used to treat cancers determined to be responsive thereto. For example, antibodies that block the

interaction between PD-L1, PD-L2, and/or CTLA-4 and their receptors (e.g., PD-L1 binding to PD-1, PD-L2 binding to PD-1, and the like) can be used to treat cancer in subjects identified as likely responding thereto.

6. Pharmaceutical Compositions

In another aspect, the present invention provides pharmaceutically acceptable compositions which comprise a therapeutically-effective amount of an agent that modulates (e.g., decreases) biomarker expression and/or activity, formulated together with one or more pharmaceutically acceptable carriers (additives) and/or diluents. As described in detail below, the pharmaceutical compositions of the present invention may be specially formulated for administration in solid or liquid form, including those adapted for the following: (1) oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, boluses, powders, granules, pastes; (2) parenteral administration, for example, by subcutaneous, intramuscular or intravenous injection as, for example, a sterile solution or suspension; (3) topical application, for example, as a cream, ointment or spray applied to the skin; (4) intravaginally or intrarectally, for example, as a pessary, cream or foam; or (5) aerosol, for example, as an aqueous aerosol, liposomal preparation or solid particles containing the compound.

The phrase "therapeutically-effective amount" as used herein means that amount of an agent that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex, or composition comprising an agent that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex, which is effective for producing some desired therapeutic effect, e.g., cancer treatment, at a reasonable benefit/risk ratio.

The phrase "pharmaceutically acceptable" is employed herein to refer to those agents, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

The phrase "pharmaceutically-acceptable carrier" as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, solvent or encapsulating material, involved in carrying or transporting the subject chemical from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be "acceptable" in the sense of being compatible with the other ingredients of the formulation and not injurious to the subject. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) phosphate buffer solutions; and (21) other non-toxic compatible substances employed in pharmaceutical formulations.

The term "pharmaceutically-acceptable salts" refers to the relatively non-toxic, inorganic and organic acid addition salts of the agents that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex encompassed by the present invention. These salts can be prepared in situ during the final isolation and purification of the respiration uncoupling agents, or by separately reacting a purified respiration uncoupling agent in its free base form with a suitable organic or inorganic acid, and isolating the salt thus formed. ReRepresentative salts include the hydrobromide, hydrochloride, sulfate, bisulfate, phosphate, nitrate, acetate, valerate, oleate, palmitate, stearate, laurate, benzoate, lactate, phosphate, tosylate, citrate, maleate, fumarate, succinate, tartrate, naphthylate, mesylate, glucoheptonate, lactobionate, and laurylsulphonate salts and the like (See, for example, Berge et al. (1977) "Pharmaceutical Salts", *J. Pharm. Sci.* 66:1-19).

In other cases, the agents useful in the methods of the present invention may contain one or more acidic functional groups and, thus, are capable of forming pharmaceutically-acceptable salts with pharmaceutically-acceptable bases. The term "pharmaceutically-acceptable salts" in these instances refers to the relatively non-toxic, inorganic and organic base addition salts of agents that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex. These salts can likewise be prepared in situ during the final isolation and purification of the respiration uncoupling agents, or by separately reacting the purified respiration uncoupling agent in its free acid form with a suitable base, such as the hydroxide, carbonate or bicarbonate of a pharmaceutically-acceptable metal cation, with ammonia, or with a pharmaceutically-acceptable organic primary, secondary or tertiary amine. ReRepresentative alkali or alkaline earth salts include the lithium, sodium, potassium, calcium, magnesium, and aluminum salts and the like. ReRepresentative organic amines useful for the formation of base addition salts include ethylamine, diethylamine, ethylenediamine, ethanolamine, diethanolamine, piperazine and the like (see, for example, Berge et al., *supra*).

Wetting agents, emulsifiers and lubricants, such as sodium lauryl sulfate and magnesium stearate, as well as coloring agents, release agents, coating agents, sweetening, flavoring and perfuming agents, preservatives and antioxidants can also be present in the compositions.

Examples of pharmaceutically-acceptable antioxidants include: (1) water soluble antioxidants, such as ascorbic acid, cysteine hydrochloride, sodium bisulfate, sodium metabisulfite, sodium sulfite and the like; (2) oil-soluble antioxidants, such as ascorbyl palmitate, butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), lecithin, propyl gallate, alpha-tocopherol, and the like; and (3) metal chelating agents, such as citric acid, ethylenediamine tetraacetic acid (EDTA), sorbitol, tartaric acid, phosphoric acid, and the like.

Formulations useful in the methods of the present invention include those suitable for oral, nasal, topical (including buccal and sublingual), rectal, vaginal, aerosol and/or parenteral administration. The formulations may conveniently be presented in unit dosage form and may be prepared by any methods well known in the art of pharmacy. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the host being treated, the particular mode of administration. The amount of active ingredient, which can be combined with a carrier material to produce a single dosage form will generally be that amount of the compound

which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 1 percent to about ninety-nine percent of active ingredient, preferably from about 5 percent to about 70 percent, most preferably from about 10 percent to about 30 percent.

Methods of preparing these formulations or compositions include the step of bringing into association an agent that modulates (e.g., inhibits) biomarker expression and/or activity, with the carrier and, optionally, one or more accessory ingredients. In general, the formulations are prepared by uniformly and intimately bringing into association a respiration uncoupling agent with liquid carriers, or finely divided solid carriers, or both, and then, if necessary, shaping the product.

Formulations suitable for oral administration may be in the form of capsules, cachets, pills, tablets, lozenges (using a flavored basis, usually sucrose and acacia or tragacanth), powders, granules, or as a solution or a suspension in an aqueous or non-aqueous liquid, or as an oil-in-water or water-in-oil liquid emulsion, or as an elixir or syrup, or as pastilles (using an inert base, such as gelatin and glycerin, or sucrose and acacia) and/or as mouth washes and the like, each containing a predetermined amount of a respiration uncoupling agent as an active ingredient. A compound may also be administered as a bolus, electuary or paste.

In solid dosage forms for oral administration (capsules, tablets, pills, dragees, powders, granules and the like), the active ingredient is mixed with one or more pharmaceutically-acceptable carriers, such as sodium citrate or dicalcium phosphate, and/or any of the following: (1) fillers or extenders, such as starches, lactose, sucrose, glucose, mannitol, and/or silicic acid; (2) binders, such as, for example, carboxymethylcellulose, alginates, gelatin, polyvinyl pyrrolidone, sucrose and/or acacia; (3) humectants, such as glycerol; (4) disintegrating agents, such as agar-agar, calcium carbonate, potato or tapioca starch, alginic acid, certain silicates, and sodium carbonate; (5) solution retarding agents, such as paraffin; (6) absorption accelerators, such as quaternary ammonium compounds; (7) wetting agents, such as, for example, acetyl alcohol and glycerol monostearate; (8) absorbents, such as kaolin and bentonite clay; (9) lubricants, such as talc, calcium stearate, magnesium stearate, solid polyethylene glycols, sodium lauryl sulfate, and mixtures thereof; and (10) coloring agents. In the case of capsules, tablets and pills, the pharmaceutical compositions may also comprise buffering agents. Solid compositions of a similar type may also be employed as fillers in soft and hard-filled gelatin capsules using such excipients as lactose or milk sugars, as well as high molecular weight polyethylene glycols and the like.

A tablet may be made by compression or molding, optionally with one or more accessory ingredients. Compressed tablets may be prepared using binder (for example, gelatin or hydroxypropylmethyl cellulose), lubricant, inert diluent, preservative, disintegrant (for example, sodium starch glycolate or cross-linked sodium carboxymethyl cellulose), surface-active or dispersing agent. Molded tablets may be made by molding in a suitable machine a mixture of the powdered peptide or peptidomimetic moistened with an inert liquid diluent.

Tablets, and other solid dosage forms, such as dragees, capsules, pills and granules, may optionally be scored or prepared with coatings and shells, such as enteric coatings and other coatings well known in the pharmaceutical-formulating art. They may also be formulated so as to provide slow or controlled release of the active ingredient therein using, for example, hydroxypropylmethyl cellulose in vary-

ing proportions to provide the desired release profile, other polymer matrices, liposomes and/or microspheres. They may be sterilized by, for example, filtration through a bacteria-retaining filter, or by incorporating sterilizing agents in the form of sterile solid compositions, which can be dissolved in sterile water, or some other sterile injectable medium immediately before use. These compositions may also optionally contain opacifying agents and may be of a composition that they release the active ingredient(s) only, or preferentially, in a certain portion of the gastrointestinal tract, optionally, in a delayed manner. Examples of embedding compositions, which can be used include polymeric substances and waxes. The active ingredient can also be in micro-encapsulated form, if appropriate, with one or more of the above-described excipients.

Liquid dosage forms for oral administration include pharmaceutically acceptable emulsions, microemulsions, solutions, suspensions, syrups and elixirs. In addition to the active ingredient, the liquid dosage forms may contain inert diluents commonly used in the art, such as, for example, water or other solvents, solubilizing agents and emulsifiers, such as ethyl alcohol, isopropyl alcohol, ethyl carbonate, ethyl acetate, benzyl alcohol, benzyl benzoate, propylene glycol, 1,3-butylene glycol, oils (in particular, cottonseed, groundnut, corn, germ, olive, castor and sesame oils), glycerol, tetrahydrofuryl alcohol, polyethylene glycols and fatty acid esters of sorbitan, and mixtures thereof.

Besides inert diluents, the oral compositions can also include adjuvants such as wetting agents, emulsifying and suspending agents, sweetening, flavoring, coloring, perfuming and preservative agents.

Suspensions, in addition to the active agent may contain suspending agents as, for example, ethoxylated isostearyl alcohols, polyoxyethylene sorbitol and sorbitan esters, microcrystalline cellulose, aluminum metahydroxide, bentonite, agar-agar and tragacanth, and mixtures thereof.

Formulations for rectal or vaginal administration may be presented as a suppository, which may be prepared by mixing one or more respiration uncoupling agents with one or more suitable nonirritating excipients or carriers comprising, for example, cocoa butter, polyethylene glycol, a suppository wax or a salicylate, and which is solid at room temperature, but liquid at body temperature and, therefore, will melt in the rectum or vaginal cavity and release the active agent.

Formulations which are suitable for vaginal administration also include pessaries, tampons, creams, gels, pastes, foams or spray formulations containing such carriers as are known in the art to be appropriate.

Dosage forms for the topical or transdermal administration of an agent that modulates (e.g., inhibits) biomarker expression and/or activity include powders, sprays, ointments, pastes, creams, lotions, gels, solutions, patches and inhalants. The active component may be mixed under sterile conditions with a pharmaceutically-acceptable carrier, and with any preservatives, buffers, or propellants which may be required.

The ointments, pastes, creams and gels may contain, in addition to a respiration uncoupling agent, excipients, such as animal and vegetable fats, oils, waxes, paraffins, starch, tragacanth, cellulose derivatives, polyethylene glycols, silicates, bentonites, silicic acid, talc and zinc oxide, or mixtures thereof.

Powders and sprays can contain, in addition to an agent that modulates (e.g., inhibits) biomarker expression and/or activity, excipients such as lactose, talc, silicic acid, aluminum hydroxide, calcium silicates and polyamide powder, or

mixtures of these substances. Sprays can additionally contain customary propellants, such as chlorofluorohydrocarbons and volatile unsubstituted hydrocarbons, such as butane and propane.

The agent that modulates (e.g., inhibits) biomarker expression and/or activity, can be alternatively administered by aerosol. This is accomplished by preparing an aqueous aerosol, liposomal preparation or solid particles containing the compound. A nonaqueous (e.g., fluorocarbon propellant) suspension could be used. Sonic nebulizers are preferred because they minimize exposing the agent to shear, which can result in degradation of the compound.

Ordinarily, an aqueous aerosol is made by formulating an aqueous solution or suspension of the agent together with conventional pharmaceutically acceptable carriers and stabilizers. The carriers and stabilizers vary with the requirements of the particular compound, but typically include nonionic surfactants (Tweens, Pluronics, or polyethylene glycol), innocuous proteins like serum albumin, sorbitan esters, oleic acid, lecithin, amino acids such as glycine, buffers, salts, sugars or sugar alcohols. Aerosols generally are prepared from isotonic solutions.

Transdermal patches have the added advantage of providing controlled delivery of a respiration uncoupling agent to the body. Such dosage forms can be made by dissolving or dispersing the agent in the proper medium. Absorption enhancers can also be used to increase the flux of the peptidomimetic across the skin. The rate of such flux can be controlled by either providing a rate controlling membrane or dispersing the peptidomimetic in a polymer matrix or gel.

Ophthalmic formulations, eye ointments, powders, solutions and the like, are also contemplated as being within the scope of this invention.

Pharmaceutical compositions of this invention suitable for parenteral administration comprise one or more respiration uncoupling agents in combination with one or more pharmaceutically-acceptable sterile isotonic aqueous or non-aqueous solutions, dispersions, suspensions or emulsions, or sterile powders which may be reconstituted into sterile injectable solutions or dispersions just prior to use, which may contain antioxidants, buffers, bacteriostats, solutes which render the formulation isotonic with the blood of the intended recipient or suspending or thickening agents.

Examples of suitable aqueous and nonaqueous carriers which may be employed in the pharmaceutical compositions of the present invention include water, ethanol, polyols (such as glycerol, propylene glycol, polyethylene glycol, and the like), and suitable mixtures thereof, vegetable oils, such as olive oil, and injectable organic esters, such as ethyl oleate. Proper fluidity can be maintained, for example, by the use of coating materials, such as lecithin, by the maintenance of the required particle size in the case of dispersions, and by the use of surfactants.

These compositions may also contain adjuvants such as preservatives, wetting agents, emulsifying agents and dispersing agents. Prevention of the action of microorganisms may be ensured by the inclusion of various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol sorbic acid, and the like. It may also be desirable to include isotonic agents, such as sugars, sodium chloride, and the like into the compositions. In addition, prolonged absorption of the injectable pharmaceutical form may be brought about by the inclusion of agents which delay absorption such as aluminum monostearate and gelatin.

In some cases, in order to prolong the effect of a drug, it is desirable to slow the absorption of the drug from subcutaneous or intramuscular injection. This may be accom-

plished by the use of a liquid suspension of crystalline or amorphous material having poor water solubility. The rate of absorption of the drug then depends upon its rate of dissolution, which, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally-administered drug form is accomplished by dissolving or suspending the drug in an oil vehicle.

Injectable depot forms are made by forming microencapsule matrices of an agent that modulates (e.g., inhibits) biomarker expression and/or activity, in biodegradable polymers such as polylactide-polyglycolide. Depending on the ratio of drug to polymer, and the nature of the particular polymer employed, the rate of drug release can be controlled. Examples of other biodegradable polymers include poly(orthoesters) and poly(anhydrides). Depot injectable formulations are also prepared by entrapping the drug in liposomes or microemulsions, which are compatible with body tissue.

When the respiration uncoupling agents of the present invention are administered as pharmaceuticals, to humans and animals, they can be given per se or as a pharmaceutical composition containing, for example, 0.1 to 99.5% (more preferably, 0.5 to 90%) of active ingredient in combination with a pharmaceutically acceptable carrier.

Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be determined by the methods of the present invention so as to obtain an amount of the active ingredient, which is effective to achieve the desired therapeutic response for a particular subject, composition, and mode of administration, without being toxic to the subject.

The nucleic acid molecules of the present invention can be inserted into vectors and used as gene therapy vectors. Gene therapy vectors can be delivered to a subject by, for example, intravenous injection, local administration (see U.S. Pat. No. 5,328,470) or by stereotactic injection (see e.g., Chen et al. (1994) *Proc. Nat. Acad. Sci. USA* 91:3054 3057). The pharmaceutical preparation of the gene therapy vector can include the gene therapy vector in an acceptable diluent, or can comprise a slow release matrix in which the gene delivery vehicle is imbedded. Alternatively, where the complete gene delivery vector can be produced intact from recombinant cells, e.g., retroviral vectors, the pharmaceutical preparation can include one or more cells which produce the gene delivery system.

The present invention also encompasses kits for detecting and/or modulating biomarkers described herein. A kit of the present invention may also include instructional materials disclosing or describing the use of the kit or an antibody of the disclosed invention in a method of the disclosed invention as provided herein. A kit may also include additional components to facilitate the particular application for which the kit is designed. For example, a kit may additionally contain means of detecting the label (e.g., enzyme substrates for enzymatic labels, filter sets to detect fluorescent labels, appropriate secondary labels such as a sheep anti-mouse-HRP, etc.) and reagents necessary for controls (e.g., control biological samples or standards). A kit may additionally include buffers and other reagents recognized for use in a method of the disclosed invention. Non-limiting examples include agents to reduce non-specific binding, such as a carrier protein or a detergent.

EXEMPLIFICATION

This invention is further illustrated by the following examples, which should not be construed as limiting.

Example 1: Materials and Methods for Examples 2-3

a. Meta-Analysis: Cohort Consolidation

Mutation annotation files, HLA types, and clinical annotations were obtained from published data. Standardized pipelines for somatic variant calling, mutational signature deconvolution, and neoantigen prediction were used. Patients were stratified into clinical benefit (CB) and no clinical benefit (NCB) as described in Van Allen et al. (2015) *Science* 350:207-211. Analyses were repeated using two other published response metrics (CB=PFS>6 months; CB=CR or PR; see at least Rizvi et al. (2015) *Science* 348:124-128 and Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199).

Specifically, pre-treatment tumors from patients who received immune checkpoint blockade for metastatic cancer at the DFCI and partner institutions were sequenced at the Broad Institute (FIG. 1). Data sources listed in FIG. 1 include data from Zaretsky et al. (2016) *New Engl. J. Med.* 375:819-829 disclosing 4 patients treated with anti-PD-1 therapy for metastatic melanoma (Zaretsky); Van Allen et al. (2015) *Science* 350:207-211 disclosing treatment of metastatic melanoma patients across Germany with anti-CTLA-4 ipilimumab therapy (Schadendorf); Hugo et al. (2016) *Cell* 165:35-44 (Hugo); Rizvi et al. (2015) *Science* 348:124-128 (Rizvi); Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199 (Synder); and samples obtained from patients treated with immune checkpoint therapies with clinical monitoring at the Dana-Farber Cancer Institute (DFCI, CANSEQ, Broad-Next10, and Rizwan Haq). For example, samples from 2 patients for WES obtained from Dana-Farber Cancer Institute attending physician, Rizwan Haq (Rizwan Haq). Cancer types included melanoma, non-small-cell lung cancer, bladder cancer, anal cancer, sarcoma, head and neck squamous cell carcinoma, including one previously published cohort in Van Allen et al. (2015) *Science* 350:207-211. Quality controls were included based on exclusion criteria and inclusion criteria (FIG. 2).

In addition, whole exome data from published clinical cohorts of patients who received immune checkpoint therapy for metastatic non-small-cell lung cancer (Rizvi et al. (2015) *Science* 348:124-128) and melanoma (Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199; Hugo et al. (2016) *Cell* 165:35-44) were used in the meta-analysis.

In separate meta-analyses described in Example 3 below, mutation annotation files, HLA types, and clinical annotations were obtained from published data. In particular, published mutation annotation files (MAFs), clinical annotations, and neoantigen files were taken from the supplemental information of Rizvi et al. (2015) *Science* 348:124-128; Van Allen et al. (2015) *Science* 350:207-211; and Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199. These files were joined to form the meta-analysis cohort. Enrichment in mutations in specific genes was assessed in this joined cohort using Fisher's exact test. The joined MAF was also processed through MutSigCV to identify genes significantly mutated throughout the cohort. Thus, standardized pipelines for somatic variant calling, mutational signature deconvolution, and neoantigen prediction were used. Patients were stratified into clinical benefit (CB) and no clinical benefit (NCB) as described in Van Allen et al. (2015) *Science* 350:207-211. Analyses were repeated using two other published response metrics (CB=PFS>6 months; CB=CR or PR).

b. Sample Preparation, DNA and RNA Extraction, and Sequencing

Sample preparation, DNA and RNA extraction, and sequencing information vary slightly between the studies described in Rizvi et al. (2015) *Science* 348:124-128; Van Allen et al. (2015) *Science* 350:207-211; and Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199, and particular details of such methods used can be found in the supplemental information files of these three publications.

For other samples, such as metastatic melanoma samples described in Example 3, or unless otherwise described, sample preparation, DNA and RNA extraction, and sequencing information proceeded as described below. Briefly, after fixation and mounting, 5-10 10 µm slices from formalin-fixed, paraffin-embedded (FFPE) tumor blocks were obtained, and tumor-enriched tissue was macrodissected. Paraffin was removed from FFPE sections and cores using CitiSolv™ (Fisher Scientific, Hampton, NH), followed by ethanol washes and tissue lysis overnight at 56° C. Samples were then incubated at 90° C. to remove DNA crosslinks, and DNA- and when possible, RNA-extraction was performed using Qiagen AllPrep DNA/RNA Mini Kit (#51306, Qiagen, Hilden, Germany). Germline DNA was obtained from adjacent PBMCs.

Whole exome and whole transcriptome sequencing of tumor and germline samples were performed as previously described (Van Allen et al. (2015) *Science* 350:207-211; Van Allen et al. (2014) *Nat. Med.* 20:682-688). All samples in the training cohort were sequenced using the Illumina exome, while a portion of the samples in the validation cohort were sequenced using the Agilent exome (Table 3A). The Illumina exome uses Illumina's in-solution DNA probe based hybrid selection method to target approximately 37.7 Mb of mainly exonic territory, using similar principles as the Broad Institute-Agilent Technologies developed in-solution RNA probe based hybrid selection method (Agilent SureSelect™ All Exon V2) (Fisher et al. (2011) *Genome Biol.* 12:R1; Gnirke et al. (2009) *Nat. Biotechnol.* 27:182-189) to generate Illumina exome sequencing libraries.

Pooled libraries were normalized to 2 nM and denatured using 0.2 N NaOH prior to sequencing. Flowcell cluster amplification and sequencing were performed according to the manufacturer's protocols using either the HiSeq 2000 v3 or HiSeq 2500. Each run was a 76 bp paired-end with a dual eight-base index barcode read. Data was analyzed using the Broad Picard Pipeline, which includes de-multiplexing and data aggregation.

Exome sequence data processing was performed using established analytical pipelines at the Broad Institute. A BAM file was produced using the Picard pipeline (at the World Wide Web address of picard.sourceforge.net), which aligns the tumor and normal sequences to the hg19 human genome build using Illumina sequencing reads. The BAM was uploaded into the Firehose pipeline (at the World Wide Web address of broadinstitute.org/cancer/cga/Firehose), which manages input and output files to be executed by GenePattern (Reich et al. (2006) *Nat. Genet.* 38:500-501). Samples with mean target coverage less than 25× in the tumor and less than 15× in matched normal were excluded. Quality control modules within Firehose were applied to all sequencing data for comparison of the origin of tumor and normal genotypes and to assess fingerprinting concordance. Cross-contamination of samples was estimated using ContEst (Cibulskis et al. (2011) *Bioinformatics* 27:2601-2602). Samples with ContEst estimates exceeding 5% were excluded from analysis.

c. Whole Exome and Whole Transcriptome Analyses

MuTect was applied to identify somatic single-nucleotide variants (Cibulskis et al. (2013) *Nat. Biotechnol.* 31:213-

219). Strelka was used to identify somatic insertions and deletions (Saunders et al. (2012) *Bioinformatics* 28:1811-1817) across the whole exome. Indelocator, which detects small insertions and deletions after local realignment of tumor and normal sequences, was additionally applied to provide further sensitivity to detect indels in PBRM1 (Cancer Genome Atlas Research (2011) *Nature* 474:609-615). The union of indels called by Strelka and Indelocator was used for final analysis. Artifacts introduced by DNA oxidation during sequencing were computationally removed using a filter-based method (Costello et al. (2013) *Nuc. Acids Res.* 41:e67). All somatic mutations detected by whole-exome sequencing were analyzed for potential false positive calls by performing a comparison to mutation calls from a panel of 2,500 germline DNA samples (Stachler et al. (2015) *Nat. Genet.* 47:1047-1055). Mutations found in germline samples were removed from analysis. Annotation of identified variants was done using Oncotator (available at the World Wide Web address of www.broadinstitute.org/cancer/cga/oncotator). All nonsynonymous alterations in PBRM1 were manually reviewed in Integrated Genomics Viewer (IGV_2.3.57) for sequencing quality (Thorvaldsdottir et al. (2013) *Brief Bioinform* 14:178-192).

Copy ratios were calculated for each captured target by dividing the tumor coverage by the median coverage

obtained in a set of reference normal samples. The resulting copy ratios were segmented using the circular binary segmentation algorithm (Olshen et al. (2004) *Biostatistics* 5:557-572). Allelic copy number alterations were called while taking into account sample-specific overall chromosomal aberrations (focality) (Brastianos et al. (2015) *Cancer Discov.* 5:1164-1177). Inference of mutational clonality, tumor purity, and tumor ploidy was accomplished with ABSOLUTE (Carter et al. (2012) *Nat Biotechnol.* 30:413-421). Samples had to have estimated tumor purity greater than 10% to be included in the final analysis. As a final quality control metric to ensure adequate sequencing coverage and tumor purity to detect relevant oncogenic events, all samples had to have at least one nonsynonymous mutation in at least one high confidence or candidate cancer driver gene to be included in the final analysis (Tamborero et al. (2013) *Sci. Rep.* 3:2650).

The 4-digit HLA type for each sample was inferred using Polysolver (Shukla et al. (2015) *Nat. Biotechnol.* 33:1152-1158). Neo-epitopes were predicted for each patient by defining all novel amino acid 9mers and 10mers resulting from each single nucleotide polymorphism and indel and determining whether the predicted binding affinity to the patient's germline HLA alleles was <500 nM using NetMHCpan (v2.4) (Hoof et al. (2009) *Immunogenetics* 61:1-13; Karosiene et al. (2013) *Immunogenetics* 65:711-724; Nielsen et al. (2007) *PLoS One* 2:e796).

Statistical analyses were also applied and are described further herein.

Example 2: Meta-Analysis of Genomic Predictors of Response to Immune Checkpoint Therapy Across a Variety of Cancers

A large cohort of whole-exome-sequenced pre-treatment tumors were gathered from subjects who received immune checkpoint therapies for various cancers, were gathered and assessed for mutations associated with differential response to immune checkpoint therapies. Tumor types included in this discovery cohort included bladder cancer, renal cell carcinoma, lung cancer, and head and neck squamous cell carcinoma. Tumor-specific (somatic) mutations were assessed in all sequenced tumors to generate mutation annotation files (MAFs), which were later joined with MAFs from previously published studies in clinical cohorts in melanoma (Van Allen et al. (2015) *Science* 350:207-211; Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199; Hugo et al. (2016) *Cell* 165:35-44) and non-small cell lung cancer (Rizvi et al. (2015) *Science* 348:124-128).

In particular, pre-treatment samples from cancer patients (N=268) were tested for their genetic sequences in relation to results of immune checkpoint therapy experienced by the cancer patients (FIG. 3). The results are shown in Table 2 as response by cancer type.

TABLE 2

	Melanoma	Lung	HNSCC	Sarcoma	Bladder
Clinical benefit	66 (38.6%)	20 (29.4%)	4 (28.6%)	1 (100%)	6 (46.2%)
Intermediate benefit	6 (3.5%)	23 (33.8%)	3 (21.4%)	0	1 (7.7%)
No clinical benefit	96 (56.1%)	25 (36.8%)	7 (50.0%)	0	4 (30.8%)
Mixed response or not evaluable	3 (1.8%)	0	0	0	2 (1.5%)
Total	171	68	14	1	13

Nonsynonymous mutational burden is a strong predictor of clinical outcomes across cancer types. As shown in FIG. 4, patients experiencing clinical benefit to immune checkpoint therapy had significantly more nonsynonymous mutations than those experiencing no clinical benefit ($p=5.52e-06$; Wilcoxon rank-sum). Such relationship is strong for patients with lung cancer ($p=5.2e-05$) and also significant for patients with bladder cancer ($p=0.019$) and melanoma ($p=0.0016$). No relationship between clinical outcome and mutational burden in HNSCC ($p=0.45$) was found. In addition, complete responders in sarcoma and anal cancer had unexceptional mutational burdens (below and near median, respectively).

In order to identify biomarkers associated with response or nonresponse to immune checkpoint blockade, the following procedure was conducted:

Split all patients into responders and non-responders

Version A: N=98 CB vs. N=132 NCB

Version B: N=98 CB vs. N=165 NCB or IB

Calculate significance value for enrichment of mutations in a given gene in responder or non-responders (Fisher's exact test)

Version A: All nonsynonymous mutations

Version B: Only truncating alterations (frame-shift insertions and deletions, nonsense single nucleotide polymorphisms, splice-site alterations)

Run MutSigCV on all 268 patients with called mutations to determine significantly mutated genes across the entire cohort, which takes into account gene size,

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patient-specific mutational rate, mutational spectra, and gene replication rates and times

Select genes that are mutated significantly more often in responders or non-responders ($p < 0.05$) and pass MutSigCV significance ($q < 0.1$; FDR).

The identified biomarkers are shown in FIG. 5. Table 3 is a summary of all genes with significantly more nonsynonymous mutations in R (N=98) or NR (N=132).

TABLE 3

Gene	N (R)	N (NR)	p-value (Fisher's exact)
CETP	4	0	0.032
COL3A1	18	10	0.015
COL5A1	22	15	0.029
DDX60	13	5	0.012
DHX8	10	2	0.005
DLEC1	16	9	0.031
DNAH2	23	14	0.011
FHOD1	8	2	0.020
HK3	7	2	0.040
KALRN	27	10	0.000093
KIF21B	15	9	0.049
KIF5A	11	5	0.036
KRAS	11	3	0.0095
MCTP1	12	6	0.045
MGAM	33	26	0.022
NARS2	6	0	0.0055
ZNF253	10	4	0.047
NF1	25	17	0.016
<i>NR1H4</i>	2	11	0.046
PBRM1	10	4	0.047
PKP1	7	2	0.040
PLCB4	21	11	0.0065
POLR2A	10	4	0.047
PREX2	25	19	0.042
<i>RB1</i>	1	10	0.026
ROCK1	8	2	0.020
SAFB2	9	3	0.032
SERPINB3	11	5	0.036
SPATA16	12	6	0.045

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TABLE 3-continued

Gene	N (R)	N (NR)	p-value (Fisher's exact)
STAB1	17	10	0.037
TGM3	11	5	0.036
TSC1	13	6	0.027
TSPAN2	7	2	0.040
ZNF207	6	1	0.044

10 *Genes in italics (i.e., *NR1H4* and *RB1*) mutated more frequently in NR vs. R. Genes in plain text mutated more frequently in R vs. NR.

**Bolded genes (i.e., DHX8, KALRN, KRAS, NARS2, and PLCB4) have $p < 0.01$.

After limiting analyses to comparing patients with objective tumor response (CR, PR, or SD by RECIST vs. PD by RECIST) in non-melanoma cancer types, it was observed that a striking association exists between mutations in one or more SWI/SNF complex subunits and response to immune checkpoint therapy (Tables 4-5 and FIGS. 6-8).

For example, truncating alterations in PBRM1 and response to immune checkpoint therapy, driven by nonsense, frameshift, and splice site mutations in bladder cancer, lung cancer, and renal cell carcinoma (9/75 responders vs. 0/41 non-responders, $p = 0.026$). Additionally, it was observed that ARID2 truncating mutations enriched in responders in melanoma across multiple clinical cohorts (6/68 responders vs. 2/96 non-responders), as well as in isolated cases in other tumor types (one frameshift deletion in lung cancer PR and one frameshift deletion in one SD RCC). Interestingly, the two ARID2 alterations occurring in non-responders occurred in patients receiving anti-CTLA4 therapies (rather than anti-PD1 therapies), though one patient with PR to anti-CTLA4 also had an ARID2 splice site mutation. Lastly, it was observed that mutations in SMARCA4 were associated with response in head and neck squamous cell carcinoma (3/6 responders vs. 0/9 non-responders, $p = 0.044$, Fisher's exact test). Thus, alterations in the SWI/SNF pathway were found to be predictive of response to immune checkpoint therapy across cancer types.

TABLE 4

Complete list of all identified SWI/SNF mutations

Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	ACTL6A	86	3	179294461	179294461	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ACTL6A	86	3	179298455	179298455	Missense_Mutation	+
Pt20	Pt20	ACTL6A	86	3	179294018	179294018	Missense_Mutation	-
CR04885	CR04885	ACTL6B	51412	7	100244379	100244379	Missense_Mutation	+
Lung-DFCI-11-104-009-Tumor-SM-5YS7O	Lung-DFCI-11-104-009-TM-NB-SM-5YS7O-SM-5YS7P	ACTL6B	51412	7	100244451	100244451	Missense_Mutation	+
MEL-IPI_Pat130-Tumor-SM-5X2R8	MEL-IPI_Pat130-TP-NT-SM-5X2R8-SM-5X2RJ	ACTL6B	51412	7	100247744	100247744	Silent	-
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ACTL6B	51412	7	100240903	100240903	Missense_Mutation	+
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	ACTL6B	51412	7	100244908	100244908	Splice_Site	-
MEL-IPI_Pat28-Tumor-SM-4DK10	MEL-IPI_Pat28-TP-NB-SM-4DK10-SM-4NFUU	ACTL6B	51412	7	100244416	100244416	Silent	-
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	ACTL6B	51412	7	100244877	100244877	Silent	+
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	ACTL6B	51412	7	100244902	100244902	Silent	+
PR4035	PR4035	ACTL6B	51412	7	100253064	100253064	Missense_Mutation	+
Pt14	Pt14	ACTL6B	51412	7	100253200	100253200	Missense_Mutation	-

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
Pt4	Pt4	ACTL6B	51412	7	100246273	100246273	Missense_Mutation	+
RH090935	RH090935	ACTL6B	51412	7	100253478	100253478	Missense_Mutation	+
SD1494	SD1494	ACTL6B	51412	7	100253045	100253045	Splice_Site	±
SU2C_Lung-SU2C-DFCI-LUAD-1011-Tumor-SM-AOL75	SU2C_Lung-SU2C-DFCI-LUAD-1011-TM-NB-SM-AOL75-SM-A46NN	ACTL6B	51412	7	100252740	100252740	Missense_Mutation	-
AL4602	AL4602	ARID1A	8289	1	27023633	27023633	Missense_Mutation	±
BLADDER-15330_CCPM_0700692-Tumor-SM-AV111	BLADDER-15330_CCPM_0700692-TM-NB-SM-AV111-SM-AVHZM	ARID1A	8289	1	27101612	27101612	Frame_Shift_Del	-
BLCA-IM01-Tumor-SM-79XD9	BLCA-IM01-TP-NB-SM-79XD9-SM-7AABJ	ARID1A	8289	1	27057664	27057664	Missense_Mutation	-
BLCA-IM01-Tumor-SM-79XD9	BLCA-IM01-TP-NB-SM-79XD9-SM-7AABJ	ARID1A	8289	1	27058092	27058092	Splice_Site	-
BLCA-IM01-Tumor-SM-79XD9	BLCA-IM01-TP-NB-SM-79XD9-SM-7AABJ	ARID1A	8289	1	27057642	27057642	Splice_Site	-
HNSCC-287-Tumor-SM-AXGEI	HNSCC-287-TP-NB-SM-AXGEI-SM-ADP7M	ARID1A	8289	1	27100352	27100355	Frame_Shift_Del	-
LO3793	LO3793	ARID1A	8289	1	27105688	27105688	Nonsense_Mutation	±
LUAD-BS-13-F33496-Tumor-SM-9J2XU	LUAD-BS-13-F33496-TP-NB-SM-9J2XU-SM-9HBZX	ARID1A	8289	1	27056342	27056343	Frame_Shift_Del	±
Lung-DFCI-11-104-009-Tumor-SM-5YS7O	Lung-DFCI-11-104-009-IM-NB-SM-5YS7O-SM-5YS7P	ARID1A	8289	1	27106621	27106621	Nonsense_Mutation	+
MEL-IPI_Pat7-Tumor-SM-4DK13	MEL-IPI_Pat07-IP-NB-SM-4DK13-SM-4NFU9	ARID1A	8289	1	27105918	27105918	Silent	+
MEL-IPI_Pat11-Tumor-SM-4DK17	MEL-IPI_Pat11-TP-NB-SM-4DK17-SM-4NFUD	ARID1A	8289	1	27101642	27101642	Missense_Mutation	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	ARID1A	8289	1	27106693	27106693	Missense_Mutation	-
MEL-IPI_Pat133-Tumor-SM-5VWJB	MEL-IPI_Pat133-TP-NB-SM-5VWJB-SM-5VWHS	ARID1A	8289	1	27089607	27089607	Missense_Mutation	-
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1A	8289	1	27023487	27023487	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1A	8289	1	27106461	27106461	Silent	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1A	8289	1	27058070	27058070	Missense_Mutation	+
MEL-IPI_Pat159-Tumor-SM-5VWK2	MEL-IPI_Pat159-TP-NB-SM-5VWK2-SM-5VWIJ	ARID1A	8289	1	27087452	27087452	Missense_Mutation	-
MEL-IPI_Pat163-Tumor-SM-5VWK6	MEL-IPI_Pat163-TP-NB-SM-5VWK6-SM-5VWIN	ARID1A	8289	1	27107176	27107176	Missense_Mutation	-
MEL-IPI_Pat37-Tumor-SM-53U3Y	MEL-IPI_Pat37-TP-NB-SM-53U3Y-SM-4NFV4	ARID1A	8289	1	27101712	27101712	Splice_Site	-
MEL-IPI_Pat37-Tumor-SM-53U3Y	MEL-IPI_Pat37-TP-NB-SM-53U3Y-SM-4NFV4	ARID1A	8289	1	27105837	27105837	Silent	-
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	ARID1A	8289	1	27087393	27087393	Missense_Mutation	+
MEL-IPI_Pat39-Tumor-SM-4DK1Z	MEL-IPI_Pat39-TP-NB-SM-4DK1Z-SM-4NFV6	ARID1A	8289	1	27057822	27057822	Silent	+
MEL-IPI_Pat62-Tumor-SM-4DK2N	MEL-IPI_Pat62-TP-NB-SM-4DK2N-SM-4NFVT	ARID1A	8289	1	27087494	27087494	Missense_Mutation	-
MEL-IPI_Pat64-Tumor-SM-4DK2P	MEL-IPI_Pat64-TP-NB-SM-4DK2P-SM-4NFVV	ARID1A	8289	1	27099940	27099940	Missense_Mutation	-
MEL-IPI_Pat85-Tumor-SM-53U2Y	MEL-IPI_Pat85-TP-NB-SM-53U2Y-SM-4NFWH	ARID1A	8289	1	27058033	27058033	Nonsense_Mutation	-
PR11217	PR11217	ARID1A	8289	1	27093053	27093053	Missense_Mutation	+
PR4077	PR4077	ARID1A	8289	1	27089494	27089494	Missense_Mutation	+
Pt15	Pt15	ARID1A	8289	1	27101090	27101090	Intron	+
SU2C_Lung-SU2C-DFCI-LUAD-1017-Tumor-SM-AOL99	SU2C_Lung-SU2C-DFCI-LUAD-1017-TM-NB-SM-AOL99-SM-A46NT	ARID1A	8289	1	27101525	27101525	Frame_Shift_Del	+
SU2C_Lung-SU2C-DFCI-LUAD-1017-Tumor-SM-AOL99	SU2C_Lung-SU2C-DFCI-LUAD-1017-TM-NB-SM-AOL99-SM-A46NT	ARID1A	8289	1	27106878	27106878	Frame_Shift_Del	+
Y2087	Y2087	ARID1A	8289	1	27023360	27023360	Missense_Mutation	±
ZA6965	ZA6965	ARID1A	8289	1	27023690	27023690	Missense_Mutation	+

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
BLCA-IM01-Tumor-SM-79XD9	BLCA-IM01-TP-NB-SM-79XD9-SM-7AABJ	ARID1B	57492	6	157100005	157100005	Silent	-
HNSCC-239-Tumor-SM-AXGCS	HNSCC-239-TP-NB-SM-AXGCS-SM-ADP7K	ARID1B	57492	6	157100005	157100005	Silent	±
HNSCC-243-Tumor-SM-CLFNS	HNSCC-243-TP-NB-SM-CLFNS-SM-AV34T	ARID1B	57492	6	157100005	157100005	Silent	-
LUAD-BS-11-R21845-Tumor-SM-9J2YH	LUAD-BS-11-R21845-TP-NT-SM-9J2YH-SM-9J2YI	ARID1B	57492	6	157100005	157100005	Silent	+
LUAD-BS-13-X14864-Tumor-SM-9J2XQ	LUAD-BS-13-X14864-TP-NB-SM-9J2XQ-SM-9HBZU	ARID1B	57492	6	157528372	157528372	Missense_Mutation	±
LUAD-BS-14-G65174-Tumor-SM-9J2YF	LUAD-BS-14-G65174-TP-NT-SM-9J2YF-SM-9J2YG	ARID1B	57492	6	157100005	157100005	Silent	-
MEL-682321-Tumor-SM-CN21G	MEL-682321-TP-NB-SM-CN21G-SM-CJP7S	ARID1B	57492	6	157099512	157099512	Missense_Mutation	+
MEL-682321-Tumor-SM-CN21G	MEL-682321-TP-NB-SM-CN21G-SM-CJP7S	ARID1B	57492	6	157099511	157099511	Missense_Mutation	+
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	ARID1B	57492	6	157517305	157517305	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1B	57492	6	157522597	157522597	Silent	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1B	57492	6	157527479	157527479	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1B	57492	6	157502265	157502265	Silent	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID1B	57492	6	157511325	157511325	Silent	+
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	ARID1B	57492	6	157527627	157527627	Silent	-
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	ARID1B	57492	6	157505558	157505558	Missense_Mutation	-
MEL-IPI_Pat174-Tumor-SM-5VOB4	MEL-IPI_Pat174-TP-NB-SM-5VOB4-SM-5VWIY	ARID1B	57492	6	157528667	157528667	Missense_Mutation	+
MEL-IPI_Pat21-Tumor-SM-4DK1H	MEL-IPI_Pat21-TP-NT-SM-4DK1H-SM-53U5G	ARID1B	57492	6	157522222	157522222	Missense_Mutation	+
MEL-IPI_Pat39-Tumor-SM-4DK1Z	MEL-IPI_Pat39-TP-NB-SM-4DK1Z-SM-4NFW6	ARID1B	57492	6	157511303	157511303	Missense_Mutation	+
MEL-IPI_Pat74-Tumor-SM-4DK2Z	MEL-IPI_Pat74-TP-NB-SM-4DK2Z-SM-4NFW6	ARID1B	57492	6	157100377	157100377	Silent	-
MEL-IPI_Pat74-Tumor-SM-4DK2Z	MEL-IPI_Pat74-TP-NB-SM-4DK2Z-SM-4NFW6	ARID1B	57492	6	157100376	157100376	Missense_Mutation	-
PR11217	PR11217	ARID1B	57492	6	157522344	157522344	Missense_Mutation	+
PR4092	PR4092	ARID1B	57492	6	157521990	157521990	Missense_Mutation	+
Pt8	Pt8	ARID1B	57492	6	157222594	157222594	Missense_Mutation	+
SA9755	SA9755	ARID1B	57492	6	157522508	157522508	Missense_Mutation	+
SD1494	SD1494	ARID1B	57492	6	157522095	157522095	Missense_Mutation	±
SU2C_Lung-SU2C-DFCI-LUAD-1006-Tumor-SM-AOL5E	SU2C_Lung-SU2C-DFCI-LUAD-1006-TP-NB-SM-AOL5E-SM-A46NI	ARID1B	57492	6	157100005	157100005	Silent	-
SU2C_Lung-SU2C-DFCI-LUAD-1011-Tumor-SM-AOL75	SU2C_Lung-SU2C-DFCI-LUAD-1011-TM-NB-SM-AOL75-SM-A46NN	ARID1B	57492	6	157222621	157222621	Missense_Mutation	-
Y2087	Y2087	ARID1B	57492	6	157099481	157099481	Missense_Mutation	±
Case1-BaselineTumor	Case1-TP-NB-Zaretsky	ARID2	196528	12	46287234	46287234	Missense_Mutation	+
Case3-BaselineTumor	Case3-TP-NB-Zaretsky	ARID2	196528	12	46243857	46243857	Nonsense_Mutation	+
HNSCC-323-Tumor-SM-CK9WS	HNSCC-323-TP-NB-SM-CK9WS-SM-AV34N	ARID2	196528	12	46240638	46240638	Splice_Site	+
LSD6819	LSD6819	ARID2	196528	12	46243857	46243857	Nonsense_Mutation	+
LUAD-BS-08-013532-Tumor-SM-9J2Y1	LUAD-BS-08-013532-TP-NT-SM-9J2Y1-SM-9J2Y2	ARID2	196528	12	46245525	46245525	Frame_Shift_Del	+
LUAD-BS-13-J60666-Tumor-SM-9J2YL	LUAD-BS-13-J60666-TP-NB-SM-9J2YL-SM-9HBZW	ARID2	196528	12	46246071	46246071	Missense_Mutation	+
MEL-650366-Tumor-SM-CN221	MEL-650366-TP-NB-SM-CN221-SM-CJP7U	ARID2	196528	12	46245857	46245857	Silent	-
MEL-IPI_Pat100-Tumor-SM-53U2D	MEL-IPI_Pat100-TP-NT-SM-53U2D-SM-53U4M	ARID2	196528	12	46230641	46230641	Missense_Mutation	-

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
MEL-IPI_Pat100-Tumor-SM-53U2D	MEL-IPI_Pat100-TP-NT-SM-53U2D-SM-53U4M	ARID2	196528	12	46242701	46242701	Nonsense_Mutation	-
MEL-IPI_Pat103-Tumor-SM-4CU6Q	MEL-IPI_Pat103-TP-NT-SM-4CU6Q-SM-53U4P	ARID2	196528	12	46243514	46243514	Missense_Mutation	+
MEL-IPI_Pat109-Tumor-SM-4CU6W	MEL-IPI_Pat109-TP-NT-SM-4CU6W-SM-4MGPN	ARID2	196528	12	46243825	46243825	Missense_Mutation	-
MEL-IPI_Pat109-Tumor-SM-4CU6W	MEL-IPI_Pat109-TP-NT-SM-4CU6W-SM-4MGPN	ARID2	196528	12	46243824	46243824	Missense_Mutation	-
MEL-IPI_Pat115-Tumor-SM-5X2QS	MEL-IPI_Pat115-TP-NT-SM-5X2QS-SM-5X2RA	ARID2	196528	12	46211474	46211474	Missense_Mutation	-
MEL-IPI_Pat117-Tumor-SM-5X2QU	MEL-IPI_Pat117-TP-NT-SM-5X2QU-SM-5X2RC	ARID2	196528	12	46244997	46244997	Nonsense_Mutation	+
MEL-IPI_Pat117-Tumor-SM-5X2QU	MEL-IPI_Pat117-TP-NT-SM-5X2QU-SM-5X2RC	ARID2	196528	12	46245843	46245843	Nonsense_Mutation	+
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	ARID2	196528	12	46243530	46243530	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	ARID2	196528	12	46242749	46242749	Missense_Mutation	+
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	ARID2	196528	12	46205217	46205217	Missense_Mutation	-
MEL-IPI_Pat151-Tumor-SM-5VWJT	MEL-IPI_Pat151-TP-NB-SM-5VWJT-SM-5VWIB	ARID2	196528	12	46287240	46287240	Missense_Mutation	-
MEL-IPI_Pat159-Tumor-SM-5VWK2	MEL-IPI_Pat159-TP-NB-SM-5VWK2-SM-5VWIJ	ARID2	196528	12	46244889	46244889	Nonsense_Mutation	-
MEL-IPI_Pat174-Tumor-SM-5VOB4	MEL-IPI_Pat174-TP-NB-SM-5VOB4-SM-5VWIY	ARID2	196528	12	46215271	46215271	Splice_Site	+
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	ARID2	196528	12	46245648	46245648	Missense_Mutation	-
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	ARID2	196528	12	46240672	46240672	Missense_Mutation	-
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	ARID2	196528	12	46287428	46287428	Missense_Mutation	-
MEL-IPI_Pat66-Tumor-SM-4DK2R	MEL-IPI_Pat66-TP-NB-SM-4DK2R-SM-4NFVX	ARID2	196528	12	46230691	46230691	Missense_Mutation	+
PR11217	PR11217	ARID2	196528	12	46233249	46233249	Nonsense_Mutation	+
PR11217	PR11217	ARID2	196528	12	46245639	46245639	Nonsense_Mutation	+
PR4077	PR4077	ARID2	196528	12	46243857	46243857	Nonsense_Mutation	+
PR4092	PR4092	ARID2	196528	12	46242619	46242619	Splice_Site	+
Pt1	Pt1	ARID2	196528	12	46123846	46123846	Missense_Mutation	-
Pt31	Pt31	ARID2	196528	12	46243467	46243467	Missense_Mutation	-
Pt37	Pt37	ARID2	196528	12	46211600	46211600	Frame_Shift_Del	+
SU2C_Lung-SU2C-DFCI-LUAD-1016-Tumor-SM-AOL8W	SU2C_Lung-SU2C-DFCI-LUAD-1016-TM-NB-SM-AOL8W-SM-A46NS	ARID2	196528	12	46244393	46244393	Silent	±
WA7899	WA7899	ARID2	196528	12	46244529	46244529	Missense_Mutation	-
DM123062	DM123062	BRD7	29117	16	50384049	50384049	Missense_Mutation	-
Lung-DFCI-11-104-009-Tumor-SM-5YS7O	Lung-DFCI-11-104-009-TM-NB-SM-5YS7O-SM-5YS7P	BRD7	29117	16	50388348	50388348	Missense_Mutation	+
MEL-IPI_Pat03-Tumor-SM-4DJZY	MEL-IPI_Pat03-TP-NB-SM-4DJZY-SM-4NFU5	BRD7	29117	16	50357497	50357497	Splice_Site	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	BRD7	29117	16	50368748	50368748	Missense_Mutation	-
MEL-IPI_Pat03-Tumor-SM-4DJZY	MEL-IPI_Pat03-TP-NB-SM-4DJZY-SM-4NFU5	DPF1	8193	19	38706825	38706825	Missense_Mutation	-
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	DPF1	8193	19	38709622	38709622	Silent	+
MEL-IPI_Pat134-Tumor-SM-7A151	MEL-IPI_Pat134-TP-NB-SM-7A151-SM-5VWHT	DPF1	8193	19	38709646	38709646	Silent	-
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	DPF1	8193	19	38702995	38702995	Silent	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	DPF1	8193	19	38704352	38704352	Missense_Mutation	+
MEL-IPI_Pat74-Tumor-SM-4DK2Z	MEL-IPI_Pat74-TP-NB-SM-4DK2Z-SM-4NFW6	DPF1	8193	19	38712998	38712998	Splice_Site	-
Pt27	Pt27	DPF1	8193	19	38709621	38709621	Missense_Mutation	+

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
LUAD-BS-12-R10269-Tumor-SM-9J2XO	LUAD-BS-12-R10269-TP-NB-SM-9J2XO-SM-9HBZT	DPF2	5977	11	65108997	65108997	Silent	+
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	DPF2	5977	11	65107914	65107914	Missense_Mutation	+
MEL-IPI_Pat159-Tumor-SM-5VWK2	MEL-IPI_Pat159-TP-NB-SM-5VWK2-SM-5VWIJ	DPF2	5977	11	65113741	65113741	Intron	-
MEL-IPI_Pat32-Tumor-SM-53U3T	MEL-IPI_Pat32-TP-NT-SM-53U3T-SM-53U67	DPF2	5977	11	65108462	65108462	Silent	-
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	DPF2	5977	11	65123565	65123565	IGR	+
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	DPF2	5977	11	65113530	65113530	Intron	-
MEL-IPI_Pat74-Tumor-SM-4DK2Z	MEL-IPI_Pat74-TP-NB-SM-4DK2Z-SM-4NFW6	DPF2	5977	11	65113812	65113812	Intron	-
Pt13	Pt13	DPF2	5977	11	65111304	65111304	Intron	+
Pt14	Pt14	DPF2	5977	11	65109007	65109007	Missense_Mutation	-
SU2C_Lung-SU2C-DFCI-LUAD-1011-Tumor-SM-AOL75	SU2C_Lung-SU2C-DFCI-LUAD-1011-TM-NB-SM-AOL75-SM-A46NN	DPF2	5977	11	65113251	65113251	Intron	-
DFCI_MM_2-Tumor-SM-BZRJA	DFCI_MM_2-TP-NB-SM-BZRJA-SM-BZRJD	DPF3	8110	14	73137945	73137945	Intron	+
DFCI_MM_2-Tumor-SM-BZRJA	DFCI_MM_2-TP-NB-SM-BZRJA-SM-BZRJD	DPF3	8110	14	73238507	73238507	Missense_Mutation	+
FR9547	FR9547	DPF3	8110	14	73140993	73140993	Missense_Mutation	+
Lung-DFCI-11-104-009-Tumor-SM-5YS7O	Lung-DFCI-11-104-009-TM-NB-SM-5YS7O-SM-5YS7P	DPF3	8110	14	73137964	73137964	Intron	+
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	DPF3	8110	14	73137905	73137905	Intron	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	DPF3	8110	14	73137904	73137904	Intron	-
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	DPF3	8110	14	73138006	73138006	Intron	-
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	DPF3	8110	14	73138005	73138005	Intron	-
MEL-IPI_Pat16-Tumor-SM-53U3E	MEL-IPI_Pat16-TP-NT-SM-53U3E-SM-53U5B	DPF3	8110	14	73220050	73220050	Missense_Mutation	-
MEL-IPI_Pat21-Tumor-SM-4DK1H	MEL-IPI_Pat21-TP-NT-SM-4DK1H-SM-53U5G	DPF3	8110	14	73190370	73190370	Missense_Mutation	+
SA9755	SA9755	DPF3	8110	14	73190391	73190391	Missense_Mutation	+
BLCA-IM07-Tumor-SM-79XDD	BLCA-IM07-TP-NB-SM-79XDD-SM-7AABP	PBRM1	55193	3	52621431	52621431	Missense_Mutation	+
CR04885	CR04885	PBRM1	55193	3	52597336	52597336	Missense_Mutation	+
M4945	M4945	PBRM1	55193	3	52696148	52696148	Splice_Site	+
MA7027	MA7027	PBRM1	55193	3	52598231	52598231	Missense_Mutation	-
MEL-IPI_Pat103-Tumor-SM-4CU6Q	MEL-IPI_Pat103-TP-NT-SM-4CU6Q-SM-53U4P	PBRM1	55193	3	52620592	52620592	Missense_Mutation	+
MEL-IPI_Pat103-Tumor-SM-4CU6Q	MEL-IPI_Pat103-TP-NT-SM-4CU6Q-SM-53U4P	PBRM1	55193	3	52620593	52620593	Missense_Mutation	+
MEL-IPI_Pat118-Tumor-SM-5X2QV	MEL-IPI_Pat118-TP-NT-SM-5X2QV-SM-5X2RD	PBRM1	55193	3	52692325	52692325	Missense_Mutation	-
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	PBRM1	55193	3	52595959	52595959	Missense_Mutation	+
MEL-IPI_Pat151-Tumor-SM-5VWJT	MEL-IPI_Pat151-TP-NB-SM-5VWJT-SM-5VWIB	PBRM1	55193	3	52643530	52643530	Missense_Mutation	-
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	PBRM1	55193	3	52643692	52643692	Missense_Mutation	+
MEL-IPI_Pat70-Tumor-SM-4DK2V	MEL-IPI_Pat70-TP-NB-SM-4DK2V-SM-4NFW2	PBRM1	55193	3	52584527	52584527	Missense_Mutation	-
MEL-IPI_Pat79-Tumor-SM-53U2S	MEL-IPI_Pat79-TP-NB-SM-53U2S-SM-4NFWB	PBRM1	55193	3	52621315	52621315	Intron	+
MEL-IPI_Pat88-Tumor-SM-4DK3E	MEL-IPI_Pat88-TP-NT-SM-4DK3E-SM-53U4C	PBRM1	55193	3	52668815	52668815	Silent	+
MEL-IPI_Pat88-Tumor-SM-4DK3E	MEL-IPI_Pat88-TP-NT-SM-4DK3E-SM-53U4C	PBRM1	55193	3	52668765	52668765	Missense_Mutation	+
PR4035	PR4035	PBRM1	55193	3	52643768	52643768	Nonsense_Mutation	+
Pt13	Pt13	PBRM1	55193	3	52643768	52643768	Nonsense_Mutation	+

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
SU2C_Lung-SU2C-DFCI-LUAD-1017-Tumor-SM-AOL99	SU2C_Lung-SU2C-DFCI-LUAD-1017-TM-NB-SM-AOL99-SM-A46NT	PBRM1	55193	3	52651406	52651406	Nonsense_Mutation	+
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	PHF10	55274	6	170112483	170112483	Splice_Site	-
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	PHF10	55274	6	170116103	170116103	Missense_Mutation	+
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	PHF10	55274	6	170117919	170117919	Splice_Site	-
Pt1	Pt1	PHF10	55274	6	170112579	170112579	Missense_Mutation	-
Pt2	Pt2	PHF10	55274	6	170116131	170116131	Missense_Mutation	+
SD1494	SD1494	PHF10	55274	6	170117924	170117924	Missense_Mutation	±
BLCA-IM10-Tumor-SM-79XDG	BLCA-IM10-TP-NB-SM-79XDG-SM-9QSPX	SMARCA2	6595	9	2181573	2181573	Missense_Mutation	+
BLCA-IM11-Tumor-SM-79XDH	BLCA-IM11-TP-NT-SM-79XDH-SM-79XDI	SMARCA2	6595	9	2033008	2033008	Missense_Mutation	+
LSD0167	LSD0167	SMARCA2	6595	9	2161819	2161819	Missense_Mutation	+
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	SMARCA2	6595	9	2039901	2039901	Splice_Site	+
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	SMARCA2	6595	9	2186134	2186134	Silent	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	SMARCA2	6595	9	2056722	2056722	Silent	+
MEL-IPI_Pat15-Tumor-SM-4DK1B	MEL-IPI_Pat15-TP-NB-SM-4DK1B-SM-4NFUH	SMARCA2	6595	9	2070473	2070473	Splice_Site	-
MEL-IPI_Pat21-Tumor-SM-4DK1H	MEL-IPI_Pat21-TP-NT-SM-4DK1H-SM-53U5G	SMARCA2	6595	9	2161845	2161845	Missense_Mutation	+
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	SMARCA2	6595	9	2039623	2039623	Silent	+
MEL-IPI_Pat90-Tumor-SM-4DK3G	MEL-IPI_Pat90-TP-NB-SM-4DK3G-SM-4NFWM	SMARCA2	6595	9	2186142	2186142	Missense_Mutation	+
PR4092	PR4092	SMARCA2	6595	9	2161836	2161836	Missense_Mutation	+
Pt31	Pt31	SMARCA2	6595	9	2104046	2104046	Missense_Mutation	-
RH090935	RH090935	SMARCA2	6595	9	2123911	2123911	Missense_Mutation	+
SD2056	SD2056	SMARCA2	6595	9	2077654	2077654	Missense_Mutation	+
Y2087	Y2087	SMARCA2	6595	9	2047355	2047355	Missense_Mutation	±
BLCA-IM07-Tumor-SM-79XDD	BLCA-IM07-TP-NB-SM-79XDD-SM-7AABP	SMARCA4	6597	19	11134267	11134267	Missense_Mutation	+
BLCA-IM09-Tumor-SM-79XDF	BLCA-IM09-TP-NB-SM-79XDF-SM-7AABN	SMARCA4	6597	19	11097617	11097617	Missense_Mutation	+
FR9547	FR9547	SMARCA4	6597	19	11136975	11136975	Splice_Site	+
HNSCC-186-Tumor-SM-AXGDN	HNSCC-186-TP-NB-SM-AXGDN-SM-ADP7L	SMARCA4	6597	19	11144853	11144853	3'UTR	+
HNSCC-186-Tumor-SM-AXGDN	HNSCC-186-TP-NB-SM-AXGDN-SM-ADP7L	SMARCA4	6597	19	11144853	11144853	3'UTR	+
HNSCC-258-Tumor-SM-AXGAI	HNSCC-258-TP-NB-SM-AXGAI-SM-ADP7G	SMARCA4	6597	19	11170556	11170556	Missense_Mutation	+
HNSCC-323-Tumor-SM-CK9WS	HNSCC-323-TP-NB-SM-CK9WS-SM-AV34N	SMARCA4	6597	19	11096069	11096069	Nonsense_Mutation	+
LUAD-BS-13-J60666-Tumor-SM-9J2YL	LUAD-BS-13-J60666-TP-NB-SM-9J2YL-SM-9HBZW	SMARCA4	6597	19	11132428	11132428	Missense_Mutation	+
LUAD-BS-14-G65174-Tumor-SM-9J2YF	LUAD-BS-14-G65174-TP-NT-SM-9J2YF-SM-9J2YG	SMARCA4	6597	19	11145805	11145805	3'UTR	-
M4945	M4945	SMARCA4	6597	19	11144072	11144072	IGR	+
MA7027	MA7027	SMARCA4	6597	19	11134207	11134207	Missense_Mutation	-
MEL-IPI_Pat08-Tumor-SM-4DK14	MEL-IPI_Pat08-TP-NB-SM-4DK14-SM-4NFUA	SMARCA4	6597	19	11121151	11121151	Missense_Mutation	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	SMARCA4	6597	19	11096986	11096986	Silent	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	SMARCA4	6597	19	11134230	11134230	Missense_Mutation	-
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	SMARCA4	6597	19	11144106	11144106	IGR	+

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	SMARCA4	6597	19	11144028	11144028	IGR	+
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	SMARCA4	6597	19	11136986	11136986	Missense_Mutation	-
MEL-IPI_Pat16-Tumor-SM-53U3E	MEL-IPI_Pat16-TP-NT-SM-53U3E-SM-53U5B	SMARCA4	6597	19	11141561	11141561	IGR	-
MEL-IPI_Pat19-Tumor-SM-4DK1F	MEL-IPI_Pat19-TP-NB-SM-4DK1F-SM-4NFUL	SMARCA4	6597	19	11144856	11144856	3'UTR	-
MEL-IPI_Pat28-Tumor-SM-4DK1O	MEL-IPI_Pat28-TP-NB-SM-4DK1O-SM-4NFUU	SMARCA4	6597	19	11137018	11137018	Nonsense_Mutation	-
MEL-IPI_Pat49-Tumor-SM-4DK2A	MEL-IPI_Pat49-TP-NT-SM-4DK2A-SM-53U5W	SMARCA4	6597	19	11121110	11121110	Missense_Mutation	±
MEL-IPI_Pat52-Tumor-SM-4DK2D	MEL-IPI_Pat52-TP-NT-SM-4DK2D-SM-53U5Z	SMARCA4	6597	19	11097614	11097614	Missense_Mutation	-
MEL-IPI_Pat54-Tumor-SM-4DK2F	MEL-IPI_Pat54-TP-NB-SM-4DK2F-SM-4NFVL	SMARCA4	6597	19	11100047	11100047	Silent	-
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	SMARCA4	6597	19	11098595	11098595	Silent	-
SA9755	SA9755	SMARCA4	6597	19	11170804	11170804	Nonsense_Mutation	-
SD1494	SD1494	SMARCA4	6597	19	11123685	11123685	Missense_Mutation	+
HN5CC-181-Tumor-SM-CK9X1	HN5CC-181-TP-NB-SM-CK9X1-SM-AV34P	SMARCB1	6598	22	24129440	24129440	Missense_Mutation	±
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	SMARCB1	6598	22	24143148	24143148	Silent	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	SMARCB1	6598	22	24143149	24143149	Intron	-
MEL-IPI_Pat130-Tumor-SM-5X2R8	MEL-IPI_Pat130-TP-NT-SM-5X2R8-SM-5X2RJ	SMARCB1	6598	22	24143281	24143281	Intron	-
MEL-IPI_Pat62-Tumor-SM-4DK2N	MEL-IPI_Pat62-TP-NB-SM-4DK2N-SM-4NFVT	SMARCB1	6598	22	24145537	24145537	Silent	-
Pt2	Pt2	SMARCB1	6598	22	24133958	24133958	Missense_Mutation	+
Lung-DFCI-11-104-009-Tumor-SM-5YS7O	Lung-DFCI-11-104-009-TM-NB-SM-5YS7O-SM-5YS7P	SMARCC1	6599	3	47823230	47823230	Missense_Mutation	+
MEL-IPI_Pat03-Tumor-SM-4DJZY	MEL-IPI_Pat03-TP-NB-SM-4DJZY-SM-4NFU5	SMARCC1	6599	3	47680267	47680267	Missense_Mutation	-
MEL-IPI_Pat03-Tumor-SM-4DJZY	MEL-IPI_Pat03-TP-NB-SM-4DJZY-SM-4NFU5	SMARCC1	6599	3	47787455	47787455	Missense_Mutation	-
MEL-IPI_Pat08-Tumor-SM-4DK14	MEL-IPI_Pat08-TP-NB-SM-4DK14-SM-4NFUA	SMARCC1	6599	3	47777539	47777539	Silent	-
MEL-IPI_Pat110-Tumor-SM-4CU6X	MEL-IPI_Pat110-TP-NT-SM-4CU6X-SM-4MGPO	SMARCC1	6599	3	47632172	47632172	Silent	-
MEL-IPI_Pat151-Tumor-SM-5VWJT	MEL-IPI_Pat151-TP-NB-SM-5VWJT-SM-5VWIB	SMARCC1	6599	3	47787430	47787430	Silent	-
MEL-IPI_Pat28-Tumor-SM-4DK1O	MEL-IPI_Pat28-TP-NB-SM-4DK1O-SM-4NFUU	SMARCC1	6599	3	47742863	47742863	Missense_Mutation	-
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	SMARCC1	6599	3	47651680	47651680	Silent	-
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	SMARCC1	6599	3	47770567	47770567	Silent	-
Pt4	Pt4	SMARCC1	6599	3	47629788	47629788	Missense_Mutation	+
CR04885	CR04885	SMARCC2	6601	12	56565627	56565627	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	SMARCC2	6601	12	56558459	56558459	Missense_Mutation	+
MEL-IPI_Pat139-Tumor-SM-5VWJH	MEL-IPI_Pat139-TP-NB-SM-5VWJH-SM-5VWHY	SMARCC2	6601	12	56558475	56558475	Missense_Mutation	-
MEL-IPI_Pat27-Tumor-SM-4DK1N	MEL-IPI_Pat27-TP-NB-SM-4DK1N-SM-4NFUT	SMARCC2	6601	12	56578857	56578857	Missense_Mutation	-
MEL-IPI_Pat32-Tumor-SM-53U3T	MEL-IPI_Pat32-TP-NT-SM-53U3T-SM-53U67	SMARCC2	6601	12	56572223	56572223	Silent	-
MEL-IPI_Pat58-Tumor-SM-4DK2J	MEL-IPI_Pat58-TP-NB-SM-4DK2J-SM-4NFVP	SMARCC2	6601	12	56563668	56563668	Missense_Mutation	-
MEL-IPI_Pat64-Tumor-SM-4DK2P	MEL-IPI_Pat64-TP-NB-SM-4DK2P-SM-4NFVV	SMARCC2	6601	12	56563227	56563230	Intron	-
MEL-IPI_Pat71-Tumor-SM-4DK2W	MEL-IPI_Pat71-TP-NB-SM-4DK2W-SM-4NFW3	SMARCC2	6601	12	56566475	56566475	Missense_Mutation	-
MEL-IPI_Pat77-Tumor-SM-4DK33	MEL-IPI_Pat77-TP-NT-SM-4DK33-SM-53U63	SMARCC2	6601	12	56575309	56575309	Missense_Mutation	+

TABLE 4-continued

Complete list of all identified SWI/SNF mutations								
Tumor_Sample_Barcode	pair_id	Hugo_Symbol	Entrez_Gene_Id	Chromosome	Start_position	End_position	Variant_Classification	response
MEL-IPI_Pat77-Tumor-SM-4DK33	MEL-IPI_Pat77-TP-NT-SM-4DK33-SM-53U63	SMARCC2	6601	12	56575308	56575308	Missense_Mutation	+
MEL-IPI_Pat62-Tumor-SM-4DK2N	MEL-IPI_Pat62-TP-NB-SM-4DK2N-SM-4NFVT	SMARCD1	6602	12	50480624	50480624	Missense_Mutation	-
Pt37	Pt37	SMARCD1	6602	12	50484135	50484135	Nonsense_Mutation	+
LUAD-BS-13-J60666-Tumor-SM-9J2YL	LUAD-BS-13-J60666-TP-NB-SM-9J2YL-SM-9HBZW	SMARCD2	6603	17	61912836	61912836	Missense_Mutation	+
MEL-IPI_Pat119-Tumor-SM-7459N	MEL-IPI_Pat119-TP-NT-SM-7459N-SM-7459Q	SMARCD2	6603	17	61914856	61914856	Nonsense_Mutation	-
MEL-IPI_Pat119-Tumor-SM-7459N	MEL-IPI_Pat119-TP-NT-SM-7459N-SM-7459Q	SMARCD2	6603	17	61914857	61914857	Silent	-
MEL-IPI_Pat151-Tumor-SM-5VWJT	MEL-IPI_Pat151-TP-NB-SM-5VWJT-SM-5VWIB	SMARCD2	6603	17	61912922	61912922	Silent	-
MEL-IPI_Pat21-Tumor-SM-4DK1H	MEL-IPI_Pat21-TP-NT-SM-4DK1H-SM-53U5G	SMARCD2	6603	17	61911039	61911039	Missense_Mutation	+
MEL-IPI_Pat38-Tumor-SM-53U3Z	MEL-IPI_Pat38-TP-NT-SM-53U3Z-SM-53U5L	SMARCD2	6603	17	61914827	61914827	Silent	+
Case3-BaselineTumor	Case3-TP-NB-Zaretsky	SMARCD3	6604	7	150939235	150939235	Silent	+
MEL-IPI_Pat11-Tumor-SM-4DK17	MEL-IPI_Pat11-TP-NB-SM-4DK17-SM-4NFUD	SMARCD3	6604	7	150939045	150939045	Missense_Mutation	-
BLADDER-15330_CCPM_0700694-Tumor-SM-AVI16	15330_CCPM_0700694-TM-NB-SM-AVI16-SM-AVHZK	SMARCE1	6605	17	38793628	38793632	Intron	+
MEL-IPI_Pat123-Tumor-SM-5X2R1	MEL-IPI_Pat123-TP-NB-SM-5X2R1-SM-5VWHL	SMARCE1	6605	17	38787856	38787856	Silent	+
MEL-IPI_Pat132-Tumor-SM-5VWJA	MEL-IPI_Pat132-TP-NB-SM-5VWJA-SM-5VWHR	SMARCE1	6605	17	38788513	38788513	Silent	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	SMARCE1	6605	17	38787103	38787103	Missense_Mutation	+
MEL-IPI_Pat138-Tumor-SM-5VWJG	MEL-IPI_Pat138-TP-NB-SM-5VWJG-SM-5VWHX	SMARCE1	6605	17	38785098	38785098	Missense_Mutation	+
MEL-IPI_Pat70-Tumor-SM-4DK2V	MEL-IPI_Pat70-TP-NB-SM-4DK2V-SM-4NFV2	SMARCE1	6605	17	38792665	38792665	Silent	-

All samples in Table 4 were from broad.mit.edu with NCBI-build no. of 37 and can be further identified based on the following information: ##Oncotator v1.2.7.0 Flat File Reference hg19|GENCODE v19|UnProt_AAxfm 2011_09|ClinVar 12.03.20|ESP 6500S1-V2|ORegAnno UCSC Track|dbSNP build 134|CCLE_By_GP 09292010|COSMIC v62_291112|1000Genome phase1|UniProt_AA 2011_09|dbNSFP v2.4|ESP 6500S1-V2|COSMIC_FusionGenes v62_291112|gencode_xref_refseq metadata

v19|CCLE_By_Gene 09292010|ACHILLES_Lineage_Results 110303|CGC full_2012-03-15|UniProt 2011_09|HumanDNARepairGenes 20110905|HGNC Nov2013|COSMIC_Tissue 291112|Familial_Cancer_Genes 20110905|TUMORScape 20100104|Ensembl ICGC MUCOPA|TCGAScape 110405|MutSig Published Results 20110905.

For responses, “+” represents having clinical benefit; “±” represents having intermediate benefit; and “-” represents having no clinical benefit.

TABLE 5

Hugo_Symbol	n_cb_truncating	n_ncb_truncating	n_truncating	n_cb_nonsyn	n_ncb_nonsyn	n_nonsyn
ARID1A	3	6	11	9	14	27
SMARCE1	0	0	0	2	0	2
ARID1B	0	0	0	11	3	17
SMARCA4	2	2	4	7	7	16
PBRM1	4	0	4	11	4	15
SMARCA2	1	1	2	9	2	12
ARID2	12	2	14	18	14	32
SMARCD3	0	0	0	0	1	1
ACTL6B	0	1	2	6	3	10
SMARCC2	0	0	0	4	4	8
DPF3	0	0	0	4	1	5
BRD7	0	1	1	1	3	4
SMARCB1	0	0	0	1	0	1
DPF2	0	0	0	1	1	2
SMARCD2	0	1	1	2	1	3
SMARCC1	0	0	0	2	3	5
DPF1	0	1	1	2	2	4

TABLE 5-continued

Hugo_Symbol	n_cb_truncating	n_ncb_truncating	n_truncating	n_cb_nonsyn	n_ncb_nonsyn	n_nonsyn
PHF10	0	2	2	2	3	6
ACTL6A	0	0	0	2	1	3
SMARCD1	1	0	1	1	1	2

All samples in Table 5 were taken from 98 patients with clinical benefit and 132 patients with no clinical benefit from immune checkpoint therapy. “n_cb_truncating” refers to the total number of patients with truncating mutation in a given gene with clinical benefit from immune checkpoint therapy; “n_ncb_truncating” refers to the total number of patients with truncating mutation in a given gene with no clinical benefit from immune checkpoint therapy; “n_truncating” refers to the total number of truncating mutations in a given gene in the cohort (includes patients with intermediate clinical benefit); “n_cb_nonsyn” refers to the total number of patients with nonsynonymous mutation in a given gene with clinical benefit from immune checkpoint therapy; “n_ncb_nonsyn” refers to the total number of patients with nonsynonymous mutation in a given gene with no clinical benefit from immune checkpoint therapy; and “n_nonsyn” refers to the total number of nonsynonymous mutations in a given gene in the cohort (includes patients with intermediate clinical benefit).

A summary of SWI/SNF complex is illustrated in FIG. 9. ARID2 and PBRM1 are two representative genes in the SWI/SNF complex, which were found in this study as relevant to sensitivity to immunotherapies such as those antagonizing immune checkpoints. SMARCA2 (also known as BRM) and ARID1B did not pass cohort-wide MutSig significance, but were mutated significantly more often in responders vs. non-responders (FIG. 10).

Alterations in PBRM1 are a common driver in clear-cell renal cell carcinoma (up to 40%), where it has a tumor suppressor function, but are rarer in other cancer types. This cohort contained 14 patients (8.3%) with nonsynonymous mutations in PBRM1 and 4 patients (1.5%, 2 with melanoma and 2 with non-small-cell lung cancer) with truncating alterations.

Similarly, ARID2 is a common driver mutation in hepatocellular carcinoma and melanoma. This cohort contained 26 patients (15.4%) with nonsynonymous mutations in ARID2 and 12 (7.1%, 10 with melanoma, 1 with head and neck squamous cell carcinoma, and 1 with non-small-cell lung cancer) with truncating alterations. Truncating (but not nonsynonymous) mutations in ARID2 were significantly associated with clinical benefit vs. no clinical benefit after controlling for nonsynonymous mutational load ($p=0.0051$; logistic regression) (FIG. 11). Nonsynonymous alterations in PBRM1 were marginally associated with clinical benefit ($p=0.058$) after controlling for mutational burden, while truncating mutations were not ($p=0.98$), perhaps due to the relative rarity of these events.

KDM6A encodes an enzyme called lysine-specific demethylase 6A that functions as a histone demethylase (FIG. 12). Truncating alterations in KDM6A were seen in 8 patients (4.8%, 5 with bladder cancer and 3 with melanoma) in this cohort. Truncating (but not nonsynonymous) alterations in KDM6A were marginally associated with clinical benefit ($p=0.089$; logistic regression) after controlling for mutational burden.

Immune checkpoint therapies can yield durable responses and long-lasting survival benefit across many cancer types,

and checkpoint therapies have been approved for use in metastatic melanoma, non-small cell lung cancer, bladder cancer, and renal cell carcinoma, including as a first-line therapy for lung cancer. While past studies have highlighted mutational load, neoantigen presentation, transcriptomic signatures, microbiome features, and immune cell infiltration as correlated with response to immune checkpoint therapies in melanoma, non-small-cell lung cancer, and bladder cancer, the results described herein indicate that nonsynonymous alterations in the SWI/SNF chromatin remodeling complex has predictive value for patient response to immune checkpoint therapies. Moreover, other biomarkers described herein, such as additional chromatin modifying genes like KDM6A and EGFR (resistance) biomarkers, were identified. In particular, EGFR showed a strong trend with intrinsic resistance to immune checkpoint therapy in lung cancer (FIG. 6). In addition, as described further in Example 4 below, cancers with hotspot mutations in EGFR are significantly less likely to respond to immune checkpoint therapies.

Thus, these results are believed to have wide-ranging implications for patient stratification for immune checkpoint therapies and those treated with other therapies, such as EGFR signaling inhibitors. Additionally, this finding drew from the largest set of clinically annotated cancer types yet collected (>200 pre-treatment patient tumors) across both well-studied and more poorly understood cancer types, lending great statistical power to detect associations. Finally, these results provide biomarkers, drug design, and combination treatment strategies across cancer types.

Example 3: Meta-Analysis of Genomic Predictors of Response to Immune Checkpoint Therapy in Metastatic Melanoma

Since immune checkpoint therapies only benefit a subset of patients with metastatic melanoma and the ability to predict clinical outcomes is limited, a meta-analysis of genomic predictors of outcomes to anti-PD1 blockade and anti-CTLA4 blockade in melanoma combining 220 sequenced tumors from 3 published cohorts was conducted in order to validate existing hypotheses regarding response to immune checkpoint therapies and discover new relationships with greater power.

Nonsynonymous mutational burden was significantly higher in clinical benefit (CB) vs. no clinical benefit (NCB) using all 3 response metrics, though the significance was less pronounced when using PFS alone ($p<0.01$ vs. $p<0.0001$; Wilcoxon rank sum), partially due to 3 patients with high mutational burden who experienced PR lasting <6 months, potentially representing early acquired rather than intrinsic resistance. In order to assess the impact of mutational processes contributing to overall mutational burden, a non-negative matrix factorization framework was used to infer mutational activity in tumors from 6 signatures previously seen in melanoma: aging (S1), T>C substitutions (S5), UV (S7), mismatch repair (S6), alkylating agents (S11), and T>G substitutions (S17). Across all samples, the proportion

of mutations in S7 or S11 was positively correlated with mutational burden (Spearman's $\rho=0.66$), while S5 and S1 were anti-correlated ($\rho=-0.62$). Additionally, in a multivariate logistic model, S7 and S11 activity were independent predictors of clinical benefit adjusting for mutational load ($p<0.05$), with the sum of S7 and S11 activity being a strong predictor ($p<0.001$). Of the patients with low mutational burden ($<\text{median}$) with CB, 79% had $>1/2$ of mutations in S7 or S11, compared to only 51% of NCB ($p<0.01$; Pearson's chi-squared). Neoantigen burden was strongly correlated with mutational burden, and did not improve ability to predict CB. In examining mutations in specific genes, >500 genes were mutated significantly more frequently in CB or NCB ($p<0.05$, Fisher's exact). Restricting analysis to recurrently mutated genes in cancer and correcting for patient mutational burden by permutation, nonsynonymous mutations in ACSL3 and MET and truncating alterations in ARID2 were significantly enriched in CB.

In this meta-analysis of 220 patients, harmonized clinical and whole exome analysis confirmed that mutational burden correlates with CB from anti-PD1 and anti-CTLA4 therapy, with mutational signatures and alterations in specific genes potentially providing additional predictive power.

Example 4: SU2C Cohort Study for Lung Cancer Immunotherapy

Analyse were also performed for a cohort of patients receiving lung cancer immunotherapy. For these patients with metastatic lung cancer treated with anti-PD1/PD-L1 therapies at the Dana-Farber Cancer Institute, whole exome sequencing was performed from the clinically annotated pre-treatment biopsies, including: 36 "pairs" of samples (pre-treatment tumor+matched germline normal tissue) and 3 "trios" of samples (LUAD-1020: 4 pre-treatment tumors (1 primary+3 metastases); LUAD-1007: 2 pre-treatment tumors; and LUAD-1011: 2 pre-treatment tumors). The baseline clinical characteristics and prior therapies is summarized in Table 6 below.

TABLE 6

Characteristic	Patients (N = 39)
Age (years) - Median (range)	60 (32-83)
Age >75 - No.	3 (7.7)
Male sex - No. (%)	15 (38.5)
Smoking status - No. (%)	
Current	11 (28.2)
Former	17 (43.6)
Never	11 (28.2)
No. of prior systemic regimens - No. (%)	
0	3 (7.7)
1-2	19 (48.7)
3-4	16 (41.0)
5-6	1 (2.6)

The resulting Kaplan-Meier analysis is compared for baseline clinical variables as predictors of PFS for SU2C cohort (N=39) (FIG. 13A-13D). The corresponding quality control processes are summarized in FIG. 14. As for clinical stratification, patients were divided into three groups according to their response to immunotherapy. The definition of "clinical benefit," as for the first group of patients, includes CR or PR by RECIST or SD with PFS >12 months. The definition of "no clinical benefit" includes PD by RECIST with PFS <3 months. The definition of "stable disease" (intermediate clinical benefit) includes SD with PFS <12 months or PD with PFS >3 months. A summary of different responses of 39 SU2C lung cancer patients to immunotherapy is shown in FIG. 15. Their mutational burden and response to immune checkpoint therapies is also compared (FIG. 16, N=31). Another cohort previously reported by Rizvi et al. (2015), supra was similarly analyzed (FIG. 17 and FIG. 18), showing that pre-treatment tumor mutational load was a strong predictor of response to immune checkpoint therapy in anti-PD1/PD-L1-treated lung cancer (FIG. 18). The current cohort of lung cancer patients were tested for any mutations to genes commonly mutated in lung cancers. As shown in FIG. 19, NF1 alterations were more frequent in responders (3/6 clinical benefit, 3/13 stable disease, 0/12 NCB). EGFR hotspot alterations were seen more frequently in nonresponders. KRAS hotspot alterations were observed more frequently in responders (1/6 clinical benefit, 4/13 SD, 1/12 NCB). The following provides additional genetic observations: SU2C-1006: splice site mutation in MET; missense mutation in LTBP1; SU2C-1066: 3 missense mutations in LEPR; SU2C-1068: 2 missense mutations in LEPR; SU2C-1067: missense mutations in STAG2 and SRCAP and an observed EGFR hotspot was L85S. Sample SU2C-1066 may be excluded since its purity=0.36. A summary of significantly mutated genes in these patients is shown in FIG. 20. Patients with hotspot mutations in EGFR uniformly did not respond to immune checkpoint therapy (FIG. 21). SAFB2 indels were likely caused by sequencing artifacts (FIG. 22).

INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned herein are hereby incorporated by reference in their entirety as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated by reference. In case of conflict, the present application, including any definitions herein, will control.

Also incorporated by reference in their entirety are any polynucleotide and polypeptide sequences which reference an accession number correlating to an entry in a public database, such as those maintained by The Institute for Genomic Research (TIGR) on the world wide web and/or the National Center for Biotechnology Information (NCBI) on the world wide web.

EQUIVALENTS

Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the present invention described herein. Such equivalents are intended to be encompassed by the following claims.

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12404557B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A method of treating melanoma, non-small cell lung cancer (NSCLC), or head and neck squamous cell carcinoma (HNSCC) in a subject likely to be responsive to immune checkpoint therapy against PD-1 or CTLA-4, the method comprising:
 - i) selecting the subject, the subject having been identified according to:
 - a) determining the copy number of ARID2 in a sample from the subject having cancer selected from the group consisting of melanoma, non-small cell lung cancer (NSCLC), and head and neck squamous cell carcinoma (HNSCC), wherein the sample comprises nucleic acid molecules from the subject's cancer; and
 - b) comparing said copy number to that of a control sample, wherein an increased copy number of ARID2 encoding a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being more likely to be responsive to the immune checkpoint therapy against PD-1 or CTLA-4; and
 - ii) administering the immune checkpoint therapy against PD-1 or CTLA-4 to the selected subject.
2. The method of claim 1, wherein the control sample is determined from a non-cancerous sample from either the subject or a member of the same species to which the subject belongs.
3. The method of claim 2, wherein the control sample is a non-cancerous sample from the subject obtained from an earlier point in time than the subject sample.
4. The method of claim 3, wherein the control sample is obtained before the subject has received immune checkpoint therapy and the subject sample is obtained after the subject has received immune checkpoint therapy.
5. The method of claim 2, wherein the control sample does not comprise cells.
6. The method of claim 2, wherein the control sample comprises cells.
7. The method of claim 6, wherein the cells are cancer cells known to be non-responsive to the immune checkpoint therapy.
8. The method of claim 1, wherein the subject sample and/or the control sample has not been contacted with a renal cell cancer treatment or an inhibitor of an immune checkpoint.
9. The method of claim 1, wherein the subject has not been administered a renal cell cancer treatment or an inhibitor of an immune checkpoint.
10. The method of claim 1, wherein the subject sample is selected from the group consisting of serum, whole blood, plasma, urine, cells, cell lines, and biopsies.
11. The method of claim 1, further comprising recommending, prescribing, or administering at least one additional anti-cancer therapeutic agent.
12. The method of claim 1, wherein the immune checkpoint therapy comprises at least one antibody selected from the group consisting of anti-PD-1 antibodies, anti-CTLA-4 antibodies, and combinations thereof.
13. The method of claim 12, wherein the immune checkpoint therapy comprises an anti-PD-1 antibody.
14. The method of claim 12, wherein the immune checkpoint therapy comprises an anti-CTLA4 antibody.
15. The method of claim 1, wherein the likelihood of the cancer in the subject being responsive to immune checkpoint therapy is indicated by at least one of cellular proliferation, tumor burden, m-stage, metastasis, progressive disease, clinical benefit rate, survival until mortality, pathological complete response, semi-quantitative measures of pathologic response, clinical complete remission, clinical partial remission, clinical stable disease, recurrence-free survival, metastasis free survival, disease free survival, circulating tumor cell decrease, circulating marker response, and RECIST criteria.
16. The method of claim 1, wherein the subject is a mammal.
17. The method of claim 16, wherein mammal is an animal model of the cancer.
18. The method of claim 16, wherein the mammal is a human.

* * * * *